

Algebraic Jacobian

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Chapter 1

The Jacobian of a smooth proper curve

Throughout, k is a field and C a *smooth curve* over k : a geometrically irreducible, smooth k -scheme of relative dimension one, proper over k . We write $C \rightarrow \operatorname{Spec} k$ for its structure morphism. This chapter states the goal of the project — the genus, the Jacobian, the Abel–Jacobi map, and the universal property that characterizes the Jacobian as the Albanese variety of C . It is the mathematical counterpart of the statement file `AlgebraicJacobian/Challenge.lean`.

Provenance. Each node below should carry a `\source{...}` anchor to the reference it is read from once that reference has been consulted; anchors are added during the rebuild, never guessed.

1.1 Genus

Definition 1 (Genus). The *genus* of a smooth proper curve C over k is

$$g(C) = \dim_k H^1(C, \mathcal{O}_C),$$

the k -dimension of the first cohomology group of the structure sheaf, viewed as a sheaf of k -modules on the small Zariski site of C (Definition ??), with cohomology the Ext-groups from the constant sheaf (Definition ??).

1.2 The Jacobian as an abelian variety

Definition 2 (Jacobian). The *Jacobian* $\operatorname{Jac}(C)$ of a smooth proper curve C over k is the identity component $\operatorname{Pic}_{C/k}^0$ of the relative Picard scheme of C over k : the k -scheme representing the degree-zero subfunctor of the étale-sheafified relative Picard functor.

Proposition 3 (Group scheme structure). $\operatorname{Jac}(C)$ is a commutative group scheme over k , with multiplication induced by the tensor product of line bundles.

Theorem 4 (Smoothness of the Jacobian). $\operatorname{Jac}(C)$ is smooth over k of relative dimension $g(C)$.

Theorem 5 (Properness of the Jacobian). $\operatorname{Jac}(C)$ is proper over k .

Theorem 6 (Geometric irreducibility of the Jacobian). *$\mathrm{Jac}(C)$ is geometrically irreducible over k .*

In particular $\mathrm{Jac}(C)$ is an abelian variety over k of dimension $g(C)$.

1.3 The Abel–Jacobi map and the universal property

Definition 7 (Abel–Jacobi map). For a k -rational point $P : \mathrm{Spec} k \rightarrow C$, the *Abel–Jacobi map* $\varphi_P : C \rightarrow \mathrm{Jac}(C)$ sends a point Q to the class of the line bundle $\mathcal{O}_C(Q - P)$.

Lemma 8 (Base point maps to the origin). *The Abel–Jacobi map φ_P sends the base point P to the neutral element of $\mathrm{Jac}(C)$: $\varphi_P \circ P = 0$.*

Theorem 9 (Universal property of the Jacobian). *$\mathrm{Jac}(C)$ is the Albanese variety of C : for every abelian variety A over k and every morphism $f : C \rightarrow A$ with $f \circ P = 0$, there is a unique homomorphism of group schemes $g : \mathrm{Jac}(C) \rightarrow A$ such that $f = g \circ \varphi_P$. Milne’s Proposition 6.1 proves this under the standing hypothesis $g(C) > 0$, with P a k -rational point. The statement here includes genus zero; that case requires a separate argument and is not supplied by the cited proposition alone.*

Chapter 2

Functoriality and base change of fields

Beyond existence and the universal property, the Jacobian is *canonical*: it is functorial in the curve and commutes with extension of the base field. This chapter states those properties. As before, k is a field and curves are smooth, proper, geometrically irreducible of relative dimension one.

2.1 Functoriality

Definition 10 (Category of curves). Let $\text{Curve}(k)$ denote the category whose objects are smooth proper geometrically irreducible curves over k and whose morphisms are morphisms of k -schemes.

Definition 11 (Jacobian functor). The Jacobian is a contravariant functor

$$\text{Jac} : \text{Curve}(k)^{\text{op}} \longrightarrow \text{Grp}(\text{Over}(\text{Spec } k)),$$

sending a curve to its Jacobian group scheme and a morphism of curves to the induced pullback of line bundles. In particular an isomorphism of curves induces an isomorphism of Jacobian group schemes.

2.2 Base change of fields

Fix a field extension $k \rightarrow L$. Base change along it is the functor $\cdot_L = (\cdot) \times_{\text{Spec } k} \text{Spec } L : \text{Over}(\text{Spec } k) \rightarrow \text{Over}(\text{Spec } L)$, which preserves properness, geometric irreducibility, and smoothness of relative dimension one.

Definition 12 (Base change of schemes). For a field extension $k \rightarrow L$, base change is the functor $X \mapsto X \times_{\text{Spec } k} \text{Spec } L$ from k -schemes to L -schemes.

Theorem 13 (Jacobians commute with base change). *For every field extension $k \rightarrow L$ and every curve C over k there is a canonical isomorphism of group schemes over L*

$$\text{Jac}(C)_L \cong \text{Jac}(C_L).$$

Proposition 14 (Identity coherence). *The base-change isomorphism for the trivial extension $k \rightarrow k$ is the identity, compatibly with the identification $C_k \cong C$.*

Proposition 15 (Cocycle coherence). *For a tower $k \rightarrow L \rightarrow M$ the base-change isomorphisms compose: the isomorphism for $k \rightarrow M$ agrees with base-changing the isomorphism for $k \rightarrow L$ along $L \rightarrow M$ and then applying the isomorphism for $L \rightarrow M$, compatibly with the identification $(C_L)_M \cong C_M$.*

Proposition 16 (Compatibility with the Abel–Jacobi map). *Base change carries the Abel–Jacobi map of (C, P) to the Abel–Jacobi map of (C_L, P_L) : under $\text{Jac}(C)_L \cong \text{Jac}(C_L)$, the base change of φ_P is φ_{P_L} .*

Chapter 3

Curves over a field

Throughout this chapter k is a field. As in Chapter ??, a *smooth curve* over k is a k -scheme which is smooth of relative dimension one, proper, and geometrically irreducible over k . This chapter provides the geometric substrate for the genus: smooth schemes over a field are reduced, so a smooth curve is geometrically integral; a proper geometrically integral scheme over a field has no global functions besides the constants; the projective line carries two affine charts with polynomial section rings and Laurent-polynomial overlap; the section rings of the standard smooth charts of a curve are Dedekind domains, so the local rings of the curve are valuation rings and its specialization order has height one; a rational function transcendental over k therefore spreads out to a locally quasi-finite morphism $C \rightarrow \mathbb{P}^1$; and Zariski's main theorem upgrades it to a finite one.

3.1 Smooth algebras over a domain are reduced

Lemma 17 (Unramified flat algebras over a domain are reduced). *Let B be an integral domain and let S be a commutative B -algebra which is formally unramified, essentially of finite type, and flat over B . Then S is reduced.*

Proof. Let $K = \text{Frac } B$ be the fraction field. The K -algebra $K \otimes_B S$ is again formally unramified and essentially of finite type, both properties being stable under base change, and a formally unramified, essentially of finite type algebra over a field is reduced. It therefore suffices to embed S into $K \otimes_B S$. The canonical map $s \mapsto 1 \otimes s$ is, up to the isomorphism $S \cong B \otimes_B S$, obtained from the injection $B \hookrightarrow K$ by tensoring with S ; since S is flat over B , it is injective. A subring of a reduced ring is reduced. $\square$

Lemma 18 (Étale algebras over a domain are reduced). *An étale algebra over an integral domain is reduced.*

Proof. An étale algebra is formally unramified and of finite presentation, in particular essentially of finite type, and it is flat. Apply ??. $\square$

Lemma 19 (Standard smooth homomorphisms out of a domain). *Let B be an integral domain and let $B \rightarrow S$ be a standard smooth ring homomorphism, that is, one admitting a presentation $S = B[x_1, \dots, x_n]/(f_1, \dots, f_c)$ in which the Jacobian matrix $(\partial f_u / \partial x_v)_{u,v \leq c}$ has invertible determinant in S . Then S is reduced.*

Proof. By the structure theorem for standard smooth algebras, S is étale over a polynomial ring over the base: there are m and a factorization $B \rightarrow B[x_1, \dots, x_m] \rightarrow S$ with the second map étale. A polynomial ring over an integral domain is an integral domain, so ?? applies. $\square$

Lemma 20 (Standard smooth algebras over a domain). *A standard smooth algebra over an integral domain is reduced.*

Proof. The structure homomorphism is a standard smooth ring homomorphism out of a domain; apply ??. $\square$

Theorem 21 (Smooth schemes over a field are reduced). *Let K be a field and let X be a scheme which is smooth over $\text{Spec } K$. Then X is reduced.*

Proof. A smooth morphism is, locally on the source and the target, standard smooth: every point $x \in X$ has an affine open neighbourhood V , mapping into an affine open U of the base, such that the induced ring homomorphism $\Gamma(U) \rightarrow \Gamma(V)$ is standard smooth. Since $\text{Spec } K$ has a single point and U is a nonempty open subset containing the image of x , we have $U = \text{Spec } K$, and $\Gamma(U) \cong K$ is a field, in particular a domain. By ?? the ring $\Gamma(V)$ is reduced, hence the affine scheme V is reduced. The opens V cover X , and a scheme covered by reduced open subschemes is reduced. $\square$

Theorem 22 (Smooth morphisms are geometrically reduced). *A smooth morphism of schemes $f : X \rightarrow Y$ is geometrically reduced: for every field K and every morphism $\text{Spec } K \rightarrow Y$, the base change $X \times_Y \text{Spec } K$ is a reduced scheme.*

Proof. Smoothness is stable under base change, so the projection $X \times_Y \text{Spec } K \rightarrow \text{Spec } K$ is smooth; apply ??. $\square$

Lemma 23 (Smooth of relative dimension one implies smooth). *A morphism of schemes which is smooth of relative dimension one is smooth.*

Proof. A morphism smooth of some relative dimension is smooth. $\square$

Lemma 24 (Smooth of a relative dimension implies geometrically reduced). *For every n , a morphism of schemes which is smooth of relative dimension n is geometrically reduced.*

Proof. Such a morphism is smooth; apply ??. $\square$

Theorem 25 (Smooth and geometrically irreducible implies geometrically integral). *A smooth, geometrically irreducible morphism of schemes is geometrically integral: every base change to the spectrum of a field is an integral scheme.*

Proof. Every base change to the spectrum of a field is reduced (??) and irreducible (geometric irreducibility), and a reduced scheme with irreducible — in particular nonempty — underlying space is integral. $\square$

Theorem 26 (Relative-dimension form). *For every n , a geometrically irreducible morphism of schemes which is smooth of relative dimension n is geometrically integral.*

Proof. Such a morphism is smooth; apply ??. $\square$

3.2 Schemes over a field: the topological substrate

In this section X is a scheme over k , that is, a scheme equipped with a structure morphism $X \rightarrow \operatorname{Spec} k$.

Lemma 27 (Irreducibility). *If X is geometrically irreducible over k , then the underlying topological space of X is irreducible; in particular X is nonempty.*

Proof. Geometric irreducibility applied to the trivial extension $k \rightarrow k$ says that $X \times_{\operatorname{Spec} k} \operatorname{Spec} k \cong X$ is irreducible. $\square$

Lemma 28 (Quasi-compactness). *If X is quasi-compact over $\operatorname{Spec} k$ (for instance proper), then the underlying topological space of X is quasi-compact.*

Proof. $\operatorname{Spec} k$ is a one-point space, hence quasi-compact, and X is the preimage of it under a quasi-compact morphism. $\square$

Lemma 29 (Quasi-separatedness). *If X is quasi-separated over $\operatorname{Spec} k$ (for instance proper), then the underlying topological space of X is quasi-separated: the intersection of any two quasi-compact opens is quasi-compact.*

Proof. $\operatorname{Spec} k$ is affine, hence a quasi-separated scheme, and a scheme quasi-separated over a quasi-separated scheme is quasi-separated. $\square$

Proposition 30 (Integrality). *If X is geometrically reduced and geometrically irreducible over k , then X is an integral scheme.*

Proof. Applying geometric reducedness to the trivial extension $k \rightarrow k$ shows that X is reduced, and X is irreducible by ???. A reduced scheme with irreducible underlying space is integral. $\square$

3.3 Global sections of a proper geometrically integral scheme

Lemma 31 (Pushout of a field extension along itself). *Let $\varphi : A \rightarrow B$ and $\psi : A \rightarrow C$ be homomorphisms of commutative rings with B nontrivial, and let D , with coprojections $\iota_B : B \rightarrow D$ and $\iota_C : C \rightarrow D$, be a pushout of (φ, ψ) (for instance $D = B \otimes_A C$). Assume that D is a field and that there is a ring isomorphism $e : C \cong B$ with $e \circ \psi = \varphi$. Then φ is an epimorphism in the category of commutative rings.*

Proof. Let $\mu : D \rightarrow B$ be the homomorphism with $\mu \circ \iota_B = \operatorname{id}_B$ and $\mu \circ \iota_C = e$, given by the universal property of the pushout (the two maps agree on A because $e \circ \psi = \varphi$). Since D is a field and B is nontrivial, μ is injective; it is surjective because $\mu \circ \iota_B = \operatorname{id}_B$. Hence μ is an isomorphism, and from $\mu \circ \iota_B = \operatorname{id}_B = \mu \circ (\iota_C \circ e^{-1})$ we conclude $\iota_B = \iota_C \circ e^{-1}$.

Now let $u, v : B \rightarrow W$ satisfy $u \circ \varphi = v \circ \varphi$. Then $u \circ \varphi = (v \circ e) \circ \psi$, so there is $w : D \rightarrow W$ with $w \circ \iota_B = u$ and $w \circ \iota_C = v \circ e$. Therefore

$$u = w \circ \iota_B = w \circ \iota_C \circ e^{-1} = v \circ e \circ e^{-1} = v,$$

which proves that φ is an epimorphism. $\square$

Lemma 32 (Global sections form a field). *Let R be a commutative ring which is a field, let X be an integral scheme, and let $f : X \rightarrow \operatorname{Spec} R$ be universally closed. Then $\Gamma(X, \mathcal{O}_X)$ is a field.*

Proof. $\Gamma(X, \mathcal{O}_X)$ is an integral domain because X is integral. Since f is universally closed, the induced ring homomorphism $R \cong \Gamma(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R}) \rightarrow \Gamma(X, \mathcal{O}_X)$ is an integral extension. A domain which is integral over a field is a field. $\square$

Theorem 33 (Constant functions on a proper geometrically integral scheme). *Let K be a field and let X be a scheme, proper and geometrically integral over K . Then the ring homomorphism $\Gamma(\operatorname{Spec} K, \mathcal{O}_{\operatorname{Spec} K}) \rightarrow \Gamma(X, \mathcal{O}_X)$ induced by the structure morphism is bijective; identifying $\Gamma(\operatorname{Spec} K, \mathcal{O}_{\operatorname{Spec} K}) = K$, this says $\Gamma(X, \mathcal{O}_X) = K$.*

Proof. Write $A = \Gamma(X, \mathcal{O}_X)$ and let $\varphi : K \rightarrow A$ be the map in question. Applying geometric integrality to the trivial extension $K \rightarrow K$ shows that X is integral; in particular X is nonempty and A is a nontrivial domain, so φ , a ring homomorphism out of a field into a nontrivial ring, is injective. By ?? (properness includes universal closedness), A is a field, and since f is proper, A is moreover a finite K -module. As a finite ring homomorphism is surjective as soon as it is an epimorphism of commutative rings, it remains to prove that φ is an epimorphism.

Let $q : \operatorname{Spec} A \rightarrow \operatorname{Spec} K$ be the morphism induced by φ and form the base change $Y = X \times_{\operatorname{Spec} K} \operatorname{Spec} A$. Since A is a field, geometric integrality of f makes Y integral, and the projection $Y \rightarrow \operatorname{Spec} A$ is universally closed as a base change of f ; hence $D := \Gamma(Y, \mathcal{O}_Y)$ is a field by ?. On the other hand, X is quasi-compact and quasi-separated as a topological space (??, ??), and q is flat because every vector space over K is flat; flat base change for global sections therefore exhibits the square of section rings

$$\begin{array}{ccc} K & \longrightarrow & A \\ \downarrow & & \downarrow \\ A & \longrightarrow & D \end{array}$$

(left column φ , top row φ transported through the canonical isomorphism $\Gamma(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}) \cong A$) as a pushout square of commutative rings, $D \cong A \otimes_K A$. Its pushout D is a field and its two legs are isomorphic copies of φ , so φ is an epimorphism by ?. $\square$

Corollary 34 (Isomorphism form). *In the situation of ??, the induced map $\Gamma(\operatorname{Spec} K, \mathcal{O}_{\operatorname{Spec} K}) \rightarrow \Gamma(X, \mathcal{O}_X)$ is an isomorphism of rings.*

Proof. A bijective ring homomorphism is an isomorphism. $\square$

Corollary 35 (Global sections of the curve). *Let C be a scheme over k which is proper, geometrically irreducible and geometrically reduced over k . Then the structure morphism induces an isomorphism $\Gamma(\operatorname{Spec} k, \mathcal{O}_{\operatorname{Spec} k}) \cong \Gamma(C, \mathcal{O}_C)$; that is, $\Gamma(C, \mathcal{O}_C) = k$.*

Proof. Every base change of C to the spectrum of a field is reduced and irreducible, hence integral; so C is geometrically integral over k and ?? applies. $\square$

Remark 36 (The smooth curve). Let C be a smooth curve over k , i.e. proper, smooth of relative dimension one and geometrically irreducible over k , as in Chapter ?. Combining the above: C is smooth over k (?), geometrically reduced (??) and geometrically integral (??); its underlying space is irreducible (??), quasi-compact (??) and quasi-separated (??); C is an integral scheme (??); and $\Gamma(C, \mathcal{O}_C) = k$ (??).

3.4 The projective line and its charts

Let $A = k[X_0, X_1]$ be the polynomial ring in two variables, graded by total degree. For a homogeneous element $f \in A$ we write $A_{(f)}$ for the degree-zero part of the localization A_f ; its elements are fractions a/f^n with a homogeneous of degree $n \deg f$.

Definition 37 (The projective line). The projective line over k is $\mathbb{P}^1 = \text{Proj } A$, the Proj of the graded ring $A = k[X_0, X_1]$.

Definition 38 (Degree zero). The inclusion of constants $k \rightarrow A_0$ into the degree-zero part of A is an isomorphism of k -algebras.

Proof. It is injective because a nonzero constant polynomial is nonzero, and surjective because a polynomial which is homogeneous of degree zero is constant. $\square$

Definition 39 (Structure morphism). The structure morphism $\mathbb{P}^1 \rightarrow \text{Spec } k$ is the canonical morphism $\text{Proj } A \rightarrow \text{Spec } A_0$ followed by the isomorphism $\text{Spec } A_0 \cong \text{Spec } k$ induced by ??.

Theorem 40 (Properness of the projective line). $\mathbb{P}^1$ is proper over k .

Proof. A is a finitely generated algebra over its degree-zero part $A_0 = k$ (generated by X_0 and X_1), and the Proj of a graded ring which is finitely generated over its degree-zero part is proper over the spectrum of the degree-zero part. Composing with the isomorphism $\text{Spec } A_0 \cong \text{Spec } k$ preserves properness. $\square$

Definition 41 (The projective line as a scheme over k). $\mathbb{P}^1$, regarded as a scheme over k via the structure morphism of ??.

Definition 42 (The standard charts). For $i \in \{0, 1\}$, the chart $U_i = D_+(X_i) \subseteq \mathbb{P}^1$ is the basic open locus where the homogeneous coordinate X_i does not vanish.

Lemma 43 (The charts are affine). Each chart U_i is an affine open of $\mathbb{P}^1$.

Proof. X_i is homogeneous of degree $1 > 0$, and the basic open of Proj at a homogeneous element of positive degree is affine. $\square$

Lemma 44 (The charts cover). $U_0 \cup U_1 = \mathbb{P}^1$.

Proof. A point of $\text{Proj } A$ is a homogeneous prime ideal $\mathfrak{p}$ of A not containing the irrelevant ideal A_+ . Every monomial of positive degree is divisible by X_0 or by X_1 , so if both $X_0 \in \mathfrak{p}$ and $X_1 \in \mathfrak{p}$ then $A_+ \subseteq \mathfrak{p}$, a contradiction. Hence every point avoids some X_i , i.e. lies in $U_0 \cup U_1$. $\square$

Lemma 45 (The chart overlap). $U_0 \cap U_1 = D_+(X_0 X_1)$.

Proof. A homogeneous prime avoids the product $X_0 X_1$ if and only if it avoids both factors. $\square$

Definition 46 (Affine models of the charts). For $i \in \{0, 1\}$, the canonical morphism $\text{Spec } A_{(X_i)} \rightarrow \mathbb{P}^1$ associated with the basic open at the degree-one element X_i ; it is an open immersion.

Lemma 47 (Image of the affine model). The image of $\text{Spec } A_{(X_i)} \rightarrow \mathbb{P}^1$ is the chart U_i ; thus $U_i \cong \text{Spec } A_{(X_i)}$.

Proof. This is the standard description of the basic opens of Proj: the homogeneous primes avoiding X_i correspond to the primes of the degree-zero localization $A_{(X_i)}$. $\square$

Lemma 48 (The affine models lie over k). The composite $\text{Spec } A_{(X_i)} \rightarrow \mathbb{P}^1 \rightarrow \text{Spec } k$ is the morphism induced by the k -algebra structure of $A_{(X_i)}$; that is, the affine models are morphisms of schemes over k .

Proof. The canonical morphism $\text{Proj } A \rightarrow \text{Spec } A_0$, restricted to the affine chart, is induced by the ring homomorphism $A_0 \rightarrow A_{(X_i)}$; composing with the isomorphism $k \cong A_0$ of ?? gives the claim. $\square$

Definition 49 (Chart coordinates). For $i, j \in \{0, 1\}$, the chart coordinate is the degree-zero fraction $t_{ij} = X_j/X_i \in A_{(X_i)}$.

Definition 50 (Dehomogenization). For $i \in \{0, 1\}$, the dehomogenization at X_i is the k -algebra homomorphism $\delta_i : k[X_0, X_1] \rightarrow k[t]$ with $\delta_i(X_i) = 1$ and $\delta_i(X_j) = t$ for $j \neq i$.

Definition 51 (From polynomials to the chart ring). For $i, j \in \{0, 1\}$, the k -algebra homomorphism $k[t] \rightarrow A_{(X_i)}$ given by evaluation at the chart coordinate, $t \mapsto t_{ij}$.

Definition 52 (From the chart ring to polynomials). For $i \in \{0, 1\}$, the k -algebra homomorphism $A_{(X_i)} \rightarrow k[t]$ sending a degree-zero fraction a/X_i^n (with a homogeneous of degree n) to $\delta_i(a)$.

Proof. Well-definedness: $\delta_i(X_i) = 1$ is a unit of $k[t]$, so δ_i extends to the localization $A_{X_i} \rightarrow k[t]$ by the universal property of localization; on the degree-zero subring its value on a/X_i^n is $\delta_i(a) \delta_i(X_i)^{-n} = \delta_i(a)$. $\square$

Lemma 53 (One composite is the identity). For $i \neq j$, the composite $k[t] \rightarrow A_{(X_i)} \rightarrow k[t]$ of ?? followed by ?? is the identity.

Proof. Both sides are k -algebra homomorphisms on $k[t]$, so it suffices to check the generator: $t \mapsto t_{ij} = X_j/X_i \mapsto \delta_i(X_j) = t$. $\square$

Lemma 54 (Monomial fractions are powers of the coordinate). Let $i \neq j$ and let $e_0, e_1 \geq 0$ with $e_0 + e_1 = a$. Then in $A_{(X_i)}$

$$\frac{X_0^{e_0} X_1^{e_1}}{X_i^a} = t_{ij}^{e_j}.$$

Proof. Since $X_0^{e_0} X_1^{e_1} = X_i^{e_i} X_j^{e_j}$ and $a = e_i + e_j$, the left-hand side is $X_i^{e_i} X_j^{e_j} / X_i^{e_i + e_j} = X_j^{e_j} / X_i^{e_j}$, which is $t_{ij}^{e_j}$. $\square$

Lemma 55 (Surjectivity onto the chart ring). For $i \neq j$, the homomorphism $k[t] \rightarrow A_{(X_i)}$ of ?? is surjective.

Proof. Since A is generated over A_0 by the degree-one elements X_0, X_1 , the degree-zero localization $A_{(X_i)}$ is spanned, as a module over $A_0 = k$ (??), by the fractions m/X_i^a where m ranges over the monomials $X_0^{e_0} X_1^{e_1}$ of degree a . By ?? each such fraction equals $t_{ij}^{e_j}$, which is the image of t^{e_j} . The image of a ring homomorphism is closed under addition and under multiplication by scalars from k , so it is all of $A_{(X_i)}$. $\square$

Lemma 56 (The other composite is the identity). For $i \neq j$, the composite $A_{(X_i)} \rightarrow k[t] \rightarrow A_{(X_i)}$ of ?? followed by ?? is the identity.

Proof. By ?? every element of $A_{(X_i)}$ is of the form $p(t_{ij})$ with $p \in k[t]$, and on such elements the claim follows from ??. $\square$

Definition 57 (The chart identification). For $i \neq j$, the isomorphism of k -algebras $\alpha_i : A_{(X_i)} \cong k[t]$ given by the mutually inverse homomorphisms of ?? and ??; the chart coordinate t_{ij} corresponds to t .

Proof. The two composites are the identity by ?? and ??. $\square$

3.5 The chart overlap and Laurent polynomials

Write $k[T, T^{-1}]$ for the ring of Laurent polynomials over k .

Lemma 58 (Laurent span). *Every Laurent polynomial $f \in R[T, T^{-1}]$ over a commutative (semi)ring R can be written as $f = p(T) + q(T^{-1})$ with $p, q \in R[t]$ ordinary polynomials.*

Proof. Split f into its monomials and collect those with nonnegative exponent into p and those with negative exponent into q . $\square$

Definition 59 (Restriction from the left chart). The canonical ring homomorphism $\lambda : A_{(X_0)} \rightarrow A_{(X_0 X_1)}$, sending a/X_0^n to $aX_1^n/(X_0 X_1)^n$.

Definition 60 (Restriction from the right chart). The canonical ring homomorphism $\rho : A_{(X_1)} \rightarrow A_{(X_0 X_1)}$, sending a/X_1^n to $aX_0^n/(X_0 X_1)^n$.

Lemma 61 (The two coordinates are mutually inverse on the overlap). *In $A_{(X_0 X_1)}$ one has $\lambda(t_{01}) \cdot \rho(t_{10}) = 1$.*

Proof. Computing with the formulas of ?? and ??: $\lambda(t_{01}) = X_1^2/(X_0 X_1)$ and $\rho(t_{10}) = X_0^2/(X_0 X_1)$, so

$$\lambda(t_{01}) \cdot \rho(t_{10}) = \frac{X_0^2 X_1^2}{(X_0 X_1)^2} = 1.$$

$\square$

Lemma 62 (The overlap ring is a localization of the chart ring). *The homomorphism λ of ?? identifies $A_{(X_0 X_1)}$ with the localization of $A_{(X_0)}$ at the powers of the chart coordinate t_{01} .*

Proof. This is a general property of degree-zero localizations: for homogeneous elements f, g of degree one, $A_{(fg)}$ is the localization of $A_{(f)}$ away from the degree-zero fraction g/f . Here $f = X_0$, $g = X_1$ and $g/f = t_{01}$. $\square$

Definition 63 (The overlap identification). The isomorphism of k -algebras $\theta : A_{(X_0 X_1)} \cong k[T, T^{-1}]$, with T the image of the coordinate t_{01} .

Proof. By ??, $A_{(X_0 X_1)}$ is the localization of $A_{(X_0)}$ at the powers of t_{01} . The isomorphism $\alpha_0 : A_{(X_0)} \cong k[t]$ of ?? carries t_{01} to t , hence induces an isomorphism of the two localizations $A_{(X_0 X_1)} \cong k[t]_t = k[T, T^{-1}]$ (localizations at corresponding multiplicative sets are canonically isomorphic). The map is k -linear because α_0 is. $\square$

Lemma 64 (Left restriction is $t \mapsto T$). *For every $z \in A_{(X_0)}$ one has $\theta(\lambda(z)) = p(T)$, where $p = \alpha_0(z) \in k[t]$. In other words, under the identifications α_0 and θ , the restriction λ is the canonical inclusion $k[t] \rightarrow k[T, T^{-1}]$, $t \mapsto T$.*

Proof. By construction (??), θ is the comparison of localizations induced by α_0 ; under such a comparison the localization map λ of ?? corresponds to the localization map $k[t] \rightarrow k[T, T^{-1}]$. $\square$

Lemma 65 (Right restriction is $t \mapsto T^{-1}$). *For every $z \in A_{(X_1)}$ one has $\theta(\rho(z)) = q(T^{-1})$, where $q = \alpha_1(z) \in k[t]$. In other words, under the identifications α_1 and θ , the restriction ρ is the homomorphism $k[t] \rightarrow k[T, T^{-1}]$, $t \mapsto T^{-1}$.*

Proof. By ?? the image of $\lambda(t_{01})$ under θ is T , and by ?? $\lambda(t_{01})\rho(t_{10}) = 1$; applying θ gives $\theta(\rho(t_{10})) = T^{-1}$. By ?? (for the indices $(1, 0)$) every $z \in A_{(X_1)}$ is a polynomial in t_{10} with coefficients in k ; both sides of the claim are k -algebra homomorphisms in z , and they agree on t_{10} . $\square$

Lemma 66 (Laurent span on the overlap ring). *Every $w \in A_{(X_0X_1)}$ can be written as $w = \lambda(z_0) + \rho(z_1)$ with $z_0 \in A_{(X_0)}$ and $z_1 \in A_{(X_1)}$.*

Proof. Write $\theta(w) = p(T) + q(T^{-1})$ by ?? and set $z_0 = \alpha_0^{-1}(p)$, $z_1 = \alpha_1^{-1}(q)$. By ?? and ??, $\theta(\lambda(z_0) + \rho(z_1)) = p(T) + q(T^{-1}) = \theta(w)$, and θ is injective. $\square$

3.6 Section rings and the Laurent chart pair

Lemma 67 (The overlap is affine). *The chart overlap $U_0 \cap U_1 = D_+(X_0X_1)$ is an affine open of $\mathbb{P}^1$.*

Proof. X_0X_1 is homogeneous of degree $2 > 0$, and the basic open of Proj at a homogeneous element of positive degree is affine. $\square$

Definition 68 (Sections on the left chart). The ring isomorphism $\gamma_0 : \Gamma(\mathbb{P}^1, U_0) \cong k[t]$: the canonical identification $\Gamma(\mathbb{P}^1, D_+(X_0)) \cong A_{(X_0)}$ (sections of Proj over a basic open; equivalently, $U_0 = \text{Spec } A_{(X_0)}$ via ??) followed by α_0 of ??.

Definition 69 (Sections on the right chart). The ring isomorphism $\gamma_1 : \Gamma(\mathbb{P}^1, U_1) \cong k[t]$, defined as in ?? with the identification $\Gamma(\mathbb{P}^1, D_+(X_1)) \cong A_{(X_1)}$ followed by α_1 .

Definition 70 (Sections on the overlap). The ring isomorphism $\gamma_{01} : \Gamma(\mathbb{P}^1, U_0 \cap U_1) \cong k[T, T^{-1}]$: the canonical identification $\Gamma(\mathbb{P}^1, D_+(X_0X_1)) \cong A_{(X_0X_1)}$ followed by θ of ??.

Lemma 71 (Restriction of sections from the left chart). *For every section $a \in \Gamma(\mathbb{P}^1, U_0)$,*

$$\gamma_{01}(a|_{U_0 \cap U_1}) = p(T), \quad p = \gamma_0(a);$$

under the identifications, restriction from the left chart to the overlap is $t \mapsto T$.

Proof. The identifications $\Gamma(\mathbb{P}^1, D_+(f)) \cong A_{(f)}$ are natural with respect to inclusions of basic opens: restriction of sections from $D_+(X_0)$ to $D_+(X_0X_1)$ corresponds to the ring homomorphism λ . Conclude by ??. $\square$

Lemma 72 (Restriction of sections from the right chart). *For every section $b \in \Gamma(\mathbb{P}^1, U_1)$,*

$$\gamma_{01}(b|_{U_0 \cap U_1}) = q(T^{-1}), \quad q = \gamma_1(b);$$

under the identifications, restriction from the right chart to the overlap is $t \mapsto T^{-1}$.

Proof. As in ??, restriction of sections corresponds to ρ ; conclude by ??. $\square$

Definition 73 (Laurent chart pair). A *Laurent chart pair* on $\mathbb{P}^1$ consists of open subsets $V_0, V_1, V_{01} \subseteq \mathbb{P}^1$ with $V_{01} \subseteq V_0$, $V_{01} \subseteq V_1$ and $V_0 \cap V_1 = V_{01}$, all three affine, covering the projective line ($V_0 \cup V_1 = \mathbb{P}^1$), together with ring isomorphisms

$$\Gamma_0 : \Gamma(\mathbb{P}^1, V_0) \cong k[t], \quad \Gamma_1 : \Gamma(\mathbb{P}^1, V_1) \cong k[t], \quad \Gamma_{01} : \Gamma(\mathbb{P}^1, V_{01}) \cong k[T, T^{-1}],$$

such that, under these identifications, restriction $V_{01} \subseteq V_0$ is $t \mapsto T$ and restriction $V_{01} \subseteq V_1$ is $t \mapsto T^{-1}$.

Theorem 74 (Laurent span for a chart pair). *Let $(V_0, V_1, V_{01}, \Gamma_0, \Gamma_1, \Gamma_{01})$ be a Laurent chart pair on $\mathbb{P}^1$. Every section $z \in \Gamma(\mathbb{P}^1, V_{01})$ is the sum of a restriction from V_0 and a restriction from V_1 : there are $a \in \Gamma(\mathbb{P}^1, V_0)$ and $b \in \Gamma(\mathbb{P}^1, V_1)$ with $z = a|_{V_{01}} + b|_{V_{01}}$.*

Proof. Write $\Gamma_{01}(z) = p(T) + q(T^{-1})$ by ?? and set $a = \Gamma_0^{-1}(p)$, $b = \Gamma_1^{-1}(q)$. By the two restriction conditions, $\Gamma_{01}(a|_{V_{01}} + b|_{V_{01}}) = p(T) + q(T^{-1}) = \Gamma_{01}(z)$, and Γ_{01} is injective. $\square$

Definition 75 (The canonical Laurent chart pair). The Laurent chart pair on $\mathbb{P}^1$ given by the standard charts: $V_0 = U_0$, $V_1 = U_1$, $V_{01} = D_+(X_0 X_1)$, with the identifications $\gamma_0, \gamma_1, \gamma_{01}$ of ??-??.

Proof. The overlap conditions and affineness are ??, ?? and ??; the covering condition is ??; the two restriction conditions are ?? and ??. $\square$

3.7 Unramified finite-type algebras over Dedekind domains

The results of this section are pure commutative algebra, stated in full generality. The first theorem strengthens the classical statement that a domain which is unramified and *module-finite* over a Dedekind domain is Dedekind: module-finiteness is relaxed to finite type. The point of the generalization is where incomparability of primes comes from — for module-finite algebras it follows from integrality (lying over/incomparability), while here it follows from quasi-finiteness, which every unramified algebra essentially of finite type enjoys.

Theorem 76 (Unramified finite-type domains over a Dedekind domain, DVR form). *Let A be a Dedekind domain and B an integral domain which is an A -algebra of finite type and formally unramified over A . Then B is Noetherian, and the localization of B at every nonzero prime ideal is a discrete valuation ring.*

Proof. B is Noetherian as a finite-type algebra over the Noetherian ring A (Hilbert basis theorem). Let $\mathfrak{q}$ be a nonzero prime of B and set $\mathfrak{p} = \mathfrak{q} \cap A$. The localization $B_{\mathfrak{q}}$ is a Noetherian local domain and not a field (its maximal ideal is nonzero because $\mathfrak{q}$ is), so it suffices to show that its maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$ is principal.

Step 1: $\mathfrak{p} \neq 0$. An algebra which is formally unramified and of finite type is quasi-finite, and over a quasi-finite algebra two comparable primes of B lying over the same prime of A coincide. If $\mathfrak{p}$ were zero, the comparable primes $0 \subseteq \mathfrak{q}$ of B would both contract to 0 in A , forcing $\mathfrak{q} = 0$, a contradiction. Since A is Dedekind, the nonzero prime $\mathfrak{p}$ is maximal, and $A_{\mathfrak{p}}$ is a discrete valuation ring; in particular $\mathfrak{p}A_{\mathfrak{p}}$ is principal.

Step 2: $\mathfrak{q}$ is minimal over $\mathfrak{p}B$. The fiber ring $B/\mathfrak{p}B$ is an algebra over the residue field $A/\mathfrak{p}$ which is again formally unramified (base change); every module over a field is free, and an unramified algebra which is free as a module is module-finite. Hence $B/\mathfrak{p}B$ is a finite module over the field $A/\mathfrak{p}$, in particular an Artinian ring, so all of its primes are minimal; applying this to the image of $\mathfrak{q}$ shows that $\mathfrak{q}$ is a minimal prime over $\mathfrak{p}B$.

Step 3: $\mathfrak{q}B_{\mathfrak{q}} = \mathfrak{p}B_{\mathfrak{q}}$. Since $\mathfrak{p}$ is maximal and B is unramified over A , the extension $\mathfrak{p}B$ is a radical ideal of B (the fiber ring $B/\mathfrak{p}B$ is reduced, being unramified and essentially of finite type over a field). Localization commutes with radicals, so $\mathfrak{p}B_{\mathfrak{q}}$ is a radical ideal of $B_{\mathfrak{q}}$; and because $\mathfrak{q}$ is minimal over $\mathfrak{p}B$ (Step 2), the radical of $\mathfrak{p}B_{\mathfrak{q}}$ is exactly the maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$. A radical ideal equals its own radical, so $\mathfrak{p}B_{\mathfrak{q}} = \mathfrak{q}B_{\mathfrak{q}}$.

Step 4: conclusion. By Step 3 and Step 1, the maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$ is the extension of the principal ideal $\mathfrak{p}A_{\mathfrak{p}} \subseteq A_{\mathfrak{p}}$ along $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$, hence principal. A Noetherian local domain which is not a field and whose maximal ideal is principal is a discrete valuation ring. $\square$

Corollary 77 (Unramified finite-type domains over a Dedekind domain). *Let A be a Dedekind domain and B an integral domain which is an A -algebra of finite type and formally unramified over A . Then B is a Dedekind domain.*

Proof. A Noetherian domain whose localizations at all nonzero primes are discrete valuation rings is a Dedekind domain; apply ??.

Lemma 78 (The polynomial ring in one variable is a principal ideal ring). *Let k be a field. The polynomial ring over k on a one-element set of variables is a principal ideal ring.*

Proof. It is isomorphic as a ring to the ordinary one-variable polynomial ring $k[t]$, which is a principal ideal domain (Euclidean division), and being a principal ideal ring transports along a surjective ring homomorphism.

Lemma 79 (The polynomial ring in one variable is Dedekind). *Let k be a field. The polynomial ring over k on a one-element set of variables is a Dedekind domain.*

Proof. A principal ideal domain is a Dedekind domain.

Theorem 80 (Structure theorem for standard smooth algebras). *Provided by Mathlib. Let R be a commutative ring and S an R -algebra which is standard smooth of relative dimension n : it admits a presentation $S = R[x_1, \dots, x_m]/(f_1, \dots, f_c)$ with $m - c = n$ in which the Jacobian matrix $(\partial f_u / \partial x_v)_{u,v \leq c}$ has invertible determinant in S . Then the structure map factors as*

$$R \longrightarrow R[y_1, \dots, y_n] \longrightarrow S$$

with the second homomorphism étale. (For such a presentation the module of Kähler differentials $\Omega_{S/R}$ is free on the images of $dx_{c+1}, \dots, dx_m$; consequently S is unramified, hence étale, over the polynomial subring on those last n coordinates.)

Theorem 81 (Standard smooth domains of relative dimension one are Dedekind). *Let k be a field and B an integral domain which is a standard smooth k -algebra of relative dimension 1. Then B is a Dedekind domain.*

Proof. By the structure theorem ??, B is étale over a polynomial ring over k in one variable, which is a Dedekind domain (?). An étale algebra is formally unramified and of finite type, so ?? applies.

Theorem 82 (Standard smooth algebras of relative dimension one contain a transcendental). *Let k be a field and B a nontrivial standard smooth k -algebra of relative dimension 1. Then B contains an element transcendental over k .*

Proof. By the structure theorem ??, there is an étale homomorphism $g : k[y] \rightarrow B$ from a one-variable polynomial ring; we claim that the étale coordinate $t = g(y)$ is transcendental over k . Note that no dimension theory enters: the argument is pure torsion-freeness.

First, g is injective. An étale algebra is flat, and a flat module over a domain is torsion-free; so B is a torsion-free $k[y]$ -module. If $g(P) = g(Q)$ for $P \neq Q$, then the nonzero element $P - Q$ of the domain $k[y]$ would satisfy $(P - Q) \cdot 1_B = g(P - Q) = 0$ with $1_B \neq 0$ (B is nontrivial), contradicting torsion-freeness.

Now suppose $P(t) = 0$ for some nonzero $P \in k[T]$. Evaluating P at $t = g(y)$ is the image under g of the evaluation of P at y , so $g(P(y)) = 0$; by injectivity $P(y) = 0$ in $k[y]$, contradicting the transcendence of the variable y over k . Hence t is transcendental over k .

3.8 Stalks, specializations and finiteness on curve-like schemes

A scheme covered by affine opens whose section rings are Dedekind domains behaves like a smooth curve: its stalks are valuation rings, and its specialization order has height one. This section proves the pointwise statements from the single hypothesis “the point lies in an affine open with Dedekind section ring”, derives the finiteness of proper closed subsets, and discharges the hypotheses for a scheme smooth of relative dimension one over a field.

Theorem 83 (Stalks over Dedekind section rings are valuation rings). *Let X be an integral scheme, $V \subseteq X$ an affine open whose ring of sections $\Gamma(X, V)$ is a Dedekind domain, and $x \in V$. Then the stalk $\mathcal{O}_{X,x}$ is a valuation ring.*

Proof. The stalk at x is the localization of $\Gamma(X, V)$ at the prime ideal $\mathfrak{q}$ corresponding to x under $V \cong \text{Spec } \Gamma(X, V)$. If $\mathfrak{q} = 0$, its complement is the set of nonzero elements of the domain $\Gamma(X, V)$, so the stalk is the fraction field — a field, and every field is a valuation ring. If $\mathfrak{q} \neq 0$, the localization of a Dedekind domain at a nonzero prime is a discrete valuation ring, and every discrete valuation ring is a valuation ring. $\square$

Theorem 84 (Height-one specialization over Dedekind section rings). *Let X be a scheme whose underlying space is irreducible, $V \subseteq X$ an affine open with $\Gamma(X, V)$ a Dedekind domain, and $x \in V$. Then every generalization of x is the generic point of X or x itself.*

Proof. Let y be a generalization of x . Open sets are stable under generalization, so $y \in V$. Under the open immersion $\text{Spec } \Gamma(X, V) \cong V \subseteq X$ the points x, y correspond to primes $\mathfrak{q}_x, \mathfrak{q}_y$ of $\Gamma(X, V)$ with $\mathfrak{q}_y \subseteq \mathfrak{q}_x$ (specialization order of the spectrum). If $\mathfrak{q}_y = 0$, then y is the image of the generic point of $\text{Spec } \Gamma(X, V)$, and the generic point of a nonempty open subscheme of an irreducible scheme is the generic point of the whole scheme. If $\mathfrak{q}_y \neq 0$, then $\mathfrak{q}_y$ is maximal (nonzero primes of a Dedekind domain are maximal), so $\mathfrak{q}_y = \mathfrak{q}_x$ and $y = x$. $\square$

Lemma 85 (Non-generic points are closed). *Let X be a scheme whose underlying space is irreducible and whose specialization order has height one: every generalization of any point is the generic point or the point itself. Then every point other than the generic point is closed.*

Proof. Let $x \neq \xi$ (ξ the generic point) and let y be a point of the closure of $\{x\}$, i.e. x is a generalization of y . By hypothesis $x = \xi$ or $x = y$; the first is excluded, so $y = x$. Hence the closure of $\{x\}$ is $\{x\}$. $\square$

Theorem 86 (Finiteness of proper closed subsets). *Let X be a scheme whose underlying space is irreducible and Noetherian and whose specialization order has height one. Then every closed subset $Z \subseteq X$ not containing the generic point is finite.*

Proof. A closed subset of a Noetherian space is a finite union of irreducible closed subsets. Schemes are sober, so each irreducible closed subset T in such a decomposition of Z is the closure of a unique point $z \in T \subseteq Z$. Since the generic point does not lie in Z , z is not the generic point, so $\{z\}$ is closed by ?? and $T = \overline{\{z\}} = \{z\}$. Hence Z is a finite union of singletons. $\square$

Theorem 87 (Dedekind charts on a curve). *Let k be a field and X an integral scheme, smooth of relative dimension one over $\text{Spec } k$. Then every point of X lies in an affine open V whose ring of sections $\Gamma(X, V)$ is a Dedekind domain.*

Proof. By definition of smoothness of relative dimension one, a point x has an affine open neighbourhood V , mapping into an affine open U of $\text{Spec } k$, such that $\Gamma(U) \rightarrow \Gamma(X, V)$ is standard smooth of relative dimension 1. The base $\text{Spec } k$ has a single point, and U is nonempty (it receives

the image of x), so $U = \operatorname{Spec} k$ and $\Gamma(U) \cong k$. The ring $\Gamma(X, V)$ is an integral domain, since V is a nonempty open of the integral scheme X . Apply ?? $\square$

Theorem 88 (The stalks of a curve are valuation rings). *Let k be a field and X an integral scheme, smooth of relative dimension one over $\operatorname{Spec} k$. Then every stalk of the structure sheaf of X is a valuation ring.*

Proof. Combine ?? with ??. $\square$

Theorem 89 (Height-one specialization on a curve). *Let k be a field and X an integral scheme, smooth of relative dimension one over $\operatorname{Spec} k$. Then every generalization of a point of X is the generic point or the point itself.*

Proof. Combine ?? with ??; the underlying space of an integral scheme is irreducible. $\square$

Theorem 90 (Existence of a transcendental rational function). *Let k be a field and X an integral scheme, smooth of relative dimension one over $\operatorname{Spec} k$. Regard the function field $K(X)$ — the stalk of the structure sheaf at the generic point — as a ring under k via the structure morphism (global sections followed by the germ at the generic point). Then $K(X)$ contains an element f_0 transcendental over k : no nonzero polynomial with coefficients in k vanishes at f_0 .*

Proof. Apply the definition of smoothness of relative dimension one at the generic point ξ : there is an affine open $V \ni \xi$ such that $k \rightarrow \Gamma(X, V)$ is standard smooth of relative dimension 1 (the base affine open is all of $\operatorname{Spec} k$, as in the proof of ??). The ring $\Gamma(X, V)$ is nontrivial (V is nonempty), so by ?? it contains an element t transcendental over k ; let f_0 be the germ of t at ξ .

On an integral scheme, the germ map $\Gamma(X, V) \rightarrow K(X)$ at the generic point is injective. Let $P \in k[T]$ be nonzero. Evaluating P at f_0 through the structure map $k \rightarrow K(X)$ is the germ of the evaluation of P at t through $k \rightarrow \Gamma(X, V)$ (the structure map factors through $\Gamma(X, V)$, and germs commute with polynomial evaluation). The latter is nonzero by transcendence of t (with coefficients moved into $\Gamma(X, V)$ along the injective map $k \rightarrow \Gamma(X, V)$), so by injectivity of the germ map $P(f_0) \neq 0$. $\square$

3.9 The curve-morphism dichotomy

A morphism from a complete nonsingular curve to a curve is either constant or a finite surjection. This section proves the bundle form of that dichotomy for the proper smooth geometrically irreducible curve bundles of the challenge, together with its two general ingredients: a morphism between proper bundles is itself proper, and a morphism whose set-range is a single closed point factors through the spectrum of the residue field of that point.

Theorem 91 (Morphisms of proper bundles are proper). *Let S be a scheme and D, E schemes over S , with $D \rightarrow S$ proper and $E \rightarrow S$ separated. Then every morphism $h : D \rightarrow E$ over S is proper; in particular h is locally of finite type and quasi-compact.*

Proof. The composite of h with the structure morphism $E \rightarrow S$ is the structure morphism $D \rightarrow S$, which is proper. Properness cancels along separated morphisms: if $g \circ f$ is proper and g is separated, then f is proper — f factors as its graph (id, f) , a closed immersion because it is a base change of the diagonal of the separated g , followed by a projection which is a base change of the proper $g \circ f$; closed immersions are proper, and properness is stable under base change and composition. Apply this with $f = h$ and g the structure morphism of E . Properness comprises finite type, so h is locally of finite type; and h is universally closed, hence quasi-compact (the preimage of a quasi-compact open under a universally closed morphism is quasi-compact). $\square$

Lemma 92 (The kernel comparison at a one-point range). *Let $f : X \rightarrow Y$ be a quasi-compact morphism of schemes with X reduced whose set-range is a single point y : $f(X) = \{y\}$. Then the kernel ideal sheaf of the canonical morphism $\mathrm{Spec} \kappa(y) \rightarrow Y$ is contained in the kernel ideal sheaf of f : for every open $U \subseteq Y$, every section $s \in \Gamma(Y, U)$ pulling back to 0 on $\mathrm{Spec} \kappa(y)$ pulls back to 0 in $\Gamma(X, f^{-1}U)$.*

Proof. Let $s \in \Gamma(Y, U)$ pull back to 0 on $\mathrm{Spec} \kappa(y)$. First, $y \notin D(s)$, the basic open of s : the point y lies in the range of $\mathrm{Spec} \kappa(y) \rightarrow Y$, and if its preimage point lay in the preimage of $D(s)$, that preimage — the basic open of the pulled-back section, i.e. of 0 — would be nonempty, which is absurd. Since the range of f is $\{y\}$ and $y \notin D(s)$, the preimage $f^{-1}D(s)$ is empty. But $f^{-1}D(s)$ is the basic open of the pulled-back section $f^\#s \in \Gamma(X, f^{-1}U)$, so $f^\#s$ has empty basic open: its image in every stalk of X lies in the maximal ideal, so on each affine chart it lies in every prime ideal, i.e. it is nilpotent. On the reduced scheme X it is therefore 0. $\square$

Theorem 93 (Factorization through the residue field of a closed point). *Let $f : X \rightarrow Y$ be a quasi-compact morphism of schemes with X reduced whose set-range is a single closed point y : $\{y\}$ is closed and $f(X) = \{y\}$. Then f factors through the canonical morphism from the spectrum of the residue field: there is a morphism $q : X \rightarrow \mathrm{Spec} \kappa(y)$ with*

$$f = (\mathrm{Spec} \kappa(y) \rightarrow Y) \circ q.$$

Proof. Since $\{y\}$ is closed, the canonical morphism $\mathrm{Spec} \kappa(y) \rightarrow Y$ is a closed immersion (it presents the reduced induced structure on the closed set $\{y\}$: a singleton is irreducible, its reduced subscheme is the spectrum of a field, namely the residue field of its point). A morphism factors through a closed immersion — necessarily uniquely — exactly when the kernel ideal sheaf of the immersion is contained in its own kernel ideal sheaf; this containment is $??$. $\square$

Theorem 94 (Proper closed subsets of the curve bundle are finite). *Let K be a field and X an integral scheme, smooth of relative dimension one over $\mathrm{Spec} K$, whose underlying space is quasi-compact. Then every closed subset $Z \subseteq X$ not containing the generic point is finite.*

Proof. The underlying space of X is irreducible, and its specialization order has height one: every generalization of a point is the generic point or the point itself ($??$). The space is moreover Noetherian: X is locally of finite type over the field K (smoothness of relative dimension one includes local finite presentation), so each affine chart is the spectrum of a finitely generated K -algebra — a Noetherian ring by the Hilbert basis theorem — and a quasi-compact space covered by finitely many Noetherian opens is Noetherian. Apply the finiteness of proper closed subsets ($??$). $\square$

Theorem 95 (The curve-morphism dichotomy). *Let K be a field and let D, E be objects of $\mathrm{Over}(\mathrm{Spec} K)$, both proper, smooth of relative dimension one and geometrically irreducible over K . Then every morphism $h : D \rightarrow E$ of $\mathrm{Over}(\mathrm{Spec} K)$ satisfies at least one of the following: either (1) h is surjective and finite, or (2) there is a point x of E with $\{x\}$ closed and $h(D) = \{x\}$. The two cases may overlap: when the space of E is a single point, both hold.*

Proof. Both bundles are integral schemes: smoothness gives geometric reducedness ($??$), which together with geometric irreducibility gives integrality ($??$); in particular the underlying spaces are irreducible ($??$) and sober, with generic points η_D, η_E .

The image is the closure of a point. By $??$ the underlying morphism of h is proper, hence a closed map, so the set-range $h(D)$ is closed in E ; and it is irreducible, being the continuous image of the irreducible space D . A nonempty irreducible closed subset of a sober space is the closure of a unique point: $h(D) = \overline{\{\zeta\}}$.

Case $\zeta \neq \eta_E$. The singleton $\{\zeta\}$ is closed (??), so $h(D) = \overline{\{\zeta\}} = \{\zeta\}$, and case (2) holds with $x = \zeta$.

Case $\zeta = \eta_E$. Then $h(D) = \overline{\{\eta_E\}} = E$, so h is surjective; and a surjection matches the generic points, $h(\eta_D) = \eta_E$ (??).

Suppose first that some point $z \neq \eta_D$ also maps to η_E . Then $\{z\}$ is closed in D (??), and h , a closed map, carries it to the closed set $\{\eta_E\}$. A closed subset containing the generic point of an irreducible space is the whole space, so E has the single point η_E ; since the range is all of $E = \{\eta_E\}$, case (2) holds with $x = \eta_E$ — this is the overlap of the two cases.

Otherwise the fiber of η_E is exactly $\{\eta_D\}$, and every fiber of h is finite: the fiber over a point $y \neq \eta_E$ is closed — $\{y\}$ is closed by ?? — and avoids η_D , whose image is η_E ; so it is finite by ??. Since h is locally of finite type and quasi-compact (??), finiteness of all fibers makes it locally quasi-finite; and a proper, locally quasi-finite morphism is finite, by Zariski's main theorem. So case (1) holds. $\square$

3.10 The topology of the projective line and morphisms into it

Two groups of results about $\mathbb{P}^1$ feed the construction of the finite morphism from the curve: the topology of $\mathbb{P}^1$ (irreducibility, height-one specialization, closed points, separatedness), and the construction of morphisms $\text{Spec } A \rightarrow \mathbb{P}^1$ from a single element of a ring A under k , with the gluing identity $[1 : u] = [u^{-1} : 1]$ for units.

Lemma 96 (The chart models are domains). *The chart rings $A_{(X_0)}$ and $A_{(X_1)}$ are integral domains, and the overlap ring $A_{(X_0 X_1)}$ is nontrivial.*

Proof. Transport along the isomorphisms $\alpha_i : A_{(X_i)} \cong k[t]$ of ?? and $\theta : A_{(X_0 X_1)} \cong k[T, T^{-1}]$ of ??: polynomial rings over a field are domains, and the Laurent polynomial ring is nontrivial. $\square$

Lemma 97 (The chart section rings are principal ideal rings). *For $i \in \{0, 1\}$, the ring $\Gamma(\mathbb{P}^1, U_i)$ is a principal ideal ring.*

Proof. Being a principal ideal ring transports along the surjective homomorphism $\gamma_i^{-1} : k[t] \rightarrow \Gamma(\mathbb{P}^1, U_i)$ (??-??), and $k[t]$ is a principal ideal domain. $\square$

Lemma 98 (The chart section rings are domains). *For $i \in \{0, 1\}$, the ring $\Gamma(\mathbb{P}^1, U_i)$ is an integral domain.*

Proof. Transport along $\gamma_i : \Gamma(\mathbb{P}^1, U_i) \cong k[t]$. $\square$

Theorem 99 (The chart section rings are Dedekind). *For $i \in \{0, 1\}$, the ring $\Gamma(\mathbb{P}^1, U_i)$ is a Dedekind domain.*

Proof. A principal ideal domain is a Dedekind domain; combine ?? and ??. $\square$

Lemma 100 (Every nonempty open meets the left chart). *Every nonempty open subset $W \subseteq \mathbb{P}^1$ contains a point of the chart U_0 .*

Proof. Let $w \in W$. The charts cover (??), so $w \in U_0$ or $w \in U_1$; in the first case we are done. In the second, work in the affine model $U_1 \cong \text{Spec } A_{(X_1)}$ (??), an irreducible space since $A_{(X_1)}$ is a domain (??). The preimage of W there is a nonempty open (it contains the preimage of w). The preimage of U_0 there is open and nonempty: the overlap $D_+(X_0 X_1)$ is contained in both charts and is nonempty, because its coordinate ring $A_{(X_0 X_1)}$ is nontrivial (??). Two nonempty opens of an irreducible space intersect; the image in $\mathbb{P}^1$ of a common point lies in $W \cap U_0$. $\square$

Theorem 101 (Irreducibility of the projective line). *The underlying topological space of $\mathbb{P}^1$ is irreducible (in particular nonempty).*

Proof. $\mathbb{P}^1$ is nonempty: the chart $U_0 \cong \operatorname{Spec} A_{(X_0)}$ is the spectrum of a nontrivial ring. Let u, v be nonempty opens of $\mathbb{P}^1$. By ?? both meet U_0 , so their preimages in the affine model $\operatorname{Spec} A_{(X_0)}$ (??) are nonempty opens; the spectrum of the domain $A_{(X_0)}$ (??) is irreducible, so the preimages intersect, and the image of a common point lies in $u \cap v$. $\square$

Lemma 102 (The generic point through the chart). *The affine model $\operatorname{Spec} A_{(X_0)} \rightarrow \mathbb{P}^1$ sends the generic point of $\operatorname{Spec} A_{(X_0)}$ to the generic point of $\mathbb{P}^1$.*

Proof. An open immersion with nonempty source into an irreducible space has dense image, and the image of the generic point of the source is a generic point of the image, hence of the closure of the image, which is the whole space. $\square$

Lemma 103 (A second point of the projective line). *$\mathbb{P}^1$ has a point distinct from its generic point.*

Proof. $A_{(X_0)} \cong k[t]$ (??) is not a field, so it has a nonzero prime ideal $\mathfrak{p}$. The corresponding point of the chart differs from the generic point: the generic point corresponds to the zero ideal (??), and the chart immersion is injective. $\square$

Theorem 104 (Height-one specialization on the projective line). *Every generalization of a point of $\mathbb{P}^1$ is the generic point or the point itself.*

Proof. Every point lies in one of the two charts (??), which are affine (??) with Dedekind section rings (??); apply ??. $\square$

Lemma 105 (Non-generic points of the projective line are closed). *Every point of $\mathbb{P}^1$ other than the generic point is closed.*

Proof. Immediate from ?? and ??. $\square$

Theorem 106 (Separatedness of the projective line). *$\mathbb{P}^1$ is a separated scheme.*

Proof. The structure morphism $\mathbb{P}^1 \rightarrow \operatorname{Spec} k$ is proper (??), hence separated, and $\operatorname{Spec} k$, being affine, is a separated scheme; a scheme separated over a separated scheme is separated. $\square$

Definition 107 (Evaluation of the chart ring). Let A be a commutative ring under k , i.e. equipped with a ring homomorphism $\sigma : k \rightarrow A$, let $i \in \{0, 1\}$ and $a \in A$. The evaluation of the chart ring at a is the ring homomorphism

$$A_{(X_i)} \longrightarrow A, \quad z \longmapsto (\text{the polynomial } \alpha_i(z) \in k[t] \text{ evaluated at } t = a \text{ with coefficients mapped by } \sigma$$

(composite of ?? with polynomial evaluation). It sends the chart coordinate t_{ij} ($j \neq i$) to a and a constant $c \in k$ to $\sigma(c)$.

Definition 108 (Morphisms to the projective line through a chart). With $\sigma : k \rightarrow A$, $i \in \{0, 1\}$ and $a \in A$ as in ??: the morphism $\operatorname{Spec} A \rightarrow \mathbb{P}^1$ obtained by applying Spec to the evaluation $A_{(X_i)} \rightarrow A$ at a and composing with the affine model $\operatorname{Spec} A_{(X_i)} \rightarrow \mathbb{P}^1$ of ??. For $i = 0$ this is the “point” $[1 : a]$; for $i = 1$ it is $[a : 1]$.

Lemma 109 (The chart morphisms lie over the base). *The composite of the morphism of ?? with the structure morphism $\mathbb{P}^1 \rightarrow \operatorname{Spec} k$ is $\operatorname{Spec} \sigma : \operatorname{Spec} A \rightarrow \operatorname{Spec} k$.*

Proof. The affine model lies over k (??), and the evaluation homomorphism of ?? restricts to σ on the constants $k \rightarrow A_{(X_i)}$; apply Spec to this equality of ring homomorphisms. $\square$

Lemma 110 (Naturality of the chart morphisms). *Let $g : A \rightarrow A'$ be a homomorphism of rings under k (so A' is under k via $g \circ \sigma$), $i \in \{0, 1\}$ and $a \in A$. Then the composite $\text{Spec } A' \xrightarrow{\text{Spec } g} \text{Spec } A \rightarrow \mathbb{P}^1$ of ?? at a is the morphism of ?? at $g(a) \in A'$.*

Proof. Polynomial evaluation commutes with g : evaluating at a and applying g is evaluating at $g(a)$ with coefficients mapped by $g \circ \sigma$. Apply Spec. $\square$

Lemma 111 (The chart restrictions factor the overlap model). *Applying Spec to the restriction homomorphism $\lambda : A_{(X_0)} \rightarrow A_{(X_0 X_1)}$ (resp. $\rho : A_{(X_1)} \rightarrow A_{(X_0 X_1)}$) of ?? (resp. ??) and composing with the affine model of the chart U_0 (resp. U_1) yields the affine model $\text{Spec } A_{(X_0 X_1)} \rightarrow \mathbb{P}^1$ of the overlap $D_+(X_0 X_1)$.*

Proof. This is the compatibility of the basic-open affine models of Proj with inclusions of basic opens: the inclusion $D_+(fg) \subseteq D_+(f)$ of basic opens corresponds, on affine models, to Spec of the canonical map $A_{(f)} \rightarrow A_{(fg)}$. $\square$

Theorem 112 (The gluing identity). *Let $\sigma : k \rightarrow A$ and let $u \in A$ be a unit. Then the morphism $[1 : u] : \text{Spec } A \rightarrow \mathbb{P}^1$ through the chart U_0 equals the morphism $[u^{-1} : 1]$ through the chart U_1 (both as in ??).*

Proof. Both morphisms factor through the chart overlap, where they are induced by one and the same ring homomorphism.

The evaluation $e_0 : A_{(X_0)} \rightarrow A$ at u (??) sends the coordinate t_{01} to the unit u . Since $A_{(X_0 X_1)}$ is the localization of $A_{(X_0)}$ at the powers of t_{01} (??), the universal property of localization extends e_0 to a ring homomorphism $\psi : A_{(X_0 X_1)} \rightarrow A$ with $\psi \circ \lambda = e_0$.

From $\lambda(t_{01}) \cdot \rho(t_{10}) = 1$ in $A_{(X_0 X_1)}$ (??) we get $u \cdot \psi(\rho(t_{10})) = 1$, i.e. $\psi(\rho(t_{10})) = u^{-1}$. Moreover $\psi \circ \rho$ agrees with the structure map on constants: the two restrictions λ and ρ restrict to the same map on the constants $k \rightarrow A_{(X_0 X_1)}$, and $\psi \circ \lambda = e_0$ restricts to $\sigma : k \rightarrow A$ on constants. Since $A_{(X_1)}$ is generated as a k -algebra by t_{10} (??), it follows that $\psi \circ \rho$ is exactly the evaluation $e_1 : A_{(X_1)} \rightarrow A$ at u^{-1} .

Now apply Spec and compose with the chart models. Writing $\iota_0, \iota_1, \iota_{01}$ for the affine models of U_0, U_1 and $D_+(X_0 X_1)$, the factorization $e_0 = \psi \circ \lambda$ and ?? give

$$[1 : u] = \iota_0 \circ \text{Spec } e_0 = \iota_0 \circ \text{Spec } \lambda \circ \text{Spec } \psi = \iota_{01} \circ \text{Spec } \psi,$$

and the same computation for $e_1 = \psi \circ \rho$ gives $[u^{-1} : 1] = \iota_1 \circ \text{Spec } e_1 = \iota_{01} \circ \text{Spec } \psi$. The two morphisms coincide. $\square$

Theorem 113 (Transcendental points land on the generic point). *Let A be an integral domain under $\sigma : k \rightarrow A$ and let $a \in A$ be transcendental in the sense that no nonzero polynomial over k , evaluated at a with coefficients mapped by σ , vanishes. Then the morphism $[1 : a] : \text{Spec } A \rightarrow \mathbb{P}^1$ of ?? sends the generic point of $\text{Spec } A$ to the generic point of $\mathbb{P}^1$.*

Proof. The evaluation $A_{(X_0)} \rightarrow A$ at a is injective: every $z \in A_{(X_0)}$ is a polynomial in the coordinate t_{01} with coefficients in k (??), its image is that polynomial evaluated at a , and by hypothesis this vanishes only for the zero polynomial. The spectrum of an injective homomorphism between domains sends the zero ideal to the zero ideal, i.e. the generic point of $\text{Spec } A$ to the generic point of $\text{Spec } A_{(X_0)}$; conclude by ??. $\square$

3.11 Spreading out a rational function to the projective line

On an integral scheme X , proper and smooth of relative dimension one over k , a rational function $f_0 \in K(X)$ transcendental over k spreads out to a k -morphism $\pi : X \rightarrow \mathbb{P}^1$ with finite fibers. This section provides the stalk-level bricks and the spreading-out argument. Classically, the corresponding morphism is produced from the inclusion of function fields $k(f_0) \subseteq K(X)$ together with the finiteness of $K(X)$ over $k(f_0)$; the route taken here never uses that finiteness — quasi-finiteness of π is obtained purely topologically from the height-one specialization order of both spaces.

Lemma 114 (Stalk morphisms are compatible with specialization). *Let X be a scheme, $U \subseteq X$ open, $x \in U$, and let y be a generalization of x (so $y \in U$ as well). Write $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{X,y}$ for the canonical map of stalks (restriction of germs). Then the composite*

$$\mathrm{Spec} \mathcal{O}_{X,y} \longrightarrow \mathrm{Spec} \mathcal{O}_{X,x} \longrightarrow U$$

of the induced morphism of spectra with the canonical morphism $\mathrm{Spec} \mathcal{O}_{X,x} \rightarrow U$ is the canonical morphism $\mathrm{Spec} \mathcal{O}_{X,y} \rightarrow U$.

Proof. The stalk of X at a point of U is canonically the stalk of the open subscheme U at that point, compatibly with the canonical maps of stalks along specializations (naturality of germs along the open immersion $U \rightarrow X$); and on X itself the analogous triangle of canonical morphisms $\mathrm{Spec} \mathcal{O}_{X,y} \rightarrow \mathrm{Spec} \mathcal{O}_{X,x} \rightarrow X$ commutes, both composites being the morphism determined by the local ring $\mathcal{O}_{X,y}$ and its ideal-theoretic image. Transport this triangle through the open immersion. $\square$

Definition 115 (Partial and rational maps). *Provided by Mathlib.* Let X, Y be schemes over a base S . A *partial map* $X \dashrightarrow Y$ is a morphism to Y from a dense open subscheme of X (its *domain*); a *rational map* is an equivalence class of partial maps, two being equivalent when they agree on a common dense open. For X irreducible with generic point ξ and function field $K(X) = \mathcal{O}_{X,\xi}$, a partial map g restricts to a morphism $\mathrm{Spec} K(X) \rightarrow Y$ (its *function-field morphism*), which determines the rational map. For X integral and Y locally of finite type over S :

1. every S -morphism $\mathrm{Spec} K(X) \rightarrow Y$ arises as the function-field morphism of a unique rational S -map $X \dashrightarrow Y$ (spreading out);
2. if moreover an S -morphism $\mathrm{Spec} \mathcal{O}_{X,x} \rightarrow Y$ restricting to the given one on $\mathrm{Spec} K(X)$ exists at a point x , then x lies in the domain of definition of that rational map (the union of the domains of its representatives);
3. if Y is separated, a rational map has a largest representative, defined on its whole domain of definition.

Lemma 116 (Restriction of a partial map to the function field). *Let X be a scheme with irreducible underlying space, $g : X \dashrightarrow Y$ a partial map and x a point of its domain. Then the function-field morphism of g is the composite*

$$\mathrm{Spec} K(X) \longrightarrow \mathrm{Spec} \mathcal{O}_{X,x} \longrightarrow Y,$$

where the first arrow is induced by the canonical map $\mathcal{O}_{X,x} \rightarrow K(X) = \mathcal{O}_{X,\xi}$ and the second is the restriction of g to $\mathrm{Spec} \mathcal{O}_{X,x}$.

Proof. The function-field morphism is by definition the restriction of g to Spec of the stalk at the generic point, which is a generalization of x ; apply ?? inside the domain of g . $\square$

Definition 117 (The structure map of a stalk). Let $f : X \rightarrow \text{Spec } k$ be a scheme over k and $x \in X$. The structure map of the stalk at x is the composite ring homomorphism $k \rightarrow \Gamma(X, \mathcal{O}_X) \rightarrow \mathcal{O}_{X,x}$ of the map induced by f on global sections with the germ at x ; it makes every stalk of X a ring under k .

Lemma 118 (Coherence of the structure maps of stalks). *Let $f : X \rightarrow \text{Spec } k$ and $x \in X$.*

1. *Applying Spec to the structure map of ?? yields the composite $\text{Spec } \mathcal{O}_{X,x} \rightarrow X \xrightarrow{f} \text{Spec } k$ of the canonical morphism with f .*
2. *For a generalization y of x , composing the structure map at x with the canonical map $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{X,y}$ yields the structure map at y .*

Proof. (1) is the compatibility of the canonical morphism $\text{Spec } \mathcal{O}_{X,x} \rightarrow X$ with global sections: its composite with f corresponds, under the anti-equivalence of affine schemes and rings, to the composite $k \rightarrow \Gamma(X, \mathcal{O}_X) \rightarrow \mathcal{O}_{X,x}$. (2) is the compatibility of germs with the canonical maps of stalks: the germ at x followed by restriction of germs to y is the germ at y . $\square$

Theorem 119 (A smooth projective curve maps locally quasi-finitely and dominantly to the line). *Let X be an integral scheme, proper and smooth of relative dimension one over the field k . Then there is a morphism $\pi : X \rightarrow \mathbb{P}^1$ over k which is locally quasi-finite and dominant.*

Proof. Write ξ for the generic point of X and $K(X) = \mathcal{O}_{X,\xi}$ for the function field, a field under k via the structure map of ??.

(a) *The generic morphism.* By ?? choose $f_0 \in K(X)$ transcendental over k ; in particular $f_0 \neq 0$ (the polynomial T does not vanish at f_0). Let $\varphi_K = [1 : f_0] : \text{Spec } K(X) \rightarrow \mathbb{P}^1$ be the chart-0 morphism of ??. It lies over k , compatibly with the canonical morphism $\text{Spec } K(X) \rightarrow X$, by ?? and ??(1).

(b) *Local regularity.* Let $x \in X$. The stalk $\mathcal{O}_{X,x}$ is a valuation ring (??) with fraction field $K(X)$, so f_0 or f_0^{-1} lies in $\mathcal{O}_{X,x}$. In the first case, with $f_0 = a$ in $\mathcal{O}_{X,x}$, set $\varphi_x = [1 : a] : \text{Spec } \mathcal{O}_{X,x} \rightarrow \mathbb{P}^1$ through chart 0; in the second, with $f_0^{-1} = b$, set $\varphi_x = [b : 1]$ through chart 1. In both cases φ_x lies over k (??, ??(1)), and its restriction along $\text{Spec } K(X) \rightarrow \text{Spec } \mathcal{O}_{X,x}$ is φ_K : in the first case by naturality (??) together with ??(2), since $a \mapsto f_0$ under $\mathcal{O}_{X,x} \rightarrow K(X)$; in the second case, naturality gives the chart-1 morphism $[f_0^{-1} : 1]$ on $\text{Spec } K(X)$, which equals $[1 : f_0]$ by the gluing identity ?? applied to the unit f_0 of the field $K(X)$.

(c) *Spreading out.* The projective line is locally of finite type and separated over k (?). By ??(1), φ_K is the function-field morphism of a rational k -map $F : X \dashrightarrow \mathbb{P}^1$. Every point x lies in the domain of definition of F : the morphism φ_x of (b) spreads out to a partial map defined at x (??(2)) whose function-field morphism is the restriction of φ_x to $\text{Spec } K(X)$ (?), i.e. φ_K — so this partial map represents F . Since X is reduced and $\mathbb{P}^1$ separated (?), F has a largest representative (??(3)), defined on the whole of X : an honest morphism $\pi : X \rightarrow \mathbb{P}^1$ whose restriction to $\text{Spec } K(X)$ is φ_K .

(d) *π lies over k .* The canonical morphism $\text{Spec } K(X) \rightarrow X$ hits the generic point, hence is dominant, and $\mathbb{P}^1$ is separated; two morphisms out of the reduced scheme X to a separated scheme agreeing after composition with a dominant morphism agree. Both π followed by the structure morphism of $\mathbb{P}^1$ and the structure morphism f of X restrict on $\text{Spec } K(X)$ to the structure morphism of $\text{Spec } K(X)$ (step (a)), so they agree.

(e) π is proper and locally of finite type. By (d), π followed by the separated structure morphism of $\mathbb{P}^1$ is the proper, locally of finite type morphism f ; properness and local finiteness descend along such factorizations (cancellation).

(f) π maps ξ to the generic point of $\mathbb{P}^1$. The point $\pi(\xi)$ is the image of the unique point of $\text{Spec } K(X)$ under φ_K , and $\varphi_K = [1 : f_0]$ sends it to the generic point of $\mathbb{P}^1$ by ?? (the hypothesis is the transcendence of f_0 , and the unique point of $\text{Spec } K(X)$ is its generic point).

(g) All fibers of π are finite. The underlying space of X is Noetherian: X is locally of finite type over a field, hence locally Noetherian, and quasi-compact (proper). Its specialization order has height one (??).

The fiber of the generic point of $\mathbb{P}^1$ is contained in $\{\xi\}$: suppose $x \neq \xi$ with $\pi(x)$ generic. Then $\{x\}$ is closed (??), and π , being proper, is a closed map, so $\{\pi(x)\}$ is closed in $\mathbb{P}^1$. But the closure of the generic point is all of $\mathbb{P}^1$, which has a second point (??) — a contradiction.

Any other fiber is finite: for y a non-generic point of $\mathbb{P}^1$, the singleton $\{y\}$ is closed (??), so the fiber $\pi^{-1}(y)$ is closed in X , and it avoids ξ because $\pi(\xi)$ is the generic point of $\mathbb{P}^1$ (step (f)), which differs from y . By ?? the fiber is finite.

(h) A morphism which is locally of finite type and has finite fibers is locally quasi-finite; by (e) and (g), π is locally quasi-finite.

(i) π is dominant. By (f) the image of π contains the generic point of $\mathbb{P}^1$, whose closure is all of $\mathbb{P}^1$; so the image of π is dense, i.e. π is dominant. $\square$

Theorem 120 (A smooth projective curve maps locally quasi-finitely to the line). *Let X be an integral scheme, proper and smooth of relative dimension one over the field k . Then there is a morphism $\pi : X \rightarrow \mathbb{P}^1$ over k which is locally quasi-finite.*

Proof. The morphism of ??, forgetting that it is dominant. $\square$

3.12 A finite morphism to the projective line

Theorem 121 (Zariski's main theorem, cancellation form). *Let $\pi : X \rightarrow Y$ and $g : Y \rightarrow S$ be morphisms of schemes with π locally quasi-finite, g separated and $g \circ \pi$ proper. Then π is finite.*

Proof. By the cancellation property of proper morphisms (using that g is separated), π is proper. A proper, locally quasi-finite morphism is finite (Zariski's main theorem). $\square$

Lemma 122 (Locally quasi-finite maps to the projective line are finite). *Let X be a scheme, proper over k , and let $\pi : X \rightarrow \mathbb{P}^1$ be a locally quasi-finite morphism over k (i.e. its composite with the structure morphism of $\mathbb{P}^1$ is the structure morphism of X). Then π is finite.*

Proof. The structure morphism of $\mathbb{P}^1$ is proper (??), in particular separated, and the composite of π with it is the structure morphism of X , which is proper. Apply ??. $\square$

Lemma 123 (Reduction to local quasi-finiteness). *Let C be a scheme, proper over k . If there exists a locally quasi-finite morphism $C \rightarrow \mathbb{P}^1$ over k , then there exists a finite morphism $C \rightarrow \mathbb{P}^1$ over k .*

Proof. Immediate from ??: the given locally quasi-finite morphism is itself finite. $\square$

Theorem 124 (Existence of a finite morphism to the projective line). *Let C be a smooth curve over k (proper, smooth of relative dimension one, geometrically irreducible). Then there exists a finite morphism $\pi : C \rightarrow \mathbb{P}^1$ over k .*

Proof. C is geometrically reduced over k (??), and being also geometrically irreducible, it is an integral scheme (??). By ?? there is a locally quasi-finite morphism $C \rightarrow \mathbb{P}^1$ over k , and by ?? it may be taken finite. $\square$

Theorem 125 (Existence of a finite dominant morphism to the projective line). *Let C be a smooth curve over k (proper, smooth of relative dimension one, geometrically irreducible). Then there exists a finite morphism $\pi : C \rightarrow \mathbb{P}^1$ over k which is moreover dominant. The dominance is part of the output of the construction — the constructed π sends the generic point of C to the generic point of $\mathbb{P}^1$ — and is not asserted as a formal consequence of finiteness alone.*

Proof. C is geometrically reduced over k (??), and being also geometrically irreducible, it is an integral scheme (??). By ?? there is a locally quasi-finite *dominant* morphism $\pi : C \rightarrow \mathbb{P}^1$ over k , and by ?? this same morphism π is finite. Since the finiteness upgrade concerns the very morphism already constructed, its dominance and its compatibility with the structure morphisms persist unchanged. $\square$

Lemma 126 (Preimages of the charts are affine). *Let $\pi : Y \rightarrow \mathbb{P}^1$ be an affine morphism (for instance a finite one). Then $\pi^{-1}(U_i)$ is an affine open of Y for $i \in \{0, 1\}$.*

Proof. The charts are affine (??), and the preimage of an affine open under an affine morphism is affine. $\square$

Lemma 127 (Preimages of the charts cover). *For any morphism $\pi : Y \rightarrow \mathbb{P}^1$, the preimages of the two standard charts cover Y : $\pi^{-1}(U_0) \cup \pi^{-1}(U_1) = Y$.*

Proof. Preimages of a cover form a cover (??). $\square$

Lemma 128 (Module-finiteness over the charts). *Let $\pi : Y \rightarrow \mathbb{P}^1$ be a finite morphism. For $i \in \{0, 1\}$, the induced ring homomorphism $\Gamma(\mathbb{P}^1, U_i) \rightarrow \Gamma(Y, \pi^{-1}(U_i))$ is module-finite. (By ??-??, the source is the polynomial ring $k[t]$.)*

Proof. A finite morphism induces module-finite maps on sections over affine opens; the charts are affine by ?? $\square$

Lemma 129 (Module-finiteness over the overlap). *Let $\pi : Y \rightarrow \mathbb{P}^1$ be a finite morphism. The induced ring homomorphism $\Gamma(\mathbb{P}^1, U_0 \cap U_1) \rightarrow \Gamma(Y, \pi^{-1}(U_0 \cap U_1))$ is module-finite. (By ??, the source is the Laurent polynomial ring $k[T, T^{-1}]$.)*

Proof. As in ??, with the overlap affine by ?? $\square$

3.13 An affine two-chart cover of the curve, and its stability under base change

The two standard charts of $\mathbb{P}^1$ pull back, along a finite morphism $C \rightarrow \mathbb{P}^1$, to a cover of the curve C by two affine opens with affine overlap. This section records that cover as a structure in its own right, together with its stability under base change of the test algebra.

Definition 130 (Affine two-chart cover). *Let Y be a scheme. An *affine two-chart cover* of Y is a pair of affine open subschemes $V_0, V_1 \subseteq Y$ with $V_0 \cup V_1 = Y$, whose overlap $V_0 \cap V_1$ is again affine.*

Theorem 131 (Existence of an affine two-chart cover of the curve). *Let C be a smooth, proper, geometrically irreducible curve over a field k . Then C admits an affine two-chart cover.*

Proof. By Theorem ?? there is a finite morphism $\pi : C \rightarrow \mathbb{P}^1$ over k . Take $V_0 := \pi^{-1}(U_0)$ and $V_1 := \pi^{-1}(U_1)$, the preimages of the two standard charts of $\mathbb{P}^1$ (Definition ??); each is affine, since π is finite, hence affine, and preimages of affine opens under affine morphisms are affine (Lemma ??). The preimages cover C (Lemma ??). For the overlap, preimage commutes with intersection, so $V_0 \cap V_1 = \pi^{-1}(U_0 \cap U_1)$; since $U_0 \cap U_1 = D_+(X_0 X_1)$ (Lemma ??) is affine (Lemma ??), its preimage $V_0 \cap V_1$ is affine as well, again because preimages of affine opens under the affine morphism π are affine. Hence (V_0, V_1) is an affine two-chart cover of C . $\square$

Theorem 132 (Base change of an affine two-chart cover). *Let k be a field, C an object of $\text{Over}(\text{Spec } k)$, and $D = (V_0, V_1)$ an affine two-chart cover of the underlying scheme of C . Let R be a commutative k -algebra, and write $C \otimes \text{Spec } R$ for the fibre product of C with the object $\text{Spec } R$ of $\text{Over}(\text{Spec } k)$ (the k -algebra structure of R making $\text{Spec } R$ an object of $\text{Over}(\text{Spec } k)$), with first projection $\text{pr}_1 : C \otimes \text{Spec } R \rightarrow C$. Then $(\text{pr}_1^{-1}(V_0), \text{pr}_1^{-1}(V_1))$ is an affine two-chart cover of $C \otimes \text{Spec } R$.*

Proof. The projection pr_1 is an affine morphism: it is the base change, along the structure morphism of C , of the structure morphism $\text{Spec } R \rightarrow \text{Spec } k$ (a morphism between affine schemes, hence affine), and affineness of morphisms is stable under base change. Preimages of affine opens under an affine morphism are affine, so $\text{pr}_1^{-1}(V_0)$ and $\text{pr}_1^{-1}(V_1)$ are affine. Preimage commutes with union and with intersection, so these two preimages cover $C \otimes \text{Spec } R$ (since $V_0 \cup V_1 = C$) and their overlap is $\text{pr}_1^{-1}(V_0 \cap V_1)$, the preimage of the affine overlap of D , hence itself affine. Thus the two preimages form an affine two-chart cover of $C \otimes \text{Spec } R$. $\square$

3.14 Product charts on a fibre product

Let X and T be objects of $\text{Over}(\text{Spec } k)$, with the two projections $\text{fst} : X \otimes T \rightarrow X$ and $\text{snd} : X \otimes T \rightarrow T$ exhibiting the underlying scheme of the fibre product $X \otimes T$ as $X \times_{\text{Spec } k} T$. For opens U of X and V of T this section builds the open $\mathfrak{W}(U, V) \subseteq X \otimes T$ cut out by the two projections and, for U, V affine, identifies its section ring with the tensor product $\Gamma(X, U) \otimes_k \Gamma(T, V)$. All k -algebra structures on section rings are those obtained by pulling back along the structure morphism to $\text{Spec } k$.

Definition 133 (The product chart). For opens U of X and V of T , the *product chart* is the open

$$\mathfrak{W}(U, V) := \text{fst}^{-1}(U) \cap \text{snd}^{-1}(V) \subseteq X \otimes T.$$

It is monotone in U and V . If U and V are affine, then $\mathfrak{W}(U, V)$ is affine.

Proof. Monotonicity follows from monotonicity of preimages and of intersections. For affineness, the two projections exhibit $X \otimes T$ as the fibre product $X \times_{\text{Spec } k} T$. Restricting this cartesian square to the opens $U \subseteq X$, $V \subseteq T$ and the whole base $\text{Spec } k$ presents the restriction of $X \otimes T$ to $\text{fst}^{-1}(U) \cap \text{snd}^{-1}(V) = \mathfrak{W}(U, V)$ as the fibre product $U \times_{\text{Spec } k} V$ of the restricted schemes. When U, V and the base $\text{Spec } k$ are affine, this is a fibre product of affine schemes over an affine scheme, hence affine; so $\mathfrak{W}(U, V)$ is an affine open. $\square$

Definition 134 (The sections identification). Let U and V be affine. The square of section rings

$$\begin{array}{ccc} \Gamma(\text{Spec } k, \mathcal{O}) & \longrightarrow & \Gamma(X, U) \\ \downarrow & & \downarrow \\ \Gamma(T, V) & \longrightarrow & \Gamma(X \otimes T, \mathfrak{W}(U, V)) \end{array}$$

— the top and left edges the pullbacks along the two structure morphisms to $\text{Spec } k$, the bottom and right edges the two projection pullbacks $\text{fst}^\sharp, \text{snd}^\sharp$ — is a pushout of commutative rings. Since $\Gamma(\text{Spec } k, \mathcal{O}) = k$, it follows that there is a k -algebra isomorphism

$$\Gamma(X, U) \otimes_k \Gamma(T, V) \xrightarrow{\sim} \Gamma(X \otimes T, \mathfrak{W}(U, V)), \quad s \otimes t \mapsto \text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t),$$

natural in restriction in both factors: for affine opens $U' \subseteq U$ and $V' \subseteq V$, restricting an identified section to $\mathfrak{W}(U', V')$ equals identifying the section obtained by restricting the two tensor factors to U' and V' .

Proof. The square is a pushout by the base-change theorem for the sections of a fibre product, in the affine case: for a cartesian square of schemes and compatible opens over which the base and the two source opens are affine, the induced square of section rings is a pushout; here the base $\text{Spec } k$ and the opens U, V are affine, and no flatness hypothesis is needed. On the other hand the tensor product $\Gamma(X, U) \otimes_k \Gamma(T, V)$ is the pushout of the span $k \rightarrow \Gamma(X, U), k \rightarrow \Gamma(T, V)$ of k -algebra structure maps. Comparing the two pushouts of this common span yields the displayed k -algebra isomorphism; it sends the two coprojections $s \otimes 1$ and $1 \otimes t$ to $\text{fst}^\sharp(s)$ and $\text{snd}^\sharp(t)$, hence a pure tensor $s \otimes t = (s \otimes 1)(1 \otimes t)$ to the product $\text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t)$. Naturality is checked on pure tensors: both sides send $s \otimes t$ to the restriction of $\text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t)$ to $\mathfrak{W}(U', V')$, because projection pullback commutes with restriction of sections. $\square$

Theorem 135 (The lift-app rule). *Let U, V be affine and let $q: Z \rightarrow X \otimes T$ be a morphism of $\text{Over}(\text{Spec } k)$ with components $a := \text{fst} \circ q: Z \rightarrow X$ and $b := \text{snd} \circ q: Z \rightarrow T$. Then on any open W of Z contained in $q^{-1}\mathfrak{W}(U, V)$, in $a^{-1}(U)$ and in $b^{-1}(V)$, the pullback $q^\sharp$ sends the identified pure tensor and its difference shape to*

$$q^\sharp(s \otimes t) = a^\sharp(s) \cdot b^\sharp(t), \quad q^\sharp(s \otimes 1 - 1 \otimes t) = a^\sharp(s) - b^\sharp(t).$$

In particular, for the morphism $\langle a, b \rangle: Z \rightarrow X \otimes T$ into the product determined by two maps $a: Z \rightarrow X, b: Z \rightarrow T$, the pullback $\langle a, b \rangle^\sharp$ sends $s \otimes t$ to $a^\sharp(s) \cdot b^\sharp(t)$.

Proof. The identification sends $s \otimes t$ to $\text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t)$ (Definition ??). Pulling back along q and using functoriality of section pullback, $q^\sharp \circ \text{fst}^\sharp = (\text{fst} \circ q)^\sharp = a^\sharp$ and $q^\sharp \circ \text{snd}^\sharp = b^\sharp$ over the opens in question, so $q^\sharp(s \otimes t) = a^\sharp(s) \cdot b^\sharp(t)$. The difference formula follows by additivity together with $s \otimes 1 \mapsto a^\sharp(s), 1 \otimes t \mapsto b^\sharp(t)$ and $a^\sharp(1) = b^\sharp(1) = 1$. For $\langle a, b \rangle$ one has $\text{fst} \circ \langle a, b \rangle = a$ and $\text{snd} \circ \langle a, b \rangle = b$, so the general rule applies with $q = \langle a, b \rangle$. $\square$

Lemma 136 (Basic open of a pure tensor). *Let U, V be affine, $s \in \Gamma(X, U)$ and $t \in \Gamma(T, V)$. The basic open of the identified pure tensor is the product chart of the two factor basic opens:*

$$D(\text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t)) = \mathfrak{W}(D(s), D(t)).$$

Proof. The basic open of a product is the intersection of the two basic opens, and the basic open of a pullback $\text{fst}^\sharp(s)$ is the preimage $\text{fst}^{-1}(D(s))$ of the basic open of s (likewise for $\text{snd}^\sharp(t)$). Hence $D(\text{fst}^\sharp(s) \cdot \text{snd}^\sharp(t)) = \text{fst}^{-1}(D(s)) \cap \text{snd}^{-1}(D(t)) = \mathfrak{W}(D(s), D(t))$, using $D(s) \subseteq U$ and $D(t) \subseteq V$. $\square$

Theorem 137 (Sections over a basic open localize the tensor ring). *Let U, V be affine and $x \in \Gamma(X, U) \otimes_k \Gamma(T, V)$; write ξ for the section of $\mathfrak{W}(U, V)$ identified with x . Then the basic open $D(\xi) \subseteq \mathfrak{W}(U, V)$ is affine, and its section ring, regarded as a $\Gamma(X, U) \otimes_k \Gamma(T, V)$ -algebra through the identification followed by restriction to $D(\xi)$, is the localization of $\Gamma(X, U) \otimes_k \Gamma(T, V)$ away from x .*

Proof. The chart $\mathfrak{W}(U, V)$ is affine (Definition ??), so a basic open of it is affine, and the section ring over $D(\xi)$ is the localization of $\Gamma(X \otimes T, \mathfrak{W}(U, V))$ away from ξ . Transporting along the ring isomorphism of Definition ??, which carries x to ξ and the powers of x to the powers of ξ , identifies this with the localization of $\Gamma(X, U) \otimes_k \Gamma(T, V)$ away from x . $\square$

3.15 The diagonal of the curve square

Fix an object C of $\text{Over}(\text{Spec } k)$ and form the fibre-product square $C \otimes C$ with its two projections $\text{fst}, \text{snd}: C \otimes C \rightarrow C$. The diagonal is the graph of the identity; when C is separated over $\text{Spec } k$ it is a closed immersion, and its image is the closed set along which the per-chart local equation of the next section cuts out the diagonal.

Definition 138 (The diagonal of the curve square). The *diagonal* of $C \otimes C$ is the morphism into the product determined by the identity in both components,

$$\delta := \langle \text{id}_C, \text{id}_C \rangle: C \longrightarrow C \otimes C, \quad \text{fst} \circ \delta = \text{id}_C, \quad \text{snd} \circ \delta = \text{id}_C.$$

For opens U, V of C its preimage of a product chart is the intersection of the factors, $\delta^{-1}\mathfrak{W}(U, V) = U \cap V$. If moreover C is separated over $\text{Spec } k$, then δ is a closed immersion; its image $\Delta := \delta(C)$ is closed, with open complement $\Delta^c := (C \otimes C) \setminus \Delta$; and the support of the kernel ideal sheaf of δ is exactly Δ .

Proof. The morphism $\langle \text{id}_C, \text{id}_C \rangle$ is the unique one with both composites equal to id_C , by the universal property of the product. A point z lies in $\delta^{-1}\mathfrak{W}(U, V)$ iff $\text{fst}(\delta z) \in U$ and $\text{snd}(\delta z) \in V$; since $\text{fst} \circ \delta = \text{snd} \circ \delta = \text{id}_C$, this means $z \in U$ and $z \in V$, i.e. $z \in U \cap V$.

Now assume C separated. The second projection $\text{snd}: C \otimes C \rightarrow C$ is the base change of the structure morphism $C \rightarrow \text{Spec } k$ along itself, hence separated because C is. The composite $\text{snd} \circ \delta = \text{id}_C$ is a closed immersion; and if a composite $g \circ f$ is a closed immersion and g is separated, then f is a closed immersion. Applying this with $f = \delta$, $g = \text{snd}$ shows δ is a closed immersion. A closed immersion is a closed embedding, so its image Δ is closed and Δ^c is open. Finally, the support of the kernel ideal sheaf of a quasi-compact morphism is the closure of its image; since Δ is closed, this support equals Δ . $\square$

3.16 The per-chart local equation of the diagonal

Fix an affine open U of C and write $B := \Gamma(C, U)$. Suppose B carries the abstract étale-coordinate data of Definition ?? and Theorem ??: a $k[X]$ -algebra structure making $k \rightarrow k[X] \rightarrow B$ a tower, an étale coordinate $u \in B$ (the image of X), and an idempotent lift $\text{elift} \in B \otimes_k B$ whose base change to $B \otimes_{k[X]} B$ is idempotent and generates the $k[X]$ -diagonal ideal. On such a chart the diagonal of $C \otimes C$ is, in one affine piece, the vanishing locus of the single diagonal equation $u \otimes 1 - 1 \otimes u$; this section makes that precise and records the regularity of the equation.

Definition 139 (The diagonal member and its equation). Let $\vartheta \in \Gamma(C \otimes C, \mathfrak{W}(U, U))$ be the section identified with $1 - \text{elift}$ (Definition ??). The *diagonal member* attached to the chart U is the affine basic open

$$\mathfrak{D}(U) := D(\vartheta) \subseteq \mathfrak{W}(U, U).$$

The *diagonal equation* $e_U \in \Gamma(C \otimes C, \mathfrak{D}(U))$ is the restriction to $\mathfrak{D}(U)$ of the section of $\mathfrak{W}(U, U)$ identified with the diagonal generator $u \otimes 1 - 1 \otimes u$. Writing ξ_x for the section of $\mathfrak{W}(U, U)$ identified with $x \in B \otimes_k B$, the pullback along the diagonal is multiplication: over $\delta^{-1}\mathfrak{W}(U, U) = U$,

$$\delta^{\sharp}(\xi_x) = \mu(x) \in \Gamma(C, U), \quad \mu: B \otimes_k B \rightarrow B, \quad x \otimes y \mapsto xy.$$

Consequently, if $\mu(\text{elift}) = 0$, then $\delta^{-1}\mathfrak{D}(U) = U$.

Proof. Affineness of $\mathfrak{D}(U)$ is affineness of a basic open of the affine chart $\mathfrak{W}(U, U)$ (Theorem ??). For the pullback formula, apply the lift-app rule (Theorem ??) to $q = \delta$, whose components are $\text{fst} \circ \delta = \text{snd} \circ \delta = \text{id}_C$: on a pure tensor $s \otimes t$ it gives $\delta^\sharp(\xi_{s \otimes t}) = \text{id}^\sharp(s) \cdot \text{id}^\sharp(t) = st = \mu(s \otimes t)$, and both sides are additive in x , so $\delta^\sharp(\xi_x) = \mu(x)$ for every x . Finally, $\mathfrak{D}(U) = D(\vartheta)$ and $\vartheta = \xi_{1-\text{elift}}$, so its δ -preimage is the basic open of $\delta^\sharp(\vartheta) = \mu(1 - \text{elift}) = 1 - \mu(\text{elift}) = 1$ when $\mu(\text{elift}) = 0$; the basic open of 1 is everything, whence $\delta^{-1}\mathfrak{D}(U) = \delta^{-1}\mathfrak{W}(U, U) = U \cap U = U$ using $\delta^{-1}\mathfrak{W}(U, U) = U$ (Definition ??). $\square$

Theorem 140 (The per-chart kernel of the diagonal is principal). *Assume C is separated over $\text{Spec } k$. On the diagonal member $\mathfrak{D}(U)$ the kernel ideal sheaf of the diagonal is principal, generated by the diagonal equation:*

$$(\ker \delta)(\mathfrak{D}(U)) = (e_U).$$

Proof. Because the base change of elift generates the $k[X]$ -diagonal ideal, the multiplication kills it, $\mu(\text{elift}) = 0$ (Theorem ??); hence $\delta^{-1}\mathfrak{D}(U) = U$ (Definition ??), and restriction of sections along this equality of opens is a ring isomorphism, in particular injective. On the affine open $\mathfrak{D}(U)$ the value of the kernel ideal sheaf is the kernel of the section map $\delta^\sharp: \Gamma(C \otimes C, \mathfrak{D}(U)) \rightarrow \Gamma(C, U)$.

By Theorem ?? the ring $\Gamma(C \otimes C, \mathfrak{D}(U))$ is the localization of $B \otimes_k B$ away from $1 - \text{elift}$, through the identification followed by restriction; write $\lambda: B \otimes_k B \rightarrow \Gamma(C \otimes C, \mathfrak{D}(U))$ for its structure map, so that $e_U = \lambda(u \otimes 1 - 1 \otimes u)$ and, by the pullback formula of Definition ??, $\delta^\sharp(\lambda(z)) = \mu(z)|_U$ for all $z \in B \otimes_k B$.

The equation lies in the kernel. One has $\delta^\sharp(e_U) = \mu(u \otimes 1 - 1 \otimes u)|_U = 0$, because $\mu(u \otimes 1 - 1 \otimes u) = u - u = 0$ (Definition ??); so $(e_U) \subseteq (\ker \delta)(\mathfrak{D}(U))$.

Every kernel element is a multiple of the equation. Let s lie in the kernel. As $\Gamma(C \otimes C, \mathfrak{D}(U))$ is a localization away from $1 - \text{elift}$, write $s \cdot \lambda(m) = \lambda(r)$ with $r \in B \otimes_k B$ and m a power of $1 - \text{elift}$, whose image $\lambda(m)$ is a unit. Applying $\delta^\sharp$, $\delta^\sharp(\lambda(r)) = \delta^\sharp(s) \cdot \delta^\sharp(\lambda(m)) = 0$; but $\delta^\sharp(\lambda(r)) = \mu(r)|_U$ and restriction along $\delta^{-1}\mathfrak{D}(U) = U$ is injective, so $\mu(r) = 0$, i.e. r lies in the k -diagonal ideal $\ker \mu$. By the localized principality of the k -diagonal ideal (Theorem ??), the extension of $\ker \mu$ along λ is generated by $\lambda(u \otimes 1 - 1 \otimes u) = e_U$; hence $\lambda(r) \in (e_U)$. Since $\lambda(m)$ is a unit and $s = \lambda(r) \cdot \lambda(m)^{-1}$, we conclude $s \in (e_U)$. $\square$

Lemma 141 (Germs of a regular section on an affine open). *Let W be an affine open of a scheme X and $s \in \Gamma(X, W)$ a nonzerodivisor. Then for every point $y \in W$ the germ of s at y is a nonzerodivisor of the stalk $\mathcal{O}_{X,y}$.*

Proof. On an affine open the stalk $\mathcal{O}_{X,y}$ is the localization of $\Gamma(X, W)$ at the prime corresponding to y , and the germ map is the localization map. A localization is flat, so the germ map is a flat ring homomorphism, and a flat ring homomorphism carries the nonzerodivisor s to a nonzerodivisor (Lemma ??). $\square$

Theorem 142 (Regularity of the diagonal equation). *Assume in addition that B is flat over $k[X]$. Then the diagonal equation e_U is a nonzerodivisor of $\Gamma(C \otimes C, \mathfrak{D}(U))$, and its germ at every point of $\mathfrak{D}(U)$ is a nonzerodivisor of the corresponding stalk.*

Proof. With B flat over $k[X]$, the diagonal generator $u \otimes 1 - 1 \otimes u$ is a nonzerodivisor of $B \otimes_k B$ (Theorem ??). The structure map $\lambda: B \otimes_k B \rightarrow \Gamma(C \otimes C, \mathfrak{D}(U))$ factors as the ring isomorphism of Definition ?? followed by the localization to the basic open $\mathfrak{D}(U)$ (Theorem ??); both are flat, so λ is flat. A flat ring homomorphism preserves nonzerodivisors (Lemma ??), and $e_U = \lambda(u \otimes 1 - 1 \otimes u)$, so e_U is a nonzerodivisor. Since $\mathfrak{D}(U)$ is affine (Definition ??), the germ regularity follows from the section regularity by Lemma ??). $\square$

Theorem 143 (The diagonal member meets the diagonal complement in a basic open). *Assume C is separated over $\mathrm{Spec} k$. The overlap of the diagonal member with the diagonal complement is the basic open of the diagonal equation,*

$$\mathfrak{D}(U) \cap \Delta^c = D(e_U),$$

and the restriction of e_U to this overlap is a unit.

Proof. On the affine $\mathfrak{D}(U)$ the kernel ideal sheaf of the diagonal is generated by e_U (Theorem ??), so the support of the kernel ideal sheaf, intersected with $\mathfrak{D}(U)$, is the zero locus $V(e_U) \cap \mathfrak{D}(U)$. That support is the diagonal Δ (Definition ??), whence $\Delta \cap \mathfrak{D}(U) = V(e_U) \cap \mathfrak{D}(U)$. Taking complements inside $\mathfrak{D}(U)$ gives $\mathfrak{D}(U) \cap \Delta^c = \mathfrak{D}(U) \setminus V(e_U) = D(e_U)$. A section is a unit on its own basic open, so the restriction of e_U to $D(e_U) = \mathfrak{D}(U) \cap \Delta^c$ is a unit. $\square$

3.17 The frozen étale-coordinate chart data

The per-chart local equation of the previous section runs on any affine chart carrying the abstract étale-coordinate data of Definition ?. To assemble a single divisor on the whole square $C \otimes C$ we must *freeze* one such chart, with its coordinate and idempotent lift, per point of the curve; this section makes the choice and records the per-chart deliverables specialised to it.

Definition 144 (The frozen chart data of a smooth curve). Let C be smooth of relative dimension one over $\mathrm{Spec} k$. Then every point p of C has an affine open $U \ni p$ whose section ring $B := \Gamma(C, U)$ carries an étale $k[X]$ -algebra structure — an étale coordinate $u \in B$ — compatible with the section k -algebra structure (making $k \rightarrow k[X] \rightarrow B$ a tower).

A *chart datum* for C is a choice, one per point p , of such a chart $\mathrm{chart}(p) \ni p$ with its étale coordinate. On each chart the coordinate is flat, formally unramified and essentially of finite type, so the packaged diagonal localisation (Theorem ??) supplies a frozen idempotent lift $\mathrm{elift}(p) \in B \otimes_k B$ with $\mu_k(\mathrm{elift}(p)) = 0$, whose base change is idempotent and generates the $k[X]$ -diagonal ideal.

Proof. Smoothness of relative dimension one over the one-point base $\mathrm{Spec} k$ yields, at each p , a standard-smooth affine chart whose section ring is standard smooth of relative dimension one over $\Gamma(\mathrm{Spec} k, \mathbb{T})$. Transport the base along the ring isomorphism $k \cong \Gamma(\mathrm{Spec} k, \mathbb{T})$ — standard-smoothness is stable under composition with a relative-dimension-zero isomorphism, dispatching the base-field-versus- $\Gamma(\mathrm{Spec} k)$ subtlety at the level of ring homomorphisms — to get $\Gamma(C, U)$ standard smooth of relative dimension one over k . The structure theorem for standard-smooth homomorphisms produces an étale factorisation $k[X_1] \rightarrow \Gamma(C, U)$ through the one-variable polynomial ring; transporting the source along $k[X_1] \cong k[X]$ gives the étale coordinate. Étaleness entails flatness, formal unramifiedness and essential finiteness, the hypotheses under which Theorem ?? produces the idempotent lift and its properties. Choosing one such chart per point gives the chart datum; every point of the curve has one, so a datum exists. $\square$

Theorem 145 (The frozen diagonal equation on a chart). *Fix a chart datum on a separated C and a point p , and write $\mathfrak{D}_p$ for the diagonal member $\mathfrak{D}(\mathrm{chart}(p))$ and e_p for its diagonal equation (Definition ??) built with the frozen coordinate. Then:*

- *the kernel ideal sheaf of the diagonal on $\mathfrak{D}_p$ is principal, $(\ker \delta)(\mathfrak{D}_p) = (e_p)$;*
- *the germ of e_p at every point of $\mathfrak{D}_p$ is a nonzerodivisor;*

- the diagonal member meets the diagonal complement in the basic open of the equation, $\mathfrak{D}_p \cap \Delta^c = D(e_p)$;
- for every affine open $W \leq \mathfrak{D}_p$, the restriction $e_p|_W$ generates the kernel ideal sheaf there, $\text{span}\{e_p|_W\} = (\ker \delta)(W)$, and $e_p|_W$ is a nonzerodivisor of $\Gamma(C \otimes C, W)$.

Proof. The first three are the per-chart deliverables (Theorems ??, ??, ??) read on the frozen chart, whose coordinate instance is baked into the definition of e_p . For the fourth, the ideal sheaf of a closed subscheme is compatible with restriction to a smaller affine open, so the principal generator on $\mathfrak{D}_p$ restricts to a generator on W : $\text{span}\{e_p|_W\} = (\ker \delta)(W)$. The section regularity of $e_p|_W$ is the propagation of germ regularity: a section whose every germ is a nonzerodivisor is itself a nonzerodivisor, because sections inject into the product of their germs; the germs of $e_p|_W$ are those of e_p , which are nonzerodivisors. $\square$

Lemma 146 (Regular generators of a principal ideal differ by a unit). *Let R be a commutative ring and $a, b \in R$ with $\text{span}\{a\} = \text{span}\{b\}$. If a is a nonzerodivisor, then a and b differ by a unit: there is a unit $u \in R^\times$ with $a = ub$.*

Proof. From $a \in \text{span}\{b\}$ and $b \in \text{span}\{a\}$ write $a = cb$ and $b = da$. Substituting, $a = cda$, so $(1 - cd)a = 0$; as a is a nonzerodivisor this forces $cd = 1$, so c is a unit with inverse d . Taking $u = c$ gives $a = ub$. $\square$

Definition 147 (The diagonal pointed cover and its equation). Fix a chart datum on a separated C . The *diagonal pointed cover* of $C \otimes C$ assigns to a point z :

$$\mathcal{U}(z) = \begin{cases} \mathfrak{D}_{\text{fst}(z)} & z \in \Delta \text{ (i.e. } z \text{ in the range of } \delta), \\ \Delta^c & z \notin \Delta, \end{cases}$$

the diagonal member of the chart at the first projection of z on the diagonal, the complement off it. The *diagonal equation* is $e_z = e_{\text{fst}(z)}|_{\mathcal{U}(z)}$ on the diagonal and $e_z = 1$ off it.

Proof. Each assignment is an open containing z : on the diagonal, $z = \delta(\text{fst}(z))$ lies in $\mathfrak{D}_{\text{fst}(z)}$ because $\delta^{-1}\mathfrak{D}(U) = U$ contains $\text{fst}(z)$ (Definition ??, using $\mu_k(\text{elift}) = 0$, and $\text{fst}(z) \in \text{chart}(\text{fst}(z))$); off the diagonal, $z \in \Delta^c$ by definition of the complement. The equation is well typed by restriction along $\mathcal{U}(z) = \mathfrak{D}_{\text{fst}(z)}$, and its stated restriction behaviour is the definition unfolded. $\square$

Theorem 148 (The diagonal as an effective divisor in local equations). *Let C be smooth of relative dimension one and separated over $\text{Spec } k$. The diagonal cover and equation of Definition ?? (for the frozen chart datum) form a system of local equations (Definition ??) on $C \otimes C$, cutting out the diagonal Δ . Its Picard class is the class of the diagonal,*

$$\text{diagonalPicClass}(C) = \mathcal{O}_{C \otimes C}(\Delta) \in \text{Pic}(C \otimes C).$$

Proof. Regularity. On a diagonal member the equation is e_p , germ-regular everywhere (Theorem ??); off the diagonal it is the constant 1, whose germs are units, hence nonzerodivisors.

Overlap ratios are units. Consider two cover members over points x, y . If both are on the diagonal, their overlap $W = \mathcal{U}(x) \cap \mathcal{U}(y)$ is affine and contained in both diagonal members; there both $e_{\text{fst}(x)}|_W$ and $e_{\text{fst}(y)}|_W$ generate the same kernel ideal sheaf $(\ker \delta)(W)$ (Theorem ??), and $e_{\text{fst}(x)}|_W$ is a nonzerodivisor, so by Lemma ?? the two differ by a unit. If one is on the diagonal and the other off, the overlap lies in $\mathfrak{D} \cap \Delta^c = D(e_p)$ (Theorem ??), where e_p restricts to a unit, and the off-diagonal equation is 1; so the ratio is a unit (or its inverse). If both are off the diagonal, both equations are 1 and the ratio is 1. Thus the data is a local-equation system.

Its Picard class (Definition ??) is by construction the class of the effective divisor it cuts out, which is the diagonal Δ ; that is, $\mathcal{O}_{C \otimes C}(\Delta)$. The diagonal is an effective (relative Cartier) divisor precisely because C is smooth, matching Kleiman's criterion. $\square$

3.18 The graph of a curve point as an effective divisor

Fix C smooth of relative dimension one and separated over $\text{Spec } k$. For a test object T and a point $t: T \rightarrow C$ (a morphism of $\text{Over}(\text{Spec } k)$), the graph $\Gamma_t \subseteq C \otimes T$ is realised as the pullback of the diagonal divisor along the graph lift $\langle \text{fst}, \text{snd} \circ t \rangle: C \otimes T \rightarrow C \otimes C$. The pullback of the local-equation system is available once the pulled equations stay regular; this is the content of the section.

Lemma 149 (The graph-lift whisker square). *For $g: T' \rightarrow T$ and $t: T \rightarrow C$, whiskering the graph lift of t by $C \triangleleft g$ is the graph lift of the restricted point $g \circ t$:*

$$(C \triangleleft g) \circ \langle \text{fst}_{C,T}, \text{snd}_{C,T} \circ t \rangle = \langle \text{fst}_{C,T'}, \text{snd}_{C,T'} \circ g \circ t \rangle.$$

Proof. Both sides are maps into the product $C \otimes C$; by the universal property it suffices to compare their two projections. Post-composing with fst : the left side gives $(C \triangleleft g) \circ \text{fst}_{C,T} = \text{fst}_{C,T'}$, the right side gives $\text{fst}_{C,T'}$. Post-composing with snd : the left side gives $(C \triangleleft g) \circ \text{snd}_{C,T} \circ t = \text{snd}_{C,T'} \circ g \circ t$ (as $\text{snd} \circ (C \triangleleft g) = g \circ \text{snd}$), matching the right side. $\square$

Theorem 150 (Regularity of the pulled-back diagonal equation). *For a chart datum on a separated C and a point $t: T \rightarrow C$, the pullback of the diagonal equation along the graph lift $q = \langle \text{fst}, \text{snd} \circ t \rangle$ is germ-regular at every point of the pullback cover.*

Proof. Fix a point y of the pullback cover and a point z of its member. Off the diagonal (i.e. $q(y) \notin \Delta$) the diagonal equation at $q(y)$ is 1, so its q -pullback is $q^\sharp(1) = 1$, whose germ is a unit, hence a nonzerodivisor.

On the diagonal, set $p_0 = \text{fst}(q(y))$, let $U = \text{chart}(p_0)$ with étale coordinate u , so the member is $q^{-1}\mathfrak{D}(U)$. Choose an affine open V of T with $(\text{snd})(z) \in V \subseteq t^{-1}U$, and pass to the affine product chart $\mathfrak{W}(U, V)$ (Definition ??) through z ; on the overlap W of the member with $\mathfrak{W}(U, V)$, the pulled equation restricts to the product-chart section of $u \otimes 1 - 1 \otimes b$, where $b := t^\sharp(u) \in \Gamma(T, V)$. This is the lift-app rule (Theorem ??) for q , whose components are $\text{fst} \circ q = \text{fst}$ and $\text{snd} \circ q = \text{snd} \circ t$: the diagonal generator $u \otimes 1 - 1 \otimes u$ (Definition ??) pulls back to $\text{fst}^\sharp(u) - (\text{snd} \circ t)^\sharp(u) = \text{fst}^\sharp(u) - \text{snd}^\sharp(b)$, i.e. the identified $u \otimes 1 - 1 \otimes b$ (Definition ??).

Now $u \otimes 1 - 1 \otimes b$ is a nonzerodivisor of $\Gamma(C, U) \otimes_k \Gamma(T, V)$ by the regularity engine (Theorem ??), which holds uniformly in the second tensor factor $\Gamma(T, V)$ and the element b — no integrality of $C \otimes T$ is used. The product-chart identification is a ring isomorphism, hence flat, so it carries this nonzerodivisor to a nonzerodivisor of $\Gamma(C \otimes T, \mathfrak{W}(U, V))$ (Lemma ??). Since $\mathfrak{W}(U, V)$ is affine, the germ at z is a nonzerodivisor (Lemma ??); matching this germ to the pullback germ across the overlap W (germs are compatible with restriction) gives the claim. $\square$

Definition 151 (The graph divisor and its Picard class). Let C be smooth of relative dimension one and separated over $\text{Spec } k$, T a test object and $t: T \rightarrow C$ a point. The graph divisor $\Gamma_t \subseteq C \otimes T$ is cut out by the pullback of the diagonal local-equation system (Theorem ??) along the graph lift $\langle \text{fst}, \text{snd} \circ t \rangle$; the pullback is a local-equation system by Theorem ?. Its Picard class $\text{graphPicClass}(C, t) = \mathcal{O}_{C \otimes T}(\Gamma_t)$ is the pullback of the diagonal class:

$$\mathcal{O}_{C \otimes T}(\Gamma_t) = \langle \text{fst}, \text{snd} \circ t \rangle^* \mathcal{O}_{C \otimes C}(\Delta).$$

Proof. The diagonal is a local-equation system (Theorem ??); its pullback along the graph lift is again a local-equation system by Definition ??, whose regularity hypothesis is discharged by Theorem ?. The Picard class of a pullback of a local-equation system is the pullback of its Picard class (Theorem ??), giving the displayed identity. This is Kleiman's graph divisor Γ_t , a relative effective divisor on $X \times T$. $\square$

Theorem 152 (Base change of the graph class). *For $g: T' \rightarrow T$ and $t: T \rightarrow C$, the class of the graph of the restricted point $g \circ t$ is the pullback of the class of the graph of t along $C \triangleleft g$:*

$$\mathcal{O}_{C \otimes T'}(\Gamma_{g \circ t}) = (C \triangleleft g)^* \mathcal{O}_{C \otimes T}(\Gamma_t).$$

Proof. By Definition ?? each side is a pullback of the diagonal class $\mathcal{O}_{C \otimes C}(\Delta)$: the left along the graph lift of $g \circ t$, the right along the graph lift of t followed by $C \triangleleft g$. Contravariant functoriality of the pullback of Picard classes turns the latter into the pullback along the composite $(C \triangleleft g) \circ \langle \text{fst}, \text{snd} \circ t \rangle$, which is the graph lift of $g \circ t$ by the whisker square (Lemma ??). Hence the two pullbacks agree. $\square$

3.19 The graph square and the graph fibre

The graph divisor of the previous section is supported, over a field test, on a single point: the graph is the base change of the diagonal, and the fibre of a one-point test is one point. This section proves the pullback square and extracts the support statement.

Lemma 153 (The graph square commutes). *For a point $t: T \rightarrow C$, the section of the projection followed by the graph lift is the point followed by the diagonal:*

$$\langle \text{fst}, \text{snd} \circ t \rangle \circ g_t = \Delta \circ t \quad \text{as morphisms } T \rightarrow C \otimes C.$$

Proof. Compare the two projections. Onto the first factor: $\text{fst} \circ q \circ g_t = \text{fst} \circ g_t = t$ and $\text{fst} \circ \Delta \circ t = t$. Onto the second: $\text{snd} \circ q \circ g_t = t \circ \text{snd} \circ g_t = t$ and $\text{snd} \circ \Delta \circ t = t$, using the projection identities of the section (Definition ??) and of the diagonal (Definition ??). $\square$

Theorem 154 (The graph square is a pullback). *For any test object T and point $t: T \rightarrow C$, the commuting square of underlying schemes*

$$\begin{array}{ccc} T_{\text{left}} & \xrightarrow{t} & C_{\text{left}} \\ \downarrow g_t & & \downarrow \Delta \\ (C \otimes T)_{\text{left}} & \xrightarrow{q} & (C \otimes C)_{\text{left}} \end{array}$$

($q = \langle \text{fst}, \text{snd} \circ t \rangle$ the graph lift) is a pullback: the graph of t is the base change of the diagonal along the graph lift — Kleiman's $\Gamma_g = (1 \times g)^{-1} \Delta$. No separatedness or properness is used.

Proof. Two paste-cancellations on the product bridge (Lemma ??). First, the middle square: the square with horizontals q and t and verticals the two second projections is a pullback, by right-cancellation of the product square of $C \otimes C$ against the product square of $C \otimes T$ (whose first projection factors as $\text{fst} \circ q$). Second, the top square: the square with verticals g_t and Δ pastes with the middle square to the square with verticals $g_t \circ \text{snd} = \text{id}$ and $\Delta \circ \text{snd} = \text{id}$, which is a pullback (vertical isomorphisms); bottom-cancellation against the middle square, using commutativity (Lemma ??), makes the top square a pullback. $\square$

Theorem 155 (The range of the section is the graph). *For any point $t: T \rightarrow C$, the topological range of the section of the projection is the graph-lift preimage of the diagonal:*

$$\text{range}(g_t) = q^{-1}(\text{range } \Delta) \quad \text{in } (C \otimes T)_{\text{left}}.$$

Proof. A pullback of schemes surjects onto the topological fibre product (Theorem ?? plus the surjectivity of scheme pullbacks onto fibre products of spaces); the fibre product of Δ and q projects onto exactly the points of $(C \otimes T)_{\text{left}}$ whose q -image lies in the range of Δ , and the projection to $(C \otimes T)_{\text{left}}$ is carried by the square onto the range of g_t . $\square$

Definition 156 (The graph point of a field test). Let K be a field extension of k and $t: \text{Spec } K \rightarrow C$ a field point. The *graph point* $x_t \in (C \otimes \text{Spec } K)_{\text{left}}$ is the image of the unique point of $\text{Spec } K$ under the section g_t . It is the entire graph fibre:

$$q(z) \in \text{range } \Delta \iff z = x_t,$$

its first projection is the image point of t , and $q(x_t) = \Delta(t(*))$.

Proof. $\text{Spec } K$ has a single point, so $\text{range}(g_t) = \{x_t\}$; by Theorem ?? the q -preimage of the diagonal is that singleton, which is the displayed equivalence. The two projection formulas follow from $\text{fst} \circ g_t = t$ and the commuting graph square (Definition ??, Lemma ??). $\square$

3.20 Rational points over a separably closed field

Over a separably closed field K the K -rational points of a smooth curve are *dense*: every nonempty open contains one. Over an imperfect separably closed field not every closed point is rational (on $\mathbb{A}_K^1$ there are closed points with residue field $K(a^{1/p})$), so the algebraically-closed argument through closed points fails; what survives is density, through the standard-smooth chart, which is étale over the affine line. There is no classical source anchored here; the statements are as formalised.

Lemma 157 (A separably closed field is infinite). *A separably closed field is infinite.*

Proof. Suppose K had exactly n elements. The polynomial $f = X^{n+1} - 1$ is separable over K : in characteristic zero this is clear, and in characteristic $p > 0$ the cardinality n is a power of p , so $n + 1 \equiv 1 \not\equiv 0 \pmod{p}$; hence the derivative $(n + 1)X^n$ is a nonzero monomial, sharing no root with f (a common root would satisfy $x = 0$ and $x^{n+1} = 1$). A separable polynomial over a separably closed field splits with distinct roots, so K contains $n + 1$ distinct roots of f — more than n elements, a contradiction. (This mirrors the classical argument that an algebraically closed field is infinite, with the splitting input replaced by separable splitting.) $\square$

Definition 158 (Rational points of the affine line). For a field K and a value $x \in K$, the K -rational point of $\mathbb{A}^1 = \text{Spec } K[X]$ at x : the maximal ideal $\ker(\text{ev}_x) = (X - x)$. It lies in a basic open $D(h)$ iff $h(x) \neq 0$.

Proof. ev_x is a surjection onto the field K , so its kernel is maximal, hence prime. Membership in $D(h)$ means $h \notin \ker(\text{ev}_x)$, i.e. $h(x) \neq 0$. $\square$

Lemma 159 (A rational base point under an étale algebra). *Let K be an infinite field and $S \neq 0$ a K -algebra étale over $K[X]$. Then the image of $\text{Spec } S \rightarrow \mathbb{A}^1$ contains a K -rational point.*

Proof. Étale maps are flat and of finite presentation, so the image of $\mathrm{Spec} S$ is open (going-down plus finite presentation makes the induced map on spectra open); it is nonempty since $S \neq 0$. A nonempty open of $\mathrm{Spec} K[X]$ contains a nonempty basic open $D(h)$, $h \neq 0$. Over the infinite field K the nonzero polynomial h has a non-vanishing value x (a nonzero polynomial function on an infinite field is not identically zero), and $\mathrm{ratPt}(x) \in D(h)$ (Definition ??) lies in the image. $\square$

Lemma 160 (Étale algebras over a separably closed field split). *A nonzero étale algebra A over a separably closed field K admits a K -algebra homomorphism $A \rightarrow K$.*

Proof. Over a separably closed field, a (formally) étale algebra splits as a finite product of copies of K ; $A \neq 0$ means the product is nonempty, and any factor projection is a K -point. $\square$

Theorem 161 (The commutative-algebra core of the density theorem). *Let K be separably closed and $S \neq 0$ a K -algebra étale over $K[X]$. Then S has a K -point: a K -algebra homomorphism $S \rightarrow K$.*

Proof. K is infinite (Lemma ??); pick a K -rational point x of $\mathbb{A}^1$ in the image of $\mathrm{Spec} S$ (Lemma ??). The fibre $S_x := K \otimes_{K[X]} S$ (base change along ev_x) is étale over K (base change of étale is étale) and nonzero: a prime q of S over $\mathrm{ratPt}(x)$ yields a $K[X]$ -algebra map $S_x \rightarrow S/q$ — lift $K \rightarrow S/q$ through the surjection $K[X] \twoheadrightarrow K$, whose kernel dies in S/q precisely because q lies over $\ker(\mathrm{ev}_x)$, and pair it with the residue map — and $S/q \neq 0$. By the splitting lemma (Lemma ??) the fibre has a K -point, which composes with $S \rightarrow K \otimes_{K[X]} S$, $s \mapsto 1 \otimes s$, to a K -point of S . $\square$

Theorem 162 (A nonempty smooth curve over a separably closed field has a rational point). *Let K be separably closed and X a nonempty scheme smooth of relative dimension 1 over K . Then X has a K -point over K : a morphism $p: \mathrm{Spec} K \rightarrow X$ with $p \circ f = \mathrm{id}$.*

Proof. Pick a point $x_0 \in X$ and a standard-smooth chart $V \ni x_0$ of relative dimension 1. Its section ring is étale over a one-variable polynomial ring (Theorem ??), and transporting along the identification $\Gamma(\mathrm{Spec} K, \top) \cong K$ makes $\Gamma(X, V)$ an étale $K[X]$ -algebra, nonzero since $V \ni x_0$ is nonempty. The core (Theorem ??) yields a K -algebra map $\varphi: \Gamma(X, V) \rightarrow K$; $\mathrm{Spec} \varphi$ composed with the canonical map of the affine open V is a scheme point $\mathrm{Spec} K \rightarrow X$. It lies over K : composing with the structure morphism corresponds, on rings, to $\varphi \circ (\text{chart map } K \rightarrow \Gamma(X, V))$, which is the identity of K since φ is a K -algebra map. $\square$

Theorem 163 (Density of rational points over a separably closed field). *Let K be separably closed, $f: X \rightarrow \mathrm{Spec} K$ smooth of relative dimension 1, and $W \subseteq X$ a nonempty open. Then there is a K -point $p: \mathrm{Spec} K \rightarrow X$ over K whose topological image lies in W . In particular, for a curve pack C over a separably closed field, every nonempty open of the underlying space of C contains a rational point.*

Proof. Smoothness of relative dimension 1 is local on the source, so the open subscheme W is itself smooth of relative dimension 1 over K , and nonempty; by Theorem ?? it has a K -point over K , whose composite with the open immersion $W \rightarrow X$ is the required point of X , with image in W . $\square$

3.21 Coheight and the Krull dimension of stalks

For a point z of a topological space, the *coheight* of z is the supremum of the lengths of strictly increasing chains starting at z in the specialisation preorder ($y \leq z$ iff z specialises to y , i.e. $y \in \overline{\{z\}}$); on a scheme it measures the codimension of the closure of z . This section identifies it with the Krull dimension of the stalk.

Lemma 164 (Coheight is local on opens). *Let X be a topological space, $U \subseteq X$ an open subset and $z \in U$. The coheight of z computed in X (for the specialisation preorder) equals the coheight of z computed in the subspace U .*

Proof. ($\geq$) The inclusion $U \rightarrow X$ is continuous, hence monotone for the specialisation preorders, and injective; a strictly increasing chain in U starting at z therefore maps to one in X of the same length (strictness is preserved: if the images were comparable both ways they would be equal).

($\leq$) Conversely, let $z = p_0 < p_1 < \dots < p_n$ be a chain in X starting at z . Each p_i is a generisation of z : $z \in \overline{\{p_i\}}$. Since U is open and $z \in U$, every generisation of a point of U lies in U (an open set containing a point of the closure of $\{p_i\}$ meets $\{p_i\}$), so all $p_i \in U$. The specialisation preorder of the subspace U agrees with that of X on points of U when U is open, so the chain lives in U with the same strict inequalities and the same length. $\square$

Lemma 165 (Coheight in an affine scheme is the height of the prime). *Let R be a commutative ring and p a point of $\operatorname{Spec} R$, with corresponding prime ideal $\mathfrak{p}$. The coheight of p in the specialisation preorder of $\operatorname{Spec} R$ equals the height of $\mathfrak{p}$ (the supremum of lengths of chains of primes descending from $\mathfrak{p}$).*

Proof. In $\operatorname{Spec} R$, a point q specialises to p exactly when $\mathfrak{p} \supseteq \mathfrak{q}$; so the specialisation preorder on $\operatorname{Spec} R$ is the opposite of the inclusion order on primes, and the order isomorphism $\operatorname{Spec} R \cong (\text{primes of } R)^{\text{op}}$ converts coheight at p into height at $\mathfrak{p}$ (chains going up from p in the specialisation preorder are chains of primes going down from $\mathfrak{p}$). $\square$

Theorem 166 (Stalk dimension equals coheight). *Let X be a scheme and $z \in X$ a point. Then the Krull dimension of the stalk $\mathcal{O}_{X,z}$ equals the coheight of z in the underlying topological space of X .*

Proof. Choose an affine open $U \ni z$, say $U \cong \operatorname{Spec} A$ with $A = \Gamma(X, U)$, and let $\mathfrak{p}$ be the prime of A corresponding to z . The stalk $\mathcal{O}_{X,z}$ is the localisation $A_{\mathfrak{p}}$, whose Krull dimension is the height of $\mathfrak{p}$ (primes of the localisation are the primes of A contained in $\mathfrak{p}$). By ??, the coheight of z in X equals its coheight in U ; the isomorphism $U \cong \operatorname{Spec} A$ is a homeomorphism, hence an isomorphism of specialisation preorders, so this equals the coheight of $\mathfrak{p}$ in $\operatorname{Spec} A$, which is the height of $\mathfrak{p}$ by ??. Chaining the equalities gives $\dim \mathcal{O}_{X,z} = \text{coheight}(z)$. $\square$

Lemma 167 (Codimension-one points have one-dimensional stalks). *Let X be a scheme and $z \in X$ with $\text{coheight}(z) = 1$. Then the stalk $\mathcal{O}_{X,z}$ has Krull dimension ≤ 1 .*

Proof. By ??, $\dim \mathcal{O}_{X,z} = \text{coheight}(z) = 1$. $\square$

3.22 The dimension of a standard-smooth algebra at a maximal ideal

This section proves the dimension half of the smooth-implies-regular pipeline at closed points: for a standard-smooth algebra of relative dimension n over a field, every maximal ideal has height at least n .

Theorem 168 (Maximal ideals of polynomial rings have full height). *Let k be a field, ι a finite index type, and M a maximal ideal of the polynomial ring $k[x_i : i \in \iota]$. Then the height of M is $\#\iota$.*

Proof. Induct on the finite type ι (along equivalences, the empty type, and adding one variable). Heights are invariant under ring isomorphisms, which handles transport along an equivalence $\iota \cong \iota'$. If ι is empty the ring is the field k , of Krull dimension 0, so every maximal ideal has height $0 = \#\iota$. For $\iota = \{*\} \sqcup \alpha$, identify $k[x_i : i \in \iota] \cong R[X]$ with $R = k[x_i : i \in \alpha]$, and let $M \subseteq R[X]$ be maximal. The ring R , a finitely generated algebra over a field, is a Jacobson ring, so the contraction $\mathfrak{p} = M \cap R$ is again *maximal*. For a maximal ideal of $R[X]$ lying over a maximal ideal $\mathfrak{p}$ of a Noetherian ring one has the height formula $\text{ht}(M) = \text{ht}(\mathfrak{p}) + 1$ (the fibre $R[X] \otimes_R R/\mathfrak{p} = (R/\mathfrak{p})[X]$ is a principal ideal domain over the field $R/\mathfrak{p}$, and M generates a nonzero — because maximal — prime there). By the inductive hypothesis $\text{ht}(\mathfrak{p}) = \#\alpha$, so $\text{ht}(M) = \#\alpha + 1 = \#\iota$. $\square$

Theorem 169 (Height lower bound at maximal ideals of standard-smooth algebras). *Let k be a field and S a standard-smooth k -algebra of relative dimension n . Then every maximal ideal $\mathfrak{m} \subseteq S$ has $\text{ht}(\mathfrak{m}) \geq n$.*

Proof. Choose a submersive presentation $S = P/\ker$, where $P = k[x_i : i \in \iota]$ is a polynomial ring on $\#\iota$ generators, the kernel of $P \twoheadrightarrow S$ is generated by $\#\sigma$ relations, and $n = \#\iota - \#\sigma$ (with $\#\sigma \leq \#\iota$). The pullback $M \subseteq P$ of $\mathfrak{m}$ is a maximal ideal (the quotient map is surjective), of height $\#\iota$ by ??, and it contains the kernel. Krull's height theorem, in the relative form $\text{ht}(M) \leq \text{ht}(M/\ker) + \mu(\ker)$ for a prime containing an ideal generated by $\mu(\ker)$ elements, gives

$$\#\iota = \text{ht}(M) \leq \text{ht}(\mathfrak{m}) + \#\sigma$$

(the first isomorphism theorem identifies $M/\ker$ with $\mathfrak{m}$, and heights are preserved by this identification). Subtracting $\#\sigma$ yields $n = \#\iota - \#\sigma \leq \text{ht}(\mathfrak{m})$. $\square$

Lemma 170 (Localised form of the height lower bound). *In the setting of ??, for any localisation $S_{\mathfrak{m}}$ of S at a maximal ideal $\mathfrak{m}$, $\dim S_{\mathfrak{m}} \geq n$.*

Proof. The Krull dimension of the localisation at a prime is the height of that prime (primes of $S_{\mathfrak{m}}$ are the primes of S contained in $\mathfrak{m}$); apply ??. $\square$

Lemma 171 (Regularity from the cotangent upper bound). *Let $(A, \mathfrak{m})$ be a Noetherian local ring with $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 \leq \dim A$. Then A is a regular local ring.*

Proof. By Nakayama, $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2$ is the minimal number $\mu(\mathfrak{m})$ of generators of $\mathfrak{m}$, and Krull's height theorem gives the reverse inequality $\dim A \leq \mu(\mathfrak{m})$ in any Noetherian local ring. So the hypothesis forces equality, which is the definition of regularity. $\square$

3.23 The trdeg–height inequality and smooth-prime regularity

This section closes the smooth-implies-regular pipeline at *arbitrary* primes of a standard-smooth algebra over a perfect field, with no recourse to Serre's homological characterisation of regularity (and hence no need for the theorem that localisations of regular local rings are regular): the cotangent dimension at a prime is computed directly from the conormal sequence, and matched against a transcendence-degree/height inequality proved by induction on the number of variables.

Lemma 172 (One-variable step of the trdeg–height inequality). *Let k be a field, R a Noetherian k -algebra, $P \subseteq R[X]$ a prime ideal and $\mathfrak{p} = P \cap R$. Then one of the following holds:*

1. $\text{ht}(\mathfrak{p}) + 1 \leq \text{ht}(P)$ and $\text{trdeg}_k(R/\mathfrak{p}) \leq \text{trdeg}_k(R[X]/P)$; or

2. $\text{ht}(P) = \text{ht}(\mathfrak{p})$ and $\text{trdeg}_k(R/\mathfrak{p}) + 1 \leq \text{trdeg}_k(R[X]/P)$.

Proof. Write $A = R/\mathfrak{p}$, a domain. The extension $R \rightarrow R[X]$ is free, hence flat, hence has the going-down property, so the dimension formula for a prime lying over $\mathfrak{p}$ applies (Stacks 00ON): $\text{ht}(P) = \text{ht}(\mathfrak{p}) + \text{ht}(\bar{P})$, where $\bar{P}$ is the image of P in the fibre ring $R[X]/\mathfrak{p}R[X] \cong A[X]$ — a prime $Q \subseteq A[X]$ contracting to (0) in A (because $P \cap R = \mathfrak{p}$).

Case $Q = (0)$. Then $\text{ht}(P) = \text{ht}(\mathfrak{p})$, and $R[X]/P \cong A[X]/Q = A[X]$ as k -algebras. Adjoining a variable to the domain A raises the transcendence degree by one ($\text{trdeg}_k A[X] \geq \text{trdeg}_k A + \text{trdeg}_A A[X] = \text{trdeg}_k A + 1$), giving case (2).

Case $Q \neq (0)$. Then $(0) \subsetneq Q$ is a chain of primes in $A[X]$, so $\text{ht}(Q) \geq 1$ and $\text{ht}(P) \geq \text{ht}(\mathfrak{p}) + 1$. Moreover Q contracts to (0) in A , so the composite $A \rightarrow A[X] \rightarrow A[X]/Q \cong R[X]/P$ is injective, whence $\text{trdeg}_k(A) \leq \text{trdeg}_k(R[X]/P)$, giving case (1). $\square$

Theorem 173 (The trdeg -height inequality in polynomial rings). *Let k be a field, ι a finite index type, and P a prime ideal of $k[x_i : i \in \iota]$. Then there is a natural number d with*

$$d \leq \text{trdeg}_k(k[x_i : i \in \iota]/P) \quad \text{and} \quad \#\iota \leq \text{ht}(P) + d.$$

Proof. Induct on the finite type ι (along equivalences, the empty type, and adding one variable); heights and transcendence degrees are invariant under k -algebra isomorphisms, which handles equivalences. For ι empty take $d = 0$. For $\iota = \{*\} \sqcup \alpha$, identify the ring with $R[X]$, $R = k[x_i : i \in \alpha]$, and let $\mathfrak{p} = P \cap R$; by the inductive hypothesis pick d with $d \leq \text{trdeg}_k(R/\mathfrak{p})$ and $\#\alpha \leq \text{ht}(\mathfrak{p}) + d$. Apply ???. In case (1) (height jump) keep the witness d : $d \leq \text{trdeg}_k(R/\mathfrak{p}) \leq \text{trdeg}_k(R[X]/P)$ and $\#\iota = \#\alpha + 1 \leq \text{ht}(\mathfrak{p}) + 1 + d \leq \text{ht}(P) + d$. In case (2) (transcendence jump) take $d + 1$: $d + 1 \leq \text{trdeg}_k(R/\mathfrak{p}) + 1 \leq \text{trdeg}_k(R[X]/P)$ and $\#\iota = \#\alpha + 1 \leq \text{ht}(\mathfrak{p}) + d + 1 = \text{ht}(P) + (d + 1)$. $\square$

Theorem 174 (The trdeg -height inequality for standard-smooth algebras). *Let k be a field and S a standard-smooth k -algebra of relative dimension n . For every prime ideal $\mathfrak{q} \subseteq S$ there is a natural number d with*

$$d \leq \text{trdeg}_k(S/\mathfrak{q}) \quad \text{and} \quad n \leq \text{ht}(\mathfrak{q}) + d.$$

Proof. As in ???: choose a submersive presentation $P = k[x_i : i \in \iota] \twoheadrightarrow S$ with kernel generated by $\#\sigma$ relations and $n = \#\iota - \#\sigma$. Let $M \subseteq P$ be the pullback of $\mathfrak{q}$, a prime containing the kernel. By ??? pick d with $d \leq \text{trdeg}_k(P/M)$ and $\#\iota \leq \text{ht}(M) + d$. The induced map $P/M \rightarrow S/\mathfrak{q}$ is an injective k -algebra map (indeed an isomorphism by the first isomorphism theorem), so $d \leq \text{trdeg}_k(S/\mathfrak{q})$. Krull's height theorem bounds the height defect by the number of relations, $\text{ht}(M) \leq \text{ht}(\mathfrak{q}) + \#\sigma$, so $\#\iota \leq \text{ht}(\mathfrak{q}) + \#\sigma + d$ and $n = \#\iota - \#\sigma \leq \text{ht}(\mathfrak{q}) + d$. $\square$

Theorem 175 (Kähler rank equals transcendence degree over a perfect field). *Let k be a perfect field and K a field extension of k that is essentially of finite type. Then*

$$\dim_K \Omega_{K/k} = \text{trdeg}_k(K).$$

Proof. Since k is perfect and K essentially of finite type over k , K admits a *separating* transcendence basis $s \subseteq K$: a transcendence basis such that K is separable algebraic over the intermediate field $F = k(s)$. A separable algebraic extension is formally étale, so base change along $F \rightarrow K$ identifies $\Omega_{K/k} \cong K \otimes_F \Omega_{F/k}$; in particular $\dim_K \Omega_{K/k} = \dim_F \Omega_{F/k}$. The field F is k -isomorphic to the fraction field of the polynomial ring $k[x_i : i \in s]$, and Kähler differentials localise: $\Omega_{F/k}$ is the localisation of the free module $\Omega_{k[x_i]/k}$ with basis $\{dx_i : i \in s\}$, hence is F -free on $\{dx_i\}$. Therefore $\dim_F \Omega_{F/k} = \#s = \text{trdeg}_k(K)$, the last equality because s is a transcendence basis. $\square$

Theorem 176 (The conormal dimension identity). *Let R be a commutative ring and $(S, \mathfrak{m})$ a local R -algebra that is formally smooth over R , whose residue field $\kappa = S/\mathfrak{m}$ is also formally smooth over R , and whose Kähler differentials $\Omega_{S/R}$ are S -free of rank $n \in \mathbb{N}$. Then*

$$\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 + \dim_{\kappa} \Omega_{\kappa/R} = n.$$

Proof. Consider the conormal (second fundamental) exact sequence of the surjection $S \twoheadrightarrow \kappa$ with kernel $\mathfrak{m}$ (Stacks 00RU):

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{c} \kappa \otimes_S \Omega_{S/R} \longrightarrow \Omega_{\kappa/R} \longrightarrow 0, \quad c(\bar{f}) = df \otimes 1.$$

Because κ is formally smooth over R , the surjection $S \twoheadrightarrow \kappa$ admits enough infinitesimal liftings to *split* the conormal map: c is (split) injective. All three terms are κ -vector spaces and the middle has dimension $\dim_{\kappa}(\kappa \otimes_S \Omega_{S/R}) = n$ (base change of a free rank- n module). The map onto $\Omega_{\kappa/R}$ is surjective with kernel exactly the image of c , which has dimension $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2$ by injectivity; rank-nullity gives the identity. (Finite-dimensionality of all terms follows from that of the middle term.) $\square$

Theorem 177 (Smooth-prime regularity over a perfect field). *Let k be a perfect field (for instance an algebraically closed field) and S a nonzero standard-smooth k -algebra. For every prime ideal $\mathfrak{q} \subseteq S$, every localisation $S_{\mathfrak{q}}$ of S at $\mathfrak{q}$ is a regular local ring.*

Proof. S is of finite type over k , hence Noetherian, and so is the localisation $A := S_{\mathfrak{q}}$, a local ring with maximal ideal $\mathfrak{m}$ and residue field κ . Fix n with S standard smooth of relative dimension n over k .

The Kähler package at A . $\Omega_{S/k}$ is S -free of rank n (standard smoothness), and Kähler differentials localise, so $\Omega_{A/k}$ is A -free of rank n . Localisations of (formally) smooth algebras are formally smooth, so A is formally smooth over k ; and κ , a field extension of the perfect field k essentially of finite type, is formally smooth over k as well. The conormal identity ?? thus gives

$$\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = n - \dim_{\kappa} \Omega_{\kappa/k}.$$

The dimension count. By ??, $\dim_{\kappa} \Omega_{\kappa/k} = \text{trdeg}_k(\kappa)$. By ?? there is d with $d \leq \text{trdeg}_k(S/\mathfrak{q})$ and $n \leq \text{ht}(\mathfrak{q}) + d$; and $S/\mathfrak{q}$ embeds into κ as k -algebras (localisation at $\mathfrak{q}$ followed by reduction is injective on $S/\mathfrak{q}$), so $d \leq \text{trdeg}_k(\kappa) = \dim_{\kappa} \Omega_{\kappa/k}$. Combining,

$$\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = n - \dim_{\kappa} \Omega_{\kappa/k} \leq (\text{ht}(\mathfrak{q}) + d) - \dim_{\kappa} \Omega_{\kappa/k} \leq \text{ht}(\mathfrak{q}) = \dim A.$$

By ??, A is a regular local ring. $\square$

Chapter 4

Lattice finiteness over the Laurent ring

This chapter isolates a general finiteness lemma about pairs of lattices in a module over the ring of Laurent polynomials. Write $k[T, T^{-1}]$ for the ring of Laurent polynomials over a commutative ring k (Chapter ??): the free k -module on the monomials T^n , $n \in \mathbb{Z}$, with multiplication $T^m \cdot T^n = T^{m+n}$, so that $T^0 = 1$ and T^{-1} is a two-sided inverse of T . For a subset s of a $k[T, T^{-1}]$ -module N , write $\langle s \rangle_{k[T, T^{-1}]}$ for the $k[T, T^{-1}]$ -submodule it generates. Call a k -module *finite* if it is finitely generated over k , and for k -submodules Q_0, Q_1 of a k -module, write $Q_0 + Q_1$ for the smallest k -submodule containing both.

The result abstracts the classical computation, for the projective line covered by its two standard affine charts, of the cohomology of a coherent sheaf as the cokernel of the difference-of-restrictions map out of the two chart section rings into the ring of sections on the overlap: two k -submodules Q_0, Q_1 of a $k[T, T^{-1}]$ -module N , one stable under multiplication by T and containing enough of N , the other stable under multiplication by T^{-1} and eventually absorbing every element of N , leave a quotient $N/(Q_0 + Q_1)$ that is a finite k -module — because only a finite window of monomials can fail to be absorbed by one lattice or the other.

4.1 Stability under powers of the generator

Throughout this section N is an abelian group carrying both a k -module structure and a $k[T, T^{-1}]$ -module structure (the two need not be related).

Lemma 178 (Stability under nonnegative powers of T). *Let $Q \subseteq N$ be a k -submodule such that $T \cdot x \in Q$ for every $x \in Q$. Then for every $x \in Q$ and every integer $m \geq 0$, $T^m \cdot x \in Q$.*

Proof. Induction on m . For $m = 0$, $T^0 \cdot x = x \in Q$. For the step, suppose $T^m \cdot x \in Q$; then

$$T^{m+1} \cdot x = T \cdot (T^m \cdot x) \in Q$$

by the hypothesis on Q applied to the element $T^m \cdot x \in Q$. □

Lemma 179 (Stability under nonnegative powers of T^{-1}). *Let $Q \subseteq N$ be a k -submodule such that $T^{-1} \cdot x \in Q$ for every $x \in Q$. Then for every $x \in Q$ and every integer $m \geq 0$, $T^{-m} \cdot x \in Q$.*

Proof. Induction on m , exactly as in the proof of Lemma ??, with T^{-1} in place of T : for $m = 0$, $T^0 \cdot x = x \in Q$; for the step, if $T^{-m} \cdot x \in Q$ then

$$T^{-(m+1)} \cdot x = T^{-1} \cdot (T^{-m} \cdot x) \in Q$$

by the hypothesis on Q . □

4.2 The two-lattice lemma

Theorem 180 (Two-lattice finiteness, spanning-set form). *Let k be a commutative ring and N a module over $k[T, T^{-1}]$ whose k -module structure is the restriction of scalars of the $k[T, T^{-1}]$ -action along the canonical ring map $k \rightarrow k[T, T^{-1}]$. Let $Q_0, Q_1 \subseteq N$ be k -submodules and $s \subset N$ a finite subset such that:*

1. $s \subseteq Q_0$;
2. s spans N over $k[T, T^{-1}]$: $\langle s \rangle_{k[T, T^{-1}]} = N$;
3. Q_0 is stable under multiplication by T : $T \cdot x \in Q_0$ for all $x \in Q_0$;
4. Q_1 is stable under multiplication by T^{-1} : $T^{-1} \cdot x \in Q_1$ for all $x \in Q_1$;
5. every $n \in N$ is eventually absorbed by Q_1 : there is an integer $m \geq 0$ with $T^{-m} \cdot n \in Q_1$.

Then $N/(Q_0 + Q_1)$ is a finite k -module: it is spanned by the classes of the monomials $T^j \cdot x$, for $x \in s$ and $-M < j < 0$, where M is any uniform bound with $T^{-M} \cdot x \in Q_1$ for every $x \in s$.

Proof. Since s is finite, hypothesis (5) applied to each $x \in s$ gives an integer $m_x \geq 0$ with $T^{-m_x} \cdot x \in Q_1$; let $M = \max_{x \in s} m_x$ (or $M = 0$ if s is empty). For each $x \in s$, since $M - m_x \geq 0$ and Q_1 is stable under T^{-1} , Lemma ?? gives

$$T^{-M} \cdot x = T^{-(M-m_x)} \cdot (T^{-m_x} \cdot x) \in Q_1,$$

so M is a uniform bound for all of s .

Let $F = \{T^j \cdot x : x \in s, -M < j < 0\}$, a finite subset of N (finite since s is finite and j ranges over the $M - 1$ integers strictly between $-M$ and 0).

Claim. For every $x \in s$ and every $j \in \mathbb{Z}$, $T^j \cdot x \in Q_0 + Q_1 + \langle F \rangle_k$, where $\langle F \rangle_k$ is the k -submodule generated by F .

Indeed, if $j \geq 0$, then since $x \in s \subseteq Q_0$ and Q_0 is stable under T , Lemma ?? gives $T^j \cdot x \in Q_0$. If $j \leq -M$, write $j = -d - M$ with $d = -j - M \geq 0$; then

$$T^j \cdot x = T^{-d} \cdot (T^{-M} \cdot x),$$

and since $T^{-M} \cdot x \in Q_1$ and Q_1 is stable under T^{-1} , Lemma ?? gives $T^j \cdot x \in Q_1$. Finally, if $-M < j < 0$, then $T^j \cdot x \in F \subseteq \langle F \rangle_k$ by definition of F . This proves the claim.

Every element $f \in k[T, T^{-1}]$ is, by definition of the Laurent polynomial ring, a k -linear combination of finitely many monomials T^j , $j \in \mathbb{Z}$. Since $Q_0 + Q_1 + \langle F \rangle_k$ is a k -submodule of N , the claim and this decomposition of f give, for every $x \in s$,

$$f \cdot x \in Q_0 + Q_1 + \langle F \rangle_k.$$

Now let $n \in N$ be arbitrary. By hypothesis (2), n is a (finite) $k[T, T^{-1}]$ -linear combination $n = \sum_{x \in s} f_x \cdot x$ of elements of s ; each summand lies in $Q_0 + Q_1 + \langle F \rangle_k$ by the previous paragraph, and this submodule is closed under addition, so $n \in Q_0 + Q_1 + \langle F \rangle_k$. Hence

$$Q_0 + Q_1 + \langle F \rangle_k = N.$$

Consequently the composite $\langle F \rangle_k \hookrightarrow N \twoheadrightarrow N/(Q_0 + Q_1)$ is surjective, so the (finite) image of F in $N/(Q_0 + Q_1)$ spans it as a k -module. Thus $N/(Q_0 + Q_1)$ is a finite k -module. $\square$

4.3 Localization forms

The spanning-set form of Theorem ?? is inconvenient when only a localization-type hypothesis on Q_0 is available, symmetric to hypothesis (5) on Q_1 : every element of N is eventually absorbed by Q_0 after multiplying by a sufficiently large power of T . The next two results derive the finiteness conclusion from this symmetric pair of hypotheses, at the cost of assuming N itself is a finite $k[T, T^{-1}]$ -module; this is exactly the shape of hypothesis produced by localizing a finite module away from an element.

Corollary 181 (Two-lattice finiteness, localization form). *Let k be a commutative ring and N a finite module over $k[T, T^{-1}]$ whose k -module structure is the restriction of scalars of the $k[T, T^{-1}]$ -action along the canonical ring map $k \rightarrow k[T, T^{-1}]$. Let $Q_0, Q_1 \subseteq N$ be k -submodules such that:*

1. Q_0 is stable under multiplication by T ;
2. Q_1 is stable under multiplication by T^{-1} ;
3. every $n \in N$ is eventually absorbed by Q_0 : there is an integer $m \geq 0$ with $T^m \cdot n \in Q_0$;
4. every $n \in N$ is eventually absorbed by Q_1 : there is an integer $m \geq 0$ with $T^{-m} \cdot n \in Q_1$.

Then $N/(Q_0 + Q_1)$ is a finite k -module.

Proof. Since N is a finite $k[T, T^{-1}]$ -module, choose a finite subset $t \subset N$ with $\langle t \rangle_{k[T, T^{-1}]} = N$. By hypothesis (3), for each $y \in t$ choose an integer $\mu(y) \geq 0$ with $T^{\mu(y)} \cdot y \in Q_0$, and set

$$s = \{ T^{\mu(y)} \cdot y : y \in t \} \subseteq Q_0,$$

a finite subset of N .

The set s again spans N over $k[T, T^{-1}]$: each element of s is a $k[T, T^{-1}]$ -multiple of an element of t , so $\langle s \rangle_{k[T, T^{-1}]} \subseteq \langle t \rangle_{k[T, T^{-1}]} = N$; and since T is invertible in $k[T, T^{-1}]$ with inverse T^{-1} , every $y \in t$ is recovered from s by

$$y = T^{-\mu(y)} \cdot (T^{\mu(y)} \cdot y),$$

a $k[T, T^{-1}]$ -multiple of the element $T^{\mu(y)} \cdot y \in s$; so $t \subseteq \langle s \rangle_{k[T, T^{-1}]}$, giving $\langle s \rangle_{k[T, T^{-1}]} \supseteq \langle t \rangle_{k[T, T^{-1}]} = N$ as well. Hence $\langle s \rangle_{k[T, T^{-1}]} = N$.

Now $s \subseteq Q_0$, s spans N over $k[T, T^{-1}]$, Q_0 is T -stable, Q_1 is T^{-1} -stable, and hypothesis (4) is exactly hypothesis (5) of Theorem ?. Applying that theorem to s , Q_0 , Q_1 gives that $N/(Q_0 + Q_1)$ is a finite k -module. $\square$

Corollary 182 (Two-lattice finiteness, localization form via powers of the generator). *Let k , N , Q_0 , Q_1 be as in Corollary ??, satisfying hypotheses (1) and (2) there, and suppose in place of (3) and (4) that:*

(3a) every $n \in N$ satisfies: there is an integer $m \geq 0$ with $T^m \cdot n \in Q_0$, where T^m denotes the ordinary m -th power of the ring element $T \in k[T, T^{-1}]$;

(4a) every $n \in N$ satisfies: there is an integer $m \geq 0$ with $T^{-m} \cdot n \in Q_1$, where T^{-m} denotes the ordinary m -th power of $T^{-1} \in k[T, T^{-1}]$.

Then $N/(Q_0 + Q_1)$ is a finite k -module.

Proof. For every integer $m \geq 0$, the m -th power of the ring element T is the monomial T^m , and the m -th power of T^{-1} is the monomial T^{-m} ; so hypotheses (3a) and (4a) here coincide, element for element, with hypotheses (3) and (4) of Corollary ???. The conclusion is immediate from that corollary. $\square$

4.4 Tensor products of localizations and the descent decompositions

This section collects two pieces of commutative-algebra scaffolding for the descent along a finite localization cover. Let $A \rightarrow B$ be a homomorphism of commutative rings. Recall that for $r \in B$, a *model of the localization of B away from r* is a B -algebra S in which the image of r is invertible and which is universal with this property, so that $S \cong B[1/r]$; we treat such a model as an A -algebra through the composite $A \rightarrow B \rightarrow S$. The first part exhibits the A -tensor product of two such localization models — taken over two possibly distinct bases B_1, B_2 , each a commutative A -algebra — as itself a localization; the second decomposes the tensor square and cube of a finite product of localizations into their pairwise and triple components.

4.4.1 The tensor product of two localizations

Definition 183 (The tensor-localization algebra structure). Let $A \rightarrow B_1$ and $A \rightarrow B_2$ be homomorphisms of commutative rings, let $r \in B_1$ and $s \in B_2$, and let S_i, S_j be models of the localization of B_1 away from r , respectively of B_2 away from s , regarded as A -algebras through $A \rightarrow B_1 \rightarrow S_i$ and $A \rightarrow B_2 \rightarrow S_j$. Tensoring the two structure maps $B_1 \rightarrow S_i$ and $B_2 \rightarrow S_j$ over A gives an A -algebra homomorphism

$$B_1 \otimes_A B_2 \longrightarrow S_i \otimes_A S_j, \quad b_1 \otimes b_2 \mapsto \bar{b}_1 \otimes \bar{b}_2,$$

whose underlying ring map endows $S_i \otimes_A S_j$ with the structure of a $(B_1 \otimes_A B_2)$ -algebra.

Theorem 184 (The tensor product of two localizations is a localization of the tensor square). *With the structure of Definition ??, $S_i \otimes_A S_j$ is the localization of $B_1 \otimes_A B_2$ away from the element $(r \otimes 1)(1 \otimes s) = r \otimes s$.*

Proof. Factor the structure map $B_1 \otimes_A B_2 \rightarrow S_i \otimes_A S_j$ through the intermediate ring $S_i \otimes_A B_2$. The first stage $B_1 \otimes_A B_2 \rightarrow S_i \otimes_A B_2$ localizes only the left factor: it is the base change of the localization $B_1 \rightarrow S_i$ at r along the ring map $B_1 \rightarrow B_1 \otimes_A B_2$, hence exhibits $S_i \otimes_A B_2$ as the localization of $B_1 \otimes_A B_2$ away from $r \otimes 1$. The second stage $S_i \otimes_A B_2 \rightarrow S_i \otimes_A S_j$ localizes the right factor: it is the base change of the localization $B_2 \rightarrow S_j$ at s along $B_2 \rightarrow S_i \otimes_A B_2$, exhibiting $S_i \otimes_A S_j$ as the localization of $S_i \otimes_A B_2$ away from $1 \otimes s$. Two successive away-localizations, first at $r \otimes 1$ then at $1 \otimes s$, compose to the away-localization at the product $(r \otimes 1)(1 \otimes s)$; and by the multiplication rule in $B_1 \otimes_A B_2$ this product equals $r \otimes s$. $\square$

Definition 185 (Comparison with an arbitrary model). Let T be any model of the localization of $B_1 \otimes_A B_2$ away from $(r \otimes 1)(1 \otimes s)$. The canonical comparison of two localizations of $B_1 \otimes_A B_2$ at the same element is a $(B_1 \otimes_A B_2)$ -algebra isomorphism

$$S_i \otimes_A S_j \xrightarrow{\sim} T.$$

Lemma 186 (Values of the comparison isomorphism). *The comparison isomorphism sends the image of $w \in B_1 \otimes_A B_2$ under the structure map to the image of w in T ; in particular, on a pure tensor of structure-map images it sends $\bar{b}_1 \otimes \bar{b}_2$ to the image of $b_1 \otimes b_2$ in T .*

Proof. Being a $(B \otimes_A B)$ -algebra homomorphism, the comparison isomorphism commutes with the two structure maps, which is the first statement. The second follows because the pure tensor $\bar{b}_1 \otimes \bar{b}_2$ is the structure-map image of $b_1 \otimes b_2$. $\square$

4.4.2 Decompositions of the tensor square and cube of a finite product

Throughout this subsection A is a commutative ring, ι a finite index set, and $(S_i)_{i \in \iota}$ a finite family of A -algebras with product $P := \prod_i S_i$.

Definition 187 (The pairwise and triple decompositions). Distributing the tensor product over the finite product in each factor gives the canonical A -algebra identification

$$P \otimes_A P \cong \prod_{(i,j) \in \iota \times \iota} S_i \otimes_A S_j,$$

which sends a pure tensor $s \otimes t$ to the family $(i, j) \mapsto s_i \otimes t_j$.

Definition 188 (The triple decomposition). Applying the pairwise decomposition to the inner factor and distributing once more gives the canonical A -algebra identification

$$P \otimes_A (P \otimes_A P) \cong \prod_{(i,j,k) \in \iota^3} S_i \otimes_A (S_j \otimes_A S_k),$$

sending $s \otimes v$ to the family $(i, j, k) \mapsto s_i \otimes (v_{(j,k)})$, where $v_{(j,k)}$ is the (j, k) -component of v under the pairwise decomposition.

Definition 189 (The index-wise simplicial maps). On the pairwise tensor components one has the index-wise simplicial maps: for indices i, j, k , the two *inclusions* $S_i \rightarrow S_i \otimes_A S_j$, $x \mapsto x \otimes 1$, and $S_j \rightarrow S_i \otimes_A S_j$, $y \mapsto 1 \otimes y$; the three *faces* into the triple tensor $S_i \otimes_A (S_j \otimes_A S_k)$, hitting the positions $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$ out of $(S_i \otimes_A S_j)$, $(S_i \otimes_A S_k)$, $(S_j \otimes_A S_k)$ respectively; and the *diagonal* $S_i \otimes_A S_i \rightarrow S_i$, $x \otimes y \mapsto xy$.

Lemma 190 (The descent inclusions become the index-wise inclusions). *Under the pairwise decomposition, the two inclusions $P \rightarrow P \otimes_A P$ become the index-wise inclusions: $s \otimes 1$ corresponds to the family $(i, j) \mapsto s_i \otimes 1$, and $1 \otimes s$ to $(i, j) \mapsto 1 \otimes s_j$.*

Proof. Evaluate at a component (i, j) : by the pure-tensor formula of Definition ??, the decomposition sends $s \otimes 1$ to $s_i \otimes 1_j = s_i \otimes 1$, which is the first index-wise inclusion of s_i , and $1 \otimes s$ to $1 \otimes s_j$, the second index-wise inclusion of s_j . $\square$

Lemma 191 (The descent faces become the index-wise faces). *Under the pairwise and triple decompositions, the three faces $P \otimes_A P \rightarrow P \otimes_A (P \otimes_A P)$ become the index-wise faces: the $\{2, 3\}$ -face carries a family $(w_{(i,j)})$ to $(i, j, k) \mapsto \text{face}_{23}(w_{(j,k)})$, and likewise the $\{1, 2\}$ - and $\{1, 3\}$ -faces carry it to $(i, j, k) \mapsto \text{face}_{12}(w_{(i,j)})$ and $(i, j, k) \mapsto \text{face}_{13}(w_{(i,k)})$.*

Proof. Each identity is checked on generators using the pure-tensor formulas of Definitions ?? and ?. For the $\{2, 3\}$ -face, the source element $1 \otimes v$ is carried to $(i, j, k) \mapsto 1 \otimes (v_{(j,k)}) = \text{face}_{23}(w_{(j,k)})$ directly. The $\{1, 2\}$ - and $\{1, 3\}$ -faces mix the two product factors; since both sides are A -algebra homomorphisms it suffices to compare them on pure tensors, where the two decompositions evaluate to the stated index-wise faces. $\square$

Lemma 192 (Multiplication becomes the index-wise diagonal). *Under the pairwise decomposition, the multiplication map $P \otimes_A P \rightarrow P$, $a \otimes b \mapsto ab$, becomes the index-wise diagonal: a family $(w_{(i,j)})$ is carried to $i \mapsto \text{diag}(w_{(i,i)})$, the product of the diagonal components.*

Proof. Both sides are A -algebra homomorphisms, so it suffices to compare on a pure tensor $s \otimes t$. Multiplication sends it to $i \mapsto s_i t_i$; the decomposition sends it to $(i, j) \mapsto s_i \otimes t_j$, whose diagonal (i, i) -component is $s_i \otimes t_i$, and $\text{diag}(s_i \otimes t_i) = s_i t_i$. The two agree. $\square$

Theorem 193 (Faithful flatness of the composite cover tower). *Let $A \rightarrow B$ be faithfully flat, let $f: \iota \rightarrow B$ be a covering family — the f_i generate the unit ideal of B — and let each S_i be a model of the localization of B away from f_i . Then $P = \prod_i S_i$ is a faithfully flat A -algebra.*

Proof. The product of the localizations at a covering family is faithfully flat over B : each $B \rightarrow S_i$ is flat, being a localization, and because the f_i generate the unit ideal the family jointly detects triviality, so $B \rightarrow \prod_i S_i$ is faithfully flat. Faithful flatness is preserved by composition, so composing with the faithfully flat $A \rightarrow B$ makes the tower $A \rightarrow B \rightarrow \prod_i S_i$ faithfully flat; hence P is a faithfully flat A -algebra. $\square$

4.5 The composite descent unit of a coherent family of units

Keep the notation of the previous section: A is a commutative ring, ι a finite index set, and $(S_i)_{i \in \iota}$ a finite family of A -algebras with product $P := \prod_i S_i$. This section assembles a family of componentwise units, one on each pairwise tensor $S_i \otimes_A S_j$, into a single unit of $P \otimes_A P$, and isolates the conditions under which that unit is an A -descent 1-cocycle (Definition ??) whose collapse (Definition ??) recovers a prescribed cover cocycle. This is the pure-algebra core of the composite descent unit assembled by the étale-separatedness campaign.

Definition 194 (The composite descent unit). Given a family of units $w_{ij} \in (S_i \otimes_A S_j)^\times$, the *composite descent unit* $\text{piAssembly}(w) \in (P \otimes_A P)^\times$ is the unit corresponding to the family (w_{ij}) under the pairwise decomposition $P \otimes_A P \cong \prod_{(i,j)} S_i \otimes_A S_j$ of Definition ??.

Theorem 195 (The assembled unit of a coherent family is a descent cocycle). *Suppose the family (w_{ij}) is:*

1. *diagonally normalized: $\text{diag}(w_{ii}) = 1$ in S_i for every i , where $\text{diag}: S_i \otimes_A S_i \rightarrow S_i$ is the multiplication (Definition ??);*
2. *index-wise cocyclic: for all i, j, k , $\text{face}_{23}(w_{jk}) \cdot \text{face}_{12}(w_{ij}) = \text{face}_{13}(w_{ik})$ in $S_i \otimes_A (S_j \otimes_A S_k)$.*

Then the composite descent unit $\text{piAssembly}(w)$ is an A -descent 1-cocycle relative to $A \rightarrow P$.

Proof. A descent 1-cocycle is a unit satisfying the normalization $\mu(u) = 1$ and the cocycle identity $\partial_{23}(u) \cdot \partial_{12}(u) = \partial_{13}(u)$ (Definition ??); both conditions are verified after transporting through the pairwise and triple decompositions.

For the normalization: under the pairwise decomposition the multiplication $\mu: P \otimes_A P \rightarrow P$ becomes the index-wise diagonal (Lemma ??), carrying $\text{piAssembly}(w)$ to $i \mapsto \text{diag}(w_{ii})$, which is 1 by hypothesis (1); since the decomposition is an isomorphism, $\mu(\text{piAssembly}(w)) = 1$.

For the cocycle identity: the triple decomposition (Definition ??) is injective, so it suffices to compare the two sides after applying it. Under the two decompositions the three faces $\partial_{23}, \partial_{12}, \partial_{13}$ become the index-wise faces $\text{face}_{23}, \text{face}_{12}, \text{face}_{13}$ (Lemma ??), so the transported left side is the family $(i, j, k) \mapsto \text{face}_{23}(w_{jk}) \cdot \text{face}_{12}(w_{ij})$ and the transported right side is $(i, j, k) \mapsto \text{face}_{13}(w_{ik})$; these agree componentwise by hypothesis (2). $\square$

For the remaining results fix, in addition, a commutative A -algebra B , a finite family $f: \iota \rightarrow B$, models S_i of the localization of B away from f_i , and *overlap models* T_{ij} of the localization of B away from $f_i f_j$, all regarded as A -algebras through $A \rightarrow B$. Write $\iota_1: S_i \rightarrow T_{ij}$ and $\iota_2: S_j \rightarrow T_{ij}$ for the two canonical B -algebra structure maps of the overlap model (f_i , resp. f_j , being invertible in T_{ij}).

Definition 196 (The index-wise collapse). The *index-wise collapse* is the A -algebra homomorphism

$$\kappa_{ij}: S_i \otimes_A S_j \longrightarrow T_{ij}, \quad x \otimes y \mapsto \iota_1(x) \cdot \iota_2(y).$$

It is the $(B \otimes_A B)$ -semilinear comparison of the two-base tensor localization $S_i \otimes_A S_j$ of $B \otimes_A B$ (Theorem ?? with $B_1 = B_2 = B$) with the one-base overlap localization T_{ij} of B .

Lemma 197 (The collapse computes componentwise). *Decompose both $P \otimes_A P \cong \prod_{ij} S_i \otimes_A S_j$ (Definition ??) and the target $P \otimes_B P \cong \prod_{ij} T_{ij}$ of the collapse pairwise. Then the collapse map $\pi: P \otimes_A P \rightarrow P \otimes_B P$ of Definition ?? becomes the index-wise collapse: it carries a family (v_{ij}) to $(i, j) \mapsto \kappa_{ij}(v_{ij})$.*

Proof. Both composites are A -algebra homomorphisms out of $P \otimes_A P$, so it suffices to compare them on a pure tensor $s \otimes t$. The pairwise decomposition sends $s \otimes t$ to the family $(i, j) \mapsto s_i \otimes t_j$; applying κ_{ij} yields $\iota_1(s_i) \cdot \iota_2(t_j)$. On the other side, the collapse sends $s \otimes t$ to the same elementary tensor in $P \otimes_B P$, whose (i, j) -component under the B -side pairwise decomposition is the image of $s_i \otimes t_j$ under the canonical comparison of $S_i \otimes_B S_j$ with T_{ij} ; by uniqueness of B -algebra maps out of a localization of B this comparison equals κ_{ij} on pure tensors, again $\iota_1(s_i) \cdot \iota_2(t_j)$. The two families coincide. $\square$

Theorem 198 (The collapse of the assembled unit is the descent unit of the cover cocycle). *Let (c_{ij}) , $c_{ij} \in T_{ij}^\times$, be a cover cocycle for f , and suppose the componentwise units w_{ij} collapse onto it: $\kappa_{ij}(w_{ij}) = c_{ij}$ for all i, j . Then the collapse of the composite descent unit along the tower $A \rightarrow B \rightarrow P$ is the descent unit of c :*

$$\pi(\text{piAssembly}(w)) = \text{cocycleUnit}(c) \quad \text{in } (P \otimes_B P)^\times,$$

where $\text{cocycleUnit}(c)$ is the unit of $P \otimes_B P$ corresponding to the family (c_{ij}) under the pairwise decomposition $P \otimes_B P \cong \prod_{ij} T_{ij}$.

Proof. Both sides are units of $P \otimes_B P$, apply the injective pairwise decomposition $P \otimes_B P \cong \prod_{ij} T_{ij}$. By Lemma ?? the decomposition of $\pi(\text{piAssembly}(w))$ is the family $(i, j) \mapsto \kappa_{ij}(u_{ij})$, where u_{ij} is the (i, j) -component of $\text{piAssembly}(w)$; by Definition ?? that component is w_{ij} , so the family is $(i, j) \mapsto \kappa_{ij}(w_{ij}) = c_{ij}$ by hypothesis. The decomposition of $\text{cocycleUnit}(c)$ is the family (c_{ij}) by definition. The two families agree, hence so do the two units. $\square$

Theorem 199 (Diagonal normalization from a collapse onto a normalized family). *Suppose the componentwise units collapse onto a family (c_{ij}) , $\kappa_{ij}(w_{ij}) = c_{ij}$, whose diagonal is normalized: the image of c_{ii} under the canonical B -algebra map $\delta_i: T_{ii} \rightarrow S_i$ (well-defined because f_i is already invertible in S_i) is 1, for every i . Then $\text{diag}(w_{ii}) = 1$ in S_i for every i .*

Proof. Fix i . The composites $\delta_i \circ \iota_1$ and $\delta_i \circ \iota_2$ are B -algebra endomorphisms of the localization S_i , hence both equal the identity by uniqueness of B -algebra maps out of S_i . Consequently, for a pure tensor $x \otimes y \in S_i \otimes_A S_i$,

$$\delta_i(\kappa_{ii}(x \otimes y)) = \delta_i(\iota_1 x) \cdot \delta_i(\iota_2 y) = x \cdot y = \text{diag}(x \otimes y),$$

and both $\delta_i \circ \kappa_{ii}$ and diag are additive, so $\delta_i \circ \kappa_{ii} = \text{diag}$ on all of $S_i \otimes_A S_i$. Applying this to w_{ii} and using the two hypotheses, $\text{diag}(w_{ii}) = \delta_i(\kappa_{ii}(w_{ii})) = \delta_i(c_{ii}) = 1$. In particular the diagonal hypothesis of Theorem ?? is automatic once the components collapse onto a cover cocycle. $\square$

Lemma 200 (The index-wise operators intertwine the two-base structure maps). *Write $m: B \otimes_A B \rightarrow S_i \otimes_A S_j$ for the two-base structure map of Definition ?? (with $B_1 = B_2 = B$) and $m_3: B \otimes_A (B \otimes_A B) \rightarrow S_i \otimes_A (S_j \otimes_A S_k)$ for its triple-tensor analogue. Then the three index-wise faces intertwine the double and triple structure maps over the descent cofaces,*

$$\text{face}_\bullet \circ m = m_3 \circ \partial_\bullet \quad (\bullet \in \{12, 13, 23\}),$$

and the index-wise collapse intertwines the double structure map with the multiplication: $\kappa_{ij} \circ m = \iota \circ \mu$, where $\mu: B \otimes_A B \rightarrow B$ is the multiplication and $\iota: B \rightarrow T_{ij}$ is the structure map.

Proof. Each identity is an equality of A -algebra homomorphisms out of $B \otimes_A B$ (respectively $B \otimes_A (B \otimes_A B)$), so it suffices to check it on a pure tensor, where both sides are read off from the pure-tensor formulas for the structure maps, faces, cofaces, collapse and multiplication. For the $\{2, 3\}$ -face, m sends $x \otimes y$ to $\bar{x} \otimes \bar{y}$ and face_{23} prepends a 1, giving $1 \otimes (\bar{x} \otimes \bar{y})$; the same element results from $\partial_{23}(x \otimes y) = 1 \otimes (x \otimes y)$ followed by m_3 , using $1 = 1$. The $\{1, 2\}$ - and $\{1, 3\}$ -faces are identical after tracking which tensor slot receives the 1. For the collapse, $\kappa_{ij}(m(x \otimes y)) = \iota_1(\bar{x}) \iota_2(\bar{y}) = \iota(x) \iota(y) = \iota(xy) = \iota(\mu(x \otimes y))$, since ι_1, ι_2 restrict to ι on the image of B . $\square$

4.6 Degree-zero Amitsur descent and unit production

This section records the two pure-algebra inputs to the faithfully flat descent that closes the étale separatedness of the relative Picard functor: the degree-zero exactness of the Amitsur complex, in the right-tensor form used to descend a section, and the production of a unit from the evaluated generator of a trivialized invertible module.

Theorem 201 (The degree-zero Amitsur equalizer). *Let $A \rightarrow B$ be faithfully flat and S_0 an A -algebra. If an element $x \in S_0 \otimes_A B$ has equal images under the two coprojection faces*

$$\text{id}_{S_0} \otimes (b \mapsto b \otimes 1), \quad \text{id}_{S_0} \otimes (b \mapsto 1 \otimes b) : S_0 \otimes_A B \longrightarrow S_0 \otimes_A (B \otimes_A B),$$

then $x = s \otimes 1$ for a unique $s \in S_0$.

Proof. This is exactness in degree 0 of the Amitsur complex of the faithfully flat extension $A \rightarrow B$, applied with coefficients in S_0 . Transport x through the commutativity and associativity isomorphisms into the coaction encoding of the tautological base-change descent datum on S_0 ; the two coprojection faces become the two legs of that coaction, so the hypothesis says exactly

that the coaction difference vanishes on the transported element. Degree-zero Amitsur exactness for a faithfully flat base change then produces $s \in S_0$ whose image $1 \otimes s$ is the transported element; applying the inverse transport gives $s \otimes 1 = x$. Uniqueness is faithful flatness: the map $s \mapsto s \otimes 1$, $S_0 \rightarrow S_0 \otimes_A B$, is injective when B is faithfully flat over A . $\square$

Theorem 202 (Unit production from an evaluated generator). *Let $A \rightarrow R \rightarrow R'$ be a tower with R' also an A -algebra and $\varphi: R \rightarrow R'$ the ring map, carried by an A -linear avatar. Let M' be an R -module presented as trivial by an R -linear isomorphism $t: M' \cong R$ together with a collapse isomorphism $c: M' \cong R \otimes_A M$. Then, for every surjective R' -linear map $F: R' \otimes_A M \rightarrow R'$, the value*

$$F(\varphi_* c(t^{-1}1)) \in R'$$

of F on the φ -pushforward of the collapsed generator $c(t^{-1}1)$ is a unit.

Proof. Write $g_0 := c(t^{-1}1) \in R \otimes_A M$. Since t is an isomorphism, every $x \in R \otimes_A M$ is the scalar multiple $x = t(c^{-1}x) \cdot g_0$; that is, g_0 generates $R \otimes_A M$ as an R -module. Pushing forward along φ : the map $\varphi_*: R \otimes_A M \rightarrow R' \otimes_A M$ is φ -semilinear, so every $z \in R' \otimes_A M$ is an R' -multiple of $g := \varphi_* g_0$. Indeed on a pure tensor $r' \otimes m = r' \cdot (1 \otimes m)$ and $1 \otimes m = \varphi_*(1 \otimes m) = \varphi_*(t(c^{-1}(1 \otimes m)) \cdot g_0) = \varphi(t(c^{-1}(1 \otimes m))) \cdot g$, and the general element is a finite sum of such. Now choose, by surjectivity of F , an element y with $F(y) = 1$, and write $y = c' \cdot g$ for some $c' \in R'$. Then $1 = F(c' \cdot g) = c' \cdot F(g)$, so $F(g)$ is a unit. $\square$

Definition 203 (Base change of a descent equivalence). Let $A \rightarrow B' \rightarrow R'$ be a tower and $dE: B' \otimes_A M \cong B'$ a B' -linear descent equivalence. Its *base change* is the R' -linear equivalence

$$\text{descentMulEval}(dE): R' \otimes_A M \xrightarrow{\sim} R', \quad r' \otimes m \mapsto dE(1 \otimes m) \cdot r',$$

obtained by cancelling the iterated base change $R' \otimes_{B'} (B' \otimes_A M) \cong R' \otimes_A M$, applying dE in the second factor, and using the right unit $R' \otimes_{B'} B' \cong R'$. Being an isomorphism, it is in particular a surjective R' -linear map, the concrete evaluation fed to Theorem ??.

4.7 Additivity of dimension along exact sequences

The colength computations of the next section, and the Euler characteristic of the Riemann–Roch ledger, rest on the additivity of the k -dimension along exact sequences of finite-dimensional vector spaces. Throughout, k is a field and all vector spaces are finite-dimensional over k ; $\dim_k$ is the dimension.

Theorem 204 (Additivity of dimension along a short exact sequence). *Let*

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be a short exact sequence of finite-dimensional k -vector spaces (f injective, g surjective, and $\ker g = \text{im } f$). Then

$$\dim_k B = \dim_k A + \dim_k C.$$

Proof. The length of a module is additive along a short exact sequence: for $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ one has $\ell(B) = \ell(A) + \ell(C)$, where ℓ denotes the length of a composition series. For a finite-dimensional vector space over the field k , a composition series is a maximal flag of subspaces, so its length equals the dimension: $\ell(V) = \dim_k V$. Substituting this identity for each of A, B, C into the length additivity gives $\dim_k B = \dim_k A + \dim_k C$. $\square$

Theorem 205 (Vanishing of the alternating sum of dimensions along a five-term sequence). *Let*

$$0 \longrightarrow V_1 \xrightarrow{f_1} V_2 \xrightarrow{f_2} V_3 \xrightarrow{f_3} V_4 \xrightarrow{f_4} V_5 \longrightarrow 0$$

be an exact sequence of finite-dimensional k -vector spaces (f_1 injective, f_4 surjective, and $\ker f_{i+1} = \operatorname{im} f_i$ for $i = 1, 2, 3$). Then the alternating sum of dimensions vanishes in $\mathbb{Z}$:

$$\dim_k V_1 - \dim_k V_2 + \dim_k V_3 - \dim_k V_4 + \dim_k V_5 = 0.$$

Proof. Apply the rank-nullity theorem to each of the four maps: for every i ,

$$\dim_k V_i = \dim_k(\ker f_i) + \dim_k(\operatorname{im} f_i).$$

Injectivity of f_1 gives $\dim_k(\ker f_1) = 0$; surjectivity of f_4 gives $\dim_k(\operatorname{im} f_4) = \dim_k V_5$; and exactness at V_{i+1} rewrites the kernel as the preceding range, $\dim_k(\ker f_{i+1}) = \dim_k(\operatorname{im} f_i)$ for $i = 1, 2, 3$. Writing $r_i = \dim_k(\operatorname{im} f_i)$, these read

$$\dim_k V_1 = r_1, \quad \dim_k V_2 = r_1 + r_2, \quad \dim_k V_3 = r_2 + r_3, \quad \dim_k V_4 = r_3 + \dim_k V_5.$$

Forming the alternating sum, every r_i and $\dim_k V_5$ cancels:

$$\dim_k V_1 - \dim_k V_2 + \dim_k V_3 - \dim_k V_4 + \dim_k V_5 = r_1 - (r_1 + r_2) + (r_2 + r_3) - (r_3 + \dim_k V_5) + \dim_k V_5 = 0.$$

□

4.8 The colength of Dedekind quotients

This section computes the k -dimension of the quotient of a Dedekind domain B , finitely generated as a module over a base field k through a fixed k -algebra structure, by a nonzero ideal. The pattern is the classical dévissage: a nonzero principal ideal $I = (\pi)$ has all its successive powers separated by copies of the residue quotient B/I , so $\dim_k(B/I^e) = e \cdot \dim_k(B/I)$; the Chinese remainder theorem then reduces an arbitrary ideal to its prime-power factors. Throughout, k is a field and B is a commutative domain with a k -algebra structure; ideals of B are written I, P .

Theorem 206 (One-step filtration jump). *Let $I = (\pi)$ be a nonzero principal ideal of B , with B/I finite-dimensional over k , and let $n \geq 0$. Multiplication by π^n induces a k -linear isomorphism*

$$B/I \xrightarrow{\sim} I^n/I^{n+1},$$

so that $\dim_k(I^n/I^{n+1}) = \dim_k(B/I)$.

Proof. Since $I = (\pi)$ is principal, $I^n = (\pi^n)$ and $I^{n+1} = (\pi^{n+1})$, and $I^{n+1} = I \cdot I^n$ is a B -submodule of I^n . Multiplication by π^n is a B -linear map $\mu : B \rightarrow I^n$, surjective because every element of $I^n = (\pi^n)$ is a B -multiple of π^n . Composing with the projection $I^n \twoheadrightarrow I^n/I^{n+1}$ yields a surjective B -linear map $B \rightarrow I^n/I^{n+1}$. Its kernel is I : for $b \in B$, $\pi^n b \in I^{n+1} = (\pi^{n+1})$ means $\pi^n b = \pi^{n+1}c$ for some c , i.e. $\pi^n(b - \pi c) = 0$; as B is a domain and $\pi^n \neq 0$ (because $I \neq 0$), this forces $b = \pi c \in (\pi) = I$, and conversely every element of I maps into I^{n+1} . Hence μ descends to a B -linear isomorphism $B/I \cong I^n/I^{n+1}$. Restriction of scalars along $k \rightarrow B$ makes it k -linear, so the two k -vector spaces have equal dimension. □

Theorem 207 (Inductive step of the colength formula). *Let I be a nonzero principal ideal of B and $n \geq 0$, with B/I^{n+1} and B/I^n finite-dimensional over k . Then*

$$\dim_k(B/I^{n+1}) = \dim_k(B/I) + \dim_k(B/I^n).$$

Proof. Because $I^{n+1} \subseteq I^n$, the identity of B induces a surjective k -linear projection $B/I^{n+1} \twoheadrightarrow B/I^n$, whose kernel is the image of I^n in B/I^{n+1} , namely I^n/I^{n+1} . This gives a short exact sequence of finite-dimensional k -vector spaces

$$0 \longrightarrow I^n/I^{n+1} \longrightarrow B/I^{n+1} \longrightarrow B/I^n \longrightarrow 0.$$

By ??, $\dim_k(B/I^{n+1}) = \dim_k(I^n/I^{n+1}) + \dim_k(B/I^n)$, and by the filtration jump ??, $\dim_k(I^n/I^{n+1}) = \dim_k(B/I)$. Combining the two gives the claim. $\square$

Theorem 208 (Prime-power (DVR) colength formula). *Let $I = (\pi)$ be a nonzero principal ideal of B and $e \geq 0$, with B/I^e finite-dimensional over k . Then*

$$\dim_k(B/I^e) = e \cdot \dim_k(B/I).$$

Proof. Induction on e . For $e = 0$, I^0 is the unit ideal $(1) = B$, so B/I^0 is the zero module and $\dim_k(B/I^0) = 0 = 0 \cdot \dim_k(B/I)$. For the step, assume the formula for $e = n$; each quotient B/I^n is a quotient of B/I^{n+1} , hence finite-dimensional, so ?? applies and

$$\dim_k(B/I^{n+1}) = \dim_k(B/I) + \dim_k(B/I^n) = \dim_k(B/I) + n \cdot \dim_k(B/I) = (n+1) \cdot \dim_k(B/I).$$

$\square$

For the remaining two results B is a Dedekind domain, so every nonzero ideal factors uniquely into height-one primes. For a nonzero ideal I we write its factorization as $I = \prod_P P^{\text{ord}_P(I)}$, the product over the finitely many prime factors P of I and $\text{ord}_P(I) \geq 1$ the multiplicity of P .

Definition 209 (The Chinese remainder decomposition of a Dedekind quotient). Let B be a Dedekind domain, a k -algebra, and $I \neq 0$ an ideal with prime factorization $I = \prod_{P|I} P^{\text{ord}_P(I)}$. The canonical map

$$B/I \xrightarrow{\sim} \prod_{P|I} B/P^{\text{ord}_P(I)}$$

is an isomorphism of k -algebras.

Proof. Distinct prime factors $P \neq P'$ of I are nonzero primes of a Dedekind domain, hence maximal, hence comaximal, and powers of comaximal ideals remain comaximal; the Chinese remainder theorem therefore identifies $B/\prod_P P^{\text{ord}_P(I)}$ with the product of the $B/P^{\text{ord}_P(I)}$, by the canonical map. Each component of the canonical map is the quotient map $B/I \rightarrow B/P^{\text{ord}_P(I)}$, a k -algebra homomorphism, so the isomorphism is k -algebraic. $\square$

Theorem 210 (Multi-prime colength via the Chinese remainder theorem). *Let B be a Dedekind domain, a k -algebra, and let $I \neq 0$ be an ideal with each prime-power quotient $B/P^{\text{ord}_P(I)}$ finite-dimensional over k . Then*

$$\dim_k(B/I) = \sum_{P|I} \dim_k(B/P^{\text{ord}_P(I)}),$$

the sum ranging over the prime factors P of I .

Proof. The Chinese remainder decomposition ?? identifies B/I with $\prod_{P|I} B/P^{\text{ord}_P(I)}$ as k -vector spaces. The dimension of a finite product of vector spaces is the sum of the dimensions of the factors, which is the stated formula. $\square$

Theorem 211 (Dedekind colength formula, ord-weighted form). *Let B be a Dedekind domain, a k -algebra, and let $f \neq 0$ be an element whose height-one prime factors P of the principal ideal (f) are all principal (for instance when B is a principal ideal domain), with each prime-power quotient finite-dimensional over k . Then*

$$\dim_k(B/(f)) = \sum_{P|(f)} \text{ord}_P(f) \cdot \dim_k(B/P),$$

where $\text{ord}_P(f) = \text{ord}_P((f))$ is the multiplicity of P in the factorization of (f) .

Proof. Since $f \neq 0$, the ideal (f) is nonzero, and ?? gives $\dim_k(B/(f)) = \sum_{P|(f)} \dim_k(B/P^{\text{ord}_P(f)})$. Each prime factor P is a nonzero principal ideal by hypothesis, so the prime-power formula ?? applies to it and yields $\dim_k(B/P^{\text{ord}_P(f)}) = \text{ord}_P(f) \cdot \dim_k(B/P)$. Substituting into the sum gives the claim. $\square$

4.9 The Dedekind colength through localization at a prime

The ord-weighted formula ?? assumes each prime factor of (f) is principal — true in a principal ideal domain, but false in a general Dedekind domain. This section removes the principality hypothesis: each prime-power block B/P^e is transported to the localization of B at P , a discrete valuation ring, where the maximal ideal *is* principal and the prime-power formula ?? applies. The bookkeeping needed for the transport — that multiplicities of prime factors are computed by membership in prime powers, and are preserved by the localization — is proved along the way. Throughout, k is a field; B is a domain with a k -algebra structure, Dedekind where stated; and for a nonzero ideal $I \subseteq B$ of a Dedekind domain, $\text{ord}_P(I)$ denotes the multiplicity of the prime P in the factorization of I , with $\text{ord}_P(f) = \text{ord}_P((f))$ for an element $f \neq 0$.

Theorem 212 (Finiteness dévissage along principal prime powers). *Let B be a domain over k and $I = (\pi)$ a nonzero principal ideal with B/I finite-dimensional over k . Then B/I^n is finite-dimensional over k for every $n \geq 0$.*

Proof. Induction on n . For $n = 0$ the quotient $B/I^0 = B/B$ is zero. For the step, the projection $B/I^{n+1} \twoheadrightarrow B/I^n$ is surjective with kernel I^n/I^{n+1} , which is k -isomorphic to B/I by the filtration jump ??, hence finite-dimensional. Thus B/I^{n+1} is an extension of the finite-dimensional B/I^n (inductive hypothesis) by the finite-dimensional I^n/I^{n+1} , and is finite-dimensional. $\square$

Lemma 213 (The adic valuation counts the multiplicity). *Let B be a Dedekind domain, P a nonzero prime of B , and $f \neq 0$. Then the P -adic valuation of f equals the multiplicity of P in the prime factorization of the principal ideal (f) :*

$$v_P(f) = \text{ord}_P(f).$$

Proof. In a Dedekind domain every nonzero ideal factors uniquely into nonzero primes, and the P -adic valuation of a nonzero element f is by definition the exponent of P in the factorization of (f) . Uniqueness of the factorization identifies that exponent with the number of occurrences of P in the multiset of prime factors of (f) , which is $\text{ord}_P(f)$. $\square$

Lemma 214 (Membership in prime powers reads off the multiplicity). *Let B be a Dedekind domain, P a nonzero prime, $f \neq 0$ and $n \geq 0$. Then*

$$f \in P^n \iff n \leq \text{ord}_P(f).$$

Proof. Membership $f \in P^n$ means $(f) \subseteq P^n$, and in a Dedekind domain to contain is to divide, so this holds exactly when $v_P(f) \geq n$ — the valuation-theoretic characterization of the powers of P . By ??, $v_P(f) = \text{ord}_P(f)$, giving the claim. $\square$

Theorem 215 (Multiplicities are preserved by localization at the prime). *Let B be a Dedekind domain, P a nonzero prime of B , and S a Dedekind local ring which is a localization of B at P — for instance the localization B_P itself, a discrete valuation ring. Write $\mathfrak{m}$ for the maximal ideal of S . Then for every $f \neq 0$ in B ,*

$$\text{ord}_{\mathfrak{m}}(f \cdot S) = \text{ord}_P(f) :$$

the multiplicity of $\mathfrak{m}$ in the factorization of the extended principal ideal equals the multiplicity of P in the factorization of (f) .

Proof. The localization map $B \rightarrow S$ is injective (its kernel consists of elements killed by some $s \notin P$, and B is a domain), so the image of f in S is nonzero and both multiplicities are defined. Under the localization, the powers of $\mathfrak{m}$ contract to the powers of P : $\mathfrak{m}^n \cap B = P^n$ (the P -primary ideal P^n is saturated with respect to the complement of P). Hence for every $n \geq 0$,

$$f \cdot S \subseteq \mathfrak{m}^n \iff f \in P^n.$$

By ??, applied in S (a Dedekind domain, with $\mathfrak{m}$ its nonzero prime) and in B , the left side holds exactly when $n \leq \text{ord}_{\mathfrak{m}}(f \cdot S)$ and the right side exactly when $n \leq \text{ord}_P(f)$. Two natural numbers with the same lower sets are equal. $\square$

Definition 216 (The localization isomorphisms, k -linearly). Let B be a k -algebra, P a maximal ideal of B , and S a local k -algebra which is a localization of B at P compatibly with the k -structures, with maximal ideal $\mathfrak{m}$. For every $n \geq 0$ the canonical ring isomorphism

$$B/P^n \xrightarrow{\sim} S/\mathfrak{m}^n,$$

induced by the localization map, is k -linear; in particular ($n = 1$) the residue isomorphism $B/P \cong S/\mathfrak{m}$ is k -linear.

Proof. The composite $B \rightarrow S \rightarrow S/\mathfrak{m}^n$ kills P^n , and the induced map $B/P^n \rightarrow S/\mathfrak{m}^n$ is bijective: every element of the complement of P acts invertibly on B/P^n (the ideal P^n is P -primary with P maximal, so B/P^n is local with maximal ideal P/P^n , and elements outside P map to units), so the localization map factors through an inverse. The maps are k -algebra maps because the k -structure of S is that of B composed with the localization; hence the isomorphism is k -linear. $\square$

Theorem 217 (Per-prime finiteness through the localization). *Let B be a Dedekind domain over k , P a nonzero maximal ideal with residue B/P finite-dimensional over k , and S a domain over k which is a localization of B at P compatibly with the k -structures. Then B/P^n is finite-dimensional over k for every $n \geq 0$.*

Proof. The localization S of the Dedekind domain B at the nonzero prime P is a discrete valuation ring; its maximal ideal $\mathfrak{m}$ is nonzero and principal. By ?? the residue $S/\mathfrak{m} \cong B/P$ is finite-dimensional, so the dévissage ??, applied in S to the nonzero principal ideal $\mathfrak{m}$, makes every $S/\mathfrak{m}^n$ finite-dimensional; transporting back along ?? finishes. $\square$

Theorem 218 (The per-prime colength collapse). *In the situation of ??,*

$$\dim_k(B/P^n) = n \cdot \dim_k(B/P) \quad (n \geq 0).$$

Proof. As in ??, S is a discrete valuation ring with nonzero principal maximal ideal $\mathfrak{m}$, the quotients $S/\mathfrak{m}^n$ are finite-dimensional (??), and the prime-power formula ?? in S gives $\dim_k(S/\mathfrak{m}^n) = n \cdot \dim_k(S/\mathfrak{m})$. The k -linear identifications $B/P^n \cong S/\mathfrak{m}^n$ and $B/P \cong S/\mathfrak{m}$ (??) transport this to the claim. $\square$

Lemma 219 (Prime factors of a principal ideal). *Let B be a Dedekind domain and $f \in B$. Every prime factor P of the ideal (f) is a nonzero prime ideal of B containing f .*

Proof. A factor occurring in the prime factorization of (f) is by definition a nonzero prime ideal, and it divides (f) ; in a Dedekind domain to divide is to contain, so $(f) \subseteq P$, i.e. $f \in P$. $\square$

Theorem 220 (Finiteness of the Dedekind colength module). *Let B be a Dedekind domain over k and $f \neq 0$ an element such that the residue B/P is finite-dimensional over k for every prime factor P of (f) . Then $B/(f)$ is finite-dimensional over k .*

Proof. The ideal (f) is nonzero, so the Chinese remainder decomposition ?? identifies $B/(f)$ with the finite product $\prod_{P|(f)} B/P^{\text{ord}_P(f)}$. Each factor is finite-dimensional: P is a nonzero prime (??), hence maximal, with finite-dimensional residue by hypothesis, so ?? applies with $S = B_P$ (a domain, being a subring of the fraction field of B). A finite product of finite-dimensional spaces is finite-dimensional. $\square$

Theorem 221 (Dedekind colength formula, localized form). *Let B be a Dedekind domain over k and $f \neq 0$ an element such that the residue B/P is finite-dimensional over k for every prime factor P of (f) . Then*

$$\dim_k(B/(f)) = \sum_{P|(f)} \text{ord}_P(f) \cdot \dim_k(B/P),$$

the sum ranging over the prime factors of (f) . No principality of the primes P is assumed: this is the generalization of ?? to arbitrary Dedekind domains.

Proof. Each prime factor P of (f) is nonzero (??), hence maximal, and ?? with $S = B_P$ makes each prime-power quotient $B/P^{\text{ord}_P(f)}$ finite-dimensional. The multi-prime formula ?? then gives $\dim_k(B/(f)) = \sum_{P|(f)} \dim_k(B/P^{\text{ord}_P(f)})$, and the per-prime collapse ?? (again through $S = B_P$) evaluates each summand as $\text{ord}_P(f) \cdot \dim_k(B/P)$. $\square$

4.10 The diagonal ideal of an étale-coordinatised algebra

Let k be a commutative ring and B a commutative k -algebra carrying a compatible structure of algebra over the polynomial ring $k[X]$, that is, a tower of algebra maps $k \rightarrow k[X] \rightarrow B$. Write $u :=$ the image of X in B for the *étale coordinate*. The two tensor squares $B \otimes_k B$ and $B \otimes_{k[X]} B$ carry the two multiplication maps $\mu_k: B \otimes_k B \rightarrow B$ and $\mu_{k[X]}: B \otimes_{k[X]} B \rightarrow B$, $x \otimes y \mapsto xy$; their kernels are the *k -diagonal ideal* and the *$k[X]$ -diagonal ideal*. The geometric picture is that $\text{Spec } B$ is a standard-smooth chart of a relative curve, u cuts it out over $\mathbb{A}^1 = \text{Spec } k[X]$, and on a basic open containing the whole diagonal of $\text{Spec } B \times_k \text{Spec } B$ that diagonal is the vanishing locus of the single equation $u \otimes 1 - 1 \otimes u$. This section is the pure commutative-algebra core computing the k -diagonal ideal after localising away from an idempotent lift.

Definition 222 (The diagonal equation and the base-change collapse). The *diagonal equation* is

$$\delta := u \otimes 1 - 1 \otimes u \in B \otimes_k B.$$

The *base-change collapse* is the k -algebra homomorphism

$$\pi: B \otimes_k B \longrightarrow B \otimes_{k[X]} B, \quad x \otimes y \mapsto x \otimes y,$$

the surjection identifying the two tensor squares. It satisfies:

- $\mu_k(\delta) = 0$: the diagonal equation lies in the k -diagonal ideal;
- $\pi(\delta) = 0$: the collapse kills the diagonal equation, because $u \otimes 1 = 1 \otimes u$ already in $B \otimes_{k[X]} B$ (here $u = X \cdot 1$ is a scalar of the tensor product);
- $\mu_{k[X]} \circ \pi = \mu_k$: the two multiplications agree through the collapse.

Proof. The three identities are checked on pure tensors. For $\mu_k(\delta)$: $\mu_k(u \otimes 1) - \mu_k(1 \otimes u) = u - u = 0$. For $\pi(\delta)$: $u \otimes 1 = (X \cdot 1) \otimes 1 = 1 \otimes (X \cdot 1) = 1 \otimes u$ in $B \otimes_{k[X]} B$, since X is a scalar of the $k[X]$ -tensor product and passes across the tensor. For the factorisation: both sides send $x \otimes y$ to xy , and $\pi, \mu_{k[X]}, \mu_k$ are additive, so agreement on pure tensors suffices. $\square$

4.10.1 The kernel of the collapse (deliverable (a))

Lemma 223 (Polynomial telescoping). *Write $g: k[X] \rightarrow B$ for the coordinate map, so $g(p) = p(u)$. For every polynomial $p \in k[X]$,*

$$g(p) \otimes 1 - 1 \otimes g(p) \in (\delta),$$

the ideal generated by the diagonal equation.

Proof. Since $g(p) = p(u)$ is the evaluation of p at u , the two tensor factors are themselves evaluations: $g(p) \otimes 1 = p(u \otimes 1)$ and $1 \otimes g(p) = p(1 \otimes u)$, each obtained by applying the algebra map $a \mapsto p(a)$ to the element $u \otimes 1$ respectively $1 \otimes u$ of $B \otimes_k B$ (using that the two inclusions $B \rightarrow B \otimes_k B$ are algebra maps). The difference of the evaluations of a polynomial at two points a, b is divisible by $a - b$; here $a - b = (u \otimes 1) - (1 \otimes u) = \delta$, so $p(u \otimes 1) - p(1 \otimes u) \in (\delta)$. $\square$

Lemma 224 (The section of the collapse). *There is a $k[X]$ -algebra homomorphism*

$$\psi: B \otimes_{k[X]} B \longrightarrow (B \otimes_k B)/(\delta), \quad x \otimes y \mapsto [x \otimes y],$$

and it is a section of the collapse in the sense that $\psi \circ \pi$ is the quotient map $B \otimes_k B \rightarrow (B \otimes_k B)/(\delta)$.

Proof. The right inclusion $b \mapsto [1 \otimes b]$ is a ring homomorphism $B \rightarrow (B \otimes_k B)/(\delta)$; it is moreover a $k[X]$ -algebra map, because the two ways of scaling by $p \in k[X]$ — through $[1 \otimes g(p)]$ and through $[g(p) \otimes 1]$ (the ambient $k[X]$ -structure) — agree in the quotient by the telescoping Lemma ???. Together with the left inclusion $x \mapsto [x \otimes 1]$, which commutes with it, the universal property of $B \otimes_{k[X]} B$ produces the $k[X]$ -algebra map ψ with $\psi(x \otimes y) = [x \otimes y]$. Both $\psi \circ \pi$ and the quotient map send a pure tensor $x \otimes y$ to $[x \otimes y]$, and both are additive, so they coincide. $\square$

Theorem 225 (The kernel of the base-change collapse). *The kernel of the collapse $\pi: B \otimes_k B \rightarrow B \otimes_{k[X]} B$ is exactly the principal ideal generated by the diagonal equation:*

$$\ker \pi = (\delta) = (u \otimes 1 - 1 \otimes u).$$

Proof. The inclusion $(\delta) \subseteq \ker \pi$ is $\pi(\delta) = 0$ (Definition ??). Conversely, if $\pi(z) = 0$ then, applying the section ψ of Lemma ??, the class of z in the quotient is $[z] = \psi(\pi(z)) = \psi(0) = 0$, so $z \in (\delta)$. $\square$

4.10.2 The idempotent generator and the localised principality (deliverables (b), (c))

Theorem 226 (Idempotent generator of the $k[X]$ -diagonal ideal). *Suppose $k[X] \rightarrow B$ is formally unramified and essentially of finite type. Then the $k[X]$ -diagonal ideal $\ker \mu_{k[X]}$ is generated by an idempotent element, and this idempotent is $\pi(\text{elift})$ for some lift $\text{elift} \in B \otimes_k B$:*

$$\ker \mu_{k[X]} = (\pi \text{elift}), \quad (\pi \text{elift})^2 = \pi \text{elift}.$$

Proof. The $k[X]$ -diagonal ideal $I' := \ker \mu_{k[X]}$ has $I'/I'^2 \cong \Omega_{B/k[X]}$, the module of Kähler differentials, which vanishes because $k[X] \rightarrow B$ is formally unramified; hence I' is idempotent, $I' = I'^2$. Essential finite type makes I' finitely generated. A finitely generated idempotent ideal is generated by an idempotent element e ($I' = (e)$ with $e^2 = e$). Finally π is surjective, so $e = \pi(\text{elift})$ for some $\text{elift} \in B \otimes_k B$. $\square$

Theorem 227 (Localised principality of the k -diagonal ideal). *Let $\text{elift} \in B \otimes_k B$ be a lift with $\pi(\text{elift})$ idempotent and $\ker \mu_{k[X]} = (\pi \text{elift})$. Then for any localisation L of $B \otimes_k B$ away from $1 - \text{elift}$, the k -diagonal ideal, extended to L , is the principal ideal generated by the diagonal equation:*

$$(\ker \mu_k) \cdot L = (\delta) \cdot L.$$

Proof. First, in L the lift is a multiple of δ . Indeed $\text{elift}(1 - \text{elift}) \in \ker \pi = (\delta)$ (by Theorem ??, since $\pi(\text{elift}(1 - \text{elift})) = e(1 - e) = 0$ as $e = \pi(\text{elift})$ is idempotent), while $1 - \text{elift}$ is a unit of L ; dividing shows $\text{elift} \in (\delta) \cdot L$.

Now $(\delta) \cdot L \subseteq (\ker \mu_k) \cdot L$ because $\delta \in \ker \mu_k$ ($\mu_k(\delta) = 0$). For the reverse inclusion it suffices, by $\ker \mu_k \subseteq (\delta) + (\text{elift})$, that both (δ) and (elift) map into $(\delta) \cdot L$; the first is clear and the second is the previous paragraph. The containment $\ker \mu_k \subseteq (\delta) + (\text{elift})$ holds because for x with $\mu_k(x) = 0$ one has $\pi(x) \in \ker \mu_{k[X]} = (\pi \text{elift})$ (using $\mu_{k[X]} \circ \pi = \mu_k$), so $\pi(x) = \pi(\text{elift} \cdot w)$ for some w ; then $x - \text{elift} \cdot w \in \ker \pi = (\delta)$ and $x = (x - \text{elift} \cdot w) + \text{elift} \cdot w \in (\delta) + (\text{elift})$. $\square$

Theorem 228 (Packaged diagonal localisation). *Suppose $k[X] \rightarrow B$ is formally unramified and essentially of finite type. Then there is a lift $\text{elift} \in B \otimes_k B$ with $\mu_k(\text{elift}) = 0$, with $\pi(\text{elift})$ idempotent generating $\ker \mu_{k[X]}$, and with the localised principality of Theorem ?? holding on the basic open away from $1 - \text{elift}$.*

Proof. Take elift from Theorem ??. Then $\mu_k(\text{elift}) = \mu_{k[X]}(\pi \text{elift}) = 0$, since $\pi(\text{elift}) \in \ker \mu_{k[X]}$ and $\mu_{k[X]} \circ \pi = \mu_k$ (Definition ??); the idempotent-generation is Theorem ??, and the localised principality is Theorem ??. $\square$

4.11 The diagonal-equation regularity engine

The regularity input needed alongside the diagonal ideal is the fact that the diagonal equation is a nonzerodivisor, and this holds uniformly in an arbitrary *second* tensor factor, with no integrality, fibre, or transcendence hypothesis — so it survives at every point of a non-integral product.

Lemma 229 (A flat ring homomorphism preserves nonzerodivisors). *Let $f: R \rightarrow S$ be a flat homomorphism of commutative rings. If $r \in R$ is a nonzerodivisor, then $f(r) \in S$ is a nonzerodivisor.*

Proof. View S as an R -module through f ; flatness makes it a flat R -module. Multiplication by a nonzerodivisor r is injective on a flat module, i.e. r acts regularly on S . Since $r \cdot s = f(r)s$ for $s \in S$, the map $s \mapsto f(r)s$ is injective, and by commutativity $f(r)$ is a nonzerodivisor of S . $\square$

Theorem 230 (The regularity engine). *Assume in addition that B is flat over $k[X]$. Then for any k -algebra A and any $b \in A$, the element*

$$u \otimes 1 - 1 \otimes b \in B \otimes_k A$$

is a nonzerodivisor. In particular (taking $A = B$, $b = u$) the diagonal equation $u \otimes 1 - 1 \otimes u$ is a nonzerodivisor of $B \otimes_k B$.

Proof. The linear polynomial $X - b \in A[X]$ is monic, hence a nonzerodivisor of $A[X]$. Consider the composite

$$\Theta: A[X] \xrightarrow{\sim} A \otimes_k k[X] \xrightarrow{\text{id} \otimes (k[X] \rightarrow B)} A \otimes_k B \xrightarrow{\sim} B \otimes_k A,$$

which sends $X - b$ to $u \otimes 1 - 1 \otimes b$. Its middle arrow is the base change along A of the flat coordinate map $k[X] \rightarrow B$, hence flat; the two outer arrows are algebra isomorphisms (the polynomial-tensor identification and the commutativity of the tensor product), hence flat; so Θ is a flat ring homomorphism. By Lemma ?? it carries the nonzerodivisor $X - b$ to a nonzerodivisor, which is $u \otimes 1 - 1 \otimes b$. $\square$

4.12 Vanishing propagation via residue fields

Let R be a commutative ring. For a prime ideal $\mathfrak{p} \subseteq R$ write $R_{\mathfrak{p}}$ for the localisation, $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ for the residue field, and $M_{\mathfrak{p}}$ for the localisation of an R -module M . The *fibre* of M at $\mathfrak{p}$ is the $\kappa(\mathfrak{p})$ -vector space $M \otimes_R \kappa(\mathfrak{p})$, and the *support* of M is $\text{Supp } M = \{\mathfrak{p} \text{ prime} : M_{\mathfrak{p}} \neq 0\}$. This section proves that for a finitely generated module the vanishing of all fibres forces the module to vanish, and that the fibre-vanishing locus is open in $\text{Spec } R$. Applied with $M = H^1$ of the relative curve (Chapter ??), these are the vanishing-propagation and openness steps of the exchange-property analysis for the Picard functor.

Lemma 231 (A module with vanishing localisations vanishes). *Let M be an R -module with $M_{\mathfrak{p}} = 0$ for every prime $\mathfrak{p}$. Then $M = 0$. (No finiteness hypothesis is needed.)*

Proof. Suppose $m \in M$ is nonzero. Its annihilator $\text{Ann}(m) = \{r \in R : rm = 0\}$ is an ideal not containing 1, hence contained in a maximal ideal $\mathfrak{p}$. If $m/1 = 0$ in $M_{\mathfrak{p}}$, some $s \notin \mathfrak{p}$ satisfies $sm = 0$, contradicting $\text{Ann}(m) \subseteq \mathfrak{p}$; so $M_{\mathfrak{p}} \neq 0$. $\square$

Lemma 232 (The Nakayama fibre criterion). *Let M be a finitely generated R -module and $\mathfrak{p}$ a prime. Then $M \otimes_R \kappa(\mathfrak{p}) = 0$ if and only if $M_{\mathfrak{p}} = 0$, i.e. $\mathfrak{p} \notin \text{Supp } M$.*

Proof. Localisation commutes with the tensor product and $\kappa(\mathfrak{p})$ is an $R_{\mathfrak{p}}$ -module, so $M \otimes_R \kappa(\mathfrak{p}) \cong M_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} \kappa(\mathfrak{p}) \cong M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}}$ (the last identification because $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ and $N \otimes_A A/I \cong N/IN$). If $M_{\mathfrak{p}} = 0$ the fibre certainly vanishes. Conversely suppose $M_{\mathfrak{p}} = \mathfrak{p}M_{\mathfrak{p}}$. The module $M_{\mathfrak{p}}$ is finitely generated over the local ring $R_{\mathfrak{p}}$ (the images of generators of M generate), whose maximal ideal is $\mathfrak{p}R_{\mathfrak{p}}$. If $M_{\mathfrak{p}} \neq 0$, choose generators $x_1, \dots, x_n$ of $M_{\mathfrak{p}}$ with $n \geq 1$ minimal. Expanding $x_n \in \mathfrak{p}M_{\mathfrak{p}}$ in the generators gives $x_n = \sum_{i \leq n} a_i x_i$ with all $a_i \in \mathfrak{p}R_{\mathfrak{p}}$, whence $(1 - a_n)x_n = \sum_{i < n} a_i x_i$. But $1 - a_n$ lies outside the maximal ideal, so it is a unit, and x_n lies in the span of $x_1, \dots, x_{n-1}$ — contradicting minimality. Hence $M_{\mathfrak{p}} = 0$. $\square$

Theorem 233 (Vanishing propagation). *Let M be a finitely generated R -module with $M \otimes_R \kappa(\mathfrak{p}) = 0$ for every prime $\mathfrak{p}$. Then $M = 0$: fibrewise vanishing over the whole base propagates to global vanishing.*

Proof. By ?? every localisation $M_{\mathfrak{p}}$ vanishes; by ??, $M = 0$. $\square$

Lemma 234 (The fibre-vanishing locus is the complement of the support). *For a finitely generated R -module M ,*

$$\{\mathfrak{p} \in \operatorname{Spec} R : M \otimes_R \kappa(\mathfrak{p}) = 0\} = \operatorname{Spec} R \setminus \operatorname{Supp} M.$$

Proof. A pointwise restatement of ??. $\square$

Theorem 235 (Openness of the fibre-vanishing locus). *For a finitely generated R -module M , the fibre-vanishing locus $\{\mathfrak{p} : M \otimes_R \kappa(\mathfrak{p}) = 0\}$ is open in $\operatorname{Spec} R$.*

Proof. By ?? the locus is the complement of $\operatorname{Supp} M$, so it suffices to prove $\operatorname{Supp} M = V(\operatorname{Ann} M)$ for M generated by $x_1, \dots, x_n$, a closed set. If $\operatorname{Ann} M \not\subseteq \mathfrak{p}$, pick $s \in \operatorname{Ann} M \setminus \mathfrak{p}$: then s kills M , so $M_{\mathfrak{p}} = 0$. Conversely if $M_{\mathfrak{p}} = 0$, then each $x_i/1 = 0$ in $M_{\mathfrak{p}}$, so $s_i x_i = 0$ for some $s_i \notin \mathfrak{p}$; the product $s = s_1 \cdots s_n$ lies outside the prime $\mathfrak{p}$ and kills every generator, so $s \in \operatorname{Ann} M \setminus \mathfrak{p}$. Thus a prime lies off the support exactly when it fails to contain $\operatorname{Ann} M$, i.e. $\operatorname{Supp} M = V(\operatorname{Ann} M)$. (The second computation is the classical localisation step: a finitely generated module vanishing at $\mathfrak{p}$ vanishes on the basic open neighbourhood $D(s)$.) $\square$

4.13 Split rigidity of a surjective two-term complex

Throughout this section R is a commutative ring and $\delta : A \rightarrow B$ a map of R -modules — a two-term complex, with $H^0 := \ker \delta$ and $H^1 := B/\delta(A)$ its cohomology. In the intended application (Chapter ??) δ is the restriction-difference map of an affine two-cover and H^1 is degree-one cohomology; everything here is pure module algebra. The theme is *rigidity*: once $H^1 = 0$ — that is, once δ is surjective — the formation of H^0 behaves as if the complex were split. Without the surjectivity, kernels do not commute with tensor products: for $\delta = p \cdot (-) : \mathbb{Z} \rightarrow \mathbb{Z}$ with p prime, $\ker \delta = 0$, yet after $\otimes \mathbb{Z}/p$ the map becomes zero and acquires the kernel $\mathbb{Z}/p$.

Lemma 236 (Right-exactness of the tensor product). *Provided by Mathlib. Let $M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of R -modules and P any R -module. Then the sequences $M' \otimes P \rightarrow M \otimes P \rightarrow M'' \otimes P \rightarrow 0$ and $P \otimes M' \rightarrow P \otimes M \rightarrow P \otimes M'' \rightarrow 0$ are exact.*

Lemma 237 (A flat cokernel keeps the kernel inclusion injective under tensoring). *Provided by Mathlib. Let $0 \rightarrow K \xrightarrow{\iota} A \xrightarrow{\delta} B \rightarrow 0$ be a short exact sequence of R -modules with B flat. Then for every R -module P the map $\operatorname{id}_P \otimes \iota : P \otimes K \rightarrow P \otimes A$ is injective.*

Lemma 238 (A vanishing cokernel means surjectivity). *If $H^1 = B/\delta(A) = 0$, then δ is surjective.*

Proof. Every $b \in B$ has class zero in the zero quotient, i.e. lies in $\delta(A)$. $\square$

Lemma 239 (The cokernel of a surjective complex base-changes to zero). *Let $\delta : A \rightarrow B$ be surjective (equivalently, $H^1 = 0$; ??). Then for every R -module P the map $\delta \otimes \operatorname{id}_P : A \otimes P \rightarrow B \otimes P$ is surjective; equivalently, the cokernel of the base-changed complex vanishes for every coefficient module: H^1 base-changes to zero.*

Proof. $B \otimes P$ is generated by the pure tensors $b \otimes p$; writing $b = \delta(a)$ exhibits each as $(\delta \otimes \operatorname{id})(a \otimes p)$. The image, a submodule containing a generating set, is everything. $\square$

Theorem 240 (Base change of H^0 with module coefficients). *Let $\delta : A \rightarrow B$ be surjective (equivalently $H^1 = 0$) with B flat, and set $K = \ker \delta$. For every R -module P , the natural map $K \otimes P \rightarrow \ker(\delta \otimes \operatorname{id}_P)$ induced by the inclusion $K \subseteq A$ is an isomorphism: the formation of H^0 commutes with $\otimes P$ for every P .*

Proof. Since δ is surjective, $0 \rightarrow K \xrightarrow{\iota} A \xrightarrow{\delta} B \rightarrow 0$ is exact. The natural map is well defined: $(\delta \otimes \text{id}) \circ (\iota \otimes \text{id}) = (\delta\iota) \otimes \text{id} = 0$, so $\iota \otimes \text{id}$ lands in the kernel. *Injectivity:* the cokernel B of ι is flat, so $\iota \otimes \text{id}_P$ remains injective (??), and the natural map is its corestriction. *Surjectivity onto the kernel:* by right-exactness (??) applied to the exact $K \rightarrow A \rightarrow B \rightarrow 0$, the sequence $K \otimes P \rightarrow A \otimes P \rightarrow B \otimes P$ is exact at $A \otimes P$; an element of $\ker(\delta \otimes \text{id})$ therefore comes from $K \otimes P$. (Flatness is not used in this half.) $\square$

Theorem 241 (Base change of H^0 along a ring map). *Let $\delta : A \rightarrow B$ be surjective with B flat, and let R' be a commutative R -algebra. Write $\delta_{R'} : R' \otimes_R A \rightarrow R' \otimes_R B$ for the base-changed R' -linear map. Then the natural R' -linear map*

$$R' \otimes_R \ker \delta \longrightarrow \ker \delta_{R'}$$

is an isomorphism: the formation of H^0 commutes with every base change $R \rightarrow R'$.

Proof. The underlying additive map is that of ?? with $P = R'$ (with the tensor factors flipped), hence bijective; and it is R' -linear because it is induced by $\text{id}_{R'} \otimes \iota$, which commutes with the R' -action carried by the left tensor factor. $\square$

Theorem 242 (Flatness of H^0 : the splitting argument). *Let $\delta : A \rightarrow B$ be surjective with A flat and B projective. Then $K = \ker \delta$ is flat.*

Proof. Projectivity of B lifts id_B through the surjection δ : there is $\sigma : B \rightarrow A$ with $\delta\sigma = \text{id}_B$, so the complex splits. The map $\pi := \text{id}_A - \sigma\delta$ lands in K ($\delta\pi = \delta - \delta\sigma\delta = 0$) and restricts to the identity on K (if $\delta x = 0$ then $\pi x = x$); moreover $A = K \oplus \sigma(B)$: every a is $\pi a + \sigma(\delta a)$, and if $x = \sigma b \in K \cap \sigma(B)$ then $0 = \delta x = b$, so $x = 0$. A direct summand of a flat module is flat: for an injective map $M' \rightarrow M$, the map $K \otimes M' \rightarrow K \otimes M$ is a direct summand of the injective map $A \otimes M' \rightarrow A \otimes M$ (the tensor product distributes over direct sums), hence injective. So K is flat. $\square$

Lemma 243 (Finite modules over a Noetherian ring are finitely presented). *Provided by Mathlib. Over a Noetherian ring R , every finitely generated module is finitely presented: the kernel of any surjection $R^n \rightarrow M$ is again finitely generated.*

Lemma 244 (Finitely presented flat modules are projective). *Provided by Mathlib. A finitely presented flat module over a commutative ring is projective.*

Theorem 245 (H^0 is finite projective). *Let R be Noetherian and $\delta : A \rightarrow B$ surjective with A flat, B projective, and $K = \ker \delta$ finitely generated. Then K is a finite projective R -module.*

Proof. K is flat (??) and finitely presented (??); finitely presented flat modules are projective (??). $\square$

Theorem 246 (Base change commutes with cokernels). *Let $f : M \rightarrow N$ be an R -linear map and S a commutative R -algebra. There is an S -linear equivalence*

$$S \otimes_R (N/\text{range } f) \cong (S \otimes_R N)/\text{range}(f_S), \quad s \otimes [n] \mapsto [s \otimes n],$$

where $f_S = \text{id}_S \otimes f$ is the base-changed map.

Proof. Apply $\text{id}_S \otimes -$ to the exact sequence $M \xrightarrow{f} N \xrightarrow{q} N/\text{range } f \rightarrow 0$ (q the quotient map): by right-exactness (??) the sequence $S \otimes M \rightarrow S \otimes N \rightarrow S \otimes (N/\text{range } f) \rightarrow 0$ is exact. So $\text{id}_S \otimes q$ is surjective with kernel $\text{range}(f_S)$, and the first isomorphism theorem gives an S -linear equivalence $(S \otimes N)/\text{range}(f_S) \cong S \otimes (N/\text{range } f)$, inverse to the displayed one; on classes of pure tensors it sends $[s \otimes n]$ to $s \otimes [n]$. $\square$

4.14 Duality for finite projective modules

For an R -module M write $M^\vee = \text{Hom}_R(M, R)$ for its dual. The *contraction map*

$$c_M : M^\vee \otimes_R N \longrightarrow \text{Hom}_R(M, N), \quad \psi \otimes n \longmapsto (x \mapsto \psi(x)n)$$

is natural in both variables. This section upgrades the classical fact that c_M is an isomorphism for finite free M to finite *projective* M , by a retract argument, and derives the duality $M \otimes P \cong \text{Hom}(M^\vee, P)$ — the form in which the representing module of the exchange isomorphism is consumed (Chapter ??).

Lemma 247 (Contraction is an equivalence for finite free modules). *Provided by Mathlib. For a finite free R -module F and every R -module N , the contraction map $c_F : F^\vee \otimes N \rightarrow \text{Hom}(F, N)$ is an isomorphism.*

Theorem 248 (Contraction is an equivalence for finite projective modules). *For a finite projective R -module M and every R -module N , the contraction map $c_M : M^\vee \otimes N \rightarrow \text{Hom}(M, N)$ is an isomorphism.*

Proof. A finite projective module is a retract of a finite free one: generators give a surjection $f : F := R^n \rightarrow M$, and projectivity lifts id_M through f to $g : M \rightarrow F$ with $f \circ g = \text{id}_M$. Contraction is natural in the module variable; concretely, on pure tensors one checks the two identities

$$c_M \circ (g^\vee \otimes \text{id}_N) = (- \circ g) \circ c_F, \quad (- \circ f) \circ c_M = c_F \circ (f^\vee \otimes \text{id}_N),$$

both sides of the first sending $\psi \otimes n$ to $x \mapsto \psi(gx)n$, and of the second to $x \mapsto \psi(fx)n$. Define $\Phi : \text{Hom}(M, N) \rightarrow M^\vee \otimes N$ by $\Phi(h) := (g^\vee \otimes \text{id}_N)(c_F^{-1}(h \circ f))$, using ??. Then $c_M(\Phi h) = c_F(c_F^{-1}(h \circ f)) \circ g = h \circ f \circ g = h$ by the first identity; and $\Phi(c_M y) = (g^\vee \otimes \text{id})((f^\vee \otimes \text{id})y) = ((g^\vee \circ f^\vee) \otimes \text{id})y = y$ by the second identity and $g^\vee \circ f^\vee = (f \circ g)^\vee = \text{id}$. So Φ is a two-sided inverse of c_M . $\square$

Theorem 249 (Duality: $M \otimes P \cong \text{Hom}(M^\vee, P)$). *For a finite projective R -module M and every R -module P , there is a natural equivalence*

$$M \otimes_R P \cong \text{Hom}_R(M^\vee, P) :$$

the functor $P \mapsto M \otimes P$ is represented, in dual-Hom form, by $M^\vee$.

Proof. Keep $f : F \rightarrow M$, $g : M \rightarrow F$ with $fg = \text{id}_M$ as in the proof of ??. First, $M^\vee$ is again finite projective: $f^\vee : M^\vee \rightarrow F^\vee$ and $g^\vee : F^\vee \rightarrow M^\vee$ satisfy $g^\vee \circ f^\vee = \text{id}$, so $M^\vee$ is a retract — hence a direct summand and a quotient — of $F^\vee \cong R^n$. Second, M is reflexive: the evaluation map $\text{ev}_M : M \rightarrow M^{\vee\vee}$, $\text{ev}_M(x)(\psi) = \psi(x)$, is an isomorphism. For finite free F this is the dual-basis computation; in general, naturality $\text{ev}_F \circ u = u^{\vee\vee} \circ \text{ev}_M$ (for $u : M \rightarrow F$) makes $\beta := f \circ \text{ev}_F^{-1} \circ g^{\vee\vee}$ a two-sided inverse: $\beta \circ \text{ev}_M = f \circ \text{ev}_F^{-1} \circ \text{ev}_F \circ g = fg = \text{id}$, and $\text{ev}_M \circ \beta = f^{\vee\vee} \circ \text{ev}_F \circ \text{ev}_F^{-1} \circ g^{\vee\vee} = (fg)^{\vee\vee} = \text{id}$. Now compose:

$$M \otimes P \cong M^{\vee\vee} \otimes P = (M^\vee)^\vee \otimes P \cong \text{Hom}(M^\vee, P),$$

the first map by ev_M , the second by ?? applied to the finite projective module $M^\vee$. $\square$

4.15 Transport of kernels and of chart-lattice finiteness

Two small transport toolkits consumed by the sheaf-level assembly of the rigid engine (Chapter ??). For an endomorphism e of an S -module M , write M_e for M viewed as an $S[t]$ -module with t acting through e — the *chart lattice* of e ; “ M_e is finite over $S[t]$ ” is the finiteness hypothesis under which the coherence theorems of two-lattice pairs fire (??, ??).

Lemma 250 (Kernel and surjectivity transport along a commuting square). *Let $\delta : A \rightarrow B$ and $\delta' : A' \rightarrow B'$ be S -linear maps and $e_1 : A \cong A'$, $e_2 : B \cong B'$ S -linear equivalences with $e_2 \circ \delta = \delta' \circ e_1$. Then e_1 restricts to an S -linear equivalence $\ker \delta \cong \ker \delta'$, and δ is surjective if and only if δ' is.*

Proof. If $\delta a = 0$ then $\delta'(e_1 a) = e_2(\delta a) = 0$, so e_1 maps $\ker \delta$ into $\ker \delta'$; symmetrically e_1^{-1} maps $\ker \delta'$ into $\ker \delta$, using $\delta = e_2^{-1} \circ \delta' \circ e_1$. The two restrictions are mutually inverse and S -linear. For surjectivity: given $b' \in B'$, pick a with $\delta a = e_2^{-1} b'$; then $\delta'(e_1 a) = e_2(\delta a) = b'$; the converse is symmetric. $\square$

Lemma 251 (Chart-lattice transport along an equivariant equivalence). *Let e, e' be endomorphisms of the S -modules M, M' and $\varphi : M \cong M'$ an S -linear equivalence intertwining them, $\varphi \circ e = e' \circ \varphi$. Then φ is an isomorphism of $S[t]$ -modules $M_e \cong M'_{e'}$; in particular, if M_e is a finite $S[t]$ -module then so is $M'_{e'}$.*

Proof. The $S[t]$ -action on M_e is generated over the S -action by the action of t , which is e ; an S -linear map intertwining e with e' therefore commutes with the action of every polynomial, i.e. is $S[t]$ -linear. Being bijective, φ is an $S[t]$ -isomorphism, and finiteness transports along it (the image of a finite generating set generates). $\square$

Lemma 252 (Chart-lattice finiteness from a finite ring map). *Let k be a commutative ring, A a commutative ring which is also a k -module, and $\varphi : k[t] \rightarrow A$ a finite ring homomorphism. Let e be a k -linear endomorphism of A acting as multiplication by $\varphi(t)$, and suppose the k -scalar action is multiplication by the constants, $c \bullet a = \varphi(c) \cdot a$. Then the polynomial action of $p \in k[t]$ on A_e is multiplication by $\varphi(p)$, and A_e is a finite $k[t]$ -module.*

Proof. For the action formula, induct on the monomials of p : the action of ct^n on a is $c \bullet e^n(a) = \varphi(c) \varphi(t)^n a = \varphi(ct^n) a$, using that e^n is multiplication by $\varphi(t)^n$; sums of monomials act by sums. Consequently the $k[t]$ -module A_e is exactly A viewed as a $k[t]$ -module through φ , which is finite by hypothesis: a finite generating set of A over $k[t]$ through φ generates A_e . $\square$

Theorem 253 (Base change of chart-lattice finiteness). *Let k be a commutative ring, R a commutative k -algebra, M a k -module and e a k -linear endomorphism of M . If M_e is a finite $k[t]$ -module, then $(R \otimes_k M)_{\text{id} \otimes e}$ is a finite $R[t]$ -module.*

Proof. First, base change of the polynomial action on pure tensors: for $p \in k[t]$ and $m \in M$,

$$(p^R) \cdot (1 \otimes m) = 1 \otimes (p \cdot m),$$

where $p^R \in R[t]$ is p with coefficients pushed into R — by induction on monomials, since $(\text{id} \otimes e)^n(1 \otimes m) = 1 \otimes e^n(m)$ and the k -scalars pass through the tensor sign. Now let $z_1, \dots, z_r$ generate M_e over $k[t]$; we claim the $1 \otimes z_i$ generate $(R \otimes_k M)_{\text{id} \otimes e}$ over $R[t]$. Every element of $R \otimes_k M$ is a finite sum of tensors $r \otimes m$, and $r \otimes m = (rt^0) \cdot (1 \otimes m)$ with $rt^0 \in R[t]$ acting as the constant r ; so it suffices to reach the $1 \otimes m$. Write $m = \sum_i p_i \cdot z_i$ with $p_i \in k[t]$; then by the displayed formula and additivity, $1 \otimes m = \sum_i p_i^R \cdot (1 \otimes z_i)$, an $R[t]$ -linear combination of the claimed generators. $\square$

4.16 The point-fiber ideal of an étale-coordinatised algebra

The diagonal-ideal package of Section ?? computes the kernel of the multiplication $B \otimes_{k[t]} B \rightarrow B$ after localising at the idempotent lift. Here the second tensor factor is specialised along a point: F a second $k[t]$ -coordinatised algebra receiving B through a $k[t]$ -algebra homomorphism $c: B \rightarrow F$ (in the application, F is the section ring of an affine field test and c the pullback along a point of the curve). Everything mirrors the diagonal computation, with the idempotent input *pushed* from the frozen diagonal package — no unramifiedness of F is invoked.

Throughout, k is a commutative ring, B and F are k -algebras carrying compatible $k[t]$ -algebra structures, $u_B := t \cdot 1_B$ and $u_F := t \cdot 1_F$ denote the coordinates, and $c: B \rightarrow F$ is a $k[t]$ -algebra homomorphism (so $c(u_B) = u_F$).

Definition 254 (The point generator and the point evaluations). The *point generator* is $g := u_B \otimes 1 - 1 \otimes u_F \in B \otimes_k F$; the *point evaluation* is the k -algebra map $\varepsilon: B \otimes_k F \rightarrow F$, $x \otimes \lambda \mapsto c(x)\lambda$, which kills g . Auxiliary to them: the base-change surjection $\beta: B \otimes_k F \rightarrow B \otimes_{k[t]} F$; the $k[t]$ -level evaluation $\varepsilon_2: B \otimes_{k[t]} F \rightarrow F$ with $\varepsilon_2 \circ \beta = \varepsilon$; and the second-factor pushes $1 \otimes c: B \otimes_k B \rightarrow B \otimes_k F$ and $B \otimes_{k[t]} B \rightarrow B \otimes_{k[t]} F$, which carry the diagonal generator to the point generator and intertwine the two base changes.

Proof. $c(u_B) = u_F$ since c is a $k[t]$ -algebra map, so $\varepsilon(g) = c(u_B) \cdot 1 - 1 \cdot u_F = 0$, and the push of the diagonal generator $u_B \otimes 1 - 1 \otimes u_B$ (Definition ??) is g . The base change is surjective because it is the identity on pure tensors; the remaining identities are checked on pure tensors. $\square$

Theorem 255 (The kernel of the point base change). *The kernel of the base-change surjection $\beta: B \otimes_k F \rightarrow B \otimes_{k[t]} F$ is exactly the principal ideal on the point generator: $\ker \beta = (g)$.*

Proof. Mirror of Theorem ??. $\beta(g) = 0$ since in $B \otimes_{k[t]} F$ the two coordinates agree across the tensor (both are $t \cdot (1 \otimes 1)$), so $(g) \subseteq \ker \beta$. Conversely the telescoping identity holds in the point setting: for every $p \in k[t]$, $p(u_B) \otimes 1 - 1 \otimes p(u_F) \in (g)$, because a difference of evaluations is divisible by the difference of the points (as in Lemma ??). Telescoping promotes the right inclusion $\lambda \mapsto 1 \otimes \lambda$ into the quotient $(B \otimes_k F)/(g)$ to a $k[t]$ -algebra map, whence a section $B \otimes_{k[t]} F \rightarrow (B \otimes_k F)/(g)$ of β with section $\circ \beta =$ quotient map (as in Lemma ??). An element of $\ker \beta$ is then killed by the quotient map, i.e. lies in (g) . $\square$

Theorem 256 (The kernel of the $k[t]$ -level point evaluation). *Suppose the diagonal ideal is principal: $\ker(\text{mul}: B \otimes_{k[t]} B \rightarrow B) = (e)$ for some $e \in B \otimes_{k[t]} B$. Then the kernel of the $k[t]$ -level point evaluation is the principal ideal on the push of e :*

$$\ker(\varepsilon_2: B \otimes_{k[t]} F \rightarrow F) = ((1 \otimes c)(e)).$$

Proof. The push kills into the kernel: $\varepsilon_2((1 \otimes c)(z)) = c(\text{mul}(z))$ on pure tensors, and $\text{mul}(e) = 0$.

Conversely, the generators $x \otimes 1 - 1 \otimes c(x)$ ($x \in B$) of the point ideal are pushes of the diagonal-ideal elements $x \otimes 1 - 1 \otimes x$, each a multiple of e by hypothesis; hence they lie in $((1 \otimes c)(e))$. Now the section trick: modulo $((1 \otimes c)(e))$, every $z \in B \otimes_{k[t]} F$ is congruent to $1 \otimes \varepsilon_2(z)$ — on a pure tensor $x \otimes y$, the difference $x \otimes y - 1 \otimes c(x)y = (x \otimes 1 - 1 \otimes c(x))(1 \otimes y)$ is a multiple of a pushed generator. So $\varepsilon_2(z) = 0$ forces $z \equiv 1 \otimes 0 = 0$, i.e. $z \in ((1 \otimes c)(e))$. $\square$

Theorem 257 (The localised point-fiber ideal). *Let $\tilde{e} \in B \otimes_k B$ be a lift of an idempotent generator of the diagonal ideal: suppose $\beta_B(\tilde{e})$ is idempotent and $\ker(\text{mul}) = (\beta_B(\tilde{e}))$, where $\beta_B: B \otimes_k B \rightarrow B \otimes_{k[t]} B$ is the diagonal base change (such a lift exists for the étale coordinate*

by Theorem ??). Write $\tilde{e}_F := (1 \otimes c)(\tilde{e}) \in B \otimes_k F$ for its push. Then for any localisation L of $B \otimes_k F$ away from $1 - \tilde{e}_F$,

$$(\ker \varepsilon) L = (g) L,$$

the extension of the point-evaluation kernel is the principal ideal on the point generator.

Proof. Mirror of Theorem ??.

Containment into the sum. $\ker \varepsilon \subseteq (g) + (\tilde{e}_F)$: for $x \in \ker \varepsilon$, the image $\beta(x)$ lies in $\ker \varepsilon_2$ (Definition ??), which is principal on the pushed generator (Theorem ??, whose hypothesis holds with $e := \beta_B(\tilde{e})$, the pushes intertwining by Definition ??); lift the factorisation through the surjection β and correct by $\ker \beta = (g)$ (Theorem ??).

The pushed lift dies in the localisation. $\beta(\tilde{e}_F)$ is idempotent (the push of an idempotent), so $\tilde{e}_F(1 - \tilde{e}_F) \in \ker \beta = (g)$; in L the factor $1 - \tilde{e}_F$ is invertible, hence $\tilde{e}_F \in (g)L$.

Combining, $(\ker \varepsilon)L \subseteq (g)L$; the reverse containment is $\varepsilon(g) = 0$ (Definition ??). $\square$

4.17 Depth and the Auslander–Buchsbaum formula

This section develops the depth theory of finite modules over a Noetherian local ring — the numerical depth, its characterisation by the vanishing of $\text{Ext}^\bullet(\kappa, -)$, and its behaviour in short exact sequences — and proves the Auslander–Buchsbaum formula $\text{pd}_R(M) + \text{depth}(M) = \text{depth}(R)$.

Definition 258 (Depth of a module). Let R be a commutative ring, $I \subseteq R$ an ideal and M an R -module. An *M -regular sequence* is a finite sequence $x_1, \dots, x_n \in R$ such that each x_i is a non-zero-divisor on $M/(x_1, \dots, x_{i-1})M$ and $(x_1, \dots, x_n)M \neq M$. The *I -depth* of M is the element $\text{depth}_I(M)$ of $\{0, 1, 2, \dots, \infty\}$ defined by:

1. if $IM = M$, then $\text{depth}_I(M) = \infty$;
2. otherwise, $\text{depth}_I(M)$ is the supremum of the lengths of M -regular sequences contained in I (the supremum of the empty set being 0).

When $(R, \mathfrak{m})$ is local, $\text{depth}(M) := \text{depth}_{\mathfrak{m}}(M)$ is called simply the *depth* of M .

Definition 259 (Projective dimension). For a ring R and an R -module M , the *projective dimension* $\text{pd}_R(M) \in \{-\infty\} \cup \{0, 1, 2, \dots, \infty\}$ is the categorical projective dimension of M in the abelian category of R -modules: the infimum of the n such that $\text{Ext}^i(M, -) = 0$ for all $i > n$, equivalently the smallest length of a projective resolution of M ; it is $-\infty$ exactly when $M = 0$.

Lemma 260 (The covariant long exact sequence of Ext). *Provided by Mathlib. Let $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ be a short exact sequence in an abelian category with Ext -groups, with extension class $\delta \in \text{Ext}^1(C, A)$ (??), and let Y be any object. For every $n \geq 0$ the sequence of abelian groups*

$$\text{Ext}^n(Y, A) \xrightarrow{f_*} \text{Ext}^n(Y, B) \xrightarrow{g_*} \text{Ext}^n(Y, C) \xrightarrow{(-) \cdot \delta} \text{Ext}^{n+1}(Y, A) \xrightarrow{f_*} \text{Ext}^{n+1}(Y, B)$$

is exact at each inner term, where f_, g_* are postcomposition (covariant functoriality of $\text{Ext}(Y, -)$) and $(-) \cdot \delta$ is the Yoneda product with the extension class. Moreover, if $f : A \rightarrow B$ is a monomorphism then postcomposition $f_* : \text{Ext}^0(Y, A) \rightarrow \text{Ext}^0(Y, B)$ is injective (in degree zero $\text{Ext}^0 = \text{Hom}$, and postcomposition with a mono is injective on Hom).*

Lemma 261 (Annihilator elements kill Ext). *Let R be a commutative ring, N, M two R -modules, $i \geq 0$, and $x \in \text{Ann}_R(N)$. Then x annihilates $\text{Ext}_R^i(N, M)$: for every class $e \in \text{Ext}_R^i(N, M)$ we have $x \cdot e = 0$.*

Proof. The R -action on $\text{Ext}_R^i(N, M)$ can be computed through the first argument: by R -bilinearity of the Yoneda composition and functoriality, $x \cdot e = (x \cdot \text{id}_N)^*(e)$, the precomposition of e with the endomorphism “multiplication by x ” of N . Since $x \in \text{Ann}_R(N)$, the endomorphism $x \cdot \text{id}_N$ is the zero map of N , and precomposition with the zero morphism is the zero map on Ext . Hence $x \cdot e = 0$. $\square$

Theorem 262 (Depth via Ext). *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $\kappa = R/\mathfrak{m}$ and let M be a nonzero finite R -module. For every $n \in \mathbb{N}$,*

$$n \leq \text{depth}(M) \iff \text{Ext}_R^i(\kappa, M) = 0 \text{ for all } i < n.$$

Equivalently, $\text{depth}(M)$ is the smallest i such that $\text{Ext}_R^i(\kappa, M) \neq 0$.

Proof. First note two facts used repeatedly. (a) Since M is nonzero and finite, Nakayama’s lemma rules out $\mathfrak{m}M = M$, so $\text{depth}(M)$ is the supremum of Definition ??; moreover, for the same reason, any *prefix* of an M -regular sequence contained in $\mathfrak{m}$ is again M -regular (weak regularity passes to prefixes, and the nondegeneracy condition $(x_1, \dots, x_j)M \neq M$ holds because $(x_1, \dots, x_j) \subseteq \mathfrak{m}$ and $\mathfrak{m}M \neq M$). So if $n < \text{depth}(M)$ there is an M -regular sequence in $\mathfrak{m}$ of length exactly n . (b) $\text{Ann}_R(\kappa) = \mathfrak{m}$, so every $x \in \mathfrak{m}$ annihilates every $\text{Ext}^i(\kappa, -)$ by ??.

We induct on n , generalising M . The case $n = 0$ is trivial (both sides always hold). Assume the statement for n and all nonzero finite modules.

($\Rightarrow$) Let $n + 1 \leq \text{depth}(M)$. By (a) extract an M -regular sequence $x, x_1, \dots, x_n$ of length $n + 1$ in $\mathfrak{m}$. Then x is a non-zero-divisor on M , the quotient M/xM is nonzero (Nakayama, $x \in \mathfrak{m}$) and finite, and $x_1, \dots, x_n$ is (M/xM) -regular, so $n \leq \text{depth}(M/xM)$; by the inductive hypothesis $\text{Ext}^j(\kappa, M/xM) = 0$ for all $j < n$. Now fix $i \leq n$ and consider the short exact sequence $0 \rightarrow M \xrightarrow{x} M \rightarrow M/xM \rightarrow 0$. If $i = 0$: postcomposition with the monomorphism $x : M \rightarrow M$ is injective on $\text{Ext}^0(\kappa, M)$ (??) and coincides with the action of x , which is zero by (b); hence $\text{Ext}^0(\kappa, M) = 0$. If $i = j + 1$ with $j < n$: in the covariant long exact sequence (??)

$$\text{Ext}^j(\kappa, M/xM) \xrightarrow{\delta} \text{Ext}^{j+1}(\kappa, M) \xrightarrow{x_*} \text{Ext}^{j+1}(\kappa, M),$$

the map x_* (postcomposition with multiplication by x) is the action of x , hence zero by (b); exactness makes δ surjective, and its source vanishes, so $\text{Ext}^{j+1}(\kappa, M) = 0$.

($\Leftarrow$) Suppose $\text{Ext}^i(\kappa, M) = 0$ for all $i \leq n$. From $\text{Ext}^0(\kappa, M) = \text{Hom}_R(\kappa, M) = 0$ it follows that some $x \in \mathfrak{m} = \text{Ann}_R(\kappa)$ is a non-zero-divisor on M (for a finite module over a Noetherian ring, if every element of $\text{Ann}(\kappa)$ were a zero-divisor on M , then $\mathfrak{m}$ would be contained in an associated prime of M , producing a nonzero map $\kappa \rightarrow M$). The quotient M/xM is nonzero (Nakayama) and finite. For $j < n$, the segment

$$\text{Ext}^j(\kappa, M) \rightarrow \text{Ext}^j(\kappa, M/xM) \xrightarrow{\delta} \text{Ext}^{j+1}(\kappa, M)$$

of the covariant long exact sequence (??) has both outer terms zero ($j < n$ and $j + 1 \leq n$), so $\text{Ext}^j(\kappa, M/xM) = 0$. By the inductive hypothesis $n \leq \text{depth}(M/xM)$; by (a) pick an (M/xM) -regular sequence $x_1, \dots, x_n$ of length n in $\mathfrak{m}$. Then $x, x_1, \dots, x_n$ is an M -regular sequence of length $n + 1$ in $\mathfrak{m}$, whence $n + 1 \leq \text{depth}(M)$.

Finally, the two formulations are equivalent: the displayed equivalence for all n says exactly that $\text{depth}(M)$ is the least index with nonvanishing $\text{Ext}(\kappa, M)$. $\square$

Theorem 263 (Depth in short exact sequences). *Let $(R, \mathfrak{m})$ be a Noetherian local ring and $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ a short exact sequence of nonzero finite R -modules. Then, with truncated subtraction in $\{0, 1, \dots, \infty\}$,*

1. $\text{depth}(N) \geq \min\{\text{depth}(N'), \text{depth}(N'')\},$
2. $\text{depth}(N'') \geq \min\{\text{depth}(N), \text{depth}(N') - 1\},$
3. $\text{depth}(N') \geq \min\{\text{depth}(N), \text{depth}(N'') + 1\}.$

Proof. Write $\kappa = R/\mathfrak{m}$. In each case it suffices, by ??, to fix a natural number a below the right-hand side and prove $\text{Ext}^i(\kappa, -) = 0$ of the middle-named module for all $i < a$; all three are read off the covariant long exact sequence of $\text{Ext}(\kappa, -)$ (??) applied to the given short exact sequence, using ?? to convert the depth bounds on the other two modules into Ext-vanishing.

(1) For $i < a \leq \min\{\text{depth}(N'), \text{depth}(N'')\}$: in $\text{Ext}^i(\kappa, N') \rightarrow \text{Ext}^i(\kappa, N) \rightarrow \text{Ext}^i(\kappa, N'')$, the outer terms vanish, so by exactness at the middle every class of $\text{Ext}^i(\kappa, N)$ lifts to $\text{Ext}^i(\kappa, N') = 0$.

(2) For $i < a \leq \min\{\text{depth}(N), \text{depth}(N') - 1\}$: in $\text{Ext}^i(\kappa, N) \rightarrow \text{Ext}^i(\kappa, N'') \xrightarrow{\delta} \text{Ext}^{i+1}(\kappa, N')$, the first term vanishes since $i < \text{depth}(N)$ and the last since $i + 1 < \text{depth}(N')$ (from $i + 1 \leq a \leq \text{depth}(N') - 1$); exactness at the middle gives $\text{Ext}^i(\kappa, N'') = 0$.

(3) For $i < a \leq \min\{\text{depth}(N), \text{depth}(N'') + 1\}$: if $i = 0$, postcomposition $\text{Ext}^0(\kappa, N') \rightarrow \text{Ext}^0(\kappa, N)$ with the monomorphism $N' \rightarrow N$ is injective (??) and lands in $\text{Ext}^0(\kappa, N) = 0$. If $i = j + 1$, then $j < \text{depth}(N'')$ (from $j + 2 \leq a \leq \text{depth}(N'') + 1$), and in $\text{Ext}^j(\kappa, N'') \xrightarrow{\delta} \text{Ext}^{j+1}(\kappa, N') \rightarrow \text{Ext}^{j+1}(\kappa, N)$ the outer terms vanish, so $\text{Ext}^{j+1}(\kappa, N') = 0$. $\square$

Lemma 264 (Depth is invariant under linear equivalence). *Let R be a commutative ring, $I \subseteq R$ an ideal, and $e : M \cong M'$ an R -linear equivalence of R -modules. Then $\text{depth}_I(M) = \text{depth}_I(M')$.*

Proof. The condition $IM = M$ transfers along e (the image of IM under e is IM' , and e is surjective), so the two depths land in the same branch of Definition ?. In the supremum branch, a sequence $x_1, \dots, x_n$ is M -regular if and only if it is M' -regular: e intertwines the multiplication maps and induces equivalences on the successive quotients $M/(x_1, \dots, x_j)M \cong M'/(x_1, \dots, x_j)M'$, so non-zero-divisor conditions and the nondegeneracy condition match term by term. Hence the two suprema agree. $\square$

Lemma 265 (Ideal action on a finite power is componentwise). *Let R be a commutative ring, $I \subseteq R$ an ideal, M an R -module and ι a finite index type. Inside the product module $\iota \rightarrow M$,*

$$I \cdot (\iota \rightarrow M) = \{ f : \iota \rightarrow M \mid f(i) \in IM \text{ for all } i \}.$$

Proof. $\subseteq$: a generator $a \cdot f$ with $a \in I$ has every component $a \cdot f(i) \in IM$, and the right-hand side is closed under sums. $\supseteq$: a function f with all components in IM decomposes as the finite sum $f = \sum_i \delta_i(f(i))$ of its one-hot components, where $\delta_i : M \rightarrow (\iota \rightarrow M)$ places a value in slot i and zero elsewhere; each δ_i is R -linear, so it carries IM into $I \cdot (\iota \rightarrow M)$. $\square$

Lemma 266 (Depth of a finite power). *Let R be a commutative ring, $I \subseteq R$ an ideal, M an R -module and ι a nonempty finite index type. Then $\text{depth}_I(\iota \rightarrow M) = \text{depth}_I(M)$. In particular $\text{depth}_I(R^k) = \text{depth}_I(R)$ for $k \geq 1$.*

Proof. The side condition agrees on both sides: by ??, $I \cdot (\iota \rightarrow M) = \iota \rightarrow M$ forces $IM = M$ (evaluate a one-hot function at its slot) and conversely (componentwise). It remains to see that a sequence $x_1, \dots, x_n \in R$ is $(\iota \rightarrow M)$ -regular iff it is M -regular; then the two suprema of Definition ?? coincide. Induct on the sequence. For the empty sequence, the nondegeneracy conditions $(\) \cdot (\iota \rightarrow M) \neq \iota \rightarrow M$, i.e. $\iota \rightarrow M \neq 0$, and $M \neq 0$ agree since ι is nonempty. For $x, x_1, \dots, x_n$: multiplication by x on $\iota \rightarrow M$ is componentwise, so x is a non-zero-divisor on $\iota \rightarrow M$ iff it is one on M (ι nonempty for the converse); and by ?? at the ideal (x) there is an

R -linear equivalence $(\iota \rightarrow M)/x(\iota \rightarrow M) \cong (\iota \rightarrow M/xM)$ (a quotient of a finite product by a componentwise submodule is the product of the quotients), so by transport of regular sequences along linear equivalences (as in ??) and the inductive hypothesis applied to M/xM , the tails match as well. $\square$

Lemma 267 (Minimal surjections from a free module). *Let $(R, \mathfrak{m})$ be a local ring with residue field κ and M a finite R -module. There exist $n = \dim_{\kappa}(\kappa \otimes_R M)$ and a surjective R -linear map $f : R^n \rightarrow M$ with $\ker f \subseteq \mathfrak{m} \cdot R^n$.*

Proof. Choose a κ -basis $b_1, \dots, b_n$ of $\kappa \otimes_R M = M/\mathfrak{m}M$ and lift each b_i to $m_i \in M$ along the surjection $M \rightarrow \kappa \otimes_R M$, $m \mapsto 1 \otimes m$. Let $f : R^n \rightarrow M$ send the i -th standard basis vector to m_i , so $f(x) = \sum_i x_i m_i$. The images $1 \otimes m_i = b_i$ span $\kappa \otimes_R M$, so by Nakayama's lemma the m_i generate M : f is surjective. If $f(x) = 0$, applying $1 \otimes -$ gives $\sum_i \bar{x}_i b_i = 0$ in $\kappa \otimes_R M$; linear independence of the b_i forces every $\bar{x}_i = 0$, i.e. every $x_i \in \mathfrak{m}$, so $x \in \mathfrak{m} \cdot R^n$. $\square$

Lemma 268 (From the dimension equation to the vanishing bound). *Let R be a ring, M an R -module and $n \in \mathbb{N}$ with $\text{pd}_R(M) = n$. Then M has projective dimension $< n + 1$, i.e. $\text{Ext}_R^i(M, -) = 0$ for all $i \geq n + 1$.*

Proof. Immediate from Definition ??: the projective dimension is the infimum of the bounds m with $\text{Ext}_R^i(M, -) = 0$ for $i > m$, and it equals $n < n + 1$. $\square$

Lemma 269 (Syzygy descent). *Let R be a commutative ring, $f : R^n \rightarrow M$ a surjective R -linear map, and $k \in \mathbb{N}$. If M has projective dimension $< k + 2$, then $K = \ker f$ has projective dimension $< k + 1$.*

Proof. Consider the short exact sequence $0 \rightarrow K \rightarrow R^n \rightarrow M \rightarrow 0$ and any test module Y . For $i \geq k + 1$, the contravariant long exact sequence (??) contains the exact segment

$$\text{Ext}^i(R^n, Y) \rightarrow \text{Ext}^i(K, Y) \rightarrow \text{Ext}^{i+1}(M, Y).$$

The first term vanishes because R^n is free, hence projective, and $i \geq 1$; the last vanishes because $i + 1 \geq k + 2$ and M has projective dimension $< k + 2$. Hence $\text{Ext}^i(K, Y) = 0$ for all $i \geq k + 1$ and all Y . $\square$

Lemma 270 (Syzygy ascent). *Let R be a commutative ring, $f : R^n \rightarrow M$ a surjective R -linear map, and $k \in \mathbb{N}$. If $K = \ker f$ has projective dimension $< k + 1$, then M has projective dimension $< k + 2$.*

Proof. For a test module Y and $i \geq k + 2$, the contravariant long exact sequence (??) for $0 \rightarrow K \rightarrow R^n \rightarrow M \rightarrow 0$ contains

$$\text{Ext}^{i-1}(K, Y) \xrightarrow{\delta} \text{Ext}^i(M, Y) \rightarrow \text{Ext}^i(R^n, Y).$$

The first term vanishes because $i - 1 \geq k + 1$, the last because R^n is projective and $i \geq 1$. Hence $\text{Ext}^i(M, Y) = 0$. $\square$

Lemma 271 (The two depth inequalities of the kernel sequence). *Let $(R, \mathfrak{m})$ be a Noetherian local ring, M a nonzero finite R -module, $n \geq 1$, and $f : R^n \rightarrow M$ a surjective R -linear map with nonzero kernel K . Then*

$$\min\{\text{depth}(R), \text{depth}(K) - 1\} \leq \text{depth}(M) \quad \text{and} \quad \min\{\text{depth}(R), \text{depth}(M) + 1\} \leq \text{depth}(K).$$

Proof. K is finite (a submodule of a finite module over a Noetherian ring), so parts (2) and (3) of ?? apply to $0 \rightarrow K \rightarrow R^n \rightarrow M \rightarrow 0$; substitute $\text{depth}(R^n) = \text{depth}(R)$ from ?? ($n \geq 1$). $\square$

Lemma 272 (A nonzero class at the depth index). *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field κ and M a nonzero finite R -module with $\text{depth}(M) = D \in \mathbb{N}$. Then $\text{Ext}_R^D(\kappa, M) \neq 0$.*

Proof. By ??, $\text{Ext}^i(\kappa, M) = 0$ for all $i < D$, while $D + 1 \leq \text{depth}(M)$ fails, so the vanishing of all $\text{Ext}^i(\kappa, M)$ with $i < D + 1$ fails; the failure can only occur at $i = D$. $\square$

Lemma 273 (Matrix collapse on Ext). *Let R be a commutative ring, N an R -module, and $A : R^m \rightarrow R^n$ an R -linear map all of whose matrix entries $a_{ij} = (Ae_j)_i$ lie in $\text{Ann}_R(N)$. Then for every $p \geq 0$ the postcomposition map $A_* : \text{Ext}_R^p(N, R^m) \rightarrow \text{Ext}_R^p(N, R^n)$ is zero.*

Proof. Write $A = \sum_{i,j} a_{ij} E_{ij}$, where $E_{ij} : R^m \rightarrow R^n$ is the elementary map sending e_j to e_i and the other basis vectors to 0 (evaluate both sides on each e_j). Postcomposition on $\text{Ext}^p(N, -)$ is additive and R -linear in the map, so for $e \in \text{Ext}^p(N, R^m)$, $A_*(e) = \sum_{i,j} a_{ij} \cdot (E_{ij})_*(e)$; each summand is the action of $a_{ij} \in \text{Ann}_R(N)$ on a class of $\text{Ext}^p(N, R^n)$, hence zero by ??. $\square$

Theorem 274 (Auslander–Buchsbaum, positive projective dimension). *Let $(R, \mathfrak{m})$ be a Noetherian local ring and M a nonzero finite R -module with $\text{pd}_R(M) = k + 1$ for some $k \in \mathbb{N}$. Then*

$$(k + 1) + \text{depth}(M) = \text{depth}(R).$$

Proof. Induct on k , generalising M .

Base case $\text{pd}_R(M) = 1$. Take a minimal surjection $f : R^n \twoheadrightarrow M$ with $K = \ker f \subseteq \mathfrak{m} \cdot R^n$ (?); $n \geq 1$ since $M \neq 0$. By ?? and ?? (at $k = 0$), K has projective dimension < 1 , i.e. K is projective; it is finite (Noetherian) and R is local, so K is free. $K \neq 0$: otherwise f is an isomorphism, M is free and $\text{pd}_R(M) = 0 \neq 1$. Fix an isomorphism $K \cong R^{k'}$ with $k' \geq 1$ and let $A : R^{k'} \rightarrow R^n$ be the composite inclusion. Every matrix entry of A lies in $\mathfrak{m}$: the columns of A lie in $K \subseteq \mathfrak{m} \cdot R^n$, which is componentwise $\mathfrak{m}$ -valued by ??; and $\mathfrak{m} = \text{Ann}_R(\kappa)$ for the residue field κ .

$\geq$: part (2) of ?? applied to $0 \rightarrow R^{k'} \xrightarrow{A} R^n \rightarrow M \rightarrow 0$, with $\text{depth}(R^{k'}) = \text{depth}(R^n) = \text{depth}(R)$ (?), gives $\text{depth}(M) \geq \min\{\text{depth}(R), \text{depth}(R) - 1\} = \text{depth}(R) - 1$, i.e. $\text{depth}(R) \leq 1 + \text{depth}(M)$.

$\leq$: if $\text{depth}(R) = \infty$ there is nothing to prove beyond the $\geq$ direction forcing $\text{depth}(M) = \infty$; so let $\text{depth}(R) = D < \infty$. Pick a nonzero class $\alpha \in \text{Ext}^D(\kappa, R^{k'})$ (?), using $\text{depth}(R^{k'}) = D$. By ??, $A_*\alpha = 0$. If $D = 0$, postcomposition with the monomorphism A is injective on Ext^0 (?), contradicting $\alpha \neq 0$; so $D = D' + 1$. The covariant long exact sequence (?) contains

$$\text{Ext}^{D'}(\kappa, M) \xrightarrow{\delta} \text{Ext}^D(\kappa, R^{k'}) \xrightarrow{A_*} \text{Ext}^D(\kappa, R^n),$$

so $\alpha = \delta(\beta)$ for some $\beta \in \text{Ext}^{D'}(\kappa, M)$, and $\beta \neq 0$. Hence $\text{Ext}^{D'}(\kappa, M) \neq 0$, so $\text{depth}(M) \leq D'$ by ??, i.e. $1 + \text{depth}(M) \leq D = \text{depth}(R)$. Together: $1 + \text{depth}(M) = \text{depth}(R)$.

Inductive step $\text{pd}_R(M) = k + 2$. Take a minimal surjection $f : R^n \twoheadrightarrow M$ (?), $n \geq 1$, and set $K = \ker f$. Then $\text{pd}_R(K) = k + 1$ exactly: $\leq$ by ?? and ??; and if K had projective dimension $< k + 1$, then ?? would give $\text{pd}_R(M) < k + 2$, a contradiction. In particular $K \neq 0$ (the zero module has projective dimension $-\infty$), and K is finite. The inductive hypothesis applied to K gives $(k+1) + \text{depth}(K) = \text{depth}(R)$, and ?? gives the two inequalities $\min\{\text{depth}(R), \text{depth}(K) - 1\} \leq \text{depth}(M)$ and $\min\{\text{depth}(R), \text{depth}(M) + 1\} \leq \text{depth}(K)$.

It remains to combine these in $\{0, 1, \dots, \infty\}$. If $\text{depth}(K) = \infty$, the hypothesis forces $\text{depth}(R) = \infty$, the first inequality forces $\text{depth}(M) = \infty$, and both sides of the claim are ∞ . If

$\text{depth}(K) = K_0 < \infty$ and $\text{depth}(M) = \infty$, the second inequality reads $\min\{(k+1) + K_0, \infty\} = (k+1) + K_0 \leq K_0$, impossible since $k+1 \geq 1$. So both are finite, say $\text{depth}(M) = M_0$, and $\text{depth}(R) = (k+1) + K_0 > K_0 - 1$; the first inequality then reads $K_0 - 1 \leq M_0$. In the second inequality the minimum cannot be $\text{depth}(R)$, since $(k+1) + K_0 \leq K_0$ is impossible; so it reads $M_0 + 1 \leq K_0$. Hence $M_0 = K_0 - 1$ with $K_0 \geq 1$, and $(k+2) + M_0 = (k+1) + K_0 = \text{depth}(R)$. $\square$

Theorem 275 (The Auslander–Buchsbaum formula). *Let $(R, \mathfrak{m})$ be a Noetherian local ring and M a nonzero finite R -module of finite projective dimension $\text{pd}_R(M) = n \in \mathbb{N}$. Then*

$$\text{pd}_R(M) + \text{depth}(M) = \text{depth}(R).$$

Proof. If $n = 0$, then M is projective (Definition ??), finite over the local Noetherian ring R , hence free: $M \cong R^k$ with $k = \text{rank } M \geq 1$ (as $M \neq 0$). By ?? and ??, $\text{depth}(M) = \text{depth}(R^k) = \text{depth}(R)$, which is the formula with $n = 0$. If $n = k + 1$, this is ??. $\square$

4.18 Cohen–Macaulay and regular local rings

This section defines the Cohen–Macaulay condition for a Noetherian local ring and proves the two classical structure theorems for regular local rings: a regular local ring is a domain, and a regular system of parameters is a regular sequence — so regular local rings are Cohen–Macaulay. Throughout, for a Noetherian local ring $(R, \mathfrak{m})$ we write $\mu(\mathfrak{m})$ for the minimal number of generators of $\mathfrak{m}$, which by Nakayama’s lemma equals $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2$, the dimension of the cotangent space; recall that R is *regular* when $\mu(\mathfrak{m}) = \dim R$, the Krull dimension (and that $\dim R \leq \mu(\mathfrak{m})$ always, by Krull’s height theorem).

Definition 276 (Cohen–Macaulay local ring). A Noetherian local ring $(R, \mathfrak{m})$ is *Cohen–Macaulay* if $\text{depth}(R) = \dim R$, its depth (Definition ??) equals its Krull dimension.

Theorem 277 (Cotangent dimension drop). *Let $(R, \mathfrak{m})$ be a Noetherian local ring and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Write $\mathfrak{m}'$ for the maximal ideal of the local ring $R/(x)$. Then*

$$\dim_{\kappa'} \mathfrak{m}'/\mathfrak{m}'^2 + 1 = \dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2,$$

where κ, κ' are the residue fields; equivalently $\mu(\mathfrak{m}') + 1 = \mu(\mathfrak{m})$.

Proof. By Nakayama both sides may be computed as minimal generator counts, so we prove $\mu(\mathfrak{m}') + 1 = \mu(\mathfrak{m})$, by two inequalities.

$\leq$ (Steinitz exchange): let $V = \{v_1, \dots, v_d\}$ be a generating set of $\mathfrak{m}$ with $d = \mu(\mathfrak{m})$; note $d \geq 1$, since $d = 0$ would give $\mathfrak{m} = 0$ and $x \in \mathfrak{m}^2$. Write $x = \sum_i c_i v_i$. If every $c_i \in \mathfrak{m}$, then $x \in \mathfrak{m} \cdot \mathfrak{m} = \mathfrak{m}^2$, a contradiction; so some c_{i_0} is a unit, and $v_{i_0} = c_{i_0}^{-1}(x - \sum_{i \neq i_0} c_i v_i)$ lies in the span of $(V \setminus \{v_{i_0}\}) \cup \{x\}$. Hence $(V \setminus \{v_{i_0}\}) \cup \{x\}$ still generates $\mathfrak{m}$; its image in $R/(x)$ generates $\mathfrak{m}' = \mathfrak{m}/(x)$, and the image of x is zero, so the $d - 1$ images of $V \setminus \{v_{i_0}\}$ generate $\mathfrak{m}'$: $\mu(\mathfrak{m}') \leq d - 1$.

$\geq$ (lift and cons): let T be a generating set of $\mathfrak{m}'$ of size $\mu(\mathfrak{m}')$ and lift each element along the surjection $R \twoheadrightarrow R/(x)$ to $\tilde{T} \subseteq \mathfrak{m}$ (the preimage of $\mathfrak{m}'$ is $\mathfrak{m}$, the unique maximal ideal). Since $\mathfrak{m}$ is the preimage of $\mathfrak{m}' = (\text{image of } \tilde{T})$ and the kernel of the surjection is (x) , we get $\mathfrak{m} = (\tilde{T}) + (x)$, so $\tilde{T} \cup \{x\}$ generates $\mathfrak{m}$: $\mu(\mathfrak{m}) \leq \mu(\mathfrak{m}') + 1$. $\square$

Lemma 278 (The Nakayama witness $x \in \mathfrak{m} \setminus \mathfrak{m}^2$). *Let $(R, \mathfrak{m})$ be a Noetherian local ring with $\mu(\mathfrak{m}) \geq 1$. Then there exists $x \in \mathfrak{m}$ with $x \notin \mathfrak{m}^2$.*

Proof. Otherwise $\mathfrak{m} \subseteq \mathfrak{m}^2 = \mathfrak{m} \cdot \mathfrak{m}$; since $\mathfrak{m}$ is finitely generated (Noetherian) and contained in the Jacobson radical, Nakayama's lemma gives $\mathfrak{m} = 0$, so $\mu(\mathfrak{m}) = 0$ — a contradiction. $\square$

Lemma 279 (The regular quotient step). *Let R be a regular local Noetherian ring with $\mu(\mathfrak{m}) = k + 1$ and let $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Then $R/(x)$ is a nonzero regular local Noetherian ring with $\mu(\mathfrak{m}') = k$, where $\mathfrak{m}'$ is its maximal ideal. (No non-zero-divisor hypothesis on x is made.)*

Proof. Since $x \in \mathfrak{m}$ is a nonunit, $(x) \neq R$ and $R/(x)$ is a nonzero local Noetherian ring. By ??, $\mu(\mathfrak{m}') = k$. For the Krull dimension: since x lies in the Jacobson radical of R , quotienting by the single element x can drop the dimension by at most one, $\dim R \leq \dim R/(x) + 1$ (the general bound $\dim R \leq \dim R/(S) + \#S$ for a subset S of the Jacobson radical, which does *not* require x to be a non-zero-divisor). Regularity of R gives $\dim R = \mu(\mathfrak{m}) = k + 1$, so $\dim R/(x) \geq k$; Krull's height theorem gives $\dim R/(x) \leq \mu(\mathfrak{m}') = k$. Hence $\dim R/(x) = k = \mu(\mathfrak{m}')$ and $R/(x)$ is regular. $\square$

Theorem 280 (Regular local rings are domains). *Every regular local Noetherian ring is an integral domain.*

Proof. Strong induction on $d = \mu(\mathfrak{m})$, generalising the ring.

Base case $d = 0$. Then $\mathfrak{m} = 0$, so R is a field, hence a domain.

Inductive step $d = k + 1$. Pick $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ (?). By ??, $R/(x)$ is regular local with $\mu = k$, hence a domain by the inductive hypothesis; so (x) is a prime ideal. Choose a minimal prime $\mathfrak{p} \subseteq (x)$ of R (every prime contains a minimal prime).

Case $x \notin \mathfrak{p}$. For $y \in \mathfrak{p} \subseteq (x)$ write $y = xz$; since $\mathfrak{p}$ is prime and $x \notin \mathfrak{p}$, we get $z \in \mathfrak{p}$. Hence $\mathfrak{p} = x\mathfrak{p}$ with x in the Jacobson radical and $\mathfrak{p}$ finitely generated; Nakayama's lemma gives $\mathfrak{p} = 0$. A minimal prime equal to (0) means (0) is prime, so R is a domain.

Case $x \in \mathfrak{p}$. Then $(x) \subseteq \mathfrak{p} \subseteq (x)$, so (x) itself is a minimal prime of R . We refute this. R is Noetherian, so it has only finitely many minimal primes. The maximal ideal $\mathfrak{m}$ is contained in none of the ideals $\mathfrak{m}^2, \mathfrak{q}_1, \dots, \mathfrak{q}_r$ ($\mathfrak{q}_i$ the minimal primes): containment in $\mathfrak{m}^2$ would give $\mathfrak{m} = 0$ by Nakayama, contradicting $d \geq 1$; and $\mathfrak{m} \subseteq \mathfrak{q}_i$ would force $\mathfrak{q}_i = \mathfrak{m}$ (primes of a local ring lie in $\mathfrak{m}$), making $\mathfrak{m}$ a minimal prime, so $\dim R = 0$ — but $\dim R = \mu(\mathfrak{m}) = k + 1 \geq 1$ by regularity. By prime avoidance (with the single non-prime member $\mathfrak{m}^2$ allowed), $\mathfrak{m} \not\subseteq \mathfrak{m}^2 \cup \mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_r$; pick $x' \in \mathfrak{m}$ outside this union. Then $x' \in \mathfrak{m} \setminus \mathfrak{m}^2$ avoids every minimal prime. Run the preceding two paragraphs with x' in place of x : $R/(x')$ is a domain by ?? and the inductive hypothesis, so (x') is prime; a minimal prime $\mathfrak{p}' \subseteq (x')$ cannot contain x' , so the previous case applies and shows $\mathfrak{p}' = 0$ and R is a domain. But in a domain the only minimal prime is (0) , so the minimal prime (x) equals (0) , giving $x = 0 \in \mathfrak{m}^2$ — contradicting $x \notin \mathfrak{m}^2$. $\square$

Theorem 281 (A regular system of parameters is a regular sequence). *Let R be a regular local Noetherian ring. Then there is an R -regular sequence contained in $\mathfrak{m}$ of length $\mu(\mathfrak{m}) = \dim R$.*

Proof. Induct on $d = \mu(\mathfrak{m})$, generalising the ring. For $d = 0$ take the empty sequence ($R \neq 0$, so it is vacuously R -regular). For $d = k + 1$: pick $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ (?); then $x \neq 0$ (zero lies in $\mathfrak{m}^2$) and R is a domain (?), so x is a non-zero-divisor on R . By ??, $R/(x)$ is regular local with $\mu(\mathfrak{m}') = k$, so by the inductive hypothesis there is an $R/(x)$ -regular sequence $\bar{y}_1, \dots, \bar{y}_k$ in $\mathfrak{m}'$. Lift each $\bar{y}_i$ to $y_i \in \mathfrak{m}$ (the preimage of $\mathfrak{m}'$ is $\mathfrak{m}$). Regularity of a sequence on the R -module R/xR is computed through the images of its entries in $R/(x)$, so $y_1, \dots, y_k$ is a regular sequence on the R -module R/xR ; conjoining, $x, y_1, \dots, y_k$ is an R -regular sequence of length $k + 1$ in $\mathfrak{m}$. $\square$

Theorem 282 (Regular local rings are Cohen–Macaulay). *Every regular local Noetherian ring R is Cohen–Macaulay: $\text{depth}(R) = \dim R$.*

Proof. The side condition of Definition ?? is in the supremum branch: $\mathfrak{m} \cdot R = \mathfrak{m} \neq R$.

Lower bound. By ?? there is an R -regular sequence in $\mathfrak{m}$ of length $\mu(\mathfrak{m}) = \dim R$, so $\text{depth}(R) \geq \dim R$.

Upper bound. Every R -regular sequence $x_1, \dots, x_\ell$ (in any ideal) satisfies $\ell \leq \dim R$: for a Noetherian local ring, quotienting by a regular sequence drops the Krull dimension by exactly its length, $\dim R/(x_1, \dots, x_\ell) + \ell = \dim R$, and the quotient is nonzero (the nondegeneracy condition $(x_1, \dots, x_\ell)R \neq R$), so its dimension is ≥ 0 . Hence the supremum defining $\text{depth}(R)$ is at most $\dim R$.

Therefore $\text{depth}(R) = \dim R = \mu(\mathfrak{m})$, and R is Cohen–Macaulay (Definition ??). $\square$

Chapter 5

Cohomology of sheaves of modules

Fix a commutative ring R ; for the genus of a curve, R will be the base field k . This chapter defines the cohomology of sheaves of R -modules on a small site as Ext-groups from the constant sheaf, and more generally the cohomology of an object of the site as Ext-groups from a free sheaf of R -modules; it identifies degree-zero cohomology with (global) sections and proves the vanishing of degree-one cohomology of the structure sheaf on an affine open of a scheme. It then develops the Mayer–Vietoris $(0, 1)$ -slice for a sheaf of R -modules on a site and specializes it to a scheme covered by two affine opens, producing an explicit description of $H^1(X, \mathcal{O}_{X/k})$ as the cokernel of a map of section modules. It then runs the two-lattice ladder of Chapter ?? on that cokernel along a finite morphism to the projective line, proving the finiteness of $H^1(C, \mathcal{O}_C)$ — so that the genus of Definition ?? is the dimension of a finite-dimensional vector space. The two closing sections glue a twisted sheaf out of a unit cocycle on a two-chart cover and prove that the two-cover H^1 of the relative curve — for the structure sheaf, and termwise for the twisted sheaf — commutes with arbitrary change of the affine test ring.

5.1 Cohomology on a site

Let $(\mathcal{C}, J)$ be a small site: a small category $\mathcal{C}$ with a Grothendieck topology J . The category of sheaves of R -modules on $(\mathcal{C}, J)$ is a Grothendieck abelian category; in particular it has enough injectives, and all Ext-groups exist and carry R -module structures.

Definition 283 (The constant sheaf). The constant sheaf R_J of R -modules on $(\mathcal{C}, J)$ with value R : the sheafification of the constant presheaf with value R .

Definition 284 (Sheaf cohomology). For a sheaf F of R -modules on $(\mathcal{C}, J)$ and $n \geq 0$, the degree- n cohomology of F is the R -module

$$H^n(F) = \operatorname{Ext}^n(R_J, F),$$

the Ext-group computed in the category of sheaves of R -modules.

Definition 285 (Functoriality). A morphism of sheaves $f : F \rightarrow G$ induces an R -linear map $H^n(F) \rightarrow H^n(G)$ for every n , by covariant functoriality of Ext in the second argument. The identity induces the identity, and a composite induces the composite.

Proof. Functoriality of $\text{Ext}^n(R_J, -)$; the R -linearity comes from the R -linear structure of the sheaf category. $\square$

Lemma 286 (No higher cohomology on injectives). *If F is an injective object of the category of sheaves of R -modules, then $H^{n+1}(F) = 0$ for every $n \geq 0$.*

Proof. Positive-degree Ext-groups into an injective object vanish. $\square$

Definition 287 (The adjunction, linearly). Assume the category $\mathcal{C}$ has a terminal object $\top$. For every R -module M and every sheaf F of R -modules there is an R -linear equivalence

$$\text{Hom}(\underline{M}, F) \cong \text{Hom}_R(M, F(\top)),$$

where $\underline{M}$ denotes the constant sheaf with value M ; it is the hom-bijection of the adjunction between the constant-sheaf functor and evaluation at $\top$.

Proof. The constant-sheaf functor is left adjoint to evaluation at the terminal object. The bijection sends φ to the composite of the unit $M \rightarrow \underline{M}(\top)$ with $\varphi_\top$; both this assignment and its inverse are additive and commute with the R -action. $\square$

Definition 288 (Morphisms out of the constant sheaf are sections). The R -linear equivalence $\text{Hom}(R_J, F) \cong F(\top)$: specialize ?? to $M = R$ and compose with the canonical evaluation $\text{Hom}_R(R, N) \cong N$, $u \mapsto u(1)$.

Lemma 289 (Naturality). *The equivalence of ?? is natural in the sheaf: for $\varphi : R_J \rightarrow F$ and $f : F \rightarrow G$, the section corresponding to $f \circ \varphi$ is $f_\top$ applied to the section corresponding to φ .*

Proof. Naturality of the adjunction hom-bijection in the second variable, and naturality of the evaluation $\text{Hom}_R(R, N) \cong N$. $\square$

Definition 290 (Degree-zero cohomology is global sections). The R -linear equivalence $H^0(F) \cong F(\top)$: the canonical identification $\text{Ext}^0(R_J, F) \cong \text{Hom}(R_J, F)$ followed by ??.

Lemma 291 (Naturality in degree zero). *The equivalence $H^0(F) \cong F(\top)$ of ?? is natural: for $f : F \rightarrow G$, applying $f_\top$ to the section corresponding to $x \in H^0(F)$ yields the section corresponding to the image of x under $H^0(f)$.*

Proof. The identification $\text{Ext}^0 \cong \text{Hom}$ turns composition with the degree-zero class of f into postcomposition with f ; conclude by ??. $\square$

5.2 Cohomology of an object of the site

For an object U of $\mathcal{C}$, this section defines the cohomology of U with coefficients in a sheaf F of R -modules as Ext-groups from a *free* sheaf of R -modules on U , generalizing the previous section (where the implicit object was the terminal object $\top$). This is the coefficient system needed to run the Mayer–Vietoris sequence of Section ?? R -linearly, and it recovers the cohomology of the site when U is terminal (Definition ?? below).

Definition 292 (The free sheaf of R -modules on an object of the site). For an object U of $\mathcal{C}$, the free sheaf of R -modules on U , written $R[U]$: the sheafification of the presheaf $V \mapsto$ the free R -module on the set $\text{Hom}(V, U)$, with restriction maps induced by precomposition.

Definition 293 (Functoriality in the object of the site). A morphism $i : U \rightarrow V$ of $\mathcal{C}$ induces a morphism of sheaves of R -modules $R[i] : R[U] \rightarrow R[V]$: apply the free- R -module-on-a-set functor and sheafification to the induced map of representable presheaves $\text{Hom}(-, U) \rightarrow \text{Hom}(-, V)$, $g \mapsto i \circ g$.

Definition 294 (Free sheaves of modules represent sections). For every object U of $\mathcal{C}$ and every sheaf F of R -modules, an R -linear equivalence

$$\text{Hom}(R[U], F) \cong F(U).$$

Proof. Write $y_U = \text{Hom}(-, U)$ for the presheaf represented by U , so that $R[U]$ is the sheafification of the presheaf of R -modules obtained by applying the free- R -module functor to y_U objectwise. Compose three natural bijections:

- sheafification is left adjoint to the inclusion of sheaves into presheaves, giving $\text{Hom}(R[U], F) \cong \text{Hom}_{\text{PSh}}(y_U \otimes R, F)$, naturally in F ;
- on each object V , the free- R -module functor is left adjoint to the forgetful functor from R -modules to sets, giving $\text{Hom}_{\text{PSh}}(y_U \otimes R, F) \cong \text{Hom}_{\text{PSh}(\text{Set})}(y_U, F)$ (F viewed as a presheaf of sets), naturally in F ;
- the Yoneda lemma identifies $\text{Hom}_{\text{PSh}(\text{Set})}(y_U, F) \cong F(U)$.

Each bijection is additive and R -linear in F (the first two are hom-sets of R -linear/additive adjunctions; the last carries the R -module structure of $F(U)$ tautologically, evaluating a natural transformation at the identity of U), so the composite is an R -linear equivalence. $\square$

Lemma 295 (Naturality of the section equivalence). *The equivalence of ?? is natural in both variables:*

1. for $\varphi : R[U] \rightarrow F$ and $f : F \rightarrow G$, the section of $G(U)$ corresponding to $f \circ \varphi$ is f_U applied to the section of $F(U)$ corresponding to φ ;
2. for $i : U \rightarrow V$ and $\varphi : R[V] \rightarrow F$, the section of $F(U)$ corresponding to the composite $R[U] \xrightarrow{R[i]} R[V] \xrightarrow{\varphi} F$ is the restriction along i of the section of $F(V)$ corresponding to φ .

Proof. Both are naturality of the composite of adjunction bijections used in ??: (1) is naturality of each of the three bijections in the target sheaf F ; (2) is naturality of the Yoneda bijection $\text{Hom}_{\text{PSh}(\text{Set})}(y_U, -) \cong \text{ev}_U$ in U (precomposition with $y_i : y_U \rightarrow y_V$ corresponds to restriction along i), transported through the other two (fixed) bijections. $\square$

Definition 296 (Cohomology of an object of the site). For an object U of $\mathcal{C}$, a sheaf F of R -modules and $n \geq 0$, the degree- n cohomology of U with coefficients in F is the R -module

$$H'^n(U, F) = \text{Ext}^n(R[U], F).$$

Definition 297 (Degree zero is sections). The R -linear equivalence $H'^0(U, F) \cong F(U)$: the canonical identification $\text{Ext}^0(R[U], F) \cong \text{Hom}(R[U], F)$ followed by ??.

Definition 298 (Restriction). For $i : U \rightarrow V$ in $\mathcal{C}$, restriction along i is the R -linear map $H'^n(V, F) \rightarrow H'^n(U, F)$, $x \mapsto x \circ R[i]$: precomposition with $R[i] : R[U] \rightarrow R[V]$ (contravariant functoriality of $\text{Ext}^n(-, F)$).

Definition 299 (Functoriality in the coefficient sheaf). A morphism of sheaves of R -modules $f : F \rightarrow G$ induces, for every object U of $\mathcal{C}$ and every n , an R -linear map $H'^n(U, F) \rightarrow H'^n(U, G)$, $x \mapsto x \cdot f$: composition (Yoneda product) with the degree-zero class of f , i.e. covariant functoriality of $\text{Ext}^n(R[U], -)$ in the coefficient variable. This is the coefficient-variable analogue of the restriction ?? (functoriality in the object of the site). The identity induces the identity, and a composite $g \circ f$ induces the composite of the induced maps.

Proof. Postcomposition with the degree-zero Ext-class of f is R -linear; it sends the degree-zero class of the identity to the identity operation, and by associativity of the Yoneda product it turns the class of a composite into the composite of the two postcompositions. $\square$

Definition 300 (Coefficient-isomorphism transport for cohomology of an object). An isomorphism $\alpha : F \cong G$ of coefficient sheaves of R -modules induces, for every object U and every n , an R -linear isomorphism $H'^n(U, F) \cong H'^n(U, G)$.

Proof. Take the induced maps of ?? for α and for α^{-1} . By functoriality their two composites are the maps induced by $\alpha^{-1} \circ \alpha = \text{id}_F$ and $\alpha \circ \alpha^{-1} = \text{id}_G$, which are the identity operations; hence the two induced maps are mutually inverse R -linear isomorphisms. $\square$

Lemma 301 (Transport of vanishing along a coefficient isomorphism). If $F \cong G$ as coefficient sheaves of R -modules and $H'^n(U, G) = 0$, then $H'^n(U, F) = 0$.

Proof. The R -linear isomorphism $H'^n(U, F) \cong H'^n(U, G)$ of ?? identifies $H'^n(U, F)$ with the zero module $H'^n(U, G)$. $\square$

Definition 302 (The free sheaf on a terminal object is the constant sheaf). If T is a terminal object of $\mathcal{C}$, a canonical isomorphism of sheaves of R -modules $R[T] \cong R_J$.

Proof. Both sides corepresent the same functor: $\text{Hom}(R[T], -) \cong \text{ev}_T$ by ??, and $\text{Hom}(R_J, -) \cong \text{ev}_T$ by ?? (valid since T is terminal), both naturally in the sheaf argument. Two objects corepresenting the same functor via natural equivalences are canonically isomorphic: the isomorphism corresponds, under each equivalence, to the section of the other object at T that corresponds to its own identity morphism, and the two composites are the identity by naturality applied to the identity morphism of each side. $\square$

Definition 303 (Cohomology of the site is cohomology of a terminal object). For T terminal in $\mathcal{C}$ and every n , an R -linear equivalence $H^n(F) \cong H^n(T, F)$.

Proof. Apply the (contravariant, R -linear) functoriality of $\text{Ext}^n(-, F)$ to the isomorphism $R[T] \cong R_J$ of ??: an isomorphism of the first Ext-argument induces, by composition (Yoneda product) with its degree-zero class, a natural R -linear isomorphism on Ext-groups in every degree, with inverse given by composition with the inverse isomorphism. $\square$

5.3 The structure sheaf as a sheaf of modules

Let k be a commutative ring and let X be a scheme over $\text{Spec } k$. The relevant site is the *small Zariski site* of X : the partially ordered set of open subsets of X , with coverings the open coverings. The whole space is a terminal object of this site.

Definition 304 (The algebra map on sections). For every open $U \subseteq X$, the canonical ring homomorphism $k \rightarrow \Gamma(X, U)$: the map induced on global sections by the structure morphism, followed by restriction to U . It commutes with the restriction maps of the structure sheaf; in particular each $\Gamma(X, U)$ is a k -algebra — hence a k -module — and all restriction maps are k -linear.

Proof. Restriction maps are ring homomorphisms and commute with restriction from the global sections; the maps from k are fixed composites of such maps, so they commute with further restriction. $\square$

Definition 305 (The structure sheaf as a sheaf of k -modules). The structure sheaf of X , viewed as a sheaf $\mathcal{O}_{X/k}$ of k -modules on the small Zariski site of X : its sections over an open U are $\Gamma(X, U)$ with the k -module structure of ??, and its restriction maps are those of the structure sheaf.

Proof. The restriction maps are k -linear by ??, so this is a presheaf of k -modules. Its underlying presheaf of sets is that of the structure sheaf, which satisfies the sheaf condition; since limits of k -modules are computed on underlying sets, the presheaf of k -modules is a sheaf. $\square$

Definition 306 (Degree-zero cohomology of the structure sheaf). The k -linear equivalence $H^0(X, \mathcal{O}_{X/k}) \cong \Gamma(X, \mathcal{O}_X)$.

Proof. The open X is a terminal object of the small Zariski site; apply ??. $\square$

5.4 Čech cobounding on affine opens

For a scheme X , a section g of the structure sheaf on an open $V \subseteq X$ and an open $U \subseteq V$, we write $D(g)$ for the basic open locus of V where g is invertible, and $g|_U$, $s|_U$ for restrictions of sections.

Lemma 307 (Denominator clearing on an affine open). *Let X be a scheme, $U \subseteq X$ an affine open, $V \subseteq X$ an open with $U \subseteq V$, and $g \in \Gamma(X, V)$ a section of the structure sheaf on V (so that $D(g) \subseteq V$ is its basic open). For every $c \in \Gamma(X, U \cap D(g))$ there exist $N \geq 0$ and $e \in \Gamma(X, U)$ such that*

$$e|_{U \cap D(g)} = (g|_U)|_{U \cap D(g)}^N \cdot c.$$

Proof. $U \cap D(g)$ is the basic open of the affine U defined by $g|_U$, and restriction identifies the sections over the basic open of an affine open with the localization: $\Gamma(X, U \cap D(g)) = \Gamma(X, U)[1/(g|_U)]$ (quasi-coherence of the structure sheaf). Writing c as a fraction $e/(g|_U)^N$ and clearing the denominator gives the claim. $\square$

Lemma 308 (Annihilation on an affine open). *Let X be a scheme, $U \subseteq X$ an affine open, $V \subseteq X$ an open with $U \subseteq V$, and $g \in \Gamma(X, V)$ a section of the structure sheaf on V (so that $D(g) \subseteq V$ is its basic open). If $t \in \Gamma(X, U)$ restricts to zero on $U \cap D(g)$, then $(g|_U)^M \cdot t = 0$ in $\Gamma(X, U)$ for some $M \geq 0$.*

Proof. As in ??, $\Gamma(X, U \cap D(g))$ is the localization of $\Gamma(X, U)$ at the powers of $g|_U$, and the kernel of the localization map consists of the elements annihilated by a power of $g|_U$. $\square$

Theorem 309 (Degree-one Čech cobounding on an affine open). *Let X be a scheme and U an affine open of X . Let $f_1, \dots, f_n \in \Gamma(X, U)$ with $D(f_1) \cup \dots \cup D(f_n) = U$. Let $a_{ij} \in \Gamma(X, D(f_i) \cap D(f_j))$ be a Čech 1-cocycle: for all i, j, l ,*

$$a_{il} = a_{ij} + a_{jl} \quad \text{on } D(f_i) \cap D(f_j) \cap D(f_l).$$

Then there are sections $b_i \in \Gamma(X, D(f_i))$ with

$$a_{ij} = b_i - b_j \quad \text{on } D(f_i) \cap D(f_j)$$

for all i, j .

Proof. All restriction maps are suppressed from the notation. Note that each $D(f_i)$ and each pairwise intersection $D(f_i) \cap D(f_j) = D(f_i f_j)$ is an affine open of U , hence of X .

Step 1: clear denominators with a uniform exponent. For each pair (i, j) apply ?? with affine open $D(f_i)$, ambient open $V = U$ and $g = f_j$: there are $N_{ij} \geq 0$ and $e_{ij}^0 \in \Gamma(X, D(f_i))$ with $e_{ij}^0 = f_j^{N_{ij}} a_{ij}$ on $D(f_i) \cap D(f_j)$. Let N be the maximum of the N_{ij} and set $e_{ij} = f_j^{N-N_{ij}} e_{ij}^0$, so that $e_{ij} = f_j^N a_{ij}$ on $D(f_i) \cap D(f_j)$.

Step 2: kill the defects. For i, j, l define, on $D(f_i) \cap D(f_j)$,

$$t_{ijl} = e_{il} - e_{jl} - f_l^N a_{ij}.$$

On the triple intersection $D(f_i) \cap D(f_j) \cap D(f_l)$ we have $e_{il} = f_l^N a_{il}$ and $e_{jl} = f_l^N a_{jl}$, so the cocycle identity gives $t_{ijl} = f_l^N (a_{il} - a_{jl} - a_{ij}) = 0$ there. Since $D(f_i) \cap D(f_j)$ is an affine open (of U , hence of X) and the triple intersection is its basic open defined by f_l , ?? (with ambient open $V = U$) yields $M_{ijl} \geq 0$ with $f_l^{M_{ijl}} t_{ijl} = 0$ on $D(f_i) \cap D(f_j)$. Let M be the maximum of the M_{ijl} ; then $f_l^M t_{ijl} = 0$ for all i, j, l .

Step 3: partition of unity. Since the $D(f_i)$ cover the affine open U , the f_i generate the unit ideal of $\Gamma(X, U)$; hence so do the powers f_i^{N+M} , and we may write $1 = \sum_l h_l f_l^{N+M}$ with $h_l \in \Gamma(X, U)$.

Step 4: assemble the cobounding. Set $b_i = \sum_l h_l f_l^M e_{il} \in \Gamma(X, D(f_i))$. On $D(f_i) \cap D(f_j)$,

$$b_i - b_j = \sum_l h_l f_l^M (e_{il} - e_{jl}) = \sum_l h_l f_l^M (f_l^N a_{ij} + t_{ijl}) = \left(\sum_l h_l f_l^{N+M} \right) a_{ij} = a_{ij},$$

using $f_l^M t_{ijl} = 0$ in the third equality. $\square$

Theorem 310 (Degree-one Čech cobounding on an affine scheme). *Let X be an affine scheme and let $f_1, \dots, f_n$ be global sections of the structure sheaf with $D(f_1) \cup \dots \cup D(f_n) = X$. Let $a_{ij} \in \Gamma(X, D(f_i) \cap D(f_j))$ be a Čech 1-cocycle as in ??. Then there are sections $b_i \in \Gamma(X, D(f_i))$ with $a_{ij} = b_i - b_j$ on $D(f_i) \cap D(f_j)$ for all i, j .*

Proof. The special case $U = X$ of ??: an affine scheme is an affine open of itself. $\square$

5.5 Vanishing of degree-one cohomology on affine opens

Lemma 311 (A vanishing criterion for Ext^1). *Let $\mathcal{A}$ be an abelian category (with Ext -groups), let $\iota : A \rightarrow I$ be a monomorphism with I injective, and let L be an object. If every morphism $L \rightarrow \text{coker } \iota$ lifts along the projection $\pi : I \rightarrow \text{coker } \iota$, then $\text{Ext}^1(L, A) = 0$.*

Proof. Consider the short exact sequence $0 \rightarrow A \rightarrow I \rightarrow \text{coker } \iota \rightarrow 0$ and the induced long exact sequence

$$\text{Hom}(L, I) \xrightarrow{\pi_*} \text{Hom}(L, \text{coker } \iota) \xrightarrow{\delta} \text{Ext}^1(L, A) \longrightarrow \text{Ext}^1(L, I).$$

Let $z \in \text{Ext}^1(L, A)$. Its image in $\text{Ext}^1(L, I)$ vanishes because I is injective, so by exactness $z = \delta(x)$ for some $x : L \rightarrow \text{coker } \iota$. By hypothesis $x = \pi \circ \psi$ for some $\psi : L \rightarrow I$, and $\delta \circ \pi_* = 0$ (composite of consecutive maps of the long exact sequence); hence $z = \delta(\pi_*(\psi)) = 0$. $\square$

Lemma 312 (Sections are left exact). *Let $\iota : F \rightarrow G$ be a monomorphism of sheaves of R -modules on a site and let U be an object of the site. If $c \in G(U)$ maps to zero in $(\text{coker } \iota)(U)$, then $c = \iota_U(a)$ for a (unique) $a \in F(U)$.*

Proof. In an abelian category, a monomorphism is a kernel of its cokernel. Evaluation of sheaves at U preserves limits: it is the composite of the inclusion of sheaves into presheaves (a right adjoint, sheafification being left adjoint to it) with evaluation of presheaves at U (limits of presheaves are computed objectwise). Hence $F(U) \rightarrow G(U)$ is a kernel of $G(U) \rightarrow (\text{coker } \iota)(U)$, which is the claim. $\square$

Lemma 313 (Monomorphisms are injective on sections). *A monomorphism of sheaves of R -modules on a site is injective on sections over every object of the site.*

Proof. Evaluation of sheaves at an object preserves limits (as in ??), hence preserves monomorphisms, and a monomorphism of R -modules is an injective map. $\square$

Theorem 314 (Sections surjectivity on an affine open). *Let k be a commutative ring, X a scheme over $\text{Spec } k$, U an affine open of X , G a sheaf of k -modules on the small Zariski site of X , and $\iota : \mathcal{O}_{X/k} \rightarrow G$ a monomorphism. Then every section of $\text{coker } \iota$ over U lifts to a section of G over U : the map $G(U) \rightarrow (\text{coker } \iota)(U)$ is surjective.*

Proof. Let $q \in (\text{coker } \iota)(U)$ and write $\pi : G \rightarrow \text{coker } \iota$ for the projection.

(a) π is an epimorphism of sheaves, hence locally surjective on sections: every point of U has an open neighbourhood on which the restriction of q lifts to G . Shrinking, and since the basic opens of U form a basis of U (as U is affine), every point of U lies in some basic open $D(g) \subseteq U$ carrying a lift of $q|_{D(g)}$.

(b) U is quasi-compact (it is affine), so there are finitely many sections $f_1, \dots, f_n \in \Gamma(X, U)$ with $D(f_1) \cup \dots \cup D(f_n) = U$ and lifts $s_i \in G(D(f_i))$ with $\pi(s_i) = q|_{D(f_i)}$.

(c) On $D(f_i) \cap D(f_j)$ the difference $s_i - s_j$ is killed by π , so by ?? there is a unique $a_{ij} \in \Gamma(X, D(f_i) \cap D(f_j))$ with $\iota(a_{ij}) = s_i - s_j$.

(d) The a_{ij} form a Čech 1-cocycle: on triple intersections, $\iota(a_{il}) = s_i - s_l = (s_i - s_j) + (s_j - s_l) = \iota(a_{ij} + a_{jl})$, and ι is injective on sections (??).

(e) By ?? (applied to the affine open U) there are $b_i \in \Gamma(X, D(f_i))$ with $a_{ij} = b_i - b_j$ on overlaps. The corrected lifts $s_i - \iota(b_i)$ then agree on all overlaps, so by the sheaf condition they glue to a section $s \in G(U)$.

(f) Finally $\pi(s) = q$: on each $D(f_i)$ we have $\pi(s) = \pi(s_i) - \pi(\iota(b_i)) = q|_{D(f_i)}$ since $\pi \circ \iota = 0$, and sections of the sheaf $\text{coker } \iota$ that agree on a cover of U agree on U . $\square$

Theorem 315 (Vanishing of $H^{1'}$ on affine opens). *Let k be a commutative ring, X a scheme over $\text{Spec } k$ and U an affine open of X . Then*

$$H^{1'}(U, \mathcal{O}_{X/k}) = 0.$$

No hypothesis on X beyond the affineness of U is needed.

Proof. The category of sheaves of k -modules on the small Zariski site of X is Grothendieck abelian, hence has enough injectives: choose a monomorphism $\iota : \mathcal{O}_{X/k} \rightarrow G$ with G injective. By ?? (with $L = k[U]$ the free sheaf of k -modules on U), it suffices to lift every morphism $k[U] \rightarrow \text{coker } \iota$ along $G \rightarrow \text{coker } \iota$. By ??, morphisms out of $k[U]$ are, compatibly with postcomposition, sections over U ; so the required lifting is exactly the surjectivity of $G(U) \rightarrow (\text{coker } \iota)(U)$, which is ??. $\square$

Theorem 316 (Vanishing of H^1 on affine schemes). *Let k be a commutative ring and X an affine scheme over $\text{Spec } k$. Then*

$$H^1(X, \mathcal{O}_{X/k}) = 0.$$

Proof. The category of sheaves of k -modules on the small Zariski site of X is Grothendieck abelian, hence has enough injectives: choose a monomorphism $\iota : \mathcal{O}_{X/k} \rightarrow G$ with G injective. By ?? (with $L = k_J$ the constant sheaf), it suffices to lift every morphism $k_J \rightarrow \text{coker } \iota$ along $G \rightarrow \text{coker } \iota$. By ?? and its naturality ??, morphisms out of k_J are global sections, compatibly with postcomposition; so the required lifting is exactly the surjectivity of $G(X) \rightarrow (\text{coker } \iota)(X)$, i.e. of $G(\top) \rightarrow (\text{coker } \iota)(\top)$, which is ?? applied to the affine open $U = \top$ (an affine scheme is an affine open of itself). $\square$

5.6 Twisted affine vanishing: quasi-coherent coefficients

The vanishing of the previous section is stated for the structure sheaf, but Serre's argument only ever uses two features of $\mathcal{O}_{X/k}$: that ambient sections act on sections, and that sections localise on affine opens. This section isolates those two features as a packaging on an arbitrary sheaf of k -modules and reruns the argument, obtaining degree-one vanishing on an affine open with those coefficients. The intended application is to line bundles trivialised on an affine chart, for which the packaging holds chart by chart.

For a sheaf F of k -modules on the small Zariski site of X and an inclusion of opens $W \subseteq V$, write $s|_W$ (*section restriction*) for the image of $s \in F(V)$ under the k -linear restriction $F(V) \rightarrow F(W)$; it is functorial in the inclusion and natural in F .

Definition 317 (Quasi-coherence packaging on an open). Let X be a scheme, U an open of X , and F a sheaf of k -modules on the small Zariski site of X . A *quasi-coherence packaging* of F on U is the data of:

- **(P1) an action** of the fixed ring $\Gamma(X, U)$: for every open $W \leq U$ a scaling operation $r \cdot m$ of $r \in \Gamma(X, U)$ on $m \in F(W)$, satisfying the module laws over $\Gamma(X, U)$ (unit, associativity, and both distributivities) and natural under section restriction, $(r \cdot m)|_{W'} = r \cdot (m|_{W'})$ for $W' \leq W$;
- **(P2) localisation on affine opens** $V \leq U$, for every $g \in \Gamma(X, U)$, in the two-lemma form of localisation away from g :
 - *denominator clearing* — every $c \in F(V \cap D(g))$ is $(g^N \cdot e)|_{V \cap D(g)}$ for some $N \geq 0$ and $e \in F(V)$;
 - *defect annihilation* — every $t \in F(V)$ with $t|_{V \cap D(g)} = 0$ is killed by a power of g : $g^M \cdot t = 0$ for some $M \geq 0$.

The action is by the single ring $\Gamma(X, U)$ (not a per-open module structure), with the inclusion $W \leq U$ an explicit argument.

Theorem 318 (The structure sheaf is quasi-coherently packaged). *For X a scheme over $\text{Spec } k$, the structure sheaf $\mathcal{O}_{X/k}$, viewed as a sheaf of k -modules, carries a quasi-coherence packaging on every open U : the action of $r \in \Gamma(X, U)$ on $s \in \Gamma(X, W)$ is $r|_W \cdot s$, the ordinary restriction-then-multiplication.*

Proof. The (P1) module laws are the ring axioms of $\Gamma(X, W)$, and naturality under restriction holds because restriction is a ring homomorphism. The two (P2) axioms, with the action spelled out, are exactly denominator clearing (??) and annihilation (??) for the structure sheaf. $\square$

Theorem 319 (Degree-one Čech cobounding, quasi-coherent coefficients). *Let U be an affine open of a scheme X , covered by basic opens $D(f_1), \dots, D(f_n)$, and let F be a sheaf of k -modules*

with a quasi-coherence packaging on U . Every Čech 1-cocycle $a_{ij} \in F(D(f_i) \cap D(f_j))$ (with $a_{il} = a_{ij} + a_{jl}$ on triple overlaps) is a coboundary: there are $b_i \in F(D(f_i))$ with $a_{ij} = b_i|_{D(f_i) \cap D(f_j)} - b_j|_{D(f_i) \cap D(f_j)}$.

Proof. Verbatim the structure-sheaf argument ??, reading the packaging axioms in place of the structure-sheaf localisation. Clear denominators with one uniform exponent N (denominator clearing of the packaging), producing $e_{ij} \in F(D(f_i))$ with $e_{ij}|_{D(f_i) \cap D(f_j)} = f_j^N \cdot a_{ij}$. The triple-overlap defects $t_{ijl} = e_{il} - e_{jl} - f_l^N \cdot a_{ij}$ vanish on the triple intersections by the cocycle identity, hence are killed by a uniform power f_l^M (defect annihilation). With a partition of unity $1 = \sum_l h_l f_l^{N+M}$ of the affine U , set $b_i = \sum_l h_l f_l^M \cdot e_{il}$; the assembly $b_i - b_j = \sum_l h_l f_l^M (e_{il} - e_{jl}) = (\sum_l h_l f_l^{N+M}) \cdot a_{ij} = a_{ij}$ uses only that the action distributes over restriction and sums, i.e. the (P1) mixin. $\square$

Theorem 320 (Sections surjectivity on an affine open, quasi-coherent coefficients). *Let U be an affine open of a scheme X , let F be a sheaf of k -modules with a quasi-coherence packaging on U , and let $\iota : F \rightarrow G$ be a monomorphism of sheaves of k -modules. Then every section of $\text{coker } \iota$ over U lifts to a section of G over U .*

Proof. Verbatim ?? with F in place of $\mathcal{O}_{X/k}$. Local lifts of $q \in (\text{coker } \iota)(U)$ exist on a finite basic cover $D(f_i)$ of U by local surjectivity of the epimorphism $\text{coker } \iota$ -projection and quasi-compactness. On overlaps the differences of the lifts are killed by the projection, so by left-exactness of sections (??) they come from F -sections a_{ij} ; these form a Čech 1-cocycle (ι is injective on sections, ??), which cobounds by ??. The corrected lifts glue to a section of G over U mapping to q . $\square$

Theorem 321 (Twisted affine vanishing of H^1). *Let U be an affine open of a scheme X and F a sheaf of k -modules with a quasi-coherence packaging on U . Then the degree-one cohomology of the object U with coefficients in F vanishes:*

$$H^1(U, F) = 0.$$

Taking $F = \mathcal{O}_{X/k}$ recovers ??.

Proof. As in ??: choose a monomorphism $\iota : F \rightarrow G$ with G injective; by the Ext^1 vanishing criterion ?? (with $L = k[U]$ the free sheaf of k -modules on U) it suffices to lift every morphism $k[U] \rightarrow \text{coker } \iota$ along $G \rightarrow \text{coker } \iota$, which by ?? is the surjectivity of $G(U) \rightarrow (\text{coker } \iota)(U)$ — Theorem ??. $\square$

Theorem 322 (Twisted affine vanishing of H^1). *Let X be an affine scheme and F a sheaf of k -modules with a quasi-coherence packaging on $\mathbb{T}$. Then the degree-one cohomology of the site vanishes, $H^1(X, F) = 0$. Taking $F = \mathcal{O}_{X/k}$ recovers ??.*

Proof. Apply ?? to the affine open $U = \mathbb{T}$ (an affine scheme is an affine open of itself) and transport the object-cohomology vanishing to the site cohomology along the equivalence ??. $\square$

The two (P2) axioms of the packaging are not merely *sufficient* for Serre's argument: they say precisely that restriction from an affine chart to a basic open of the chart is a localisation of modules. The remaining nodes of this section record that bridge, which lets any packaged sheaf on an affine chart be read through the general theory of localised modules — in particular, it is how the two-lattice description of the two-cover complex consumes the packaging.

Definition 323 (The chart-ring module structure on packaged sections). Let F carry a quasi-coherence packaging on U and let $V \leq U$. The (P1) action makes the k -module $F(V)$ a module over the chart ring $\Gamma(X, U)$: the unit, associativity and distributivity laws are the (P1) axioms, and the remaining law $0 \cdot m = 0$ follows by adding $0 \cdot m$ to both sides of $(0 + 0) \cdot m = 0 \cdot m$ and cancelling.

Definition 324 (Restriction is linear over the chart ring). For opens $V \leq W \leq U$, section restriction $F(W) \rightarrow F(V)$ is $\Gamma(X, U)$ -linear for the module structures of ?? : it is additive, being a map of k -modules, and it commutes with the action by the (P1) restriction-naturality $(r \cdot m)|_V = r \cdot (m|_V)$.

Theorem 325 (The packaging axioms are the localisation axioms). *Let U be an affine open of a scheme X , let F be a sheaf of k -modules with a quasi-coherence packaging on U , and let $g \in \Gamma(X, U)$ act invertibly on $F(U \cap D(g))$ (for the module structure of ??). Then the restriction map $F(U) \rightarrow F(U \cap D(g))$ exhibits $F(U \cap D(g))$ as the localisation of the $\Gamma(X, U)$ -module $F(U)$ at the multiplicative set $S = \{1, g, g^2, \dots\}$ of powers of g ; that is:*

1. every element of S acts invertibly on $F(U \cap D(g))$;
2. every $c \in F(U \cap D(g))$ is of the form $g^{-N} \cdot (e|_{U \cap D(g)})$ for some $N \geq 0$ and $e \in F(U)$;
3. any two sections of $F(U)$ with equal restriction to $U \cap D(g)$ differ by an element killed by a power of g .

These three properties characterise the localisation map $M \rightarrow S^{-1}M$ up to unique isomorphism under $F(U)$.

Proof. Each axiom is the term-for-term image of one piece of the packaging data. (1) Powers of an invertibly-acting element act invertibly. (2) Denominator clearing on the affine open U itself produces, for $c \in F(U \cap D(g))$, an exponent N and a section $e \in F(U)$ with $e|_{U \cap D(g)} = g^N \cdot c$; acting by the inverse of g^N gives $c = g^{-N} \cdot (e|_{U \cap D(g)})$. (3) If $x_1|_{U \cap D(g)} = x_2|_{U \cap D(g)}$ then $x_1 - x_2$ restricts to 0 on $U \cap D(g)$, so defect annihilation yields $M \geq 0$ with $g^M \cdot (x_1 - x_2) = 0$. $\square$

Theorem 326 (Sections over a basic open are the localised sections). *Let X be a scheme over $\text{Spec } k$, U an affine open, and $g \in \Gamma(X, U)$. Then restriction exhibits $\Gamma(X, U \cap D(g))$ as the localisation of $\Gamma(X, U)$ at the powers of g — the classical $\Gamma(X, D(g)) = \Gamma(X, U)_g$, with no hypothesis on g .*

Proof. The structure sheaf carries the packaging on U (??), with the action of $r \in \Gamma(X, U)$ given by restriction-then-multiplication. The unit hypothesis of ?? is discharged for free: g restricts to a unit on $D(g)$, hence on $U \cap D(g)$, so multiplication by $g|_{U \cap D(g)}$ — which is the packaging action of g — is invertible on $\Gamma(X, U \cap D(g))$. Apply ??. $\square$

5.7 The Mayer–Vietoris $(0, 1)$ -slice

For a Mayer–Vietoris square S (typically: opens $X_1 = U \cap V$, $X_2 = U$, $X_3 = V$, $X_4 = U \cup V$ of a topological space) and a sheaf of R -modules F on the site, this section produces the exactness of

$$0 \rightarrow F(X_4) \rightarrow F(X_2) \times F(X_3) \rightarrow F(X_1) \xrightarrow{\delta} H^1(X_4, F) \rightarrow H^1(X_2, F) \times H^1(X_3, F)$$

and derives the two-cover cokernel computation: if $H^1(X_2, F)$ and $H^1(X_3, F)$ vanish, then $F(X_1)/\text{range}(\text{difference of restrictions}) \cong H^1(X_4, F)$, the quotient map being the connecting homomorphism δ .

Definition 327 (Mayer–Vietoris square). *Provided by Mathlib.* Let $(\mathcal{C}, J)$ be a small site. A Mayer–Vietoris square consists of four objects X_1, X_2, X_3, X_4 of $\mathcal{C}$ and morphisms $f_{12} : X_1 \rightarrow X_2$, $f_{13} : X_1 \rightarrow X_3$, $f_{24} : X_2 \rightarrow X_4$, $f_{34} : X_3 \rightarrow X_4$ such that the two composites $X_1 \rightarrow X_2 \rightarrow X_4$ and $X_1 \rightarrow X_3 \rightarrow X_4$ agree, f_{13} is a monomorphism, and, after applying the representable-presheaf-then-sheafification functor $\mathcal{C} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Set})$, the resulting square of sheaves of sets is a pushout square: the image of X_4 is the pushout, over the image of X_1 , of the images of X_2 and X_3 . The prototypical example is $X_1 = U \cap V$, $X_2 = U$, $X_3 = V$, $X_4 = U \cup V$ for two opens U, V of a topological space, with the evident inclusions.

Definition 328 (The sheaf condition of a Mayer–Vietoris square). *Provided by Mathlib.* Let $S = (X_1, X_2, X_3, X_4, f_{12}, f_{13}, f_{24}, f_{34})$ be a Mayer–Vietoris square on $(\mathcal{C}, J)$ and let P be a presheaf on $\mathcal{C}$. Say P satisfies the sheaf condition for S when the square $P(X_4) \rightarrow P(X_2) \rightarrow P(X_1)$, $P(X_4) \rightarrow P(X_3) \rightarrow P(X_1)$ (restriction along f_{24}, f_{12} and along f_{34}, f_{13}) is a pullback square: equivalently, restriction identifies $P(X_4)$ with the pairs $(u, v) \in P(X_2) \times P(X_3)$ of equal restriction to X_1 . Every sheaf satisfies the sheaf condition for every Mayer–Vietoris square of its site.

Definition 329 (The Mayer–Vietoris short complex of free sheaves). For a Mayer–Vietoris square S on $(\mathcal{C}, J)$, the complex of sheaves of R -modules

$$R[X_1] \xrightarrow{(R[f_{12}], -R[f_{13}])} R[X_2] \oplus R[X_3] \xrightarrow{R[f_{24}] + R[f_{34}]} R[X_4],$$

whose composite is zero by the commutativity of S 's square.

Theorem 330 (The Mayer–Vietoris short complex is short exact). *For every Mayer–Vietoris square S , the complex of ?? is a short exact sequence of sheaves of R -modules:*

$$0 \rightarrow R[X_1] \rightarrow R[X_2] \oplus R[X_3] \rightarrow R[X_4] \rightarrow 0.$$

Proof. The square of free sheaves is a pushout. The free- R -module-sheaf functor $U \mapsto R[U]$ (?? on morphisms) is a left adjoint of the forgetful functor to sheaves of sets (it factors the free- R -module-on-a-set functor, itself a left adjoint, through sheafification, also a left adjoint). Left adjoints preserve colimits, in particular pushout squares; applying this functor to the defining pushout square of S (Definition ???: the square of sheafified representables at X_1, X_2, X_3, X_4) shows that the square $R[X_1] \rightarrow R[X_2], R[X_1] \rightarrow R[X_3], R[X_2] \rightarrow R[X_4], R[X_3] \rightarrow R[X_4]$ is again a pushout, now in the category of sheaves of R -modules.

Pushout along a mono is short exact. In an additive category with biproducts, a commuting square with legs $p : A \rightarrow B$, $q : A \rightarrow C$, $r : B \rightarrow D$, $s : C \rightarrow D$ is a pushout square exactly when D , together with the map $(r, s) : B \oplus C \rightarrow D$, is a cokernel of the difference map $(p, -q) : A \rightarrow B \oplus C$. Applied to the pushout square above, this shows that $R[X_4]$, with the sum map, is a cokernel of the difference map $R[X_1] \rightarrow R[X_2] \oplus R[X_3]$; that is, the complex of ?? is exact at $R[X_2] \oplus R[X_3]$ and at $R[X_4]$, with the map to $R[X_4]$ an epimorphism.

The difference map is a monomorphism. By definition of a Mayer–Vietoris square, $f_{13} : X_1 \rightarrow X_3$ is a monomorphism of $\mathcal{C}$; postcomposition with a monomorphism is injective on morphism sets, so the representable-presheaf map it induces is a monomorphism of presheaves of sets, and both the free- R -module functor (on a set, mapping along an injection of index sets is injective on finitely supported functions) and sheafification preserve monomorphisms. Hence $R[f_{13}] : R[X_1] \rightarrow R[X_3]$ is a monomorphism of sheaves of R -modules, and so is its negative. The second component of the difference map $(R[f_{12}], -R[f_{13}]) : R[X_1] \rightarrow R[X_2] \oplus R[X_3]$ (its composite with the projection to $R[X_3]$) is exactly $-R[f_{13}]$; since a composite $g \circ h$ being a monomorphism forces h to be a monomorphism, the difference map itself is a monomorphism.

Together, the three paragraphs give short exactness. $\square$

Definition 331 (The restriction-difference map). For a Mayer–Vietoris square S and a sheaf F of R -modules, the R -linear map

$$F(X_2) \times F(X_3) \longrightarrow F(X_1), \quad (s_2, s_3) \mapsto s_2|_{X_1} - s_3|_{X_1},$$

restricting along f_{12} and f_{13} respectively and taking the difference.

Lemma 332 (Degree-zero exactness). For a Mayer–Vietoris square S and a sheaf F of R -modules, the sequence

$$0 \rightarrow F(X_4) \xrightarrow{(\text{res}, \text{res})} F(X_2) \times F(X_3) \xrightarrow{\text{moduleDiff}} F(X_1)$$

is exact: the pair of restrictions to X_2, X_3 is injective, and its image is exactly the kernel of ??.

Proof. This is the sheaf condition of S for (the sheaf of sets underlying) F (Definition ??): the map $F(X_4) \rightarrow F(X_2) \times F(X_3)$ has image exactly the pairs with equal restriction to X_1 (i.e. $\ker(\text{moduleDiff})$), and two elements of $F(X_4)$ with the same pair of restrictions are equal. $\square$

Lemma 333 (Contravariant long exact sequence of Ext). Provided by Mathlib. Let $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ be a short exact sequence in an abelian category with Ext -groups and let Y be any object. There is a canonical extension class $\delta \in \text{Ext}^1(C, A)$ (built from the connecting homomorphism of the sequence), and for every $n \geq 0$ the sequence of abelian groups

$$\text{Ext}^n(C, Y) \xrightarrow{g^*} \text{Ext}^n(B, Y) \xrightarrow{f^*} \text{Ext}^n(A, Y) \xrightarrow{\delta \cdot (-)} \text{Ext}^{n+1}(C, Y) \xrightarrow{g^*} \text{Ext}^{n+1}(B, Y) \xrightarrow{f^*} \text{Ext}^{n+1}(A, Y)$$

is exact at each of the four inner terms, where f^*, g^* are precomposition (contravariant functoriality of $\text{Ext}(-, Y)$) and $\delta \cdot (-)$ is composition (Yoneda product) with δ . (Convention: $\text{Ext}^0 = \text{Hom}$.) If $B = X_2 \oplus X_3$ is a direct sum, $\text{Ext}^n(B, Y) \cong \text{Ext}^n(X_2, Y) \times \text{Ext}^n(X_3, Y)$ naturally, compatibly with f^*, g^* .

Definition 334 (The connecting map). For a Mayer–Vietoris square S and a sheaf F of R -modules, the connecting map $F(X_1) \rightarrow H^1(X_4, F)$: the identification $F(X_1) \cong H^0(X_1, F) = \text{Ext}^0(R[X_1], F)$ of ??, followed by the map $\delta \cdot (-) : \text{Ext}^0(R[X_1], F) \rightarrow \text{Ext}^1(R[X_4], F) = H^1(X_4, F)$ of ?? for the short exact sequence of ?? ($A = R[X_1]$, $B = R[X_2] \oplus R[X_3]$, $C = R[X_4]$).

Lemma 335 (Exactness at $F(X_1)$). For a Mayer–Vietoris square S and a sheaf F of R -modules, $\ker(\text{moduleDelta}) = \text{range}(\text{moduleDiff})$ as submodules of $F(X_1)$: the connecting class of a difference of restrictions vanishes, and conversely every section of $F(X_1)$ with vanishing connecting class is such a difference.

Proof. This is exactness of the sequence of ?? (for $n = 0$, the short exact sequence of ??, $Y = F$) at the term $\text{Ext}^0(A, F) = F(X_1)$, transported along the identifications $\text{Ext}^0(B, F) \cong F(X_2) \times F(X_3)$ (??, direct sum) and $\text{Ext}^0(A, F) \cong F(X_1)$ (??): under these identifications $f^* : \text{Ext}^0(B, F) \rightarrow \text{Ext}^0(A, F)$ becomes moduleDiff and $\delta \cdot (-) : \text{Ext}^0(A, F) \rightarrow \text{Ext}^1(C, F)$ becomes moduleDelta , by naturality of the section equivalence in the site variable (??), which identifies precomposition at the level of morphisms out of a free sheaf with restriction at the level of sections. $\square$

Lemma 336 (Exactness at $H^1(X_4, F)$). For a Mayer–Vietoris square S and a sheaf F of R -modules, $\ker(\text{res}_{f_{24}}) \cap \ker(\text{res}_{f_{34}}) = \text{range}(\text{moduleDelta})$ as submodules of $H^1(X_4, F)$, where $\text{res}_{f_{24}}, \text{res}_{f_{34}} : H^1(X_4, F) \rightarrow H^1(X_2, F), H^1(X_3, F)$ are the restrictions of ??: the connecting class of any section of $F(X_1)$ restricts to zero on both X_2 and X_3 , and conversely every class restricting to zero on both is a connecting class.

Proof. As in ??, this is exactness of the sequence of ?? (for $n = 0$, i.e. at the term $\text{Ext}^1(C, F)$ between $\delta \cdot (-)$ and g^*) transported along the same identifications, together with the splitting $\text{Ext}^1(B, F) \cong H^1(X_2, F) \times H^1(X_3, F)$ under which g^* becomes the pair $(\text{res}_{f_{24}}, \text{res}_{f_{34}})$. $\square$

Theorem 337 (Surjectivity of the connecting map). *If $H^1(X_2, F)$ and $H^1(X_3, F)$ are both zero, then $\text{moduleDelta} : F(X_1) \rightarrow H^1(X_4, F)$ is surjective.*

Proof. Immediate from ??: $\ker(\text{res}_{f_{24}}) \cap \ker(\text{res}_{f_{34}}) = H^1(X_4, F)$ once both targets vanish, so $\text{range}(\text{moduleDelta}) = H^1(X_4, F)$. $\square$

Definition 338 (The connecting map on the cokernel). Since $\text{range}(\text{moduleDiff}) \subseteq \ker(\text{moduleDelta})$ (??), moduleDelta descends to an R -linear map

$$F(X_1)/\text{range}(\text{moduleDiff}) \longrightarrow H^1(X_4, F).$$

Theorem 339 (The Mayer–Vietoris two-cover cokernel computation). *If $H^1(X_2, F)$ and $H^1(X_3, F)$ are both zero, the map of ?? is an R -linear isomorphism*

$$F(X_1)/\text{range}(\text{moduleDiff}) \cong H^1(X_4, F).$$

Proof. Injectivity: if the class of $s \in F(X_1)$ maps to 0, then $\text{moduleDelta}(s) = 0$, so by ?? $s \in \text{range}(\text{moduleDiff})$, i.e. the class of s is already 0. Surjectivity is ??. $\square$

Lemma 340 (Compatibility with the connecting map). *Under the isomorphism of ??, the class of $s \in F(X_1)$ corresponds to $\text{moduleDelta}(s)$.*

Proof. Immediate from the construction of ?? and ??. $\square$

5.8 The two-affine-cover cokernel bridge

For a scheme X over $\text{Spec } k$ covered by two affine opens $U_0 \cup U_1 = X$, this section computes the degree-one cohomology of the structure sheaf as

$$H^1(X, \mathcal{O}_{X/k}) \cong \Gamma(X, U_0 \cap U_1) / \text{range}((s_0, s_1) \mapsto s_0|_{U_0 \cap U_1} - s_1|_{U_0 \cap U_1}),$$

the computational access point for the genus of a curve. No hypothesis on the intersection $U_0 \cap U_1$ is needed: the derived Mayer–Vietoris sequence of Section ?? is exact regardless, and affineness of the two pieces alone kills their degree-one cohomology (??) — the classical affine-intersection hypothesis is needed only for the Čech-complex identification of H^1 , which this route bypasses.

Definition 341 (Mayer–Vietoris square of two opens). *Provided by Mathlib.* For a topological space T and opens $X_1, X_2, X_3, X_4 \subseteq T$ with $X_1 = X_2 \cap X_3$ and $X_4 = X_2 \cup X_3$, the square (X_1, X_2, X_3, X_4) with the inclusion morphisms is a Mayer–Vietoris square (Definition ??) on the small Zariski site of T .

Definition 342 (The two-cover square). For a scheme X and opens $U_0, U_1 \subseteq X$ with $U_0 \cup U_1 = X$, the Mayer–Vietoris square $(U_0 \cap U_1, U_0, U_1, X)$ on the small Zariski site of X (Definition ??, specialized to $X_4 = X$), with the inclusion morphisms.

Theorem 343 (The two-cover H^1 computation, general coefficients). *Let k be a commutative ring, X a scheme over $\text{Spec } k$, $U_0, U_1 \subseteq X$ opens with $U_0 \cup U_1 = X$, and F a sheaf of k -modules on the small Zariski site of X with $H^1(U_0, F) = H^1(U_1, F) = 0$. Then*

$$H^1(X, F) \cong F(U_0 \cap U_1) / \text{range}((s_0, s_1) \mapsto s_0|_{U_0 \cap U_1} - s_1|_{U_0 \cap U_1}).$$

Proof. Compose the isomorphism $H^1(X, F) \cong H^1(X, F)$ of ?? (X is a terminal object of its own small Zariski site) with the inverse of the isomorphism of ?? for the two-cover square of ??, whose moduleDiff is exactly the displayed restriction-difference map. $\square$

Definition 344 (The two-cover restriction-difference map). Let k be a commutative ring, X a scheme over $\text{Spec } k$, and $U_0, U_1 \subseteq X$ opens. The k -linear map

$$\Gamma(X, U_0) \times \Gamma(X, U_1) \longrightarrow \Gamma(X, U_0 \cap U_1), \quad (s_0, s_1) \longmapsto s_0|_{U_0 \cap U_1} - s_1|_{U_0 \cap U_1}.$$

When $U_0 \cup U_1 = X$, this is the map ?? of the two-cover square (Definition ??) for the structure sheaf $\mathcal{O}_{X/k}$.

Definition 345 (The two-cover cokernel carrier). The k -module $\Gamma(X, U_0 \cap U_1)/\text{range}(\text{diff})$, the cokernel of the map of ??.

Lemma 346 (Sections extending to one piece vanish in the cokernel). *For $t \in \Gamma(X, U_0)$, the class of $t|_{U_0 \cap U_1}$ in ?? is zero; symmetrically, for $t \in \Gamma(X, U_1)$ the class of $t|_{U_0 \cap U_1}$ is zero.*

Proof. The first is the class of $\text{diff}(t, 0)$, the second of $\text{diff}(0, -t)$ (Definition ??), both in the range of diff . $\square$

Definition 347 (The two-cover connecting map). Assume $U_0 \cup U_1 = X$. The k -linear map $\Gamma(X, U_0 \cap U_1) \rightarrow H^1(X, \mathcal{O}_{X/k})$: the connecting map ?? of the two-cover square (Definition ??) for $F = \mathcal{O}_{X/k}$, followed by the inverse of the isomorphism $H^1(X, \mathcal{O}_{X/k}) \cong H^1(X, \mathcal{O}_{X/k})$ of ??.

Lemma 348 (The connecting class of a difference vanishes). *For $t \in \Gamma(X, U_0) \times \Gamma(X, U_1)$, $\text{delta}(\text{diff}(t)) = 0$.*

Proof. $\text{diff}(t)$ equals the two-cover square's moduleDiff(t) (Definition ??), whose connecting class vanishes by ??; transport along the (linear, hence class-zero-preserving) isomorphism of ?? used in the definition of delta . $\square$

Lemma 349 (The connecting class of an extending section vanishes). *For $t \in \Gamma(X, U_0)$, $\text{delta}(t|_{U_0 \cap U_1}) = 0$; symmetrically, for $t \in \Gamma(X, U_1)$, $\text{delta}(t|_{U_0 \cap U_1}) = 0$.*

Proof. $t|_{U_0 \cap U_1} = \text{diff}(t, 0)$, resp. $-\text{diff}(0, t)$ (Definition ??), so this is ?? (using delta 's linearity in the second case). $\square$

Theorem 350 (Surjectivity of the two-cover connecting map). *If U_0 and U_1 are affine opens of X (still with $U_0 \cup U_1 = X$), then $\text{delta} : \Gamma(X, U_0 \cap U_1) \rightarrow H^1(X, \mathcal{O}_{X/k})$ is surjective.*

Proof. By ??, $H^1(U_0, \mathcal{O}_{X/k}) = H^1(U_1, \mathcal{O}_{X/k}) = 0$; by ?? applied to the two-cover square, its connecting map moduleDelta is surjective onto $H^1(X, \mathcal{O}_{X/k})$, and transporting along the isomorphism of ?? used to define delta preserves surjectivity. $\square$

Theorem 351 (The two-affine-cover H^1 cokernel bridge). *Let k be a commutative ring, X a scheme over $\text{Spec } k$, and U_0, U_1 affine opens of X with $U_0 \cup U_1 = X$. There is a k -linear isomorphism*

$$H^1(X, \mathcal{O}_{X/k}) \cong \Gamma(X, U_0 \cap U_1)/\text{range}(\text{diff}),$$

where $\text{diff}(s_0, s_1) = s_0|_{U_0 \cap U_1} - s_1|_{U_0 \cap U_1}$ (??). No hypothesis on $U_0 \cap U_1$ (affineness, separatedness, or even quasi-compactness) is needed: the derived Mayer–Vietoris sequence of ?? is exact regardless, and affineness of U_0, U_1 alone kills their degree-one cohomology (??); the classical hypothesis on the intersection is needed only for the Čech-complex identification of H^1 , which this route bypasses.

Proof. By ??, $H^1(U_0, \mathcal{O}_{X/k}) = H^1(U_1, \mathcal{O}_{X/k}) = 0$. Apply ?? to the two-cover square of ?? with $F = \mathcal{O}_{X/k}$, giving $\Gamma(X, U_0 \cap U_1)/\text{range}(\text{diff}) \cong H^1(X, \mathcal{O}_{X/k})$, and compose with the inverse of the isomorphism $H^1(X, \mathcal{O}_{X/k}) \cong H^1(X, \mathcal{O}_{X/k})$ of ??. $\square$

Lemma 352 (Compatibility with the connecting map). *Under the isomorphism of ??, the class $\text{delta}(s)$ (Definition ??) corresponds, for every $s \in \Gamma(X, U_0 \cap U_1)$, to the class of s in $\Gamma(X, U_0 \cap U_1)/\text{range}(\text{diff})$.*

Proof. Unwind the definitions of ?? and ?? (both built from the same two ingredients, ?? and ??) and apply the compatibility ?? of the Mayer–Vietoris cokernel isomorphism with its connecting map. $\square$

Lemma 353 (Compatibility with the connecting map, inverse form). *The inverse of the isomorphism of ?? sends the class of $s \in \Gamma(X, U_0 \cap U_1)$ to $\text{delta}(s)$.*

Proof. Apply the isomorphism of ?? to both sides of ?? and use injectivity. $\square$

5.9 Base change of the two-cover cohomology over an affine test ring

Fix a field k and an object C of $\text{Over}(\text{Spec } k)$. For a commutative k -algebra R — the *affine test ring* — base change produces the relative curve $C_R = C \otimes \text{Spec } R$ over $\text{Spec } R$; an affine two-chart cover of C (Section ?? and Chapter ??) pulls back to one of C_R , and the two-affine-cover bridge of the previous section then computes $H^1(C_R, \mathcal{O}_{C_R/R})$ as an explicit two-cover cokernel of R -modules. When C is a smooth curve, the degree-zero cohomology of the relative curve is the test ring itself, $H^0(C_R, \mathcal{O}_{C_R/R}) = R$.

Definition 354 (The relative curve over the test ring). For $C \in \text{Over}(\text{Spec } k)$ and a commutative k -algebra R , the *relative curve* C_R is the underlying scheme of the fibre product $C \otimes \text{Spec } R$ of C with the object $\text{Spec } R$ of $\text{Over}(\text{Spec } k)$, regarded as a scheme over $\text{Spec } R$ via the second projection $\text{pr}_2 : C \otimes \text{Spec } R \rightarrow \text{Spec } R$.

Definition 355 (The base-changed affine two-cover). Let $D = (V_0, V_1)$ be an affine two-chart cover (??) of the underlying scheme of C . The *base-changed cover* of the relative curve C_R is the pair of preimages $(\text{pr}_1^{-1}V_0, \text{pr}_1^{-1}V_1)$ of the charts of D under the first projection $\text{pr}_1 : C_R \rightarrow C$; by ?? it is again an affine two-chart cover, now of C_R .

Lemma 356 (The base-changed charts are affine and cover the relative curve). *The two charts $\text{pr}_1^{-1}V_0$ and $\text{pr}_1^{-1}V_1$ of the base-changed cover are affine open, and their union is all of C_R .*

Proof. These are the affineness of the two charts and the covering property recorded in the affine-two-chart-cover structure $\text{pr}_1^{-1}D$ of ??. $\square$

Lemma 357 (The overlap of the base-changed charts). *The overlap of the two base-changed charts is the preimage of the overlap of D :*

$$\text{pr}_1^{-1}V_0 \cap \text{pr}_1^{-1}V_1 = \text{pr}_1^{-1}(V_0 \cap V_1).$$

Proof. Taking preimages under pr_1 commutes with intersection of opens. $\square$

Definition 358 (The relative two-cover cohomology carrier). For the structure sheaf of the relative curve C_R viewed as a sheaf of R -modules, its degree-one cohomology over R is the two-cover cokernel of the base-changed cover: an R -linear isomorphism

$$H^1(C_R, \mathcal{O}_{C_R/R}) \cong \Gamma(C_R, \text{pr}_1^{-1}V_0 \cap \text{pr}_1^{-1}V_1) / \text{range}(\text{diff}),$$

where $\text{diff}(s_0, s_1) = s_0|_{\cap} - s_1|_{\cap}$ (??).

Proof. The base-changed charts are affine and cover C_R (??), so this is the two-affine-cover H^1 cokernel bridge ?? applied over the base ring R to the scheme C_R with its base-changed cover; the affine-vanishing inputs required by that bridge are exactly the affineness of the two charts. $\square$

Theorem 359 (Degree-zero cohomology of the relative curve is the test ring). *Let C be a smooth curve over k (proper, smooth of relative dimension one, geometrically irreducible), so that C is in particular geometrically reduced. For every commutative k -algebra R , the degree-zero cohomology of the structure sheaf of the relative curve is the test ring itself:*

$$H^0(C_R, \mathcal{O}_{C_R/R}) = \Gamma(C_R, \top) \cong R,$$

the isomorphism being pullback of global sections along the second projection $\text{pr}_2 : C_R \rightarrow \text{Spec } R$. This is the degree-zero case of “cohomology commutes with base change”, universal in the test ring R .

Proof. Degree-zero cohomology is global sections, $H^0(C_R, \mathcal{O}_{C_R/R}) \cong \Gamma(C_R, \top)$ (??, over the base ring R). Specialize the base-change isomorphism ?? to the test object $T = \text{Spec } R$ and its affine open $V = \top$: pullback along the second projection is an isomorphism $\Gamma(\text{Spec } R, \top) \xrightarrow{\sim} \Gamma(C_R, \text{pr}_2^{-1}\top) = \Gamma(C_R, \top)$ (the hypotheses of ?? — C proper, geometrically irreducible and geometrically reduced — hold for a smooth curve). Composing with the identification $\Gamma(\text{Spec } R, \top) \cong R$ gives $\Gamma(C_R, \top) \cong R$. $\square$

5.10 The projective line: inputs for the finiteness argument

This section collects the facts about $\mathbb{P}^1$ and about morphisms over $\text{Spec } k$ that the finiteness argument of the next section consumes. It builds on the description of $\mathbb{P}^1$, its standard charts U_0, U_1 , the chart coordinates $t_{ij} = X_j/X_i \in A_{(X_i)}$, and the section-ring identifications $\gamma_0, \gamma_1, \gamma_{01}$ developed in Chapter ?? (Definitions ??, ??, ??–??); the statements here are recorded in this chapter because they are private inputs of the finiteness glue rather than general theory of the projective line. Throughout, for a homogeneous f of positive degree we write $A_{(f)} \rightarrow \Gamma(\mathbb{P}^1, D_+(f))$ for the canonical map from the degree-zero localization to the sections over the basic open (an isomorphism).

Lemma 360 (Morphisms over the base intertwine the structure algebra maps). *Let k be a commutative ring, let X, Y be schemes over $\text{Spec } k$ and let $\pi : X \rightarrow Y$ be a morphism compatible with the structure morphisms. For opens $U \subseteq Y$ and $V \subseteq X$ with $V \subseteq \pi^{-1}(U)$, the two composite ring homomorphisms*

$$\Gamma(\text{Spec } k, \top) \rightarrow \Gamma(Y, \top) \rightarrow \Gamma(Y, U) \xrightarrow{\pi^\sharp} \Gamma(X, V) \quad \text{and} \quad \Gamma(\text{Spec } k, \top) \rightarrow \Gamma(X, \top) \rightarrow \Gamma(X, V)$$

agree, where the unlabeled maps are induced by the structure morphisms and by restriction.

Proof. Pulling back sections along π commutes with restriction, so the first composite is the map on sections induced by π composed with the structure morphism of Y , restricted to V ; by compatibility this composite of morphisms is the structure morphism of X . $\square$

Lemma 361 (Pulling back the structure algebra map). *In the situation of ??, for every $r \in k$ the map $\pi^\sharp : \Gamma(Y, U) \rightarrow \Gamma(X, V)$ sends the image of r under the structure algebra map of Y (Definition ??) to the image of r under the structure algebra map of X .*

Proof. Evaluate the equality of ring homomorphisms of ?? at the section corresponding to r . $\square$

Lemma 362 (The basic-open models lie over the base). *Let $f \in k[X_0, X_1]$ be homogeneous of positive degree. The composite of the affine model $\text{Spec } A_{(f)} \rightarrow \mathbb{P}^1$ of the basic open $D_+(f)$ with the structure morphism $\mathbb{P}^1 \rightarrow \text{Spec } k$ is the morphism induced by the algebra map $k \rightarrow A_{(f)}$. (This generalizes ?? from the two standard charts to all standard basic opens.)*

Proof. As for ??: the canonical morphism $\text{Proj } A \rightarrow \text{Spec } A_0$, restricted to the affine model of a basic open, is induced by the ring homomorphism $A_0 \rightarrow A_{(f)}$, and composing with the isomorphism $k \cong A_0$ of ?? gives the claim. $\square$

Lemma 363 (The basic-open inclusion factors through its affine model). *Let $f \in k[X_0, X_1]$ be homogeneous of positive degree. The inclusion $D_+(f) \rightarrow \mathbb{P}^1$ is the composite of the canonical identification $D_+(f) \cong \text{Spec } A_{(f)}$ with the affine model $\text{Spec } A_{(f)} \rightarrow \mathbb{P}^1$.*

Proof. The affine model is by construction an open immersion with image $D_+(f)$, and the canonical identification is the induced isomorphism onto the image; composing the two recovers the inclusion. $\square$

Lemma 364 (The structure algebra map on a basic open). *Let $f \in k[X_0, X_1]$ be homogeneous of positive degree. The composite*

$$\Gamma(\text{Spec } k, \top) \rightarrow \Gamma(\mathbb{P}^1, \top) \rightarrow \Gamma(\mathbb{P}^1, D_+(f))$$

(structure morphism on global sections, then restriction) is the composite of the identification $\Gamma(\text{Spec } k, \top) \cong k$, the algebra map $k \rightarrow A_{(f)}$, and the canonical map $A_{(f)} \rightarrow \Gamma(\mathbb{P}^1, D_+(f))$.

Proof. By ?? and ??, the composite of the inclusion $D_+(f) \rightarrow \mathbb{P}^1$ with the structure morphism is the identification $D_+(f) \cong \text{Spec } A_{(f)}$ followed by the morphism induced by $k \rightarrow A_{(f)}$. Taking global sections of this equality of scheme morphisms, and identifying the global sections of the open subscheme $D_+(f)$ with $\Gamma(\mathbb{P}^1, D_+(f))$ — under which the map on sections induced by the identification with $\text{Spec } A_{(f)}$ becomes the canonical map $A_{(f)} \rightarrow \Gamma(\mathbb{P}^1, D_+(f))$ — yields the claim. $\square$

Lemma 365 (The distinguished fraction of a chart pair is the chart coordinate). *For $i, j \in \{0, 1\}$, the canonical degree-zero fraction $X_j^{\deg X_i} / X_i^{\deg X_j}$ attached to the pair (X_i, X_j) of homogeneous elements equals the chart coordinate $t_{ij} = X_j / X_i \in A_{(X_i)}$.*

Proof. Both degrees are one, and a first power is the element itself. $\square$

Theorem 366 (The chart coordinate cuts out the overlap). *For $i, j \in \{0, 1\}$, regard the chart coordinate $t_{ij} \in A_{(X_i)}$ as a section of the structure sheaf over the chart $U_i = D_+(X_i)$, via the canonical map $A_{(X_i)} \rightarrow \Gamma(\mathbb{P}^1, U_i)$. Then the basic open of this section in the scheme $\mathbb{P}^1$ — the locus of U_i where it is invertible — is the chart overlap:*

$$D(t_{ij}) = U_i \cap U_j.$$

Proof. Both sides are open subsets of U_i (the basic open of a section over U_i is contained in U_i), so it suffices to show that they have the same preimage under the identification $U_i \cong \operatorname{Spec} A_{(X_i)}$, and to transport the equality forward along this open immersion. No stalks are involved: everything is transported through the affine model.

On the one hand, the preimage of the basic open $D(t_{ij})$ is the basic open of $\operatorname{Spec} A_{(X_i)}$ at the ring element t_{ij} : under the identification of an affine open with the spectrum of its section ring, scheme-level basic opens of sections correspond to spectrum-level basic opens of ring elements.

On the other hand, the preimage of $U_j = D_+(X_j)$ under the affine model $\operatorname{Spec} A_{(X_i)} \rightarrow \mathbb{P}^1$ (through which the inclusion of U_i factors, ??) is the basic open at the canonical degree-zero fraction attached to the pair (X_i, X_j) : a homogeneous prime avoiding X_i avoids X_j exactly when the degree-zero fraction X_j/X_i is invertible in the corresponding localization. By ?? this fraction is t_{ij} , so the preimage of $U_i \cap U_j$ is again the basic open at t_{ij} . $\square$

Lemma 367 (The Laurent generators as restrictions from the charts). *Under the identification $\gamma_{01} : \Gamma(\mathbb{P}^1, U_0 \cap U_1) \cong k[T, T^{-1}]$ of ??:*

1. T corresponds to the restriction to the overlap of the chart coordinate t_{01} , regarded as a section over U_0 ;
2. T^{-1} corresponds to the restriction to the overlap of the chart coordinate t_{10} , regarded as a section over U_1 ;
3. each scalar $r \in k$ corresponds to the section over the overlap given by the image of r under the algebra map $k \rightarrow A_{(X_0 X_1)}$.

Proof. (1) and (2): by ?? (resp. ??), restriction from the chart U_0 (resp. U_1) to the overlap corresponds, under the identifications $\gamma_0, \gamma_1, \gamma_{01}$, to $t \mapsto T$ (resp. $t \mapsto T^{-1}$); apply this to the sections corresponding to the polynomial t , which are the chart coordinates. (3): γ_{01} is built from the k -algebra isomorphism θ of ??, hence matches the two algebra maps from k . $\square$

5.11 Finiteness of the first cohomology

Combining the two-affine-cover cokernel bridge with the two-lattice ladder of Chapter ?? along a finite morphism to the projective line yields the finiteness of H^1 . Classically this computation goes back to the explicit Čech computation of the cohomology of projective space — for $\mathbb{P}^1$ itself, the description of H^1 as the cokernel of the two-chart restriction-difference map into the Laurent ring, spanned by the finitely many “negative” monomials — extended to a curve by pushing forward along a finite morphism; the two-lattice ladder is exactly the abstraction of that monomial bookkeeping. Hartshorne’s cited calculation is for projective space over a Noetherian base; it is the model for the explicit two-chart argument below, whose hypotheses are stated independently.

Theorem 368 (Finiteness of H^1 along a finite morphism to the projective line). *Let k be a field, X a scheme over $\operatorname{Spec} k$, and $\pi : X \rightarrow \mathbb{P}^1$ a finite morphism over k (its composite with the structure morphism of $\mathbb{P}^1$ is the structure morphism of X). Then $H^1(X, \mathcal{O}_{X/k})$ is a finite k -module.*

Proof. Write U_0, U_1 for the standard charts of $\mathbb{P}^1$ and set $V_i = \pi^{-1}(U_i)$.

(a) *The cover and the cokernel.* Since π is finite, hence affine, the V_i are affine opens of X (??), and they cover X (??). By the two-affine-cover bridge ??,

$$H^1(X, \mathcal{O}_{X/k}) \cong N / \operatorname{range}(\operatorname{diff}), \quad N := \Gamma(X, V_0 \cap V_1),$$

with $\text{diff}(s_0, s_1) = s_0|_{V_0 \cap V_1} - s_1|_{V_0 \cap V_1}$ (??). It therefore suffices to prove that this quotient is a finite k -module.

(b) *The Laurent module structure.* Preimages commute with intersections, and $U_0 \cap U_1 = D_+(X_0 X_1)$ (??), so $V_0 \cap V_1 = \pi^{-1}(U_0 \cap U_1)$. Make N an algebra over the Laurent polynomial ring via the composite

$$k[T, T^{-1}] \xrightarrow{\gamma_{01}^{-1}} \Gamma(\mathbb{P}^1, U_0 \cap U_1) \xrightarrow{\pi^\#} \Gamma(X, \pi^{-1}(U_0 \cap U_1)) = N$$

(??). This structure is compatible with the k -module structure of N : a scalar $r \in k[T, T^{-1}]$ corresponds under γ_{01} to the image of r under the algebra map into the overlap ring (??(3)), which is the structure algebra map of $\mathbb{P}^1$ on $U_0 \cap U_1$ (??), and π pulls the structure algebra map of $\mathbb{P}^1$ back to that of X (??). Hence the k -module structure of N is the restriction of scalars of the $k[T, T^{-1}]$ -structure along the canonical map $k \rightarrow k[T, T^{-1}]$.

(c) *Module-finiteness over the Laurent ring.* Since π is finite and $U_0 \cap U_1$ is an affine open of $\mathbb{P}^1$, the map $\pi^\# : \Gamma(\mathbb{P}^1, U_0 \cap U_1) \rightarrow N$ is module-finite (??); composing with the isomorphism γ_{01}^{-1} , N is a finite $k[T, T^{-1}]$ -module.

(d) *The two lattices, and the range of the difference map.* Let $Q_0, Q_1 \subseteq N$ be the images of the (k -linear) restriction maps $\Gamma(X, V_0) \rightarrow N$ and $\Gamma(X, V_1) \rightarrow N$. Then $\text{range}(\text{diff}) = Q_0 + Q_1$: every value $s_0| - s_1|$ of diff is the sum of $s_0| \in Q_0$ and $-(s_1|) = (-s_1)| \in Q_1$; conversely $s_0| = \text{diff}(s_0, 0)$ and $s_1| = \text{diff}(0, -s_1)$ exhibit every element of Q_0 and of Q_1 as a value of diff — the sign is absorbed because Q_1 is the image of a linear map.

(e) *Lattice stability.* Let $\tau_0 = \pi^\#(t_{01}) \in \Gamma(X, V_0)$ and $\tau_1 = \pi^\#(t_{10}) \in \Gamma(X, V_1)$ be the pullbacks of the chart coordinates, regarded as sections over the charts (Definition ??). By ??(1) and the compatibility of $\pi^\#$ with restriction, T acts on N as multiplication by $\tau_0|_{V_0 \cap V_1}$; symmetrically T^{-1} acts as multiplication by $\tau_1|_{V_0 \cap V_1}$ (??(2)). Hence Q_0 is stable under multiplication by T : for $s \in \Gamma(X, V_0)$,

$$T \cdot s|_{V_0 \cap V_1} = \tau_0| \cdot s| = (\tau_0 s)|_{V_0 \cap V_1} \in Q_0,$$

and symmetrically Q_1 is stable under multiplication by T^{-1} .

(f) *Absorption up to powers.* The overlap is a basic open inside each chart preimage: pulling back the identity $D(t_{ij}) = U_i \cap U_j$ of ?? along π (basic opens of sections pull back to basic opens of the pulled-back sections) gives

$$V_0 \cap V_1 = D(\tau_0) \subseteq V_0, \quad V_0 \cap V_1 = D(\tau_1) \subseteq V_1.$$

Since V_0 is affine and $V_0 \cap V_1$ is its basic open defined by τ_0 , restriction identifies N with the localization of $\Gamma(X, V_0)$ at the powers of τ_0 (quasi-coherence of the structure sheaf). Hence for every $n \in N$ there are $m \geq 0$ and $a \in \Gamma(X, V_0)$ with $n \cdot (\tau_0|)^m = a|$, i.e. $T^m \cdot n \in Q_0$. Symmetrically, every $n \in N$ satisfies $T^{-m} \cdot n \in Q_1$ for some $m \geq 0$.

(g) *The ladder.* By (b), (c), (e) and (f), the module N over $k[T, T^{-1}]$ with the k -submodules Q_0, Q_1 satisfies all hypotheses of the two-lattice finiteness theorem in its localization form via powers of the generator (??); hence $N/(Q_0 + Q_1)$ is a finite k -module. By (d) this quotient is $N/\text{range}(\text{diff})$, and by (a) it is $H^1(X, \mathcal{O}_{X/k})$. $\square$

Theorem 369 (Finiteness of H^1 of the curve). *Let k be a field and C a smooth curve over k (proper, smooth of relative dimension one, geometrically irreducible). Then $H^1(C, \mathcal{O}_{C/k})$ is a finite k -module; equivalently, it is a finite-dimensional k -vector space.*

Proof. By ?? there is a finite morphism $\pi : C \rightarrow \mathbb{P}^1$ over k ; apply ??. $\square$

Remark 370 (The genus is the dimension of a finite-dimensional space). For a smooth curve C over k , the genus of Definition ?? is the k -dimension of $H^1(C, \mathcal{O}_C)$, which by ?? is a finite-dimensional k -vector space; the genus is therefore a well-defined natural number, attained as the cardinality of any basis.

5.12 The cocycle-glued twisted sheaf on a two-cover

Let k be a commutative ring and X a scheme over $\text{Spec } k$, its structure sheaf viewed as a sheaf of k -modules (??). Fix two opens $V_0, V_1 \subseteq X$ and a *unit cocycle*: an invertible section $g \in \Gamma(X, V_0 \cap V_1)^\times$. (On a two-chart cover a Čech 1-cocycle has a single component and the cocycle condition on triple overlaps is vacuous.) This section glues from g a sheaf F_g of k -modules — for $V_0 \cup V_1 = X$, the line bundle with transition function g , trivialised on each chart — and computes its degree-one cohomology as a two-cover cokernel, by transporting the quasi-coherence packaging of Section ?? through the chart trivialisations.

Definition 371 (Transport of the packaging along a sections-level trivialisation). Let F and G be sheaves of k -modules on the small Zariski site of X , let U be an open, and suppose given k -linear equivalences $e_W : F(W) \cong G(W)$ for every open $W \leq U$, commuting with section restriction. If G carries a quasi-coherence packaging on U (??), then so does F , with the conjugated action $r \cdot m := e_W^{-1}(r \cdot e_W(m))$.

Proof. Each (P1) law for F follows by conjugating the corresponding law for G through the equivalences, and naturality under restriction combines the (P1) naturality for G with the compatibility of e with restriction. For (P2): given $c \in F(V \cap D(g))$ on an affine $V \leq U$, denominator clearing for G applied to $e(c)$ produces N and $e' \in G(V)$ with $e'|_{V \cap D(g)} = g^N \cdot e(c)$; then $e_V^{-1}(e')$ clears the denominator for c , since e commutes with restriction and the action is conjugated. Defect annihilation transports the same way. $\square$

Definition 372 (Twisted sections of a unit cocycle). For an open $W \subseteq X$, the *defect map* of the cocycle g over W is the k -linear map

$$\Gamma(X, W \cap V_0) \times \Gamma(X, W \cap V_1) \longrightarrow \Gamma(X, W \cap V_0 \cap V_1), \quad (s_0, s_1) \longmapsto s_0| - g \cdot s_1|,$$

where both restrictions are to $W \cap V_0 \cap V_1$ and g acts through its restriction there. The module of *twisted sections* $F_g(W)$ is the kernel of the defect map: the pairs (s_0, s_1) that match through g on the overlap.

Lemma 373 (Twisted sections form a presheaf). *For $W' \leq W$, componentwise restriction carries $F_g(W)$ into $F_g(W')$, and the resulting k -linear maps are functorial in the inclusion; so $W \mapsto F_g(W)$ is a presheaf of k -modules.*

Proof. Restrict the matching identity $s_0| = g \cdot s_1|$ from $W \cap V_0 \cap V_1$ to $W' \cap V_0 \cap V_1$: restriction of the structure sheaf is a ring homomorphism commuting with further restriction, so the restricted pair again matches. Functoriality is functoriality of restriction in each component. $\square$

Theorem 374 (The twisted sheaf). *The presheaf F_g of twisted sections is a sheaf of k -modules on the small Zariski site of X .*

Proof. Let $\{U_i\}$ be an open cover of W and $(s_0^i, s_1^i) \in F_g(U_i)$ sections agreeing on overlaps. The first components are sections of the structure sheaf over the cover $\{U_i \cap V_0\}$ of $W \cap V_0$, compatible on its overlaps; by the sheaf property of the structure sheaf they glue to a unique $t_0 \in \Gamma(X, W \cap V_0)$, and likewise the second components glue to a unique $t_1 \in \Gamma(X, W \cap V_1)$.

The matching $t_0| = g \cdot t_1|$ on $W \cap V_0 \cap V_1$ may be checked on the cover $\{U_i \cap V_0 \cap V_1\}$, where it restricts to the matching of (s_0^i, s_1^i) ; by separation of the structure sheaf it holds on the union. So $(t_0, t_1) \in F_g(W)$ restricts to every (s_0^i, s_1^i) , and it is the unique such element because each of its components is determined by separation. $\square$

Definition 375 (Chart trivialisations of the twisted sheaf). Let $W \leq V_0$ (so $W \cap V_0 = W$ and $W \cap V_0 \cap V_1 = W \cap V_1$). Projection to the first component is a k -linear equivalence

$$\text{triv}_0 : F_g(W) \cong \Gamma(X, W), \quad (s_0, s_1) \mapsto s_0,$$

with inverse $s \mapsto (s, g^{-1} \cdot s|_{W \cap V_1})$. Symmetrically, for $W \leq V_1$, projection to the second component is a k -linear equivalence $\text{triv}_1 : F_g(W) \cong \Gamma(X, W)$ with inverse $s \mapsto (g \cdot s|_{W \cap V_0}, s)$.

Proof. For $W \leq V_0$ the matching relation of a pair $(s_0, s_1) \in F_g(W)$ reads $s_0| = g \cdot s_1|$ on $W \cap V_1$, i.e. $s_1 = g^{-1} \cdot s_0|$: the second component is rebuilt from the first, so projection and the displayed inverse are mutually inverse; the pair $(s, g^{-1} \cdot s|)$ matches by construction. Both maps are k -linear, multiplication by the fixed sections $g^{\pm 1}$ being k -linear. The second chart is symmetric. $\square$

Lemma 376 (The trivialisations commute with restriction). *For $W' \leq W \leq V_0$, trivialising after restricting equals restricting after trivialising: $\text{triv}_0(p|_{W'}) = (\text{triv}_0 p)|_{W'}$; likewise on the second chart.*

Proof. Both sides are the first (resp. second) component of p restricted to W' ; componentwise restriction is functorial. $\square$

Theorem 377 (The twisted sheaf is quasi-coherently packaged on each chart). *The twisted sheaf F_g carries a quasi-coherence packaging on V_0 and on V_1 .*

Proof. The chart trivialisations give k -linear equivalences $F_g(W) \cong \Gamma(X, W)$ for every $W \leq V_0$ (resp. $W \leq V_1$), compatible with restriction (??); the structure sheaf is packaged on every open (??); transport the packaging along the trivialisation (??). $\square$

Theorem 378 (Twisted affine vanishing). *If V_0 (resp. V_1) is an affine open of X , then $H^1(V_0, F_g) = 0$ (resp. $H^1(V_1, F_g) = 0$).*

Proof. The twisted affine vanishing theorem ??, applied to the packaging of ??. $\square$

Definition 379 (The two-cover H^1 carrier of the twisted sheaf). Let V_0, V_1 be affine opens with $V_0 \cup V_1 = X$. Then the degree-one cohomology of the twisted sheaf is the two-cover cokernel of twisted sections: a k -linear isomorphism

$$H^1(X, F_g) \cong F_g(V_0 \cap V_1) / \text{range}((p_0, p_1) \mapsto p_0|_{V_0 \cap V_1} - p_1|_{V_0 \cap V_1}),$$

the restriction-difference map being that of the two-cover square (??) with coefficients in F_g . (The overlap sections $F_g(V_0 \cap V_1)$ are further identified with $\Gamma(X, V_0 \cap V_1)$ by the chart trivialisation at $V_0 \cap V_1 \leq V_0$, ??.)

Proof. The two charts are affine, so $H^1(V_i, F_g) = 0$ (??); apply the two-cover H^1 computation with general coefficients (??) to $F = F_g$. $\square$

Definition 380 (The relative twisted sheaf). Let k be a field, $C \in \text{Over}(\text{Spec } k)$, R a commutative k -algebra, and $D = (V_0, V_1)$ an affine two-chart cover of C . A unit cocycle $g_k \in \Gamma(C, V_0 \cap V_1)^\times$ pulls back along the first projection to a unit cocycle on the base-changed cover of the relative curve: $\text{pr}_1^\sharp(g_k) \in \Gamma(C_R, \text{pr}_1^{-1}V_0 \cap \text{pr}_1^{-1}V_1)^\times$ (pullback of sections is a ring homomorphism, so it preserves units; the overlap of the base-changed charts is the preimage of $V_0 \cap V_1$ by ??). The *relative twisted sheaf* of a unit cocycle g on the base-changed cover is the twisted sheaf F_g on C_R , a sheaf of R -modules.

Definition 381 (The relative twisted H^1 carrier). In the setting of ??, the degree-one cohomology of the relative twisted sheaf is the explicit two-cover cokernel of R -modules: an R -linear isomorphism

$$H^1(C_R, F_g) \cong F_g(\text{pr}_1^{-1}V_0 \cap \text{pr}_1^{-1}V_1) / \text{range}(\text{diff}).$$

Proof. The base-changed charts are affine and cover C_R (?); apply ?? over the base ring R . $\square$

5.13 Base change of the two-term complex and of H^1 on the relative curve

Fix a field k , an object $C \in \text{Over}(\text{Spec } k)$, and a homomorphism $R \rightarrow R'$ of commutative k -algebras — a change of affine test rings. This section identifies the sections of the relative curve as the free base change of the sections of C , builds the comparison morphism $C_{R'} \rightarrow C_R$, base-changes the two-term two-cover complex of Section ?? along $R \rightarrow R'$ termwise, and concludes that the degree-one cohomology of the structure sheaf of the relative curve commutes with *every* change of test ring — no flatness of $R \rightarrow R'$ is needed, because the two-cover H^1 is a cokernel and cokernels commute with arbitrary base change (Chapter ??).

Theorem 382 (Sections of the relative curve are the base change of sections). *Let k be a field, $C \in \text{Over}(\text{Spec } k)$, R a commutative k -algebra, and $V \subseteq C$ a quasi-compact quasi-separated open (for instance any affine open, or a finite intersection or union of affine opens of a quasi-separated scheme). Write $V_R := \text{pr}_1^{-1}V \subseteq C_R$. Then there is a ring isomorphism*

$$\Gamma(C, V) \otimes_k R \cong \Gamma(C_R, V_R), \quad s \otimes a \mapsto \text{pr}_1^\sharp(s) \cdot \text{pr}_2^\sharp(a),$$

where $\text{pr}_1^\sharp$ is pullback of sections along the first projection, and $\text{pr}_2^\sharp(a)$ is the pullback along the second projection of the global section $a \in R = \Gamma(\text{Spec } R, \top)$. (The k -algebra structure on $\Gamma(C, V)$ is restriction of scalars along the structure morphism.)

Proof. The displayed assignment is a well-defined k -algebra map χ_V out of the tensor product: both pullbacks are k -algebra homomorphisms into the commutative ring $\Gamma(C_R, V_R)$, and the tensor product of commutative k -algebras is their pushout.

Step 1: V affine. The open $V_R = \text{pr}_1^{-1}(V)$ of the fibre product $C_R = C \times_{\text{Spec } k} \text{Spec } R$ is the fibre product $V \times_{\text{Spec } k} \text{Spec } R$ of the open subscheme V , a fibre product of affine schemes, hence the affine scheme $\text{Spec}(\Gamma(V) \otimes_k R)$: Spec converts the pushout of commutative rings into the fibre product of affine schemes. Under this identification χ_V is the canonical isomorphism.

Step 2: V quasi-compact quasi-separated. Choose a finite affine cover $V = U_1 \cup \dots \cup U_n$ (quasi-compactness); each intersection $U_i \cap U_j$ is quasi-compact (quasi-separatedness), so it in turn has a finite affine cover $\{W_l^{ij}\}$. The sheaf axiom writes $\Gamma(C, V)$ as the equaliser of the two restriction maps between the finite products $\prod_i \Gamma(U_i)$ and $\prod_{i,j,l} \Gamma(W_l^{ij})$. The preimages $\text{pr}_1^{-1}U_i$ form an affine cover of V_R (Step 1) whose pairwise intersections are covered by the affine

$\text{pr}_1^{-1}W_l^{ij}$ (preimage commutes with unions and intersections), so $\Gamma(C_R, V_R)$ is the equaliser of the corresponding restriction maps for the preimage cover. Since R is a free, hence flat, k -module (k is a field), the functor $-\otimes_k R$ is exact and commutes with finite products, so it carries the first equaliser diagram to an equaliser diagram. The maps χ on the terms commute with the two restriction maps (pullback commutes with restriction) and are isomorphisms on the product terms by Step 1; hence the induced map on equalisers, which is χ_V , is an isomorphism. $\square$

Definition 383 (Relative sections as a free R -module base change). In the setting of ??, endow $\Gamma(C_R, V_R)$ with the R -module structure $r \cdot y := \text{pr}_2^\sharp(r) \cdot y$ given by pullback along the structure morphism $\text{pr}_2 : C_R \rightarrow \text{Spec } R$. The isomorphism of ??, read with the tensor factors flipped, is an R -linear equivalence

$$R \otimes_k \Gamma(C, V) \cong \Gamma(C_R, V_R), \quad a \otimes s \mapsto \text{pr}_2^\sharp(a) \cdot \text{pr}_1^\sharp(s),$$

with R acting on the left tensor factor; in particular $1 \otimes r \mapsto \text{pr}_2^\sharp(r)$.

Proof. Only R -linearity needs comment: on a pure tensor, $r \cdot (a \otimes s) = (ra) \otimes s$ maps to $\text{pr}_2^\sharp(ra) \cdot \text{pr}_1^\sharp(s) = \text{pr}_2^\sharp(r) \cdot (\text{pr}_2^\sharp(a) \cdot \text{pr}_1^\sharp(s))$, because $\text{pr}_2^\sharp$ is a ring homomorphism — and the right-hand side is r acting on the image. $\square$

Definition 384 (The comparison morphism of relative curves). Let $\varphi : R \rightarrow R'$ be a homomorphism of commutative k -algebras. Applying Spec gives a morphism $\text{Spec } R' \rightarrow \text{Spec } R$ over $\text{Spec } k$, and the induced morphism of fibre products

$$\rho = \text{id}_C \times \text{Spec}(\varphi) : C_{R'} \longrightarrow C_R$$

is the *comparison morphism* between the relative curves.

Lemma 385 (Projection identities of the comparison morphism). *The comparison morphism satisfies $\text{pr}_1 \circ \rho = \text{pr}_1$ and $\text{pr}_2 \circ \rho = \text{Spec}(\varphi) \circ \text{pr}_2$; consequently $\rho^{-1}(V_R) = V_{R'}$ for every open $V \subseteq C$.*

Proof. The first two identities are the compatibility of the induced morphism of fibre products with the two projections; the third follows from the first by taking preimages of V . $\square$

Definition 386 (The sections comparison map). For $V \subseteq C$ open, pullback of sections along ρ (allowed since $\rho^{-1}(V_R) = V_{R'}$, ??) is a ring homomorphism

$$\rho^\sharp : \Gamma(C_R, V_R) \longrightarrow \Gamma(C_{R'}, V_{R'}).$$

Lemma 387 (Governing identities of the sections comparison map). *The sections comparison map $\rho^\sharp$:*

1. *commutes with the curve pullbacks:* $\rho^\sharp(\text{pr}_1^\sharp s) = \text{pr}_1^\sharp s$ for $s \in \Gamma(C, V)$;
2. *intertwines the structure pullbacks:* $\rho^\sharp(\text{pr}_2^\sharp r) = \text{pr}_2^\sharp(\varphi(r))$ for $r \in R$;
3. *hence is semilinear along φ :* $\rho^\sharp(r \cdot y) = \varphi(r) \cdot \rho^\sharp(y)$ for the module structures of ??;
4. *commutes with restriction along $W \leq V$.*

Proof. (1) is pullback along $\text{pr}_1 \circ \rho = \text{pr}_1$ (??); (2) is pullback along $\text{pr}_2 \circ \rho = \text{Spec}(\varphi) \circ \text{pr}_2$, and the pullback of the global section r along $\text{Spec}(\varphi)$ is $\varphi(r)$; (3) is (2) together with multiplicativity of $\rho^\sharp$; (4) is functoriality of pullback of sections with respect to inclusions of opens. $\square$

Theorem 388 (CBC-1: termwise base change of relative sections). *Let $R \rightarrow R'$ be a homomorphism of test rings and $V \subseteq C$ a quasi-compact quasi-separated open. There is an R' -linear equivalence*

$$R' \otimes_R \Gamma(C_R, V_R) \cong \Gamma(C_{R'}, V_{R'}), \quad r' \otimes y \mapsto r' \cdot \rho^\sharp(y).$$

Proof. Compose the free identifications of ?? over R and over R' with the tensor cancellation $R' \otimes_R (R \otimes_k M) \cong R' \otimes_k M$:

$$R' \otimes_R \Gamma(C_R, V_R) \cong R' \otimes_R (R \otimes_k \Gamma(C, V)) \cong R' \otimes_k \Gamma(C, V) \cong \Gamma(C_{R'}, V_{R'}).$$

For the formula, both sides are additive in y and R' -linear in r' , so it suffices to treat $y = \text{pr}_2^\sharp(a) \cdot \text{pr}_1^\sharp(s)$ (the images of pure tensors, which span). Chasing $r' \otimes y$ through the composite gives $\text{pr}_2^\sharp(r' \varphi(a)) \cdot \text{pr}_1^\sharp(s) = r' \cdot (\text{pr}_2^\sharp(\varphi(a)) \cdot \text{pr}_1^\sharp(s))$, which by (1) and (2) of ?? is $r' \cdot \rho^\sharp(y)$. $\square$

Lemma 389 (CBC-1: the two-cover complex base-changes). *Let $D = (V_0, V_1)$ be an affine two-chart cover of C , and write diff_R for the two-cover restriction-difference map (??) of C_R with its base-changed cover (??). Under the term equivalences of ?? — the product of the two chart equivalences on the domain, the overlap equivalence on the codomain — the base change $\text{diff}_R \otimes_R R'$ corresponds to $\text{diff}_{R'}$: the evident square commutes. In particular the overlap term equivalence carries $\text{range}(\text{diff}_R \otimes_R R')$ onto $\text{range}(\text{diff}_{R'})$.*

Proof. Both composites are additive, so it suffices to compare them on $r' \otimes (s_0, s_1)$. Along the top and right, the result is $r' \cdot \rho^\sharp(s_0|_\cap - s_1|_\cap)$; along the left and bottom, it is $\text{diff}_{R'}(r' \cdot \rho^\sharp s_0, r' \cdot \rho^\sharp s_1) = (r' \cdot \rho^\sharp s_0)|_\cap - (r' \cdot \rho^\sharp s_1)|_\cap$. These agree because $\rho^\sharp$ commutes with restriction (??(4)) and the R' -action commutes with restriction (the structure pullback restricts to the structure pullback). The range statement follows because the domain equivalence is surjective. $\square$

Theorem 390 (CBC-2, cokernel form: the two-cover H^1 carrier base-changes). *In the setting of ?? there is an R' -linear equivalence*

$$R' \otimes_R \left(\Gamma(C_R, \cap) / \text{range}(\text{diff}_R) \right) \cong \Gamma(C_{R'}, \cap) / \text{range}(\text{diff}_{R'}), \quad r' \otimes [s] \mapsto [r' \cdot \rho^\sharp s],$$

where $\cap$ denotes the overlap of the respective base-changed covers (??). No hypothesis on $R \rightarrow R'$ is needed.

Proof. Base change commutes with cokernels (??, with $N = \Gamma(C_R, \cap)$, $f = \text{diff}_R$, $S = R'$): $R' \otimes_R (N / \text{range } f) \cong (R' \otimes_R N) / \text{range}(f \otimes_R R')$. Then pass along the overlap term equivalence, which identifies $\text{range}(\text{diff}_R \otimes_R R')$ with $\text{range}(\text{diff}_{R'})$ (??), hence descends to the quotients. The composite sends $r' \otimes [s]$ to $[r' \cdot \rho^\sharp s]$ by ??. $\square$

Theorem 391 (CBC-2: H^1 of the relative curve commutes with change of the test ring). *For every homomorphism $R \rightarrow R'$ of commutative k -algebras there is an R' -linear equivalence*

$$R' \otimes_R H^1(C_R, \mathcal{O}_{C_R/R}) \cong H^1(C_{R'}, \mathcal{O}_{C_{R'}/R'}) :$$

degree-one cohomology of the structure sheaf of the relative curve commutes with every base change of the test ring. No flatness of $R \rightarrow R'$ is required: the two-cover H^1 is the cokernel of a two-term complex whose terms base-change freely, and cokernels commute with arbitrary base change.

Proof. Conjugate the cokernel equivalence of ?? by the relative two-cover identifications $H^1(C_R, \mathcal{O}_{C_R/R}) \cong \Gamma(C_R, \cap) / \text{range}(\text{diff}_R)$ of ??, over R and over R' . $\square$

Definition 392 (Base change of the unit cocycle). For a unit cocycle g on the base-changed cover of C_R , the image $\rho^\sharp(g)$ is a unit cocycle on the base-changed cover of $C_{R'}$: the ring homomorphism $\rho^\sharp$ preserves units, and the overlaps correspond under ρ (??).

Definition 393 (CBC-1 for the twisted sheaf: term equivalences). Let g be a unit cocycle on the base-changed cover of C_R and $g' = \rho^\sharp(g)$ its base change (??). For each term of the twisted two-cover complex — the two charts and their overlap — the composite of the chart trivialisation over R (??), the structure-sheaf term base change of ?? (at the overlap: the codomain equivalence of ??), and the inverse chart trivialisation over R' is an R' -linear equivalence

$$R' \otimes_R F_g(\text{term over } R) \cong F_{g'}(\text{term over } R').$$

On the overlap, the first-chart trivialisation (at $\text{pr}_1^{-1}V_0 \cap \text{pr}_1^{-1}V_1 \leq \text{pr}_1^{-1}V_0$) is used.

5.14 The base-field transition and its sections base change

Fix a field k and an object $C \in \text{Over}(\text{Spec } k)$. For a field extension K/k write $C_K := (C \otimes \text{Spec } K)_{\text{left}}$ for the base-changed curve, a scheme over $\text{Spec } K$ through the second projection. A k -algebra homomorphism $\varphi : K_1 \rightarrow K_2$ between field extensions of k induces the *base-field transition*

$$\pi = (C \triangleleft \text{Spec}(\varphi))_{\text{left}} : C_{K_2} \longrightarrow C_{K_1},$$

the whiskering of C with $\text{Spec } \varphi$ between the test objects (??). This section provides the transition’s geometric toolkit and identifies the sections of C_{K_2} over the preimage of an affine open with the base change of the sections of C_{K_1} .

A design point governs everything that follows: the “base-field shuffle” $C_{K_1} \otimes_{K_1} K_2 \cong C_{K_2}$ is *never constructed as an isomorphism of schemes*. All statements are made against the transition square below, which exhibits π as a base change of $\text{Spec } \varphi$ directly. In particular, the sections base change of this section is deliberately *not* obtained by instantiating Theorem ?? (whose tensor factor is anchored at the base field k) over the base K_1 and transporting along a shuffle isomorphism — the short pushout argument is re-run against the pasted square instead, with the K_1 -algebra structure on sections carried by the second projection.

Lemma 394 (The product bridge). *Let S be a scheme and $X, T \in \text{Over}(S)$. The underlying scheme of the cartesian-monoidal product, with its two projections, is the fibre product of the structure morphisms:*

$$\begin{array}{ccc} (X \otimes T)_{\text{left}} & \xrightarrow{\text{fst}} & X_{\text{left}} \\ \downarrow \text{snd} & & \downarrow X.\text{hom} \\ T_{\text{left}} & \xrightarrow{T.\text{hom}} & S \end{array}$$

is a pullback square of schemes. This is the single point where the product of the over category is unfolded to a scheme-level pullback.

Proof. The cartesian-monoidal product of the over category is constructed as the chosen pullback of the structure morphisms, with the two projections as its legs; the square is that chosen pullback square. $\square$

Theorem 395 (The pasted square of a morphism of test objects). *Let S be a scheme, $X \in \text{Over}(S)$, and $t : T' \rightarrow T$ a morphism of $\text{Over}(S)$. The square*

$$\begin{array}{ccc} (X \otimes T')_{\text{left}} & \xrightarrow{(X \triangleleft t)_{\text{left}}} & (X \otimes T)_{\text{left}} \\ \downarrow \text{pr}'_2 & & \downarrow \text{pr}_2 \\ T'_{\text{left}} & \xrightarrow{t_{\text{left}}} & T_{\text{left}} \end{array}$$

(verticals the second projections) is a pullback of schemes. This generalises the base-change square of the product (??) from the base $\text{Spec } k$ to an arbitrary base S ; it is stated for an arbitrary morphism of test objects so that later campaigns can re-instantiate it without re-proving.

Proof. Paste the two cartesian squares expressing $X \otimes T$ and $X \otimes T'$ as products over S . The first-projection triangle $(X \triangleleft t)_{\text{left}} \circ \text{fst} = \text{fst}'$ and the base identity $t_{\text{left}} \circ (T.\text{hom}) = T'.\text{hom}$ exhibit the outer rectangle over $X_{\text{left}} \times S$ as the product square of $X \otimes T'$; the right-cancellation property of pullback squares applied to the product square of $X \otimes T$ then makes the displayed square a pullback. $\square$

Theorem 396 (The base-field transition square). *Let k be a commutative ring, $C \in \text{Over}(\text{Spec } k)$, and $\varphi : K_1 \rightarrow K_2$ a k -algebra homomorphism. The square*

$$\begin{array}{ccc} C_{K_2} & \xrightarrow{\pi} & C_{K_1} \\ \downarrow \text{pr}_2 & \searrow \text{Spec } \varphi & \downarrow \text{pr}_2 \\ \text{Spec } K_2 & \xrightarrow{\quad} & \text{Spec } K_1 \end{array}$$

is a pullback of schemes: the base-field transition is the base change of $\text{Spec } \varphi$ along the second projection of the K_1 -curve.

Proof. Instantiate the pasted square (??) at the morphism of test objects $\text{Spec } \varphi : \text{Spec } K_2 \rightarrow \text{Spec } K_1$ over $\text{Spec } k$ (??). $\square$

The pasted square above compares two tests over one base. The next squares cross a morphism of bases $f : S' \rightarrow S$: the slice product of the base change f^*X with a test over S' agrees, as a scheme, with the slice product of X itself with the test pushed forward to S . Throughout, $f^*X \in \text{Over}(S')$ denotes the base change of $X \in \text{Over}(S)$ along f — its underlying scheme is the fibre product $X_{\text{left}} \times_S S'$, its structure morphism the second projection — and $f_!T \in \text{Over}(S)$ denotes a test $T \in \text{Over}(S')$ regarded over S through f : the same underlying scheme, with structure morphism $f \circ T.\text{hom}$.

Theorem 397 (The cross-base pasted square). *Let $f : S' \rightarrow S$ be a morphism of schemes, $X \in \text{Over}(S)$, and $T \in \text{Over}(S')$. The square*

$$\begin{array}{ccc} (f^*X \otimes T)_{\text{left}} & \xrightarrow{\text{pr}_1 \circ \text{fst}} & X_{\text{left}} \\ \downarrow \text{snd} & \searrow f \circ T.\text{hom} & \downarrow X.\text{hom} \\ T_{\text{left}} & \xrightarrow{\quad} & S \end{array}$$

*is a pullback of schemes; here the top edge is the first slice projection $\text{fst} : (f^*X \otimes T)_{\text{left}} \rightarrow (f^*X)_{\text{left}} = X_{\text{left}} \times_S S'$ followed by the first projection of the fibre product, and the bottom edge $f \circ T.\text{hom}$ is the structure morphism of $f_!T$. Thus the slice product $f^*X \otimes T$, formed over S' , is at the same time a pullback of the cross-base cospan $X_{\text{left}} \rightarrow S \leftarrow T_{\text{left}}$.*

Proof. Paste horizontally the slice-product square of f^*X and T over S' (??) — horizontals fst and $T.\text{hom}$, verticals snd and the structure morphism of f^*X — with the defining base-change square of f^*X ,

$$\begin{array}{ccc} X_{\text{left}} \times_S S' & \xrightarrow{\text{pr}_1} & X_{\text{left}} \\ \downarrow \text{pr}_2 & \searrow f & \downarrow X.\text{hom} \\ S' & \xrightarrow{\quad} & S \end{array}$$

whose left vertical pr_2 is that structure morphism. The two pullback squares share this middle edge, and a horizontal paste of two pullback squares is a pullback; the outer rectangle is the displayed square, its top the composite $\text{pr}_1 \circ \text{fst}$ and its bottom the composite $f \circ T.\text{hom}$. $\square$

Definition 398 (The cross-base comparison isomorphism). In the situation of ??, the *cross-base comparison isomorphism* is the isomorphism of schemes

$$(f^*X \otimes T)_{\text{left}} \cong (X \otimes f_!T)_{\text{left}}$$

identifying the two pullbacks of the common cospan $X_{\text{left}} \rightarrow S \leftarrow T_{\text{left}}$: the left-hand side by the cross-base pasted square (??), the right-hand side by the product bridge for X and $f_!T$ over S (??). It is compatible with the projections: composed with the first projection of the right-hand side it is the top edge $\text{pr}_1 \circ \text{fst}$ of the pasted square, and composed with the second projection it is snd .

Proof. Two pullbacks of one cospan are canonically isomorphic: each limiting cone factors uniquely through the other, and the two factorizations compose to endomorphisms of limiting cones commuting with the projections, hence to the identities. The stated projection compatibilities are the defining property of the factorization. $\square$

Theorem 399 (Naturality of the cross-base isomorphism in the test). *For a morphism $t : T' \rightarrow T$ of $\text{Over}(S')$, the comparison isomorphisms of ?? intertwine the two whiskered morphisms: the square*

$$\begin{array}{ccc} (f^*X \otimes T')_{\text{left}} & \xrightarrow{(f^*X \triangleleft t)_{\text{left}}} & (f^*X \otimes T)_{\text{left}} \\ \downarrow \cong & & \downarrow \cong \\ (X \otimes f_!T')_{\text{left}} & \xrightarrow{(X \triangleleft f_!t)_{\text{left}}} & (X \otimes f_!T)_{\text{left}} \end{array}$$

commutes, where $f_!t$ is t regarded as a morphism $f_!T' \rightarrow f_!T$ over S .

Proof. Two morphisms into the pullback $(X \otimes f_!T)_{\text{left}}$ (??) are equal once they agree after composing with both projections. With the first projection: whiskering leaves the first factor untouched, and the comparison isomorphisms identify the first projections with the top edges of the pasted squares (??), so both composites equal the top edge $\text{pr}_1 \circ \text{fst}'$ of the square of T' . With the second projection: whiskering acts on the second factor by t — whose underlying morphism of schemes is also that of $f_!t$ — and the comparison isomorphisms identify the second projections, so both composites equal $t_{\text{left}} \circ \text{snd}'$. Hence the square commutes. $\square$

Theorem 400 (The cross-base square at a base-changed curve). *Let $k \rightarrow L$ be a field extension, $\sigma : \text{Spec } L \rightarrow \text{Spec } k$ the induced morphism of spectra, and $C \in \text{Over}(\text{Spec } k)$, with base change $\sigma^*C \in \text{Over}(\text{Spec } L)$ along L/k (??). Then for every test object $T \in \text{Over}(\text{Spec } L)$: the slice product $(\sigma^*C \otimes T)_{\text{left}}$, formed over $\text{Spec } L$, is a pullback of C .hom against the structure morphism of $\sigma_!T$, and the comparison isomorphism of schemes*

$$(\sigma^*C \otimes T)_{\text{left}} \cong (C \otimes \sigma_!T)_{\text{left}}$$

is natural in T .

Proof. Base change along L/k is the slice pullback along σ (??), so this is ??, ?? and ?? instantiated at $f = \sigma$ and $X = C$. $\square$

Definition 401 (Pushing an affine test down a field extension). Let $k \rightarrow L$ be a field extension, $\sigma : \text{Spec } L \rightarrow \text{Spec } k$ the induced morphism, and A an L -algebra whose k -algebra structure is the restriction through L (a scalar tower $k \rightarrow L \rightarrow A$). Write $\text{Spec } A / \text{Spec } F$ for the affine test attached to an F -algebra A : the object $\text{Spec } A$ of $\text{Over}(\text{Spec } F)$ with structure morphism $\text{Spec } F$ of

the algebra map. Then pushing the affine test of A over L forward along σ gives the affine test of A over k : there is an isomorphism

$$\sigma_!(\mathrm{Spec} A / \mathrm{Spec} L) \cong \mathrm{Spec} A / \mathrm{Spec} k \quad \text{in } \mathrm{Over}(\mathrm{Spec} k),$$

whose underlying morphism of schemes is the identity of $\mathrm{Spec} A$.

Proof. The push-forward keeps the underlying scheme $\mathrm{Spec} A$ and composes the structure morphism with σ . The algebra map $k \rightarrow A$ is the composite $k \rightarrow L \rightarrow A$ (the tower identity), so by contravariant functoriality of Spec the composed structure morphism $\sigma \circ \mathrm{Spec}(L \rightarrow A)$ is Spec of the algebra map $k \rightarrow A$ — which is exactly the structure morphism of the affine test of A over k . Hence the identity of $\mathrm{Spec} A$ is an isomorphism between the two objects of $\mathrm{Over}(\mathrm{Spec} k)$. $\square$

The cross-base comparison and the affine matching combine at an affine test: for an algebra A in a scalar tower $k \rightarrow L \rightarrow A$, the base-changed curve tensored with the affine test of A over L and the original curve tensored with the affine test of A over k have canonically isomorphic underlying schemes — a *same-carrier* comparison, compatible with the two projections onto the common carrier $\mathrm{Spec} A$ and natural in A .

Definition 402 (The same-carrier comparison at an affine test). Let $k \rightarrow L$ be a field extension, $\sigma : \mathrm{Spec} L \rightarrow \mathrm{Spec} k$ the induced morphism, $C \in \mathrm{Over}(\mathrm{Spec} k)$ with base change $\sigma^*C \in \mathrm{Over}(\mathrm{Spec} L)$ (??), and A an algebra in a scalar tower $k \rightarrow L \rightarrow A$. The *same-carrier comparison* is the isomorphism of schemes

$$(\sigma^*C \otimes (\mathrm{Spec} A / \mathrm{Spec} L))_{\mathrm{left}} \cong (C \otimes (\mathrm{Spec} A / \mathrm{Spec} k))_{\mathrm{left}}$$

(the left slice product formed over $\mathrm{Spec} L$, the right one over $\mathrm{Spec} k$) defined as the composite of the cross-base comparison at the test $T = \mathrm{Spec} A / \mathrm{Spec} L$ (??),

$$(\sigma^*C \otimes (\mathrm{Spec} A / \mathrm{Spec} L))_{\mathrm{left}} \cong (C \otimes \sigma_!(\mathrm{Spec} A / \mathrm{Spec} L))_{\mathrm{left}},$$

with the underlying morphism of C whiskered with the affine matching $\sigma_!(\mathrm{Spec} A / \mathrm{Spec} L) \cong \mathrm{Spec} A / \mathrm{Spec} k$ (??).

Proof. Both factors are isomorphisms of schemes: the first by ??; the second because whiskering with C is a functor on $\mathrm{Over}(\mathrm{Spec} k)$ and taking underlying schemes is a functor to schemes, and functors carry the isomorphism of ?? to an isomorphism. The composite of the two is therefore an isomorphism, and by construction its underlying morphism is the composite of the underlying morphisms of the two factors. $\square$

Lemma 403 (The same-carrier comparison lies over $\mathrm{Spec} A$). *Both slice products of ?? project onto the same scheme $\mathrm{Spec} A$ by their second projections, and the same-carrier comparison c commutes with them:*

$$\mathrm{snd}_k \circ c = \mathrm{snd}_L, \quad \mathrm{snd}_L \circ c^{-1} = \mathrm{snd}_k,$$

where snd_L and snd_k denote the underlying morphisms of the second projections of $\sigma^*C \otimes (\mathrm{Spec} A / \mathrm{Spec} L)$ and of $C \otimes (\mathrm{Spec} A / \mathrm{Spec} k)$. In particular c is an isomorphism of schemes over $\mathrm{Spec} A$.

Proof. Write $c = w \circ b$ with b the cross-base comparison and w the whiskered affine matching (??). Whiskering acts on the second factor by the whiskered morphism: $\mathrm{snd}_k \circ w$ is the underlying morphism of the affine matching composed with the second projection of $C \otimes \sigma_!(\mathrm{Spec} A / \mathrm{Spec} L)$, and the underlying morphism of the affine matching is the identity of $\mathrm{Spec} A$ (??). The cross-base

comparison identifies the second projections (??): the second projection of $C \otimes \sigma_!(\text{Spec } A / \text{Spec } L)$ composed with b is snd_L . Chaining, $\text{snd}_k \circ c = \text{snd}_L$; composing this identity with c^{-1} on the right gives $\text{snd}_L \circ c^{-1} = \text{snd}_k$. Finally, the two second projections are the structure morphisms of the two products regarded over $\text{Spec } A$, so c is an isomorphism over $\text{Spec } A$. $\square$

Theorem 404 (Naturality of the same-carrier comparison in the algebra). *Let A, B be algebras in scalar towers $k \rightarrow L \rightarrow A$, $k \rightarrow L \rightarrow B$ and $f : A \rightarrow B$ a homomorphism of L -algebras; through the towers, f is then also a homomorphism of k -algebras. Writing c_A, c_B for the same-carrier comparisons (??) and $\text{Spec}_L f : \text{Spec } B / \text{Spec } L \rightarrow \text{Spec } A / \text{Spec } L$, $\text{Spec}_k f : \text{Spec } B / \text{Spec } k \rightarrow \text{Spec } A / \text{Spec } k$ for the morphisms of affine tests induced by f over the two bases (??), the square*

$$\begin{array}{ccc} (\sigma^* C \otimes (\text{Spec } B / \text{Spec } L))_{\text{left}} & \xrightarrow{c_B} & (C \otimes (\text{Spec } B / \text{Spec } k))_{\text{left}} \\ \downarrow (\sigma^* C \triangleleft \text{Spec}_L f)_{\text{left}} & & \downarrow (C \triangleleft \text{Spec}_k f)_{\text{left}} \\ (\sigma^* C \otimes (\text{Spec } A / \text{Spec } L))_{\text{left}} & \xrightarrow{c_A} & (C \otimes (\text{Spec } A / \text{Spec } k))_{\text{left}} \end{array}$$

commutes.

Proof. The affine matchings intertwine the two $\text{Spec } f$. The push-forward $\sigma_!(\text{Spec}_L f)$ and the morphism $\text{Spec}_k f$ have the same underlying morphism of schemes $\text{Spec } f$, and the affine matchings $\text{match}_A, \text{match}_B$ have underlying morphism the identity of the respective carrier (??). Hence the two composites $\text{match}_A \circ \sigma_!(\text{Spec}_L f)$ and $\text{Spec}_k f \circ \text{match}_B$, both morphisms $\sigma_!(\text{Spec } B / \text{Spec } L) \rightarrow \text{Spec } A / \text{Spec } k$ of $\text{Over}(\text{Spec } k)$, have equal underlying morphisms $\text{Spec } f$; morphisms of a slice category with equal underlying morphisms are equal, so the matchings intertwine $\sigma_!(\text{Spec}_L f)$ with $\text{Spec}_k f$.

Pasting. Factor $c_A = w_A \circ b_A$ and $c_B = w_B \circ b_B$ as in ???. Whiskering with C and passing to underlying schemes is functorial, so the intertwining above whiskers to $(C \triangleleft \text{Spec}_k f)_{\text{left}} \circ w_B = w_A \circ (C \triangleleft \sigma_!(\text{Spec}_L f))_{\text{left}}$. Naturality of the cross-base comparison in the test (??, at the frozen instantiation of ??) applied to $\text{Spec}_L f$ gives $(C \triangleleft \sigma_!(\text{Spec}_L f))_{\text{left}} \circ b_B = b_A \circ (\sigma^* C \triangleleft \text{Spec}_L f)_{\text{left}}$. Chaining,

$$(C \triangleleft \text{Spec}_k f)_{\text{left}} \circ c_B = w_A \circ (C \triangleleft \sigma_!(\text{Spec}_L f))_{\text{left}} \circ b_B = w_A \circ b_A \circ (\sigma^* C \triangleleft \text{Spec}_L f)_{\text{left}} = c_A \circ (\sigma^* C \triangleleft \text{Spec}_L f)_{\text{left}}.$$

$\square$

Lemma 405 (The transition is flat, affine and surjective). *Let K_1 be a field and K_2 a nonzero commutative ring. Then $\text{Spec } \varphi : \text{Spec } K_2 \rightarrow \text{Spec } K_1$ is flat and surjective, and it is affine since both schemes are affine. Consequently, for field extensions K_1, K_2 of k , the base-field transition $\pi : C_{K_2} \rightarrow C_{K_1}$ is flat, affine and surjective.*

Proof. Flatness of $\text{Spec } \varphi$ is flatness of K_2 as a K_1 -module, and every module over a field is flat (free). Surjectivity: $\text{Spec } K_1$ is a one-point space and $\text{Spec } K_2$ is nonempty (as $K_2 \neq 0$), so the unique point is hit. Each of the three properties — flatness, affineness, surjectivity — is stable under base change of schemes, and the transition square (??) exhibits π as a base change of $\text{Spec } \varphi$; so all three transport to π . $\square$

Lemma 406 (Generic points under surjective morphisms). *A surjective morphism $f : X \rightarrow Y$ of schemes with irreducible underlying spaces maps the generic point to the generic point: $f(\eta_X) = \eta_Y$.*

Proof. Choose $z \in X$ with $f(z) = \eta_Y$. The generic point η_X specialises to every point of X , in particular $\eta_X \rightsquigarrow z$; applying the continuous map f gives $f(\eta_X) \rightsquigarrow \eta_Y$. Conversely η_Y specialises to every point of Y , so $\eta_Y \rightsquigarrow f(\eta_X)$. The underlying space of a scheme is sober, hence T_0 , so the two specialisations force $f(\eta_X) = \eta_Y$. $\square$

Lemma 407 (The transition matches generic points). *Let C be smooth of relative dimension one and geometrically irreducible over the field k , and let K_1, K_2 be field extensions of k . Then the base-field transition satisfies $\pi(\eta_{C_{K_2}}) = \eta_{C_{K_1}}$.*

Proof. Both base-changed curves are integral (??), in particular irreducible, and π is surjective (??); apply ??.

Definition 408 (Pullback of rational functions). Let $f : X \rightarrow Y$ be a morphism of schemes with irreducible underlying spaces such that $f(\eta_X) = \eta_Y$. The *function-field map*

$$f^\# : K(Y) \longrightarrow K(X)$$

is the stalk map of f at η_X , preceded by the canonical identification of the stalk of Y at $f(\eta_X)$ with the stalk at η_Y . It induces a group homomorphism $K(Y)^\times \rightarrow K(X)^\times$ on units. For the base-field transition (with ?? matching the generic points) this is the pullback of rational functions $\pi^\# : K(C_{K_1}) \rightarrow K(C_{K_2})$.

Lemma 409 (Germ naturality and injectivity of the function-field map). *In the setting of ??: for an open $U \subseteq Y$ containing η_Y and a section $s \in \Gamma(Y, U)$,*

$$f^\#(\text{germ}_{\eta_Y}(s)) = \text{germ}_{\eta_X}(f^\#s) \quad \text{in } K(X),$$

where $f^\#s \in \Gamma(X, f^{-1}U)$ is the pullback of the section (note $\eta_X \in f^{-1}U$ since $f(\eta_X) = \eta_Y \in U$). If moreover X and Y are integral, the function-field map is injective.

Proof. Germ naturality: the stalk identification carries $\text{germ}_{\eta_Y}(s)$ to the germ of s at the point $f(\eta_X) \in U$, and the stalk map of f at a point sends the germ of a section to the germ of its pullback — the defining compatibility of the stalk map with germs. Injectivity: on integral schemes both function fields are fields, and a ring homomorphism out of a field is injective. $\square$

Theorem 410 (Base change of sections along the transition, pushout form). *Let K_1, K_2 be field extensions of k , $\varphi : K_1 \rightarrow K_2$ a k -algebra homomorphism, and $V \subseteq C_{K_1}$ an affine open. The square of section rings*

$$\begin{array}{ccc} \Gamma(\text{Spec } K_1, \top) & \xrightarrow{\text{pr}_2^\#} & \Gamma(C_{K_1}, V) \\ \downarrow (\text{Spec } \varphi)^\# & & \downarrow \pi^\# \\ \Gamma(\text{Spec } K_2, \top) & \xrightarrow{\text{pr}_2^\#} & \Gamma(C_{K_2}, \pi^{-1}V) \end{array}$$

is a pushout of commutative rings. No flatness enters in the affine case.

Proof. Restrict the transition square (??) over the affine opens: $\pi^{-1}V$ is the fibre product $V \times_{\text{Spec } K_1} \text{Spec } K_2$ of the open subscheme V , a fibre product of affine schemes, hence affine with $\Gamma(\pi^{-1}V) = \Gamma(V) \otimes_{\Gamma(\text{Spec } K_1)} \Gamma(\text{Spec } K_2)$: Spec converts pushouts of commutative rings into fibre products of affine schemes. The four maps of the displayed square are the section-level maps of that restricted square, so the square is the tensor-product pushout. $\square$

Definition 411 (The second-projection algebra structure on sections). Let K/k be a field extension and $V \subseteq C_K$ an open. The base-changed curve, bundled over $\text{Spec } K$ by its second projection (??), endows $\Gamma(C_K, V)$ with a K -algebra structure: the structure map is the pullback of global sections of $\text{Spec } K$ along pr_2 , followed by restriction to V ,

$$K = \Gamma(\text{Spec } K, \top) \xrightarrow{\text{pr}_2^\#} \Gamma(C_K, V).$$

All K -linear statements about sections of C_K below refer to this structure.

Lemma 412 (Restriction of sections is K -algebraic). *For opens $W \leq V$ of C_K , the structure maps of ?? intertwine restriction — restricting the pullback of a constant is the pullback of the constant — so restriction $\Gamma(C_K, V) \rightarrow \Gamma(C_K, W)$ is a homomorphism of K -algebras.*

Proof. Both maps $K \rightarrow \Gamma(C_K, W)$ — structure map of W , and structure map of V followed by restriction — are the pullback of global sections of $\text{Spec } K$ along pr_2 restricted to W , since pullback of sections commutes with restriction. A ring homomorphism commuting with the structure maps is a map of K -algebras. $\square$

Theorem 413 (The algebra-corner pushout square). *In the setting of ??, the square*

$$\begin{array}{ccc} K_1 & \longrightarrow & \Gamma(C_{K_1}, V) \\ \downarrow \varphi & & \downarrow \pi^\sharp \\ K_2 & \longrightarrow & \Gamma(C_{K_2}, \pi^{-1}V) \end{array}$$

— top and bottom the structure maps of the second-projection algebra structures (??), left the homomorphism φ itself — is a pushout of commutative rings. It is stated for an arbitrary k -algebra homomorphism φ ; no K_1 -algebra structure on K_2 is chosen.

Proof. Identify the corners $\Gamma(\text{Spec } K_i, \top) = K_i$ of the pushout square of ?? by the canonical isomorphisms. Under these, the left edge $(\text{Spec } \varphi)^\sharp$ becomes φ (naturality of the identification in the ring), and the two horizontal edges become the structure maps of ?. A square isomorphic to a pushout square is a pushout square. $\square$

Definition 414 (Sections base change along the base-field transition). Let $K_1 \subseteq K_2$ be a tower of field extensions of k (that is, a K_1 -algebra structure on K_2 compatible with the k -structures), with φ the inclusion, and let $V \subseteq C_{K_1}$ be an affine open. There is an isomorphism of rings

$$\Gamma(C_{K_1}, V) \otimes_{K_1} K_2 \cong \Gamma(C_{K_2}, \pi^{-1}V),$$

the K_1 -module structure on $\Gamma(C_{K_1}, V)$ being the second-projection one (??).

Proof. The tensor product $\Gamma(C_{K_1}, V) \otimes_{K_1} K_2$ with its two canonical maps is a pushout of the same span $K_1 \rightarrow \Gamma(C_{K_1}, V)$, $K_1 \rightarrow K_2$ — the tensor product of commutative algebras is their pushout — as the square of ?. Two pushouts of one span are canonically isomorphic under the corners. $\square$

Lemma 415 (The base change on pure tensors). *The isomorphism of ?? sends*

$$s \otimes 1 \mapsto \pi^\sharp(s), \quad 1 \otimes a \mapsto \text{pr}_2^\sharp(a), \quad s \otimes a \mapsto \pi^\sharp(s) \cdot \text{pr}_2^\sharp(a),$$

for $s \in \Gamma(C_{K_1}, V)$ and $a \in K_2$.

Proof. The comparison isomorphism of two pushouts of one span commutes with the two co-projections. On the tensor-product side the coprojections are $s \mapsto s \otimes 1$ and $a \mapsto 1 \otimes a$; on the sections side they are $\pi^\sharp$ and $\text{pr}_2^\sharp$ (?). This gives the first two rules, and the third follows since $s \otimes a = (s \otimes 1) \cdot (1 \otimes a)$ and the comparison is a ring homomorphism. $\square$

Lemma 416 (Naturality of the base change in the open). *For affine opens $W \leq V$ of C_{K_1} , restricting the base-changed section is base-changing the restriction: the square*

$$\begin{array}{ccc} \Gamma(C_{K_1}, V) \otimes_{K_1} K_2 & \cong & \Gamma(C_{K_2}, \pi^{-1}V) \\ \downarrow \text{res} \otimes \text{id} & & \downarrow \text{res} \\ \Gamma(C_{K_1}, W) \otimes_{K_1} K_2 & \cong & \Gamma(C_{K_2}, \pi^{-1}W) \end{array}$$

commutes, the left vertical being the $(K_1\text{-algebraic, ??})$ restriction tensored with the identity of K_2 .

Proof. Both composites are additive, so it suffices to compare them on a pure tensor $s \otimes a$. By ?? the top-right composite gives $(\pi^\sharp(s) \cdot \text{pr}_2^\sharp(a))|_{\pi^{-1}W} = \pi^\sharp(s)|_{\pi^{-1}W} \cdot \text{pr}_2^\sharp(a)|_{\pi^{-1}W}$ (restriction is a ring homomorphism), while the left-bottom composite gives $\pi^\sharp(s|_W) \cdot \text{pr}_2^\sharp(a)$ read over $\pi^{-1}W$. These agree because pullback of sections commutes with restriction on both factors. $\square$

5.15 Two-lattice pairs and their cohomology

The remaining sections of this chapter build the module-level core of the two-term pushforward engine. All schemes disappear: a *two-lattice pair* axiomatises, in pure commutative algebra, exactly the data that an affine two-cover extracts from a quasi-coherent sheaf — the two chart section modules, the overlap module, the two chart coordinates, and the two restriction maps, each exhibiting the overlap as a localization of its chart (the module counterpart of denominator clearing and annihilation on affine opens, ?? and ??). The Čech complex of the two-cover (??, ??) becomes a two-term complex of modules; this section studies its cohomology in degrees zero and one and its functoriality, the next section proves the six-term exact sequence of a short exact sequence of pairs, and the following two sections compute the line-bundle models on the two-chart projective line and deduce the coherence theorems: finiteness of H^1 over an *arbitrary* commutative base ring, and of H^0 over a Noetherian one. The fibrewise Nakayama arguments and the split-rigidity theorems of Chapter ?? (Sections ?? and ??) consume the finiteness produced here.

Throughout, R is a commutative ring.

Definition 417 (Two-lattice pair). A *two-lattice pair* P over R consists of:

- R -modules M_0, M_1 (the *chart lattices*) and N (the *overlap module*);
- R -linear endomorphisms t_0 of M_0 and t_1 of M_1 (the *coordinate actions*) and an *invertible* R -linear endomorphism u of N ;
- R -linear maps $\iota_0 : M_0 \rightarrow N$ and $\iota_1 : M_1 \rightarrow N$ (the *restrictions to the overlap*) intertwining the actions: $\iota_0(t_0x) = u(\iota_0x)$ for all $x \in M_0$ and $\iota_1(t_1y) = u^{-1}(\iota_1y)$ for all $y \in M_1$;

subject to the localization axioms:

1. *denominator clearing*: every $n \in N$ admits an integer $m \geq 0$ and $x \in M_0$ with $u^m n = \iota_0 x$, and an integer $m' \geq 0$ and $y \in M_1$ with $u^{-m'} n = \iota_1 y$;
2. *annihilation*: if $\iota_0 x = 0$ then $t_0^m x = 0$ for some $m \geq 0$, and if $\iota_1 y = 0$ then $t_1^m y = 0$ for some $m \geq 0$.

Regarding M_0 and N as modules over the polynomial ring $R[t]$, with t acting as t_0 on M_0 and as u on N , the three conditions — invertibility of u on N , clearing, annihilation — say precisely that ι_0 exhibits N as the localization of M_0 at the powers of t ; likewise ι_1 exhibits N as the localization of M_1 at the powers of t , for the structures through t_1 and u^{-1} . In the geometric application M_0, M_1 are the sections of a quasi-coherent sheaf on the two charts of an affine two-cover, N its sections on the overlap, and t_0, t_1 the two chart coordinates, mutually inverse on the overlap.

Definition 418 (The two-term complex of a pair). The *difference map* of a two-lattice pair P is the R -linear map

$$\delta_P : M_0 \times M_1 \longrightarrow N, \quad \delta_P(x, y) = \iota_0 x - \iota_1 y.$$

The two-term complex $M_0 \times M_1 \xrightarrow{\delta_P} N$ is the module avatar of the two-cover Čech complex.

Definition 419 (H^0 of a pair). The degree-zero cohomology of the pair P is the kernel of the difference map,

$$H^0(P) = \ker \delta_P = \{ (x, y) \in M_0 \times M_1 : \iota_0 x = \iota_1 y \},$$

an R -submodule of $M_0 \times M_1$: the pairs of chart sections agreeing on the overlap.

Definition 420 (H^1 of a pair). The *image lattice* of the pair P is the R -submodule $Q(P) = \iota_0(M_0) + \iota_1(M_1) \subseteq N$, and the degree-one cohomology of P is the quotient

$$H^1(P) = N/Q(P),$$

with quotient projection $N \twoheadrightarrow H^1(P)$, $n \mapsto [n]$.

Lemma 421 (H^1 is the cokernel of the difference map). *The image of δ_P is the image lattice: $\delta_P(M_0 \times M_1) = Q(P)$. In particular $H^1(P)$ is the cokernel of δ_P , so $H^0(P)$ and $H^1(P)$ are the two cohomology modules of the two-term complex of ??.*

Proof. Each value $\delta_P(x, y) = \iota_0 x - \iota_1 y$ is a difference of an element of $\iota_0(M_0)$ and an element of $\iota_1(M_1)$, hence lies in $Q(P)$. Conversely $\iota_0 x = \delta_P(x, 0)$ and $\iota_1 y = \delta_P(0, -y)$, so both summands of $Q(P)$ lie in the image, and the image is a submodule. $\square$

Definition 422 (Map of two-lattice pairs). A *map of two-lattice pairs* $f : P \rightarrow P'$ is a triple of R -linear maps $f_0 : M_0 \rightarrow M'_0$, $f_1 : M_1 \rightarrow M'_1$, $f_N : N \rightarrow N'$ commuting with the coordinate actions and the restrictions:

$$f_0 \circ t_0 = t'_0 \circ f_0, \quad f_1 \circ t_1 = t'_1 \circ f_1, \quad f_N \circ u = u' \circ f_N, \quad f_N \circ \iota_0 = \iota'_0 \circ f_0, \quad f_N \circ \iota_1 = \iota'_1 \circ f_1.$$

Compatibility with the *inverse* overlap action is automatic: applying the u -compatibility to $u^{-1}n$ gives $f_N(n) = u'(f_N(u^{-1}n))$, whence $f_N(u^{-1}n) = u'^{-1}(f_N n)$.

Lemma 423 (Naturality of the difference map). *For a map of pairs $f : P \rightarrow P'$ and $(x, y) \in M_0 \times M_1$,*

$$\delta_{P'}(f_0 x, f_1 y) = f_N(\delta_P(x, y)).$$

Proof. Expand both sides: $\delta_{P'}(f_0 x, f_1 y) = \iota'_0(f_0 x) - \iota'_1(f_1 y) = f_N(\iota_0 x) - f_N(\iota_1 y) = f_N(\iota_0 x - \iota_1 y)$, using the two restriction compatibilities and linearity of f_N . $\square$

Definition 424 (The induced map on H^0). A map of pairs $f : P \rightarrow P'$ induces an R -linear map $H^0(f) : H^0(P) \rightarrow H^0(P')$, $(x, y) \mapsto (f_0 x, f_1 y)$.

Proof. If $\delta_P(x, y) = 0$ then $\delta_{P'}(f_0 x, f_1 y) = f_N(\delta_P(x, y)) = 0$ by naturality (??), so the restriction of $f_0 \times f_1$ to $H^0(P)$ lands in $H^0(P')$; it is R -linear as a restriction of an R -linear map. $\square$

Definition 425 (The induced map on H^1). A map of pairs $f : P \rightarrow P'$ induces an R -linear map $H^1(f) : H^1(P) \rightarrow H^1(P')$, $[n] \mapsto [f_N n]$.

Proof. f_N carries the image lattice into the image lattice: $f_N(\iota_0 x) = \iota'_0(f_0 x)$ and $f_N(\iota_1 y) = \iota'_1(f_1 y)$, so $f_N(Q(P)) \subseteq Q(P')$ and f_N descends to the quotients. $\square$

Lemma 426 (Right exactness of H^1 in the pair). *If a map of pairs $f : P \rightarrow P'$ is surjective on overlap modules, then $H^1(f) : H^1(P) \rightarrow H^1(P')$ is surjective.*

Proof. A class in $H^1(P')$ is $[n']$ for some $n' \in N'$; write $n' = f_N(n)$ by surjectivity; then $H^1(f)[n] = [f_N n] = [n']$. $\square$

5.16 The six-term exact sequence of two-lattice pairs

A short exact sequence of two-lattice pairs induces the entire long exact cohomology sequence of the two-term complexes; since a two-term complex has no cohomology beyond degree one, the sequence has six terms and closes on both ends:

$$0 \rightarrow H^0(P) \rightarrow H^0(P') \rightarrow H^0(P'') \xrightarrow{\partial} H^1(P) \rightarrow H^1(P') \rightarrow H^1(P'') \rightarrow 0.$$

The connecting map and the six exactness statements are proved by explicit element chases on the two-term complexes.

Definition 427 (Short exact sequence of pairs). A pair of maps of two-lattice pairs $\varphi : P \rightarrow P'$ and $\psi : P' \rightarrow P''$ is a *short exact sequence* $0 \rightarrow P \rightarrow P' \rightarrow P'' \rightarrow 0$ if it is componentwise short exact: on each of the three carriers (the two chart lattices and the overlap module), φ is injective, ψ is surjective, and the image of φ is the kernel of ψ .

Definition 428 (The connecting homomorphism). For a short exact sequence $0 \rightarrow P \xrightarrow{\varphi} P' \xrightarrow{\psi} P'' \rightarrow 0$ of two-lattice pairs, the *connecting homomorphism*

$$\partial : H^0(P'') \rightarrow H^1(P)$$

is the R -linear map characterized by: whenever $z = (z_0, z_1) \in H^0(P'')$, $e_0 \in M'_0$ and $e_1 \in M'_1$ satisfy $\psi_0 e_0 = z_0$ and $\psi_1 e_1 = z_1$, and $n \in N$ satisfies $\varphi_N n = \delta_{P'}(e_0, e_1)$, then $\partial z = [n]$.

Proof. Existence of the chase data. Given $z \in H^0(P'')$, componentwise surjectivity of ψ provides lifts e_0, e_1 . Then $\psi_N(\delta_{P'}(e_0, e_1)) = \delta_{P''}(\psi_0 e_0, \psi_1 e_1) = \delta_{P''}(z) = 0$ by naturality (??) and $z \in H^0(P'')$, so $\delta_{P'}(e_0, e_1)$ lies in $\ker \psi_N = \varphi_N(N)$, and n exists; it is unique because φ_N is injective.

Independence of the lifts. If e'_0, e'_1 is a second componentwise lift of z , with $\varphi_N n' = \delta_{P'}(e'_0, e'_1)$, then $\psi_0(e'_0 - e_0) = 0$ and $\psi_1(e'_1 - e_1) = 0$, so by componentwise exactness $e'_0 - e_0 = \varphi_0 a$ and $e'_1 - e_1 = \varphi_1 b$ for some $a \in M_0, b \in M_1$. Then

$$\varphi_N(n' - n) = \delta_{P'}(\varphi_0 a, \varphi_1 b) = \varphi_N(\delta_P(a, b))$$

by naturality, and injectivity of φ_N gives $n' - n = \delta_P(a, b) \in Q(P)$ (??); hence $[n'] = [n]$ in $H^1(P)$.

Linearity. The chase data may be chosen additively — lifts of a sum as the sum of lifts, and similarly for scalar multiples — and n depends linearly on the chosen data; since the class $[n]$ is independent of the choices, ∂ is R -linear. $\square$

Theorem 429 (Six-term exactness, left end). For a short exact sequence $0 \rightarrow P \xrightarrow{\varphi} P' \xrightarrow{\psi} P'' \rightarrow 0$, the map $H^0(\varphi) : H^0(P) \rightarrow H^0(P')$ is injective.

Proof. $H^0(\varphi)$ acts componentwise as (φ_0, φ_1) , and both φ_0 and φ_1 are injective. $\square$

Theorem 430 (Six-term exactness at $H^0(P')$). In the situation of ??, the image of $H^0(\varphi)$ equals the kernel of $H^0(\psi)$.

Proof. $\subseteq$: $\psi \circ \varphi$ vanishes componentwise (image of φ inside kernel of ψ), so $H^0(\psi) \circ H^0(\varphi) = 0$. $\supseteq$: let $(z_0, z_1) \in H^0(P')$ with $\psi_0 z_0 = 0$ and $\psi_1 z_1 = 0$. By componentwise exactness $z_0 = \varphi_0 a$ and $z_1 = \varphi_1 b$. The pair (a, b) lies in $H^0(P)$: by naturality (??) $\varphi_N(\delta_P(a, b)) = \delta_{P'}(z_0, z_1) = 0$, and φ_N is injective. Then $H^0(\varphi)(a, b) = (z_0, z_1)$. $\square$

Theorem 431 (Six-term exactness at $H^0(P'')$). *In the situation of ??, the image of $H^0(\psi)$ equals the kernel of the connecting homomorphism ∂ .*

Proof. $\subseteq$: for $z = H^0(\psi)(w)$ with $w \in H^0(P')$, take the components of w themselves as the lifts in the characterization of ∂ (?); their difference is $\delta_{P'}(w) = 0 = \varphi_N(0)$, so $\partial z = [0] = 0$. $\supseteq$: let $z \in H^0(P'')$ with $\partial z = 0$; choose lifts e_0, e_1 and n with $\varphi_N n = \delta_{P'}(e_0, e_1)$, so that $\partial z = [n]$. Vanishing of $[n]$ means $n \in Q(P) = \delta_P(M_0 \times M_1)$ (?); write $n = \delta_P(a, b)$. Then $w := (e_0 - \varphi_0 a, e_1 - \varphi_1 b)$ satisfies $\delta_{P'}(w) = \delta_{P'}(e_0, e_1) - \varphi_N(\delta_P(a, b)) = 0$ by naturality, so $w \in H^0(P')$; and since $\psi \circ \varphi = 0$ componentwise, $H^0(\psi)(w) = (\psi_0 e_0, \psi_1 e_1) = z$. $\square$

Theorem 432 (Six-term exactness at $H^1(P)$). *In the situation of ??, the image of the connecting homomorphism ∂ equals the kernel of $H^1(\varphi)$.*

Proof. $\subseteq$: if $\partial z = [n]$ with $\varphi_N n = \delta_{P'}(e_0, e_1)$, then $H^1(\varphi)[n] = [\varphi_N n] = [\delta_{P'}(e_0, e_1)] = 0$, the class of an element of the image lattice (?). $\supseteq$: let $[n] \in H^1(P)$ with $H^1(\varphi)[n] = 0$, i.e. $\varphi_N n \in Q(P') = \delta_{P'}(M'_0 \times M'_1)$; write $\varphi_N n = \delta_{P'}(e_0, e_1)$. Then $z := (\psi_0 e_0, \psi_1 e_1)$ lies in $H^0(P'')$: $\delta_{P''}(z) = \psi_N(\delta_{P'}(e_0, e_1)) = \psi_N(\varphi_N n) = 0$ by naturality and $\psi \circ \varphi = 0$. By the characterization of ∂ (?), $\partial z = [n]$. $\square$

Theorem 433 (Six-term exactness at $H^1(P')$). *In the situation of ??, the image of $H^1(\varphi)$ equals the kernel of $H^1(\psi)$.*

Proof. $\subseteq$: $H^1(\psi)(H^1(\varphi)[n]) = [\psi_N(\varphi_N n)] = 0$. $\supseteq$: let $[n'] \in H^1(P')$ with $[\psi_N n'] = 0$, i.e. $\psi_N n' \in Q(P'') = \delta_{P''}(M''_0 \times M''_1)$; write $\psi_N n' = \delta_{P''}(z_0, z_1)$ and lift $z_0 = \psi_0 e_0, z_1 = \psi_1 e_1$. Then $\psi_N(n' - \delta_{P'}(e_0, e_1)) = \psi_N n' - \delta_{P''}(z_0, z_1) = 0$ by naturality, so by exactness on the overlap $n' - \delta_{P'}(e_0, e_1) = \varphi_N w$ for some $w \in N$. Since $[\delta_{P'}(e_0, e_1)] = 0$ in $H^1(P')$ (?), $[n'] = [\varphi_N w] = H^1(\varphi)[w]$. $\square$

Theorem 434 (Six-term exactness, right end). *In the situation of ??, the map $H^1(\psi) : H^1(P') \rightarrow H^1(P'')$ is surjective.*

Proof. ψ is surjective on overlap modules; apply the right exactness of H^1 in the pair (?). $\square$

5.17 The line-bundle models and the finite surjection

The coherence theorems are proved by comparing an arbitrary pair with *models*: the two-lattice pairs of the direct sums $\bigoplus_i \mathcal{O}(m_i)$ of line bundles on the two-chart projective line over R , whose cohomology is the classical monomial window. Say a two-lattice pair P has *finite lattices* if M_0 is a finitely generated module over the polynomial ring $R[t]$, with t acting as t_0 , and likewise M_1 over $R[t]$ through t_1 . Every pair with finite lattices receives a componentwise surjection from a model — the two-cover form of the finite free approximation used by Mumford in the coherence dévissage on abelian varieties, here terminating in one step because generation is measured over $R[t]$ rather than over R .

Definition 435 (The family line-bundle model). Let I be a finite index set and $m : I \rightarrow \mathbb{Z}$ a family of twists. The *model* $E(m)$ is the two-lattice pair over R with chart lattices $M_0 = M_1 = (R[t])^I$, overlap module $N = (R[t, t^{-1}])^I$, coordinate actions $t_0 = t_1 =$ componentwise multiplication by t , invertible overlap action $u =$ componentwise multiplication by t , and restrictions

$$\iota_0(p)_i = p_i(t), \quad \iota_1(q)_i = t^{m_i} \cdot q_i(t^{-1}),$$

the componentwise inclusion of polynomials into Laurent polynomials on the chart-0 side, and the twisted substitution $t \mapsto t^{-1}$ on the chart-1 side. It is the pair of the sheaf $\bigoplus_{i \in I} \mathcal{O}(m_i)$ on the two standard charts of the projective line over R .

Proof. The axioms of a two-lattice pair hold. *Equivariance:* on the chart-0 side, multiplying a polynomial by t multiplies its Laurent image by t ; on the chart-1 side, $t^{m_i} \cdot (tq)(t^{-1}) = t^{m_i} \cdot t^{-1} \cdot q(t^{-1})$, so $\iota_1(t_1q) = u^{-1}(\iota_1q)$. *Denominator clearing:* a Laurent vector n has finitely many components, each with exponents bounded below, so a single power u^M (componentwise multiplication by t^M) makes every component a polynomial and lands n in $\iota_0((R[t])^I)$. Symmetrically, the exponents of each n_i are bounded above, say by D_i ; for any $M' \geq \max_i(D_i - m_i)$ with $M' \geq 0$, every exponent of $t^{-M'-m_i} n_i$ is nonpositive, so there is a polynomial q_i with $q_i(t^{-1}) = t^{-M'-m_i} n_i$, and then $(u^{-M'} n)_i = t^{-M'} n_i = t^{m_i} q_i(t^{-1})$, i.e. $u^{-M'} n = \iota_1(q)$. *Annihilation:* both ι_0 and ι_1 are injective — the inclusion of polynomials is injective, the substitution $t \mapsto t^{-1}$ is an automorphism of the Laurent ring, and multiplication by the unit t^{m_i} is injective — so a section vanishing on the overlap is already zero. $\square$

Theorem 436 (The monomial window: H^1 of a model over any ring). *Let R be any commutative ring, I finite and $m : I \rightarrow \mathbb{Z}$. Then $H^1(E(m))$ is a finite R -module. (It is spanned by the classes of the monomials t^j in the components i with $m_i < j < 0$ — the classical negative-monomial window of the projective line.)*

Proof. Apply the two-lattice finiteness ladder (??) with $k = R$, the Laurent module $N = (R[t, t^{-1}])^I$ — finitely generated over $R[t, t^{-1}]$, being free on I — and the R -submodules $Q_0 = \iota_0((R[t])^I)$, $Q_1 = \iota_1((R[t])^I)$. Q_0 is stable under multiplication by t and Q_1 under multiplication by t^{-1} , by the equivariance of ι_0 and ι_1 (??); and every element of N is absorbed into Q_0 by a large power of t and into Q_1 by a large power of t^{-1} , by the denominator-clearing axioms verified in ?? . The ladder concludes that $N/(Q_0 + Q_1) = H^1(E(m))$ is a finite R -module. $\square$

Theorem 437 (The degree window: H^0 of a model over a Noetherian ring). *Let R be a Noetherian commutative ring, I finite and $m : I \rightarrow \mathbb{Z}$. Then $H^0(E(m))$ is a finite R -module.*

Proof. A class in $H^0(E(m))$ is a pair (p, q) of polynomial vectors with $p_i(t) = t^{m_i} q_i(t^{-1})$ in $R[t, t^{-1}]$ for every i . Comparing coefficients: the right side has no exponent above m_i (the exponents of $q_i(t^{-1})$ are ≤ 0), so $\deg p_i \leq m_i$ — in particular $p_i = 0$ when $m_i < 0$. Moreover the first component determines the class: ι_1 is injective (??) and $\iota_1 q = \iota_0 p$. Hence $H^0(E(m))$ embeds R -linearly into $\bigoplus_i \{p \in R[t] : \deg p \leq m_i\}$, a finitely generated free R -module (of rank $\sum_i \max(m_i + 1, 0)$). A submodule of a finitely generated module over the Noetherian ring R is finitely generated. $\square$

Definition 438 (The Laurent action on the overlap module). For a two-lattice pair P , the R -algebra homomorphism

$$R[t, t^{-1}] \longrightarrow \text{End}_R(N), \quad t \mapsto u,$$

extending the assignment $t^j \mapsto u^j$ ($j \in \mathbb{Z}$, negative powers through the inverse of u) by R -linearity. It exists because u is invertible, so $j \mapsto u^j$ is a homomorphism from $\mathbb{Z}$ to the units of $\text{End}_R(N)$, and the Laurent ring is the monoid algebra of $\mathbb{Z}$ over R . We write $g(u)$ for the action of $g \in R[t, t^{-1}]$ on N .

Lemma 439 (The Laurent action intertwines the chart actions). *For a polynomial $g \in R[t]$, an element $x \in M_0$ and an element $y \in M_1$,*

$$g(u)(\iota_0 x) = \iota_0(g(t_0) x) \quad \text{and} \quad g(u^{-1})(\iota_1 y) = \iota_1(g(t_1) y),$$

where $g(t_0)$ and $g(t_1)$ denote the evaluations of g at the coordinate actions.

Proof. Both sides are additive in g , so it suffices to treat a monomial $g = ct^k$. Iterating the equivariance $\iota_0(t_0x) = u(\iota_0x)$ gives $\iota_0(t_0^kx) = u^k(\iota_0x)$, and multiplying by c gives the chart-0 identity; iterating $\iota_1(t_1y) = u^{-1}(\iota_1y)$ gives the chart-1 identity with u^{-k} . $\square$

Definition 440 (The model map determined by matched generators). Let P be a two-lattice pair, I a finite index set, $m : I \rightarrow \mathbb{Z}$, and let $a : I \rightarrow M_0$ and $b : I \rightarrow M_1$ be families whose overlap images match up to the twist:

$$u^{m_i}(\iota_0 a_i) = \iota_1 b_i \quad (i \in I).$$

The *model map* $f : E(m) \rightarrow P$ is the map of pairs with components

$$f_0(p) = \sum_{i \in I} p_i(t_0) a_i, \quad f_1(q) = \sum_{i \in I} q_i(t_1) b_i, \quad f_N(n) = \sum_{i \in I} n_i(u) (\iota_0 a_i),$$

the polynomial actions on the generators on the charts and the Laurent action (??) on their overlap images.

Proof. The five compatibilities of a map of pairs (??) hold term by term. Compatibility with the coordinate actions: multiplying p_i (resp. q_i , n_i) by t multiplies $p_i(t_0)$ (resp. $q_i(t_1)$, $n_i(u)$) by t_0 (resp. t_1 , u), which is exactly the model action carried to P . Compatibility with ι_0 : for a polynomial vector p , $f_N(\iota_0 p) = \sum_i p_i(u)(\iota_0 a_i) = \sum_i \iota_0(p_i(t_0)a_i) = \iota_0(f_0 p)$ by the chart-0 intertwining (??). Compatibility with ι_1 : for a polynomial vector q , the i -th term of $f_N(\iota_1 q)$ is $(t^{m_i} q_i(t^{-1}))(u)(\iota_0 a_i) = q_i(u^{-1})(u^{m_i}(\iota_0 a_i)) = q_i(u^{-1})(\iota_1 b_i) = \iota_1(q_i(t_1)b_i)$, using the matching hypothesis and the chart-1 intertwining; summing gives $f_N(\iota_1 q) = \iota_1(f_1 q)$. $\square$

Lemma 441 (Models have finite lattices). *The chart lattices of a model $E(m)$ are finite: $(R[t])^I$, with t acting by componentwise multiplication, is a finitely generated $R[t]$ -module.*

Proof. The $R[t]$ -module structure through the coordinate action coincides with the standard one on the free module $(R[t])^I$, which is generated by the $|I|$ standard basis vectors. $\square$

Theorem 442 (The finite model surjection). *Let P be a two-lattice pair with finite lattices over a commutative ring R . Then there exist a finite index set I , twists $m : I \rightarrow \mathbb{Z}$, and a map of pairs $f : E(m) \rightarrow P$ surjective on the chart-0 lattice, on the chart-1 lattice, and on the overlap module.*

Proof. Choose finite generating families $(a_j)_{j \in J_0}$ of M_0 over $R[t]$ (acting through t_0) and $(b_j)_{j \in J_1}$ of M_1 over $R[t]$ (through t_1). Match each generator across the overlap by denominator clearing (??): for $j \in J_0$ there are $k_j \geq 0$ and $y_j \in M_1$ with $u^{-k_j}(\iota_0 a_j) = \iota_1 y_j$; set $m_j = -k_j$, so $u^{m_j}(\iota_0 a_j) = \iota_1 y_j$. For $j \in J_1$ there are $k_j \geq 0$ and $x_j \in M_0$ with $u^{k_j}(\iota_1 b_j) = \iota_0 x_j$; set $m_j = -k_j$, so $u^{m_j}(\iota_0 x_j) = u^{-k_j} u^{k_j}(\iota_1 b_j) = \iota_1 b_j$. Take $I = J_0 \sqcup J_1$, with generator pairs (a_j, y_j) for $j \in J_0$ and (x_j, b_j) for $j \in J_1$, and let $f : E(m) \rightarrow P$ be the model map of ??.

Chart surjectivity. Any $x \in M_0$ is an $R[t]$ -combination $x = \sum_{j \in J_0} g_j(t_0) a_j$; the vector with components g_j on J_0 and 0 on J_1 maps to x under f_0 . Symmetrically for M_1 .

Overlap surjectivity. Given $n \in N$, clear denominators: $u^k n = \iota_0 x$ for some $k \geq 0$, $x \in M_0$; pick p with $f_0(p) = x$. Since f commutes with the overlap actions and their inverses and with ι_0 ,

$$f_N(u_E^{-k}(\iota_0^E p)) = u^{-k}(\iota_0(f_0 p)) = u^{-k}(u^k n) = n,$$

where u_E and ι_0^E denote the overlap action and restriction of the model. $\square$

5.18 The coherence theorems for two-lattice pairs

The two finiteness keystones of the engine: for a pair with finite lattices, H^1 is a finite R -module over *every* commutative ring R , and H^0 is a finite R -module over every *Noetherian* R . Noetherianity enters only through H^0 ; every downstream consequence that only needs H^1 — vanishing propagation and the openness of the vanishing locus (Section ??) — holds relative to an arbitrary base ring.

Definition 443 (The kernel pair of a map of pairs). For a map of two-lattice pairs $f : P \rightarrow P'$, the *kernel pair* $K(f)$ is the two-lattice pair with carriers $\ker f_0 \subseteq M_0$, $\ker f_1 \subseteq M_1$, $\ker f_N \subseteq N$, the restricted coordinate and overlap actions, and the restricted maps ι_0, ι_1 ; the componentwise inclusions form a map of pairs $K(f) \rightarrow P$.

Proof. The kernels are stable under the actions and the restrictions because f commutes with them (for the inverse overlap action, by the automatic compatibility of ??), so all the structure restricts; the annihilation axioms restrict verbatim. Denominator clearing must be re-proved: let $n \in \ker f_N$. Clearing in P gives $k \geq 0$ and $x \in M_0$ with $u^k n = \iota_0 x$. Then $\iota'_0(f_0 x) = f_N(u^k n) = u'^k(f_N n) = 0$, so by the annihilation axiom of P' there is $j \geq 0$ with $t_0'^j(f_0 x) = 0$, i.e. $f_0(t_0^j x) = 0$: the element $t_0^j x$ lies in $\ker f_0$, and

$$u^{j+k} n = u^j(\iota_0 x) = \iota_0(t_0^j x)$$

exhibits clearing inside the kernel pair. The chart-1 side is symmetric. The componentwise inclusions commute with all the structure by construction. $\square$

Lemma 444 (The kernel pair sits in a short exact sequence). *If a map of pairs $f : P \rightarrow P'$ is surjective on all three carriers, then*

$$0 \longrightarrow K(f) \longrightarrow P \xrightarrow{f} P' \longrightarrow 0$$

is a short exact sequence of two-lattice pairs.

Proof. On each carrier, the inclusion of the kernel is injective, its image is exactly the kernel of the corresponding component of f , and that component is surjective by hypothesis. $\square$

Lemma 445 (Kernel lattices are finite over a Noetherian ring). *Let R be Noetherian and let $f : P \rightarrow P'$ be a map of pairs where P has finite lattices. Then the kernel pair $K(f)$ has finite lattices.*

Proof. $\ker f_0$ is an $R[t]$ -submodule of M_0 for the structure through t_0 (it is stable under t_0). By Hilbert's basis theorem $R[t]$ is Noetherian, so the finitely generated $R[t]$ -module M_0 is a Noetherian module, and every submodule — in particular $\ker f_0$ — is finitely generated. Symmetrically for $\ker f_1$. $\square$

Theorem 446 (COH-1: coherence of H^1 over any base ring). *Let R be any commutative ring — no Noetherian hypothesis — and let P be a two-lattice pair with finite lattices. Then $H^1(P)$ is a finite R -module.*

Proof. Choose a finite model surjection $f : E(m) \rightarrow P$ (?). It is surjective on overlap modules, so $H^1(f) : H^1(E(m)) \rightarrow H^1(P)$ is surjective (?). The source is a finite R -module — the monomial window (?) — and a quotient of a finite module is finite. $\square$

Theorem 447 (COH-0: coherence of H^0 over a Noetherian base ring). *Let R be a Noetherian commutative ring and let P be a two-lattice pair with finite lattices. Then $H^0(P)$ is a finite R -module.*

Proof. Choose a finite model surjection $f : E(m) \rightarrow P$ (??), surjective on all three carriers, and form the short exact sequence $0 \rightarrow K(f) \rightarrow E(m) \rightarrow P \rightarrow 0$ of ??. Its connecting homomorphism $\partial : H^0(P) \rightarrow H^1(K(f))$ (??) has:

- kernel equal to the image of $H^0(E(m)) \rightarrow H^0(P)$ (??), a quotient of the finite R -module $H^0(E(m))$ of the degree window (??, using that R is Noetherian), hence finite;
- image a submodule of $H^1(K(f))$. The model has finite lattices (??), so the kernel pair has finite lattices over the Noetherian R (??), and COH-1 applied to $K(f)$ (??) makes $H^1(K(f))$ a finite R -module; over the Noetherian R it is a Noetherian module, so the submodule $\partial(H^0(P))$ is finite.

Thus $H^0(P)$ is an extension of the finite module $\partial(H^0(P))$ by the finite module $\ker \partial$, hence finite. $\square$

5.19 The rigid engine on a packaged two-cover

This section assembles the module-level rigid engine — the coherence theorems of two-lattice pairs (??, ??), the fibrewise Nakayama propagation (Section ??) and the split rigidity (Section ??) — at the level of sheaves. For a sheaf F of R -modules on the small Zariski site of a scheme X , packaged quasi-coherently (??) on both charts of an affine two-cover $U_0 \cup U_1 = X$, a small amount of coordinate data produces a two-lattice pair whose cohomology *is* the cohomology of the site with coefficients in F ; firing the engine on the pair yields the curve-lite form of Kleiman’s theorem on the module $\mathcal{Q}$: vanishing of the degree-one fibres forces $H^1 = 0$ and makes H^0 finite projective, with formation of H^0 commuting with all coefficients.

The degree-one carrier is the landed two-cover cokernel (??); the missing degree-zero carrier — the kernel twin of the cokernel bridge — is built first.

Theorem 448 (The Mayer–Vietoris degree-zero kernel bridge). *For a Mayer–Vietoris square S and a sheaf F of R -modules, the pair of restrictions, corestricted to the kernel of the restriction-difference map (??), is an R -linear equivalence*

$$F(X_4) \cong \ker(\text{moduleDiff} : F(X_2) \times F(X_3) \rightarrow F(X_1)), \quad s \mapsto (s|_{X_2}, s|_{X_3}).$$

Proof. The map lands in the kernel and is injective and surjective onto it by degree-zero exactness (??): injectivity is the separation half of the sheaf condition of the square, surjectivity onto the kernel is the gluing half. Linearity is linearity of the two restrictions. $\square$

Theorem 449 (The two-cover H^0 kernel bridge). *Let X be a scheme, $U_0 \cup U_1 = X$ a two-cover by opens, and F any sheaf of R -modules on the small Zariski site. Then degree-zero cohomology of the site is the kernel of the two-cover restriction-difference map: an R -linear equivalence*

$$H^0(X, F) \cong \ker(F(U_0) \times F(U_1) \rightarrow F(U_0 \cap U_1)).$$

No affineness is needed: degree-zero exactness is the sheaf condition alone.

Proof. Identify $H^0(X, F)$ with $H^0(\top, F) = F(\top)$ (??, ??), then apply the kernel bridge ?? to the two-cover square (??), whose restriction-difference map is exactly the displayed one. $\square$

Definition 450 (The two-cover pair data of a packaged sheaf). Let F be a sheaf of R -modules on the small Zariski site of X , packaged quasi-coherently (??) on two opens U_0, U_1 . *Two-cover pair data* for F consists of chart coordinates $g_0 \in \Gamma(X, U_0)$, $g_1 \in \Gamma(X, U_1)$ such that:

- the overlap is the basic open of either coordinate: $U_0 \cap U_1 = D(g_0) = D(g_1)$;
- the packaged actions of g_0 and of g_1 commute with the R -scalars on sections of every open $W \leq U_i$;
- on the overlap the two packaged actions are mutually inverse (the relation $g_0 g_1 = 1$, expressed on the actions: acting by g_0 and then by g_1 , in either order, is the identity of $F(U_0 \cap U_1)$).

Definition 451 (The two-lattice pair of a packaged sheaf). Let U_0, U_1 be *affine* opens and d two-cover pair data for a packaged sheaf F . Then the data

$$M_0 = F(U_0), \quad M_1 = F(U_1), \quad N = F(U_0 \cap U_1),$$

with coordinate actions the packaged actions of g_0, g_1 (made R -linear endomorphisms by the R -linearity field of d), overlap automorphism the action of g_0 on N (invertible with inverse the action of g_1 , by the mutual-inverse field), and restrictions ι_0, ι_1 the section restrictions to the overlap, is a two-lattice pair (??) over R . Its difference map is the Mayer–Vietoris restriction-difference map of the two-cover square with coefficients in F , on the nose.

Proof. The intertwining laws $\iota_i(g_i \cdot x) = g_i \cdot \iota_i(x)$ are the (P1) naturality of the packaged action under restriction. The localization axioms are the (P2) axioms of the packaging, transported to the overlap along the identification $U_0 \cap U_1 = U_i \cap D(g_i)$: denominator clearing for the pair asks that every $n \in N$ admit $m \geq 0$ and $x \in M_i$ with $g_i^m \cdot n = \iota_i(x)$, which is (P2) denominator clearing on the affine open U_i ; and annihilation for the pair asks that $\iota_i(x) = 0$ force $g_i^m \cdot x = 0$, which is (P2) defect annihilation — exactly the three-axiom match by which the packaging exhibits the overlap sections as the localization of the chart sections (??). Powers of the action are the actions of powers (by (P1) associativity), which is used implicitly throughout. The difference map $(x, y) \mapsto \iota_0 x - \iota_1 y$ is by construction the restriction-difference map of ?? for the two-cover square (??). $\square$

Definition 452 (The cohomology carriers of the pair). Let U_0, U_1 be affine opens with $U_0 \cup U_1 = X$ and d two-cover pair data for the packaged sheaf F , with pair P (??). There are R -linear equivalences

$$H^0(X, F) \cong H^0(P), \quad H^1(X, F) \cong H^1(P) :$$

the cohomology of the site with coefficients in F is the cohomology of the two-term complex of the pair.

Proof. Degree zero: the kernel bridge (??) lands in the kernel of the restriction-difference map, which is the kernel of the pair’s difference map since the two maps are equal (??), i.e. $H^0(P)$ (??). Degree one: the packaging on the two affine charts gives the twisted affine vanishing $H^1(U_i, F) = 0$ (??), so the two-cover computation (??) identifies $H^1(X, F)$ with the cokernel of the restriction-difference map; and the cokernel of the pair’s difference map is $H^1(P)$ (??, ??). $\square$

Lemma 453 (Vanishing of H^1 of the pair makes the differential surjective). *If H^1 of the pair of ?? is trivial (a subsingleton), then the difference map $\delta : F(U_0) \times F(U_1) \rightarrow F(U_0 \cap U_1)$ is surjective.*

Proof. H^1 of the pair is the cokernel of δ (??), and a map with trivial cokernel is surjective (??). $\square$

Theorem 454 (The rigid engine, sheaf level). *Let R be Noetherian, U_0, U_1 affine opens covering X , and d two-cover pair data for the packaged sheaf F whose pair P has finite lattices (the chart sections $F(U_i)$ are finite over $R[t]$ through the coordinate actions). Assume the section modules are tame: $F(U_0) \times F(U_1)$ is a flat R -module and $F(U_0 \cap U_1)$ is a projective R -module (in the intended applications all three are free). If every residue-field fibre of H^1 of the pair vanishes,*

$$H^1(P) \otimes_R \kappa(\mathfrak{p}) = 0 \quad \text{for every prime } \mathfrak{p} \subseteq R,$$

then

$$H^1(X, F) = 0, \quad \text{and } H^0(X, F) \text{ is a finite projective } R\text{-module.}$$

This is the curve-lite form of Kleiman's criterion for the module $\mathcal{Q}$: vanishing of the degree-one cohomology of the fibres ((v) of the source) forces the representing module to be free/projective ((i)) — here delivered on a two-chart cover with the two-term Čech complex playing the role of the pushforward. The fibre hypothesis is deliberately consumed in this complex form (fibres of H^1 of the pair); its sheaf-level discharge — identifying the fibre of the complex with the cohomology of a fibre sheaf — is a seam of a later campaign and is not part of this statement.

Proof. $H^1(P)$ is a finite R -module by COH-1 (??), so the fibrewise vanishing hypothesis propagates to $H^1(P) = 0$ by Nakayama (??), and along the degree-one carrier (??) this gives $H^1(X, F) = 0$. The differential is then surjective (??). $H^0(P) = \ker \delta$ is finite over the Noetherian R by COH-0 (??), and the split rigidity of a surjective two-term complex with flat domain and projective codomain makes the finite kernel projective (??). Transport both along the degree-zero carrier (??). $\square$

Theorem 455 (The openness export). *In the setting of ??, with finite lattices but no Noetherian hypothesis on R , the fibrewise-vanishing locus*

$$\{ \mathfrak{p} \in \operatorname{Spec} R : H^1(P) \otimes_R \kappa(\mathfrak{p}) = 0 \}$$

is open in $\operatorname{Spec} R$. (In the source the analogous openness is obtained from Serre's finiteness theorem over a Noetherian base; here it falls out of the coherence of H^1 over an arbitrary base ring, Noetherian-free.)

Proof. $H^1(P)$ is a finite R -module over any commutative R (??); for a finite module the fibrewise-vanishing locus is the complement of the support, which is closed (??). $\square$

Definition 456 (The module-coefficient clause). *In the setting of ??, assume $H^1(P)$ is trivial and the overlap module $F(U_0 \cap U_1)$ is flat over R . Then for every R -module P' there is an R -linear equivalence*

$$H^0(X, F) \otimes_R P' \cong \ker(\delta \otimes_R P') :$$

formation of H^0 commutes with all module coefficients. This is the two-chart form of the isomorphism (eq. Q of the source) $\operatorname{Hom}(\mathcal{Q}, \mathcal{N}) \cong f_(\mathcal{F} \otimes f^*\mathcal{N})$ in its right-exact regime: on the vanishing locus the functor $P' \mapsto \ker(\delta \otimes_R P')$ is represented by H^0 .*

Proof. The differential is surjective (??) with flat codomain, so the kernel-tensor rigidity (??) gives $\ker(\delta) \otimes P' \cong \ker(\delta \otimes P')$; compose with the degree-zero carrier (??) tensored with P' . $\square$

Definition 457 (The ring-map clause, abstract half). In the setting of ??, for every commutative R -algebra R' there is an R' -linear equivalence

$$R' \otimes_R H^0(X, F) \cong \ker(\delta \otimes_R R'),$$

the module-coefficient clause upgraded to carry the R' -structure. Reaching H^0 of a base-changed *sheaf* on the nose is a separate step: it threads this equivalence through term identifications of the base-changed complex (?? below, for the relative twisted sheaf).

Proof. As in ??, with the R' -linear kernel base-change form of the rigidity (??) in place of the module form. $\square$

5.20 The pair data of the twisted sheaf

The Layer-2 discharge of the pair data for the cocycle-glued twisted sheaf F_g of Section ??, generic in the base ring k : every field of ?? reduces, through the chart trivialisations, to section algebra in the structure sheaf.

Definition 458 (Multiplication by a section as an endomorphism). Let X be a scheme over $\text{Spec } k$ and U an open. Multiplication by a fixed section $c \in \Gamma(X, U)$ is a k -linear endomorphism of $\Gamma(X, U)$ (for the k -structure through the structure morphism, ??): the comparison endomorphism through which the twisted pair's coordinate actions are identified with honest section multiplication.

Lemma 459 (The packaged action of the twisted sheaf, and its k -linearity). *Fix opens V_0, V_1 and a unit cocycle g , with twisted sheaf F_g . For $W \leq V_i$, the packaged action (??) of $r \in \Gamma(X, V_i)$ on $F_g(W)$ is trivialise–multiply–untrivialise:*

$$r \cdot m = \text{triv}_i^{-1}(r|_W \cdot \text{triv}_i(m)),$$

and it commutes with the k -scalars.

Proof. The packaging of F_g on the chart V_i is by definition the transport (??) of the structure-sheaf packaging along the chart trivialisation (??), whose action is exactly the displayed conjugate; the first claim is definitional. For k -linearity: multiplication of sections commutes with the k -scalars (both are multiplications in a commutative ring), and the trivialisations are k -linear, so the conjugate commutes with the scalars too. $\square$

Lemma 460 (The two trivialisations differ by the cocycle on the overlap). *On the overlap sections $F_g(V_0 \cap V_1)$,*

$$\text{triv}_0(p) = g \cdot \text{triv}_1(p), \quad \text{triv}_1(p) = g^{-1} \cdot \text{triv}_0(p).$$

Proof. A twisted section $p = (s_0, s_1)$ over $V_0 \cap V_1$ satisfies the matching relation $s_0| = g \cdot s_1|$ on $(V_0 \cap V_1) \cap V_0 \cap V_1 = V_0 \cap V_1$ (??); the two trivialisations are the two components (??), so the relation reads $\text{triv}_0(p) = g \cdot \text{triv}_1(p)$. Multiply by g^{-1} for the second form. $\square$

Lemma 461 (The mutual-inverse law of the coordinate actions). *Let $r_0 \in \Gamma(X, V_0)$, $r_1 \in \Gamma(X, V_1)$ satisfy $r_0|_{\cap} \cdot r_1|_{\cap} = 1$ on the overlap. Then the packaged actions of r_0 and r_1 on $F_g(V_0 \cap V_1)$ are mutually inverse.*

Proof. Compute the composite on m through the chart-0 trivialisation. Acting by r_1 is $\text{triv}_1^{-1}(r_1| \cdot \text{triv}_1(m))$ (??); by ?? its chart-0 trivialisation is $g \cdot r_1| \cdot g^{-1} \cdot \text{triv}_0(m) = r_1| \cdot \text{triv}_0(m)$ (the ring is commutative). Acting then by r_0 multiplies by $r_0|$, giving $r_0| r_1| \cdot \text{triv}_0(m) = \text{triv}_0(m)$; untrivialising returns m . The other order is symmetric, through the chart-1 trivialisation. $\square$

Definition 462 (The pair data of the twisted sheaf). Let $r_0 \in \Gamma(X, V_0)$ and $r_1 \in \Gamma(X, V_1)$ satisfy $V_0 \cap V_1 = D(r_0) = D(r_1)$ and $r_0|_{\cap} \cdot r_1|_{\cap} = 1$. Then (r_0, r_1) is two-cover pair data (??) for the twisted sheaf F_g on (V_0, V_1) : the basic-open identifications are the hypotheses, k -linearity of the packaged actions is ??, and the mutual-inverse law is ??.

Theorem 463 (Chart-lattice finiteness of the twisted pair). *In the setting of ?? with V_0, V_1 affine: if the chart sections $\Gamma(X, V_i)$ are finite over $k[t]$ with t acting as multiplication by r_i (??), then the pair of the twisted sheaf (??) has finite lattices.*

Proof. The chart trivialisation $\text{triv}_i : F_g(V_i) \cong \Gamma(X, V_i)$ (??) intertwines the pair's coordinate action — which is trivialise–multiply–untrivialise (??) — with multiplication by r_i . Chart-lattice finiteness transports along an equivariant equivalence (??). $\square$

Theorem 464 (Freeness of the twisted section modules). *For $W \leq V_i$, if $\Gamma(X, W)$ is a free k -module then so is $F_g(W)$: transport a basis along the chart trivialisation (??).*

Proof. A module equivalent to a free module is free. $\square$

5.21 The pinned relative cover: coordinates, finiteness and freeness

The geometric half of the engine's Layer-2 discharge: the pinned affine two-cover of the curve — the chart preimages of a finite morphism π to $\mathbb{P}^1$ — carries polynomial chart algebras through which the finiteness of π becomes chart-lattice finiteness, first over the field k and then, by base change, over every test ring R . Throughout, k is a field and $\pi : Y \rightarrow \mathbb{P}^1$ an affine morphism (finite where stated).

Definition 465 (The pinned affine two-cover). For an affine morphism $\pi : Y \rightarrow \mathbb{P}^1$, the chart preimages $V_0 = \pi^{-1}D_+(X_0)$, $V_1 = \pi^{-1}D_+(X_1)$ form an *affine two-cover* of Y (??): both charts are affine (??), they cover Y (??), and the overlap $V_0 \cap V_1 = \pi^{-1}D_+(X_0X_1)$ (??) is affine, being the preimage of an affine open under an affine morphism. This is the cover of ??, upgraded with the affineness of the overlap; the whole engine runs on it and on its base change.

Theorem 466 (The overlap is the basic open of the chart-1 coordinate). $V_0 \cap V_1 = D(t_1)$, where $t_1 = \pi^\sharp(X_0/X_1) \in \Gamma(Y, V_1)$ is the *pulled-back chart-1 coordinate* (??) — the *chart-1 twin* of the identification $V_0 \cap V_1 = D(t_0)$ of ??.

Proof. On $\mathbb{P}^1$ the basic open of the chart-1 coordinate X_0/X_1 in $D_+(X_1)$ is the chart overlap (??); basic opens commute with pullback of sections, so applying π^{-1} gives $V_1 \cap V_0 = D(\pi^\sharp(X_0/X_1))$. $\square$

Definition 467 (The chart polynomial algebras). The *chart- i polynomial algebra* is the ring homomorphism

$$k[t] \cong \Gamma(\mathbb{P}^1, D_+(X_i)) \xrightarrow{\pi^\sharp} \Gamma(Y, V_i),$$

the chart identification (??, ??) followed by pullback along π . It sends t to the pulled-back chart coordinate: $t \mapsto t_0 = \pi^\sharp(X_1/X_0)$ on chart 0 (??) and $t \mapsto t_1 = \pi^\sharp(X_0/X_1)$ on chart 1 (??).

Proof. Only the values at t need comment: the chart identification matches t with the chart coordinate $X_{1-i}/X_i \in \Gamma(\mathbb{P}^1, D_+(X_i))$, whose pullback is t_i by definition. $\square$

Theorem 468 (E-i: the chart polynomial algebras are finite ring maps). *If π is finite, both chart polynomial algebras $k[t] \rightarrow \Gamma(Y, V_i)$ are finite ring homomorphisms.*

Proof. For a finite morphism, the induced map on sections over an affine chart of the target is a finite ring map (??); precomposing with the ring isomorphism $k[t] \cong \Gamma(\mathbb{P}^1, D_+(X_i))$ preserves finiteness. $\square$

Theorem 469 (Scalar compatibility of the chart polynomial algebras). *Let $Y = C$ be the challenge curve and suppose π factors the structure morphism, $\pi \circ (\text{structure map of } \mathbb{P}^1) = (C \rightarrow \text{Spec } k)$. Then the chart polynomial algebras carry constants to the structure algebra map (??): for $c \in k$,*

$$\text{fiberPolyHom}_i(c) = \text{the image of } c \text{ under } k \rightarrow \Gamma(C, V_i).$$

Proof. The chart identification carries the constant $c \in k[t]$ to the restriction to $D_+(X_i)$ of the structure algebra map of $\mathbb{P}^1$ (??). Pullback along π carries the structure algebra map of $\mathbb{P}^1$ to that of C , because π commutes with the two structure morphisms (??, applied to the factorization hypothesis). $\square$

Theorem 470 (E-i in chart-lattice form, field level). *In the setting of ?? with π finite: the chart sections $\Gamma(C, V_i)$ are finite over $k[t]$, with t acting as multiplication by the pulled-back chart coordinate t_i .*

Proof. Apply the finite-ring-map criterion (??) to the chart polynomial algebra: it is finite (??), sends t to t_i (??), and sends constants to the structure algebra map, which is the k -scalar action on sections (??). $\square$

Definition 471 (The relative chart coordinates). Let R be a commutative k -algebra and consider the base-changed pinned cover $V_i^R = \text{pr}_1^{-1} V_i$ of the relative curve C_R (??, ??). The *relative chart coordinates* are the pullbacks along the first projection,

$$t_0^R = \text{pr}_1^\sharp(t_0) \in \Gamma(C_R, V_0^R), \quad t_1^R = \text{pr}_1^\sharp(t_1) \in \Gamma(C_R, V_1^R)$$

(??); pullback of sections is multiplicative.

Theorem 472 (The relative overlap identifications). *On the base-changed pinned cover,*

$$V_0^R \cap V_1^R = D(t_0^R) = D(t_1^R).$$

Proof. The overlap of the base-changed cover is the preimage of the overlap (??); the overlap downstairs is $D(t_0) = D(t_1)$ (??, ??); and basic opens commute with pullback of sections along pr_1 . $\square$

Theorem 473 (The relative coordinates are mutually inverse on the overlap). *$t_0^R|_\cap \cdot t_1^R|_\cap = 1$ in $\Gamma(C_R, V_0^R \cap V_1^R)$.*

Proof. The identity $t_0|_\cap \cdot t_1|_\cap = 1$ holds downstairs (??); pull it back along $\text{pr}_1^\sharp$ over the overlap, a ring homomorphism commuting with restriction, using $\text{pr}_1^{-1}(V_0 \cap V_1) = V_0^R \cap V_1^R$ (??). $\square$

Theorem 474 (The relative E-i: chart-lattice finiteness over the test ring). *In the setting of ??, for every commutative k -algebra R : the relative chart sections $\Gamma(C_R, V_i^R)$ are finite over $R[t]$, with t acting as multiplication by the relative coordinate t_i^R (??).*

Proof. The field-level chart lattice is finite over $k[t]$ (??); its base change to R is finite over $R[t]$ (??). The sections base change $R \otimes_k \Gamma(C, V_i) \cong \Gamma(C_R, V_i^R)$ (??) intertwines the base-changed multiplication endomorphism $\text{id} \otimes (t_i \cdot)$ with multiplication by t_i^R : on pure tensors, $a \otimes t_i s \mapsto \text{pr}_2^\sharp(a) \text{pr}_1^\sharp(t_i s) = t_i^R \cdot (\text{pr}_2^\sharp(a) \text{pr}_1^\sharp(s))$, since $\text{pr}_1^\sharp$ is multiplicative (??). Chart-lattice finiteness transports along the equivariant equivalence (??). $\square$

Theorem 475 (Freeness of the relative section modules). *For every quasi-compact quasi-separated open $V \subseteq C$ and every commutative k -algebra R , the R -module $\Gamma(C_R, \text{pr}_1^{-1}V)$ is free: a k -basis of $\Gamma(C, V)$ becomes an R -basis through the sections base change (??). This feeds the flatness and projectivity hypotheses of the engine: all three section modules of the pinned relative cover are free.*

Proof. $\Gamma(C, V)$ is a free k -module (k is a field), so $R \otimes_k \Gamma(C, V)$ is a free R -module, and freeness transports along the R -linear equivalence of ??). $\square$

5.22 The rigid engine on the relative twisted sheaf

The keystones: the engine of Section ?? fired on the relative twisted sheaf F_g on the base change of the pinned cover, with every hypothesis discharged from the preceding two sections — and the base-change clause threaded, through the term identifications of Section ??, onto H^0 of the base-changed sheaf on the nose.

One staging note, recorded here because it is the honest state of the statements: the engine's vanishing hypothesis is consumed in *complex form* — vanishing of the residue-field fibres of H^1 of the two-lattice pair of F_g . Its sheaf-level discharge (identifying the glued fibre sheaf with the divisor sheaf of a witness and citing the fibrewise large-twist vanishing of Chapter ??), and the rank export $\dim_{\kappa(\mathfrak{p})} H^0 \otimes \kappa(\mathfrak{p}) = \deg + \chi$, are deliberately staged behind a later seam and are *not* claimed by the statements below; consumers supply the fibre hypothesis.

Throughout: k a field, C the challenge curve, $\pi : C \rightarrow \mathbb{P}^1$ finite, R a commutative k -algebra, and g a unit cocycle on the overlap of the base-changed pinned cover (V_0^R, V_1^R) of C_R , with relative twisted sheaf F_g (??).

Definition 476 (The pair data of the relative twisted sheaf). The relative twisted sheaf is packaged quasi-coherently on both base-changed charts (??, read on the cover of C_R), and the relative chart coordinates (t_0^R, t_1^R) satisfy the two hypotheses of ?? on the base-changed pinned cover: the overlap-as-basic-open identifications (??) and the mutual-inverse relation (??). This yields the two-cover pair data of F_g , and with it the two-lattice pair of the relative twisted sheaf that the engine consumes.

Theorem 477 (Chart-lattice finiteness of the relative twisted pair). *If R acts through the structure factorization of π (the hypothesis of ??), the pair of the relative twisted sheaf (??) has finite lattices over $R[t]$.*

Proof. The relative E-i gives finiteness of the structure-sheaf chart lattices at the relative coordinates (??); the twisted pair's lattices are trivialisation-transports of these (??). $\square$

Theorem 478 (Freeness of the twisted section modules on the pinned relative cover). *The three section modules of the pair of the relative twisted sheaf — the two charts and the overlap — are free R -modules.*

Proof. The charts and the overlap of the pinned cover are affine (??), in particular quasi-compact and quasi-separated, so the structure-sheaf sections of their base changes are free (??); the twisted sections are trivialisation-equivalent to them (??). $\square$

Theorem 479 (The rigid engine on the relative twisted sheaf). *Let R be Noetherian. If every residue-field fibre of H^1 of the pair of F_g vanishes,*

$$H^1(\text{pair}) \otimes_R \kappa(\mathfrak{p}) = 0 \quad \text{for every prime } \mathfrak{p} \subseteq R,$$

then

$$H^1(C_R, F_g) = 0, \quad \text{and } H^0(C_R, F_g) \text{ is a finite projective } R\text{-module.}$$

Every other hypothesis of the abstract engine is discharged: this is the curve-lite Kleiman criterion, $(v) \Rightarrow (i)$, on the relative twisted sheaf. The fibre hypothesis remains in complex form (see the staging note at the head of this section).

Proof. The base-changed pinned cover is an affine two-cover of C_R (??); the pair data is ??; the chart lattices are finite (??); the section modules are free (??), so the product of the chart modules is flat and the overlap module is projective. Fire the sheaf-level engine (??). $\square$

Lemma 480 (Fibrewise vanishing propagates to the pair). *With no Noetherian hypothesis on R : if every residue-field fibre of H^1 of the pair of F_g vanishes, then H^1 of the pair vanishes. (Consumers holding the fibrewise hypothesis use this to feed the two clauses below, which are stated on the vanishing locus.)*

Proof. H^1 of the pair is a finite R -module by COH-1 (??, with the finite lattices of ??); fibrewise vanishing of a finite module propagates by Nakayama (??). $\square$

Theorem 481 (The openness export on the relative twisted sheaf). *With no Noetherian hypothesis on R , the fibrewise-vanishing locus $\{\mathfrak{p} \in \text{Spec } R : H^1(\text{pair of } F_g) \otimes_R \kappa(\mathfrak{p}) = 0\}$ is open in $\text{Spec } R$ — the strata over which the engine fires.*

Proof. The openness export of the abstract engine (??), with the finite lattices of ??). $\square$

Definition 482 (The module-coefficient clause on the relative twisted sheaf). *On the vanishing locus (H^1 of the pair of F_g trivial), for every R -module P there is an R -linear equivalence*

$$H^0(C_R, F_g) \otimes_R P \cong \ker(\delta_{F_g} \otimes_R P),$$

δ_{F_g} the twisted two-cover differential: formation of the twisted H^0 commutes with all module coefficients (the curve-lite eq. Q).

Proof. The abstract module-coefficient clause (??), the flatness of the overlap module coming from its freeness (??). $\square$

Definition 483 (The ring-map clause on the relative twisted sheaf, abstract half). *On the vanishing locus, for every commutative R -algebra R' there is an R' -linear equivalence*

$$R' \otimes_R H^0(C_R, F_g) \cong \ker(\delta_{F_g} \otimes_R R').$$

Proof. The abstract ring-map clause (??), with flatness of the overlap module from freeness (??). $\square$

Lemma 484 (The twisted differential in trivialised coordinates). *For any twisted sheaf F_g on opens V_0, V_1 : under the chart-0 trivialisation of the overlap, the twisted restriction-difference map $(p_0, p_1) \mapsto p_0|_{\cap} - p_1|_{\cap}$ becomes*

$$(s_0, s_1) \mapsto s_0|_{\cap} - g \cdot s_1|_{\cap}, \quad s_i = \text{triv}_i(p_i) :$$

the defect-map shape of ??, with the cocycle carrying the chart-1 term across.

Proof. The trivialisations commute with restriction (??), so $\text{triv}_0(p_0|_\cap) = s_0|_\cap$; and on the overlap the chart-0 trivialisation of $p_1|_\cap$ is g times its chart-1 trivialisation (??), which is $s_1|_\cap$, again by ???. Subtract. $\square$

Definition 485 (CBC-1 for the twisted complex, assembled). Let $R \rightarrow R'$ be a homomorphism of commutative k -algebras, D an affine two-chart cover of C , g a unit cocycle on the base-changed cover of C_R and $g' = \rho^\sharp(g)$ its base change. The product of the two chart term equivalences of ?? is an R' -linear equivalence

$$R' \otimes_R (F_g(V_0^R) \times F_g(V_1^R)) \cong F_{g'}(V_0^{R'}) \times F_{g'}(V_1^{R'}),$$

the degree-zero term of the twisted complex base-changed. In trivialised coordinates each term equivalence computes, on a pure tensor $r' \otimes p$, as $r' \cdot \rho^\sharp(\text{triv}(p))$ — the structure-sheaf term base change of ?? (at the overlap, the codomain equivalence of ??) read through the trivialisations.

Proof. Each twisted term equivalence is by construction (??) the conjugate of the corresponding structure-sheaf term equivalence by the chart trivialisations over R and R' ; the displayed formula on pure tensors is the formula of ?? for the structure sheaf, conjugated. The product of two R' -linear equivalences is one. $\square$

Theorem 486 (The twisted δ -naturality square). *In the setting of ??, write δ_g for the twisted restriction-difference map over R and $\delta_{g'}$ for the one over R' . Under the CBC-1 term identifications — the domain equivalence of ?? and the overlap equivalence of ?? — the base change of the twisted differential is carried to the twisted differential of the base-changed cocycle:*

$$\text{overlap} \circ (\delta_g \otimes_R R') = \delta_{g'} \circ \text{dom}.$$

This is the cocycle-conjugated analogue of the untwisted δ -naturality (??); on the pinned cover, δ_g is on the nose the differential of the pair of the relative twisted sheaf, so the square also reads on the pairs.

Proof. Both composites are additive, so compare them on $r' \otimes (p_0, p_1)$. Read everything in chart-0 trivialised coordinates. By ?? the twisted differential is $(s_0, s_1) \mapsto s_0|_\cap - g \cdot s_1|_\cap$, so along the top and right the tensor goes to

$$r' \cdot \rho^\sharp(s_0|_\cap) - r' \cdot \rho^\sharp(g) \rho^\sharp(s_1|_\cap),$$

by ?? and multiplicativity of $\rho^\sharp$. Along the left and bottom it goes to $(r' \cdot \rho^\sharp s_0)|_\cap - g' \cdot (r' \cdot \rho^\sharp s_1)|_\cap$, where $g' = \rho^\sharp(g)$ (??). The two agree because $\rho^\sharp$ commutes with restriction (??), the R' -action commutes with restriction, and the g' -factor commutes with the R' -scalar (commutative ring). The pair form on the pinned cover is the same identity, the pair's differential being definitionally the twisted restriction-difference map. $\square$

Theorem 487 (H^1 of the twisted sheaf vanishes after base change). *On the vanishing locus (H^1 of the pair of F_g over R trivial), for every homomorphism $R \rightarrow R'$ of commutative k -algebras: the twisted differential over R' at the base-changed cocycle g' is surjective, and*

$$H^1(C_{R'}, F_{g'}) = 0.$$

Proof. Over R the twisted differential is surjective (??); tensoring with R' preserves surjectivity (right exactness), and surjectivity transports across the δ -naturality square (??) by ??. So $\delta_{g'}$ is surjective; hence H^1 of the pair of $F_{g'}$ — its cokernel (??) — is trivial, and the degree-one carrier over R' (??, for the pair data of $F_{g'}$, ??) transports the vanishing to $H^1(C_{R'}, F_{g'})$. $\square$

Definition 488 (H^0 base change on the nose). On the vanishing locus, for every homomorphism $R \rightarrow R'$ of commutative k -algebras there is an R' -linear equivalence

$$R' \otimes_R H^0(C_R, F_g) \cong H^0(C_{R'}, F_{g'}), \quad g' = \rho^\sharp(g) :$$

formation of the twisted H^0 commutes on the nose with every change of the test ring — the curve-lite form of Kleiman’s clause (iii) at ring maps.

Proof. Compose three equivalences: the abstract ring-map clause $R' \otimes_R H^0(C_R, F_g) \cong \ker(\delta_g \otimes_R R')$ (??); the kernel transport $\ker(\delta_g \otimes_R R') \cong \ker(\delta_{g'})$ along the twisted δ -naturality square (??, ??); and the inverse of the degree-zero carrier of the pair of $F_{g'}$ (??, ??), $\ker(\delta_{g'}) \cong H^0(C_{R'}, F_{g'})$. $\square$

5.23 Base-field invariance of H^0 , H^1 and the genus

Fix a field k and an object $C \in \text{Over}(\text{Spec } k)$. Section ?? base-changed the two-cover complex along a homomorphism $R \rightarrow R'$ of test rings; this section runs the same mechanism once more with the *absolute* curve as source, for an arbitrary commutative k -algebra R : the two-term Čech complex of an affine two-cover of C base-changes termwise by the free identification ??, and H^1 , being its cokernel, commutes with any base change by right-exactness. Specialised to a field extension $R = K$, this yields $H^i(C, \mathcal{O}_C) \otimes_k K \cong H^i(C_K, \mathcal{O}_{C_K})$ for $i = 0, 1$ and the base-field invariance of the genus. No flatness, semicontinuity or exchange argument is used.

Lemma 489 (Pullback of sections commutes with restriction). *Let R be a commutative k -algebra and $W \leq V$ opens of C . For a section $s \in \Gamma(C, V)$, restricting on C and then pulling back along the first projection $\text{pr}_1 : C_R \rightarrow C$ agrees with pulling back and then restricting on the relative curve:*

$$\text{pr}_1^\sharp(s|_W) = (\text{pr}_1^\sharp s)|_{\text{pr}_1^{-1}W}.$$

Proof. Both sides are the image of s under the map of sections induced by pr_1 from V to $\text{pr}_1^{-1}W \subseteq \text{pr}_1^{-1}V$: pullback along a morphism of schemes is compatible with restriction on source and target. $\square$

Definition 490 (Termwise base change of the two-cover complex of the curve). Let $D = (V_0, V_1)$ be an affine two-cover of C and R a commutative k -algebra. The two terms of the two-cover complex of C base-change freely by ??: an R -linear equivalence

$$R \otimes_k (\Gamma(C, V_0) \times \Gamma(C, V_1)) \cong \Gamma(C_R, V_0^R) \times \Gamma(C_R, V_1^R)$$

on the degree-zero term (product of the two chart identifications; the charts are affine, hence qcqs), and

$$R \otimes_k \Gamma(C, V_0 \cap V_1) \cong \Gamma(C_R, V_0^R \cap V_1^R)$$

on the overlap term (the overlap of affines is qcqs, and $V_0^R \cap V_1^R = \text{pr}_1^{-1}(V_0 \cap V_1)$ by ??), where $V_i^R = \text{pr}_1^{-1}V_i$. On a pure tensor the overlap identification is $a \otimes s \mapsto a \cdot \text{pr}_1^\sharp(s)$.

Theorem 491 (The base-changed difference map is the relative difference map). *Under the two term equivalences of ??, the base change $\text{id}_R \otimes \text{diff}$ of the two-cover restriction-difference map of C (??) is carried to the restriction-difference map of the relative curve C_R with its base-changed cover. In particular the overlap-term equivalence maps $\text{range}(\text{id}_R \otimes \text{diff})$ onto $\text{range}(\text{diff}_R)$.*

Proof. Both sides are R -linear, so it suffices to check on a pure tensor $a \otimes (s_0, s_1)$. The left leg gives $a \cdot \text{pr}_1^\sharp(s_0|_\cap - s_1|_\cap)$. The right leg first sends $a \otimes (s_0, s_1)$ to $(a \cdot \text{pr}_1^\sharp s_0, a \cdot \text{pr}_1^\sharp s_1)$, then takes the difference of restrictions on C_R . By ??, $\text{pr}_1^\sharp(s_i|_\cap) = (\text{pr}_1^\sharp s_i)|_{\cap_R}$; restriction on C_R is R -linear (the R -action is pullback of constants along pr_2 , itself compatible with restriction), so the right leg gives $a \cdot ((\text{pr}_1^\sharp s_0)|_{\cap_R} - (\text{pr}_1^\sharp s_1)|_{\cap_R})$, the same element. The range statement follows since both term equivalences are surjective. $\square$

Theorem 492 (Base change of the two-cover H^1 cokernel of the curve). *For every commutative k -algebra R there is an R -linear equivalence*

$$R \otimes_k \text{H1Cok}(C; V_0, V_1) \cong \text{H1Cok}(C_R; V_0^R, V_1^R),$$

sending $a \otimes [s]$ to $[a \cdot \text{pr}_1^\sharp(s)]$. No flatness of R is needed.

Proof. H1Cok is the cokernel of diff (?), and base change commutes with cokernels (?): $R \otimes_k \text{H1Cok}(C) \cong (R \otimes_k \Gamma(C, V_0 \cap V_1))/\text{range}(\text{id}_R \otimes \text{diff})$. Now transport along the overlap-term equivalence of ??, which by ?? carries the range of $\text{id}_R \otimes \text{diff}$ onto the range of the relative difference map; the induced map of quotients is the displayed equivalence, and its effect on the class of a pure tensor is read off the overlap identification. $\square$

Theorem 493 (Cohomology commutes with base change in degree one, from the absolute curve). *Let D be an affine two-cover of C and R a commutative k -algebra. There is an R -linear equivalence*

$$R \otimes_k H^1(C, \mathcal{O}_C) \cong H^1(C_R, \mathcal{O}_{C_R})$$

on the cohomology carriers.

Proof. Compose: the base change of the absolute cokernel bridge $H^1(C, \mathcal{O}_C) \cong \text{H1Cok}(C)$ (?), tensored with R ; the cokernel base change $R \otimes_k \text{H1Cok}(C) \cong \text{H1Cok}(C_R)$ (?); and the inverse of the relative cokernel bridge $H^1(C_R, \mathcal{O}_{C_R}) \cong \text{H1Cok}(C_R)$ (?). $\square$

Definition 494 (A pinned affine two-cover of the challenge curve). For the challenge curve C (proper, smooth of relative dimension one, geometrically irreducible over k), fix once and for all an affine two-chart cover of C , which exists by ??; all base-change equivalences below are built on this one cover.

Theorem 495 (Cohomology commutes with base change in degree zero). *Let C be the challenge curve and R a commutative k -algebra. There is an R -linear equivalence*

$$R \otimes_k H^0(C, \mathcal{O}_C) \cong H^0(C_R, \mathcal{O}_{C_R}).$$

Proof. Both carriers are global sections (??, over k and over R). The total space $\mathbb{T} \subseteq C$ is quasi-compact (C is proper over a field) and quasi-separated, so the free sections base change ?? applies at $V = \mathbb{T}$: $R \otimes_k \Gamma(C, \mathbb{T}) \cong \Gamma(C_R, \mathbb{T})$. Conjugating by the two degree-zero identifications gives the claim. $\square$

Theorem 496 (H^1 base-field invariance). *Let C be the challenge curve over k and K/k a field extension. Then*

$$H^1(C, \mathcal{O}_C) \otimes_k K \cong H^1(C_K, \mathcal{O}_{C_K})$$

as K -vector spaces, where C_K is the base-changed curve regarded over $\text{Spec } K$ (?).

Proof. Specialise ?? to the algebra $R = K$ and the pinned cover of ??; the relative curve C_K over $\text{Spec } K$ is the base-changed curve of ??. $\square$

Theorem 497 (H^0 base-field invariance). *In the setting of ??, $H^0(C, \mathcal{O}_C) \otimes_k K \cong H^0(C_K, \mathcal{O}_{C_K})$ as K -vector spaces.*

Proof. Specialise ?? to $R = K$. □

Theorem 498 (The scalar identities of base-field invariance). *For the challenge curve C over k and any field extension K/k ,*

$$\dim_K H^1(C_K, \mathcal{O}_{C_K}) = \dim_k H^1(C, \mathcal{O}_C) = \text{genus}(C), \quad \dim_K H^0(C_K, \mathcal{O}_{C_K}) = \dim_k H^0(C, \mathcal{O}_C).$$

Proof. Extension of scalars multiplies no dimensions: $\dim_K(V \otimes_k K) = \dim_k V$ for any k -vector space V . Apply this across the equivalences of ?? and ??; the $\text{genus}(C)$ identification is Definition ?. □

Theorem 499 (Base-field invariance of the genus). *Let C be the challenge curve over k and K/k a field extension. The base-changed curve C_K , a legal challenge curve over K (??), satisfies*

$$\text{genus}(C_K) = \text{genus}(C).$$

Proof. By Definition ?? applied over K , $\text{genus}(C_K) = \dim_K H^1(C_K, \mathcal{O}_{C_K})$, which equals $\text{genus}(C)$ by ?. □

Chapter 6

Closed points, divisors and skyscraper cohomology

This chapter assembles the local and divisorial foundations of the Riemann–Roch ledger of a smooth curve. On an integral scheme X smooth of relative dimension one over a field K , every point other than the generic point is closed, and the stalk of the structure sheaf there is a discrete valuation ring; this yields, on the function field $K(X)$, the order-of-vanishing valuation ord_x at each closed point x . Each closed point also carries a residue degree $[\kappa(x) : K]$, the K -dimension of its residue field. A Weil divisor is then a finitely supported $\mathbb{Z}$ -family on the closed points, and its degree is the residue-degree weighted sum of its coefficients — the correct notion over a non-closed base field. Finally, to each K -module M and point x is attached the skyscraper sheaf of K -modules concentrated at x ; its degree-zero cohomology recovers M and its degree-one cohomology vanishes, the keystone input to the local contribution of a point in the cohomological Riemann–Roch computation.

Throughout, K is a field, X is an integral scheme smooth of relative dimension one over $\text{Spec } K$ — the curve bundle of Chapter ?? — and η denotes its generic point. We write $\mathcal{O}_{X,x}$ for the stalk at x , $\kappa(x)$ for its residue field, and $K(X)$ for the function field (the stalk at η).

6.1 Discrete valuations at closed points

Theorem 500 (Discrete valuation ring at a closed point of an affine Dedekind chart). *Let X be an integral scheme, $V \subseteq X$ an affine open whose ring of sections $\Gamma(X, V)$ is a Dedekind domain, and $x \in V$ a point which is not the generic point of X . Then the stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring.*

Proof. Under the isomorphism $V \cong \text{Spec } \Gamma(X, V)$ the point x corresponds to a prime ideal $\mathfrak{q}$ of $\Gamma(X, V)$, and the stalk $\mathcal{O}_{X,x}$ is the localization of $\Gamma(X, V)$ at $\mathfrak{q}$. The prime $\mathfrak{q}$ is nonzero: if $\mathfrak{q} = 0$ then x is the image of the generic point of $\text{Spec } \Gamma(X, V)$, which is the generic point of the nonempty open V , hence the generic point of the irreducible scheme X — contrary to hypothesis. The localization of a Dedekind domain at a nonzero prime ideal is a discrete valuation ring, which is the claim. $\square$

Theorem 501 (Discrete valuation ring at a closed point of a curve). *Let X be an integral scheme smooth of relative dimension one over a field K , and let $x \in X$ be a point other than the generic point. Then the stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring.*

Proof. By ?? the point x lies in an affine open V whose ring of sections $\Gamma(X, V)$ is a Dedekind domain. Apply ??. $\square$

Theorem 502 (Dedekind stalk at a closed point of a curve). *In the situation of ??, the stalk $\mathcal{O}_{X,x}$ is a Dedekind domain.*

Proof. A discrete valuation ring is a local principal ideal domain, in particular a Dedekind domain; apply this to the stalk, which is a discrete valuation ring by ??. $\square$

Theorem 503 (Non-generic points of a curve are closed). *Let X be an integral scheme smooth of relative dimension one over a field K , and let $x \in X$ be a point other than the generic point. Then the singleton $\{x\}$ is closed.*

Proof. By ?? the specialization order of X has height one: every generalization of any point is the generic point or the point itself. A scheme with irreducible underlying space and height-one specialization order has all its non-generic points closed (?); apply this to x . $\square$

Definition 504 (The order valuation at a closed point). For a closed point x of X (a point other than the generic point), the *order valuation* ord_x is the discrete valuation of the function field $K(X)$ with values in $\mathbb{Z} \cup \{+\infty\}$ determined by the discrete valuation ring $\mathcal{O}_{X,x}$: it assigns to a nonzero rational function its order of vanishing at x , and $+\infty$ to 0.

Proof. The stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring (?), hence a Dedekind domain (??) which is not a field, so its maximal ideal is a nonzero prime, i.e. a height-one prime of $\mathcal{O}_{X,x}$. On an integral scheme the function field $K(X)$ is the fraction field of every stalk, in particular of $\mathcal{O}_{X,x}$; and a height-one prime of a Dedekind domain B determines a discrete valuation on $\text{Frac } B$ — the $\mathfrak{p}$ -adic valuation counting the multiplicity of $\mathfrak{p}$ in the factorization of a fractional ideal. Taking $B = \mathcal{O}_{X,x}$ and $\mathfrak{p}$ its maximal ideal gives the order valuation on $K(X)$. $\square$

6.2 The residue degree

The curve X is a scheme over $\text{Spec } K$; as in Chapter ?? the base field acts on the sections through the structure morphism, and the same discipline supplies the module structure on each residue field.

Definition 505 (The base-field algebra map on a residue field). For a scheme X over $\text{Spec } K$ and a point $x \in X$, the canonical ring homomorphism

$$K \longrightarrow \kappa(x)$$

through the structure morphism: the composite of the global algebra map $K \rightarrow \Gamma(X, \mathcal{O}_X)$ of ??, the germ $\Gamma(X, \mathcal{O}_X) \rightarrow \mathcal{O}_{X,x}$ at x , and the residue map $\mathcal{O}_{X,x} \rightarrow \kappa(x)$. By restriction of scalars along it, the residue field $\kappa(x)$ is a K -module (indeed a K -algebra); as with the module structure on the structure sheaf, this is used as an explicit map rather than a global instance.

Proof. Each of the three maps is a ring homomorphism, and $K \rightarrow \Gamma(X, \mathcal{O}_X)$ is the base algebra map of ??; their composite is the stated ring homomorphism. $\square$

Definition 506 (The residue degree). For a scheme X over $\text{Spec } K$ and a point $x \in X$, the *residue degree* at x is

$$\deg_K(x) = [\kappa(x) : K] = \dim_K \kappa(x),$$

the K -dimension of the residue field, taken with the K -module structure of ??.

6.3 Weil divisors and their degree

A Weil divisor on the curve records a finite prescription of orders at closed points. Its degree must weight each point by its residue degree over the base field: this is the correct notion when K is not algebraically closed. For example, on $\mathbb{P}_{\mathbb{R}}^1$ the rational function t^2+1 has a single closed zero x , whose residue field is $\mathbb{C}$, and a double pole at ∞ , with residue field $\mathbb{R}$; the unweighted point count $1-2 = -1$ is nonzero, whereas the residue-degree weighted degree $1 \cdot [\mathbb{C} : \mathbb{R}] - 2 \cdot [\mathbb{R} : \mathbb{R}] = 2-2 = 0$ vanishes, as the degree of a principal divisor must.

Definition 507 (Weil divisors on a curve). For an integral scheme X with generic point η , a *Weil divisor* is a finitely supported $\mathbb{Z}$ -valued function on the non-generic points, i.e. an element of the free abelian group

$$\mathrm{Div}(X) = (\{x \in X : x \neq \eta\} \rightarrow_{\mathrm{fin}} \mathbb{Z})$$

on the set of non-generic points. It carries the pointwise abelian-group structure and the pointwise partial order, both independent of any base field. For the curve X the non-generic points are exactly the closed points (??), so a Weil divisor is a finite formal $\mathbb{Z}$ -combination of closed points.

Definition 508 (The degree of a Weil divisor). For a scheme X over $\mathrm{Spec} K$ and a Weil divisor $D \in \mathrm{Div}(X)$, the *degree* of D , weighted by the residue degrees over the base field, is

$$\deg_K D = \sum_x D_x \cdot [\kappa(x) : K] \in \mathbb{Z},$$

the finite sum over the support of D of the coefficients D_x times the residue degrees $[\kappa(x) : K]$ of ??.

Theorem 509 (The degree is a group homomorphism). *The degree map $\deg_K : \mathrm{Div}(X) \rightarrow \mathbb{Z}$ is a homomorphism of abelian groups: it satisfies $\deg_K 0 = 0$, $\deg_K(D + D') = \deg_K D + \deg_K D'$ and $\deg_K(-D) = -\deg_K D$.*

Proof. The degree is the finitely supported sum $D \mapsto \sum_x D_x \cdot [\kappa(x) : K]$. The zero divisor has empty support, so $\deg_K 0 = 0$. For additivity, the summand $n \mapsto n \cdot [\kappa(x) : K]$ is, at each point x , additive in the coefficient n and vanishes at $n = 0$; hence the finitely supported sum of the pointwise sums $(D + D')_x = D_x + D'_x$ splits as $\sum_x D_x [\kappa(x) : K] + \sum_x D'_x [\kappa(x) : K]$, giving $\deg_K(D + D') = \deg_K D + \deg_K D'$. Finally, applying additivity to D and $-D$ and using $\deg_K 0 = 0$ yields $\deg_K(-D) = -\deg_K D$. $\square$

Lemma 510 (Degree of a single-point divisor). *For a closed point x and an integer n , the degree of the single-point divisor $n \cdot x$ is*

$$\deg_K(n \cdot x) = n \cdot [\kappa(x) : K].$$

Proof. The divisor $n \cdot x$ is supported at the single point x with coefficient n , so the defining sum of ?? has the single term $n \cdot [\kappa(x) : K]$ (the summand vanishes at coefficient 0, so no other point contributes). $\square$

6.4 Skyscraper sheaves and the vanishing of H^1

Fix a scheme X over $\mathrm{Spec} K$ and work on its small Zariski site (the poset of opens with open coverings), whose terminal object is the whole space $\top = X$. For a point $x \in X$ and a K -module M the skyscraper sheaf concentrates M at x .

Definition 511 (The skyscraper sheaf of K -modules). For $x \in X$ and a K -module M , the *skyscraper sheaf* $\text{sky}_x(M)$ of K -modules on the small Zariski site of X is the sheaf whose sections over an open U are

$$\text{sky}_x(M)(U) = \begin{cases} M & x \in U, \\ 0 & x \notin U, \end{cases}$$

with restriction between two opens containing x the identity of M , and to an open avoiding x the zero map. Its restriction map between any two opens that both contain x is an isomorphism, and its sections over any open avoiding x are the zero module.

Definition 512 (Degree-zero cohomology of a skyscraper is its value). For $x \in X$ and a K -module M , there is a K -linear equivalence

$$H^0(X, \text{sky}_x(M)) \cong M.$$

Proof. Global sections live over the terminal open $\mathbb{T} = X$, which contains x , so $\text{sky}_x(M)(\mathbb{T}) = M$ by ???. Compose with the identification of degree-zero cohomology with global sections, $H^0(F) \cong F(\mathbb{T})$ of ??, applied to $F = \text{sky}_x(M)$. $\square$

Theorem 513 (Global lifting along the cokernel of a skyscraper embedding). *Let $x \in X$, let M be a K -module, and let $\iota : \text{sky}_x(M) \rightarrow G$ be a monomorphism of sheaves of K -modules. Then every global section of $\text{coker } \iota$ lifts to a global section of G : the map $G(\mathbb{T}) \rightarrow (\text{coker } \iota)(\mathbb{T})$ is surjective.*

Proof. Write $\pi : G \rightarrow \text{coker } \iota$ for the projection and let $q \in (\text{coker } \iota)(\mathbb{T})$. The proof glues corrected local lifts, the correction being an *elementwise* coboundary supplied by flasqueness of the skyscraper — so no quasi-compactness or finite subcover is needed. Restriction maps of a sheaf F are written $t|_W$.

(a) *Local lifts.* As an epimorphism of sheaves, π is locally surjective on sections: for each point $p \in X$ there is an open $W_p \ni p$ and a section $s_p \in G(W_p)$ with $\pi(s_p) = q|_{W_p}$. The opens W_p cover X , since each contains its point p .

(b) *The differences come from the skyscraper.* On $W_p \cap W_{p'}$, the difference $s_p|_{W_p \cap W_{p'}} - s_{p'}|_{W_p \cap W_{p'}}$ maps to $q|_{W_p \cap W_{p'}} - q|_{W_p \cap W_{p'}} = 0$ under π . Since ι is a monomorphism it is the kernel of its cokernel on sections (??), so there is a unique $a_{pp'} \in \text{sky}_x(M)(W_p \cap W_{p'})$ with $\iota(a_{pp'}) = s_p - s_{p'}$ there.

(c) *Cocycle identity.* On a triple overlap $W_p \cap W_{p'} \cap W_l$,

$$\iota(a_{pl}) = s_p - s_l = (s_p - s_{p'}) + (s_{p'} - s_l) = \iota(a_{pp'} + a_{p'l}),$$

and ι is injective on sections (??), so $a_{pl} = a_{pp'} + a_{p'l}$ there.

(d) *Pivot on the support point.* Because $\text{sky}_x(M)$ is a skyscraper sheaf it is *flasque*: every restriction map of $\text{sky}_x(M)$ is surjective. In particular the restriction $\text{sky}_x(M)(W_p) \rightarrow \text{sky}_x(M)(W_p \cap W_x)$ is surjective, so choose $b_p \in \text{sky}_x(M)(W_p)$ with $b_p|_{W_p \cap W_x} = a_{px}$, where the index x -th cover member $W_x \ni x$ is the neighbourhood of the support point itself.

(e) *Restriction to W_x is injective.* For every open A , the restriction $\text{sky}_x(M)(A) \rightarrow \text{sky}_x(M)(A \cap W_x)$ is injective: if $x \in A$ then both opens contain x and the restriction is an isomorphism (??); if $x \notin A$ then $\text{sky}_x(M)(A) = 0$ is trivial, so any map out of it is injective.

(f) *The cocycle is the coboundary of b .* We claim $a_{pp'} = b_p|_{W_p \cap W_{p'}} - b_{p'}|_{W_p \cap W_{p'}}$. By (e) it suffices to check this after restriction to $(W_p \cap W_{p'}) \cap W_x$. There the cocycle identity (c) with $l = x$ gives $a_{pp'} = a_{px} - a_{p'x}$, and $a_{px} = b_p$, $a_{p'x} = b_{p'}$ after restriction to $\cap W_x$ by the choice in (d); hence $a_{pp'} = b_p - b_{p'}$ restricted to $(W_p \cap W_{p'}) \cap W_x$, as required.

(g) *Corrected local sections glue.* Put $t_p = s_p - \iota(b_p) \in G(W_p)$. On $W_p \cap W_{p'}$,

$$t_p - t_{p'} = (s_p - s_{p'}) - \iota(b_p - b_{p'}) = \iota(a_{pp'}) - \iota(a_{pp'}) = 0,$$

using (b) and (f). So the t_p agree on overlaps and, by the sheaf condition, glue to a global section $s \in G(\mathbb{T})$ with $s|_{W_p} = t_p$.

(h) *The glued section lifts q .* On each W_p , since $\pi \circ \iota = 0$,

$$\pi(s)|_{W_p} = \pi(t_p) = \pi(s_p) - \pi(\iota(b_p)) = q|_{W_p}.$$

Sections of the sheaf $\text{coker } \iota$ that agree on the cover $\{W_p\}$ are equal, so $\pi(s) = q$. Thus s is the required lift. $\square$

Theorem 514 (Degree-one cohomology of a skyscraper vanishes). *For every point $x \in X$ and every K -module M ,*

$$H^1(X, \text{sky}_x(M)) = 0.$$

Proof. The category of sheaves of K -modules on the small Zariski site of X is Grothendieck abelian, hence has enough injectives: embed $\text{sky}_x(M)$ into an injective sheaf by a monomorphism $\iota : \text{sky}_x(M) \rightarrow G$. By the Ext^1 -vanishing criterion ?? (with $L = K_J$ the constant sheaf), it suffices to lift every morphism $K_J \rightarrow \text{coker } \iota$ along $\pi : G \rightarrow \text{coker } \iota$. By ?? and its naturality ??, morphisms out of K_J are global sections, compatibly with postcomposition; so the required lifting is exactly the surjectivity of $G(\mathbb{T}) \rightarrow (\text{coker } \iota)(\mathbb{T})$, which is ??. $\square$

6.5 Finiteness of the residue degree

Over an arbitrary base field the residue field at a closed point of the curve is a finite extension of K ; this finiteness is the arithmetic gate through which every point contributes a finite local term to the ledger.

Theorem 515 (Finiteness of the residue degree). *Let X be an integral scheme smooth of relative dimension one and locally of finite type over a field K , and let x be a non-generic (hence closed) point. Then the residue field $\kappa(x)$ is a finite K -module; equivalently the residue degree $[\kappa(x) : K]$ is finite.*

Proof. By ?? the point x lies in an affine open V whose ring of sections $B = \Gamma(X, V)$ is a Dedekind domain. Under $V \cong \text{Spec } B$ the point x corresponds to a prime $\mathfrak{q}$ of B , which is nonzero — otherwise x would be the generic point, as in ?? — and hence maximal, so $B/\mathfrak{q}$ is a field. The stalk $\mathcal{O}_{X,x}$ is the localization $B_{\mathfrak{q}}$, whose residue field is canonically $B/\mathfrak{q}$; this identification is K -linear for the residue K -algebra structure of ??, so $\kappa(x) \cong B/\mathfrak{q}$ as K -modules. Now $K \rightarrow B$ is of finite type (X is locally of finite type over K , restricted to the chart V), hence so is the quotient $K \rightarrow B/\mathfrak{q}$; a finite-type algebra over a field which is itself a field is a finite extension (Zariski's lemma, valid because a field is a Jacobson ring). Therefore $B/\mathfrak{q} \cong \kappa(x)$ is finite over K . $\square$

Theorem 516 (Positivity of the residue degree). *In the situation of ??, the residue degree is positive: $0 < [\kappa(x) : K]$.*

Proof. By ?? the residue field $\kappa(x)$ is a finite K -module; being a field extension of K it is nonzero and free over K . A nonzero finite free module over a field has strictly positive rank, so $[\kappa(x) : K] = \dim_K \kappa(x) > 0$. $\square$

6.6 Principal divisors

A nonzero rational function on the curve has a principal divisor recording its zeros and poles with multiplicities. The construction rests on the $\mathbb{Z}$ -valued order at each closed point and on the finiteness of the set of points where a given function has a zero or a pole.

Definition 517 (The order homomorphism on function-field units). For a closed point x , the *order homomorphism*

$$\text{ord}_x : K(X)^\times \longrightarrow \mathbb{Z}$$

sends a nonzero rational function to its order of vanishing at x (??): positive at a zero, negative at a pole, with a uniformizer of the discrete valuation ring $\mathcal{O}_{X,x}$ having order $+1$. It is a homomorphism from the multiplicative group $K(X)^\times$ to the additive group $\mathbb{Z}$.

Proof. The order valuation of ?? is a valuation of $K(X)$, hence multiplicative on nonzero elements; restricting it to units and reading its value in $\mathbb{Z}$ with the classical sign (a uniformizer, whose valuation generates the value group, is assigned $+1$) yields a group homomorphism $K(X)^\times \rightarrow \mathbb{Z}$. $\square$

Lemma 518 (Trivial order where a section is invertible). *Let $s \in \Gamma(X, U)$ represent the rational function $g = \text{germ}_\eta(s)$ at the generic point (with $\eta \in U$), and let x be a closed point of the basic open set where s is invertible, i.e. the germ of s at x is a unit of $\mathcal{O}_{X,x}$. Then $\text{ord}_x g = 0$.*

Proof. Since η specializes to x , the function g is the image of the germ $\text{germ}_x(s)$ under the localization map $\mathcal{O}_{X,x} \rightarrow K(X)$. By hypothesis this germ is a unit of the discrete valuation ring $\mathcal{O}_{X,x}$ (??), and a unit of the valuation ring has order of vanishing 0; hence $\text{ord}_x g = 0$. $\square$

Theorem 519 (Finiteness of the order support). *For $g \in K(X)^\times$, the set of closed points at which g has nonzero order, $\{x : \text{ord}_x g \neq 0\}$, is finite. (Here X is additionally assumed quasi-compact over K , so that its underlying space is Noetherian.)*

Proof. Represent g by a section s on a nonempty open $U \ni \eta$ (the generic point lies in every nonempty open). Then g is a unit at η , so η lies in the basic open set $D(s)$ where s is invertible. At every closed point of $D(s)$ the order of g is 0 (??), so the locus where $\text{ord}_x g \neq 0$ is contained in the complement $X \setminus D(s)$, a closed set not containing η . On the curve — an irreducible Noetherian space whose specialization order has height one (??) — a proper closed set avoiding the generic point is finite (??). Quasi-compactness over K supplies the Noetherian hypothesis: finite type alone gives only a locally Noetherian space. $\square$

Definition 520 (The principal divisor of a rational function). For $g \in K(X)^\times$, the *principal divisor*

$$\text{div}(g) = \sum_x \text{ord}_x(g) x$$

is the Weil divisor whose coefficient at a closed point x is the order of vanishing $\text{ord}_x(g)$: positive at zeros, negative at poles. It is finitely supported (??), hence a well-defined element of $\text{Div}(X)$.

Theorem 521 (The principal divisor is a homomorphism). *The principal divisor is a homomorphism from $K(X)^\times$ to $\text{Div}(X)$: for all $g, g' \in K(X)^\times$,*

$$\text{div}(gg') = \text{div}(g) + \text{div}(g'), \quad \text{div}(1) = 0.$$

Proof. At each closed point x the order ord_x is a homomorphism $K(X)^\times \rightarrow \mathbb{Z}$ (??), so $\text{ord}_x(gg') = \text{ord}_x g + \text{ord}_x g'$ and $\text{ord}_x(1) = 0$. Since the coefficients of $\text{div}(g)$ are these orders (??), the two identities hold coefficientwise, hence as divisors. $\square$

6.7 The divisor sheaf of a Weil divisor

To a Weil divisor D is attached the sheaf $\mathcal{O}(D)$ of rational functions with poles bounded by D . Its sections over an open U are the classical Riemann–Roch space of D restricted to U ; the sheaf structure is available because on the irreducible curve a compatible family of such functions shares a single underlying rational function.

Definition 522 (The K -structure on the function field). For X over $\text{Spec } K$, the function field $K(X)$ carries a canonical K -algebra structure through the structure morphism: the composite

$$K \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow K(X)$$

of the global algebra map ?? with the germ at the generic point. By restriction of scalars along it, $K(X)$ is a K -module (as with the residue field, this is used as an explicit map rather than a global instance).

Lemma 523 (Integral elements have no pole). *The image in $K(X)$ of any element of the stalk $\mathcal{O}_{X,x}$ at a closed point x has no pole at x : its order of vanishing is ≥ 0 .*

Proof. The stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring (??), and it is exactly the valuation ring of ord_x — the order valuation is the adic valuation of its maximal ideal (??). Every element of a valuation ring has nonnegative order, so the image of any stalk element has $\text{ord}_x \geq 0$. $\square$

Lemma 524 (Constants have no pole). *The image in $K(X)$ of a constant $r \in K$ has no pole at any closed point x : $\text{ord}_x(r) \geq 0$.*

Proof. The image of r is the germ at the generic point of the global section $r \cdot 1$ (??); its germ at x realises it as an element of the stalk $\mathcal{O}_{X,x}$, so it has no pole at x by ??. $\square$

Definition 525 (Sections of the divisor sheaf). For a Weil divisor D and a nonempty open U , the sections of $\mathcal{O}(D)$ over U are

$$\mathcal{O}(D)(U) = \{g \in K(X) : \text{at every closed point } x \in U, g \text{ has pole order } \leq D_x\},$$

the rational functions with poles bounded by D on U ; over the empty open the sections are the zero module (forced by the sheaf condition, since the empty sieve covers $\perp$). This is a K -submodule of $K(X)$: it contains 0; it is closed under addition because a valuation is subadditive, $\text{ord}_x(g_1 + g_2) \geq \min(\text{ord}_x g_1, \text{ord}_x g_2)$, so the pole bound is preserved; and closed under K -scaling because constants have no poles (??).

Lemma 526 (Nonempty opens of an irreducible space meet). *Any two nonempty open subsets of the irreducible space X have nonempty intersection.*

Proof. The underlying space of the integral scheme X is irreducible, so no two nonempty opens are disjoint. $\square$

Definition 527 (The divisor sheaf $\mathcal{O}(D)$). The presheaf $U \mapsto \mathcal{O}(D)(U)$ of ?? is a sheaf of K -modules on the small Zariski site of X , denoted $\mathcal{O}(D)$.

Proof. Restriction between nonempty opens is the inclusion of subspaces of $K(X)$, so a section is determined by its underlying rational function. Given a compatible family over a cover of U : any two members with nonempty domain agree as rational functions, since their domains meet (??) and they restrict equally on the overlap; hence the family shares a single $g \in K(X)$. Every closed point of U lies in some cover member, where g respects the bound D ; so $g \in \mathcal{O}(D)(U)$ and restricts to the given sections. Over the empty open all sections are 0. Uniqueness is again the determinacy of a section by its underlying rational function. $\square$

Theorem 528 (Monotone inclusion of divisor sheaves). *For $D \leq D'$ the identity on rational functions gives a monomorphism of sheaves $\mathcal{O}(D) \hookrightarrow \mathcal{O}(D')$.*

Proof. For $D \leq D'$ one has $D_x \leq D'_x$ at every closed point, so a pole of order $\leq D_x$ is a fortiori a pole of order $\leq D'_x$; hence $\mathcal{O}(D)(U) \subseteq \mathcal{O}(D')(U)$ as submodules of $K(X)$ for every U . These section-wise inclusions are injective and natural in U , so they assemble into a morphism of sheaves $\mathcal{O}(D) \hookrightarrow \mathcal{O}(D')$ which, being injective on every section, is a monomorphism. $\square$

Lemma 529 (No pole means integral). *If a rational function g has no pole at a closed point x (order of vanishing ≥ 0), then g is integral at x : it lies in the image of the localization map $\mathcal{O}_{X,x} \rightarrow K(X)$.*

Proof. The stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring (??), a Dedekind domain (??) with a single nonzero prime — its maximal ideal — whose adic valuation is ord_x (??). Since $K(X)$ is the fraction field of $\mathcal{O}_{X,x}$ and there is only this one valuation to test, the single condition $\text{ord}_x g \geq 0$ places g in the valuation ring $\mathcal{O}_{X,x}$. $\square$

6.8 Comparison of $\mathcal{O}(0)$ with the structure sheaf

The divisor sheaf of the zero divisor is the structure sheaf: a rational function with no pole anywhere is a global regular section, and conversely. This identification transports the Euler characteristic of $\mathcal{O}(0)$ to that of $\mathcal{O}_X$.

Lemma 530 (The germ/stalk seam). *For a section $s \in \Gamma(X, U)$ over an open containing both the generic point η and a point x , the germ at η equals the image of the germ at x under the localization map $\mathcal{O}_{X,x} \rightarrow K(X)$.*

Proof. The generic point specializes to x , and the germ maps of the structure sheaf are compatible with the specialization map between stalks; applying this compatibility to s gives the stated identity. $\square$

Lemma 531 (Regular germs have poles bounded by zero). *For a section $s \in \Gamma(X, U)$ over a nonempty open U , its germ at the generic point $\text{germ}_\eta(s) \in K(X)$ has poles bounded by the zero divisor on U , i.e. lies in $\mathcal{O}(0)(U)$.*

Proof. At every closed point $x \in U$, $\text{germ}_\eta(s)$ is the image of the germ $\text{germ}_x(s) \in \mathcal{O}_{X,x}$ (??), hence has no pole at x (??). Thus it satisfies the zero-divisor bound at every point of U . $\square$

Lemma 532 (Local regular representatives). *A rational function $g \in \mathcal{O}(0)(U)$ is, near every point $x \in U$, the germ at η of a regular section over a neighbourhood of x inside U .*

Proof. At $x = \eta$, the function g is directly represented by a regular section on some neighbourhood. At a closed point x , the zero-divisor bound says g has no pole at x , so g is the image of some $y \in \mathcal{O}_{X,x}$ (??). Represent y by a regular section s on a neighbourhood $W \ni x$ inside U ; then $\text{germ}_\eta(s)$ is the image of $\text{germ}_x(s) = y$, which is g (??). $\square$

Lemma 533 (Global representability). *Every $g \in \mathcal{O}(0)(U)$ over a nonempty open U is the germ at η of a section $s \in \Gamma(X, U)$.*

Proof. Cover U by the neighbourhoods W of ?? carrying regular sections s_W with $\text{germ}_\eta(s_W) = g$. On overlaps the s_W have equal germ at η (which lies in every nonempty overlap of the irreducible space), and since X is integral a section is determined by its germ at η , so the s_W agree there. They therefore glue to a section $s \in \Gamma(X, U)$ with $\text{germ}_\eta(s) = g$. $\square$

Theorem 534 ($\mathcal{O}(0) \cong \mathcal{O}_X$). *The divisor sheaf of the zero divisor is the structure sheaf of X as a sheaf of K -modules:*

$$\mathcal{O}(0) \cong \mathcal{O}_X.$$

Proof. The section-wise map $\Gamma(X, U) \rightarrow \mathcal{O}(0)(U)$, $s \mapsto \text{germ}_\eta(s)$, is well defined (??) and K -linear. On a nonempty open it is injective, because X is integral so a section is determined by its germ at η , and surjective by ??; over the empty open both sides are the zero module. Being bijective on every open, these maps assemble into an isomorphism of sheaves of K -modules. This is the identification through which $\chi(\mathcal{O}(0))$ equals $\chi(\mathcal{O}_X)$. $\square$

6.9 The dévissage complex at a closed point

Removing a single closed point from a divisor changes the divisor sheaf by a skyscraper at that point. The dévissage short complex makes this precise: it relates $\mathcal{O}(D - x)$, $\mathcal{O}(D)$ and a skyscraper sheaf built from the one-point jump module, the associated-graded piece of the pole-order filtration at x .

Definition 535 (The one-point lattice). For a closed point x and $n \in \mathbb{Z}$, the *one-point lattice* L_n is the K -submodule of $K(X)$ of rational functions whose pole order at x is at most n (equivalently $\text{ord}_x g \geq -n$). It is a submodule by the same argument as ?? — the valuation is subadditive and K -multiples create no poles (??) — and $L_m \subseteq L_n$ whenever $m \leq n$.

Definition 536 (The jump module). The *jump module* at x is the quotient

$$J = L_{D_x} / L_{D_x - 1},$$

the K -space of functions with pole order $\leq D_x$ at x modulo those with pole order $\leq D_x - 1$ — the associated-graded piece of the pole-order filtration at the single point x .

Definition 537 (The projection to the jump module). For an open $U \ni x$, the *jump projection* $\text{proj} : \mathcal{O}(D)(U) \rightarrow J$, $g \mapsto [g]$. It is well defined and K -linear: a section of $\mathcal{O}(D)$ over U has pole order $\leq D_x$ at the point $x \in U$, so it lies in L_{D_x} and determines a class in J ; the class depends only on the underlying rational function.

Definition 538 (The divisor minus a point). For a closed point x , the divisor $D - x$ is obtained by decreasing the coefficient of D at x by one and leaving every other coefficient unchanged. Thus $(D - x)_x = D_x - 1$ and $D - x \leq D$.

Definition 539 (The skyscraper projection π). The morphism of sheaves $\pi : \mathcal{O}(D) \rightarrow \text{sky}_x(J)$ sending a section g over an open $U \ni x$ to its jump class $[g] \in J$, and every section over an open avoiding x to 0. Naturality holds because the jump projection depends only on the underlying rational function, and the restriction of the skyscraper between two opens containing x is the identity of J .

Lemma 540 (The dévissage complex vanishes). *The composite $\mathcal{O}(D - x) \hookrightarrow \mathcal{O}(D) \xrightarrow{\pi} \text{sky}_x(J)$ is zero.*

Proof. Over an open $U \ni x$, a section of $\mathcal{O}(D - x)$ has pole order $\leq (D - x)_x = D_x - 1$ at x , so it lies in $L_{D_x - 1}$ and its jump class in $J = L_{D_x} / L_{D_x - 1}$ vanishes. Over an open avoiding x the target $\text{sky}_x(J)$ has zero sections. Hence the composite is zero on every open. $\square$

Definition 541 (The dévissage short complex). The *dévissage short complex*

$$0 \longrightarrow \mathcal{O}(D - x) \longrightarrow \mathcal{O}(D) \xrightarrow{\pi} \text{sky}_x(J) \longrightarrow 0$$

of sheaves of K -modules: the left map is the inclusion ?? for $D - x \leq D$ and the right map is π ; their composite vanishes (??). Its left leg is a monomorphism.

Proof. The left leg is the inclusion of ??, which is a monomorphism of sheaves. $\square$

6.10 Short exactness of the dévissage sequence

The dévissage short complex is short exact. The monomorphism is the inclusion of ??; the epimorphism is local surjectivity of π , which at the support point excises a finite bad locus to lift a jump class; middle exactness is checked on sections and transported to the sheaf level by reflection of limits.

Lemma 542 (Local lifting of a jump class). *Let $x \in U$ and let t be a section of $\text{sky}_x(J)$ over U , i.e. a class in J . There is an open W with $x \in W \subseteq U$ and a section of $\mathcal{O}(D)$ over W whose jump class is $t|_W$.*

Proof. Choose a representative $g \in L_{D_x}$ of the class t (the quotient $L_{D_x} \twoheadrightarrow J$ is surjective). If $t = 0$, the zero section over U itself has jump class 0. Otherwise $g \neq 0$, so $g \in K(X)^\times$. The closed points z where g violates the D -bound (pole order at z exceeding D_z) form a finite set: outside the support of $\text{div}(g)$ and the support of D , the function g has neither zero nor pole and $D_z = 0$, so the bound holds; both supports are finite (?? and finiteness of a divisor's support). This bad set is a finite union of closed points (??), hence closed, and does not contain x , where $g \in L_{D_x}$ already respects the bound. Let $W = U \setminus (\text{bad set})$; then $x \in W \subseteq U$ and g satisfies the D -bound at every closed point of W , so $g \in \mathcal{O}(D)(W)$, and by construction its jump class is $t|_W$. $\square$

Theorem 543 (The projection is an epimorphism). *The dévissage projection $\pi : \mathcal{O}(D) \rightarrow \text{sky}_x(J)$ is an epimorphism of sheaves.*

Proof. It suffices that π is locally surjective. Over an open avoiding x the target skyscraper sections are zero, so the zero section lifts any section there; over a neighbourhood of x every jump section lifts by ??. A locally surjective morphism of sheaves of K -modules is an epimorphism. $\square$

Lemma 544 (Sectionwise middle exactness). *Over every open W , the sequence of sections $\mathcal{O}(D - x)(W) \rightarrow \mathcal{O}(D)(W) \xrightarrow{\pi_W} \text{sky}_x(J)(W)$ is exact in K -modules.*

Proof. Let s be a section of $\mathcal{O}(D)$ over W with $\pi_W(s) = 0$. If $x \in W$: the vanishing of $\pi_W(s)$ means the jump class of s is zero, i.e. s has pole order $\leq D_x - 1$ at x ; together with the D -bound at the other points — where D and $D - x$ have equal coefficients — this says $s \in \mathcal{O}(D - x)(W)$. If $x \notin W$: the divisors D and $D - x$ agree on W , so $\mathcal{O}(D)(W) = \mathcal{O}(D - x)(W)$ and s already lies in the subspace. In either case s is the image of a $(D - x)$ -section under the inclusion, which is the kernel of π_W . $\square$

Theorem 545 (Middle exactness of the dévissage complex). *The dévissage short complex is exact at $\mathcal{O}(D)$.*

Proof. Over each open W the sections complex is exact (??) and its left leg is injective (??), so the left map realises $\mathcal{O}(D-x)(W)$ as the kernel of π_W in K -modules. The forgetful functor from sheaves to presheaves creates limits, and evaluation at the opens jointly reflects them; hence these section-wise kernel forks assemble into a kernel fork of sheaves, exhibiting the left map as the kernel of π . This is exactness in the middle. $\square$

Theorem 546 (The dévissage sequence is short exact). *The dévissage sequence*

$$0 \longrightarrow \mathcal{O}(D-x) \longrightarrow \mathcal{O}(D) \xrightarrow{\pi} \text{sky}_x(J) \longrightarrow 0$$

is a short exact sequence of sheaves of K -modules.

Proof. The left leg is a monomorphism (??), the right leg π is an epimorphism (??), and the complex is exact in the middle (??); these are exactly the three certificates of a short exact sequence. $\square$

6.11 The jump module and the residue field

The jump module at a closed point is the residue field, viewed as a K -space; its dimension is therefore the residue degree. This is the local input identifying the degree-zero cohomology of the skyscraper term with the residue degree of the point.

Lemma 547 (A uniformizer at a closed point). *There is a rational function $t \in K(X)$ with $\text{ord}_x t = 1$: a uniformizer of the discrete valuation ring $\mathcal{O}_{X,x}$ (??), a function with a simple zero at x .*

Proof. The stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring, so its maximal ideal has a generator t , a uniformizer; the adic valuation ord_x (??) sends the generator to 1, so $\text{ord}_x t = 1$. $\square$

Theorem 548 (The shift isomorphism). *Let t be a uniformizer at x . Multiplication by t^a is a K -linear isomorphism $L_a \xrightarrow{\sim} L_0$, carrying L_{a-1} onto L_{-1} .*

Proof. Since $\text{ord}_x(t^a g) = a + \text{ord}_x g$, a function g has pole order $\leq a$ at x iff $t^a g$ has pole order ≤ 0 ; so multiplication by t^a maps L_a into L_0 (and L_{a-1} into L_{-1}), with two-sided inverse multiplication by t^{-a} , hence is a bijection. It is K -linear because the K -action on $K(X)$ is multiplication in the field, which commutes with multiplication by t^a . $\square$

Theorem 549 (The base map to the residue field). *The residue map induces a K -linear surjection $L_0 \twoheadrightarrow \kappa(x)$ with kernel L_{-1} .*

Proof. A function $g \in L_0$ has no pole at x , hence is the image of a unique element of the stalk $\mathcal{O}_{X,x}$ (??); send g to the residue of that element in $\kappa(x)$. This is K -linear for the residue K -structure of ?? (using injectivity of $\mathcal{O}_{X,x} \rightarrow K(X)$) and surjective, because the residue map $\mathcal{O}_{X,x} \rightarrow \kappa(x)$ is (??). Its kernel is exactly L_{-1} : the residue vanishes iff the stalk preimage lies in the maximal ideal, i.e. has strictly positive order of vanishing, i.e. g has pole order ≤ -1 — a zero — at x . $\square$

Theorem 550 (The jump module is the residue field). *There is a K -linear isomorphism*

$$J = L_{D_x} / L_{D_x-1} \cong \kappa(x).$$

Proof. Compose the shift isomorphism $L_{D_x} \xrightarrow{\sim} L_0$ (??, multiplication by t^{D_x}) with the residue surjection $L_0 \twoheadrightarrow \kappa(x)$ (??). The composite $L_{D_x} \rightarrow \kappa(x)$ is surjective, and its kernel is the preimage of L_{-1} under the shift, namely L_{D_x-1} (the shift carries L_{D_x-1} onto L_{-1}). By the first isomorphism theorem, $J = L_{D_x}/L_{D_x-1} \cong \kappa(x)$. $\square$

Theorem 551 (Dimension of the jump module). *The K -dimension of the jump module is the residue degree:*

$$\dim_K J = [\kappa(x) : K] = \deg_K(x).$$

Proof. Immediate from the isomorphism $J \cong \kappa(x)$ (??) and the definition of the residue degree as $\dim_K \kappa(x)$ (??). $\square$

Corollary 552 (The jump module is finite). *The jump module J is a finite K -module.*

Proof. By ??, $J \cong \kappa(x)$, which is finite over K (??). $\square$

6.12 The multiplication isomorphism

Multiplication by a nonzero rational function g identifies $\mathcal{O}(D)$ with $\mathcal{O}(D - \operatorname{div} g)$. This is the engine forcing the degree of a principal divisor to vanish: the two sheaves being isomorphic, their Euler characteristics agree.

Theorem 553 (The bound-shift). *Let $g \in K(X)^\times$ and $h \in K(X)$. For any open U , the product gh has poles bounded by $D - \operatorname{div}(g)$ on U if and only if h has poles bounded by D on U .*

Proof. At each closed point $z \in U$, the valuation is multiplicative, $\operatorname{ord}_z(gh) = \operatorname{ord}_z g + \operatorname{ord}_z h$, and $(D - \operatorname{div} g)_z = D_z - \operatorname{ord}_z g$ since $\operatorname{div}(g)_z = \operatorname{ord}_z g$ (??). Hence the bound $\operatorname{ord}_z(gh) \geq -(D - \operatorname{div} g)_z$ reads $\operatorname{ord}_z g + \operatorname{ord}_z h \geq -D_z + \operatorname{ord}_z g$, i.e. $\operatorname{ord}_z h \geq -D_z$: the finite value $\operatorname{ord}_z g$ cancels, leaving exactly the D -bound for h . As this equivalence holds at every closed point of U , it holds for the whole section (??). $\square$

Theorem 554 (Multiplication isomorphism of divisor sheaves). *For $g \in K(X)^\times$ and any Weil divisor D , multiplication by g is an isomorphism of sheaves of K -modules*

$$\mathcal{O}(D) \cong \mathcal{O}(D - \operatorname{div} g).$$

Proof. Multiplication by the unit g is a K -linear automorphism of $K(X)$, with inverse multiplication by g^{-1} . By the bound-shift ?? it restricts, on every nonempty open U , to a bijection $\mathcal{O}(D)(U) \rightarrow \mathcal{O}(D - \operatorname{div} g)(U)$ (the inverse being restriction of multiplication by g^{-1}), and both sides are the zero module over the empty open. Being bijective on every open, these maps assemble into an isomorphism of sheaves. Together with the additivity of div this is the engine of $\deg(\operatorname{div} g) = 0$: it identifies $\mathcal{O}(D)$ with $\mathcal{O}(D - \operatorname{div} g)$, forcing their Euler characteristics to agree. $\square$

6.13 The cohomological ledger: h^0 , h^1 and χ

The Euler characteristic of the Riemann–Roch ledger is a numerical invariant of a sheaf of modules. On an arbitrary small site $(\mathcal{C}, J)$ with coefficients in a commutative ring R , any sheaf F of R -modules has cohomology $H^n(F) = \operatorname{Ext}^n(R_J, F)$ (Definition ??); this section attaches to F its dimension counts in degrees 0 and 1 and their alternating combination, and records their invariance under sheaf isomorphism. Nothing here is scheme-specific — the curve enters only later, where all higher cohomology vanishes and the truncation below is the honest Euler characteristic.

Definition 555 (Transport of cohomology along a sheaf isomorphism). Let F, G be sheaves of R -modules on $(\mathcal{C}, J)$ and $e : F \cong G$ an isomorphism of sheaves. For every n , e induces an R -linear equivalence

$$H^n(F) \cong H^n(G),$$

with forward map the functorial $H^n(e)$ of the isomorphism and inverse the functorial $H^n(e^{-1})$ of its inverse.

Proof. Take the forward map to be $H^n(e)$ and the backward map $H^n(e^{-1})$ (Definition ??). By functoriality of H^n , their composites are $H^n(e^{-1}) \circ H^n(e) = H^n(e^{-1} \circ e) = H^n(\text{id}_F) = \text{id}$ and symmetrically $H^n(e) \circ H^n(e^{-1}) = \text{id}$; so the two maps are mutually inverse R -linear bijections, an R -linear equivalence. $\square$

Definition 556 (The ledger invariants h^0 , h^1 and χ). For a sheaf F of R -modules on $(\mathcal{C}, J)$, set

$$h^0(F) = \dim_R H^0(F), \quad h^1(F) = \dim_R H^1(F) \in \mathbb{N},$$

the R -dimensions of the degree-zero and degree-one cohomology, and

$$\chi(F) = h^0(F) - h^1(F) \in \mathbb{Z},$$

the (truncated) Euler characteristic. The dimensions are read by `finrank`, which returns 0 on a module that is not finite-dimensional; no finiteness hypothesis is imposed here, the finiteness bookkeeping being discharged downstream. On a curve, where all higher cohomology vanishes, χ is the honest Euler characteristic.

Theorem 557 (Isomorphism-invariance of the ledger). *Isomorphic sheaves have equal ledger invariants: if $F \cong G$ then $h^0(F) = h^0(G)$, $h^1(F) = h^1(G)$ and $\chi(F) = \chi(G)$.*

Proof. An isomorphism $F \cong G$ induces an R -linear equivalence $H^n(F) \cong H^n(G)$ in each degree (Definition ??); linearly equivalent modules have equal dimension, so $h^0(F) = h^0(G)$ and $h^1(F) = h^1(G)$. Subtracting gives $\chi(F) = \chi(G)$. $\square$

Theorem 558 (Degenerate ledger when H^1 vanishes). *If R is nontrivial and $H^1(F) = 0$, then $h^1(F) = 0$ and $\chi(F) = h^0(F)$.*

Proof. A zero module has dimension 0 over a nontrivial ring, so $h^1(F) = \dim_R H^1(F) = 0$; substituting into $\chi(F) = h^0(F) - h^1(F)$ gives $\chi(F) = h^0(F)$. $\square$

6.14 The six-term cohomology slice of a short exact sequence

A short exact sequence $0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow 0$ of sheaves of R -modules gives a long exact cohomology sequence; its degree- ≤ 1 slice

$$0 \rightarrow H^0 X_1 \rightarrow H^0 X_2 \rightarrow H^0 X_3 \xrightarrow{\delta} H^1 X_1 \rightarrow H^1 X_2 \rightarrow H^1 X_3$$

is the exactness input to both the finiteness bookkeeping and the additivity of χ . This section packages the slice — its connecting map, its exactness, and its two ends — and, over a field, the finiteness transfer and the additivity of χ that ride on it. The additivity rests on the vanishing of the alternating sum of dimensions along the five-term exact sequence (Theorem ??).

Theorem 559 (Extension of finite dimensionality along an exact pair). *Let K be a division ring and let $V_1 \xrightarrow{f} V_2 \xrightarrow{g} V_3$ be a pair of K -linear maps, exact at V_2 ($\ker g = \operatorname{im} f$), with V_1 and V_3 finite-dimensional. Then V_2 is finite-dimensional.*

Proof. Corestrict g to its range, $g' : V_2 \rightarrow \operatorname{im} g$; this is surjective, and the pair (f, g') has the same kernel of second map as (f, g) , so it is still exact at V_2 . The range $\operatorname{im} g$ is a subspace of the finite-dimensional V_3 , hence finite-dimensional. Thus V_2 sits in an exact sequence $V_1 \rightarrow V_2 \rightarrow \operatorname{im} g \rightarrow 0$ with surjective right map and finite-dimensional ends: its kernel $\operatorname{im} f$ is a quotient of the finite V_1 and its coimage is the finite $\operatorname{im} g$, so V_2 is finite-dimensional. $\square$

Definition 560 (The connecting homomorphism). For a short exact sequence $0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow 0$ of sheaves of R -modules and $n_0 + 1 = n_1$, the *connecting homomorphism*

$$\delta : H^{n_0}(X_3) \longrightarrow H^{n_1}(X_1)$$

is postcomposition of a class in $\operatorname{Ext}^{n_0}(R_J, X_3)$ with the extension class $\operatorname{Ext}^1(X_3, X_1)$ of the sequence.

Theorem 561 (Exactness of the cohomology slice). *For a short exact sequence $0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow 0$ of sheaves of R -modules and $n_0 + 1 = n_1$, the cohomology sequence is exact at each of its three interior joints:*

$$H^n(X_1) \rightarrow H^n(X_2) \rightarrow H^n(X_3), \quad H^{n_0}(X_2) \rightarrow H^{n_0}(X_3) \xrightarrow{\delta} H^{n_1}(X_1), \quad H^{n_0}(X_3) \xrightarrow{\delta} H^{n_1}(X_1) \rightarrow H^{n_1}(X_2).$$

Proof. These are the three interior slots of the long exact sequence of the covariant functor $\operatorname{Ext}^\bullet(R_J, -)$ applied to the short exact sequence, read elementwise. The maps $H^n(X_i \rightarrow X_j)$ are the functorial maps of Definition ?? and the connecting map is δ of Definition ??; the identification of the covariant Ext-sequence's maps with these is by definition of postcomposition on both sides. $\square$

Theorem 562 (Left end of the slice: injectivity at H^0). *For a short exact sequence $0 \rightarrow X_1 \xrightarrow{f} X_2 \rightarrow X_3 \rightarrow 0$ of sheaves of R -modules, the map $H^0(f) : H^0(X_1) \rightarrow H^0(X_2)$ is injective.*

Proof. The map $f : X_1 \rightarrow X_2$ is a monomorphism, and postcomposition with a monomorphism is injective on $\operatorname{Ext}^0(R_J, -) = \operatorname{Hom}(R_J, -)$; this is exactly injectivity of $H^0(f)$. $\square$

Theorem 563 (Right end of the slice: surjectivity into a vanishing term). *For a short exact sequence $0 \rightarrow X_1 \xrightarrow{f} X_2 \rightarrow X_3 \rightarrow 0$ of sheaves of R -modules and a degree n with $H^n(X_3) = 0$, the map $H^n(f) : H^n(X_1) \rightarrow H^n(X_2)$ is surjective.*

Proof. Exactness at $H^n(X_2)$ (Theorem ??) identifies the image of $H^n(f)$ with the kernel of $H^n(X_2) \rightarrow H^n(X_3)$. Since $H^n(X_3) = 0$, that map is zero and its kernel is all of $H^n(X_2)$; hence $H^n(f)$ is surjective. $\square$

Theorem 564 (Finiteness transfer along the slice). *Let R be a field and $0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow 0$ a short exact sequence of sheaves of R -modules, with $n_0 + 1 = n_1$. Then:*

1. *if $H^n(X_1)$ and $H^n(X_3)$ are finite-dimensional, so is $H^n(X_2)$;*
2. *if $H^0(X_2)$ is finite-dimensional, so is $H^0(X_1)$;*
3. *if $H^{n_0}(X_3)$ and $H^{n_1}(X_2)$ are finite-dimensional, so is $H^{n_1}(X_1)$.*

Proof. (1) The sequence $H^n(X_1) \rightarrow H^n(X_2) \rightarrow H^n(X_3)$ is exact at $H^n(X_2)$ (Theorem ??) with finite-dimensional ends, so $H^n(X_2)$ is finite-dimensional by the extension lemma ?. (2) The map $H^0(X_1) \rightarrow H^0(X_2)$ is injective (Theorem ??), so $H^0(X_1)$ embeds in the finite-dimensional $H^0(X_2)$ and is finite-dimensional. (3) The sequence $H^{n_0}(X_3) \xrightarrow{\delta} H^{n_1}(X_1) \rightarrow H^{n_1}(X_2)$ is exact at $H^{n_1}(X_1)$ (Theorem ??, with δ of Definition ??) and has finite-dimensional ends, so $H^{n_1}(X_1)$ is finite-dimensional by ?. $\square$

Theorem 565 (Additivity of χ along a short exact sequence). *Let R be a field and $0 \rightarrow X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow 0$ a short exact sequence of sheaves of R -modules with $H^1(X_3) = 0$ and all of $H^0(X_1), H^0(X_2), H^0(X_3), H^1(X_1), H^1(X_2)$ finite-dimensional. Then*

$$\chi(X_2) = \chi(X_1) + \chi(X_3).$$

Proof. Since $H^1(X_3) = 0$, the degree ≤ 1 slice truncates to the five-term sequence

$$0 \rightarrow H^0 X_1 \rightarrow H^0 X_2 \rightarrow H^0 X_3 \xrightarrow{\delta} H^1 X_1 \rightarrow H^1 X_2 \rightarrow 0,$$

exact throughout: the left map is injective (Theorem ??), the three interior joints are exact (Theorem ??), and the right map is surjective because $H^1 X_3 = 0$ (Theorem ??). All five terms being finite-dimensional, the alternating sum of their dimensions vanishes (Theorem ??):

$$h^0 X_1 - h^0 X_2 + h^0 X_3 - h^1 X_1 + h^1 X_2 = 0.$$

Moreover $h^1(X_3) = 0$ (Theorem ??). Rearranging the vanishing alternating sum,

$$(h^0 X_2 - h^1 X_2) = (h^0 X_1 - h^1 X_1) + (h^0 X_3 - h^1 X_3),$$

which is $\chi(X_2) = \chi(X_1) + \chi(X_3)$. $\square$

6.15 Finiteness of the cohomology of divisor sheaves

We return to the curve bundle X : integral, smooth of relative dimension one and quasi-compact over the field K . Assume the structure sheaf, as a sheaf of K -modules, has finite-dimensional H^0 and H^1 — the two properness inputs, taken here as standing hypotheses. This section proves that then *every* divisor sheaf $\mathcal{O}(D)$ has finite-dimensional H^0 and H^1 . The engine is dévissage: a one-point induction that transfers finiteness through the six-term slice of the dévissage short exact sequence $0 \rightarrow \mathcal{O}(D - x) \rightarrow \mathcal{O}(D) \rightarrow \text{sky}_x(J) \rightarrow 0$, whose skyscraper term has finite cohomology.

Definition 566 (The one-point divisor). For a closed point x (a point other than the generic point η) and $n \in \mathbb{Z}$, the *one-point divisor* $n \cdot x$ is the Weil divisor supported at x with coefficient n .

Lemma 567 (Arithmetic of the one-point divisor). *The one-point divisor at a fixed closed point x is additive in its multiplicity and compatible with the group operations and the degree:*

$$0 \cdot x = 0, \quad m \cdot x + n \cdot x = (m+n) \cdot x, \quad n \cdot (1 \cdot x) = n \cdot x \quad (n \in \mathbb{N}), \quad \deg_K(n \cdot x) = n \cdot [\kappa(x) : K].$$

Proof. The one-point divisor $n \cdot x$ is the finitely supported function sending x to n and every other point to 0; such functions are additive in the value and vanish at value 0, giving the first two identities, and the third follows by induction on n from the second. The degree identity is the single-point degree formula (Lemma ??) with the residue degree $[\kappa(x) : K]$ of Definition ?. $\square$

Lemma 568 (The dévissage twist is subtraction of a point). *The dévissage twist $D - x$ (Definition ??) is the subtraction of the one-point divisor $1 \cdot x$:*

$$D - x = D - 1 \cdot x, \quad (D - x) + 1 \cdot x = D.$$

Proof. The twist decreases the coefficient of D at x by one and leaves the others unchanged, which is exactly subtracting the one-point divisor $1 \cdot x$; adding $1 \cdot x$ back restores the original coefficient, giving D . $\square$

Lemma 569 (Degree drop along the twist). *The degree drops by the residue degree along the dévissage twist:*

$$\deg_K(D - x) = \deg_K D - [\kappa(x) : K].$$

Proof. The degree is a homomorphism (Theorem ??), so $\deg_K((D - x) + 1 \cdot x) = \deg_K(D - x) + \deg_K(1 \cdot x)$; the left side is $\deg_K D$ since $(D - x) + 1 \cdot x = D$ (Lemma ??), and $\deg_K(1 \cdot x) = [\kappa(x) : K]$ (Lemma ??). Subtracting gives the claim. $\square$

Theorem 570 (Dévissage induction on Weil divisors). *Let P be a predicate on Weil divisors of X that holds for the zero divisor and, for every closed point x , transfers in both directions along the one-point dévissage: $P(D) \iff P(D - x)$ for all D . Then $P(D)$ holds for every D .*

Proof. A Weil divisor is a finitely supported $\mathbb{Z}$ -family on the closed points, so it suffices to prove the following single-bump induction: a predicate on such families that holds at 0 and satisfies $P(F + 1 \cdot x) \iff P(F)$ for every point x and family F holds everywhere. First, from the ± 1 -bump equivalence one gets $P(F + n \cdot x) \iff P(F)$ for every $n \in \mathbb{Z}$, by induction on n : the case $n = 0$ is trivial, and the step from n to $n \pm 1$ composes one more bump equivalence, using $F + (n \pm 1) \cdot x = (F + n \cdot x) \pm 1 \cdot x$. Now argue by the finitely-supported induction principle: $P(0)$ holds by hypothesis, and adding a single generator $b \cdot x$ to a family F preserves P by the equivalence just established. Hence P holds for every family, i.e. every Weil divisor. The hypothesis $P(D) \iff P(D - x)$ is the bump equivalence in the form $P((D - x) + 1 \cdot x) \iff P(D - x)$, using $(D - x) + 1 \cdot x = D$ (Lemma ??). $\square$

Theorem 571 (Cohomology of a skyscraper is finite). *For a point $x \in X$ and a finite-dimensional K -module M , the cohomology of the skyscraper $\text{sky}_x(M)$ is finite-dimensional in degrees 0 and 1: H^0 is M and H^1 vanishes.*

Proof. In degree 0, $H^0(\text{sky}_x(M)) \cong M$ (Definition ??), which is finite-dimensional by hypothesis. In degree 1, $H^1(\text{sky}_x(M)) = 0$ (Theorem ??), and the zero module is finite-dimensional. $\square$

Theorem 572 (Finiteness of the cohomology of every divisor sheaf). *On the curve bundle X , quasi-compact over K , assume $H^0(X, \mathcal{O}_{X/K})$ and $H^1(X, \mathcal{O}_{X/K})$ are finite-dimensional over K . Then for every Weil divisor D , both $H^0(\mathcal{O}(D))$ and $H^1(\mathcal{O}(D))$ are finite-dimensional.*

Proof. Apply the dévissage induction (Theorem ??) to the predicate “ $H^0(\mathcal{O}(D))$ and $H^1(\mathcal{O}(D))$ are both finite-dimensional”.

Base case. For $D = 0$, the identification $\mathcal{O}(0) \cong \mathcal{O}_{X/K}$ (Theorem ??) transports finite-dimensionality of H^0 and H^1 from the structure sheaf, which holds by the standing hypothesis.

Step. Fix a closed point x . The dévissage short exact sequence $0 \rightarrow \mathcal{O}(D - x) \rightarrow \mathcal{O}(D) \rightarrow \text{sky}_x(J) \rightarrow 0$ (Theorem ??) has third term the skyscraper on the jump module J (Definition ??), which is finite-dimensional (Corollary ??); hence $H^0(\text{sky}_x(J))$ is finite-dimensional and $H^1(\text{sky}_x(J)) = 0$ (Theorem ??). We must show the finiteness of the cohomology of $\mathcal{O}(D)$ is equivalent to that of $\mathcal{O}(D - x)$; both directions use the finiteness transfer of Theorem ?? for this sequence, whose terms X_1, X_2, X_3 are $\mathcal{O}(D - x), \mathcal{O}(D), \text{sky}_x(J)$.

- *From $D - x$ to D .* If H^0, H^1 of $\mathcal{O}(D - x)$ are finite-dimensional, then so are those of $\mathcal{O}(D)$, by the middle-term transfer (1) applied in degrees 0 and 1: the two outer terms $\mathcal{O}(D - x)$ and $\text{sky}_x(J)$ have finite cohomology.
- *From D to $D - x$.* If H^0, H^1 of $\mathcal{O}(D)$ are finite-dimensional, then $H^0(\mathcal{O}(D - x))$ is finite-dimensional as it embeds in $H^0(\mathcal{O}(D))$ (left-term transfer (2)), and $H^1(\mathcal{O}(D - x))$ is finite-dimensional as an extension controlled by the finite $H^0(\text{sky}_x(J))$ and $H^1(\mathcal{O}(D))$ (left-term transfer (3)).

Both directions established, the induction concludes. $\square$

6.16 The χ -ledger and the Riemann inequality

The Euler characteristic of a divisor sheaf is now computed. Under the standing finiteness hypotheses of the previous section, subtracting a closed point drops χ by the residue degree; iterating from $\mathcal{O}(0) \cong \mathcal{O}_{X/K}$ closes the ledger into $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_K D$. Two consequences follow: principal divisors have degree zero, and h^0 grows without bound along multiples of any closed point.

Theorem 573 (The Euler characteristic of a skyscraper). *For a point $x \in X$ and a K -module M (with K a field), the degree-zero dimension and the Euler characteristic of the skyscraper are both the dimension of the value module:*

$$h^0(\text{sky}_x(M)) = \dim_K M, \quad \chi(\text{sky}_x(M)) = \dim_K M.$$

Proof. Since $H^0(\text{sky}_x(M)) \cong M$ (Definition ??), $h^0(\text{sky}_x(M)) = \dim_K M$. The degree-one cohomology vanishes (Theorem ??), so $\chi(\text{sky}_x(M)) = h^0(\text{sky}_x(M))$ (Theorem ??), which is $\dim_K M$. $\square$

Theorem 574 (The dévissage step of the χ -ledger). *Under the standing finiteness hypotheses, subtracting a closed point x from a Weil divisor D drops the Euler characteristic by the residue degree:*

$$\chi(\mathcal{O}(D)) = \chi(\mathcal{O}(D - x)) + [\kappa(x) : K].$$

Proof. Apply the additivity of χ (Theorem ??) to the dévissage short exact sequence $0 \rightarrow \mathcal{O}(D - x) \rightarrow \mathcal{O}(D) \rightarrow \text{sky}_x(J) \rightarrow 0$ (Theorem ??). Its hypotheses hold: the skyscraper term has $H^1 = 0$ (Theorem ??), and all five relevant cohomology modules are finite-dimensional — those of $\mathcal{O}(D - x)$ and $\mathcal{O}(D)$ by Theorem ??, and $H^0(\text{sky}_x(J))$ because J is finite (Corollary ??). Hence $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}(D - x)) + \chi(\text{sky}_x(J))$. Finally $\chi(\text{sky}_x(J)) = \dim_K J$ (Theorem ??) and $\dim_K J = [\kappa(x) : K]$ (Theorem ??), giving the stated identity. $\square$

Theorem 575 (The χ -ledger, closed). *Under the standing finiteness hypotheses, for every Weil divisor D ,*

$$\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_K D.$$

Proof. Dévissage induction (Theorem ??) on the predicate “ $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_K D$ ”. For $D = 0$: $\chi(\mathcal{O}(0)) = \chi(\mathcal{O}_{X/K})$ by isomorphism-invariance (Theorem ??) along $\mathcal{O}(0) \cong \mathcal{O}_{X/K}$ (Theorem ??), and $\deg_K 0 = 0$ (Theorem ??). For the step at a closed point x , combine the ledger step $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}(D - x)) + [\kappa(x) : K]$ (Theorem ??) with the degree drop $\deg_K(D - x) = \deg_K D - [\kappa(x) : K]$ (Lemma ??): the statement for $D - x$ and the statement for D differ on both sides by $[\kappa(x) : K]$, so each implies the other (using $D - x = D - 1 \cdot x$, Lemma ??). $\square$

Theorem 576 (Principal divisors have degree zero). *For a nonzero rational function $g \in K(X)^\times$,*

$$\deg_K(\operatorname{div} g) = 0.$$

Proof. Multiplication by g is an isomorphism of sheaves $\mathcal{O}(0) \cong \mathcal{O}(0 - \operatorname{div} g)$ (Theorem ??), so the two sheaves have equal Euler characteristic (Theorem ??). Evaluating both with the closed ledger (Theorem ??):

$$\chi(\mathcal{O}_{X/K}) + \deg_K 0 = \chi(\mathcal{O}_{X/K}) + \deg_K(0 - \operatorname{div} g).$$

Cancelling $\chi(\mathcal{O}_{X/K})$ and using $\deg_K 0 = 0$ and $\deg_K(-\operatorname{div} g) = -\deg_K(\operatorname{div} g)$ (Theorem ??) leaves $-\deg_K(\operatorname{div} g) = 0$, i.e. $\deg_K(\operatorname{div} g) = 0$. $\square$

Theorem 577 (The Riemann inequality). *Under the standing finiteness hypotheses, for every Weil divisor D ,*

$$\deg_K D + \chi(\mathcal{O}_{X/K}) \leq h^0(\mathcal{O}(D)).$$

Proof. By the closed ledger (Theorem ??), $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_K D$. On the other hand $\chi(\mathcal{O}(D)) = h^0(\mathcal{O}(D)) - h^1(\mathcal{O}(D)) \leq h^0(\mathcal{O}(D))$, since $h^1 \geq 0$. Combining the two gives $\deg_K D + \chi(\mathcal{O}_{X/K}) \leq h^0(\mathcal{O}(D))$. $\square$

Theorem 578 (Unbounded growth of h^0 along multiples of a point). *Under the standing finiteness hypotheses, for every closed point x and every bound N there is $n \in \mathbb{N}$ with*

$$N \leq h^0(\mathcal{O}(n \cdot x)).$$

No rational point is needed: it suffices that $[\kappa(x) : K] \geq 1$.

Proof. For each n , the closed ledger (Theorem ??) and the degree of a one-point divisor (Lemma ??) give $\chi(\mathcal{O}(n \cdot x)) = \chi(\mathcal{O}_{X/K}) + n[\kappa(x) : K]$. The residue degree is positive, $[\kappa(x) : K] \geq 1$ (Theorem ??), so $\chi(\mathcal{O}(n \cdot x)) \geq \chi(\mathcal{O}_{X/K}) + n$. Since $\chi(\mathcal{O}(n \cdot x)) \leq h^0(\mathcal{O}(n \cdot x))$ (as $h^1 \geq 0$) and $\chi(\mathcal{O}_{X/K}) = h^0(\mathcal{O}_{X/K}) - h^1(\mathcal{O}_{X/K}) \geq -h^1(\mathcal{O}_{X/K})$, we obtain

$$h^0(\mathcal{O}(n \cdot x)) \geq n - h^1(\mathcal{O}_{X/K}).$$

Taking $n = N + h^1(\mathcal{O}_{X/K})$ yields $h^0(\mathcal{O}(n \cdot x)) \geq N$. $\square$

6.17 Riemann–Roch on the challenge curve

Finally, instantiate the ledger on the bundled challenge curve C over the field k : smooth of relative dimension one, proper, and geometrically irreducible. Properness supplies both missing ingredients — the finiteness of $H^1(C, \mathcal{O}_C)$, which is the genus, and the identification $H^0(C, \mathcal{O}_C) \cong k$, which makes $h^0(\mathcal{O}_C) = 1$. The conditional ledger then reads as Riemann–Roch-lite: $\chi(\mathcal{O}(D)) = 1 - g(C) + \deg_k D$.

Definition 579 (The k -linear structure map on global sections). For a scheme X over $\operatorname{Spec} k$, the structure map $k \rightarrow \Gamma(X, \mathbb{T})$ of Definition ?? is k -linear for the k -module structure it induces on $\Gamma(X, \mathbb{T})$ (Definition ??).

Proof. The k -module structure on $\Gamma(X, \mathbb{T})$ is restriction of scalars along the very same ring map $k \rightarrow \Gamma(X, \mathbb{T})$, so additivity and k -homogeneity of that map are the defining module axioms. $\square$

Theorem 580 (The k -linear $\Gamma(X, \mathcal{T}) \cong k$ seam). *Let X be a scheme over $\operatorname{Spec} k$ whose structure morphism is bijective on global sections (the induced map $\Gamma(\operatorname{Spec} k, \mathcal{O}_{\operatorname{Spec} k}) \rightarrow \Gamma(X, \mathcal{O}_X)$ is bijective). Then the structure map $k \rightarrow \Gamma(X, \mathcal{T})$ is a k -linear equivalence*

$$k \cong \Gamma(X, \mathcal{T}).$$

Proof. The structure map $k \rightarrow \Gamma(X, \mathcal{T})$ factors as the composite of the inverse of the canonical isomorphism $\Gamma(\operatorname{Spec} k, \mathcal{O}_{\operatorname{Spec} k}) \cong k$, the map on global sections induced by the structure morphism, and the identity restriction $\Gamma(X, \mathcal{T}) \rightarrow \Gamma(X, \mathcal{T})$. The first factor is bijective (an isomorphism), the last is the identity, and the middle is bijective by hypothesis; hence the composite is bijective. A bijective k -linear map (Definition ??) is a k -linear equivalence. $\square$

Theorem 581 ($h^0 = 1$ when global sections are constant). *Let X be a scheme over $\operatorname{Spec} k$ whose structure morphism is bijective on global sections. Then $H^0(X, \mathcal{O}_{X/k})$ is one-dimensional over k ; in particular it is finite-dimensional and $h^0(\mathcal{O}_{X/k}) = 1$.*

Proof. Degree-zero cohomology is global sections, $H^0(X, \mathcal{O}_{X/k}) \cong \Gamma(X, \mathcal{T})$ (Definition ??), and the seam identifies $\Gamma(X, \mathcal{T}) \cong k$ (Theorem ??). Thus $H^0(X, \mathcal{O}_{X/k}) \cong k$ is one-dimensional, so it is finite-dimensional and $h^0(\mathcal{O}_{X/k}) = \dim_k k = 1$. $\square$

Theorem 582 (Global sections of the challenge curve). *For the challenge curve C — proper, smooth of relative dimension one and geometrically irreducible over k — the cohomology $H^0(C, \mathcal{O}_C)$ is a one-dimensional (hence finite-dimensional) k -vector space, and $h^0(\mathcal{O}_C) = 1$.*

Proof. Being smooth, C is geometrically reduced, and together with geometric irreducibility this makes C geometrically integral; being also proper, its structure morphism is bijective on global sections (Theorem ??). Apply Theorem ??. $\square$

Theorem 583 (Euler characteristic of the structure sheaf of the curve). *For the challenge curve C ,*

$$\chi(\mathcal{O}_C) = 1 - g(C),$$

where $g(C) = \dim_k H^1(C, \mathcal{O}_C)$ is the genus.

Proof. By Theorem ??, $h^0(\mathcal{O}_C) = 1$. By definition of the genus (Definition ??), $h^1(\mathcal{O}_C) = \dim_k H^1(C, \mathcal{O}_C) = g(C)$. Therefore $\chi(\mathcal{O}_C) = h^0(\mathcal{O}_C) - h^1(\mathcal{O}_C) = 1 - g(C)$. $\square$

Theorem 584 (Riemann–Roch-lite for the challenge curve). *For every Weil divisor D on the challenge curve C ,*

$$\chi(\mathcal{O}(D)) = 1 - g(C) + \deg_k D.$$

Proof. The two standing finiteness hypotheses are met on the curve: $H^0(C, \mathcal{O}_C)$ is finite-dimensional (Theorem ??) and $H^1(C, \mathcal{O}_C)$ is finite-dimensional (Theorem ??). So the closed ledger applies (Theorem ??): $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_C) + \deg_k D$. Substituting $\chi(\mathcal{O}_C) = 1 - g(C)$ (Theorem ??) gives the claim. $\square$

Theorem 585 (The Riemann inequality for the challenge curve). *For every Weil divisor D on the challenge curve C ,*

$$\deg_k D + 1 - g(C) \leq h^0(\mathcal{O}(D)).$$

Proof. The standing finiteness hypotheses hold on C (Theorem ?? and Theorem ??), so the general inequality $\deg_k D + \chi(\mathcal{O}_C) \leq h^0(\mathcal{O}(D))$ applies (Theorem ??). Substituting $\chi(\mathcal{O}_C) = 1 - g(C)$ (Theorem ??) gives the stated form. This is the classical Riemann inequality $\dim L(D) \geq \deg D + 1 - g$, read off the genus as the supremum $g = \sup_D \{\deg D - \dim L(D) + 1\}$. $\square$

Theorem 586 (Unbounded h^0 along a closed point of the curve). *For every closed point x of the challenge curve C and every bound N , there is $n \in \mathbb{N}$ with $N \leq h^0(\mathcal{O}(n \cdot x))$. No rational point is needed.*

Proof. The standing finiteness hypotheses hold on C (Theorem ?? and Theorem ??), so the general unboundedness (Theorem ??) applies verbatim. $\square$

6.18 The chart colength dictionary

This section proves the dictionary between the algebra of an affine chart of the curve and the divisor data at its closed points. Let V be an affine open of X containing the generic point η , and write $B = \Gamma(X, V)$, a K -algebra through the structure morphism (??); all dimensions over K below are taken for that structure. Under the isomorphism $V \cong \operatorname{Spec} B$, a point $x \in V$ corresponds to a prime ideal $\mathfrak{p}_x \subseteq B$, and conversely a prime $\mathfrak{q}$ of B determines a point $x_{\mathfrak{q}} \in V$; the first four lemmas record this correspondence between closed points of V and nonzero primes of B . The section ring B is then shown to be a Dedekind domain for *every* chart, and the colength $\dim_K B/(f)$ of a nonzero section f is computed in divisor terms: the primes dividing (f) are the primes of the closed points where f vanishes, the multiplicity of $\mathfrak{p}_x$ in (f) is the order of vanishing at x of the rational function determined by f , and the residue $B/\mathfrak{p}_x$ is the residue field $\kappa(x)$. The engine behind the colength formula is the localized Dedekind colength of Section ??.

Lemma 587 (The point of a prime lies in the chart). *Let X be a scheme and V an affine open. For every prime $\mathfrak{q}$ of $B = \Gamma(X, V)$, the point $x_{\mathfrak{q}}$ lies in V .*

Proof. The canonical morphism $\operatorname{Spec} B \rightarrow X$ is an open immersion with image V , and $x_{\mathfrak{q}}$ is by definition the image of the point $\mathfrak{q}$ under it. $\square$

Lemma 588 (The correspondences are mutually inverse). *Let X be a scheme and V an affine open. For every prime $\mathfrak{q}$ of B ,*

$$\mathfrak{p}_{x_{\mathfrak{q}}} = \mathfrak{q}.$$

Proof. The assignment $x \mapsto \mathfrak{p}_x$ is defined so that the point of V determined by $\mathfrak{p}_x$ is x again; since the morphism $\operatorname{Spec} B \rightarrow X$ is an open immersion, it is injective on points, and the two primes $\mathfrak{p}_{x_{\mathfrak{q}}}$ and $\mathfrak{q}$, having the same image $x_{\mathfrak{q}}$, coincide. $\square$

Lemma 589 (Nonzero prime at a non-generic point). *Let X be an integral scheme and V an affine open. For a point $x \in V$ which is not the generic point of X , the prime $\mathfrak{p}_x$ is nonzero.*

Proof. Suppose $\mathfrak{p}_x = 0$. Since X is integral and V is a nonempty open, B is a domain, so the zero ideal is the generic point of $\operatorname{Spec} B$. An open immersion between irreducible schemes with dense image carries the generic point to the generic point, and V is dense in the irreducible X ; hence x , the image of the generic point of $\operatorname{Spec} B$, is the generic point of X — contrary to hypothesis. $\square$

Lemma 590 (The point of a nonzero prime is non-generic). *Let X be an integral scheme and V an affine open. For a prime $\mathfrak{q} \neq 0$ of B , the point $x_{\mathfrak{q}}$ is not the generic point of X .*

Proof. If $x_{\mathfrak{q}}$ were the generic point of X , then — as in ??, the open immersion $\operatorname{Spec} B \rightarrow X$ matches generic points, and is injective — the prime $\mathfrak{q}$ would be the generic point of $\operatorname{Spec} B$, which is the zero ideal of the domain B ; contradiction. $\square$

Theorem 591 (Finite type of the chart sections). *Let K be a commutative ring and X a scheme locally of finite type over $\operatorname{Spec} K$. For every affine open V of X , the algebra map $K \rightarrow \Gamma(X, V)$ of ?? is of finite type.*

Proof. The algebra map is the composite of the identification $K \cong \Gamma(\text{Spec } K)$ with the map on sections $\Gamma(\text{Spec } K) \rightarrow \Gamma(X, V)$ induced by the structure morphism between the affine opens V and $\text{Spec } K$. For a morphism locally of finite type, the induced map between the section rings of an affine open of the source contained in the preimage of an affine open of the target is a ring map of finite type; apply this to V and $\text{Spec } K$. $\square$

Theorem 592 (Dedekind sections on every affine chart). *Let X be an integral scheme, smooth of relative dimension one and locally of finite type over the field K , and let V be an affine open containing the generic point (equivalently, nonempty). Then $B = \Gamma(X, V)$ is a Dedekind domain. (This upgrades ??, which produces some Dedekind chart around each point, to every affine chart.)*

Proof. B is a domain: V is a nonempty open of the integral scheme X . B is Noetherian: the algebra map $K \rightarrow B$ is of finite type (??), and a finite-type algebra over a field is Noetherian by Hilbert's basis theorem. It remains to see that the localization of B at each nonzero prime $\mathfrak{q}$ is a discrete valuation ring; a Noetherian domain, not a field, with this property is Dedekind. The point $x := x_{\mathfrak{q}}$ lies in V (??) and is not the generic point (??), so the stalk $\mathcal{O}_{X,x}$ is a discrete valuation ring (??). The stalk at a point of an affine open is the localization of the section ring at the corresponding prime, and $\mathfrak{p}_x = \mathfrak{q}$ (??); hence $B_{\mathfrak{q}} \cong \mathcal{O}_{X,x}$ is a discrete valuation ring. $\square$

Theorem 593 (Finiteness of the chart residue). *Let X be as in ??, V an affine open, and $x \in V$ a point other than the generic point. Then the residue $B/\mathfrak{p}_x$ is a finite K -module.*

Proof. The singleton $\{x\}$ is closed in X (??), so x is a closed point of the chart $V \cong \text{Spec } B$ and $\mathfrak{p}_x$ is a maximal ideal; $B/\mathfrak{p}_x$ is a field. The composite $K \rightarrow B \rightarrow B/\mathfrak{p}_x$ is of finite type (?? followed by a surjection), and a field which is a finite-type algebra over a field is a finite extension (Zariski's lemma). Hence $B/\mathfrak{p}_x$ is a finite K -module. $\square$

Theorem 594 (The residue leg: the chart residue is the residue field). *Let X be an integral scheme smooth of relative dimension one over the field K , V an affine open, and $x \in V$ a point other than the generic point. Then the residue of the chart at $\mathfrak{p}_x$ is the residue field of x , K -linearly; in particular*

$$\dim_K(B/\mathfrak{p}_x) = [\kappa(x) : K] = \deg_K(x).$$

Proof. As in ??, $\mathfrak{p}_x$ is maximal (??). The stalk $\mathcal{O}_{X,x}$ is the localization of B at $\mathfrak{p}_x$, and the residue field of the localization of a ring at a maximal ideal is the quotient by that ideal: $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x \cong B/\mathfrak{p}_x$, by the map induced by $B \rightarrow \mathcal{O}_{X,x}$. This isomorphism is K -linear: the K -structure of $\kappa(x)$ is the composite of the algebra map into the sections, the germ, and the residue map (??), and the germ of the algebra map through V agrees with the germ of the global algebra map. Taking dimensions gives $\dim_K(B/\mathfrak{p}_x) = \dim_K \kappa(x) = \deg_K(x)$ (??). $\square$

Theorem 595 (The multiplicity leg: order of vanishing is the multiplicity). *Let X be an integral scheme smooth of relative dimension one over the field K , V an affine open containing the generic point η , and assume $B = \Gamma(X, V)$ is a Dedekind domain (??). Let $x \in V$ be a point other than η , let $f \in B$ be nonzero, and let $g \in K(X)^\times$ be a unit whose value is the germ of f at η . Then the order of vanishing of g at x is the multiplicity of $\mathfrak{p}_x$ in the prime factorization of $(f) \subseteq B$:*

$$\text{ord}_x(g) = \text{ord}_{\mathfrak{p}_x}(f).$$

Proof. The prime $\mathfrak{p}_x$ is nonzero (??), and the stalk $\mathcal{O}_{X,x}$ is the localization of B at $\mathfrak{p}_x$ — a discrete valuation ring (??) with maximal ideal $\mathfrak{m}_x$. The germ of f at x is its image under $B \rightarrow \mathcal{O}_{X,x}$; it is nonzero, because on an integral scheme the germ map is injective on sections and $f \neq 0$. By

the germ/stalk seam (??), g — the germ of f at η — is the image of the germ of f at x under $\mathcal{O}_{X,x} \rightarrow K(X)$.

The order valuation ord_x is the $\mathfrak{m}_x$ -adic valuation of the discrete valuation ring $\mathcal{O}_{X,x}$ (??), so its value on g is the $\mathfrak{m}_x$ -adic valuation of the germ of f at x , which is the multiplicity of $\mathfrak{m}_x$ in the factorization of the principal ideal generated by that germ (??, applied in the Dedekind domain $\mathcal{O}_{X,x}$). Finally, multiplicities are preserved by localization at the prime (??, with $S = \mathcal{O}_{X,x}$, a Dedekind local ring which is the localization of B at $\mathfrak{p}_x$): $\text{ord}_{\mathfrak{m}_x}(f \cdot \mathcal{O}_{X,x}) = \text{ord}_{\mathfrak{p}_x}(f)$. Chaining the identities gives $\text{ord}_x(g) = \text{ord}_{\mathfrak{p}_x}(f)$. $\square$

Theorem 596 (Finiteness of the chart colength module). *Let X be an integral scheme, smooth of relative dimension one and locally of finite type over the field K , and V an affine open containing η . For every nonzero $f \in B = \Gamma(X, V)$, the quotient $B/(f)$ is a finite K -module.*

Proof. B is a Dedekind domain (??). Each prime factor P of (f) is a nonzero prime (??); its point x_P lies in V (??), is not the generic point (??), and satisfies $\mathfrak{p}_{x_P} = P$ (??); so the residue B/P is a finite K -module by ???. The finiteness of the Dedekind colength module (??) concludes. $\square$

Theorem 597 (The chart colength dictionary). *Let X be an integral scheme, smooth of relative dimension one and locally of finite type over the field K , and V an affine open containing η . Let $f \in B = \Gamma(X, V)$, let $g \in K(X)^\times$ be a unit whose value is the germ of f at η , and let S be a finite set of non-generic points of X , contained in V , such that at every non-generic point of V outside S the germ of f is a unit of the stalk. Then, in $\mathbb{Z}$,*

$$\dim_K(B/(f)) = \sum_{x \in S} \text{ord}_x(g) \cdot \deg_K(x).$$

Proof. B is a Dedekind domain (??), and $f \neq 0$: its germ at η is the value of the unit g . As in ??, every prime factor of (f) has finite residue over K , so the localized Dedekind colength formula (??) gives

$$\dim_K(B/(f)) = \sum_{P|(f)} \text{ord}_P(f) \cdot \dim_K(B/P).$$

It remains to identify the right side with the sum over S . Note first that a section is a non-unit at a point $z \in V$ exactly when it lies in $\mathfrak{p}_z$: the stalk at z is the localization of B at $\mathfrak{p}_z$, where the image of an element of B is a unit precisely when the element avoids the prime.

Terms of the S -sum outside the factor set vanish. If $x \in S$ and the germ of f at x is a unit, then x lies in the basic open of V where f is invertible, so $\text{ord}_x(g) = 0$ (??) and the term vanishes. Otherwise $f \in \mathfrak{p}_x$; then $\mathfrak{p}_x$ is a prime factor of (f) : it is nonzero (??), and membership $f \in \mathfrak{p}_x = \mathfrak{p}_x^1$ forces multiplicity ≥ 1 (??).

The assignment $x \mapsto \mathfrak{p}_x$ matches the two index sets. It is injective on V , the inverse being $\mathfrak{q} \mapsto x_{\mathfrak{q}}$ (??). It is surjective onto the prime factors: a prime factor P of (f) contains f (??), its point x_P lies in V (??), is non-generic (??) and satisfies $\mathfrak{p}_{x_P} = P$ (??); and $x_P \in S$, for otherwise the outside- S hypothesis would make the germ of f at x_P a unit, while $f \in P = \mathfrak{p}_{x_P}$.

The terms match. For $x \in S$ whose prime is a factor of (f) , the multiplicity leg gives $\text{ord}_x(g) = \text{ord}_{\mathfrak{p}_x}(f)$ (??) and the residue leg gives $\dim_K(B/\mathfrak{p}_x) = \deg_K(x)$ (??). Hence the sum over the prime factors of (f) coincides, term by term, with the sum over S of the nonzero terms, and the formula follows. $\square$

6.19 The degree homomorphism on the Čech Picard group

The degree of a Weil divisor is invariant under linear equivalence, so it descends from $\text{Div}(X)$ to the Čech Picard group $\text{Pic}(X) \cong \text{Div}(X)/\text{Princ}(X)$ (??). This section records the degree homo-

morphism $\deg_{\text{Pic}} : \text{Pic}(X) \rightarrow \mathbb{Z}$ on classes and its interface: the normalization $\deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D$, additivity, and the connection to the χ -ledger $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_{\text{Pic}} \mathcal{O}_X(D)$. The same linear-equivalence argument makes the Euler characteristic of a class well-defined, this time with no finiteness hypothesis at all.

Lemma 598 (Linearly equivalent divisors have equal degree). *If two Weil divisors have the same Čech Picard class, $\mathcal{O}_X(D) = \mathcal{O}_X(D')$, then $\deg_K D = \deg_K D'$.*

Proof. Equal classes differ by a principal divisor (??): there is $g \in K(X)^\times$ with $D - D' = \text{div } g$. Principal divisors have degree zero (Theorem ??), so $\deg_K(D - D') = 0$; and $\deg_K(D - D') = \deg_K D - \deg_K D'$ by the degree homomorphism (Theorem ??). Hence $\deg_K D = \deg_K D'$. $\square$

Theorem 599 (The Euler characteristic of a class is well-defined). *If two Weil divisors have the same Čech Picard class, $\mathcal{O}_X(D) = \mathcal{O}_X(D')$, then their divisor sheaves have the same Euler characteristic, $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}(D'))$. No finiteness hypothesis is needed.*

Proof. Equal classes differ by a principal divisor (??): $D - \text{div } g = D'$ for some $g \in K(X)^\times$. Multiplication by g is an isomorphism of divisor sheaves $\mathcal{O}(D) \cong \mathcal{O}(D - \text{div } g) = \mathcal{O}(D')$ (Theorem ??), and isomorphic sheaves have equal ledger invariants (Theorem ??), so $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}(D'))$. The argument uses only the isomorphism and its transport, never the finiteness of any cohomology. $\square$

Definition 600 (The degree homomorphism on the Picard group). The degree descends to a group homomorphism

$$\deg_{\text{Pic}} : \text{Pic}(X) \rightarrow \mathbb{Z}, \quad \deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D \quad (D \in \text{Div}(X)),$$

the *degree of a Picard class*: the degree of any Weil divisor realizing the class. The displayed normalization $\deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D$ holds for every Weil divisor D , and is the defining property.

Proof. Every class $L \in \text{Pic}(X)$ is realized by a Weil divisor, $L = \mathcal{O}_X(D)$ (??); set $\deg_{\text{Pic}} L = \deg_K D$. This is independent of the chosen realization: if $\mathcal{O}_X(D) = \mathcal{O}_X(D')$ then $\deg_K D = \deg_K D'$ (Lemma ??). Applying this to a divisor's own class gives the normalization $\deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D$. It is a homomorphism: the trivial class is $\mathcal{O}_X(0)$, of degree $\deg_K 0 = 0$; and for classes realized by D, D' the product is $\mathcal{O}_X(D) \cdot \mathcal{O}_X(D') = \mathcal{O}_X(D + D')$, whose degree is $\deg_K(D + D') = \deg_K D + \deg_K D'$ (the class map and the degree are both homomorphisms, ??). $\square$

Theorem 601 (Additivity of the class degree). *The degree of a product of Picard classes is the sum of the degrees: for $L, L' \in \text{Pic}(X)$,*

$$\deg_{\text{Pic}}(L \cdot L') = \deg_{\text{Pic}} L + \deg_{\text{Pic}} L'.$$

Proof. Realize the classes by Weil divisors, $L = \mathcal{O}_X(D)$ and $L' = \mathcal{O}_X(D')$ (??). Their product is $L \cdot L' = \mathcal{O}_X(D + D')$ by additivity of the class map (Theorem ??), so by the normalization (??) and additivity of the divisor degree (Theorem ??),

$$\deg_{\text{Pic}}(L \cdot L') = \deg_K(D + D') = \deg_K D + \deg_K D' = \deg_{\text{Pic}} L + \deg_{\text{Pic}} L'.$$

$\square$

Theorem 602 (The χ -ledger in class form). *Under the standing finiteness hypotheses, for every Weil divisor D ,*

$$\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_{\text{Pic}} \mathcal{O}_X(D).$$

Proof. By the normalization $\deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D$ (??), the claim is the closed ledger $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_{X/K}) + \deg_K D$ (Theorem ??). $\square$

Theorem 603 (The point-product class map has the same degree). *For every Weil divisor D , the degree of the point-product class equals the degree of D :*

$$\deg_{\text{Pic}}(\text{divisorClass}(D)) = \deg_K D.$$

Proof. The point-product class map and the anchored class map coincide, $\text{divisorClass}(D) = \mathcal{O}_X(D)$ (Theorem ??), so the degree of $\text{divisorClass}(D)$ is the degree of $\mathcal{O}_X(D)$, which is $\deg_K D$ by the normalization (??). $\square$

Theorem 604 (Riemann–Roch-lite in class form on the challenge curve). *For every Weil divisor D on the challenge curve C ,*

$$\chi(\mathcal{O}(D)) = 1 - g(C) + \deg_{\text{Pic}} \mathcal{O}_X(D).$$

The standing finiteness hypotheses are discharged on the curve, so there is no side condition.

Proof. On the curve the closed ledger reads $\chi(\mathcal{O}(D)) = 1 - g(C) + \deg_K D$ (Theorem ??). Substituting the normalization $\deg_{\text{Pic}} \mathcal{O}_X(D) = \deg_K D$ (??) gives the claim. $\square$

6.20 Cohomology of a Picard class is well-defined

Throughout this section X is a curve bundle over a field K : integral, smooth of relative dimension one, and quasi-compact over $\text{Spec } K$. The Euler characteristic of a divisor sheaf is a Čech Picard class invariant (??); the same holds *degree by degree* for h^0 and h^1 , and for the vanishing and finite-dimensionality of the individual cohomology modules. All of these transport along one isomorphism of divisor sheaves, so the arguments are one line each on the ledger API.

The common input is: if two Weil divisors D, D' have equal class, $[\mathcal{O}_X(D)] = [\mathcal{O}_X(D')]$, then the extraction lemma ?? writes $D - D' = \text{div } g$ for a rational function g , and multiplication by g (??) realises an isomorphism of sheaves of K -modules $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$. Every statement below is transport along this isomorphism.

Theorem 605 (h^0 and h^1 are Picard-class invariants). *Let D, D' be Weil divisors on X with equal class, $[\mathcal{O}_X(D)] = [\mathcal{O}_X(D')]$. Then*

$$h^0(\mathcal{O}_X(D)) = h^0(\mathcal{O}_X(D')), \quad h^1(\mathcal{O}_X(D)) = h^1(\mathcal{O}_X(D')).$$

No finiteness hypothesis is needed. This is the witness-independence that makes a per-class quantifier “for all D with $[\mathcal{O}_X(D)] = \lambda$ ” well-posed.

Proof. The class isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ described above induces, in each fixed degree n , an isomorphism of cohomology modules $H^n(\mathcal{O}_X(D)) \cong H^n(\mathcal{O}_X(D'))$; passing to dimensions (h^0 -, h^1 -congruence, ??) gives the two equalities. $\square$

Theorem 606 (Vanishing of H^1 is a Picard-class invariant). *Let D, D' be Weil divisors on X with $[\mathcal{O}_X(D)] = [\mathcal{O}_X(D')]$. If the degree-one cohomology $H^1(\mathcal{O}_X(D))$ vanishes (in the strong sense of being a subsingleton), then so does $H^1(\mathcal{O}_X(D'))$.*

Proof. The class isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ induces a K -linear equivalence $H^1(\mathcal{O}_X(D)) \simeq H^1(\mathcal{O}_X(D'))$ (??), and being a subsingleton transports across an equivalence. This is the form the consumers need: it survives isomorphisms and does not require finite-dimensionality, unlike an equality $h^1 = 0$. $\square$

Theorem 607 (Finite-dimensionality of cohomology is a Picard-class invariant). *Let D, D' be Weil divisors on X with $[\mathcal{O}_X(D)] = [\mathcal{O}_X(D')]$. If $H^0(\mathcal{O}_X(D))$ (respectively $H^1(\mathcal{O}_X(D))$) is a finite-dimensional K -module, then so is $H^0(\mathcal{O}_X(D'))$ (respectively $H^1(\mathcal{O}_X(D'))$).*

Proof. The class isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ induces, in each degree, a K -linear equivalence of cohomology modules (??); finite dimensionality transports across a linear equivalence. $\square$

6.21 The degree homomorphism over an extension field

Everything above applies to the challenge curve C over its base field k . We now record that the same degree apparatus is available over *every* field extension K/k , on the base-changed curve $C_K := (C \otimes \text{Spec } K)_{\text{left}}$, regarded as a K -scheme via the second projection $\text{pr}_2: C_K \rightarrow \text{Spec } K$. The point is that the geometric half of the standing hypotheses — proper, smooth of relative dimension one, geometrically irreducible — transports along base change, from which integrality and the finiteness of H^0 and H^1 re-derive over K ; so the degree homomorphism $\deg_{\text{Pic}}$ is defined on $\text{Pic}(C_K)$ with no further input.

Theorem 608 (The base-changed curve is a smooth proper curve over K). *Let K/k be a field extension. The base-changed curve C_K , as a scheme over $\text{Spec } K$ via the second projection, is smooth of relative dimension one, proper, quasi-compact, and geometrically irreducible over K .*

Proof. The second projection $C_K \rightarrow \text{Spec } K$ is the base change of the structure morphism $C \rightarrow \text{Spec } k$ along $\text{Spec } K \rightarrow \text{Spec } k$ (the curve product exhibits C_K as a fibre product, as for the relative curve of Definition ??). Smoothness of relative dimension one, properness, and geometric irreducibility are each stable under base change, so all three transport from C/k to C_K/K ; properness gives quasi-compactness. $\square$

Theorem 609 (The base-changed curve is integral). *For every field extension K/k , the base-changed curve C_K is an integral scheme.*

Proof. Being smooth, C_K is geometrically reduced over K ; it is geometrically irreducible over K by Theorem ???. A geometrically reduced and geometrically irreducible scheme over a field is integral (Proposition ??). This is the substance of the base change: geometric irreducibility and geometric reducedness of C/k yield integrality of $C \otimes_k K$ for *every* field extension K , a property the bare fibre product $C \otimes_k K$ need not have without them. $\square$

Theorem 610 (Finiteness of the cohomology of the base-changed curve). *For every field extension K/k , both $H^0(C_K, \mathcal{O}_{C_K/K})$ and $H^1(C_K, \mathcal{O}_{C_K/K})$ are finite-dimensional K -vector spaces.*

Proof. By Theorem ??, C_K is a smooth curve over K — proper, smooth of relative dimension one, geometrically irreducible. The curve-level finiteness results therefore apply over K : H^0 is finite-dimensional (Theorem ??) and H^1 is finite-dimensional (Theorem ??). $\square$

Remark 611 (The degree homomorphism over every extension field). Theorems ??, ?? and ?? discharge every hypothesis of the degree stack for C_K over K . Hence the degree homomorphism $\deg_{\text{Pic}}: \text{Pic}(C_K) \rightarrow \mathbb{Z}$ (Definition ??) is defined on the base-changed curve, for every field extension K/k .

6.22 Invariance of the degree under the base-field transition

Let k be the base field of the challenge curve C (proper, smooth of relative dimension one, geometrically irreducible), and let $\varphi : K_1 \rightarrow K_2$ be a k -algebra homomorphism between field extensions of k , with base-field transition $\pi : C_{K_2} \rightarrow C_{K_1}$ (Chapter ??, Section ??). Both ends carry the full degree apparatus (??). This section proves that the degree of a Čech Picard class is invariant under the transition,

$$\deg_{\text{Pic}}(\pi^*L) = \deg_{\text{Pic}}(L) \quad (L \in \text{Pic}(C_{K_1})),$$

and descends the degree to the relative Picard group over a field. The invariance is the algebraic form of the classical fact that the degree of a divisor does not change under extension of the base field: a closed point x' of C_{K_1} pulls back to the fibre divisor $\pi^*(x') = \sum_{x'' \mapsto x'} e_{x''} \cdot x''$, and the multiplicities and residue degrees rebalance so that $\sum e_{x''} [\kappa(x'') : K_2] = [\kappa(x') : K_1]$ — the *fiber-degree identity* proved below by comparing the colength of one chart algebra at both ends of the transition.

Lemma 612 (The degree of a power of a class). *Under the standing hypotheses of ??, for every class L and every integer $n \in \mathbb{Z}$,*

$$\deg_{\text{Pic}}(L^n) = n \cdot \deg_{\text{Pic}}(L).$$

Proof. The trivial class has degree zero: $1 = \mathcal{O}_X(0)$ and $\deg_K 0 = 0$ (??). By additivity (??), $\deg_{\text{Pic}}(L) + \deg_{\text{Pic}}(L^{-1}) = \deg_{\text{Pic}}(1) = 0$, so inversion negates the degree. Induct on n in both directions from $n = 0$, using additivity at each step. $\square$

Theorem 613 (The fiber-degree identity). *Let V be an affine chart of C_{K_1} containing the generic point, let $t \in \Gamma(C_{K_1}, V)$ be a section whose germ at η is the value of a unit $g \in K(C_{K_1})^\times$ and which is a unit at every point of V other than a fixed non-generic point $x' \in V$, and let S be a finite set of non-generic points of C_{K_2} , contained in $\pi^{-1}V$, outside which the pulled-back section $\pi^\sharp t$ is a unit on $\pi^{-1}V$. Then, in $\mathbb{Z}$,*

$$\sum_{x'' \in S} \text{ord}_{x''}(\pi^\sharp g) \cdot [\kappa(x'') : K_2] = \text{ord}_{x'}(g) \cdot [\kappa(x') : K_1].$$

(Mathematically, the left side is the sum over the fibre of x' : the hypotheses force S to exhaust the non-unit locus of $\pi^\sharp t$ on the chart, which is contained in $\pi^{-1}(x')$ since unit germs pull back to unit germs; points of S outside that locus contribute order zero. The statement is parameterised by such a finite S exactly so that the consumer may take for S the support of a divisor already in hand.)

Proof. Write $B_1 = \Gamma(C_{K_1}, V)$ and $B_2 = \Gamma(C_{K_2}, \pi^{-1}V)$. The chart $\pi^{-1}V$ is affine (the transition is an affine morphism, ??) and contains the generic point of C_{K_2} (the transition matches generic points, ??); both curves carry the standing bundle hypotheses over their fields (??, ??). By germ naturality (??), the germ of $\pi^\sharp t$ at the generic point of C_{K_2} is the value of the unit $\pi^\sharp g$.

The two colength dictionaries. The chart colength dictionary (??) applies at both ends. Over K_2 , with the finite set S (whose members lie in $\pi^{-1}V$, and outside which $\pi^\sharp t$ is a unit, by hypothesis):

$$\dim_{K_2}(B_2/(\pi^\sharp t)) = \sum_{x'' \in S} \text{ord}_{x''}(\pi^\sharp g) \cdot [\kappa(x'') : K_2].$$

Over K_1 , with the singleton $S = \{x'\}$ (the section t is a unit at every other point of V , by hypothesis):

$$\dim_{K_1}(B_1/(t)) = \text{ord}_{x'}(g) \cdot [\kappa(x') : K_1].$$

The colength bridge. Regard K_2 as a K_1 -algebra through φ . The sections base change (??) is a ring isomorphism $B_1 \otimes_{K_1} K_2 \cong B_2$ carrying $t \otimes 1$ to $\pi^\sharp t$ (??) and commuting with the K_2 -structures, hence carrying the principal ideal $(t \otimes 1)$ onto $(\pi^\sharp t)$ and inducing a K_2 -linear isomorphism of the quotients. By right-exactness of the tensor product, tensoring the exact sequence $B_1 \xrightarrow{t} B_1 \rightarrow B_1/(t) \rightarrow 0$ with K_2 gives $(B_1/(t)) \otimes_{K_1} K_2 \cong (B_1 \otimes_{K_1} K_2)/(t \otimes 1)$. Chaining,

$$(B_1/(t)) \otimes_{K_1} K_2 \cong B_2/(\pi^\sharp t) \quad (K_2\text{-linearly}),$$

and since extension of scalars along a field extension preserves dimension,

$$\dim_{K_2}(B_2/(\pi^\sharp t)) = \dim_{K_2}\left((B_1/(t)) \otimes_{K_1} K_2\right) = \dim_{K_1}(B_1/(t)).$$

Chaining the three displays gives the identity. $\square$

Lemma 614 (Pulled local equations are regular on integral schemes). *Let $f : Y \rightarrow X$ be a morphism of integral schemes with $f(\eta_Y) = \eta_X$, and let E be a local-equation system on X . Then the pulled-back equations of E (??) are automatically regular: their germs at every point are nonzerodivisors. No flatness of f is needed — integrality of both schemes discharges the regularity hypothesis of the pullback construction. (For the base-field transition this is strictly more general than the flat-pullback route: flatness of π is never consumed.)*

Proof. Fix a pulled equation $f^\sharp(e)$, e an equation of E on a member U , and a point z of $f^{-1}U$. The stalks of an integral scheme are domains, so it suffices to show the germ of $f^\sharp(e)$ at z is nonzero. Suppose it were zero. On an integral scheme, a section of a nonempty open is determined by its germ at any one point (the restriction maps between nonempty opens and the germ maps are injective, all sections embedding in the function field), so $f^\sharp(e) = 0$, and in particular its germ at η_Y vanishes. But by germ naturality (??) that germ is the image under the *injective* function-field map of the germ of e at η_X , which is a nonzerodivisor of the field $K(X)$ by regularity of E (??), i.e. nonzero. Contradiction. $\square$

Definition 615 (The pulled tracked point system). Let x' be a non-generic point of C_{K_1} , with tracked point equations at x' (??, from the tracked uniformizer data of ??). The *pulled tracked point system* is the pullback of this local-equation system along the transition π (??), with regularity discharged by integrality of both curves (??, ??, generic points matching by ??). Its *trivialising unit* is the pullback $\pi^\sharp(t) \in K(C_{K_2})^\times$ of the chosen uniformizer t at x' (??, ??).

Lemma 616 (The trivialising elements of the pulled point presentation). *The meromorphic presentation (??) of the pulled tracked point system has trivialising elements*

$$f_z = \begin{cases} \pi^\sharp(t), & \pi(z) = x', \\ 1, & \pi(z) \neq x', \end{cases}$$

at each point z of C_{K_2} .

Proof. The trivialising element at z is the germ at $\eta_{C_{K_2}}$ of the pulled equation indexed by z , which is the π -pullback of the equation of the point system at $\pi(z)$ (??). If $\pi(z) \neq x'$, that equation is the constant 1 (??), and 1 pulls back to 1, with germ 1. If $\pi(z) = x'$, the equation is the tracked section s , whose germ at $\eta_{C_{K_1}}$ is the chosen uniformizer t (??); by germ naturality (??) the germ at $\eta_{C_{K_2}}$ of $\pi^\sharp s$ is $\pi^\sharp(t)$. $\square$

Theorem 617 (E-iv: the degree is invariant under the base-field transition). *For every k -algebra homomorphism $\varphi : K_1 \rightarrow K_2$ of field extensions of k and every Čech Picard class L on C_{K_1} ,*

$$\deg_{\text{Pic}}(\pi^* L) = \deg_{\text{Pic}}(L),$$

where $\pi^* : \text{Pic}(C_{K_1}) \rightarrow \text{Pic}(C_{K_2})$ is pullback of Čech Picard classes along the base-field transition. No finiteness or separability of the extensions is assumed.

Proof. Reduction to a one-point divisor. The class map $\text{Div}(C_{K_1}) \rightarrow \text{Pic}(C_{K_1})$ is surjective (??), so $L = \mathcal{O}_{C_{K_1}}(D)$ for a Weil divisor D , a finite $\mathbb{Z}$ -combination of one-point divisors. Both sides of the claimed identity are additive in L : the class map and the pullback π^* are homomorphisms (??) and $\deg_{\text{Pic}}$ is additive with $\deg_{\text{Pic}}(L^n) = n \deg_{\text{Pic}}(L)$ (??). It therefore suffices to prove the identity for $L = \mathcal{O}_{C_{K_1}}(1 \cdot x')$, x' a non-generic point.

The pulled point class as a divisor class. The one-point class is the class of the tracked point equations at x' (??). Pulling back a local-equation system computes the pullback of its class (??, with regularity discharged as in ??), so $\pi^* L$ is the Picard class of the pulled tracked point system; and the class of any local-equation system is the class of the divisor of its presentation (??, ??). Writing D'' for the presentation divisor of the pulled system (??), the normalization of ?? turns both sides into divisor degrees:

$$\deg_{\text{Pic}}(\pi^* L) = \deg_{K_2}(D''), \quad \deg_{\text{Pic}}(L) = \deg_{K_1}(1 \cdot x') = [\kappa(x') : K_1]$$

(??).

The degree of the pulled point divisor. The coefficient of D'' at a point z is the order at z of the trivialising element f_z (??). Off the fibre of x' the element is 1 (??), of order zero; so D'' is supported in the finite set $S := \text{supp}(D'') \subseteq \pi^{-1}(x')$, and

$$\deg_{K_2}(D'') = \sum_{z \in S} \text{ord}_z(\pi^\sharp t) \cdot [\kappa(z) : K_2],$$

the coefficient on the fibre being the order of $\pi^\sharp(t)$ (??).

Closing with the fiber-degree identity. Choose an affine chart $V \ni x'$ inside the tracked neighbourhood of the uniformizer data (affine opens form a basis), containing the generic point, and restrict the tracked section s to it; the restriction still has germ t at η and is still a unit at every point of V other than x' (??). The hypotheses of the fiber-degree identity (??) hold with this chart, the unit $g = t$, and the finite set S : its members lie in $\pi^{-1}V$ (they lie in the fibre of $x' \in V$), and at a point z of $\pi^{-1}V$ outside S the pulled section is a unit — off the fibre because s is a unit at $\pi(z) \neq x'$ and unit germs pull back to unit germs; on the fibre because a non-unit germ at z would give $\text{ord}_z(\pi^\sharp t) \neq 0$ (a rational function that is the generic germ of a section non-invertible at z has nontrivial order there — the contrapositive of ??, through the discrete valuation of the stalk at z), putting z in the support S . The identity gives

$$\sum_{z \in S} \text{ord}_z(\pi^\sharp t) \cdot [\kappa(z) : K_2] = \text{ord}_{x'}(t) \cdot [\kappa(x') : K_1] = [\kappa(x') : K_1],$$

the uniformizer having order one (??). Chaining the displays proves the one-point case, and with it the theorem. $\square$

Lemma 618 (The degree vanishes on classes from the base field). *Let K/k be a field extension. Every Čech Picard class on C_K pulled back from $\text{Spec } K$ along the second projection is trivial; in particular its degree is zero.*

Proof. The underlying space of $\text{Spec } K$ is a single point, and the Čech Picard group of a scheme with one-point underlying space is trivial: every member of a pointed cover is an open containing the point, hence the whole space, so a unit cocycle (g_{xy}) lives on the cover all of whose members and overlaps are the total space; fixing a base point x_0 and setting $b_x := g_{xx_0}$, the cocycle identity gives $g_{xy} = g_{xx_0} g_{x_0y} = b_x b_y^{-1}$, a coboundary. So the pulled-back class is the pullback of the trivial class, which is trivial, of degree $\deg_{\text{Pic}}(1) = 0$ (??). $\square$

Definition 619 (The degree on the relative Picard group over a field). For a field extension K/k , recall the relative Picard group $\text{relPic}(C, \text{Spec } K) = \text{Pic}(C_K)/(\text{classes from Spec } K)$ (the quotient recalled at the head of Chapter ??). The degree homomorphism kills the classes from the base (??), so it descends along the quotient projection to the *relative degree*

$$\deg_{\text{rel}} : \text{relPic}(C, \text{Spec } K) \longrightarrow \mathbb{Z}, \quad \deg_{\text{rel}}([L]) = \deg_{\text{Pic}}(L),$$

a homomorphism of groups; the displayed anchor holds for every $L \in \text{Pic}(C_K)$.

Proof. A group homomorphism trivial on a subgroup factors uniquely through the quotient by it; the anchor is the factorization identity. $\square$

Theorem 620 (E-iv descends to the relative Picard group). *For every k -algebra homomorphism $\varphi : K_1 \rightarrow K_2$ of field extensions of k and every relative Picard class $x \in \text{relPic}(C, \text{Spec } K_1)$,*

$$\deg_{\text{rel}}(\varphi^*x) = \deg_{\text{rel}}(x),$$

where φ^* is restriction of relative Picard classes along $\text{Spec } \varphi$ (??): the degree of a relative Picard class does not depend on the field over which it is presented. This is the well-definedness core of the degree of an étale-locally-trivial class: a class defined over K_1 and read over the larger field K_2 has the same degree — its points split, but order times residue degree rebalances exactly (??).

Proof. Represent $x = [L]$ with $L \in \text{Pic}(C_{K_1})$. Restriction along $\text{Spec } \varphi$ is induced on the quotients by pullback along the whiskered transition (??), so $\varphi^*[L] = [\pi^*L]$. By the anchor of ?? and the invariance (??),

$$\deg_{\text{rel}}(\varphi^*[L]) = \deg_{\text{Pic}}(\pi^*L) = \deg_{\text{Pic}}(L) = \deg_{\text{rel}}([L]). \quad \square$$

6.23 Invariance of the degree under isomorphisms

The base-field invariance has a lighter sibling over a *single* field: the degree of a Čech Picard class is invariant under pullback along an isomorphism of curve bundles over one field K . The proof is the ladder of Section ?? with the transition replaced by the isomorphism, and every leg degenerates: the fibre of a closed point is a single point, the coefficient of the pulled point divisor there is +1 — the maximal-adic multiplicity transports along the stalk isomorphism, with no colength computation — and the residue degree transports along the induced isomorphism of residue fields.

Throughout this section K is a field and X is a curve bundle over K — integral, smooth of relative dimension one and quasi-compact over $\text{Spec } K$ — with generic point η_X ; the hypotheses on the second scheme W are stated per node.

Lemma 621 (Naturality of the algebra map on sections). *Let W, X be schemes over $\text{Spec } K$ and $f : W \rightarrow X$ a morphism of schemes over $\text{Spec } K$. For every open $U \subseteq X$ and every scalar*

$a \in K$, the pullback of sections along f carries the canonical image of a in $\Gamma(X, U)$ (??) to the canonical image of a in $\Gamma(W, f^{-1}U)$:

$$f^\sharp(\iota_U^X(a)) = \iota_{f^{-1}U}^W(a),$$

where ι denotes the algebra map on sections.

Proof. The algebra map into $\Gamma(X, U)$ is the composite of the map induced on global sections by the structure morphism of X with restriction to U (??). Pullback along f commutes with restriction (naturality of $f^\sharp$), so $f^\sharp(\iota_U^X(a))$ is the restriction to $f^{-1}U$ of $f^\sharp(\iota_X^X(a))$. And f lies over $\text{Spec } K$: the structure morphism of X composed with f is the structure morphism of W , so on global sections $f^\sharp \circ \iota_X^X = \iota_W^W$. Restricting to $f^{-1}U$ gives the claim. $\square$

Lemma 622 (Residue degrees transport along an isomorphism). *Let W, X be schemes over $\text{Spec } K$ and $f : W \rightarrow X$ an isomorphism of schemes over $\text{Spec } K$. Then for every point z of W ,*

$$[\kappa(z) : K] = [\kappa(f(z)) : K].$$

Proof. The stalk map of f at z is an isomorphism of local rings $\mathcal{O}_{X, f(z)} \rightarrow \mathcal{O}_{W, z}$, and a ring isomorphism of local rings carries the maximal ideal onto the maximal ideal (units correspond to units in both directions, and the maximal ideal of a local ring is the set of non-units); it therefore induces a ring isomorphism of residue fields $r : \kappa(f(z)) \rightarrow \kappa(z)$.

The isomorphism r intertwines the two K -structures (??): the K -structure of a residue field is the composite of the algebra map on global sections, the germ, and the residue map. By naturality of the algebra map on sections (??, at the full open) followed by the defining compatibility of the stalk map with germs, the stalk map of f carries the germ at $f(z)$ of the canonical image of $a \in K$ to the germ at z of its canonical image; passing to the residues, r carries the canonical image of a in $\kappa(f(z))$ to its canonical image in $\kappa(z)$.

An additive bijection between two K -vector spaces which is multiplicative and matches the two canonical copies of K is K -linear: $r(a \cdot m) = r(\bar{a} m) = r(\bar{a}) r(m) = \bar{a} r(m) = a \cdot r(m)$. So r is a K -linear equivalence, and the K -dimensions of $\kappa(f(z))$ and $\kappa(z)$ agree (??). $\square$

Definition 623 (The pulled tracked point system along a generic-to-generic morphism). Let x' be a non-generic point of X , with tracked point equations at x' (??, from the tracked uniformizer data of ??), and let W be an integral scheme over $\text{Spec } K$ and $f : W \rightarrow X$ a morphism with $f(\eta_W) = \eta_X$. The *pulled tracked point system* is the pullback of this local-equation system along f (??), with regularity discharged by integrality of both schemes (??). Its *trivialising unit* is the pullback $f^\sharp(t) \in K(W)^\times$ of the chosen uniformizer t at x' (??, ??). This is the construction of the pulled tracked point system of the base-field transition (??), for an arbitrary morphism mapping generic point to generic point.

Lemma 624 (The trivialising elements of the generic pulled point presentation). *In the situation of ??, the meromorphic presentation (??) of the pulled tracked point system has trivialising elements*

$$f_z = \begin{cases} f^\sharp(t), & f(z) = x', \\ 1, & f(z) \neq x', \end{cases}$$

at each point z of W .

Proof. The trivialising element at z is the germ at η_W of the pulled equation indexed by z , which is the f -pullback of the equation of the point system at $f(z)$ (??). If $f(z) \neq x'$, that equation is the constant 1 (??), and 1 pulls back to 1, with germ 1. If $f(z) = x'$, the equation is the tracked section s , whose germ at η_X is the chosen uniformizer t (??); by germ naturality (??) the germ at η_W of $f^\sharp s$ is $f^\sharp(t)$. $\square$

Lemma 625 (The pulled uniformizer has order one at the preimage point). *Let W be a further curve bundle over K — integral and smooth of relative dimension one over $\text{Spec } K$ — and let $f : W \rightarrow X$ be an isomorphism of schemes with $f(\eta_W) = \eta_X$. Let x' be a non-generic point of X and z a non-generic point of W with $f(z) = x'$. Then the trivialising unit of the pulled tracked point system has order one at z :*

$$\text{ord}_z(f^\sharp(t)) = 1.$$

Proof. Let s be the tracked section of the uniformizer data at x' , on its tracked neighbourhood $V \ni x'$, with germ $_{\eta_X}(s) = t$ (??); write $c \in \mathcal{O}_{X,x'}$ for the germ of s at x' , a nonzerodivisor of the stalk by regularity of the tracked data, hence nonzero.

Both orders are adic multiplicities of matched stalk elements. The order valuation at a closed point is the maximal-adic valuation of the stalk, a discrete valuation ring and a Dedekind domain (??, ??, ??); its value on a rational function which is the image of a nonzero germ under the localization of the stalk into the function field is the multiplicity of the maximal ideal in the prime factorization of the principal ideal generated by that germ (??, applied in the local Dedekind stalk). The uniformizer t is such an image: t is the image of c under $\mathcal{O}_{X,x'} \rightarrow K(X)$ (??), so $\text{ord}_{x'}(t)$ is the multiplicity of $\mathfrak{m}_{x'}$ in the factorization of (c) . On the other side, by germ naturality (??) the pullback $f^\sharp(t)$ is the germ at η_W of the pulled section $f^\sharp(s)$ on $f^{-1}V \ni z$, which in turn is the image under $\mathcal{O}_{W,z} \rightarrow K(W)$ of the germ of $f^\sharp(s)$ at z (?? again); and that germ is the image of c under the stalk map $\mathcal{O}_{X,x'} \rightarrow \mathcal{O}_{W,z}$ of f at z — the defining compatibility of the stalk map with germs. So $\text{ord}_z(f^\sharp t)$ is the multiplicity of $\mathfrak{m}_z$ in the factorization of the principal ideal generated by the image of c .

The multiplicity is an isomorphism invariant. Since f is an isomorphism, its stalk map at z is a ring isomorphism $e : \mathcal{O}_{X,x'} \rightarrow \mathcal{O}_{W,z}$ of local Dedekind domains. It carries the maximal ideal onto the maximal ideal (units correspond to units in both directions), hence carries $\mathfrak{m}_{x'}^n$ onto $\mathfrak{m}_z^n$ for every n , so $c \in \mathfrak{m}_{x'}^n$ if and only if $e(c) \in \mathfrak{m}_z^n$. Membership in the powers of the maximal ideal characterizes the multiplicity (??, applied in each stalk with the maximal ideal as the prime), so the two multiplicities agree.

Chaining, $\text{ord}_z(f^\sharp t) = \text{ord}_{x'}(t) = 1$, the uniformizer having order one (??). $\square$

Theorem 626 (The degree is invariant under isomorphisms: scheme form). *Let W and X both be curve bundles over the field K — integral, smooth of relative dimension one and quasi-compact over $\text{Spec } K$ — with H^0 and H^1 of their structure sheaves finite-dimensional over K , and let $f : W \rightarrow X$ be an isomorphism of schemes over $\text{Spec } K$. Then for every Čech Picard class L on X ,*

$$\deg_{\text{Pic}}(f^*L) = \deg_{\text{Pic}}(L).$$

Proof. This is the ladder of the base-field invariance (??) with the transition replaced by the isomorphism f , every leg degenerate. As an isomorphism, f is a homeomorphism on underlying spaces — in particular injective and surjective on points — and both spaces are irreducible, so it matches the generic points: $f(\eta_W) = \eta_X$ (??).

Reduction to a one-point divisor. The class map $\text{Div}(X) \rightarrow \text{Pic}(X)$ is surjective (??), so $L = \mathcal{O}_X(D)$ for a Weil divisor D , a finite $\mathbb{Z}$ -combination of one-point divisors. Both sides of the claimed identity are additive in L : the class map and the pullback f^* are homomorphisms (??) and $\deg_{\text{Pic}}$ is additive with $\deg_{\text{Pic}}(L^n) = n \deg_{\text{Pic}}(L)$ (??, ??). It therefore suffices to prove the identity for $L = \mathcal{O}_X(1 \cdot x')$, x' a non-generic point.

The pulled point class as a divisor class. The one-point class is the class of the tracked point equations at x' (??). Pulling back a local-equation system computes the pullback of its class (??, with regularity discharged as in ??), so f^*L is the Picard class of the pulled tracked point system; and the class of any local-equation system is the class of the divisor of its presentation

(??, ??). Writing D'' for the presentation divisor of the pulled system (??), the normalization of ?? turns both sides into divisor degrees:

$$\deg_{\text{Pic}}(f^*L) = \deg_K(D''), \quad \deg_{\text{Pic}}(L) = \deg_K(1 \cdot x') = [\kappa(x') : K]$$

(??).

The pulled point divisor is a single point. The coefficient of D'' at a non-generic point z of W is the order at z of the trivialising element f_z (??). Since f is injective on points, the only point of W mapping to x' is $z_0 := f^{-1}(x')$, which is non-generic: its image x' differs from $\eta_X = f(\eta_W)$. At $z \neq z_0$, $f(z) \neq x'$ and the trivialising element is 1 (??), of order zero; at z_0 the element is the pulled uniformizer $f^\sharp(t)$ (??), of order 1 (??). Hence $D'' = 1 \cdot z_0$ (??), of degree $\deg_K(D'') = [\kappa(z_0) : K]$ (??).

Closing with the residue transport. By ??, $[\kappa(z_0) : K] = [\kappa(f(z_0)) : K] = [\kappa(x') : K]$. Chaining the displays proves the one-point case, and with it the theorem. $\square$

Theorem 627 (The degree is invariant under isomorphisms of curve bundles). *Let K be a field and X, Y objects of $\text{Over}(\text{Spec } K)$ which are proper, smooth of relative dimension one and geometrically irreducible over K . Then for every isomorphism $e : X \cong Y$ of $\text{Over}(\text{Spec } K)$ and every Čech Picard class L on the underlying scheme of Y ,*

$$\deg_{\text{Pic}}(e^*L) = \deg_{\text{Pic}}(L),$$

where e^* is pullback of Čech Picard classes along the underlying isomorphism of schemes.

Proof. The hypotheses of the scheme form are discharged on both bundles by the curve pack: smoothness gives geometric reducedness (??), which together with geometric irreducibility gives integrality (??); properness includes quasi-compactness; and H^0 and H^1 of the structure sheaf are finite-dimensional over K (??, ??). The underlying morphism of e is an isomorphism of schemes over $\text{Spec } K$: the forgetful functor preserves isomorphisms, and every morphism of $\text{Over}(\text{Spec } K)$ commutes with the structure morphisms. Apply ?? to it and to L . $\square$

6.24 The degree seam of the same-carrier comparison

The base-field shuffle of the relative Picard group at an affine test (Section ??) reads one class on two presentations of one fiber: at a *field* test K in a scalar tower $k \rightarrow L \rightarrow K$, the fiber of the k -curve at K and the fiber of the base-changed L -curve at K are identified by the same-carrier comparison (??), an isomorphism of schemes over $\text{Spec } K$. This section shows that the degree over K reads the same integer on the two presentations — the invariance under isomorphisms of Section ?? fired at the comparison.

Throughout, C is the challenge curve over k — proper, smooth of relative dimension one and geometrically irreducible — and $k \rightarrow L \rightarrow K$ is a scalar tower of fields, with $\sigma : \text{Spec } L \rightarrow \text{Spec } k$ the induced morphism and $C_L := \sigma^*C$ the base-changed curve over L (??). We write $\text{Spec } K / \text{Spec } F$ for the affine test of K over a field F , and $\text{relPic}_L(C_L, K)$ for the relative Picard group of C_L at the affine test of K over L , as in Section ??.

Theorem 628 (Degree invariance along the same-carrier comparison). *Let $c : (C_L \otimes (\text{Spec } K / \text{Spec } L))_{\text{left}} \rightarrow (C \otimes (\text{Spec } K / \text{Spec } k))_{\text{left}}$ be the same-carrier comparison at the field test K (??). Then for every Čech Picard class Λ on the fiber $C_K = (C \otimes (\text{Spec } K / \text{Spec } k))_{\text{left}}$,*

$$\deg_{\text{Pic}}(c^*\Lambda) = \deg_{\text{Pic}}(\Lambda),$$

both degrees taken over K .

Proof. The two fibers satisfy every hypothesis of the scheme-form invariance engine. For the k -side fiber C_K : it is smooth of relative dimension one and quasi-compact over K through the second projection, and integral, with H^0 and H^1 of its structure sheaf finite-dimensional over K (??, ??, ??, at the extension K/k). For the L -side fiber: the base-changed curve C_L is itself proper, smooth of relative dimension one and geometrically irreducible over L — its underlying data is the fiber of C at L with the second projection (??, ?? at the extension L/k) — so the same three theorems, applied over the base field L to the curve C_L at the extension K/L , give the identical package for $(C_L \otimes (\text{Spec } K / \text{Spec } L))_{\text{left}}$ over K .

The comparison c is an isomorphism of schemes commuting with the two second projections (??), which are the structure morphisms of the two fibers over $\text{Spec } K$; so c is an isomorphism of curve bundles over $\text{Spec } K$. Apply the scheme-form invariance (??) to c and Λ . $\square$

Theorem 629 (The relative degree is invariant under the base-field shuffle). *Let $\text{sh}_K : \text{relPic}(C, \text{Spec } K) \rightarrow \text{relPic}_L(C_L, K)$ be the base-field shuffle at the field test K (??), and let the relative degree on the target be that of Definition ?? applied over the base field L to the curve C_L at the extension K/L (its hypotheses discharged as in ??). Then for every relative Picard class $x \in \text{relPic}(C, \text{Spec } K)$,*

$$\deg_{\text{rel}}(\text{sh}_K(x)) = \deg_{\text{rel}}(x) :$$

a relative Picard class of the k -curve at the field test K and its shuffled presentation on the base-changed curve read the same integer.

Proof. Represent $x = \pi(\Lambda)$ with $\Lambda \in \text{Pic}(C_K)$, the projection being surjective. The shuffle computes on cosets: $\text{sh}_K(x) = \pi(c^*\Lambda)$ (??). The anchor of Definition ??, on each side, reads the relative degree off the representative: $\deg_{\text{rel}}(\text{sh}_K(x)) = \deg_{\text{Pic}}(c^*\Lambda)$ and $\deg_{\text{rel}}(x) = \deg_{\text{Pic}}(\Lambda)$. These agree by ??.

6.25 Multiplicativity of the degree under a finite morphism

The invariance ladders of the two preceding sections generalize to the classical multiplicativity of the degree: pulling back a Čech Picard class along a finite morphism $f : X \rightarrow Y$ of curve bundles over one field K multiplies its degree by $n = [K(X) : K(Y)]$, the degree of the induced extension of function fields. This is Hartshorne's $\deg f^*D = \deg f \cdot \deg D$ for complete nonsingular curves, in bundle form. The proof is again the ladder of Section ??, with two new algebraic inputs on a Dedekind chart pair: the *generic-rank dictionary* identifying n with the rank of the chart extension, and the *fiber-degree identity* $\sum_z e_z f_z = n$ over each closed point, which replaces the colength bridge of the base-field transition.

Throughout, X and Y carry the hypotheses stated per node; η_X, η_Y denote the generic points.

Definition 630 (The chart square of a dominant morphism). Let $f : X \rightarrow Y$ be a morphism of schemes and $V \subseteq Y$ an open. Pullback of sections along f makes $\Gamma(X, f^{-1}V)$ an algebra over $\Gamma(Y, V)$, the *chart extension* of f at V . If moreover X and Y are integral with $f(\eta_X) = \eta_Y \in V$, the function-field map (??) makes $K(X)$ an algebra over $K(Y)$, and the chart extension followed by the germ at η_X makes $K(X)$ an algebra over $\Gamma(Y, V)$; these structures fit in a commuting square

$$\begin{array}{ccc} \Gamma(Y, V) & \longrightarrow & \Gamma(X, f^{-1}V) \\ \downarrow & & \downarrow \\ K(Y) & \longrightarrow & K(X) \end{array}$$

whose vertical maps are the germs at the generic points.

Proof. Note first $\eta_X \in f^{-1}V$, since $f(\eta_X) = \eta_Y \in V$. The right-then-down composite of the square is the chart extension followed by the germ at η_X ; the down-then-right composite is the germ at η_Y followed by the function-field map. They agree by germ naturality (??): the germ at η_X of a pulled-back section is the function-field image of the germ of the section at η_Y . $\square$

Lemma 631 (Module finiteness of the chart extension). *Let $f : X \rightarrow Y$ be a finite morphism of schemes and $V \subseteq Y$ an affine open. Then $\Gamma(X, f^{-1}V)$ is a finite module over $\Gamma(Y, V)$.*

Proof. Finiteness of a morphism is affine-local on the target, so the chart extension $\Gamma(Y, V) \rightarrow \Gamma(X, f^{-1}V)$ is an integral ring map and of finite type. An integral ring map of finite type is module-finite: each of the finitely many algebra generators is integral, so the image algebra is generated as a module by the finitely many monomials in the generators of degree below the degrees of their integral equations. $\square$

Lemma 632 (Injectivity of the chart extension). *Let $f : X \rightarrow Y$ be a morphism of integral schemes with $f(\eta_X) = \eta_Y$, and let $V \subseteq Y$ be an open containing η_Y . Then the chart extension $\Gamma(Y, V) \rightarrow \Gamma(X, f^{-1}V)$ is injective.*

Proof. Let $a, b \in \Gamma(Y, V)$ have equal pullbacks. Taking germs at η_X and applying germ naturality (??), the function-field map $K(Y) \rightarrow K(X)$ takes the germs of a and b at η_Y to equal values; the function-field map is injective (??), so the germs of a and b at η_Y agree; and on an integral scheme the germ map at the generic point is injective on sections of a nonempty open, so $a = b$. $\square$

Theorem 633 (The generic-rank dictionary). *Let $f : X \rightarrow Y$ be a finite morphism of integral schemes with $f(\eta_X) = \eta_Y$, and let $V \subseteq Y$ be an affine open containing η_Y . Then the degree of the function-field extension equals the rank of the chart extension:*

$$[K(X) : K(Y)] = \text{rank}_{\Gamma(Y, V)} \Gamma(X, f^{-1}V),$$

the rank of the finite torsion-free module $\Gamma(X, f^{-1}V)$ over the domain $\Gamma(Y, V)$ being the maximal number of linearly independent elements.

Proof. Write $B = \Gamma(Y, V)$ and $A = \Gamma(X, f^{-1}V)$, domains, since nonempty opens of integral schemes have integral section rings; A is a finite B -module (??) and $B \rightarrow A$ is injective (??). The germ maps at the generic points identify $K(Y)$ with the fraction field of B and $K(X)$ with the fraction field of A — on an integral scheme, the function field is the fraction field of the sections on any nonempty affine open — and under these identifications the square of ?? commutes, so the extension $K(Y) \subseteq K(X)$ is obtained from $B \subseteq A$ by passing to fractions.

Let $S = B \setminus \{0\}$, a multiplicative set of nonzerodivisors of A (its image avoids 0 by injectivity, and A is a domain). The localization $S^{-1}A = K(Y) \otimes_B A$ is a finite-dimensional $K(Y)$ -algebra, and it is a domain, being a subring of $K(X)$; a finite-dimensional algebra over a field which is a domain is a field (multiplication by a nonzero element is injective, hence surjective, by dimension). A field lying between A and its fraction field contains the inverse of every nonzero element of A , so $S^{-1}A = K(X)$.

Finally, rank passes to fractions: a family of elements of A is linearly independent over B if and only if it is linearly independent over $K(Y)$ in $S^{-1}A$ — one direction clears denominators, the other restricts scalars — so the rank of A over B is the dimension of $S^{-1}A = K(X)$ over $K(Y)$. $\square$

Theorem 634 (The fiber-degree identity for a finite morphism). *Let K be a field, X and Y integral schemes smooth of relative dimension one over $\text{Spec } K$, and $f : X \rightarrow Y$ a finite morphism over $\text{Spec } K$ with $f(\eta_X) = \eta_Y$. Let $V \ni \eta_Y$ be an affine chart of Y , let $t \in \Gamma(Y, V)$ be a section*

whose germ at η_Y is the value of a unit $g \in K(Y)^\times$, let $y \in V$ be a non-generic point with $\text{ord}_y(g) = 1$ at which the tracked section is normalized, t being a unit at every point of V other than y ; and let S be a finite set of non-generic points of X , contained in $f^{-1}V$, outside which the pulled section $f^\sharp t$ is a unit on $f^{-1}V$. Then, in $\mathbb{Z}$,

$$\sum_{z \in S} \text{ord}_z(f^\sharp g) \cdot [\kappa(z) : K] = [K(X) : K(Y)] \cdot [\kappa(y) : K].$$

(The hypotheses force S to exhaust the non-unit locus of $f^\sharp t$, which is contained in the fiber $f^{-1}(y)$; members of S outside that locus contribute order zero. The statement is parameterised by such a finite S so that the consumer may take for S the support of a divisor already in hand.)

Proof. Write $B = \Gamma(Y, V)$ and $A = \Gamma(X, f^{-1}V)$. The chart $f^{-1}V$ is affine (finite morphisms are affine) and contains η_X , so both B and A are Dedekind domains (??); A is a finite B -module (??) along the injective chart extension (??), hence torsion-free — A is a domain on which the nonzero elements of B act injectively. The chart extension commutes with the two K -algebra structures, since f lies over $\text{Spec } K$ (??). By germ naturality (??), the germ of $f^\sharp t$ at η_X is the value of the unit $f^\sharp g$. Under $V \cong \text{Spec } B$, a point z of the chart corresponds to a prime with the germ of a section at z becoming its image in the localization at that prime; so a section lies in the prime of z if and only if its germ at z is a non-unit. We use this dictionary on both charts (??, ??).

The tracked section cuts out the tracked prime. The point y corresponds to a nonzero prime $\mathfrak{p}$ of B (??), maximal since nonzero primes of a Dedekind domain are maximal. In the Dedekind domain B the nonzero ideal (t) factors into primes. Every prime factor $\mathfrak{P}$ of (t) contains t (??), so at the point of V corresponding to $\mathfrak{P}$ the germ of t is a non-unit; by the unit hypothesis that point is y , so $\mathfrak{P} = \mathfrak{p}$. The multiplicity of $\mathfrak{p}$ in (t) is $\text{ord}_y(g) = 1$ (??), so $(t) = \mathfrak{p}$. Extending along $B \rightarrow A$, which carries t to $f^\sharp t$,

$$(f^\sharp t) = \mathfrak{p}A \neq 0.$$

Each term is $e \cdot f$ times $[\kappa(y) : K]$. Let $z \in S$ contribute a nonzero term; then the germ of $f^\sharp t$ at z is a non-unit — a unit germ at z forces $\text{ord}_z(f^\sharp g) = 0$ (?: the order is the multiplicity of the prime of z in $(f^\sharp t)$, and a unit generates an ideal not contained in that prime; concretely, $f^\sharp t \notin \mathfrak{q}_z$). So the nonzero prime $\mathfrak{q} = \mathfrak{q}_z$ of A at z (??) contains $f^\sharp t$, hence $\mathfrak{q} \supseteq \mathfrak{p}A$, hence $\mathfrak{q} \cap B \supseteq \mathfrak{p}$; maximality of $\mathfrak{p}$ gives $\mathfrak{q} \cap B = \mathfrak{p}$: the prime $\mathfrak{q}$ lies over $\mathfrak{p}$. Write $e_{\mathfrak{q}}$ for the multiplicity of $\mathfrak{q}$ in the factorization of $\mathfrak{p}A$ (the ramification index) and $f_{\mathfrak{q}} = \dim_{B/\mathfrak{p}} A/\mathfrak{q}$ (the inertia degree). Then:

$$\text{ord}_z(f^\sharp g) = \text{multiplicity of } \mathfrak{q} \text{ in } (f^\sharp t) = e_{\mathfrak{q}} \quad (??),$$

and, using the residue $\text{leg } [\kappa(z) : K] = \dim_K A/\mathfrak{q}$ on both charts (??) and the tower $K \rightarrow B/\mathfrak{p} \rightarrow A/\mathfrak{q}$ — whose K -structures match by ?? —

$$[\kappa(z) : K] = \dim_K A/\mathfrak{q} = f_{\mathfrak{q}} \cdot \dim_K B/\mathfrak{p} = f_{\mathfrak{q}} \cdot [\kappa(y) : K].$$

Reindexing the sum over the primes over $\mathfrak{p}$. The assignment $z \mapsto \mathfrak{q}_z$ is a bijection from the members of S with nonzero term onto the primes of A over $\mathfrak{p}$: it is injective because the points–primes correspondence on the chart is (??); and every prime $\mathfrak{q}$ over $\mathfrak{p}$ arises — it is nonzero (its contraction is $\mathfrak{p} \neq 0$), so it corresponds to a non-generic point z of $f^{-1}V$ (??, ??) at which $f^\sharp t \in \mathfrak{p}A \subseteq \mathfrak{q}$ is a non-unit germ, so $z \in S$ by the exhaustion hypothesis, with term $e_{\mathfrak{q}} \cdot f_{\mathfrak{q}} \cdot [\kappa(y) : K] \neq 0$, the ramification index being at least 1 and the residue degree positive (??). Members of S with zero term contribute nothing. Hence

$$\sum_{z \in S} \text{ord}_z(f^\sharp g) \cdot [\kappa(z) : K] = \left(\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} \right) \cdot [\kappa(y) : K].$$

The *fundamental identity* $\sum ef = n$. Localize at $\mathfrak{p}$: over the discrete valuation ring $B_{\mathfrak{p}}$ the module $A_{\mathfrak{p}}$ is finite and torsion-free, hence free, of rank $r = \text{rank}_B A$ (rank is preserved by localization). Thus $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is an r -dimensional vector space over $B/\mathfrak{p}$. It is an Artinian ring whose primes are the (finitely many) primes of A over $\mathfrak{p}$, so it is the product of its localizations at them; the localization at $\mathfrak{q}$ is $A_{\mathfrak{q}}/\mathfrak{p}A_{\mathfrak{q}}$, and in the discrete valuation ring $A_{\mathfrak{q}}$ we have $\mathfrak{p}A_{\mathfrak{q}} = \mathfrak{q}^{e_{\mathfrak{q}}}A_{\mathfrak{q}}$, whose quotient filtration $A_{\mathfrak{q}} \supseteq \mathfrak{q}A_{\mathfrak{q}} \supseteq \cdots \supseteq \mathfrak{q}^{e_{\mathfrak{q}}}A_{\mathfrak{q}}$ has $e_{\mathfrak{q}}$ successive quotients, each one-dimensional over $\kappa(\mathfrak{q}) = A/\mathfrak{q}$, hence of dimension $f_{\mathfrak{q}}$ over $B/\mathfrak{p}$. Summing dimensions, $\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}} = r$. By the generic-rank dictionary (??), $r = [K(X) : K(Y)]$. Chaining the displays gives the identity. $\square$

Theorem 635 (Degree multiplicativity under pullback: scheme form). *Let X and Y be curve bundles over the field K — integral, smooth of relative dimension one and quasi-compact over $\text{Spec } K$ — with H^0 and H^1 of their structure sheaves finite-dimensional over K , and let $f : X \rightarrow Y$ be a finite morphism over $\text{Spec } K$ with $f(\eta_X) = \eta_Y$. Then for every Čech Picard class L on Y ,*

$$\deg_{\text{Pic}}(f^*L) = [K(X) : K(Y)] \cdot \deg_{\text{Pic}}(L).$$

Proof. Write $n = [K(X) : K(Y)]$.

Reduction to a one-point divisor. The class map $\text{Div}(Y) \rightarrow \text{Pic}(Y)$ is surjective (??), so $L = \mathcal{O}_Y(D)$ for a Weil divisor D , a finite $\mathbb{Z}$ -combination of one-point divisors. Both sides of the claimed identity are additive in L : the class map and the pullback f^* are homomorphisms (??) and $\deg_{\text{Pic}}$ is additive with $\deg_{\text{Pic}}(L^m) = m \deg_{\text{Pic}}(L)$ (??, ??). It therefore suffices to prove the identity for $L = \mathcal{O}_Y(1 \cdot x')$, x' a non-generic point of Y .

The pulled point class as a divisor class. The one-point class is the class of the tracked point equations at x' (??). Pulling back a local-equation system computes the pullback of its class (??, with regularity discharged as in ??), so f^*L is the Picard class of the pulled tracked point system along f (??, for the generic-to-generic morphism f); and the class of any local-equation system is the class of the divisor of its presentation (??, ??). Writing D'' for the presentation divisor of the pulled system (??), the normalization of ?? turns both sides into divisor degrees:

$$\deg_{\text{Pic}}(f^*L) = \deg_K(D''), \quad \deg_{\text{Pic}}(L) = \deg_K(1 \cdot x') = [\kappa(x') : K]$$

(??).

The degree of the pulled point divisor. The coefficient of D'' at a non-generic point z of X is the order at z of the trivialising element f_z (??). Off the fiber of x' the element is 1 (??), of order zero; so D'' is supported in the finite set $S := \text{supp}(D'') \subseteq f^{-1}(x')$, and on the fiber the element is the pulled uniformizer (??), so

$$\deg_K(D'') = \sum_{z \in S} \text{ord}_z(f^{\sharp}t) \cdot [\kappa(z) : K].$$

Closing with the fiber-degree identity. Choose an affine chart $V \ni x'$ inside the tracked neighbourhood of the uniformizer data (affine opens form a basis), containing the generic point (a nonempty open of an irreducible space), and restrict the tracked section s to it; the restriction still has germ t at η_Y and is still a unit at every point of V other than x' (??), and $\text{ord}_{x'}(t) = 1$ (??). The hypotheses of the fiber-degree identity (??) hold with this chart, the unit $g = t$, and the finite set S : its members lie in $f^{-1}V$ (they lie in the fiber of $x' \in V$), and at a non-generic point z of $f^{-1}V$ outside S the pulled section is a unit — off the fiber because s is a unit at $f(z) \neq x'$ and unit germs pull back to unit germs; on the fiber because a non-unit germ at z would give $\text{ord}_z(f^{\sharp}t) \neq 0$ (a rational function which is the generic germ of a section non-invertible

at z has nontrivial order there — the contrapositive of ??, through the discrete valuation of the stalk at z), putting z in the support S . The identity gives

$$\sum_{z \in S} \text{ord}_z(f^\sharp t) \cdot [\kappa(z) : K] = n \cdot [\kappa(x') : K].$$

Chaining the displays proves the one-point case, and with it the theorem. $\square$

Theorem 636 (Degree multiplicativity under pullback of curve bundles). *Let K be a field and D, E objects of $\text{Over}(\text{Spec } K)$ which are proper, smooth of relative dimension one and geometrically irreducible over K , and let $h : D \rightarrow E$ be a morphism of $\text{Over}(\text{Spec } K)$ whose underlying morphism is surjective and finite. Then for every Čech Picard class L on the underlying scheme of E ,*

$$\deg_{\text{Pic}}(h^*L) = [K(D) : K(E)] \cdot \deg_{\text{Pic}}(L).$$

Proof. The hypotheses of the scheme form are discharged on both bundles by the curve pack: smoothness gives geometric reducedness (?), which together with geometric irreducibility gives integrality (?); properness includes quasi-compactness; and H^0 and H^1 of the structure sheaves are finite-dimensional over K (?). The underlying morphism of h lies over $\text{Spec } K$ (every morphism of $\text{Over}(\text{Spec } K)$ commutes with the structure morphisms) and, being surjective, matches the generic points (?). Apply ??. $\square$

Theorem 637 (The constant leg: pullbacks along one-point morphisms have degree zero). *Let K be a field, D an object of $\text{Over}(\text{Spec } K)$ which is proper, smooth of relative dimension one and geometrically irreducible over K , and E an object of $\text{Over}(\text{Spec } K)$ with proper structure morphism. Let $h : D \rightarrow E$ be a morphism of $\text{Over}(\text{Spec } K)$ whose set-range is a single closed point x of E . Then for every Čech Picard class L on the underlying scheme of E ,*

$$\deg_{\text{Pic}}(h^*L) = 0.$$

Proof. The bundle D is integral (?), in particular reduced, and its H^0 and H^1 are finite-dimensional over K (?), so the degree homomorphism of ?? is available on D . The underlying morphism of h is proper (?: the structure morphism of D is proper and that of E is separated, properness including separatedness), in particular quasi-compact. By ??, $h^*L = 1$; and the degree of the trivial class is zero, $\deg_{\text{Pic}}$ being a homomorphism (?). $\square$

6.26 The fibre twist of a map to the projective line

Throughout this section Y is an integral scheme, smooth of relative dimension one and quasi-compact over a field K , equipped with a *dominant* morphism $\pi : Y \rightarrow \mathbb{P}^1$. Pulling the two standard charts of $\mathbb{P}^1$ back along π gives an affine two-cover of Y ; the pullback of the chart coordinate cuts out, on this cover, the divisor of a fibre of π . The class of n copies of that fibre is the *fibre twist* Θ_n , and the purpose of the section is to construct it as a class trivialised on the two-cover by construction and to record that its degree scales linearly in n .

Definition 638 (The pinned two-cover and its coordinate). Write $V_0 = \pi^{-1}(U_0)$ and $V_1 = \pi^{-1}(U_1)$ for the preimages of the two standard charts $U_i = D_+(X_i)$ of $\mathbb{P}^1$ (?); they cover Y (?). The pinned two-cover assigns to a point of Y the chart V_0 if the point lies in V_0 , and V_1 otherwise. Let

$$t_0 = \pi^\sharp(t_{01}) \in \Gamma(Y, V_0)$$

be the pullback along π of the chart-0 coordinate $t_{01} = X_1/X_0$ (??). Then the overlap of the two-cover is exactly the locus where t_0 is invertible:

$$V_0 \cap V_1 = D(t_0),$$

the basic open of t_0 ; consequently the restriction of t_0 to $V_0 \cap V_1$ is a unit of $\Gamma(Y, V_0 \cap V_1)$.

Proof. Since $U_0 \cup U_1 = \mathbb{P}^1$, the preimages cover Y (??), so a point outside V_0 lies in V_1 ; the assignment is therefore a pointed cover. On $\mathbb{P}^1$ the coordinate $t_{01} = X_1/X_0$ is a regular function on U_0 that is invertible precisely where X_1 also fails to vanish, that is, on $U_0 \cap U_1 = D_+(X_0 X_1)$ (??); in other words the basic open of t_{01} in U_0 is $U_0 \cap U_1$. Forming a basic open commutes with pullback along a morphism, so the locus where $t_0 = \pi^\sharp(t_{01})$ is invertible is $\pi^{-1}(U_0 \cap U_1) = V_0 \cap V_1$; that is, $V_0 \cap V_1 = D(t_0)$. By definition of the basic open a section is invertible at every point of its basic open, so t_0 restricts to a unit on $V_0 \cap V_1 = D(t_0)$. $\square$

Definition 639 (The fibre divisor of $n \cdot (\text{fibre})$). For $n \in \mathbb{N}$, the *fibre divisor of $n \cdot (\text{fibre of } \pi)$* is the local-equation system on the pinned two-cover (??) whose equation is t_0^n on the chart V_0 and the constant 1 on V_1 . It is a genuine system of local equations: each equation is regular, and on the overlap the two equations differ by the unit factor t_0^n .

Proof. We check regularity and the overlap condition. Regularity on V_1 is trivial, the equation being 1. On V_0 it suffices to show that the germ of t_0 at every point $y \in V_0$ is a nonzerodivisor of the stalk $\mathcal{O}_{Y,y}$, for then so is the germ of its power t_0^n . Since Y is integral, each stalk embeds in the function field $K(Y)$ through the localisation map, so a germ is a nonzerodivisor as soon as it is nonzero. The generic point η lies in the overlap $V_0 \cap V_1$: as π is dominant its image is dense, and the $\mathbb{P}^1$ -overlap $U_0 \cap U_1$ is a nonempty open (it is the range of an affine chart), so the preimage $V_0 \cap V_1$ is a nonempty open of the irreducible space Y and hence contains η . On $V_0 \cap V_1 = D(t_0)$ the germ of t_0 at η is a unit (??), in particular nonzero. For any $y \in V_0$, the germ of t_0 at η is the image of the germ at y under the localisation map $\mathcal{O}_{Y,y} \rightarrow K(Y)$ (??); were the germ at y zero, its image — the nonzero unit germ at η — would be zero, a contradiction. Hence the germ of t_0 at each $y \in V_0$ is nonzero, thus a nonzerodivisor. For the overlap condition, whenever a pair of cover members overlap the ratio of their equations is a unit: two V_0 -members or two V_1 -members give ratio 1, while a V_0 – V_1 overlap lies in $V_0 \cap V_1$, where the ratio is $t_0^n/1 = t_0^n$ (or its inverse), a unit since t_0 is (??). $\square$

Definition 640 (The fibre twist and its power law). The *fibre twist* $\Theta_n \in \text{Pic}(Y)$ is the Picard class of the fibre divisor of $n \cdot (\text{fibre})$:

$$\Theta_n = \mathcal{O}_Y(n \cdot \text{fibre}) = \text{picClass}(\text{fibre divisor } n)$$

(??). It obeys the power law $\Theta_{n+1} = \Theta_n \cdot \Theta_1$, with $\Theta_0 = 1$; hence $\Theta_n = \Theta_1^n$.

Proof. By construction Θ_n is the class of the unit Čech cocycle of transition units of the fibre divisor (??). Write $g_{xy}^{(n)}$ for its transition unit between the members at points x, y , the unique unit with $f_x^{(n)}|_\cap = g_{xy}^{(n)} \cdot f_y^{(n)}|_\cap$ on the overlap (??), where $f^{(n)}$ denotes the fibre equation of exponent n . Pointwise $f^{(n+1)} = f^{(n)} \cdot f^{(1)}$, the equation being $t_0^{n+1} = t_0^n t_0$ on V_0 and $1 = 1 \cdot 1$ on V_1 . Restricting to an overlap and using the defining equations,

$$g_{xy}^{(n+1)} f_y^{(n+1)}|_\cap = f_x^{(n+1)}|_\cap = f_x^{(n)}|_\cap f_x^{(1)}|_\cap = g_{xy}^{(n)} g_{xy}^{(1)} f_y^{(n)}|_\cap f_y^{(1)}|_\cap = g_{xy}^{(n)} g_{xy}^{(1)} f_y^{(n+1)}|_\cap;$$

the restriction of $f_y^{(n+1)}$ to the overlap is a nonzerodivisor (regularity), so cancelling it gives $g_{xy}^{(n+1)} = g_{xy}^{(n)} g_{xy}^{(1)}$. Thus the cocycle of exponent $n+1$ is the product of those of exponents n and

1; passing to classes, and using that the class map from unit cocycles to $\text{Pic}(Y)$ is multiplicative, $\Theta_{n+1} = \Theta_n \cdot \Theta_1$. The exponent-0 equation is the constant 1, with trivial cocycle, so $\Theta_0 = 1$. Induction on n yields $\Theta_n = \Theta_1^n$. $\square$

Theorem 641 (The degree of the fibre twist scales linearly). *Assume in addition that Y is smooth of relative dimension one and quasi-compact over K with finite-dimensional H^0 and H^1 , so that the degree homomorphism $\deg_{\text{Pic}}$ of ?? is available. Then the degree of the fibre twist scales linearly in n :*

$$\deg_{\text{Pic}} \Theta_n = n \cdot \deg_{\text{Pic}} \Theta_1.$$

Proof. By the power law $\Theta_n = \Theta_1^n$ (??) and additivity of the class degree (??), we argue by induction on n . The base case is $\deg_{\text{Pic}} \Theta_0 = \deg_{\text{Pic}} 1 = 0$. For the step,

$$\deg_{\text{Pic}} \Theta_{n+1} = \deg_{\text{Pic}} (\Theta_n \cdot \Theta_1) = \deg_{\text{Pic}} \Theta_n + \deg_{\text{Pic}} \Theta_1 = n \cdot \deg_{\text{Pic}} \Theta_1 + \deg_{\text{Pic}} \Theta_1 = (n+1) \cdot \deg_{\text{Pic}} \Theta_1. \quad \square$$

6.27 The fibre unit and the fibre Weil divisor

We now build the counterpart of the fibre twist on the side of the function field $K(Y)$ and of Weil divisors. The standing hypotheses are as before, with Y in addition locally of finite type over K ; the coordinate t_0 , read at the generic point, becomes a unit $u \in K(Y)^\times$, and the effective divisor of its zeros is the fibre Weil divisor F . Its zeros lie on the fibre of π over the point $[1 : 0]$, the locus $V_0 \setminus V_1$; this fibre is nonempty whenever π is surjective, in particular for a finite dominant π .

Definition 642 (The fibre unit). Let $t_1 = \pi^\sharp(t_{10}) \in \Gamma(Y, V_1)$ be the pullback of the chart-1 coordinate $t_{10} = X_0/X_1$ (??), a regular section on V_1 . On the overlap the two coordinates are mutually inverse,

$$t_0|_\cap \cdot t_1|_\cap = 1 \quad \text{in } \Gamma(Y, V_0 \cap V_1).$$

Consequently the germ $u := \text{germ}_\eta(t_0) \in K(Y)$ is a unit of the function field, the *fibre unit*; its inverse is $\text{germ}_\eta(t_1)$.

Proof. On $\mathbb{P}^1$ the two chart coordinates satisfy $t_0 t_{10} = (X_1/X_0)(X_0/X_1) = 1$ on the overlap $U_0 \cap U_1 = D_+(X_0 X_1)$ (??, ??). Pulling this identity back along π 's section map over the overlap gives $t_0|_\cap \cdot t_1|_\cap = 1$ in $\Gamma(Y, V_0 \cap V_1)$. The generic point η lies in the overlap (??), so taking germs at η — a ring homomorphism, compatible with restriction — yields $\text{germ}_\eta(t_0) \cdot \text{germ}_\eta(t_1) = 1$ in $K(Y)$. Hence $u = \text{germ}_\eta(t_0)$ is invertible with inverse $\text{germ}_\eta(t_1)$, a unit of $K(Y)$. $\square$

Theorem 643 (The order table of the fibre unit). *The principal divisor $\text{div}(u)$ of the fibre unit (??) has, at a closed point x , the following orders:*

- if $x \in V_0$, then $\text{ord}_x(u) \geq 0$ (no pole);
- if $x \in V_1$, then $\text{ord}_x(u) \leq 0$ (no zero);
- if $x \in V_0$ but $x \notin V_1$, then $\text{ord}_x(u) \geq 1$ (a genuine zero).

In particular u is a unit of the stalk $\mathcal{O}_{Y,x}$ exactly on the overlap, and its zeros lie on $V_0 \setminus V_1$, the fibre of π over $[1 : 0]$.

Proof. For $x \in V_0$, the value $u = \text{germ}_\eta(t_0)$ is the image, under the localisation map $\mathcal{O}_{Y,x} \rightarrow K(Y)$, of the germ of t_0 at x (?). The image in $K(Y)$ of a stalk element has no pole at x (?), so $\text{ord}_x(u) \geq 0$.

For $x \in V_1$, apply the same to $u^{-1} = \text{germ}_\eta(t_1)$ and the regular section t_1 on V_1 : $\text{ord}_x(u^{-1}) \geq 0$. As ord_x is a homomorphism $K(Y)^\times \rightarrow \mathbb{Z}$ (?), $\text{ord}_x(u) = -\text{ord}_x(u^{-1}) \leq 0$.

For $x \in V_0$ with $x \notin V_1$, the germ of t_0 at x is not a unit, since x lies outside $D(t_0) = V_0 \cap V_1$ (?); it is nonzero, its image in $K(Y)$ being the unit u . The stalk $\mathcal{O}_{Y,x}$ is a discrete valuation ring (?), in which a nonzero non-unit has strictly positive order; hence $\text{ord}_x(u) \geq 1$. $\square$

Definition 644 (The fibre Weil divisor). The *fibre Weil divisor* F is the effective part of $\text{div}(u)$: the Weil divisor (?) with coefficient $F_x = \max(\text{ord}_x(u), 0)$ at each closed point x . It is effective ($F \geq 0$); it vanishes on V_1 ($F_x = 0$ for $x \in V_1$); it agrees with $\text{div}(u)$ on V_0 ($F_x = \text{ord}_x(u)$ for $x \in V_0$); and its degree is nonnegative. If moreover some closed point $x_0 \in V_0$ lies outside V_1 — a point of the fibre over $[1 : 0]$ — then F is nonzero and has strictly positive degree.

Proof. Effectivity is immediate from $F_x = \max(\text{ord}_x(u), 0) \geq 0$. On V_1 one has $\text{ord}_x(u) \leq 0$ (?), so $F_x = \max(\text{ord}_x(u), 0) = 0$; on V_0 one has $\text{ord}_x(u) \geq 0$, so $F_x = \text{ord}_x(u)$. The degree $\deg_K F = \sum_x F_x [\kappa(x) : K]$ (?) is a finite sum of nonnegative terms, each $F_x \geq 0$ and each residue degree nonnegative, hence $\deg_K F \geq 0$. Given a witness $x_0 \in V_0 \setminus V_1$: there $F_{x_0} = \text{ord}_{x_0}(u) \geq 1$ (?), so $F \neq 0$; and the degree, again a sum of nonnegative terms, is bounded below by the single term $F_{x_0} [\kappa(x_0) : K] \geq 1 \cdot [\kappa(x_0) : K] > 0$, using positivity of the residue degree (?). Hence $\deg_K F > 0$. $\square$

6.28 The twist lattices on the two-cover

Fix a Weil divisor D on Y . The twisted divisors $D + nF$ have divisor sheaves whose sections over the two charts and their overlap are K -submodules of the function field $K(Y)$. We record how these section lattices vary with n : fixed over V_1 and over the overlap, growing over V_0 ; from this the two-cover denominator lattices form an increasing chain that exhausts the fixed overlap module $N = \mathcal{O}_Y(D)(V_0 \cap V_1)$, by clearing poles at the finite fibre over $[1 : 0]$.

Theorem 645 (The twist leaves the lattice fixed off V_0). *For every Weil divisor D and every $n \in \mathbb{N}$, the sections of $\mathcal{O}_Y(D + nF)$ coincide with those of $\mathcal{O}_Y(D)$ over V_1 and over the overlap $V_0 \cap V_1$:*

$$\mathcal{O}_Y(D + nF)(V_1) = \mathcal{O}_Y(D)(V_1), \quad \mathcal{O}_Y(D + nF)(V_0 \cap V_1) = \mathcal{O}_Y(D)(V_0 \cap V_1).$$

Proof. The fibre divisor F vanishes at every point of V_1 (?), so at each closed point $x \in V_1$ the coefficient of the twist is $(D + nF)_x = D_x + nF_x = D_x$. The overlap $V_0 \cap V_1$ is contained in V_1 , so the same equality holds there. The sections of $\mathcal{O}_Y(A)$ over an open U are the rational functions whose pole order at each closed point of U is bounded by A (?); since the bounds imposed by $D + nF$ and by D agree at every closed point of V_1 (respectively of the overlap), the two section modules coincide. $\square$

Theorem 646 (The twist grows the lattice over V_0). *For every Weil divisor D and every $n \in \mathbb{N}$, multiplication by u^{-n} carries $\mathcal{O}_Y(D)(V_0)$ isomorphically onto $\mathcal{O}_Y(D + nF)(V_0)$; that is,*

$$\mathcal{O}_Y(D + nF)(V_0) = u^{-n} \cdot \mathcal{O}_Y(D)(V_0)$$

as K -submodules of $K(Y)$.

Proof. On V_0 the fibre divisor agrees with $\text{div}(u)$ (??), so at each closed point $x \in V_0$ one has $(D+nF)_x = D_x + n \text{ord}_x(u)$. Since $\text{div}(u^{-n}) = -n \text{div}(u)$ (the principal divisor is a homomorphism, ??), this reads $(D+nF)_x = D_x - \text{ord}_x(u^{-n}) = (D - \text{div}(u^{-n}))_x$. Hence $\mathcal{O}_Y(D+nF)(V_0) = \mathcal{O}_Y(D - \text{div}(u^{-n}))(V_0)$. By the bound-shift (??) with $g = u^{-n}$, a rational function h has poles bounded by D on V_0 if and only if $u^{-n}h$ has poles bounded by $D - \text{div}(u^{-n})$ on V_0 ; equivalently, multiplication by u^{-n} is a K -linear bijection from $\mathcal{O}_Y(D)(V_0)$ onto $\mathcal{O}_Y(D - \text{div}(u^{-n}))(V_0) = \mathcal{O}_Y(D+nF)(V_0)$. As F is effective, these submodules increase with n . $\square$

Definition 647 (The denominator lattice and its exhaustion). The *denominator lattice* A_n of $\mathcal{O}_Y(D+nF)$ is the join, inside $K(Y)$, of the section lattices over the two charts:

$$A_n = \mathcal{O}_Y(D+nF)(V_0) + \mathcal{O}_Y(D+nF)(V_1) \quad (\text{as } K\text{-submodules of } K(Y)).$$

It lies in the fixed ambient $N := \mathcal{O}_Y(D)(V_0 \cap V_1)$, the lattices form an increasing chain, and they exhaust N :

$$A_n \leq N, \quad A_n \leq A_{n+1}, \quad \bigsqcup_n A_n = N.$$

Proof. Containment in N . For an inclusion of nonempty opens $W \subseteq W'$, the section module $\mathcal{O}_Y(A)(W')$ is contained in $\mathcal{O}_Y(A)(W)$, a bound over more points being more restrictive (??). Applying this to $V_0 \cap V_1 \subseteq V_0$ and $V_0 \cap V_1 \subseteq V_1$, and using $\mathcal{O}_Y(D+nF)(V_0 \cap V_1) = N$ (??), both $\mathcal{O}_Y(D+nF)(V_0)$ and $\mathcal{O}_Y(D+nF)(V_1)$ are contained in N ; hence so is their join A_n .

The chain is increasing. F is effective (??), so $D+nF \leq D+(n+1)F$; the section lattice of a larger divisor contains that of a smaller one over any fixed open (??), over both V_0 and V_1 , whence $A_n \leq A_{n+1}$.

Exhaustion. The inclusion $\bigsqcup_n A_n \leq N$ is the containment just proved. For the reverse, let $s \in N$; we find n with $s \in \mathcal{O}_Y(D+nF)(V_0) \subseteq A_n$. If $s = 0$ it lies in every A_n . Otherwise s is a unit of $K(Y)$, and both D and $\text{div}(s)$ are supported on a finite set of closed points (??, ??). Consider a closed point $x \in V_0$. If also $x \in V_1$, then $F_x = 0$ (??) and membership $s \in N$ gives $-\text{ord}_x(s) \leq D_x = (D+nF)_x$. If $x \notin V_1$, then $F_x \geq 1$ (??, via ??); choosing n at least the maximum, over the finitely many exceptional points, of the pole deficit $-\text{ord}_x(s) - D_x$ makes $nF_x \geq -\text{ord}_x(s) - D_x$, that is $-\text{ord}_x(s) \leq D_x + nF_x = (D+nF)_x$; at the remaining points both D_x and $\text{ord}_x(s)$ vanish and the bound is automatic. Thus for this n , s has poles bounded by $D+nF$ at every closed point of V_0 , so $s \in \mathcal{O}_Y(D+nF)(V_0) \subseteq A_n \subseteq \bigsqcup_m A_m$. Hence $N \leq \bigsqcup_n A_n$, and equality holds. $\square$

6.29 Quasi-coherence of the divisor sheaves

The two-cover computation of H^1 with general coefficients (Theorem ??) runs on any sheaf of K -modules whose degree-one cohomology vanishes over each affine chart of the cover. For the divisor sheaves $\mathcal{O}_X(D)$ this vanishing follows, exactly as for the structure sheaf, from a quasi-coherence packaging (Definition ??). This section supplies that packaging on every open, so the twisted affine vanishing fires on every $\mathcal{O}_X(D)$.

Theorem 648 (The divisor sheaves are quasi-coherently packaged). *On the curve bundle X , for every Weil divisor D the divisor sheaf $\mathcal{O}_X(D)$, viewed as a sheaf of K -modules, carries a quasi-coherence packaging on every open U : an ambient section $r \in \Gamma(X, U)$ acts on $\mathcal{O}_X(D)(W)$, $W \leq U$, by multiplication with its germ at the generic point η .*

Proof. When $\eta \notin U$ all section modules over $W \leq U$ are zero (the sheaf is guarded by the generic point) and every packaging axiom is trivial. Assume $\eta \in U$.

(P1) *action*. For $r \in \Gamma(X, U)$ the germ $\text{germ}_\eta(r)$ is integral at every closed point $x \in U$ — its order is ≤ 1 , no pole (Lemma ??) — so multiplying a section of $\mathcal{O}_X(D)$ by it can only improve the pole bounds: the product again lies in $\mathcal{O}_X(D)(W)$ (Definition ??). The module laws over $\Gamma(X, U)$ and naturality under restriction hold because restriction of $\mathcal{O}_X(D)$ -sections is the inclusion of submodules of the function field $K(X)$, on which the action is ordinary multiplication.

(P2a) *denominator clearing*. Let $g \in \Gamma(X, U)$ and $c \in \mathcal{O}_X(D)(V \cap D(g))$ with $V \cap D(g)$ nonempty. On $D(g)$ the germ $\text{germ}_\eta(g)$ has order 0 (Lemma ??), while off $D(g)$, at a zero of g , it has order ≥ 1 ; both D and $\text{div}(c)$ are finitely supported (Definitions ??, ??). Choosing N at least the finite pole-deficit budget over the exceptional points, the section $\text{germ}_\eta(g)^N \cdot c$ has poles bounded by D on all of V : at points of $D(g)$ the factor is a unit and the bound is inherited from the overlap, at zeros of g the N -fold vanishing clears the pole, elsewhere all orders vanish. Thus $c = (g^N \cdot e)|_{V \cap D(g)}$ with $e = \text{germ}_\eta(g)^N \cdot c \in \mathcal{O}_X(D)(V)$.

(P2b) *defect annihilation*. Restrictions between nonempty opens are injective on the underlying rational functions, so a section killed on $V \cap D(g)$ is already zero when $V \cap D(g)$ is nonempty; and when $V \cap D(g) = \emptyset$ while V is nonempty, the germ $\text{germ}_\eta(g)$ is a non-unit of the function field, hence 0, so g itself annihilates. Either way a power of g kills the defect. $\square$

Theorem 649 (Twisted affine vanishing for divisor sheaves). *For every affine open U of the curve bundle X and every Weil divisor D , the degree-one cohomology of U with coefficients in $\mathcal{O}_X(D)$ vanishes: $H^1(U, \mathcal{O}_X(D)) = 0$.*

Proof. Apply the twisted affine vanishing ?? to the quasi-coherence packaging of $\mathcal{O}_X(D)$ on U supplied by Theorem ??. $\square$

6.30 Fibrewise large-twist vanishing

We now fix a dominant affine morphism $\pi: Y \rightarrow \mathbb{P}^1$ from the curve bundle Y and its fibre divisor $F = \text{fiberWeilDivisor}(\pi)$ (Definition ??), with the affine two-cover $V_0 = \pi^{-1}D_+(X_0)$, $V_1 = \pi^{-1}D_+(X_1)$. The main theorem computes the twisted cohomology $H^1(\mathcal{O}_Y(D + nF))$ as a quotient of the fixed overlap lattice by the fibre lattice, then runs the Noetherian stabilisation of the exhausting chain to conclude vanishing for all large n .

Definition 650 (The fibre lattice inside the overlap, and the twisted H^1). Write $N = \mathcal{O}_Y(D)(V_0 \cap V_1)$ for the fixed ambient overlap module and A_n for the denominator lattice of $\mathcal{O}_Y(D + nF)$ (Definition ??). Viewed inside N through the inclusion $N \hookrightarrow K(Y)$, the lattices A_n form an increasing chain of submodules $\overline{A}_n \leq N$ exhausting N , $\bigsqcup_n \overline{A}_n = N$. Moreover the degree-one cohomology of the twisted sheaf is the quotient by this chain:

$$H^1(\mathcal{O}_Y(D + nF)) \cong N / \overline{A}_n \quad (K\text{-linearly}).$$

Proof. Containment, monotonicity and exhaustion of $\overline{A}_n$ inside N are those of A_n (Definition ??) read through the inclusion of N , the comap of a submodule along the subtype inclusion. For the isomorphism, the two-cover computation (Theorem ??) applies to $\mathcal{O}_Y(D + nF)$ on the cover V_0, V_1 : its two affine vanishing hypotheses $H^1(V_i, \mathcal{O}_Y(D + nF)) = 0$ are supplied by Theorem ??, giving

$$H^1(\mathcal{O}_Y(D + nF)) \cong \mathcal{O}_Y(D + nF)(V_0 \cap V_1) / \text{range}(\text{restriction-difference}).$$

The restriction-difference map (Definition ??) reads on underlying rational functions as the literal difference in $K(Y)$, so its range is exactly the join $A_n = \mathcal{O}_Y(D + nF)(V_0) + \mathcal{O}_Y(D + nF)(V_1)$ inside the overlap module. Finally the overlap module is constant in n : $\mathcal{O}_Y(D + nF)(V_0 \cap V_1) = \mathcal{O}_Y(D)(V_0 \cap V_1) = N$ (Theorem ??). Substituting identifies the cokernel with $N / \overline{A}_n$. $\square$

Theorem 651 (Noetherian stabilisation of an exhausting chain). *Let M be a module over a ring R and $(A_n)_{n \in \mathbb{N}}$ an increasing chain of submodules with $\bigsqcup_n A_n = M$. If the quotient M/A_0 is Noetherian, then the chain is eventually everything: there is n_0 with $A_n = M$ for all $n \geq n_0$.*

Proof. Push the chain into the Noetherian quotient M/A_0 : the images $\overline{A_n} = A_n/A_0$ form an increasing chain of submodules of a Noetherian module, hence stabilise at some n_0 , $\overline{A_n} = \overline{A_{n_0}}$ for $n \geq n_0$. Their join is the image of $\bigsqcup_n A_n = M$, namely all of M/A_0 ; since the stabilised value dominates every term, it equals the whole quotient, $\overline{A_{n_0}} = M/A_0$. Comapping back along the quotient map $M \rightarrow M/A_0$ and using $A_0 \leq A_n$ gives $A_n = M$ for all $n \geq n_0$. $\square$

Theorem 652 (Fibrewise large-twist vanishing). *Assume $H^0(Y, \mathcal{O}_{Y/K})$ and $H^1(Y, \mathcal{O}_{Y/K})$ are finite-dimensional over K . Then for every Weil divisor D on Y there is n_0 with*

$$H^1(\mathcal{O}_Y(D + nF)) = 0 \quad \text{for all } n \geq n_0,$$

where F is the fibre divisor of the dominant affine $\pi: Y \rightarrow \mathbb{P}^1$. The bound n_0 is per-class; the argument uses no duality and no higher direct image.

Proof. Through the identification $H^1(\mathcal{O}_Y(D + nF)) \cong N/\overline{A_n}$ (Definition ??) it suffices to show $\overline{A_n} = N$ for all large n . The chain $\overline{A_n}$ is increasing and exhausts N (the exhaustion of Definition ??, by pole clearing at the finite fibre over $[1 : 0]$). Its base quotient $N/\overline{A_0} \cong H^1(\mathcal{O}_Y(D))$ is finite-dimensional: every divisor sheaf has finite H^1 under the standing structure-sheaf finiteness hypotheses, by dévissage (Theorem ??). A finite-dimensional K -module is Noetherian, so the stabilisation Theorem ?? gives n_0 with $\overline{A_n} = N$ for $n \geq n_0$; then $N/\overline{A_n} = 0$ and the twisted H^1 vanishes. $\square$

Theorem 653 (Fibrewise large-twist vanishing along a finite map). *Let $\pi: Y \rightarrow \mathbb{P}^1$ be a finite dominant morphism compatible with the structure morphisms, and assume $H^0(Y, \mathcal{O}_{Y/K})$ is finite-dimensional. Then for every Weil divisor D there is n_0 with $H^1(\mathcal{O}_Y(D + nF)) = 0$ for all $n \geq n_0$.*

Proof. Finiteness of π makes $H^1(Y, \mathcal{O}_{Y/K})$ finite-dimensional (Theorem ??), discharging the remaining finiteness hypothesis of Theorem ?? $\square$

6.31 Effective witnesses and peeling

This section supplies the three remaining inputs of the uniform vanishing theorems: the strict positivity of the fiber divisor's degree, the realisation of every class of large degree by an *effective* divisor, and the transport of H^1 -vanishing up an effective divisor. Throughout, Y is integral, smooth of relative dimension one, locally of finite type and quasi-compact over a field K , with finite-dimensional H^0 and H^1 where degrees are used.

Lemma 654 (Class arithmetic). *For a Weil divisor D and $n \in \mathbb{N}$: $\mathcal{O}_Y(nD) = \mathcal{O}_Y(D)^n$, and for a Picard class L : $\deg_{\text{Pic}}(L^n) = n \cdot \deg_{\text{Pic}}(L)$. Moreover the divisor-bound comparison at a point is the pointwise coefficient comparison in $\mathbb{Z}$.*

Proof. Induction on n from additivity of the class map (Theorem ??) and of the degree (Theorem ??); the bound comparison unfolds the definition of the valuation bound to its integer coefficient. $\square$

Lemma 655 (A point of $\mathbb{P}^1$ off the second chart). *For any field k there is a point of $\mathbb{P}_k^1$ lying in the chart $D_+(X_0)$ but not in $D_+(X_1)$ — the origin $[1 : 0]$.*

Proof. On the chart $D_+(X_0) = \text{Spec } k[X_1/X_0]$ (Definitions ??, ??), the preimage of $D_+(X_1)$ is the basic open of the chart coordinate X_1/X_0 (Definition ??). The chart coordinate is not a unit (it corresponds to the polynomial variable), so the ideal it generates is proper and is contained in a maximal ideal M . The image of M under the chart embedding (Definition ??, with range $D_+(X_0)$ by Lemma ??) lies in $D_+(X_0)$ but not in the basic open of the coordinate, hence not in $D_+(X_1)$. $\square$

Lemma 656 (A finite dominant morphism is surjective). *A finite dominant morphism of schemes is surjective.*

Proof. Finite morphisms are closed maps, so the range is closed; dominance makes it dense; a closed dense subset is everything. $\square$

Theorem 657 (Strict positivity of the fiber divisor's degree). *For a finite dominant $\pi: Y \rightarrow \mathbb{P}^1$, the fiber Weil divisor $F = \text{fiberWeilDivisor}(\pi)$ has strictly positive degree: $0 < \deg F$.*

Proof. Choose a point $z \in D_+(X_0) \setminus D_+(X_1)$ of $\mathbb{P}^1$ (Lemma ??). By surjectivity of the finite dominant π (Lemma ??) there is $x_0 \in Y$ over z ; it lies in the fiber chart V_0 but not in V_1 , and it is not the generic point (the generic point maps into the overlap). Such a point is a pole witness: the fiber divisor has strictly positive coefficient at some point off V_1 , which forces $0 < \deg F$ (the pole-existence estimate packaged with Definition ??). $\square$

Lemma 658 (An effective witness of a class of large degree). *If a Weil divisor W satisfies $1 \leq \deg W + \chi(\mathcal{O}_Y)$, then its Picard class is realised by an effective divisor: there is $E \geq 0$ with $\mathcal{O}_Y(E) = \mathcal{O}_Y(W)$.*

Proof. By the Riemann inequality (Theorem ??), $h^0(\mathcal{O}_Y(W)) \geq \deg W + \chi(\mathcal{O}_Y) \geq 1$, so $H^0(\mathcal{O}_Y(W))$ is nontrivial. A nonzero global section is, through the degree-zero identification (Definition ??) and the section description of the divisor sheaf (Definitions ??, ??), a nonzero rational function $f \in K(Y)^\times$ with $\text{ord}_x(f) \geq -\text{coeff}_x(W)$ at every point (Definition ??). Set $E := W + \text{div}(f)$ (Definition ??): the order bound says exactly $E \geq 0$ coefficientwise, and principal divisors have trivial class (Theorem ??), so $\mathcal{O}_Y(E) = \mathcal{O}_Y(W)$. $\square$

Lemma 659 (Peeling one point). *Let x be a closed point and B a Weil divisor with $H^1(\mathcal{O}_Y(B)) = 0$ (i.e. a subsingleton). Then $H^1(\mathcal{O}_Y(B + x)) = 0$, and by iteration $H^1(\mathcal{O}_Y(B + nx)) = 0$ for all $n \in \mathbb{N}$.*

Proof. Set $D := B + x$; the dévissage divisor of D at x is B (Lemma ??). The dévissage short exact sequence (Definition ??, Theorem ??) $0 \rightarrow \mathcal{O}_Y(B) \rightarrow \mathcal{O}_Y(D) \rightarrow \text{sky}_x \rightarrow 0$ has skyscraper cokernel, whose H^1 vanishes (Theorem ??); the long exact sequence therefore gives a surjection $H^1(\mathcal{O}_Y(B)) \twoheadrightarrow H^1(\mathcal{O}_Y(D))$ (Theorem ??), and a quotient of a subsingleton is a subsingleton. Iterate for nx . $\square$

Lemma 660 (Effective peeling). *Let A be a Weil divisor with $H^1(\mathcal{O}_Y(A)) = 0$ and $E \geq 0$ an effective divisor. Then $H^1(\mathcal{O}_Y(A + E)) = 0$.*

Proof. Induction on the size of the (finite) support of E . If the support is empty, $E = 0$. Otherwise pick p in the support, with coefficient $b > 0$; write $E = bp + E'$ where $E' = E - bp \geq 0$ has support one point smaller. By the induction hypothesis $H^1(\mathcal{O}_Y(A + E')) = 0$, and peeling the multiplicity block bp (Lemma ??) gives $H^1(\mathcal{O}_Y(A + E' + bp)) = H^1(\mathcal{O}_Y(A + E)) = 0$. $\square$

Theorem 661 (The class form of the fibrewise large-twist vanishing). *Let $\pi: Y \rightarrow \mathbb{P}^1$ be finite dominant, compatible with the structure morphisms, and l, θ Picard classes with $1 \leq \deg_{\text{Pic}} \theta$. Then there is $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ and every Weil divisor D of class $\mathcal{O}_Y(D) = l \cdot \theta^n$,*

$$H^1(\mathcal{O}_Y(D)) = 0.$$

Proof. Realise l by a divisor D_0 (Definition ??). Run the fibrewise vanishing at D_0 (Theorem ??): there is m_0 with $H^1(\mathcal{O}_Y(D_0 + mF)) = 0$ for all $m \geq m_0$, F the fiber divisor; fix $m = m_0$. Let $d' := \deg F > 0$ (Theorem ??) and $\varphi := \mathcal{O}_Y(F)$. Take $n_0 := \max(0, m_0 d' + 1 - \chi(\mathcal{O}_Y))$.

For $n \geq n_0$, the residual class $c := \theta^n \varphi^{-m_0}$ has degree $n \deg \theta - m_0 d' \geq n - m_0 d' \geq 1 - \chi(\mathcal{O}_Y)$ (Lemma ??, Theorem ??, Lemma ??, using $\deg \theta \geq 1$). Realise c by a divisor W ; then $1 \leq \deg W + \chi(\mathcal{O}_Y)$, so c has an effective witness E (Lemma ??). The divisor $D_0 + m_0 F + E$ has class $l \cdot \varphi^{m_0} \cdot \theta^n \varphi^{-m_0} = l \theta^n$; its H^1 vanishes by effective peeling from the base vanishing at $D_0 + m_0 F$ (Lemma ??). Finally, witness-independence (Theorem ??) transports the vanishing from this one witness to *every* divisor D of the class $l \theta^n$. $\square$

Theorem 662 (The rank anchor). *For a Weil divisor D with $H^1(\mathcal{O}_Y(D)) = 0$,*

$$h^0(\mathcal{O}_Y(D)) = \deg D + \chi(\mathcal{O}_Y).$$

(Riemann–Roch with vanishing h^1 ; with $\chi(\mathcal{O}_Y) = 1 - g$ this is the classical $h^0(D) = \deg D + 1 - g$.)

Proof. Under the vanishing, $\chi(\mathcal{O}_Y(D)) = h^0(\mathcal{O}_Y(D))$ (Theorem ??), and the χ -ledger reads $\chi(\mathcal{O}_Y(D)) = \deg D + \chi(\mathcal{O}_Y)$ (Theorem ??). $\square$

6.32 The uniform vanishing bound

Theorem 663 (The uniform H^1 -vanishing bound). *Let $\pi: Y \rightarrow \mathbb{P}^1$ be finite dominant, compatible with the structure morphisms. There is a single threshold $b \in \mathbb{Z}$, depending only on (Y, π) , such that for every Weil divisor D ,*

$$b \leq \deg D \implies H^1(\mathcal{O}_Y(D)) = 0.$$

Explicitly $b = n_1 \deg F + 1 - \chi(\mathcal{O}_Y)$, where F is the fiber Weil divisor and n_1 is the fibrewise-vanishing threshold at the divisor 0.

Proof. Run the fibrewise vanishing at $D_0 = 0$ (Theorem ??): there is n_1 with $H^1(\mathcal{O}_Y(nF)) = 0$ for all $n \geq n_1$; fix the base twist $n_1 F$. The threshold $b := n_1 \deg F + 1 - \chi(\mathcal{O}_Y)$ is legal since $0 < \deg F$ (Theorem ??).

Let D have $\deg D \geq b$. The residual class $c := \mathcal{O}_Y(D) \cdot \mathcal{O}_Y(F)^{-n_1}$ has degree $\deg D - n_1 \deg F \geq 1 - \chi(\mathcal{O}_Y)$ (Lemma ??, Theorem ??, Lemma ??); realise it by a divisor W (Definition ??) and take an effective witness $E \geq 0$ of its class (Lemma ??). Then $n_1 F + E$ is in the class of D . Vanishing holds at $n_1 F$ by the base choice; effective peeling (Lemma ??) transports it to $n_1 F + E$; and witness-independence (Theorem ??) upgrades it to $\mathcal{O}_Y(D)$ itself. Because the base twist is fixed once, the effective witness absorbs the whole degree slack and b depends only on (Y, π) . $\square$

6.33 The degree of the fiber twist

Theorem 664 (The presentation divisor of the unit fiber divisor). *For a dominant $\pi: Y \rightarrow \mathbb{P}^1$, the presentation divisor of the meromorphic presentation of the unit fiber divisor system $\text{fiberDivisor}(\pi, 1)$ is the fiber Weil divisor:*

$$\text{presentationDivisor}(\text{fiberDivisor}(\pi, 1). \text{presentation}) = F = \text{fiberWeilDivisor}(\pi).$$

Proof. Compare coefficients point by point. The presentation (Lemma ??) of the pinned two-cover system (t_0 on V_0 , 1 on V_1) (Definition ??) has trivialising element the fiber coordinate unit $u = t_0$ at points of V_0 (Definition ??) and the constant 1 at points off V_0 (which lie in V_1 , the two charts covering Y). At $x \in V_0$ the coefficient of the presentation divisor (Definition ??) is $\text{ord}_x(u)$, which the fiber order table identifies with the coefficient of $F = (\text{div } u)^+$ at x (Theorem ??, Definition ??); at $x \notin V_0$ both coefficients are zero — the element is 1, and F is supported in V_0 (Definition ??). $\square$

Theorem 665 (Θ -positivity of the unit fiber twist). *For a dominant $\pi: Y \rightarrow \mathbb{P}^1$, the unit fiber twist is the class of the fiber Weil divisor, $\Theta_1 = \mathcal{O}_Y(F)$, and*

$$\deg_{\text{Pic}}(\Theta_1) = \deg F.$$

When π is moreover finite, the degree is strictly positive: $1 \leq \deg_{\text{Pic}}(\Theta_1)$. (Only positivity is needed downstream; the identity $\deg \Theta_1 = \deg \pi$ is never used.)

Proof. Θ_1 is the Picard class of the unit fiber divisor system (Definition ??); the class of a local-equation system is the class of the presentation divisor of its meromorphic presentation (Lemma ??, Theorem ?? via the anchored class map of Definition ??), which is F (Theorem ??). Hence $\Theta_1 = \mathcal{O}_Y(F)$, and the degree of an anchored class reads the divisor degree (Definition ??). Positivity is Theorem ??). $\square$

Theorem 666 (The positivity constant transports across base-field transitions). *For the challenge curve C over k , a base-field transition $\varphi: K_1 \rightarrow K_2$ of field extensions of k , and a Picard class L on C_{K_1} with $1 \leq \deg_{\text{Pic}} L$: the pulled-back class on C_{K_2} retains the threshold, $1 \leq \deg_{\text{Pic}}(\pi^*L)$, where π is the base-field transition of the curve.*

Proof. The degree is invariant under base-field transitions (Theorem ??); the threshold transfers verbatim. $\square$

6.34 Evaluation at the graph section

For a field point $t: \text{Spec } K \rightarrow C$ of the challenge curve, the section g_t of the projection $C_K := (C \otimes \text{Spec } K)_{\text{left}} \rightarrow \text{Spec } K$ evaluates sections of any open containing the graph point into a copy of K . This is the residue half of the degree-one certificate.

Definition 667 (The evaluation at the graph section). Let W be an open of C_K containing the graph point x_t . The preimage of W under the section g_t is all of $\text{Spec } K$ (a one-point space whose point maps to $x_t \in W$), so pullback of sections along g_t defines a ring homomorphism

$$\varepsilon_t: \Gamma(C_K, W) \longrightarrow \Gamma(\text{Spec } K, \top),$$

compatible with restriction of the open.

Theorem 668 (The evaluation splits the K -structure). *The composite of the structure map $K \rightarrow \Gamma(C_K, W)$ (sections carry the K -structure through the second projection) with the evaluation ε_t is the canonical identification $K \cong \Gamma(\text{Spec } K, \top)$ (Definition ??) — in particular it is bijective, the target is a field, and ε_t is surjective.*

Proof. The structure map of the sections is the pullback along the second projection snd , pre-composed with the identification; and $g_t \circ \text{snd} = \text{id}$ (Definition ??), so the composite pullback $\varepsilon_t \circ \text{snd}^\sharp$ is the pullback along the identity — the canonical identification, a ring isomorphism onto $\Gamma(\text{Spec } K, \top) \cong K$. Surjectivity of ε_t follows since already the composite is surjective. $\square$

Theorem 669 (The point–evaluation dictionary). *On an affine open $W \ni x_t$, the kernel of the evaluation at the graph section is the prime ideal of the graph point:*

$$\ker \varepsilon_t = \mathfrak{p}_{x_t} \subseteq \Gamma(C_K, W).$$

Proof. The stalk of C_K at x_t is the localisation of $\Gamma(C_K, W)$ at $\mathfrak{p}_{x_t}$, so a section f lies in $\mathfrak{p}_{x_t}$ iff its germ at x_t is a non-unit. Chase germs through the section: the germ of $\varepsilon_t(f)$ at the unique point of $\text{Spec } K$ is the stalk map of g_t applied to the germ of f at x_t .

If $\varepsilon_t(f) = 0$ but the germ of f at x_t were a unit, its image under the stalk map would be a unit and also the germ of 0 — impossible. Conversely if $f \in \mathfrak{p}_{x_t}$ but $\varepsilon_t(f) \neq 0$, then $\varepsilon_t(f)$ is a unit (the target is a field, Theorem ??), so its germ is a unit; stalk maps are local ring homomorphisms, so reflecting units along the stalk map of g_t makes the germ of f at x_t a unit — contradicting $f \in \mathfrak{p}_{x_t}$. $\square$

Theorem 670 (The quotient by the kernel is a copy of K). *For any open $W \ni x_t$, the kernel of ε_t is a maximal ideal and*

$$\dim_K(\Gamma(C_K, W) / \ker \varepsilon_t) = 1.$$

Proof. ε_t is surjective onto a field (Theorem ??), so the quotient by its kernel is isomorphic to that field — whence maximality — and as a K -vector space the field $\Gamma(\text{Spec } K, \top)$ is one-dimensional: the K -structure map onto it is bijective (Theorem ?? again). $\square$

6.35 The graph chart

The multiplicity half of the degree-one certificate: an affine chart of the fibre curve on which the ideal of the graph section is principal on the transported point generator.

Definition 671 (The graph chart and the graph equation). Let $t: \text{Spec } K \rightarrow C$ be a field point with image point p_0 , and let $U := \text{chart}(p_0)$ be the frozen étale-coordinate chart of the curve at p_0 (Definition ??), with coordinate u , and $F := \Gamma(\text{Spec } K, \top)$. Define:

- the *point pullback* $c := t^\sharp: \Gamma(C, U) \rightarrow F$ (the whole of $\text{Spec } K$ maps into U);
- the *point generator* $g := u \otimes 1 - 1 \otimes c(u) \in \Gamma(C, U) \otimes_k F$ (Definition ??);
- the *pushed idempotent lift* $\tilde{e}_F$, the image of the frozen diagonal idempotent lift under $1 \otimes c$;
- the *graph chart* $\mathfrak{G} := D(1 - \tilde{e}_F) \subseteq \mathfrak{W}(U, \top)$, the basic open of the product chart (Definitions ??, ??) at the pushed complementary idempotent — an affine open of C_K , and a localisation of $\Gamma(C, U) \otimes_k F$ away from $1 - \tilde{e}_F$ (Theorem ??);
- the *graph equation* $\gamma := g|_{\mathfrak{G}} \in \Gamma(C_K, \mathfrak{G})$, the restriction of the identified point generator.

Lemma 672 (The tensor-level evaluation and the graph point). *Write $\text{ev}: \Gamma(C, U) \otimes_k F \rightarrow F$, $a \otimes \lambda \mapsto c(a)\lambda$, for the tensor-level point evaluation (Definition ??). Then: the graph point lies in the product chart $\mathfrak{W}(U, \top)$; the evaluation of an identified tensor at the graph section is the tensor-level evaluation, $\varepsilon_t(\text{pcs}(y)) = \text{ev}(y)$ for $y \in \Gamma(C, U) \otimes_k F$; and the pushed idempotent lift evaluates to zero, $\text{ev}(\tilde{e}_F) = 0$.*

Proof. **Membership:** the first projection of the graph point is the image point $p_0 \in U$ (Definition ??), and the second-factor condition is vacuous at $\top$. **Evaluation:** on a pure tensor $a \otimes b$, the lift-app rule for the section $g_t = \langle t, \text{id} \rangle$ (Theorem ??) computes $\varepsilon_t(\text{pcs}(a \otimes b)) = t^\sharp(a) \cdot b = c(a) b = \text{ev}(a \otimes b)$; extend additively. **Idempotent:** the tensor evaluation of a second-factor push is the point pullback of the multiplication, and the frozen package satisfies $\text{mul}(\tilde{e}) = 0$ (Definition ??), so $\text{ev}(\tilde{e}_F) = c(\text{mul}(\tilde{e})) = 0$. $\square$

Theorem 673 (The graph point lies in the graph chart). $x_t \in \mathfrak{G}$: the pushed complementary idempotent $1 - \tilde{e}_F$ does not vanish at the graph point.

Proof. Suppose it vanished at x_t , i.e. its section were a non-unit in the stalk. On the affine product chart the point–evaluation dictionary (Theorem ??) reads: the section lies in $\ker \varepsilon_t$. But its evaluation is $\text{ev}(1 - \tilde{e}_F) = 1 - 0 = 1 \neq 0$ in the field F (Lemma ??, Theorem ??) — a contradiction. $\square$

Theorem 674 (The kernel presentation of the graph ideal). *On the graph chart, the kernel of the evaluation at the graph section is the principal ideal on the graph equation:*

$$\ker(\varepsilon_t: \Gamma(C_K, \mathfrak{G}) \rightarrow F) = (\gamma).$$

Proof. The chart sections are a localisation of the tensor ring $\Gamma(C, U) \otimes_k F$ away from $1 - \tilde{e}_F$ (Theorem ??), and on the localisation the evaluation ε_t restricts, along the structure map, to the tensor-level evaluation ev (Lemma ??, compatible with the restriction to the basic open). The frozen diagonal package supplies the idempotent lift and the kernel presentation of the diagonal ideal (Definition ??), so the localised point–fiber keystone (Theorem ??) computes the extension of $\ker(\text{ev})$ to the chart as the principal ideal on the image of the point generator — which is γ (Definition ??).

It remains to see $\ker \varepsilon_t$ equals that extension. One inclusion is $\varepsilon_t(\gamma) = \text{ev}(g) = 0$. For the other, an element $z \in \ker \varepsilon_t$ clears denominators: $z \cdot s = \beta(b)$ with s a power of $1 - \tilde{e}_F$ and b in the tensor ring; s evaluates to 1 and z to 0, so $\text{ev}(b) = 0$, and b 's image lies in the extension of $\ker(\text{ev})$; since s is invertible in the localisation, z does too. $\square$

Theorem 675 (The colength of the graph equation is one).

$$\dim_K(\Gamma(C_K, \mathfrak{G}) / (\gamma)) = 1.$$

Proof. $(\gamma) = \ker \varepsilon_t$ (Theorem ??), and the quotient by the kernel is a copy of K (Theorem ??). $\square$

6.36 The degree-one certificate of the graph class

Lemma 676 (The graph equation is nonzero). *The point generator g is a nonzerodivisor of the tensor ring $\Gamma(C, U) \otimes_k F$, and the graph equation $\gamma \neq 0$ in $\Gamma(C_K, \mathfrak{G})$.*

Proof. The regularity engine (Theorem ??) makes $u \otimes 1 - 1 \otimes b$ a nonzerodivisor of $\Gamma(C, U) \otimes_k F$, uniformly in the second tensor factor and the element b — here $b = c(u)$. If $\gamma = 0$ in the localisation, then some power s of $1 - \tilde{e}_F$ satisfies $g \cdot s = 0$ in the tensor ring, so $s = 0$ (nonzerodivisor); but then the localisation itself is the zero ring, contradicting that it evaluates onto the field F (Lemma ??, Theorem ??, Theorem ??), where $1 \neq 0$. $\square$

Theorem 677 (The graph point is closed). *The graph point is not the generic point of the fibre curve: $x_t \neq \eta$.*

Proof. On the affine graph chart, the prime of the graph point is $\ker \varepsilon_t = (\gamma)$ (Theorems ??, ??), a *nonzero* ideal since $\gamma \neq 0$ (Lemma ??). Points of nonzero primes of an affine open are not the generic point (Lemmas ??, ??). $\square$

Lemma 678 (The graph equation is a unit away from the graph point). *For every point $z \in \mathfrak{G}$ with $z \neq x_t$, the germ of the graph equation γ at z is a unit.*

Proof. If the germ were a non-unit, γ would lie in the prime $\mathfrak{p}_z$ of z on the chart, so $(\gamma) = \mathfrak{p}_{x_t} \subseteq \mathfrak{p}_z$ (Theorems ??, ??); but $\mathfrak{p}_{x_t}$ is maximal (Theorem ??), forcing $\mathfrak{p}_z = \mathfrak{p}_{x_t}$, hence $z = x_t$ (points of an affine open are recovered from their primes, Lemma ??) — contradiction. $\square$

Lemma 679 (The trivialising element away from the graph point is 1). *For $x \neq x_t$, the trivialising element of the meromorphic presentation of the graph system at x is the constant 1.*

Proof. The graph system is the pullback of the diagonal system along the graph lift q (Definition ??). At $x \neq x_t$, the image $q(x)$ misses the diagonal (Definition ??), so the diagonal equation at $q(x)$ is the constant 1 (Definition ??), whose pullback (Definition ??) is $q^\sharp(1) = 1$; the presentation element (Lemma ??) is its germ, i.e. 1. $\square$

Theorem 680 (The overlap identity at the graph point). *The rational function trivialising the graph divisor at the graph point — the germ at the generic point η of the pulled diagonal equation — is the germ at η of the graph equation:*

$$\text{elem}_{x_t}(\Gamma_t) = \gamma_\eta \text{ in } K(C_K).$$

Proof. Both sides restrict to the same section on the overlap $W := (\text{pullback member at } x_t) \cap \mathfrak{W}(U, \top)$, an open containing η ; since germs at η are computed through any open containing it, the two germs agree.

The left side on W . The pullback member at x_t carries the pulled diagonal equation $q^\sharp(e_{p_0})$ (Definitions ??, ??), where the diagonal equation e_{p_0} on the diagonal chart is the identified diagonal generator $u \otimes 1 - 1 \otimes u$ (Definitions ??, ??). The lift-app rule for $q = \langle \text{fst}, \text{snd} \circ t \rangle$ (Theorem ??) computes its restriction to W as $\text{fst}^\sharp(u) - (\text{snd} \circ t)^\sharp(u)$. The argument is run at a generalised image point ($\forall p, \text{fst}(q(x_t)) = p \Rightarrow \dots$), instantiated at $p := t(*)$, so that the rewrite along $\text{fst}(q(x_t)) = p_0$ never crosses the statement.

The right side on W . The graph equation is the identified point generator $u \otimes 1 - 1 \otimes c(u)$ restricted to the graph chart (Definition ??); by the product-chart section calculus (Definition ??), its restriction to W is $\text{fst}^\sharp(u) - \text{snd}^\sharp(c(u)) = \text{fst}^\sharp(u) - (\text{snd} \circ t)^\sharp(u)$, the same section. Finally $\eta \in W$: W is a nonempty open of the irreducible space C_K (it contains x_t : Lemma ?? for the member, Theorem ?? for the chart side of the overlap). $\square$

Theorem 681 (The presentation divisor of the graph is a one-point divisor). *The presentation divisor of the graph system is supported at the graph point:*

$$\text{presentationDivisor}(\Gamma_t) = \text{ord}_{x_t}(\text{elem}_{x_t}) \cdot x_t,$$

a single-point divisor (Definition ??, legal since $x_t \neq \eta$ by Theorem ??).

Proof. Compare coefficients pointwise (Definition ??). At x_t both sides read $\text{ord}_{x_t}(\text{elem}_{x_t})$ (Definition ??). At $x \neq x_t$ the trivialising element is 1 (Lemma ??), of order zero, and the single-point divisor has coefficient zero there. $\square$

Theorem 682 (The graph class of a field point has degree one). *For every field point $t: \text{Spec } K \rightarrow C$ of the challenge curve,*

$$\deg_{\text{Pic}}(\mathcal{O}_{C_K}(\Gamma_t)) = 1.$$

This is the degree certificate of the Abel element: Kleiman's graph divisor $\Gamma_g = (1 \times g)^{-1}\Delta$ is a relative effective divisor of fibre degree one.

Proof. The class of the graph system is the anchored class of its presentation divisor (Lemma ??, Theorem ??, Definition ??), which is the one-point divisor $m \cdot x_t$ with $m := \text{ord}_{x_t}(\text{elem}_{x_t})$ (Theorem ??). The anchored degree of a one-point divisor is $\deg(m \cdot x_t) = m \cdot [\kappa(x_t) : K]$ (Definition ??, Lemma ??).

This product collapses in one shot through the chart colength dictionary (Theorem ??) on the graph chart, applied to the rational function elem_{x_t} with section representative γ (the overlap identity, Theorem ??) and the singleton point set $\{x_t\}$: the hypotheses are that x_t lies in the chart (Theorem ??) and that γ is a unit at every other closed point of the chart (Lemma ??); the dictionary then reads

$$m \cdot [\kappa(x_t) : K] = \dim_K(\Gamma(C_K, \mathfrak{G})/(\gamma)) = 1$$

by the colength computation (Theorem ??). □

Chapter 7

The étale plus construction of the relative Picard functor

For a curve C over k and an affine test scheme $\mathrm{Spec} A$, the étale-sheafified relative Picard group is computed by a single *plus construction*: a class is represented by a relative Picard class on an étale cover of $\mathrm{Spec} A$ satisfying a descent condition, and two representatives are identified when they agree on a common refinement. This chapter constructs that plus, on affine test schemes indexed by a k -algebra A ; we recall from elsewhere in this project the ambient functor it sheafifies. For an object T of $\mathrm{Over}(\mathrm{Spec} k)$ write $\mathrm{relPic}(C, T)$ for the relative Picard group of C at T — the quotient of the Picard group of $C \otimes T$ by the subgroup of classes pulled back from T along the second projection — a commutative group, with restriction homomorphisms $g^*: \mathrm{relPic}(C, T) \rightarrow \mathrm{relPic}(C, T')$ for a morphism $g: T' \rightarrow T$ of $\mathrm{Over}(\mathrm{Spec} k)$, satisfying $(\mathrm{id}_T)^* = \mathrm{id}$ and $(g \circ h)^* = h^* \circ g^*$. For a commutative k -algebra A , write $\mathrm{Spec} A$ also for the object of $\mathrm{Over}(\mathrm{Spec} k)$ it determines via the structure morphism $\mathrm{Spec} A \rightarrow \mathrm{Spec} k$ induced by the k -algebra structure of A ; a homomorphism of k -algebras $A \rightarrow A'$ induces, via Spec , a morphism $\mathrm{Spec} A' \rightarrow \mathrm{Spec} A$ of $\mathrm{Over}(\mathrm{Spec} k)$. We abbreviate $\mathrm{relPic}(C, A) := \mathrm{relPic}(C, \mathrm{Spec} A)$.

7.1 The relative Picard group as a quotient

The group $\mathrm{relPic}(C, T)$ of the preamble is realised concretely as a quotient of the Čech Picard group of the product: this section records the quotient presentation and its exact coset relation, the calculus every later rigidification and degree statement cascades through.

Definition 683 (The relative Picard group). Let C be an object of $\mathrm{Over}(\mathrm{Spec} k)$ and T a test object. Write $H_T \leq \mathrm{Pic}(C \otimes T)$ for the subgroup of Čech Picard classes *pulled back from the base*: the range of $\mathrm{pr}_2^* = \mathrm{snd}^*: \mathrm{Pic}(T) \rightarrow \mathrm{Pic}(C \otimes T)$, so $L \in H_T$ iff $L = \mathrm{snd}^* N$ for some $N \in \mathrm{Pic}(T)$. The *relative Picard group* is the quotient

$$\mathrm{relPic}(C, T) := \mathrm{Pic}(C \otimes T) / H_T,$$

a commutative group, with surjective projection $\pi_T: \mathrm{Pic}(C \otimes T) \rightarrow \mathrm{relPic}(C, T)$. The *exact coset relation* holds: $\pi_T L = \pi_T L'$ iff $L/L' \in H_T$ — a genuine range condition, involving no choice.

Proof. H_T is the range of a group homomorphism, hence a subgroup, and $\mathrm{Pic}(C \otimes T)$ is commutative, so H_T is normal and the quotient is a commutative group with a surjective projection

homomorphism. The coset relation is the defining property of a quotient group: two elements have equal images exactly when their ratio lies in the subgroup. $\square$

Definition 684 (Restriction of relative Picard classes). For a morphism $g: T' \rightarrow T$ of test objects, pullback of Čech classes along $C \triangleleft g: C \otimes T' \rightarrow C \otimes T$ descends to the quotients: a homomorphism $g^*: \text{relPic}(C, T) \rightarrow \text{relPic}(C, T')$ with $g^*(\pi_T L) = \pi_{T'}((C \triangleleft g)^* L)$. It satisfies $(\text{id}_T)^* = \text{id}$ and $(h \circ g)^* = g'^*$ -composition contravariantly, so $T \mapsto \text{relPic}(C, T)$ is a presheaf of commutative groups on $\text{Over}(\text{Spec } k)$ — the relative Picard functor of Grothendieck–Kleiman.

Proof. Well-definedness on cosets: for $L = \text{snd}_T^* N \in H_T$,

$$(C \triangleleft g)^* \text{snd}_T^* N = (\text{snd}_T \circ (C \triangleleft g))^* N = (g \circ \text{snd}_{T'})^* N = \text{snd}_{T'}^* (g^* N) \in H_{T'},$$

using $\text{snd}_T \circ (C \triangleleft g) = g \circ \text{snd}_{T'}$ (the whiskering identity of the cartesian product). So the composite $\text{Pic}(C \otimes T) \rightarrow \text{relPic}(C, T')$ kills H_T and factors through the quotient. The functor laws follow from $C \triangleleft \text{id} = \text{id}$ and $C \triangleleft (h \circ g) = (C \triangleleft h) \circ (C \triangleleft g)$ together with functoriality of pullback of Čech classes, checked on representatives via the computation rule. $\square$

Lemma 685 (Naturality in the curve). *A morphism of curves $g: C' \rightarrow C$ over $\text{Spec } k$ induces, by pullback along $g \triangleright T$ on representatives, a natural transformation of relative Picard functors $\text{relPic}(C, -) \rightarrow \text{relPic}(C', -)$; the assignment is contravariantly functorial in g on the nose. In particular, at every test object the transports satisfy the functor laws, and they commute with restriction along any test morphism — in particular along $\text{Spec } f$ for a k -algebra map f of affine tests.*

Proof. Well-definedness on cosets: $(g \triangleright T)^* \text{snd}_{C, T}^* N = \text{snd}_{C', T}^* N$ since $\text{snd}_{C, T} \circ (g \triangleright T) = \text{snd}_{C', T}$. Naturality against a test morphism $t: T' \rightarrow T$ is the exchange identity $(C' \triangleleft t) \circ (g \triangleright T') = (g \triangleright T) \circ (C \triangleleft t)$ of whiskerings, applied to representatives. The functor laws follow from $\text{id} \triangleright T = \text{id}$ and $(g' \circ g) \triangleright T = (g \triangleright T) \circ (g' \triangleright T)$. $\square$

Lemma 686 (Constant curve morphisms kill relative Picard classes). *Let $g: C' \rightarrow C$ be a morphism of curve bundles over $\text{Spec } k$ that factors through the terminal object $\text{Spec } k$ of $\text{Over}(\text{Spec } k)$. Then for every test object T the induced map $\text{relPic}(C, T) \rightarrow \text{relPic}(C', T)$ is trivial.*

Proof. Write $g = e \circ u$ with $u: C' \rightarrow \text{Spec } k$ the unique morphism to the terminal object and $e: \text{Spec } k \rightarrow C$. Pullback along $g \triangleright T$ then factors through pullback along $u \triangleright T: C' \otimes T \rightarrow \text{Spec } k \otimes T$, so it suffices to show that every Picard class on $\text{Spec } k \otimes T$ pulls back along $u \triangleright T$ into the subgroup H_T of classes pulled back from T . The second projection $\text{snd}: \text{Spec } k \otimes T \rightarrow T$ is the left unitor, hence an isomorphism, so an arbitrary class X on $\text{Spec } k \otimes T$ is $\text{snd}^* N$ for $N := (\text{snd}^{-1})^* X$. Since $\text{snd} \circ (u \triangleright T) = \text{snd}_{C', T}$, we get $(u \triangleright T)^* X = (u \triangleright T)^* \text{snd}^* N = \text{snd}_{C', T}^* N \in H_T$. Every pulled-back class therefore lies in H_T and dies in the quotient $\text{relPic}(C', T)$. $\square$

At the Čech level the constant-morphism collapse is sharper than Lemma ???: a morphism of bundles whose set-range is a single closed point kills every Čech Picard class outright — before passing to any quotient — because it factors through the spectrum of a residue field, whose Čech Picard group is trivial.

Theorem 687 (Morphisms factoring through a residue field kill Čech classes). *Let K be a field, D, E objects of $\text{Over}(\text{Spec } K)$, and $h: D \rightarrow E$ a morphism of $\text{Over}(\text{Spec } K)$ whose underlying morphism factors as $D \rightarrow \text{Spec } \kappa(x) \rightarrow E$ through the canonical morphism from the spectrum of the residue field of a point x of E . Then for every Čech Picard class $L \in \text{Pic}(E)$,*

$$h^* L = 1.$$

Proof. Write the factorization as $h = \iota \circ q$ with $q : D \rightarrow \operatorname{Spec} \kappa(x)$ and $\iota : \operatorname{Spec} \kappa(x) \rightarrow E$. Pullback of Čech Picard classes is functorial, so $h^*L = q^*(\iota^*L)$. The underlying space of $\operatorname{Spec} \kappa(x)$ is a single point (the spectrum of a field), and the Čech Picard group of a scheme whose space has at most one point is trivial: any two members of a pointed cover coincide as opens — each is determined by whether it is empty, and members contain their base points — so every unit Čech 1-cocycle lives on a cover by a single open, where it is a coboundary. Hence $\iota^*L = 1$, and q^* , a homomorphism, gives $h^*L = q^*(1) = 1$. $\square$

Corollary 688 (Morphisms with one-point range kill Čech classes). *Let K be a field, D, E objects of $\operatorname{Over}(\operatorname{Spec} K)$ with D reduced, and $h : D \rightarrow E$ a morphism of $\operatorname{Over}(\operatorname{Spec} K)$ whose underlying morphism is quasi-compact and whose set-range is a single closed point x of E . Then $h^*L = 1$ for every Čech Picard class $L \in \operatorname{Pic}(E)$.*

Proof. By the factorization through the residue field of a closed point (??), the underlying morphism of h factors as $D \rightarrow \operatorname{Spec} \kappa(x) \rightarrow E$. Apply ?? $\square$

7.2 Presented étale covers of an affine scheme

Throughout this section A is a commutative ring (no field or k -algebra structure is needed here). An étale cover of $\operatorname{Spec} A$ in the sense consumed by the plus construction is a faithfully flat étale A -algebra; to keep the index of covers of a fixed base small — which is what makes the plus construction below a legitimate colimit, rather than one indexed by a proper class of isomorphism types — we realize such algebras through an explicit finite-variable presentation, at the cost of carrying a witness of the presentation as data.

Definition 689 (Presented étale cover). A *presented étale cover* E of A consists of a natural number n , an ideal I of the polynomial ring $A[x_1, \dots, x_n]$, a proof that the quotient algebra $B_E := A[x_1, \dots, x_n]/I$ is étale over A , and a proof that the induced map of spectra $\operatorname{Spec} B_E \rightarrow \operatorname{Spec} A$ is surjective. We call B_E the *carrier* of E . Since étale algebras are flat, surjectivity of $\operatorname{Spec} B_E \rightarrow \operatorname{Spec} A$ together with flatness makes B_E faithfully flat over A .

Definition 690 (Presented covers from an étale algebra). Let B be an étale A -algebra such that $\operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is surjective. Since étale algebras are of finite presentation, B admits a presentation as a quotient of a polynomial ring in finitely many variables over A ; choosing one such presentation produces a presented étale cover E of A , denoted $E = \operatorname{of}(B)$.

Lemma 691 (Carrier of a presented cover from an algebra). *In the situation of Definition ??, the carrier B_E of $E = \operatorname{of}(B)$ is isomorphic to B as an A -algebra.*

Proof. Immediate from the construction: E is built from a chosen presentation $A[x_1, \dots, x_n] \twoheadrightarrow B$, and the carrier of the resulting cover is the quotient of $A[x_1, \dots, x_n]$ by the kernel of that presentation, which is isomorphic to B by the first isomorphism theorem for algebras. $\square$

Definition 692 (Refinement). For presented étale covers E, E' of A , we say E' *refines* E if there exists a homomorphism of A -algebras $B_E \rightarrow B_{E'}$ (equivalently, a morphism $\operatorname{Spec} B_{E'} \rightarrow \operatorname{Spec} B_E$ over $\operatorname{Spec} A$). This is a mere existence statement: no particular refinement map is remembered as data.

Lemma 693 (Refinement is a preorder). *Every presented étale cover E of A refines itself, and if E'' refines E' and E' refines E then E'' refines E .*

Proof. Reflexivity: the identity homomorphism of B_E witnesses that E refines itself. Transitivity: composing a homomorphism $B_{E'} \rightarrow B_{E''}$ witnessing that E'' refines E' with a homomorphism $B_E \rightarrow B_{E'}$ witnessing that E' refines E gives a homomorphism $B_E \rightarrow B_{E''}$ witnessing that E'' refines E . $\square$

Definition 694 (Common refinement). For presented étale covers E, E' of A , the *common refinement* $E \times E'$ is the presented étale cover of A with carrier $B_E \otimes_A B_{E'}$ (étale over A because it is étale over the étale algebra B_E , which is étale over A ; and faithfully flat by transitivity of faithfully flat extensions), obtained by applying Definition ?? to $B_E \otimes_A B_{E'}$.

Definition 695 (Coprojections of the common refinement). The two canonical homomorphisms of A -algebras $B_E \rightarrow B_{E \times E'} \cong B_E \otimes_A B_{E'}$, $b \mapsto b \otimes 1$, and $B_{E'} \rightarrow B_{E \times E'}$, $b' \mapsto 1 \otimes b'$, transported along the carrier identification of Definition ??.

Lemma 696 (The common refinement refines both factors). $E \times E'$ refines both E and E' .

Proof. The two coprojections of Definition ?? are homomorphisms of A -algebras out of B_E , respectively $B_{E'}$, witnessing the two refinements. $\square$

Definition 697 (Universal map out of the common refinement). Given homomorphisms of A -algebras $f: B_E \rightarrow B_F$ and $g: B_{E'} \rightarrow B_F$, the induced homomorphism of A -algebras $B_{E \times E'} \rightarrow B_F$, obtained from the universal property of the tensor product $B_E \otimes_A B_{E'}$ applied to f and g (whose images commute in the commutative ring B_F), transported along the carrier identification of Definition ??.

Lemma 698 (The universal map is a common section). *In the situation of Definition ??, the induced map composed with each coprojection recovers the corresponding map: the composite $B_E \rightarrow B_{E \times E'} \rightarrow B_F$ equals f , and the composite $B_{E'} \rightarrow B_{E \times E'} \rightarrow B_F$ equals g .*

Proof. This is the defining property of the map induced by the universal property of the tensor product: on elementary tensors, $b \otimes 1 \mapsto f(b)g(1) = f(b)$ and $1 \otimes b' \mapsto f(1)g(b') = g(b')$. $\square$

Theorem 699 (Directedness). *Any two presented étale covers E, E' of A admit a common refinement: there is a presented étale cover E'' of A which refines both E and E' .*

Proof. Take $E'' = E \times E'$; it refines both factors by Lemma ??.

Definition 700 (The trivial cover). The presented étale cover of A by itself, obtained by applying Definition ?? to $B = A$ (with structure morphism $\text{Spec } A \rightarrow \text{Spec } A$ the identity, trivially surjective). Its carrier is isomorphic to A .

Definition 701 (Canonical map from the trivial cover). For a presented étale cover E of A , the canonical homomorphism of A -algebras from the carrier of the trivial cover to B_E : under the identification of the carrier of the trivial cover with A (Definition ??), the structure homomorphism $A \rightarrow B_E$.

Lemma 702 (Every cover refines the trivial cover). *Every presented étale cover E of A refines the trivial cover.*

Proof. The canonical map of Definition ?? witnesses the refinement. $\square$

Definition 703 (Base change of a cover). Let A' be a commutative A -algebra and E a presented étale cover of A . The *base change* $E_{A'}$ is the presented étale cover of A' with carrier $A' \otimes_A B_E$ (étale over A' and faithfully flat over A' , both by base change of the corresponding properties of B_E over A), obtained by applying Definition ?? to $A' \otimes_A B_E$.

Definition 704 (Canonical inclusion into the base change). The canonical homomorphism of A -algebras $\kappa_E: B_E \rightarrow B_{E_{A'}}$, which under the carrier identification $B_{E_{A'}} \cong A' \otimes_A B_E$ of Definition ?? is $b \mapsto 1 \otimes b$.

Definition 705 (Base change of a refinement map). For presented étale covers E, F of A and a homomorphism of A -algebras $h: B_E \rightarrow B_F$, the induced homomorphism of A' -algebras $B_{E_{A'}} \rightarrow B_{F_{A'}}$ which, under the carrier identifications of Definition ??, is $\text{id}_{A'} \otimes h$.

Lemma 706 (Naturality of the canonical inclusion). *For $h: B_E \rightarrow B_F$ as in Definition ??, the square of A -algebra homomorphisms commutes: the composite $B_E \rightarrow B_{E_{A'}} \rightarrow B_{F_{A'}}$ (canonical inclusion followed by the base-changed map) equals the composite $B_E \rightarrow B_F \rightarrow B_{F_{A'}}$ (h followed by the canonical inclusion).*

Proof. Both composites send $b \in B_E$ to $1 \otimes h(b) \in A' \otimes_A B_F$ under the carrier identifications: the first computes $b \mapsto 1 \otimes b \mapsto (\text{id}_{A'} \otimes h)(1 \otimes b) = 1 \otimes h(b)$, and the second computes $b \mapsto h(b) \mapsto 1 \otimes h(b)$. $\square$

Lemma 707 (Nontrivial carrier). *If A is a nontrivial ring, then the carrier B_E of any presented étale cover E of A is nontrivial.*

Proof. Since A is nontrivial, $\text{Spec } A$ is nonempty; since $\text{Spec } B_E \rightarrow \text{Spec } A$ is surjective, $\text{Spec } B_E$ is nonempty, and a ring with nonempty spectrum is nontrivial. $\square$

Theorem 708 (Field cofinality). *Let K be a field and E a presented étale cover of K . Then there exist a field L , finite and separable over K , and a homomorphism of K -algebras $B_E \rightarrow L$.*

Proof. By the structure theorem for étale algebras over a field, B_E is isomorphic, as a K -algebra, to a finite product $\prod_{i \in I} L_i$ of finite separable field extensions L_i of K . By Lemma ??, B_E is nontrivial (K being a field is nontrivial), so the product is nonempty: $I \neq \emptyset$, for the empty product is the zero ring. Choose $i_0 \in I$ and set $L = L_{i_0}$; the composite of the structure isomorphism with the projection onto the i_0 -th factor is the required homomorphism $B_E \rightarrow L$. $\square$

A finite separable field extension L of a field K is automatically étale over K : it is formally étale because it is formally unramified (separability) and formally smooth (finiteness plus the algebra being a field, hence regular), and it is of finite presentation because it is a finite-type algebra over a field which is moreover finite as a module.

Definition 709 (Field extension covers). Let K be a field and L a finite separable field extension of K . Since $\text{Spec } K$ is a single point, the spectrum map $\text{Spec } L \rightarrow \text{Spec } K$ is trivially surjective, so Definition ?? applied to L produces a presented étale cover of K with carrier isomorphic to L .

7.3 The algebra-indexed relative Picard functor

In this section k is a field and C is an object of $\text{Over}(\text{Spec } k)$.

Definition 710 (Spec of an algebra map, as a morphism of test objects). For commutative k -algebras A, B and a homomorphism of k -algebras $f: A \rightarrow B$, the induced morphism $\text{Spec } B \rightarrow \text{Spec } A$ of $\text{Over}(\text{Spec } k)$: $\text{Spec } f$, which lies over $\text{Spec } k$ because f is a homomorphism of k -algebras.

Lemma 711 (Functoriality of Spec on algebra maps). *Spec of the identity homomorphism of A is the identity morphism of $\text{Spec } A$; and for homomorphisms of k -algebras $f: A \rightarrow B$ and $g: B \rightarrow C'$, $\text{Spec}(g \circ f) = \text{Spec } f \circ \text{Spec } g$ as morphisms $\text{Spec } C' \rightarrow \text{Spec } A$.*

Proof. Spec is a contravariant functor from commutative rings to schemes, so it sends the identity to the identity and a composite $g \circ f$ to $\text{Spec } f \circ \text{Spec } g$; the morphisms of $\text{Over}(\text{Spec } k)$ so obtained are equal to the corresponding morphisms of schemes on their underlying data, so the identities transport. $\square$

Definition 712 (Restriction of relative Picard classes along an algebra map). For a homomorphism of k -algebras $f: A \rightarrow B$, the induced homomorphism of groups $f^*: \text{relPic}(C, A) \rightarrow \text{relPic}(C, B)$: restriction of relative Picard classes (Definition ??) along the morphism $\text{Spec } f: \text{Spec } B \rightarrow \text{Spec } A$ of Definition ??.

Lemma 713 (Functoriality in the algebra). *For every k -algebra A , $(\text{id}_A)^* = \text{id}$ on $\text{relPic}(C, A)$; and for homomorphisms of k -algebras $f: A \rightarrow B$ and $g: B \rightarrow C'$, $(g \circ f)^* = g^* \circ f^*$ on $\text{relPic}(C, A)$. Thus $A \mapsto \text{relPic}(C, A)$, $f \mapsto f^*$, is a covariant functor from k -algebras to commutative groups.*

Proof. Restriction of relative Picard classes along Spec of a scheme morphism satisfies the functor laws $\text{id}^* = \text{id}$ and $(\varphi \circ \psi)^* = \psi^* \circ \varphi^*$ in the morphism of $\text{Over}(\text{Spec } k)$. Combined with Lemma ??, which shows Spec of the identity algebra map is the identity morphism and Spec of a composite $g \circ f$ is $\text{Spec } f \circ \text{Spec } g$, the two claimed identities for $f^* := \text{Spec } f^*$ follow. $\square$

7.4 The base-field shuffle of the relative Picard group at an affine test

In this section $k \rightarrow L$ is a field extension, $\sigma: \text{Spec } L \rightarrow \text{Spec } k$ the induced morphism, C an object of $\text{Over}(\text{Spec } k)$ with base change $C_L := \sigma^* C \in \text{Over}(\text{Spec } L)$ (??), and A an algebra in a scalar tower $k \rightarrow L \rightarrow A$. We write $\text{relPic}_L(C_L, A)$ for the relative Picard group of C_L at the affine test of A over L , formed over the base field L exactly as in Definition ?. The two slice products $C \otimes \text{Spec } A$ (over k) and $C_L \otimes \text{Spec } A$ (over L) share the carrier $\text{Spec } A$ of their second projections, and their underlying schemes are identified by the same-carrier comparison c of Definition ?? (Chapter ??). This section descends c to an isomorphism of the relative Picard groups, natural in A .

Lemma 714 (The comparison transports base-pulled classes to base-pulled classes). *Write snd_k , snd_L for the second projections of the two slice products onto $\text{Spec } A$. For every Čech Picard class N on $\text{Spec } A$,*

$$c^*(\text{snd}_k^* N) = \text{snd}_L^* N, \quad (c^{-1})^*(\text{snd}_L^* N) = \text{snd}_k^* N.$$

In particular pullback along c carries the subgroup of $\text{Pic}(C \otimes \text{Spec } A)$ of classes pulled back from the base (??) into the corresponding subgroup of $\text{Pic}(C_L \otimes \text{Spec } A)$, and pullback along c^{-1} carries the latter into the former.

Proof. The comparison commutes with the second projections: $\text{snd}_k \circ c = \text{snd}_L$ and $\text{snd}_L \circ c^{-1} = \text{snd}_k$ (??). Pullback of Čech Picard classes is contravariantly functorial in the morphism, so $c^*(\text{snd}_k^* N) = (\text{snd}_k \circ c)^* N = \text{snd}_L^* N$, and likewise along c^{-1} . $\square$

Definition 715 (The base-field shuffle of relative Picard classes). Pullback along c and along c^{-1} descend to the quotients of Definition ?? : there are homomorphisms of groups

$$\mathrm{sh}_A : \mathrm{relPic}(C, A) \longrightarrow \mathrm{relPic}_L(C_L, A), \quad \mathrm{sh}'_A : \mathrm{relPic}_L(C_L, A) \longrightarrow \mathrm{relPic}(C, A),$$

with computation rules on cosets $\mathrm{sh}_A(\pi\Lambda) = \pi(c^*\Lambda)$ for $\Lambda \in \mathrm{Pic}(C \otimes \mathrm{Spec} A)$ and $\mathrm{sh}'_A(\pi M) = \pi((c^{-1})^*M)$ for $M \in \mathrm{Pic}(C_L \otimes \mathrm{Spec} A)$.

Proof. Pullback of Čech Picard classes along a morphism of schemes is a homomorphism of Picard groups. By Lemma ?? it carries the subgroup of base-pulled classes into the subgroup of base-pulled classes, in each direction. A homomorphism carrying the distinguished subgroup into the distinguished subgroup induces a homomorphism of the quotient groups computing on cosets by the stated rule — the universal property of the quotient (??). $\square$

Theorem 716 (The base-field shuffle is an isomorphism). *The two descended homomorphisms of Definition ?? are mutually inverse: $\mathrm{sh}'_A \circ \mathrm{sh}_A = \mathrm{id}$ and $\mathrm{sh}_A \circ \mathrm{sh}'_A = \mathrm{id}$. Hence the base-field shuffle is an isomorphism of groups $\mathrm{relPic}(C, A) \cong \mathrm{relPic}_L(C_L, A)$.*

Proof. The projections are surjective (??), so it suffices to check on cosets. By the computation rules and contravariant functoriality of pullback,

$$\mathrm{sh}'_A(\mathrm{sh}_A(\pi\Lambda)) = \pi((c^{-1})^* c^* \Lambda) = \pi((c \circ c^{-1})^* \Lambda) = \pi(\Lambda),$$

and symmetrically $\mathrm{sh}_A(\mathrm{sh}'_A(\pi M)) = \pi((c^{-1} \circ c)^* M) = \pi(M)$. A group homomorphism with a two-sided inverse homomorphism is an isomorphism. $\square$

Theorem 717 (Naturality of the shuffle against restriction along algebra maps). *Let B be a further algebra in a scalar tower $k \rightarrow L \rightarrow B$ and $f : A \rightarrow B$ a homomorphism of L -algebras (hence of k -algebras, through the towers). Then restriction and shuffle commute: for every $x \in \mathrm{relPic}(C, A)$,*

$$\mathrm{sh}_B(f_k^* x) = f_L^*(\mathrm{sh}_A(x)),$$

where $f_k^* : \mathrm{relPic}(C, A) \rightarrow \mathrm{relPic}(C, B)$ is restriction along $\mathrm{Spec} f$ over k (??) and $f_L^* : \mathrm{relPic}_L(C_L, A) \rightarrow \mathrm{relPic}_L(C_L, B)$ is the same construction over the base field L , at the curve C_L .

Proof. Check on a coset $x = \pi(\Lambda)$, $\Lambda \in \mathrm{Pic}(C \otimes \mathrm{Spec} A)$, the projection being surjective (??). By the computation rules of restriction (??, ??) and of the shuffle (??), the left side is $\pi(c_B^*((C \triangleleft \mathrm{Spec}_k f)_{\mathrm{left}})^* \Lambda)$, the pullback of Λ along the composite $(C \triangleleft \mathrm{Spec}_k f)_{\mathrm{left}} \circ c_B$, and the right side is $\pi(((C_L \triangleleft \mathrm{Spec}_L f)_{\mathrm{left}})^* c_A^* \Lambda)$, the pullback of Λ along $c_A \circ (C_L \triangleleft \mathrm{Spec}_L f)_{\mathrm{left}}$. The two composites are equal — the naturality square of the same-carrier comparison (??) — so the two pullbacks agree, and so do their cosets. $\square$

7.5 Descent classes on an étale cover

Throughout this section k is a field, C is an object of $\mathrm{Over}(\mathrm{Spec} k)$, and A is a commutative k -algebra. For a presented étale cover E of A (Definition ??) with carrier B_E , the self-tensor product $B_E \otimes_A B_E$ is the carrier of the common refinement $E \times E$ of E with itself (Definition ??): it is the algebra of the *double cover* $\mathrm{Spec}(B_E \otimes_A B_E) \rightarrow \mathrm{Spec} B_E$, whose two projections to $\mathrm{Spec} B_E$ record the overlap data of the cover E .

Definition 718 (Coprojections of the double cover). For a presented étale cover E of A , the two homomorphisms of k -algebras $B_E \rightarrow B_E \otimes_A B_E$, $b \mapsto b \otimes 1$ and $b \mapsto 1 \otimes b$ — homomorphisms of A -algebras, in particular of k -algebras since A is a k -algebra.

Definition 719 (Descent classes). For a presented étale cover E of A , the subgroup $\text{Desc}(C, E) \subseteq \text{relPic}(C, B_E)$ of classes λ satisfying the *descent condition*: the two pullbacks of λ along the coprojections of the double cover (Definition ??) agree,

$$\text{Desc}(C, E) := \{ \lambda \in \text{relPic}(C, B_E) : \text{inl}_E^* \lambda = \text{inr}_E^* \lambda \text{ in } \text{relPic}(C, B_E \otimes_A B_E) \},$$

the equalizer of the two homomorphisms $\text{inl}_E^*, \text{inr}_E^* : \text{relPic}(C, B_E) \rightarrow \text{relPic}(C, B_E \otimes_A B_E)$ of Definition ??, hence itself a subgroup.

Lemma 720 (Naturality of the descent condition). *Let E, F be presented étale covers of A and $h : B_E \rightarrow B_F$ a homomorphism of A -algebras. If $\lambda \in \text{Desc}(C, E)$, then $h^* \lambda \in \text{Desc}(C, F)$.*

Proof. The homomorphism $h \otimes h : B_E \otimes_A B_E \rightarrow B_F \otimes_A B_F$ intertwines the coprojections: $\text{inl}_F \circ h = (h \otimes h) \circ \text{inl}_E$ and $\text{inr}_F \circ h = (h \otimes h) \circ \text{inr}_E$, since both sides send $b \mapsto h(b) \otimes 1$, respectively $b \mapsto 1 \otimes h(b)$. By functoriality (Lemma ??),

$$\text{inl}_F^*(h^* \lambda) = (h \otimes h)^*(\text{inl}_E^* \lambda) = (h \otimes h)^*(\text{inr}_E^* \lambda) = \text{inr}_F^*(h^* \lambda),$$

using $\lambda \in \text{Desc}(C, E)$ in the middle equality. Hence $h^* \lambda \in \text{Desc}(C, F)$. $\square$

Definition 721 (Transport of descent classes). For $h : B_E \rightarrow B_F$ a homomorphism of A -algebras between the carriers of two presented étale covers of A , the induced homomorphism of groups $h_* : \text{Desc}(C, E) \rightarrow \text{Desc}(C, F)$, $\lambda \mapsto h^* \lambda$, well-defined by Lemma ??.

Lemma 722 (Transport along the identity). *For every presented étale cover E of A , transport along the identity homomorphism of B_E is the identity on $\text{Desc}(C, E)$.*

Proof. Immediate from $(\text{id}_{B_E})^* = \text{id}$ (Lemma ??). $\square$

Lemma 723 (Transport is functorial). *For homomorphisms of A -algebras $h : B_E \rightarrow B_F$ and $h' : B_F \rightarrow B_G$ between carriers of presented étale covers of A , transport along $h' \circ h$ equals transport along h' followed by transport along h : $(h' \circ h)_* = h'_* \circ h_*$ on $\text{Desc}(C, E)$.*

Proof. Immediate from $(h' \circ h)^* = h'^* \circ h^*$ (Lemma ??), restricted to $\text{Desc}(C, E)$. $\square$

Theorem 724 (Refinement independence, the keystone). *Let E be a presented étale cover of A with carrier $B = B_E$, let R be a commutative k -algebra equipped with a compatible A -algebra structure (so that $k \rightarrow A \rightarrow R$ is a tower of algebras), and let $j_1, j_2 : B \rightarrow R$ be two homomorphisms of A -algebras. If $\lambda \in \text{Desc}(C, E)$, then $j_1^* \lambda = j_2^* \lambda$ in $\text{relPic}(C, R)$.*

Proof. Let $H : B \otimes_A B \rightarrow R$ be the homomorphism of A -algebras induced by the universal property of the tensor product from the pair (j_1, j_2) (their images commute in the commutative ring R): $H(b \otimes b') = j_1(b)j_2(b')$. Then $H \circ \text{inl}_E = j_1$ (since $H(b \otimes 1) = j_1(b)j_2(1) = j_1(b)$) and $H \circ \text{inr}_E = j_2$ (since $H(1 \otimes b) = j_1(1)j_2(b) = j_2(b)$). By functoriality (Lemma ??),

$$j_1^* \lambda = H^*(\text{inl}_E^* \lambda) = H^*(\text{inr}_E^* \lambda) = j_2^* \lambda,$$

using $\lambda \in \text{Desc}(C, E)$, i.e. $\text{inl}_E^* \lambda = \text{inr}_E^* \lambda$, in the middle equality. $\square$

Theorem 725 (Refinement-map independence). *Let E, F be presented étale covers of A and $h_1, h_2 : B_E \rightarrow B_F$ two homomorphisms of A -algebras. Then $h_{1,*} = h_{2,*}$ as maps $\text{Desc}(C, E) \rightarrow \text{Desc}(C, F)$: transport of a descent class along h_1 and along h_2 always agree.*

Proof. Apply Theorem ?? with $R = B_F$ (an A -algebra which is also a k -algebra compatibly, since F is a cover of the k -algebra A), $j_1 = h_1$, $j_2 = h_2$. $\square$

7.6 The one-step étale plus

Definition 726 (The refinement equivalence). Consider pairs (E, λ) with E a presented étale cover of A and $\lambda \in \text{Desc}(C, E)$. Declare $(E, \lambda) \sim (F, \mu)$ if there exist a presented étale cover G of A and homomorphisms of A -algebras $f: B_E \rightarrow B_G$, $g: B_F \rightarrow B_G$ with $f_*\lambda = g_*\mu$ in $\text{Desc}(C, G)$. This is an equivalence relation.

Proof. Reflexivity. $(E, \lambda) \sim (E, \lambda)$ via $G = E$, $f = g = \text{id}_{B_E}$.

Symmetry. If $(E, \lambda) \sim (F, \mu)$ via (G, f, g) , then $(F, \mu) \sim (E, \lambda)$ via (G, g, f) .

Transitivity. Suppose $(E, \lambda) \sim (F, \mu)$ via (G, f, g) , so $f_*\lambda = g_*\mu$, and $(F, \mu) \sim (H, \nu)$ via (G', f', g') , so $f'_*\mu = g'_*\nu$. Use the common refinement $G \times G'$ with the maps $\text{inl}_{G, G'} \circ f: B_E \rightarrow B_{G \times G'}$ and $\text{inr}_{G, G'} \circ g': B_H \rightarrow B_{G \times G'}$. Using functoriality of transport (Lemma ??) and refinement-map independence (Theorem ??, applied to the two maps $\text{inl}_{G, G'} \circ g$ and $\text{inr}_{G, G'} \circ f'$ out of B_F),

$$\begin{aligned} (\text{inl}_{G, G'} \circ f)_*\lambda &= \text{inl}_{G, G', *}(f_*\lambda) = \text{inl}_{G, G', *}(g_*\mu) = (\text{inl}_{G, G'} \circ g)_*\mu \\ &= (\text{inr}_{G, G'} \circ f')_*\mu = \text{inr}_{G, G', *}(f'_*\mu) = \text{inr}_{G, G', *}(g'_*\nu) = (\text{inr}_{G, G'} \circ g')_*\nu, \end{aligned}$$

so $(E, \lambda) \sim (H, \nu)$ via $(G \times G', \text{inl}_{G, G'} \circ f, \text{inr}_{G, G'} \circ g')$. $\square$

Definition 727 (The one-step étale plus). The set of pairs (E, λ) as in Definition ??, modulo the equivalence relation there, is denoted $\text{PicEtAff}(C, A)$: the one-step étale plus of the relative Picard functor at A . On affine test schemes this is the étale sheafification of $\text{relPic}(C, -)$, one plus construction sufficing because that presheaf is separated for the étale topology (established elsewhere).

Definition 728 (The plus class of a descent class). For a presented étale cover E of A and $\lambda \in \text{Desc}(C, E)$, write $[E, \lambda] \in \text{PicEtAff}(C, A)$ for the class of the pair (E, λ) .

Lemma 729 (Every plus class is so represented). *Every element of $\text{PicEtAff}(C, A)$ is of the form $[E, \lambda]$ for some presented étale cover E of A and $\lambda \in \text{Desc}(C, E)$.*

Proof. Immediate: $\text{PicEtAff}(C, A)$ is a quotient of the set of such pairs, and every class in a quotient is the class of some representative. $\square$

Theorem 730 (Equality of plus classes). $[E, \lambda] = [F, \mu]$ in $\text{PicEtAff}(C, A)$ if and only if there exist a presented étale cover G of A and homomorphisms of A -algebras $f: B_E \rightarrow B_G$, $g: B_F \rightarrow B_G$ with $f_*\lambda = g_*\mu$.

Proof. This is exactly the defining relation of Definition ??, restated as an equality of classes in the quotient. $\square$

Lemma 731 (Transport to a finer cover preserves the plus class). *For $h: B_E \rightarrow B_F$ a homomorphism of A -algebras between carriers of presented étale covers of A and $\lambda \in \text{Desc}(C, E)$, $[F, h_*\lambda] = [E, \lambda]$.*

Proof. Take $G = F$ with $f = \text{id}_{B_F}$ and $g = h$ in Theorem ??: $f_*(h_*\lambda) = h_*\lambda = g_*\lambda$ by Lemma ??. $\square$

7.7 The group structure and the unit

Lemma 732 (Transport through the universal map collapses). *Let E, F, H be presented étale covers of A , $f: B_E \rightarrow B_H$ and $g: B_F \rightarrow B_H$ homomorphisms of A -algebras, and let $\langle f, g \rangle: B_{E \times F} \rightarrow B_H$ be the induced map of Definition ???. Then for $\lambda \in \text{Desc}(C, E)$ and $\mu \in \text{Desc}(C, F)$:*

$$\langle f, g \rangle_*(\text{inl}_{E,F,*}\lambda) = f_*\lambda, \quad \langle f, g \rangle_*(\text{inr}_{E,F,*}\mu) = g_*\mu.$$

Proof. By functoriality of transport (Lemma ??), the left-hand sides equal $(\langle f, g \rangle \circ \text{inl}_{E,F})_*\lambda$, respectively $(\langle f, g \rangle \circ \text{inr}_{E,F})_*\mu$; by Lemma ??, $\langle f, g \rangle \circ \text{inl}_{E,F} = f$ and $\langle f, g \rangle \circ \text{inr}_{E,F} = g$. $\square$

Theorem 733 (Multiplication). *$\text{PicEtAff}(C, A)$ carries a multiplication, defined on representatives by transport to a common refinement product cover: for presented étale covers E, F of A and $\lambda \in \text{Desc}(C, E)$, $\mu \in \text{Desc}(C, F)$,*

$$[E, \lambda] \cdot [F, \mu] = [E \times F, \text{inl}_{E,F,*}\lambda \cdot \text{inr}_{E,F,*}\mu].$$

Proof. It remains to check that the formula is independent of the chosen representatives, i.e. that if $(E, \lambda) \sim (E', \lambda')$ via (H, f, f') (so $f_*\lambda = f'_*\lambda'$) and $(F, \mu) \sim (F', \mu')$ via (G, g, g') (so $g_*\mu = g'_*\mu'$), then the two resulting products agree. Use the common refinement $H \times G$ of $E \times F$ and $E' \times F'$, with the maps $\langle \text{inl}_{H,G} \circ f, \text{inr}_{H,G} \circ g \rangle: B_{E \times F} \rightarrow B_{H \times G}$ and $\langle \text{inl}_{H,G} \circ f', \text{inr}_{H,G} \circ g' \rangle: B_{E' \times F'} \rightarrow B_{H \times G}$ (Definition ??). Since transport is a homomorphism of groups (Definition ??), transporting the product $\text{inl}_{E,F,*}\lambda \cdot \text{inr}_{E,F,*}\mu$ along the first map yields, by Lemma ?? applied twice, $\text{inl}_{H,G,*}(f_*\lambda) \cdot \text{inr}_{H,G,*}(g_*\mu)$; transporting the primed product along the second map yields $\text{inl}_{H,G,*}(f'_*\lambda') \cdot \text{inr}_{H,G,*}(g'_*\mu')$; these agree term by term since $f_*\lambda = f'_*\lambda'$ and $g_*\mu = g'_*\mu'$. Hence the two products transport to the same class on $H \times G$, so by Theorem ?? they define the same element of $\text{PicEtAff}(C, A)$. $\square$

Theorem 734 (Multiplication of classes on the same cover). *For a presented étale cover E of A and $\lambda, \mu \in \text{Desc}(C, E)$, $[E, \lambda] \cdot [E, \mu] = [E, \lambda\mu]$.*

Proof. By Theorem ??, $[E, \lambda][E, \mu] = [E \times E, \text{inl}_{E,E,*}\lambda \cdot \text{inr}_{E,E,*}\mu]$. By refinement-map independence (Theorem ??, applied to the two maps $\text{inr}_{E,E}$ and $\text{inl}_{E,E}$ out of B_E), $\text{inr}_{E,E,*}\mu = \text{inl}_{E,E,*}\mu$. Since transport is a homomorphism of groups, $\text{inl}_{E,E,*}\lambda \cdot \text{inl}_{E,E,*}\mu = \text{inl}_{E,E,*}(\lambda\mu)$. Hence $[E, \lambda][E, \mu] = [E \times E, \text{inl}_{E,E,*}(\lambda\mu)] = [E, \lambda\mu]$, the last equality by Lemma ???. $\square$

Definition 735 (Identity). The identity element of $\text{PicEtAff}(C, A)$ is $1 := [\text{self}(A), 1]$, the class of the identity descent datum on the trivial cover.

Theorem 736 (The identity is represented on every cover). *For every presented étale cover E of A , $[E, 1] = 1$ in $\text{PicEtAff}(C, A)$.*

Proof. Apply Theorem ?? with $G = E$, $f = \text{id}_{B_E}$, and g the canonical map from the trivial cover (Definition ??): since transport is a homomorphism of groups, both f_*1 and g_*1 equal 1, regardless of f and g , so they agree. $\square$

Definition 737 (Inverse). $\text{PicEtAff}(C, A)$ carries an inversion, defined on representatives by $[E, \lambda]^{-1} = [E, \lambda^{-1}]$; well-defined because if $(E, \lambda) \sim (F, \mu)$ via (G, f, g) , so $f_*\lambda = g_*\mu$, then applying the group homomorphism f_* , respectively g_* , to inverses gives $f_*(\lambda^{-1}) = (f_*\lambda)^{-1} = (g_*\mu)^{-1} = g_*(\mu^{-1})$, so $(E, \lambda^{-1}) \sim (F, \mu^{-1})$ via the same (G, f, g) .

Theorem 738 ($\text{PicEtAff}(C, A)$ is a commutative group). *With the multiplication of Theorem ??, the identity of Definition ??, and the inversion of Definition ??, $\text{PicEtAff}(C, A)$ is a commutative group.*

Proof. By Lemma ?? every element is of the form $[E, \lambda]$, so it suffices to check the axioms on such representatives, re-expressing every class involved as a class on one common cover by Lemma ?? (transporting a representative up along a chosen refinement map does not change its plus class) and then applying Theorem ?? to compute products of classes living on the same cover.

Associativity. For $a = [E, \lambda]$, $b = [F, \mu]$, $c = [G, \nu]$, rewrite a , b , c as classes on the common cover $(E \times F) \times G$, transporting λ along $\text{inl}_{E \times F, G} \circ \text{inl}_{E, F}$, μ along $\text{inl}_{E \times F, G} \circ \text{inr}_{E, F}$, and ν along $\text{inr}_{E \times F, G}$ (Lemma ??); the three transported representatives $\lambda', \mu', \nu' \in \text{Desc}(C, (E \times F) \times G)$ now satisfy, by Theorem ??, $(ab)c = [(E \times F) \times G, (\lambda' \mu') \nu']$ and $a(bc) = [(E \times F) \times G, \lambda'(\mu' \nu')]$; these agree by associativity of multiplication in the group $\text{Desc}(C, (E \times F) \times G)$.

Identity. For $a = [E, \lambda]$, $1 \cdot a = [\text{self}(A) \times E, \text{inl}_* 1 \cdot \text{inr}_* \lambda] = [\text{self}(A) \times E, \text{inr}_* \lambda] = [E, \lambda] = a$, using that transport preserves the identity and Lemma ??; similarly $a \cdot 1 = a$.

Inverses. For $a = [E, \lambda]$, $a^{-1}a = [E, \lambda^{-1}][E, \lambda] = [E, \lambda^{-1}\lambda] = [E, 1] = 1$, by Theorem ?? and Theorem ??.

Commutativity. For $a = [E, \lambda]$, $b = [F, \mu]$, transport both to the common cover $E \times F$ via $\text{inl}_{E, F}$, respectively $\text{inr}_{E, F}$ (Lemma ??): $a = [E \times F, \text{inl}_* \lambda]$, $b = [E \times F, \text{inr}_* \mu]$; by Theorem ??, $ab = [E \times F, \text{inl}_* \lambda \cdot \text{inr}_* \mu]$ and $ba = [E \times F, \text{inr}_* \mu \cdot \text{inl}_* \lambda]$, which agree because $\text{Desc}(C, E \times F)$, a subgroup of the commutative group $\text{relPic}(C, B_E \otimes_A B_F)$, is commutative. $\square$

Lemma 739 (Tautological descent datum). *Let E be a presented étale cover of A and $\iota_E: A \rightarrow B_E$ the structure homomorphism. For every $x \in \text{relPic}(C, A)$, the class $\iota_E^* x \in \text{relPic}(C, B_E)$ satisfies the descent condition: $\iota_E^* x \in \text{Desc}(C, E)$.*

Proof. The two composites $\text{inl}_E \circ \iota_E$ and $\text{inr}_E \circ \iota_E$, both maps $A \rightarrow B_E \otimes_A B_E$, coincide: each sends $a \mapsto \iota_E(a) \otimes 1 = 1 \otimes \iota_E(a)$, the two descriptions of the structure homomorphism of the tensor product $B_E \otimes_A B_E$ over A . By functoriality (Lemma ??), $\text{inl}_E^*(\iota_E^* x) = (\text{inl}_E \circ \iota_E)^* x = (\text{inr}_E \circ \iota_E)^* x = \text{inr}_E^*(\iota_E^* x)$. $\square$

Definition 740 (The unit of the plus construction). The homomorphism of groups $\eta_A: \text{relPic}(C, A) \rightarrow \text{PicEtAff}(C, A)$ sending x to the class $[\text{self}(A), \iota^* x]$ of x pulled back to the trivial cover with its tautological descent datum (Lemma ??).

Proof. η_A preserves the identity: the tautological descent datum of $1 \in \text{relPic}(C, A)$ is $1 \in \text{Desc}(C, \text{self}(A))$ (transport preserves the identity), so $\eta_A(1) = [\text{self}(A), 1] = 1$ by Theorem ??.

It preserves products: the tautological descent datum of xy is the product of the tautological descent data of x and y (transport is a homomorphism), so $\eta_A(xy) = [\text{self}(A), \iota^* x \cdot \iota^* y] = [\text{self}(A), \iota^* x][\text{self}(A), \iota^* y] = \eta_A(x)\eta_A(y)$ by Theorem ??.

7.8 Functoriality in the test algebra

Throughout this section k is a field, C is an object of $\text{Over}(\text{Spec } k)$, A is a commutative k -algebra, and A' is a commutative k -algebra additionally equipped with an A -algebra structure compatible with the tower $k \rightarrow A \rightarrow A'$ (following the homomorphism-along-instances idiom, this exhibits a homomorphism of k -algebras $A \rightarrow A'$, extracted from the tower as $\tau: A \rightarrow A'$). For a presented étale cover E of A , write $E_{A'}$ for its base change (Definition ??) and $\kappa_E: B_E \rightarrow B_{E_{A'}}$ for the canonical inclusion (Definition ??).

Lemma 741 (Descent survives base change of the cover). *Let E be a presented étale cover of A and $\lambda \in \text{Desc}(C, E)$. Then $\kappa_E^* \lambda \in \text{Desc}(C, E_{A'})$.*

Proof. The two coprojections of the double cover of $E_{A'}$ (Definition ??), composed with κ_E , are two homomorphisms of A -algebras $B_E \rightarrow B_{E_{A'}} \otimes_{A'} B_{E_{A'}}$ (both factor through the A -algebra

structure of the target inherited from $A \rightarrow A'$: namely $\text{inl} \circ \kappa_E$ and $\text{inr} \circ \kappa_E$, sending $b \mapsto (1 \otimes b) \otimes 1$, respectively $b \mapsto 1 \otimes (1 \otimes b)$, under the carrier identifications. By the keystone (Theorem ??), applied with $R = B_{E_{A'}} \otimes_{A'} B_{E_{A'}}$ and $j_1 = \text{inl} \circ \kappa_E$, $j_2 = \text{inr} \circ \kappa_E$, the class $\lambda \in \text{Desc}(C, E)$ satisfies $j_1^* \lambda = j_2^* \lambda$; by functoriality this reads $\text{inl}^*(\kappa_E^* \lambda) = \text{inr}^*(\kappa_E^* \lambda)$, the descent condition for $\kappa_E^* \lambda$ on $E_{A'}$. $\square$

Definition 742 (Transport along base change of the cover). The induced homomorphism of groups $\text{Desc}(C, E) \rightarrow \text{Desc}(C, E_{A'})$, $\lambda \mapsto \kappa_E^* \lambda$.

Lemma 743 (Naturality of base-change transport). *Let E, F be presented étale covers of A and $h: B_E \rightarrow B_F$ a homomorphism of A -algebras. Then base-change transport commutes with refinement transport: for $\lambda \in \text{Desc}(C, E)$,*

$$\text{baseChangeMap}(h)_*(\kappa_E^* \lambda) = \kappa_F^*(h_* \lambda)$$

in $\text{Desc}(C, F_{A'})$, where $\text{baseChangeMap}(h): B_{E_{A'}} \rightarrow B_{F_{A'}}$ is the base change of h (Definition ??).

Proof. By Lemma ??, $\text{baseChangeMap}(h) \circ \kappa_E = \kappa_F \circ h$ as homomorphisms of A -algebras $B_E \rightarrow B_{F_{A'}}$. Both sides of the claimed identity equal, by functoriality of transport, the transport of λ along this common composite. $\square$

Definition 744 (Restriction along a base extension). The homomorphism of groups $\tau^*: \text{PicEtAff}(C, A) \rightarrow \text{PicEtAff}(C, A')$, $[E, \lambda] \mapsto [E_{A'}, \kappa_E^* \lambda]$.

Proof. Well-defined on plus classes. If $(E, \lambda) \sim (F, \mu)$ via (G, f, g) (so $f_* \lambda = g_* \mu$), then base-changing gives $(E_{A'}, \kappa_E^* \lambda) \sim (F_{A'}, \kappa_F^* \mu)$ via $(G_{A'}, \text{baseChangeMap}(f), \text{baseChangeMap}(g))$: by Lemma ??, $\text{baseChangeMap}(f)_*(\kappa_E^* \lambda) = \kappa_G^*(f_* \lambda) = \kappa_G^*(g_* \mu) = \text{baseChangeMap}(g)_*(\kappa_F^* \mu)$.

Multiplicativity. For $a = [E, \lambda]$, $b = [F, \mu]$, first transport a and b to the common refinement $E \times F$ (Lemma ??): $a = [E \times F, \text{inl}_* \lambda]$, $b = [E \times F, \text{inr}_* \mu]$, so $\tau^* a = [(E \times F)_{A'}, \kappa_{E \times F}^*(\text{inl}_* \lambda)]$ and $\tau^* b = [(E \times F)_{A'}, \kappa_{E \times F}^*(\text{inr}_* \mu)]$; by Theorem ??,

$$\tau^* a \cdot \tau^* b = [(E \times F)_{A'}, \kappa_{E \times F}^*(\text{inl}_* \lambda) \cdot \kappa_{E \times F}^*(\text{inr}_* \mu)].$$

On the other hand, $ab = [E \times F, \text{inl}_* \lambda \cdot \text{inr}_* \mu]$ (Theorem ??), so, since $\kappa_{E \times F}^*$ is a homomorphism of groups, $\tau^*(ab) = [(E \times F)_{A'}, \kappa_{E \times F}^*(\text{inl}_* \lambda) \cdot \kappa_{E \times F}^*(\text{inr}_* \mu)]$ as well. Hence $\tau^*(ab) = \tau^* a \cdot \tau^* b$. $\square$

Theorem 745 (Naturality of the unit). *For $x \in \text{relPic}(C, A)$, $\tau^*(\eta_A(x)) = \eta_{A'}(\tau^* x)$ in $\text{PicEtAff}(C, A')$, where on the right $\tau^*: \text{relPic}(C, A) \rightarrow \text{relPic}(C, A')$ is restriction along the algebra map τ (Definition ??).*

Proof. Let $\sigma: B_{\text{self}(A')} \rightarrow B_{\text{self}(A)_{A'}}$ be the canonical map from the trivial cover of A' (Definition ??, for the cover $\text{self}(A)_{A'}$ of A'). The two composite A -algebra homomorphisms $A \rightarrow B_{\text{self}(A)_{A'}}$ given by $\kappa_{\text{self}(A)} \circ \iota_{\text{self}(A)}$ and by $\sigma \circ \iota_{\text{self}(A')} \circ \tau$ both equal the structure homomorphism of $B_{\text{self}(A)_{A'}}$ as an A -algebra: the first, because $\kappa_{\text{self}(A)}$ and $\iota_{\text{self}(A)}$ are structure homomorphisms; the second, because algebra-map formation is compatible with the tower $A \rightarrow A' \rightarrow B_{\text{self}(A)_{A'}}$ (the structure map of an A -algebra obtained by restriction of scalars along $A \rightarrow A'$ factors through A'). Hence the two composites coincide, and by functoriality (Lemma ??), for $x \in \text{relPic}(C, A)$,

$$\kappa_{\text{self}(A)}^*(\iota_{\text{self}(A)}^* x) = \sigma^*(\iota_{\text{self}(A')}^*(\tau^* x)).$$

The left side represents $\tau^*(\eta_A(x)) = [\text{self}(A)_{A'}, \kappa_{\text{self}(A)}^*(\iota_{\text{self}(A)}^* x)]$ (Definition ?? applied to the representative $[\text{self}(A), \iota_{\text{self}(A)}^* x]$ of $\eta_A(x)$). By the displayed equality this equals $[\text{self}(A)_{A'}, \sigma^*(\iota_{\text{self}(A')}^*(\tau^* x))]$, which by Lemma ?? (transport along σ) equals $[\text{self}(A'), \iota_{\text{self}(A')}^*(\tau^* x)] = \eta_{A'}(\tau^* x)$. $\square$

Theorem 746 (Restriction along the trivial extension is the identity). *When $A' = A$ with $\tau = \text{id}_A$, the induced homomorphism $\tau^*: \text{PicEtAff}(C, A) \rightarrow \text{PicEtAff}(C, A)$ is the identity.*

Proof. For $[E, \lambda] \in \text{PicEtAff}(C, A)$, base-changing E along $A \rightarrow A$ gives a cover with canonical inclusion $\kappa_E: B_E \rightarrow B_{E_A}$ itself an A -algebra refinement map, and by construction transport of λ along κ_E computes the same class as $\kappa_E^* \lambda$ in the sense of Definition ???. Hence $\tau^*[E, \lambda] = [E_A, \kappa_E^* \lambda] = [E, \lambda]$ by Lemma ??, applied to the refinement map $\kappa_E: B_E \rightarrow B_{E_A}$. $\square$

Theorem 747 (Restriction is transitive). *Let A'' be a commutative k -algebra with compatible algebra structures forming a tower $k \rightarrow A \rightarrow A' \rightarrow A''$. Then, writing $\tau': A' \rightarrow A''$ and $\tau'': A \rightarrow A''$ for the induced maps, restriction along $A \rightarrow A' \rightarrow A''$ equals restriction along $A \rightarrow A''$ directly: $\tau'^* \circ \tau''^* = \tau''^*$ as homomorphisms $\text{PicEtAff}(C, A) \rightarrow \text{PicEtAff}(C, A'')$.*

Proof. Fix $[E, \lambda] \in \text{PicEtAff}(C, A)$, and write $E_{A''}$ for the direct base change along $A \rightarrow A''$ and $(E_{A'})_{A''}$ for the iterated base change along $A \rightarrow A' \rightarrow A''$. Both $\tau'^*(\tau''^*[E, \lambda])$ and $\tau''^*[E, \lambda]$ restrict, respectively, to classes on $(E_{A'})_{A''}$ and on $E_{A''}$; the two canonical inclusions of B_E into these two carriers, mapped further into the common refinement $(E_{A'})_{A''} \times E_{A''}$ via its two coprojections (Definition ???), are two homomorphisms of A -algebras out of B_E landing in the same ring. By the keystone (Theorem ???), since $\lambda \in \text{Desc}(C, E)$, these two composite pullbacks of λ agree; unwinding this equality through functoriality of transport identifies the transports of $\tau'^*(\tau''^*[E, \lambda])$ and $\tau''^*[E, \lambda]$ to the common refinement $(E_{A'})_{A''} \times E_{A''}$, so by Theorem ???, $\tau'^*(\tau''^*[E, \lambda]) = \tau''^*[E, \lambda]$. $\square$

7.9 The transport core: comparing the plus across curves and base fields

Throughout this section k_D, k_E and k_T are fields, with k_T an extension of both k_D and k_E ; D is an object of $\text{Over}(\text{Spec } k_D)$ and E an object of $\text{Over}(\text{Spec } k_E)$. A *test algebra* is a commutative k_T -algebra B , regarded as a k_D - and k_E -algebra through the two embeddings; every construction performed on test algebras below (étale covers and their carriers, products, double covers, base changes along maps of test algebras) again yields test algebras. The two consumers of this section are the curve-direction functoriality of the plus (the case $k_D = k_E = k_T$, treated at the end) and the base-field shuffle $\text{PicEtAff}(C, A) \simeq \text{PicEtAff}(C_L, A)$ (the cases $\{k_D, k_E\} = \{k, L\}$, $k_T = L$).

Definition 748 (Transport family). A *transport family* from D to E assigns to every test algebra B a morphism

$$h_B: D \otimes \text{Spec } B \longrightarrow E \otimes \text{Spec } B$$

of the underlying schemes of the two product bundles (the product on the left taken over $\text{Spec } k_D$, on the right over $\text{Spec } k_E$), such that:

1. (*projection compatibility*) $\text{snd}_E \circ h_B = \text{snd}_D$ as morphisms to the shared scheme $\text{Spec } B$;
2. (*naturality*) for every homomorphism $f: B \rightarrow B'$ of test algebras that is simultaneously k_D -linear and k_E -linear, $(E \triangleleft \text{Spec } f) \circ h_{B'} = h_B \circ (D \triangleleft \text{Spec } f)$ as morphisms $D \otimes \text{Spec } B' \rightarrow E \otimes \text{Spec } B$.

Definition 749 (Relative Picard transport). Let h be a transport family from D to E . For a test algebra B , the homomorphism of groups $h_B^*: \text{relPic}(E, \text{Spec } B) \rightarrow \text{relPic}(D, \text{Spec } B)$ induced by pullback of Čech Picard classes along h_B on representatives.

Proof. Well-definedness on the cosets: if $X = \text{snd}_E^* N$ is pulled back from $\text{Spec } B$, then $h_B^* X = h_B^* \text{snd}_E^* N = \text{snd}_D^* N$ by projection compatibility, so h_B^* carries the subgroup H of classes pulled back from the test into the corresponding subgroup on the D -side and descends to the quotients. $\square$

Lemma 750 (Bilateral naturality of the relative Picard transport). *Let h be a transport family from D to E and $f: B \rightarrow B'$ a homomorphism of test algebras that is both k_D - and k_E -linear. Then $f^* \circ h_B^* = h_{B'}^* \circ f^*$ as maps $\text{relPic}(E, \text{Spec } B) \rightarrow \text{relPic}(D, \text{Spec } B')$, where the outer restrictions are taken along f over k_D and over k_E respectively.*

Proof. On a representative Čech class the two sides pull back along $h_B \circ (D \triangleleft \text{Spec } f)$ and $(E \triangleleft \text{Spec } f) \circ h_{B'}$ respectively, and these morphisms are equal by the naturality axiom of the family; conclude by functoriality of pullback of Čech classes and the computation rules of both transports. $\square$

Lemma 751 (Pointwise-inverse families invert the transport). *Let h be a transport family from D to E and h' a transport family from E to D such that $h_B \circ h'_B = \text{id}$ on $E \otimes \text{Spec } B$ for a test algebra B . Then $h'_B \circ h_B^* = \text{id}$ on $\text{relPic}(E, \text{Spec } B)$.*

Proof. On a representative, the round trip pulls back along $h_B \circ h'_B = \text{id}$, and pullback along the identity is the identity. $\square$

Lemma 752 (Descent classes transport). *Let h be a transport family from D to E , A a test algebra and F a presented étale cover of A with carrier B_F . Then $h_{B_F}^*(\text{Desc}(E, F)) \subseteq \text{Desc}(D, F)$, and the induced homomorphism of groups $h_F^*: \text{Desc}(E, F) \rightarrow \text{Desc}(D, F)$ commutes with refinement transport: $h_G^*(r_* \lambda) = r_*(h_F^* \lambda)$ for every refinement map $r: B_F \rightarrow B_G$ of covers of A .*

Proof. The two coprojections $B_F \rightrightarrows B_F \otimes_A B_F$ into the double cover are homomorphisms of test algebras linear over both base fields, so by Lemma ?? the descent condition (equal pullbacks along the two coprojections) is preserved by $h_{B_F}^*$. Refinement transport is restriction along the refinement map, which is again linear over both base fields, so the commutation is a second instance of Lemma ?. $\square$

Definition 753 (Transport of the plus). Let h be a transport family from D to E and A a test algebra. The homomorphism of groups $h^*: \text{PicEtAff}(E, A) \rightarrow \text{PicEtAff}(D, A)$, $[F, \lambda] \mapsto [F, h_F^* \lambda]$, over the literally shared index of presented étale covers of A .

Proof. Well-defined. If $(F, \lambda) \sim (F', \mu)$ via a common refinement (G, r, r') , so that $r_* \lambda = r'_* \mu$ in $\text{Desc}(E, G)$, then applying h_G^* and commuting it past the two refinement transports (Lemma ??) gives $r_*(h_F^* \lambda) = r'_*(h_{F'}^* \mu)$, i.e. $(F, h_F^* \lambda) \sim (F', h_{F'}^* \mu)$.

Multiplicative. For $a = [F, \lambda]$ and $b = [F', \mu]$, the product is represented on the product cover $F \times F'$ by $\text{inl}_* \lambda \cdot \text{inr}_* \mu$ (Theorem ??); since $h_{F \times F'}^*$ is a homomorphism of groups commuting with the two coprojection transports (Lemma ??), its value on this representative is $\text{inl}_*(h_F^* \lambda) \cdot \text{inr}_*(h_{F'}^* \mu)$, which represents $h^* a \cdot h^* b$ (Theorem ?? again). $\square$

Theorem 754 (The transport is compatible with the unit). *For $x \in \text{relPic}(E, \text{Spec } A)$, $h^*(\eta_A(x)) = \eta_A(h_A^* x)$ in $\text{PicEtAff}(D, A)$.*

Proof. Both sides are represented on the trivial cover $\text{self}(A)$; the representatives are the pullbacks of x along the two composites of h with the structure map $A \rightarrow B_{\text{self}(A)}$, which agree by the bilateral naturality square (Lemma ??) applied to the structure map, a homomorphism of test algebras linear over both base fields. $\square$

Theorem 755 (The transport is compatible with restriction of the test). *Let $\tau: A \rightarrow A'$ be a homomorphism of test algebras linear over both base fields. Then $\tau^* \circ h^* = h^* \circ \tau^*$ as maps $\text{PicEtAff}(E, A) \rightarrow \text{PicEtAff}(D, A')$.*

Proof. On a representative $[F, \lambda]$, both sides are represented on the base-changed cover $F_{A'}$: one by transporting λ along the canonical inclusion $\kappa_F: B_F \rightarrow B_{F_{A'}}$ and then applying the family transport, the other in the opposite order. The canonical inclusion is a homomorphism of test algebras linear over both base fields, so the two representatives agree by Lemma ??.

Theorem 756 (Pointwise-inverse families invert the transport of the plus). *Let h be a transport family from D to E and h' a transport family from E to D with $h_B \circ h'_B = \text{id}$ for every test algebra B . Then $h'^* \circ h^* = \text{id}$ on $\text{PicEtAff}(E, A)$ for every test algebra A . In particular a pair of transport families that are pointwise mutually inverse induces an isomorphism of the plus groups.*

Proof. On a representative $[F, \lambda]$, the round trip transports λ by $h'^*_{B_F} \circ h^*_{B_F} = \text{id}$ (Lemma ??) on the same cover F .

7.10 Functoriality of the plus in the curve

Throughout this section k is a field, D, E are objects of $\text{Over}(\text{Spec } k)$, and A is a commutative k -algebra.

Definition 757 (Curve-direction transport of the plus). Let $g: D \rightarrow E$ be a morphism of $\text{Over}(\text{Spec } k)$. The assignment $B \mapsto g \triangleright \text{Spec } B$ is a transport family from D to E in the sense of Definition ?? (with $k_D = k_E = k_T = k$); its induced relative Picard transport is the curve-direction transport of Lemma ??, and the induced homomorphism of the plus groups is denoted

$$g^*: \text{PicEtAff}(E, A) \longrightarrow \text{PicEtAff}(D, A), \quad [F, \lambda] \longmapsto [F, (g \triangleright \text{Spec } B_F)^* \lambda],$$

transporting each descent-class representative on its own cover.

Proof. Projection compatibility is the whiskering identity $\text{snd}_E \circ (g \triangleright T) = \text{snd}_D$; naturality against $\text{Spec } f$ for a k -algebra map f is the exchange identity $(E \triangleleft \text{Spec } f) \circ (g \triangleright \text{Spec } B') = (g \triangleright \text{Spec } B) \circ (D \triangleleft \text{Spec } f)$ of whiskerings.

Theorem 758 (Laws of the curve-direction transport). *The curve-direction transport of the plus is contravariantly functorial in the curve: $\text{id}_E^* = \text{id}$, and $(g' \circ g)^* = g^* \circ g'^*$ for $g: D \rightarrow E$ and $g': E \rightarrow F$. It is compatible with the unit of the plus, $g^* \circ \eta_A = \eta_A \circ (g \triangleright \text{Spec } A)^*$, and it commutes with restriction along any base extension and along any k -algebra map of affine tests.*

Proof. The functor laws hold on each descent-class representative by the pointwise functor laws of the relative Picard curve transport (Lemma ??), since the cover index is untouched. Unit compatibility is Theorem ?? and the restriction squares are Theorem ??, both applied to the whisker family.

7.11 The base-field shuffle of the plus

Throughout this section $k \rightarrow L$ is a field extension, C is an object of $\text{Over}(\text{Spec } k)$ with base change $C_L \in \text{Over}(\text{Spec } L)$ (??), and A is an algebra in a scalar tower $k \rightarrow L \rightarrow A$. The presented étale covers of A (Definition ??) do not mention the base field, so the plus groups $\text{PicEtAff}(C, A)$ (formed over k) and $\text{PicEtAff}(C_L, A)$ (formed over L) are indexed by the *same*

covers. This section transports one to the other along the same-carrier comparison $c_B: C_L \otimes \text{Spec } B \rightarrow C \otimes \text{Spec } B$ of Definition ?? (Chapter ??), by instantiating the transport core of Definition ?? in both directions.

Definition 759 (The base-field shuffle of the plus). The same-carrier comparisons $B \mapsto c_B$ and $B \mapsto c_B^{-1}$ are transport families in the sense of Definition ?? — from C_L to C with $(k_D, k_E, k_T) = (L, k, L)$, and from C to C_L with $(k_D, k_E, k_T) = (k, L, L)$, respectively. The induced transports of the plus (Definition ??) are mutually inverse, and the resulting isomorphism of groups is the *base-field shuffle*

$$\text{Sh}_A: \text{PicEtAff}(C, A) \xrightarrow{\cong} \text{PicEtAff}(C_L, A), \quad [F, \lambda] \mapsto [F, c_{B_F}^* \lambda],$$

transporting each descent-class representative on its own cover; its inverse is $[F, \mu] \mapsto [F, (c_{B_F}^{-1})^* \mu]$.

Proof. Projection compatibility of both families is the compatibility of the comparison with the two second projections onto the shared $\text{Spec } B$ (?). Naturality of the forward family against a map $f: B \rightarrow B'$ of test algebras that is L -linear (hence k -linear through the towers) is the naturality square of the comparison (?); naturality of the backward family follows by composing that square with $c_{B'}^{-1}$ on the left and c_B^{-1} on the right. The two families are pointwise mutually inverse, so the induced transports of the plus are mutually inverse isomorphisms by Theorem ??. $\square$

Theorem 760 (The shuffle restricts the relative Picard shuffle cover-wise). *The relative Picard transports induced by the two families of Definition ?? are the base-field shuffle sh of relative Picard classes and its inverse (Definition ??): for every presented étale cover F of A with carrier B_F and every descent class $\lambda \in \text{Desc}(C, F)$, $c_{B_F}^* \lambda = \text{sh}_{B_F}(\lambda)$ in $\text{relPic}_L(C_L, B_F)$, and symmetrically $(c_{B_F}^{-1})^* \mu = \text{sh}'_{B_F}(\mu)$ for $\mu \in \text{Desc}(C_L, F)$. In particular every statement about the relative Picard shuffle — notably its degree invariance at field tests — applies verbatim to the descent-class representatives of the plus shuffle.*

Proof. Both transports are defined as descent of pullback along the same morphism c_{B_F} (resp. $c_{B_F}^{-1}$) of the same product carriers to the same quotients (Definition ?? at the comparison family, and Definition ??); they agree on every coset by their shared computation rule. $\square$

Theorem 761 (Laws of the base-field shuffle). *The base-field shuffle transports identifications of representatives: if (F, λ) and (F', λ') represent the same class of $\text{PicEtAff}(C, A)$, then $(F, c_{B_F}^* \lambda)$ and $(F', c_{B_F}^* \lambda')$ represent the same class of $\text{PicEtAff}(C_L, A)$, and symmetrically for the inverse.*

It is compatible with the units of the two plus constructions, $\text{Sh}_A \circ \eta_A = \eta_A \circ \text{sh}_A$ and $\text{Sh}_A^{-1} \circ \eta_A = \eta_A \circ \text{sh}'_A$, and it commutes with restriction of the test: for every homomorphism $\tau: A \rightarrow A'$ of algebras in scalar towers $k \rightarrow L \rightarrow A$, $k \rightarrow L \rightarrow A'$ that is L -linear (hence k -linear), $\tau^ \circ \text{Sh}_A = \text{Sh}_{A'} \circ \tau^*$ and $\tau^* \circ \text{Sh}_A^{-1} = \text{Sh}_{A'}^{-1} \circ \tau^*$, where the restrictions are formed over L on the C_L -side and over k on the C -side.*

Proof. Transport of identifications holds because the shuffle is well defined on the plus and computes cover-wise on representatives (Definition ??): apply Sh_A to the equal classes and read off the representatives. Unit compatibility is Theorem ?? at the two comparison families, with the relative Picard transports identified by Theorem ??. The restriction squares are Theorem ?? at the two families, applied to τ regarded simultaneously as a k - and an L -algebra map; on relative Picard representatives they restrict to the naturality square of the relative Picard shuffle (?). $\square$

7.12 Separatedness I: pullback along the projection is injective

The plus construction of the preceding sections is a legitimate computation of the étale sheafification only because *one* plus suffices, and that rests on the relative Picard functor being separated for the étale topology. This section establishes the geometric heart of that separatedness: pullback of Picard classes along the projection $\text{pr}: C \otimes T \rightarrow T$ is *injective*, for every test object T .

Throughout, k is a field and C is an object of $\text{Over}(\text{Spec } k)$ which is proper, geometrically irreducible and geometrically reduced over k . We recall from elsewhere in this project the definitional Picard layer on a scheme X : a *pointed cover* $\mathcal{U}$ of X assigns to each point $x \in X$ an open $\mathcal{U}(x) \ni x$; pointed covers are ordered by pointwise inclusion (a *refinement* $\mathcal{W} \leq \mathcal{U}$), with pointwise intersection as meet; a *unit Čech 1-cocycle* γ on $\mathcal{U}$ assigns to each pair x, y a unit $\gamma_{xy} \in \mathcal{O}_X(\mathcal{U}(x) \cap \mathcal{U}(y))^\times$ satisfying $\gamma_{xy}\gamma_{yz} = \gamma_{xz}$ on triple intersections; $\check{H}^1(\mathcal{U})$ is the group of such cocycles modulo coboundaries $(\delta\alpha)_{xy} = \alpha_x\alpha_y^{-1}$ for families of units $\alpha_x \in \mathcal{O}_X(\mathcal{U}(x))^\times$; restriction along a refinement is a group homomorphism $\check{H}^1(\mathcal{U}) \rightarrow \check{H}^1(\mathcal{W})$; and the Picard group $\text{Pic}(X)$ is the colimit of the $\check{H}^1(\mathcal{U})$ along the (directed) refinement order, with structure maps written $[\mathcal{U}, a] \in \text{Pic}(X)$, and with contravariant pullback $f^*: \text{Pic}(Y) \rightarrow \text{Pic}(X)$ along a morphism $f: X \rightarrow Y$, computed by pulling back the cover pointwise and the cocycle by $f^\#$.

Lemma 762 (Separation for unit sections). *Let X be a scheme, $V \subseteq X$ open, and $(W_i)_{i \in I}$ a family of opens with $W_i \subseteq V$ and $V \subseteq \bigcup_i W_i$. If two units $s, t \in \mathcal{O}_X(V)^\times$ satisfy $s|_{W_i} = t|_{W_i}$ for all i , then $s = t$.*

Proof. Units are equal once their underlying sections are, and the underlying sections agree on a cover of V ; conclude by the separation half of the sheaf axiom for $\mathcal{O}_X$. $\square$

Lemma 763 (Gluing for unit sections). *Let X be a scheme, $V \subseteq X$ open, and $(W_i)_{i \in I}$ a family of opens with $W_i \subseteq V$ and $V \subseteq \bigcup_i W_i$. Any family of units $u_i \in \mathcal{O}_X(W_i)^\times$ with $u_i|_{W_i \cap W_j} = u_j|_{W_i \cap W_j}$ for all i, j glues to a unit $s \in \mathcal{O}_X(V)^\times$ with $s|_{W_i} = u_i$.*

Proof. The underlying sections (u_i) form a compatible family, as do the inverses (u_i^{-1}) ; by the sheaf axiom they glue to sections $s, s' \in \mathcal{O}_X(V)$ respectively. The products ss' and $s's$ restrict to 1 on every W_i , hence equal 1 by the separation half of the sheaf axiom; so s is a unit, and it restricts to u_i because its underlying section does. $\square$

Lemma 764 (Restriction of unit sections is transitive). *For opens $W \subseteq V \subseteq U$ of a scheme X and $u \in \mathcal{O}_X(U)^\times$, restricting u to V and then to W is restricting u to W .*

Proof. Functoriality of the structure presheaf, applied to the underlying sections. $\square$

Theorem 765 (Refinement injectivity in degree one). *Let X be a scheme and $\mathcal{W} \leq \mathcal{U}$ pointed covers of X . Then restriction $\check{H}^1(\mathcal{U}) \rightarrow \check{H}^1(\mathcal{W})$ is injective.*

Proof. Restriction is a group homomorphism, so it suffices to show a cocycle γ on $\mathcal{U}$ whose restriction to $\mathcal{W}$ is a coboundary, say $\gamma_{zz'}|_{\mathcal{W}(z) \cap \mathcal{W}(z')} = \alpha_z\alpha_{z'}^{-1}$ with $\alpha_z \in \mathcal{O}_X(\mathcal{W}(z))^\times$, is itself a coboundary. Fix a point i and consider, for each point z , the *comparison unit*

$$s_{i,z} := \alpha_z \cdot \gamma_{iz}^{-1} \in \mathcal{O}_X(\mathcal{U}(i) \cap \mathcal{W}(z))^\times,$$

which makes sense since $\mathcal{W}(z) \subseteq \mathcal{U}(z)$. On the overlap $(\mathcal{U}(i) \cap \mathcal{W}(z)) \cap (\mathcal{U}(i) \cap \mathcal{W}(z'))$ the cocycle identity $\gamma_{iz}\gamma_{zz'} = \gamma_{iz'}$ at (i, z, z') and the coboundary relation $\alpha_z\gamma_{zz'} = \alpha_{z'}$ give $s_{i,z'} =$

$\alpha_z \gamma_{zz'} \cdot (\gamma_{iz} \gamma_{zz'})^{-1} = s_{i,z}$. Since every point $w \in \mathcal{U}(i)$ lies in $\mathcal{U}(i) \cap \mathcal{W}(w)$, these opens cover $\mathcal{U}(i)$, and by gluing (Lemma ??) there is $\beta_i \in \mathcal{O}_X(\mathcal{U}(i))^\times$ with $\beta_i|_{\mathcal{U}(i) \cap \mathcal{W}(z)} = s_{i,z}$ for all z . Finally, for any i, j , on each member $\mathcal{U}(i) \cap \mathcal{U}(j) \cap \mathcal{W}(z)$ of a cover of $\mathcal{U}(i) \cap \mathcal{U}(j)$ we compute, using the cocycle identity at (i, j, z) ,

$$\beta_i \gamma_{ij} = \alpha_z \gamma_{iz}^{-1} \gamma_{ij} = \alpha_z \gamma_{jz}^{-1} = \beta_j,$$

so $\beta_i \gamma_{ij} = \beta_j$ on $\mathcal{U}(i) \cap \mathcal{U}(j)$ by separation (Lemma ??); that is, $\gamma = \delta(\beta^{-1})$ is a coboundary. $\square$

Corollary 766 (Triviality is detected on the representing cover). *For a pointed cover $\mathcal{U}$ of a scheme X and $a \in \check{H}^1(\mathcal{U})$, the Picard class $[\mathcal{U}, a] \in \text{Pic}(X)$ is trivial if and only if $a = 1$.*

Proof. If $[\mathcal{U}, a] = 1$ then a and 1 restrict to the same class on some common refinement $\mathcal{W} \leq \mathcal{U}$; by refinement injectivity (Theorem ??) already $a = 1$. The converse is immediate. $\square$

Corollary 767 ($\check{H}^1$ of a cover embeds in the Picard group). *For every pointed cover $\mathcal{U}$ of a scheme X , the canonical map $\check{H}^1(\mathcal{U}) \rightarrow \text{Pic}(X)$, $a \mapsto [\mathcal{U}, a]$, is injective.*

Proof. The map is a group homomorphism, and its kernel is trivial by Corollary ??). $\square$

Lemma 768 (Everything is flat over a field). *Every morphism of schemes $X \rightarrow \text{Spec } K$, K a field, is flat.*

Proof. Flatness is Zariski-local on the source, so it suffices to treat $\text{Spec } B \rightarrow \text{Spec } K$, i.e. a ring homomorphism $K \rightarrow B$; every module over a field is free, hence flat. $\square$

Lemma 769 (Affine refinement of pointed covers). *Every pointed cover of a scheme admits a refinement all of whose members are affine open.*

Proof. The affine opens form a basis of the topology of a scheme: for each point x choose an affine open $V(x)$ with $x \in V(x) \subseteq \mathcal{U}(x)$. $\square$

Theorem 770 (Sections over an affine open pull back bijectively). *Let T be an object of $\text{Over}(\text{Spec } k)$ and $V \subseteq T$ an affine open. Then pullback of sections along the second projection $\text{pr}: C \otimes T \rightarrow T$ is an isomorphism*

$$\Gamma(T, V) \xrightarrow{\sim} \Gamma(C \otimes T, \text{pr}^{-1}V).$$

Proof. The underlying scheme of $C \otimes T$ is the fibre product $C \times_{\text{Spec } k} T$. Since C is proper over the field k , its underlying space is quasi-compact and quasi-separated, and $T \rightarrow \text{Spec } k$ is flat (Lemma ??); therefore the canonical square of section rings

$$\begin{array}{ccc} \Gamma(\text{Spec } k, \mathcal{O}) & \longrightarrow & \Gamma(C, \mathcal{O}_C) \\ \downarrow & & \downarrow \\ \Gamma(T, V) & \longrightarrow & \Gamma(C \otimes T, \text{pr}^{-1}V) \end{array}$$

— with the affine slots at $\text{Spec } k$ (all of it) and at $V \subseteq T$, and the quasi-compact quasi-separated slot at all of C — is a pushout of commutative rings: the flat-base-change theorem for sections of fibre products. The top edge is an isomorphism, $\Gamma(C, \mathcal{O}_C) = k$ (Corollary ??); a pushout of an isomorphism is an isomorphism, so the bottom edge is one too. $\square$

Corollary 771 (Sections pull back injectively over every open). *Let T be an object of $\text{Over}(\text{Spec } k)$ and $W \subseteq T$ any open. Then pullback of sections along the second projection, $\Gamma(T, W) \rightarrow \Gamma(C \otimes T, \text{pr}^{-1}W)$, is injective.*

Proof. Let $s, t \in \Gamma(T, W)$ have equal pullbacks. For every affine open $V \subseteq W$, restriction to $\text{pr}^{-1}V$ followed by the bijection of Theorem ?? shows $s|_V = t|_V$; the affine opens contained in W cover W , so $s = t$ by separation for $\mathcal{O}_T$. $\square$

Corollary 772 (Unit sections over an affine open pull back bijectively). *Let T be an object of $\text{Over}(\text{Spec } k)$ and $V \subseteq T$ an affine open. Then pullback along the second projection is a bijection $\Gamma(T, V)^\times \rightarrow \Gamma(C \otimes T, \text{pr}^{-1}V)^\times$ of unit groups.*

Proof. Injectivity on units follows from injectivity on sections (Corollary ??). For surjectivity, a unit $y \in \Gamma(C \otimes T, \text{pr}^{-1}V)^\times$ and its inverse are pullbacks of sections $a, b \in \Gamma(T, V)$ by Theorem ??; the products ab and ba pull back to 1, hence equal 1 by injectivity, so a is a unit pulling back to y . $\square$

Lemma 773 (The projection is surjective). *For every object T of $\text{Over}(\text{Spec } k)$, the second projection $C \otimes T \rightarrow T$ is surjective on points.*

Proof. The space of C is irreducible (geometric irreducibility over the one-point base), in particular nonempty, and $\text{Spec } k$ is a single point; so $C \rightarrow \text{Spec } k$ is surjective. Surjectivity of morphisms of schemes is stable under base change, and the second projection is the base change of $C \rightarrow \text{Spec } k$ along $T \rightarrow \text{Spec } k$. $\square$

Theorem 774 (Pullback along the projection is injective on Picard groups). *For every object T of $\text{Over}(\text{Spec } k)$, pullback of Picard classes along the second projection, $\text{pr}^*: \text{Pic}(T) \rightarrow \text{Pic}(C \otimes T)$, is injective.*

Proof. Pullback is a group homomorphism, so let $\lambda \in \text{Pic}(T)$ satisfy $\text{pr}^*\lambda = 1$. By Lemma ?? and invariance of the class under refinement we may represent $\lambda = [\mathcal{U}, \gamma]$ with $\mathcal{U}$ a pointed cover of T all of whose members $\mathcal{U}(t)$ are affine. The pullback class is represented on the pulled-back pointed cover $z \mapsto \text{pr}^{-1}\mathcal{U}(\text{pr}(z))$ by the pulled-back cocycle $\text{pr}^\# \gamma$, and it is trivial; by Corollary ??, $\text{pr}^\# \gamma$ is a coboundary on the pulled-back cover itself: there are units $w_z \in \mathcal{O}_{C \otimes T}(\text{pr}^{-1}\mathcal{U}(\text{pr}(z)))^\times$ with

$$w_z \cdot \text{pr}^\#(\gamma_{\text{pr}z, \text{pr}z'}) = w_{z'} \quad \text{on } \text{pr}^{-1}(\mathcal{U}(\text{pr}z) \cap \mathcal{U}(\text{pr}z')).$$

The projection is surjective (Lemma ??); choose for each point $t \in T$ a point $\sigma(t) \in C \otimes T$ over it. Since $\mathcal{U}(t)$ is affine, the unit $w_{\sigma(t)} \in \mathcal{O}_{C \otimes T}(\text{pr}^{-1}\mathcal{U}(t))^\times$ is the pullback of a unique unit $v_t \in \mathcal{O}_T(\mathcal{U}(t))^\times$ (Corollary ??). For any t, t' , applying the injectivity of pullback of unit sections on $\mathcal{U}(t) \cap \mathcal{U}(t')$ (Corollary ??, applied to the underlying sections of units) to the displayed coboundary relation at $(\sigma(t), \sigma(t'))$ yields $v_t \cdot \gamma_{tt'} = v_{t'}$ on $\mathcal{U}(t) \cap \mathcal{U}(t')$. Hence γ is a coboundary, $\lambda = [\mathcal{U}, \gamma] = 1$. $\square$

7.13 The Čech–Picard dictionary

For a commutative ring A , write $\text{Pic}(A)$ for its *module* Picard group: the group, under $\otimes_A$, of isomorphism classes of invertible A -modules — those A -modules N admitting an A -module N' with $N \otimes_A N' \cong A$. This section identifies, for an affine scheme X with global ring $A := \Gamma(X, \mathcal{O}_X)$, the Čech Picard group $\text{Pic}(X)$ recalled at the head of the previous section — the colimit of the groups $\check{H}^1(\mathcal{U})$ of unit Čech 1-cocycles on pointed covers $\mathcal{U}$ of X — with $\text{Pic}(A)$.

For $f \in A$, write $D(f) \subseteq X$ for its basic open, the largest open on which the image of f in every stalk is a unit; $D(f) \cap D(g) = D(fg)$, and, X being affine, a family $(D(f))_{f \in S}$ covers X exactly when S generates the unit ideal of A . A *finite basic refinement* of a pointed cover $\mathcal{U}$ is a finite pointed cover $\mathcal{W}$ refining $\mathcal{U}$ all of whose members are basic opens $D(r_i)$, $i \in I$,

of elements $r_i \in A$; since X is quasi-compact, being affine, every pointed cover admits one. A unit Čech 1-cocycle δ on such a $\mathcal{W}$ determines a *descent datum* for the finite faithfully flat algebra $A \rightarrow \prod_{i \in I} \Gamma(X, D(r_i))$, by twisting the tautological gluing data of the free rank-one modules $\Gamma(X, D(r_i))$ by the transition units δ_{ij} ; faithfully flat descent produces from this datum an A -module, well defined up to isomorphism, the *module descended from δ* .

Theorem 775 (Trivializing elements span the unit ideal). *Let A be a commutative ring and N an invertible A -module. The ideal generated by*

$$\{f \in A : S \otimes_A N \text{ is a free } S\text{-module for every localization } S \text{ of } A \text{ away from } f\}$$

is the unit ideal of A .

Proof. An invertible module is finitely presented, and for every prime ideal $\mathfrak{p}$ of A the localization of N at $\mathfrak{p}$ is a finite flat module over the local ring $A_{\mathfrak{p}}$, hence free; by finite presentation, freeness at $\mathfrak{p}$ spreads out to freeness of $S \otimes_A N$ for S a localization away from some single element $f \notin \mathfrak{p}$. If the ideal generated by all such f were proper, it would be contained in some maximal ideal $\mathfrak{m}$, contradicting the existence of a trivializing element outside $\mathfrak{m}$. Hence the ideal is all of A . $\square$

Lemma 776 (Existence of a trivializing family). *Let X be an affine scheme with global ring $A = \Gamma(X, \mathcal{O}_X)$ and N an invertible A -module. There exists a trivializing family for N : an assignment, to each point $x \in X$, of a section $s_x \in A$ with $x \in D(s_x)$, together with a trivialization*

$$\Gamma(X, D(s_x)) \otimes_A N \cong \Gamma(X, D(s_x))$$

of the base change of N to $D(s_x)$.

Proof. By Theorem ?? the elements $f \in A$ over which N becomes free after localizing away from f generate the unit ideal, so their basic opens $D(f)$ cover X . For each point $x \in X$ choose one such f with $x \in D(f)$ and set $s_x := f$, together with the resulting trivialization of $\Gamma(X, D(s_x)) \otimes_A N$. $\square$

Definition 777 (The cocycle of a trivializing family). Let $F = (s_x, \text{triv}_x)_{x \in X}$ be a trivializing family for N as in Lemma ??, and let $\mathcal{U}_F(x) := D(s_x)$ be the associated pointed cover. For points $x, y \in X$, pushing triv_x and triv_y forward along the restriction maps to $D(s_x s_y) = \mathcal{U}_F(x) \cap \mathcal{U}_F(y)$ produces two trivializations of N over $D(s_x s_y)$; their *transition unit* is the unique unit $\gamma_{xy} \in \Gamma(X, D(s_x s_y))^\times$ with

$$\gamma_{xy} \cdot (\text{pushforward of } \text{triv}_x)(1 \otimes n) = (\text{pushforward of } \text{triv}_y)(1 \otimes n) \quad \text{for all } n \in N.$$

The assignment $(x, y) \mapsto \gamma_{xy}$ is a unit Čech 1-cocycle on $\mathcal{U}_F$.

Proof. The diagonal identity $\gamma_{xx} = 1$ holds because the two pushforwards of triv_x to $D(s_x^2)$ coincide, both being the unique A -algebra homomorphism out of the localization $\Gamma(X, D(s_x))$ composed with triv_x . For the cocycle identity, push the trivializations $\text{triv}_x, \text{triv}_y, \text{triv}_z$ of a triple x, y, z forward to the triple overlap $D(s_x s_y s_z)$: since the transition unit between two trivializations composes along a chain of further pushforwards — pushing γ_{xy} and γ_{yz} both forward to $D(s_x s_y s_z)$ compares triv_x to triv_y and triv_y to triv_z there — their product equals the transition unit comparing triv_x directly to triv_z , which restricts to γ_{xz} . $\square$

Theorem 778 (Descent identification for the trivializing cocycle). *In the situation of Definition ??, let $\mathcal{W} \leq \mathcal{U}_F$ be any finite basic refinement of the trivializing cover. Then the class in $\text{Pic}(A)$ of the module descended from the cocycle $\gamma|_{\mathcal{W}}$ obtained by restricting the trivializing cocycle to $\mathcal{W}$ equals the class of N .*

Proof. Write $\mathcal{W}(i) = D(r_i)$, $i \in I$, with r_i a multiple of $s_{p(i)}$ for a chosen point $p(i)$ with $\mathcal{W}(i) \subseteq \mathcal{U}_F(p(i))$. Restricting a transition unit is compatible with pushing forward its defining trivializations further, so the restricted cocycle $\gamma|_{\mathcal{W}}$ is exactly the transition cocycle, on $\mathcal{W}$, of the trivializations of N obtained by pushing $\text{triv}_{p(i)}$ forward from $D(s_{p(i)})$ to $D(r_i)$ for each i . These pushed-forward trivializations assemble into one trivialization of N over the cover algebra $\prod_i \Gamma(X, D(r_i))$, and it intertwines the descent datum twisted by $\gamma|_{\mathcal{W}}$ with the canonical descent datum of N base-changed to the cover algebra, by the defining property of the transition units on each factor of the twisted double-overlap algebra $\prod_i \Gamma(X, D(r_i)) \otimes_A \prod_i \Gamma(X, D(r_i))$. Since the cover algebra is faithfully flat over A (a finite family of basic opens covering the affine X), uniqueness of descended modules identifies the module descended from $\gamma|_{\mathcal{W}}$ with N itself, so their classes in $\text{Pic}(A)$ agree. $\square$

Definition 779 (The Čech–Picard descent homomorphism). For an affine scheme X with global ring $A = \Gamma(X, \mathcal{O}_X)$, the homomorphism of groups

$$\text{toPic}: \text{Pic}(X) \rightarrow \text{Pic}(A), \quad [\mathcal{U}, \gamma] \mapsto (\text{class of the module descended from } \gamma|_{\mathcal{W}}),$$

where $\mathcal{W}$ is any finite basic refinement of $\mathcal{U}$ and $\gamma|_{\mathcal{W}}$ is the cocycle obtained by restricting γ along $\mathcal{W} \leq \mathcal{U}$.

Proof. The class of the descended module does not depend on the choice of finite basic refinement $\mathcal{W}$ of $\mathcal{U}$: any two such refinements are dominated by a common one, along which the two descended modules become identified. Nor does it depend on the representative $(\mathcal{U}, \gamma)$ of the class $[\mathcal{U}, \gamma] \in \text{Pic}(X)$: a further common refinement of two cohomologous representing cocycles carries cohomologous restricted cocycles, hence isomorphic descended modules. So toPic is a well-defined function; it is a homomorphism because descending along a common refinement is compatible with the pointwise product of cocycles and with tensor product of the descended modules. $\square$

Theorem 780 (Injectivity of the descent homomorphism). *For every affine scheme X , the descent homomorphism $\text{toPic}: \text{Pic}(X) \rightarrow \text{Pic}(A)$ of Definition ?? is injective.*

Proof. It suffices to show that a class $[\mathcal{U}, \gamma]$ with $\text{toPic}[\mathcal{U}, \gamma] = 1$ is itself trivial. Represent $[\mathcal{U}, \gamma]$ along a finite basic refinement $\mathcal{W} \leq \mathcal{U}$ as in Definition ??: since the module descended from $\gamma|_{\mathcal{W}}$ is isomorphic to A , its twisted descent datum is isomorphic to the trivial descent datum on $\prod_i \Gamma(X, \mathcal{W}(i))$, so the transition units of γ restricted to $\mathcal{W}$ differ from those of the trivial cocycle by an index-wise coboundary. Since the opens of $\mathcal{W}$ refine those of $\mathcal{U}$ and, member by member, cover them, this coboundary glues, by the sheaf property of $\mathcal{O}_X$, to a coboundary for γ itself on $\mathcal{U}$: $\gamma = 1$ in $\check{H}^1(\mathcal{U})$. By Corollary ??, $[\mathcal{U}, \gamma] = 1$ in $\text{Pic}(X)$. $\square$

Theorem 781 (Surjectivity of the descent homomorphism). *For every affine scheme X , the descent homomorphism $\text{toPic}: \text{Pic}(X) \rightarrow \text{Pic}(A)$ is surjective.*

Proof. Let $c \in \text{Pic}(A)$ be represented by an invertible A -module N . By Lemma ??, N admits a trivializing family, giving a pointed cover $\mathcal{U}_F$ together with the unit Čech 1-cocycle γ_F of Definition ?? . Since X is affine, hence quasi-compact, $\mathcal{U}_F$ admits a finite basic refinement $\mathcal{W}$. By Theorem ??, $\text{toPic}[\mathcal{U}_F, \gamma_F] = c$. As c was arbitrary, toPic is surjective. $\square$

Theorem 782 (The Čech–Picard dictionary). *For every affine scheme X with global ring $A = \Gamma(X, \mathcal{O}_X)$, the descent homomorphism of Definition ?? is an isomorphism of groups*

$$\text{toPic}: \text{Pic}(X) \xrightarrow{\sim} \text{Pic}(A).$$

In particular the definitional Čech Picard group $\text{Pic}(X)$ on pointed Zariski covers of X is canonically isomorphic to mathlib's Picard group of invertible A -modules.

Proof. toPic is a bijective group homomorphism by Theorem ?? (injectivity) and Theorem ?? (surjectivity), hence an isomorphism. $\square$

7.14 The cocycle presentation of the dictionary, and its naturality in the base

The Čech–Picard dictionary of the previous section passes through a *cocycle* presentation of the descended module, algebraically equivalent to but more computable than the trivializing-family presentation used there: a descent 1-cocycle is a single unit of $B \otimes_A B$, for B the (faithfully flat) cover algebra, satisfying a normalization and a cocycle identity. This section records that presentation and its main structural property, *naturality in the base ring*: the descended module, and hence its class in Pic , is compatible with base change of A along a further ring map $A \rightarrow A'$. This is the algebraic counterpart of the compatibility, in the test object, of the functors compared by Kleiman's Comparison Theorem for the relative Picard functor.

Throughout, $A \rightarrow B$ is a homomorphism of commutative rings, exhibiting B as a commutative A -algebra. Write $\mu: B \otimes_A B \rightarrow B$ for the multiplication homomorphism, $x \otimes y \mapsto xy$, and, in the triple tensor product $B \otimes_A (B \otimes_A B)$, write

$$\partial_{23}(w) := 1 \otimes w, \quad \partial_{12}(x \otimes y) := x \otimes (y \otimes 1), \quad \partial_{13}(x \otimes y) := x \otimes (1 \otimes y)$$

for the three A -algebra homomorphisms $B \otimes_A B \rightarrow B \otimes_A (B \otimes_A B)$ recording the three faces of the 2-simplex of the cover B of A .

Definition 783 (Descent 1-cocycle). A *descent 1-cocycle* relative to $A \rightarrow B$ is a unit $u \in (B \otimes_A B)^\times$ such that:

1. (normalization) $\mu(u) = 1$;
2. (cocycle identity) $\partial_{23}(u) \cdot \partial_{12}(u) = \partial_{13}(u)$ in $B \otimes_A (B \otimes_A B)$.

When $B = \prod_i A_{f_i}$ is a finite product of localizations covering $\text{Spec } A$, this is exactly a Čech 1-cocycle of units on the cover by the basic opens $D(f_i)$.

Definition 784 (The descended module of a cocycle). Let u be a descent 1-cocycle relative to $A \rightarrow B$. The *descended* A -module of u is the A -submodule

$$\text{descended}(u) := \{ x \in B : u \cdot (x \otimes 1) = 1 \otimes x \} \subseteq B,$$

the equalizer of the two A -linear maps $B \rightarrow B \otimes_A B$, $x \mapsto u \cdot (x \otimes 1)$ and $x \mapsto 1 \otimes x$; equivalently, the descended module of the descent datum on the B -module B whose coaction is $x \mapsto u \cdot (x \otimes 1)$ (counitality is the normalization condition and coassociativity is the cocycle identity).

Definition 785 (The Picard class of a cocycle). Assume moreover that B is faithfully flat over A , and let u be a descent 1-cocycle relative to $A \rightarrow B$. The *Picard class* of u is the class $\text{picClass}(u) \in \text{Pic}(A)$ of the descended module $\text{descended}(u)$.

Proof. B , as a module over itself, is free of rank one, hence invertible, and the coaction $x \mapsto u \cdot (x \otimes 1)$ is a descent datum on B relative to $A \rightarrow B$ (Definition ??) whose descended module is $\text{descended}(u)$ by definition. Since invertibility of modules descends along faithfully flat ring maps together with a descent datum, $\text{descended}(u)$ is an invertible A -module, so it has a well-defined class in $\text{Pic}(A)$. $\square$

Definition 786 (Base change of the tensor square and tensor cube of a cover). Let $A \rightarrow A'$ be a homomorphism of commutative rings and B a commutative A -algebra. The *base change of the tensor square* of the cover B along $A \rightarrow A'$ is the canonical A -algebra homomorphism

$$B \otimes_A B \longrightarrow (A' \otimes_A B) \otimes_{A'} (A' \otimes_A B), \quad x \otimes y \mapsto (1 \otimes x) \otimes (1 \otimes y).$$

Likewise, the *base change of the tensor cube* is the canonical A -algebra homomorphism on triple tensor products

$$B \otimes_A (B \otimes_A B) \longrightarrow (A' \otimes_A B) \otimes_{A'} ((A' \otimes_A B) \otimes_{A'} (A' \otimes_A B)), \quad x \otimes (y \otimes z) \mapsto (1 \otimes x) \otimes ((1 \otimes y) \otimes (1 \otimes z)).$$

Theorem 787 (Base change of a descent cocycle). *Let $A \rightarrow A'$ and $A \rightarrow B$ be homomorphisms of commutative rings and let u be a descent 1-cocycle relative to $A \rightarrow B$. Then the image of u under the base-change homomorphism of Definition ?? is a descent 1-cocycle relative to $A' \rightarrow A' \otimes_A B$.*

Proof. Write $\varphi: B \otimes_A B \rightarrow (A' \otimes_A B) \otimes_{A'} (A' \otimes_A B)$ for the base-change homomorphism and ψ for its analogue on triple tensor products (Definition ??). On elementary tensors,

$$\mu'(\varphi(x \otimes y)) = (1 \otimes x)(1 \otimes y) = 1 \otimes xy = \iota(\mu(x \otimes y)),$$

where $\iota: B \rightarrow A' \otimes_A B$, $b \mapsto 1 \otimes b$, is the canonical inclusion and μ' is the multiplication of $A' \otimes_A B$; and, using $1 \otimes_{A'} 1 = 1$ under the identification $A' \otimes_A B \cong A' \otimes_{A'} (A' \otimes_A B)$,

$$\partial'_{23}(\varphi(x \otimes y)) = \psi(\partial_{23}(x \otimes y)), \quad \partial'_{12}(\varphi(x \otimes y)) = \psi(\partial_{12}(x \otimes y)), \quad \partial'_{13}(\varphi(x \otimes y)) = \psi(\partial_{13}(x \otimes y)),$$

where $\partial'_{23}, \partial'_{12}, \partial'_{13}$ are the face maps of $A' \rightarrow A' \otimes_A B$. Both sides of each displayed identity are A -bilinear in x, y , so the identities extend from elementary tensors to all of $B \otimes_A B$ by linearity.

Applying μ' to $\varphi(u)$ and using the first identity together with the normalization $\mu(u) = 1$ gives $\mu'(\varphi(u)) = \iota(\mu(u)) = \iota(1) = 1$, the normalization of $\varphi(u)$. Applying ψ to the cocycle identity $\partial_{23}(u) \cdot \partial_{12}(u) = \partial_{13}(u)$ and using the remaining three identities gives

$$\partial'_{23}(\varphi(u)) \cdot \partial'_{12}(\varphi(u)) = \psi(\partial_{23}(u) \cdot \partial_{12}(u)) = \psi(\partial_{13}(u)) = \partial'_{13}(\varphi(u)),$$

the cocycle identity for $\varphi(u)$. Hence $\varphi(u)$ is a descent 1-cocycle relative to $A' \rightarrow A' \otimes_A B$. $\square$

Theorem 788 (Base change of the descended module). *Let $A \rightarrow A'$ and $A \rightarrow B$ be homomorphisms of commutative rings with B faithfully flat over A , and let u be a descent 1-cocycle relative to $A \rightarrow B$, with base change $u' := \varphi(u)$ relative to $A' \rightarrow A' \otimes_A B$ (Theorem ??). Then there is a canonical isomorphism of A' -modules*

$$A' \otimes_A \text{descended}(u) \cong \text{descended}(u').$$

Proof. Write $B' := A' \otimes_A B$; since faithful flatness is preserved by base change, B' is faithfully flat over A' , so $\text{descended}(u')$ is the descended module of the tautological descent datum on the B' -module B' twisted by u' (Definition ??), and by the uniqueness property of descended modules it suffices to produce a B' -linear isomorphism $B' \otimes_{A'} (A' \otimes_A \text{descended}(u)) \cong B'$ intertwining the base change of the coaction $x \mapsto u \cdot (x \otimes 1)$ on $\text{descended}(u)$ with the coaction $y \mapsto u' \cdot (y \otimes 1)$ on B' .

Cancelling the iterated base change identifies B' -modules

$$B' \otimes_{A'} (A' \otimes_A \text{descended}(u)) \cong B' \otimes_A \text{descended}(u) \cong B' \otimes_B (B \otimes_A \text{descended}(u)),$$

and tensoring the descent isomorphism $B \otimes_A \text{descended}(u) \cong B$ attached to u with B' over B turns the right side into $B' \otimes_B B \cong B'$. On elementary tensors, the resulting isomorphism sends $b' \otimes (a' \otimes n)$, for $b' \in B'$, $a' \in A'$, $n \in \text{descended}(u)$, to $(a'b') \cdot n$ (viewing $n \in \text{descended}(u) \subseteq B$ inside B' via ι); using the membership identity $u \cdot (n \otimes 1) = 1 \otimes n$, this identification carries the base-changed coaction on the left to the coaction $y \mapsto u' \cdot (y \otimes 1)$ on the right. Hence the isomorphism intertwines the two coactions, and uniqueness of descended modules produces the canonical isomorphism $A' \otimes_A \text{descended}(u) \cong \text{descended}(u')$. $\square$

Theorem 789 (Naturality of the Picard class in the base ring). *Let $A \rightarrow A'$ and $A \rightarrow B$ be homomorphisms of commutative rings with B faithfully flat over A and $A' \otimes_A B$ faithfully flat over A' , and let u be a descent 1-cocycle relative to $A \rightarrow B$ with base change u' relative to $A' \rightarrow A' \otimes_A B$ (Theorem ??). Then*

$$\text{picClass}(u') = [A' \otimes_A \text{descended}(u)] \quad \text{in } \text{Pic}(A'),$$

that is, the Picard class of the base-changed cocycle is the image of $\text{picClass}(u)$ under the base-change homomorphism $\text{Pic}(A) \rightarrow \text{Pic}(A')$, $[N] \mapsto [A' \otimes_A N]$.

Proof. By Theorem ??, $A' \otimes_A \text{descended}(u) \cong \text{descended}(u')$ as A' -modules, so these two invertible A' -modules have the same class in $\text{Pic}(A')$. The left side represents the image of $\text{picClass}(u) = [\text{descended}(u)]$ under the base-change homomorphism $[N] \mapsto [A' \otimes_A N]$, and the right side represents $\text{picClass}(u')$ by definition; hence $\text{picClass}(u') = [A' \otimes_A \text{descended}(u)]$ is the image of $\text{picClass}(u)$ under this homomorphism. $\square$

7.15 Base change of a finite localization cover cocycle

Fix a homomorphism of commutative rings $A \rightarrow A'$ and a finite family $f: \iota \rightarrow A$. For each index we fix localization models

$$S_i = A_{f_i}, \quad T_{ij} = A_{f_i f_j}, \quad W_{ijk} = A_{f_i f_j f_k},$$

and the analogous A' -side models S'_i, T'_{ij}, W'_{ijk} for the pushed family $f'_i := \varphi(f_i)$, where $\varphi: A \rightarrow A'$ is the structure map; each primed model also carries the A -algebra structure of the tower $A \rightarrow A' \rightarrow \cdot$. This section transports the whole Čech picture of the cover f to the cover f' .

Definition 790 (Base-change maps between localization models). Because f_i becomes invertible in S'_i through the tower $A \rightarrow A' \rightarrow S'_i$, there is a unique A -algebra homomorphism $S_i = A_{f_i} \rightarrow S'_i$; denote it by mapAway_i . Likewise, $f_i f_j$ is invertible in T'_{ij} and $f_i f_j f_k$ in W'_{ijk} , giving unique A -algebra homomorphisms

$$\text{mapOverlap}_{ij}: T_{ij} \rightarrow T'_{ij}, \quad \text{mapTriple}_{ijk}: W_{ijk} \rightarrow W'_{ijk}.$$

Theorem 791 (Base change of a cover cocycle). *Let $\gamma = (\gamma_{ij})$ with $\gamma_{ij} \in T_{ij}^\times$ be a cover cocycle for f : its diagonal restrictions γ_{ii} map to 1 in T_{ii} , and on triple overlaps the cocycle identity $\partial_{23}\gamma_{jk} \cdot \partial_{12}\gamma_{ij} = \partial_{13}\gamma_{ik}$ holds in W_{ijk} . Then the pushed family $\gamma'_{ij} := \text{mapOverlap}_{ij}(\gamma_{ij}) \in (T'_{ij})^\times$ is a cover cocycle for f' .*

Proof. Each restriction map in the Čech picture — the diagonal identification $T_{ii} \rightarrow T_{ij}$ and the three face maps $\partial_{12}, \partial_{13}, \partial_{23}$ from T -overlaps to W -triples — fits into a square with its primed analogue and the maps of Definition ???. Both composites in each such square are A -algebra homomorphisms out of a localization of A , which admits at most one such homomorphism to

any fixed target; hence the squares commute. Applying the diagonal square to γ_{ii} and using $\gamma_{ii} \mapsto 1$ shows $\gamma'_{ii} \mapsto 1$; applying the three face squares to the cocycle identity for γ and using multiplicativity of mapTriple_{ijk} yields the cocycle identity for γ' . Thus γ' is a cover cocycle for f' . $\square$

Definition 792 (Base-change map of cover algebras). The maps mapAway_i assemble into a componentwise A -algebra homomorphism $\prod_i S_i \rightarrow \prod_i S'_i$, and hence, by the universal property of base change, into a canonical A' -algebra homomorphism

$$j: A' \otimes_A \left(\prod_i S_i \right) \longrightarrow \prod_i S'_i, \quad a' \otimes s \mapsto a' \cdot (\text{mapAway}_i(s_i))_i,$$

the *base-change map of cover algebras*.

Theorem 793 (Base change of the descent unit). *Under the canonical identification of the double product $\prod_{i,j} T_{ij}$ with the tensor square $(\prod_i S_i) \otimes_A (\prod_i S_i)$, a cover cocycle γ determines a descent unit $c(\gamma) \in ((\prod_i S_i) \otimes_A (\prod_i S_i))^\times$. Then the image of $c(\gamma)$ under the base change of the tensor square (Definition ??) along $A \rightarrow A'$, followed by the tensor square of the base-change map j (Definition ??), is the descent unit $c(\gamma')$ of the pushed cocycle γ' (Theorem ??):*

$$(j \otimes j)(\text{tensorSqBaseChange}(c(\gamma))) = c(\gamma') \quad \text{in } \left((\prod_i S'_i) \otimes_{A'} (\prod_i S'_i) \right)^\times.$$

Proof. Both sides are units, so it suffices to compare their values under the identification of the tensor square $(\prod_i S'_i) \otimes_{A'} (\prod_i S'_i)$ with the double product $\prod_{i,j} T'_{ij}$. The composite

$$\left(\prod_{i,j} T_{ij} \right) \longrightarrow (\prod_i S_i) \otimes_A (\prod_i S_i) \xrightarrow{\text{tensorSqBaseChange}} (\prod_i S'_i) \otimes_{A'} (\prod_i S'_i) \xrightarrow{j \otimes j} \left(\prod_{i,j} T'_{ij} \right)$$

is an A -algebra homomorphism out of the product of localizations $\prod_{i,j} T_{ij}$; by the pi-extensionality principle it is determined by its values on the coordinate idempotents e_{ij} . A direct computation on e_{ij} — unfolding $\text{tensorSqBaseChange}$, the tensor product $j \otimes j$, and the componentwise maps mapAway — shows the composite agrees coordinatewise with $(\text{mapOverlap}_{ij})_{i,j}$. Applying this identity of homomorphisms to the unit representing γ gives, coordinatewise, $\gamma'_{ij} = \text{mapOverlap}_{ij}(\gamma_{ij})$, i.e. the descent unit $c(\gamma')$. $\square$

Lemma 794 (Base change preserves covering families). *If the family f is covering, $\text{span}\{f_i\} = A$, then the pushed family $f'_i = \varphi(f_i)$ is covering as well, $\text{span}\{f'_i\} = A'$.*

Proof. The ideal of A' generated by the pushed family is the extension along $\varphi: A \rightarrow A'$ of the ideal generated by f , that is $\text{span}\{\varphi(f_i)\}$ is the image ideal $\varphi_*(\text{span}\{f_i\})$. By hypothesis $\text{span}\{f_i\} = A$ is the unit ideal, and the extension of the unit ideal is again the unit ideal A' ; hence $\text{span}\{f'_i\} = A'$. $\square$

Lemma 795 (Base change of a covering product of localizations is faithfully flat). *If f is a covering family, $\text{span}\{f_i\} = A$, then the base change $A' \otimes_A (\prod_i S_i)$ of the cover algebra is faithfully flat as an A' -module.*

Proof. For each index the base change $A' \otimes_A S_i$ of the localization $S_i = A_{f_i}$ is the localization $A'_{f'_i}$ of A' at the pushed element $f'_i = \varphi(f_i)$. Since the pushed family f' is again covering (Lemma ??), the finite product $\prod_i (A' \otimes_A S_i)$ of localizations at a covering family is faithfully flat over A' . Finally base change commutes with finite products, $A' \otimes_A (\prod_i S_i) \cong \prod_i (A' \otimes_A S_i)$ as A' -modules, and faithful flatness transports along this isomorphism. $\square$

Theorem 796 (Naturality of the descent Picard class in the base ring). *Assume the localization covers are faithfully flat, $\prod_i S_i$ over A and $\prod_i S'_i$ over A' , and that f is a covering family, $\text{span}\{f_i\} = A$. Then the Picard class of the descended module of the pushed cocycle γ' is the image of the Picard class of γ under the base-change homomorphism $\text{Pic}(A) \rightarrow \text{Pic}(A')$, $[N] \mapsto [A' \otimes_A N]$:*

$$\text{picClass}(c(\gamma')) = [A' \otimes_A \text{descended}(c(\gamma))] \quad \text{in } \text{Pic}(A').$$

Proof. Since $\text{span}\{f_i\} = A$, the pushed family f' again spans A' (the ideal it generates is the image of the unit ideal under $A \rightarrow A'$), so $\prod_i (A' \otimes_A S_i) \cong A' \otimes_A \prod_i S_i$ is faithfully flat over A' , each factor being a localization $A'_{f'_i}$. By Theorem ?? the descent unit $c(\gamma')$ is the image of the base-changed unit $\text{tensorSqBaseChange}(c(\gamma))$ under the algebra map j of cover algebras (Definition ??); as the Picard class of a descent unit is unchanged by transport along an algebra map of the cover, $\text{picClass}(c(\gamma'))$ equals the Picard class of $\text{tensorSqBaseChange}(c(\gamma))$, which is the descent unit of the base change of the module cocycle $c(\gamma)$. By Theorem ?? this last class is $[A' \otimes_A \text{descended}(c(\gamma))]$, the image of $\text{picClass}(c(\gamma)) = [\text{descended}(c(\gamma))]$ (Definition ??) under $[N] \mapsto [A' \otimes_A N]$. $\square$

7.16 The projection-descent equivalence for unit sections

Throughout this section C is a curve — proper, geometrically irreducible and geometrically reduced — over a field k , and T is a test object of $\text{Over}(\text{Spec } k)$. Write $\text{pr}: (C \otimes T)_{\text{left}} \rightarrow T_{\text{left}}$ for the second projection.

Definition 797 (The projection-descent equivalence for unit sections). For an affine open $V \subseteq T_{\text{left}}$, pullback of unit sections along the second projection is a group isomorphism

$$\text{unitsSndEquiv}: \Gamma(T_{\text{left}}, V)^\times \xrightarrow{\sim} \Gamma((C \otimes T)_{\text{left}}, \text{pr}^{-1}V)^\times,$$

the packaging as a multiplicative equivalence of the bijection (Corollary ??) that pullback of unit sections over an affine open is bijective. Its inverse is the *descent* of a unit on the product to a unit on the base.

Proof. Pullback of unit sections is a group homomorphism, and it is bijective over the affine open V by Corollary ??; a bijective group homomorphism is a group isomorphism. $\square$

Lemma 798 (Uniqueness of the descended unit). *Let $V \subseteq T_{\text{left}}$ be an affine open, $v \in \Gamma(T_{\text{left}}, V)^\times$ and $w \in \Gamma((C \otimes T)_{\text{left}}, \text{pr}^{-1}V)^\times$. If v pulls back to w along the projection, then v is the descent of w , that is $v = \text{unitsSndEquiv}^{-1}(w)$.*

Proof. By Definition ??, $\text{unitsSndEquiv}(v)$ is the pullback of v , which by hypothesis is w ; applying the inverse isomorphism gives $v = \text{unitsSndEquiv}^{-1}(w)$. $\square$

Lemma 799 (Naturality of the descent equivalence in the affine open). *For affine opens $V' \subseteq V \subseteq T_{\text{left}}$, pullback of unit sections commutes with restriction:*

$$\text{unitsSndEquiv}(v|_{V'}) = (\text{unitsSndEquiv}(v))|_{\text{pr}^{-1}V'}, \quad v \in \Gamma(T_{\text{left}}, V)^\times,$$

and dually the descent commutes with restriction.

Proof. Both sides are the value of pr on unit sections composed with restriction; the two restriction squares of the structure sheaf commute, so the pullback of the restriction equals the restriction of the pullback. Applying $\text{unitsSndEquiv}^{-1}$ and using the round-trips (Definition ??) gives the dual identity. $\square$

Lemma 800 (The base square of the second projection). *For a morphism $g: T' \rightarrow T$ of test objects, whiskering $C \triangleleft g$ on the product intertwines the two projections:*

$$(C \triangleleft g)_{\text{left}} \circ (\text{snd}_{C,T})_{\text{left}} = (\text{snd}_{C,T'})_{\text{left}} \circ g_{\text{left}}.$$

Consequently the two legs of this square have equal open preimages: for every open $V \subseteq T_{\text{left}}$, $(C \triangleleft g)^{-1} \text{pr}^{-1} V = \text{pr}'^{-1} g_{\text{left}}^{-1} V$.

Proof. The identity is the projection-compatibility of the cartesian monoidal structure, $(C \triangleleft g) \circ \text{snd}_{C,T} = \text{snd}_{C,T'} \circ g$, transported through the forgetful functor $\text{Over}(\text{Spec } k) \rightarrow \text{Sch}$. Taking open preimages of V through both composites and using functoriality of preimage gives the second equality. $\square$

Theorem 801 (Naturality of the descent equivalence in the test object). *Let $g: T' \rightarrow T$ be a morphism of test objects and $V \subseteq T_{\text{left}}$ an affine open whose preimage $g_{\text{left}}^{-1} V \subseteq T'_{\text{left}}$ is also affine. For $v \in \Gamma(T_{\text{left}}, V)^\times$, descending v on T and then pulling back along $C \triangleleft g$ equals first pulling v back along g and then descending on T' ; both are units over $\text{pr}'^{-1}(g_{\text{left}}^{-1} V)$.*

Proof. Unfolding both sides as pullbacks of v along the two legs of the base square of Lemma ??, the two composite pullbacks agree because the two legs are equal morphisms and the source opens are identified by the preimage identity of that lemma. The equality of pullbacks along equal morphisms then closes the goal. $\square$

7.17 Descent in stages for unit 1-cocycles

Let $A \rightarrow B \rightarrow P$ be a tower of commutative rings. This section relates A -descent along the composite $A \rightarrow P$ with B -descent along $B \rightarrow P$, the cover being factored through the intermediate ring B .

Definition 802 (The collapse maps on tensor squares and cubes). The *collapse map* is the canonical A -algebra homomorphism

$$\pi: P \otimes_A P \longrightarrow P \otimes_B P, \quad x \otimes y \mapsto x \otimes y,$$

identifying the two A -bilinear factors as B -bilinear. Its analogue on tensor cubes is the A -algebra homomorphism

$$\pi_3: P \otimes_A (P \otimes_A P) \longrightarrow P \otimes_B (P \otimes_B P), \quad x \otimes (y \otimes z) \mapsto x \otimes (y \otimes z),$$

the receptacle for the collapse of the three face maps.

Theorem 803 (Collapse of a descent cocycle). *Let $v \in (P \otimes_A P)^\times$ be an A -descent 1-cocycle relative to $A \rightarrow P$. Then its collapse $\pi(v)$ is a B -descent 1-cocycle relative to $B \rightarrow P$.*

Proof. On elementary tensors, the multiplication μ_B of $P \otimes_B P$ satisfies $\mu_B(\pi(x \otimes y)) = xy = \mu_A(x \otimes y)$, and each B -face map factors through the collapse: $\partial_{12}^B(\pi(w)) = \pi_3(\partial_{12}^A(w))$ and likewise for $\partial_{13}, \partial_{23}$; both sides of each identity are A -bilinear, so the identities hold on all of $P \otimes_A P$. Applying μ_B to $\pi(v)$ and using the normalization $\mu_A(v) = 1$ gives $\mu_B(\pi(v)) = 1$; applying π_3 to the cocycle identity $\partial_{23}^A(v) \cdot \partial_{12}^A(v) = \partial_{13}^A(v)$ and using multiplicativity of π_3 gives $\partial_{23}^B(\pi(v)) \cdot \partial_{12}^B(\pi(v)) = \partial_{13}^B(\pi(v))$. Hence $\pi(v)$ is a B -descent 1-cocycle. $\square$

Lemma 804 (The descended module lies in the collapsed descended module). *For an A -descent 1-cocycle v , the A -descended module of v is contained, as a subset of P , in the B -descended module of the collapse $\pi(v)$.*

Proof. An element $m \in P$ lies in the A -descended module of v iff $v \cdot (m \otimes_A 1) = 1 \otimes_A m$ in $P \otimes_A P$. Applying the collapse π and using $\pi(m \otimes_A 1) = m \otimes_B 1$, $\pi(1 \otimes_A m) = 1 \otimes_B m$ and multiplicativity gives $\pi(v) \cdot (m \otimes_B 1) = 1 \otimes_B m$, i.e. m lies in the B -descended module of $\pi(v)$. $\square$

Theorem 805 (Descent in stages for the descended module). *Assume $A \rightarrow B$ and $B \rightarrow P$ are faithfully flat (so $A \rightarrow P$ is faithfully flat as well). For an A -descent 1-cocycle v , the base change to B of the A -descended module of v is canonically isomorphic to the B -descended module of the collapse:*

$$B \otimes_A \text{descended}(v) \cong \text{descended}(\pi(v)) \quad (B\text{-linear}), \quad b \otimes m \mapsto b \cdot m.$$

Proof. As $B \rightarrow P$ is faithfully flat, $\text{descended}(\pi(v))$ is recognized by the uniqueness property of descended modules from any B -module N equipped with a P -linear isomorphism $P \otimes_B N \cong P$ intertwining the tautological descent datum twisted by $\pi(v)$. Take $N = B \otimes_A \text{descended}(v)$ and the composite

$$P \otimes_B (B \otimes_A \text{descended}(v)) \cong P \otimes_A \text{descended}(v) \cong P,$$

the first isomorphism cancelling the iterated base change, the second the A -descent isomorphism attached to v . A computation on elementary tensors, using the membership identity $v \cdot (m \otimes_A 1) = 1 \otimes_A m$, shows this composite intertwines the base change of the coaction $x \mapsto v \cdot (x \otimes 1)$ with the coaction $y \mapsto \pi(v) \cdot (y \otimes 1)$. Uniqueness of descended modules then yields the stated isomorphism. $\square$

Theorem 806 (Naturality of the Picard class under descent in stages). *Under the hypotheses of Theorem ??, the Picard class over B of the collapsed cocycle is the image of the Picard class over A of v under the base-change homomorphism $\text{Pic}(A) \rightarrow \text{Pic}(B)$:*

$$\text{picClass}(\pi(v)) = [B \otimes_A \text{descended}(v)] \quad \text{in } \text{Pic}(B).$$

Proof. By Theorem ??, $B \otimes_A \text{descended}(v) \cong \text{descended}(\pi(v))$ as B -modules, so the two invertible B -modules have the same class in $\text{Pic}(B)$. The right side is $\text{picClass}(\pi(v))$ by definition (Definition ??), and the left side is the image of $\text{picClass}(v) = [\text{descended}(v)]$ under $[N] \mapsto [B \otimes_A N]$. $\square$

7.18 Naturality of the Čech–Picard dictionary in the affine scheme

Theorem 807 (Naturality of the Čech–Picard descent homomorphism in the affine scheme). *For a morphism $g: X \rightarrow Y$ of affine schemes, base change of Picard classes along the ring map $g^\#: \Gamma(Y, \mathbb{T}) \rightarrow \Gamma(X, \mathbb{T})$ intertwines the Čech–Picard descent homomorphism (Definition ??) with the pullback of Čech classes; that is, for every $L \in Y.\text{CechPic}$,*

$$\text{toPic}_X(g^*L) = \text{Pic}(g^\#)(\text{toPic}_Y(L)) \quad \text{in } \text{Pic}(\Gamma(X, \mathbb{T})).$$

This is the separatedness half of Kleiman, The Picard Scheme, Theorem 2.5(1), in its Čech incarnation.

Proof. Represent L by a unit cocycle γ on a pointed cover $\mathcal{U}$ of Y and choose a finite basic refinement P of $\mathcal{U}$. Then $\text{toPic}_Y(L)$ is the Picard class descended from the cover cocycle of γ on the finite localization cover $P.r: P.\iota \rightarrow \Gamma(Y, \top)$. Base-changing this whole Čech picture along $g^\#$ (Theorem ??) computes $\text{Pic}(g^\#)(\text{toPic}_Y(L))$ as the Picard class of the pushed cover cocycle on the pushed family $f'_i = g^\#(P.r_i)$, whose basic opens $X.\text{basicOpen}(f'_i) = g^{-1}Y.\text{basicOpen}(P.r_i)$ are the pullbacks of P 's basic opens.

On the other side, $\text{toPic}_X(g^*L)$ is the Picard class descended from the cover cocycle of $g^*\gamma$ on *any* finite basic refinement of the pulled-back cover $\mathcal{U}^{\text{pb}}$. As g need not be surjective, the pushed family f' is not itself a basic refinement of $\mathcal{U}^{\text{pb}}$; take the product refinement R indexed by $P.\iota \times Q.\iota$, for an auxiliary basic refinement Q of $\mathcal{U}^{\text{pb}}$, with sections $g^\#(P.r_i) \cdot Q.r_j$. The pushed cover cocycle, refined to R by the first projection, differs from the cover cocycle of $g^*\gamma$ on R by the index-wise coboundary of the cross evaluations γ pulled back through g ; two point-selections on a common refinement always differ by such a coboundary. Since the descended Picard class is unchanged by a coboundary, the two Picard classes coincide, which is the asserted identity. $\square$

Theorem 808 (Algebra form of the naturality). *With $\Gamma(X, \top)$ regarded as a $\Gamma(Y, \top)$ -algebra through $g^\#$, the identity of Theorem ?? reads $\text{toPic}_X(g^*L) = \text{mapAlgebra}(\text{toPic}_Y(L))$ for the base-change homomorphism of Picard groups attached to the algebra structure.*

Proof. The base-change homomorphism attached to the $\Gamma(Y, \top)$ -algebra $\Gamma(X, \top)$ is by construction $\text{Pic}(g^\#)$; the statement is therefore Theorem ?? rephrased. $\square$

7.19 The class-level coherence seed for étale separatedness

Let k be a field and $A \rightarrow B$ a homomorphism of k -algebras, and let C be a curve over k . Write $\text{Spec } A, \text{Spec } B$ for the corresponding test objects of $\text{Over}(\text{Spec } k)$. This section records the class-level coherence seed of the (C1) étale-separatedness theorem — the separatedness half of Kleiman, *The Picard Scheme*, Theorem 2.5(1).

Definition 809 (The two coprojections of a double cover of a test algebra). The two *coprojections* of the double cover $B \otimes_A B$ of B are the k -algebra homomorphisms

$$\text{tensorInl}: B \rightarrow B \otimes_A B, \quad b \mapsto b \otimes 1, \quad \text{tensorInr}: B \rightarrow B \otimes_A B, \quad b \mapsto 1 \otimes b.$$

Lemma 810 (Diagonal coherence of the two coprojections). *Precomposed with the base inclusion $A \rightarrow B$, the two coprojections agree: $\text{tensorInl} \circ (A \rightarrow B) = \text{tensorInr} \circ (A \rightarrow B)$ as k -algebra homomorphisms $A \rightarrow B \otimes_A B$.*

Proof. For $a \in A$ with image $\bar{a} \in B$, both sides send a to $\bar{a} \otimes 1$ respectively $1 \otimes \bar{a}$; over A these are equal, $\bar{a} \otimes 1 = a \cdot (1 \otimes 1) = 1 \otimes \bar{a}$, because tensoring is over A . $\square$

Theorem 811 (The class-level coherence seed). *Let L be a Čech Picard class on the curve-product $C \otimes \text{Spec } A$ and N a Čech Picard class on $\text{Spec } B$. Suppose that over the test algebra $A \rightarrow B$ the class L restricts to the pullback of N from the base:*

$$\text{pr}_B^* N = (C \triangleleft \text{Spec}(A \rightarrow B))^* L.$$

Then the two base changes of N to $\text{Spec}(B \otimes_A B)$ along the coprojections coincide: $\text{Spec}(\text{tensorInl})^ N = \text{Spec}(\text{tensorInr})^* N$.*

Proof. For each coprojection $j \in \{\text{tensorInl}, \text{tensorInr}\}$, pull back $\text{Spec}(j)^*N$ along the projection $\text{pr}_{B \otimes_A B}$. Using functoriality of Čech pullback (Lemma ??), the base square of the projection (Lemma ??) and the hypothesis, this equals $(C \triangleleft \text{Spec}(j \circ (A \rightarrow B)))^*L$. By the diagonal coherence (Lemma ??), $\text{tensorInl} \circ (A \rightarrow B) = \text{tensorInr} \circ (A \rightarrow B)$, so the two composites $\text{Spec}(\text{tensorInl})^*N$ and $\text{Spec}(\text{tensorInr})^*N$ have the same pullback along $\text{pr}_{B \otimes_A B}$. Pullback along the curve projection is injective on Picard classes (Theorem ??), hence the two base changes agree. $\square$

7.20 The Amitsur toolkit for étale separatedness

This section collects three reusable groups of statements consumed by the coherent-witness step of the (C1) étale-separatedness assembly, all revolving around the fact that only global units cross the curve projection pr .

Theorem 812 (Global gluing of a compatible unit family). *Let $\mathcal{U}$ be a pointed cover of a scheme X . A family of units $\beta_x \in \Gamma(X, \mathcal{U}_x)^\times$ whose members agree on pairwise overlaps glues to a single global unit $w \in \Gamma(X, \mathbb{T})^\times$ restricting to each β_x ; and a global unit that restricts to 1 on every member of a pointed cover — more precisely, two global units agreeing on every member — is uniquely determined.*

Proof. The members $\mathcal{U}_x$ cover $\mathbb{T}$ because $x \in \mathcal{U}_x$. The gluing statement is the sheaf gluing for unit sections (Lemma ??) at the tautological cover $V = \mathbb{T}$, $W = \mathcal{U}$; the uniqueness statement is the separation for unit sections (Lemma ??) at the same cover. $\square$

Definition 813 (The three cofaces on the triple tensor). Extending the coprojection idiom of Definition ??, the three *cofaces* on the triple tensor are the k -algebra homomorphisms $B \otimes_A B \rightarrow B \otimes_A (B \otimes_A B)$

$$\text{tensorFace}_{12}(x \otimes y) = x \otimes (y \otimes 1), \quad \text{tensorFace}_{13}(x \otimes y) = x \otimes (1 \otimes y), \quad \text{tensorFace}_{23}(w) = 1 \otimes w,$$

the k -restrictions of the A -linear descent cofaces.

Lemma 814 (Simplicial coincidences of the cofaces). *The six composites $\text{tensorFace} \circ \text{coprojection}: B \rightarrow B \otimes_A (B \otimes_A B)$ collapse onto the three insertions $x \mapsto x \otimes (1 \otimes 1)$, $x \mapsto 1 \otimes (x \otimes 1)$, $x \mapsto 1 \otimes (1 \otimes x)$. In particular $\text{tensorFace}_{12} \circ \text{tensorInl} = \text{tensorFace}_{13} \circ \text{tensorInl}$, $\text{tensorFace}_{12} \circ \text{tensorInr} = \text{tensorFace}_{23} \circ \text{tensorInl}$, and $\text{tensorFace}_{13} \circ \text{tensorInr} = \text{tensorFace}_{23} \circ \text{tensorInr}$.*

Proof. Each identity is checked on a generator $b \in B$: both sides insert b into the same tensor position and fill the remaining two positions with 1, by the definitions of the cofaces and coprojections. $\square$

Definition 815 (The top-open descent equivalence for global units). Specializing the projection-descent equivalence (Definition ??) to the affine open $V = \mathbb{T}$, pullback of global units along the second projection is a group isomorphism

$$\Gamma((\text{Spec } R)_{\text{left}}, \mathbb{T})^\times \xrightarrow{\sim} \Gamma((C \otimes \text{Spec } R)_{\text{left}}, \mathbb{T})^\times$$

for a k -algebra R ; its forward map is pullback of global units along pr .

Proof. Both $(\text{Spec } R)_{\text{left}}$ and $(C \otimes \text{Spec } R)_{\text{left}}$ have affine top open, and the projection preimage of $\mathbb{T}$ is $\mathbb{T}$; so Definition ?? applies at $V = \mathbb{T}$, and its forward map is pullback of global units. $\square$

Theorem 816 (Global-unit pullback is bijective). *Pullback of global units along the second projection $\text{pr}: (C \otimes \text{Spec } R)_{\text{left}} \rightarrow (\text{Spec } R)_{\text{left}}$ is bijective: every global unit on the product descends to a unique global unit on the base.*

Proof. This is the bijectivity of the group isomorphism of Definition ??: surjectivity gives the descent of any global unit as the image of the inverse isomorphism, and injectivity is separation of global units by the projection. $\square$

Theorem 817 (Naturality of the top-open descent). *For a k -algebra homomorphism $\varphi: R \rightarrow R'$, descending a global unit on $\text{Spec } R$ and pulling it back along $C \triangleleft \text{Spec } \varphi$ equals first pulling back along $\text{Spec } \varphi$ and then descending on $\text{Spec } R'$.*

Proof. This is the naturality of the projection-descent equivalence in the test object (Theorem ??) specialized to $V = \top$ along the morphism $\text{Spec } \varphi$; all the opens occurring there are $\top$ on the nose, so the general naturality square applies directly. $\square$

7.21 The pullback calculus for unit cochains

The coherent-witness construction of the next section manipulates unit Čech cochains by pulling them back along morphisms of schemes and combining the results multiplicatively. This section records the elementary calculus of those operations over abstract schemes, so that the concrete instances in the construction are single applications.

Throughout, for a morphism $f: X \rightarrow Y$, a unit section $s \in \mathcal{O}_Y(V)^\times$ on an open $V \subseteq Y$, and an open $O \subseteq X$ contained in the preimage $f^{-1}(V)$, write $f_O^\# s \in \mathcal{O}_X(O)^\times$ for the pullback of s along f , restricted to O . This operation is a group homomorphism on units, is functorial in the morphism, $(r \circ f)_O^\# s = f_O^\#(r^\# s)$ whenever the composite is defined, and commutes with shrinking O . When $u \in \mathcal{O}_Y(\top)^\times$ is a *global* unit, its pullback $f_O^\#(u|_\top)$ is the restriction to O of the global-section pullback $f_\top^\# u \in \mathcal{O}_X(\top)^\times$.

Lemma 818 (Congruence in the morphism). *Let $g, g': X \rightarrow Y$ be two morphisms with $g = g'$, let V assign to each point of Y an open of Y , and let s assign to each point y a unit section $s_y \in \mathcal{O}_Y(V(y))^\times$. Then for every $x \in X$ and every open $O \subseteq g^{-1}(V(g(x)))$,*

$$g_O^\# s_{g(x)} = (g')_O^\# s_{g'(x)}.$$

The same holds for doubly point-indexed families $V(y, y')$, $s_{y, y'}$, such as the pair evaluations of a unit Čech cocycle.

Proof. Since $g = g'$, the two sides are identical: the equality of morphisms transports the source base point, the open $V(g(x)) = V(g'(x))$, and the section $s_{g(x)} = s_{g'(x)}$ simultaneously, so the pullbacks agree. $\square$

Lemma 819 (Pullback of a coboundary relation). *Let $\varphi: X \rightarrow Y$, $r_1: Y \rightarrow Z_1$, $r_2: Y \rightarrow Z_2$. Suppose a unit 0-cochain c on a pointed cover $\mathcal{C}$ of Y cobounds the comparison of two pair-indexed unit families E_1 on Z_1 and E_2 on Z_2 , pulled back along r_1 and r_2 : for all $y, y' \in Y$, on the overlap $\mathcal{C}(y) \cap \mathcal{C}(y')$,*

$$c_y \cdot (r_1^\# E_1)_{yy'} = (r_2^\# E_2)_{yy'} \cdot c_{y'}.$$

Then for every $x, x' \in X$ and every open $O \subseteq \varphi^{-1}(\mathcal{C}(\varphi x) \cap \mathcal{C}(\varphi x'))$, the pullback of the relation along φ holds in composite normal form:

$$\varphi_O^\# c_{\varphi x} \cdot ((r_1 \varphi)_O^\# E_1)_{xx'} = ((r_2 \varphi)_O^\# E_2)_{xx'} \cdot \varphi_O^\# c_{\varphi x'},$$

where $r_i\varphi$ denotes the composite $X \xrightarrow{\varphi} Y \xrightarrow{r_i} Z_i$.

Proof. Apply the pullback homomorphism $\varphi_O^\#$ to both sides of the given relation at the points $\varphi x, \varphi x'$. Being a group homomorphism on units it distributes over the products; and by functoriality it merges each pullback, $\varphi_O^\#(r_i^\#E_i) = (r_i\varphi)_O^\#E_i$, while leaving $\varphi_O^\#c$ unchanged. The two normal forms result. $\square$

Lemma 820 (The global-unit pullback in restriction form). *Let $f: X \rightarrow Y$ and $u \in \mathcal{O}_Y(\top)^\times$ a global unit. The restriction of u along $\top \leq \top$ is u itself; and for any point-indexed open V and any open $O \subseteq f^{-1}(V)$, pulling the restricted unit back along f equals the restriction to O of the global-section pullback of u :*

$$f_O^\#(u|_V) = (f_\top^\#u)|_O.$$

In particular, at $V = \top$ the open-indexed and global spellings of pulling back a global unit coincide.

Proof. The restriction along the identity inclusion $\top \leq \top$ is the identity map on sections. For the second identity, both sides are the image of u under one and the same ring map $\mathcal{O}_Y(\top) \rightarrow \mathcal{O}_X(O)$ — the composite of the comorphism of f on global sections with restriction to O — so they agree; the $V = \top$ case is the same statement read backwards. $\square$

Lemma 821 (Functoriality of the global-unit pullback). *For morphisms $f: X \rightarrow Y$, $g: Y \rightarrow Z$ and a global unit $u \in \mathcal{O}_Z(\top)^\times$, pulling u back along g and then along f equals pulling it back along the composite $g \circ f$:*

$$f_\top^\#(g_\top^\#u) = (g \circ f)_\top^\#u.$$

Proof. The comorphism of a composite of morphisms is the composite of the comorphisms in the reverse order; applying this on global sections and passing to units gives the claim. $\square$

Lemma 822 (Pushing a pullback through a product of cochain values). *Let $\varphi: X \rightarrow Y$ and let a, b, c be pulled-back cochain values along morphisms $g_1, g_2, g_3: Y \rightarrow Z$, each a unit section on a common open V of Y . For any open $O \subseteq \varphi^{-1}(V)$,*

$$\varphi_O^\#\left(\frac{a \cdot b}{c}\right) = \frac{(g_1\varphi)_O^\#a \cdot (g_2\varphi)_O^\#b}{(g_3\varphi)_O^\#c},$$

and likewise for a combination of the shape $t \cdot (a/b)$; pushing the pullback through the combination merges each factor into a composite pullback.

Proof. The pullback $\varphi_O^\#$ is a group homomorphism on units, so it distributes over the products and quotients; each individual pullback then merges by functoriality, $\varphi_O^\#(g^\#x) = (g\varphi)_O^\#x$. $\square$

Lemma 823 (Pullback of a glued identity in composite normal form). *Let $\varphi: X \rightarrow Y$ and morphisms $m_1, m_2, m_3: Y \rightarrow S$ with composites $t_i = m_i\varphi: X \rightarrow S$. If a global unit W on Y restricts on an open V to a combination $\frac{(m_1^\#\theta)(m_2^\#\theta)}{m_3^\#\theta}$ of pullbacks of a cochain θ , then the global-section pullback of W along φ restricts on any $O \subseteq \varphi^{-1}(V)$ to the corresponding combination $\frac{(t_1^\#\theta)(t_2^\#\theta)}{t_3^\#\theta}$ of the composite pullbacks. The analogous statement holds for a comparison combination of the shape $t \cdot (a/b)$.*

Proof. By Lemma ?? the restriction of the global-section pullback of W to O equals $\varphi_O^\#$ of the V -restriction of W , which by hypothesis is the given combination of m_i -pullbacks; pushing $\varphi_O^\#$ through that combination and merging the compositions (Lemma ??) produces the combination of $t_i = m_i\varphi$ pullbacks. The output composites are the named t_i by hypothesis. $\square$

7.22 The Amitsur-coherent Čech witness

Let k be a field, $A \rightarrow B$ a homomorphism of k -algebras, and C an object of $\text{Over}(\text{Spec } k)$ which is proper, geometrically irreducible and geometrically reduced over k . Retain the coprojections $q_1 = \text{Spec}(\text{tensorInl})$, $q_2 = \text{Spec}(\text{tensorInr})$: $\text{Spec } B \rightarrow \text{Spec}(B \otimes_A B)$ of Definition ?? and the three cofaces $f_{12}, f_{13}, f_{23} = \text{Spec}(\text{tensorFace}_\bullet)$: $\text{Spec}(B \otimes_A B) \rightarrow \text{Spec}(B \otimes_A (B \otimes_A B))$ of Definition ?. This section implements the coherent-witness step of the étale-separatedness argument of Kleiman, *The Picard Scheme*: starting from the class equality $q_1^*N = q_2^*N$ supplied by the class-level coherence seed (Theorem ??), it produces a Čech witness cochain that is moreover *Amitsur-coherent* over the triple tensor, correcting an arbitrary witness by a single global unit.

Lemma 824 (The coface coincidences at Spec level and on the curve product). *The three coface–coprojection coincidences of Lemma ?? persist after applying Spec and after whiskering with C : as morphisms of the corresponding curve products,*

$$f_{12} \circ q_1 = f_{13} \circ q_1, \quad f_{12} \circ q_2 = f_{23} \circ q_1, \quad f_{13} \circ q_2 = f_{23} \circ q_2,$$

and the same three identities hold with each map replaced by its whiskering $C \triangleleft (-)$.

Proof. Each coincidence is the image of the corresponding tensor-algebra coincidence of Lemma ?? under the functor Spec (on test objects), which is contravariant on algebra maps, respectively under the further whiskering functor $C \triangleleft (-)$; both functors preserve composition, so the equal composites of algebra maps become equal composites of morphisms. $\square$

Lemma 825 (The diagonal coincidence on the curve product). *Composed with the whiskered base inclusion $C \triangleleft \text{Spec}(A \rightarrow B)$, the two whiskered coprojections agree:*

$$(C \triangleleft q_1) \circ (C \triangleleft \text{Spec}(A \rightarrow B)) = (C \triangleleft q_2) \circ (C \triangleleft \text{Spec}(A \rightarrow B)).$$

Proof. This is the image of the diagonal coherence $\text{tensorInl} \circ (A \rightarrow B) = \text{tensorInr} \circ (A \rightarrow B)$ of Lemma ?? under Spec and the whiskering functor $C \triangleleft (-)$, both of which preserve composition. $\square$

Definition 826 (The common refinement of the three coface pullbacks). For a pointed cover $\mathcal{V}$ of $\text{Spec}(B \otimes_A B)$, its *Amitsur refinement* is the pointwise intersection

$$\mathcal{V}_{123} := f_{23}^* \mathcal{V} \cap f_{12}^* \mathcal{V} \cap f_{13}^* \mathcal{V}$$

of the pullbacks of $\mathcal{V}$ along the three cofaces, a pointed cover of the triple tensor $\text{Spec}(B \otimes_A (B \otimes_A B))$.

Definition 827 (An Amitsur-coherent Čech witness). Let γ be a unit Čech cocycle on a pointed cover $\mathcal{N}$ of $\text{Spec } B$. An *Amitsur-coherent Čech witness* for γ consists of a pointed cover $\mathcal{V}$ of $\text{Spec}(B \otimes_A B)$ refining both coprojection pullbacks $q_1^* \mathcal{N}$ and $q_2^* \mathcal{N}$, together with a unit 0-cochain θ on $\mathcal{V}$ such that:

1. (*witness*) θ cobounds the comparison of the two coprojection pullbacks of γ : on the overlaps of $\mathcal{V}$, $\theta_x \cdot (q_1^\sharp \gamma)_{xy} = (q_2^\sharp \gamma)_{xy} \cdot \theta_y$;
2. (*coherent*) on the Amitsur refinement $\mathcal{V}_{123}$, the three coface pullbacks of θ satisfy the Amitsur cocycle relation $(f_{23}^\sharp \theta) \cdot (f_{12}^\sharp \theta) = f_{13}^\sharp \theta$.

The witness relation records, at cocycle level, the class equality $q_1^*N = q_2^*N$ for $N = [\mathcal{N}, \gamma]$; the coherence relation is the triple-overlap cocycle identity that lets θ itself descend.

Lemma 828 (Global-unit correction of a witness). *Let θ_0 be a witness cochain on a common refinement $\mathcal{V}$ of the two coprojection pullbacks of $\mathcal{N}$ (satisfying the witness relation for γ), and let χ be a global unit on $\text{Spec}(B \otimes_A B)$ whose Amitsur coboundary $(f_{23}^\sharp \chi)(f_{12}^\sharp \chi)/(f_{13}^\sharp \chi)$ restricts, on $\mathcal{V}_{123}$, to the Amitsur defect $(f_{23}^\sharp \theta_0)(f_{12}^\sharp \theta_0)/(f_{13}^\sharp \theta_0)$ of θ_0 . Then the corrected cochain $\theta := \theta_0/\chi$ is an Amitsur-coherent Čech witness for γ .*

Proof. Take the cover of the witness to be $\mathcal{V}$ and the cochain to be $\theta = \theta_0/(\chi|)$, where $\chi|$ is the restriction of the global unit χ to the members of $\mathcal{V}$. For the witness relation: dividing both sides of the witness relation for θ_0 by the central unit χ , which restricts the same way at x and y , preserves it, so θ is again a witness. For coherence: the Amitsur defect of θ is the Amitsur defect of θ_0 divided by the Amitsur combination $(f_{23}^\sharp \chi)(f_{12}^\sharp \chi)/(f_{13}^\sharp \chi)$ of the coface restrictions of χ . Each coface pullback of the restricted global unit χ is the restriction of its global-section pullback (Lemma ??), so this Amitsur combination is exactly the restriction of the global unit displayed in the hypothesis; by hypothesis it equals the Amitsur defect of θ_0 . The quotient is therefore 1, i.e. $(f_{23}^\sharp \theta)(f_{12}^\sharp \theta) = f_{13}^\sharp \theta$. $\square$

Definition 829 (The Amitsur defect of a witness cochain). The *Amitsur defect* of a unit 0-cochain θ_0 on a pointed cover $\mathcal{V}$ of $\text{Spec}(B \otimes_A B)$ is the unit 0-cochain

$$\partial\theta_0 := \frac{(f_{23}^\sharp \theta_0) \cdot (f_{12}^\sharp \theta_0)}{f_{13}^\sharp \theta_0}$$

on the Amitsur refinement $\mathcal{V}_{123}$. A witness is Amitsur-coherent precisely when its defect is the constant unit 1.

Theorem 830 (The Amitsur defect of a witness glues to a global unit). *If θ_0 is a witness cochain for γ on a common refinement $\mathcal{V}$ (satisfying the witness relation), then its Amitsur defect $\partial\theta_0$ has trivial Čech coboundary and glues to a global unit $\bar{\omega}$ on the triple tensor $\text{Spec}(B \otimes_A (B \otimes_A B))$, restricting to $\partial\theta_0$ on each member of $\mathcal{V}_{123}$.*

Proof. Pull the witness relation for θ_0 back along each of the three cofaces f_{23}, f_{12}, f_{13} (Lemma ??), obtaining on the Amitsur refinement three coboundary relations between the coface pullbacks of θ_0 and the pulled-back coprojections of γ . By the coface coincidences (Lemma ??) the six composite maps $f_\bullet \circ q_\bullet$ collapse onto the three insertions, so — after rewriting the point-indexed γ -terms accordingly (Lemma ??) — the three relations share matching γ -factors. Multiplying the f_{23} - and f_{12} -relations and dividing by the f_{13} -relation, the γ -factors cancel telescopically, leaving that the defect values $\partial\theta_0$ at any two points of $\mathcal{V}_{123}$ agree on their overlap. Thus $\partial\theta_0$ is a compatible unit family, and by global gluing of a compatible family (Theorem ??) it glues to a global unit $\bar{\omega}$. $\square$

Lemma 831 (Compatibility of a ratio cochain from two coboundary relations). *Let $\varphi_t: X \rightarrow Y_c$ and $\varphi_1, \varphi_2: X \rightarrow Y_a$. Suppose a unit 0-cochain c on Y_c cobounds the comparison of a family E_Z along two maps $Y_c \rightarrow Z$, and a unit 0-cochain a on Y_a cobounds the comparison of E_Z (along a map $Y_a \rightarrow Z$) with a family E_W (along a map $Y_a \rightarrow W$). If the pulled-back comparison composites coincide pairwise and the two maps φ_1, φ_2 agree after the map to W (the diagonal coincidence), then the ratio cochain $\varphi_t^\sharp c \cdot (\varphi_2^\sharp a / \varphi_1^\sharp a)$ on X is Čech-compatible: its values at any two points agree on a common open.*

Proof. Pull the two coboundary relations back to X : the relation for c along φ_t , and the relation for a along each of φ_1, φ_2 (Lemma ??). The pairwise coincidence of composites lets us identify the E_Z -terms of the c -relation with those of the a -relations, and the diagonal coincidence identifies the two E_W -terms of the φ_1 - and φ_2 -pullbacks (Lemma ??). Writing Q_1, Q_2 for the common E_Z -terms and L for the common E_W -term, the three relations read $c_x Q_1 = Q_2 c_{x'}$, $a_i Q_i = L a'_i$; a direct computation in the unit group then gives $c_x(a_2/a_1) = c_{x'}(a'_2/a'_1)$, which is compatibility of the ratio cochain. $\square$

Definition 832 (The comparison cover on the curve product). On the curve product $X_{B \otimes B} := C \otimes \text{Spec}(B \otimes_A B)$, the *comparison cover* of a witness cover $\mathcal{V}$ of $\text{Spec}(B \otimes_A B)$ and an upstairs cover $\mathcal{A}$ of $X_B := C \otimes \text{Spec} B$ is the pointwise intersection

$$\text{pr}^* \mathcal{V} \sqcap (u_1^* \mathcal{A} \sqcap u_2^* \mathcal{A}),$$

where $\text{pr}: X_{B \otimes B} \rightarrow \text{Spec}(B \otimes_A B)$ is the projection and $u_1, u_2 = C \triangleleft q_1, C \triangleleft q_2: X_{B \otimes B} \rightarrow X_B$ are the whiskered coprojections.

Definition 833 (The comparison cochain). Given a witness cochain θ_0 on $\mathcal{V}$ and an upstairs witness cochain α on $\mathcal{A}$, their *comparison cochain* on the comparison cover is the unit 0-cochain

$$\text{pr}^\# \theta_0 \cdot \frac{u_2^\# \alpha}{u_1^\# \alpha}.$$

Theorem 834 (The comparison cochain is Čech-compatible). *Let θ_0 be a witness cochain for γ downstairs and α a witness cochain upstairs for the lift — that is, α cobounds the comparison of the projection pullback of γ with the pullback along $C \triangleleft \text{Spec}(A \rightarrow B)$ of a unit cocycle λ representing the class L . Then the comparison cochain of θ_0 against α is Čech-compatible, hence has trivial coboundary.*

Proof. Apply the compatibility principle for ratio cochains (Lemma ??) with $\varphi_t = \text{pr}$ and $\varphi_1 = u_1, \varphi_2 = u_2$: the witness relation for θ_0 downstairs and the witness relation for α upstairs are the two coboundary relations. The base squares of the projection (Lemma ??) identify pr followed by each coprojection q_i with u_i followed by the projection pr_B , giving the pairwise coincidence of the γ -composites; and the diagonal coincidence (Lemma ??) identifies u_1 and u_2 after the whiskered base inclusion, cancelling the λ -terms carrying the class L . The hypotheses of Lemma ?? are met, and its conclusion is exactly compatibility of the comparison cochain. $\square$

Theorem 835 (The comparison cochain glues and descends to the correcting unit). *In the setting of Theorem ??, the comparison cochain glues to a global unit on the curve product $X_{B \otimes B}$, which descends through the projection to a global unit χ on $\text{Spec}(B \otimes_A B)$ whose projection pullback restricts to the comparison cochain.*

Proof. By Theorem ?? the comparison cochain is a compatible unit family, so it glues to a global unit ψ on $X_{B \otimes B}$ (Theorem ??). Global-unit pullback along the projection is a group isomorphism (Definition ??); let χ be the image of ψ under its inverse. Then the projection pullback of χ is ψ , which restricts to the comparison cochain on each member of the comparison cover. $\square$

Theorem 836 (The glued defect is the Amitsur coboundary of the comparison unit). *Suppose the glued Amitsur defect $\bar{\omega}$ of a witness cochain θ_0 restricts to the Amitsur defect $\partial\theta_0$, and the comparison unit χ pulls back on the curve product to the comparison cochain of θ_0 against an upstairs witness α . Then $\bar{\omega}$ is the Amitsur coboundary of χ :*

$$\bar{\omega} = \frac{(f_{23}^\# \chi) \cdot (f_{12}^\# \chi)}{f_{13}^\# \chi}.$$

Proof. Both sides are global units on the triple tensor; since global-unit pullback along the curve projection is injective (Theorem ??), it suffices to compare their pullbacks to the triple-tensor curve product. Pull $\bar{\omega}$ back along the projection: by Lemma ?? and the base squares of the projection (Lemma ??) it restricts to the Amitsur combination of the three whiskered-coface pullbacks of θ_0 . Pull each coface pullback of χ back along the projection: by functoriality of the global-unit pullback (Lemma ??, Lemma ??) and the base squares it becomes the corresponding whiskered-coface pullback of the descended comparison unit, which by Lemma ?? together with the coface coincidences (Lemma ??) restricts to the same θ_0 -term times a telescoping combination of correction factors built from α . Forming the Amitsur combination of the three, the α -telescope cancels exactly, leaving precisely the Amitsur combination of θ_0 -terms found for $\bar{\omega}$. The two global units thus have equal projection pullbacks, hence are equal. $\square$

Theorem 837 (Existence of the Amitsur-coherent Čech witness). *Let C be proper, geometrically irreducible and geometrically reduced over the field k , let $k \rightarrow A \rightarrow B$ be a tower of algebras, let L be a Čech Picard class on $C \otimes \text{Spec } A$, and let γ be a unit cocycle on a pointed cover $\mathcal{N}$ of $\text{Spec } B$ whose class N pulls back, on the curve product, from L : $\text{pr}_B^* N = (C \triangleleft \text{Spec}(A \rightarrow B))^* L$. Then there exists an Amitsur-coherent Čech witness for γ — a witness for the class equality $q_1^* N = q_2^* N$ over $\text{Spec}(B \otimes_A B)$ whose coface pullbacks are Amitsur-coherent over the triple tensor.*

Proof. By the class-level coherence seed (Theorem ??) the hypothesis gives the equality $q_1^* N = q_2^* N$; unwinding it through the equality criterion for Čech Picard classes yields a common refinement $\mathcal{V}$ of the two coprojection pullbacks of $\mathcal{N}$ and a witness cochain θ_0 on it satisfying the witness relation. Likewise, the hypothesis $\text{pr}_B^* N = (C \triangleleft \text{Spec}(A \rightarrow B))^* L$ yields an upstairs witness cochain α on a cover $\mathcal{A}$ of X_B , cobounding the comparison of the projection pullback of γ with the pullback of a representative λ of L . The Amitsur defect of θ_0 glues to a global unit $\bar{\omega}$ (Theorem ??); the comparison of θ_0 against α glues and descends to a global unit χ on $\text{Spec}(B \otimes_A B)$ (Theorem ??); and $\bar{\omega}$ is identified with the Amitsur coboundary of χ (Theorem ??). Thus χ satisfies the hypothesis of the correction lemma (Lemma ??), and the corrected cochain θ_0/χ is the required Amitsur-coherent Čech witness. $\square$

7.23 The simplicial degeneracies of the tensor tower

Complementing the cofaces of Definition ?? and their Spec-level coincidences (Lemma ??), the composite descent unit pulls back along the two *degeneracies* of the tensor tower: the multiplication, whose Spec is the diagonal, and the first-two-positions multiplication, whose Spec is a degeneracy of the triple tensor.

Definition 838 (The multiplication and the first degeneracy). The *multiplication* is the k -algebra homomorphism

$$\text{tensorMul}: B \otimes_A B \rightarrow B, \quad x \otimes y \mapsto xy,$$

whose Spec is the diagonal $\Delta: \text{Spec } B \rightarrow \text{Spec}(B \otimes_A B)$. The *first degeneracy* is the k -algebra homomorphism multiplying the first two tensor positions,

$$\text{tensorMul}_{12}: B \otimes_A (B \otimes_A B) \rightarrow B \otimes_A B, \quad x \otimes (y \otimes z) \mapsto xy \otimes z,$$

whose Spec is the degeneracy $\sigma: \text{Spec}(B \otimes_A B) \rightarrow \text{Spec}(B \otimes_A (B \otimes_A B))$.

Lemma 839 (Degeneracy identities at ring level). *The degeneracies satisfy the simplicial identities against the coprojections and cofaces:*

$$\text{tensorMul} \circ \text{tensorInl} = \text{id}_B = \text{tensorMul} \circ \text{tensorInr},$$

and, on the triple tensor,

$$\text{tensorMul}_{12} \circ \text{tensorFace}_{23} = \text{id}, \quad \text{tensorMul}_{12} \circ \text{tensorFace}_{13} = \text{id}, \quad \text{tensorMul}_{12} \circ \text{tensorFace}_{12} = \text{tensorInl} \circ \text{tensorMul}$$

Proof. Each identity is an equality of k -algebra homomorphisms, checked on a generator. For the first pair, $\text{tensorMul}(\text{tensorInl}b) = \text{tensorMul}(b \otimes 1) = b$ and $\text{tensorMul}(\text{tensorInr}b) = \text{tensorMul}(1 \otimes b) = b$. For the degeneracy against the cofaces, compute on $x \otimes y$: $\text{tensorMul}_{12}(1 \otimes (x \otimes y)) = x \otimes y$; $\text{tensorMul}_{12}(x \otimes (1 \otimes y)) = x \otimes y$; and $\text{tensorMul}_{12}(x \otimes (y \otimes 1)) = xy \otimes 1 = \text{tensorInl}(xy) = \text{tensorInl}(\text{tensorMul}(x \otimes y))$. $\square$

Lemma 840 (The degeneracy coincidences at Spec level). *Applying Spec to Lemma ?? gives the degeneracy coincidences of the affine test objects: as morphisms $\text{Spec } B \rightarrow \text{Spec } B$ and $\text{Spec}(B \otimes_A B) \rightarrow \text{Spec}(B \otimes_A B)$,*

$$\Delta \circ q_1 = \text{id}, \quad \Delta \circ q_2 = \text{id}, \quad \sigma \circ f_{23} = \text{id}, \quad \sigma \circ f_{13} = \text{id}, \quad \sigma \circ f_{12} = q_1 \circ \Delta.$$

Proof. Each is the image of the corresponding ring-level identity of Lemma ?? under the contravariant functor Spec on test objects (Definition ??), which turns each equal composite of algebra maps into the reversed equal composite of morphisms and sends the identity algebra map to the identity morphism. $\square$

7.24 The trivialization-corrected witness cochain

The values θ_x of an Amitsur-coherent Čech witness (Definition ??) do not by themselves glue over a basic open of the tensor square: their coboundary is the comparison of the two coprojection pullbacks of γ . The composite descent unit corrects them by cocycle values of γ anchored at two distinguished points, after which they glue, satisfy the Amitsur cocycle identity, and collapse onto the Zariski cover cocycle on the diagonal. This section develops that calculus over abstract schemes; the concrete tensor-tower instantiation is Section ??.

Throughout, $r_1, r_2: Y \rightarrow Z$ are two morphisms (playing the coprojections $\text{Spec}(B \otimes_A B) \rightarrow \text{Spec } B$); $\mathcal{C}$ is a pointed cover of Y with a unit 0-cochain c (the witness cover and cochain); $\mathcal{N}$ is a pointed cover of Z with a unit cocycle γ_Z (the trivializing cover of $\text{Spec } B$); $a, b \in Z$ are two points; and $D \subseteq Y$ is an open on which r_1 lands in $\mathcal{N}_a$ and r_2 in $\mathcal{N}_b$.

Definition 841 (The corrected witness cochain). *The corrected witness value at a point $y \in Y$ is the unit on $\mathcal{C}_y \cap D$*

$$(\text{corr } c)_y := r_1^\# \gamma_Z(a, r_1 y) \cdot (c_y)^{-1} \cdot r_2^\# \gamma_Z(r_2 y, b),$$

the two correction factors being the coprojection pullbacks of the cocycle values of γ_Z anchored $a \rightarrow r_1 y$ on the left and $r_2 y \rightarrow b$ on the right, with the witness entering inverted. This is the unique orientation whose coboundary cancels the coboundary of θ^{-1} , and which on the diagonal — where the coherent witness is 1 — telescopes to the cocycle value $\gamma_Z(a, b)$.

Theorem 842 (The corrected values are Čech-compatible). *Assume the witness relation for c : on the overlaps of $\mathcal{C}$, $c_x \cdot r_1^\# \gamma_Z(r_1 x, r_1 y) = r_2^\# \gamma_Z(r_2 x, r_2 y) \cdot c_y$. Then the corrected values agree on overlaps: the restrictions of $(\text{corr } c)_y$ and $(\text{corr } c)_{y'}$ to $(\mathcal{C}_y \cap D) \cap (\mathcal{C}_{y'} \cap D)$ coincide.*

Proof. Restrict all data to the overlap $O := (\mathcal{C}_y \cap D) \cap (\mathcal{C}_{y'} \cap D)$. Three relations hold on O : the cocycle identity of γ_Z at $(a, r_1 y, r_1 y')$ pulled back along r_1 , which expresses the left factor at y times the r_1 -comparison term $r_1^\sharp \gamma_Z(r_1 y, r_1 y')$ as the left factor at y' ; the witness relation for c at (y, y') , relating the two witness values through the r_1 - and r_2 -comparison terms; and the cocycle identity at $(r_2 y, r_2 y', b)$ pulled back along r_2 , which relates the two right factors symmetrically. Writing the three as $A_y Q_1 = A_{y'}$, $t_y Q_1 = Q_2 t_{y'}$, $Q_2 C_{y'} = C_y$ in the commutative group of units on O (with A the left factors, t the witnesses, C the right factors, Q_1, Q_2 the comparison terms), substituting $Q_2 = t_y Q_1 t_{y'}^{-1}$ and cancelling $t_{y'}^{-1} t_y$ turns $A_y t_y^{-1} C_y$ into $A_{y'} t_{y'}^{-1} C_{y'}$, which is the asserted equality of corrected values. $\square$

Theorem 843 (The corrected values glue to a unit). *Under the witness relation of Theorem ??, the corrected values glue to a single unit $u \in \Gamma(Y, D)^\times$ whose restriction to each $\mathcal{C}_y \cap D$ is $(\text{corr } c)_y$.*

Proof. The opens $\{\mathcal{C}_y \cap D\}_y$ cover D , since $y \in \mathcal{C}_y$. The corrected values agree on overlaps by Theorem ??, so the sheaf gluing for unit sections (Lemma ??) produces the glued unit u on D . $\square$

Theorem 844 (Pullback expansion of a glued corrected unit). *Let $\varphi: X' \rightarrow Y$ and let $O \subseteq X'$ be an open inside $\varphi^{-1}(\mathcal{C}_{\varphi x} \cap D)$. Then the φ -pullback of a glued corrected unit u expands into its three factors, the correction factors pulled back along the composites $\varphi \circ r_1$ and $\varphi \circ r_2$:*

$$\varphi^\sharp u = (\varphi r_1)^\sharp \gamma_Z(a, (\varphi r_1)x) \cdot (\varphi^\sharp c_{\varphi x})^{-1} \cdot (\varphi r_2)^\sharp \gamma_Z((\varphi r_2)x, b).$$

Proof. By the defining property of u (Theorem ??), its restriction to $\mathcal{C}_{\varphi x} \cap D$ is $(\text{corr } c)_{\varphi x}$. Pull this equality back along φ into O and distribute the pullback over the product and inverse; each point-indexed factor pulled back along φ then along r_i becomes its pullback along the composite $\varphi \circ r_i$, which is the displayed normal form. $\square$

Theorem 845 (Coherence kills the witness along a coface section). *Let $g_{23}, g_{12}, g_{13}: T \rightarrow Y$ satisfy the Amitsur coherence relation for a witness cochain c on a pointed cover $\mathcal{C}$ of Y : $(g_{23}^\sharp c) \cdot (g_{12}^\sharp c) = g_{13}^\sharp c$ on the common refinement of the three coface pullbacks. If $\sigma: Y \rightarrow T$ and $\delta_r: Y \rightarrow Y$ satisfy $\sigma \circ g_{23} = \text{id}$, $\sigma \circ g_{12} = \delta_r$ and $\sigma \circ g_{13} = \text{id}$, then the δ_r -pullback of c is trivial: $\delta_r^\sharp c = 1$.*

Proof. Pull the coherence relation at the point σy back along σ onto a suitable open. By the composite-section identities $\sigma \circ g_{23} = \text{id}$, $\sigma \circ g_{12} = \delta_r$, $\sigma \circ g_{13} = \text{id}$, the three pulled back witness terms become c , $\delta_r^\sharp c$ and c respectively, so the relation reads $c \cdot \delta_r^\sharp c = c$. Cancelling the unit c gives $\delta_r^\sharp c = 1$. $\square$

Theorem 846 (Diagonal triviality of a coherent witness). *Let $\delta: Z \rightarrow Y$ and $\delta_r: Y \rightarrow Y$ satisfy $\delta \circ \delta_r = \delta$, and suppose the δ_r -pullback of c is trivial. Then the δ -pullback of c is trivial: $\delta^\sharp c = 1$.*

Proof. Pull the identity $\delta_r^\sharp c = 1$ (at the point δv) back along δ ; by $\delta \circ \delta_r = \delta$ the composite $\delta \circ \delta_r$ reduces to δ , so the pullback of $\delta_r^\sharp c$ along δ is $\delta^\sharp c$, while the pullback of the constant 1 is 1. Hence $\delta^\sharp c = 1$. $\square$

Theorem 847 (The glued corrected units satisfy the Amitsur cocycle identity). *Let $g_{23}, g_{12}, g_{13}: T \rightarrow Y$ satisfy the simplicial coincidences $g_{13} \circ r_1 = g_{12} \circ r_1$, $g_{23} \circ r_1 = g_{12} \circ r_2$, $g_{23} \circ r_2 = g_{13} \circ r_2$, and let c be Amitsur-coherent. Let u_{23}, u_{12}, u_{13} be the glued corrected units for the point pairs (b, c') , (a, b) , (a, c') on opens D_{23}, D_{12}, D_{13} . Then, on any D_T inside the three preimages,*

$$g_{23}^\sharp u_{23} \cdot g_{12}^\sharp u_{12} = g_{13}^\sharp u_{13}.$$

Proof. Check the identity on each member of the cover $\{(\mathcal{C}\text{-triple-overlap})_z \cap D_T\}$ of D_T . On such a member, expand the three pulled back glued units by Theorem ??; the six correction factors, pulled back along the composites $g_\bullet \circ r_\bullet$, collapse by the three simplicial coincidences onto matching cocycle values. Grouping, the left correction factor of u_{23} and the right correction factor of u_{12} are the two γ_Z -values anchored at b ; by the cocycle identity of γ_Z at $(b, \cdot, b)$ and $\gamma_Z(b, b) = 1$ they cancel, leaving the remaining correction factors as the correction factors of u_{13} . The three witness terms combine by the coherence relation $(g_{23}^\sharp c)(g_{12}^\sharp c) = g_{13}^\sharp c$. What remains is exactly $g_{13}^\sharp u_{13}$; this is the commutative-group telescope of the paired correction factors against the coherence relation. $\square$

Theorem 848 (The glued corrected unit collapses onto the cocycle value on the diagonal). *Let $\delta: Z \rightarrow Y$ be a common section of r_1, r_2 ($\delta \circ r_1 = \text{id} = \delta \circ r_2$) along which the witness is trivial ($\delta^\sharp c = 1$). Then, on an open E with $\delta(E) \subseteq D$ and $E \subseteq \mathcal{N}_a \cap \mathcal{N}_b$, the δ -pullback of the glued corrected unit for the pair (a, b) is the restriction of the cocycle value: $\delta^\sharp u = \gamma_Z(a, b)$.*

Proof. Check the identity on each member of the cover $\{\delta^{-1}\mathcal{C}_{\delta v} \cap E\}$ of E . On such a member, expand $\delta^\sharp u$ by Theorem ??; since $\delta \circ r_1 = \text{id} = \delta \circ r_2$, the two correction factors become $\gamma_Z(a, \delta v)$ and $\gamma_Z(\delta v, b)$, and the witness factor is 1 by hypothesis. The product of the two correction factors telescopes through the cocycle identity of γ_Z at $(a, \delta v, b)$ to $\gamma_Z(a, b)$. $\square$

7.25 The glued witness components on the tensor square

We now instantiate the corrected-witness calculus on the tensor towers $\text{Spec } B \leftarrow \text{Spec}(B \otimes_A B) \leftarrow \text{Spec}(B \otimes_A (B \otimes_A B))$. Fix an Amitsur-coherent Čech witness $\theta': \text{CoherentCechWitness}$ for γ (Definition ??) and a *finite basic refinement* P of the trivializing cover $\mathcal{N}$: a finite family of sections $P.r_i \in \Gamma(\text{Spec } B, \top)$ whose basic opens $D(P.r_i)$ refine $\mathcal{N}$, each with a distinguished point $P.\text{pt}_i \in \mathcal{N}$. Write q_1, q_2 for the coprojections and f_{12}, f_{13}, f_{23} for the cofaces.

Definition 849 (The double and triple basic sections). The *double basic section* cutting out the double basic open of $\text{Spec}(B \otimes_A B)$ is

$$\text{pairSection}(i, j) := q_1^\sharp(P.r_i) \cdot q_2^\sharp(P.r_j) \in \Gamma(\text{Spec}(B \otimes_A B), \top),$$

and the *triple basic section* cutting out the triple basic open of $\text{Spec}(B \otimes_A (B \otimes_A B))$ is

$$\text{tripleSection}(i, j, l) := (f_{12}q_1)^\sharp(P.r_i) \cdot ((f_{12}q_2)^\sharp(P.r_j) \cdot (f_{13}q_2)^\sharp(P.r_l)),$$

spelled through the three canonical insertions $f_{12} \circ q_1, f_{12} \circ q_2, f_{13} \circ q_2$.

Definition 850 (The glued witness component). The *glued witness component* $\text{witnessBasicSection}(i, j)$ is the glued unit on the double basic open $D(\text{pairSection}(i, j))$ obtained from Theorem ?? applied to the coprojections q_1, q_2 , the witness cover and cochain of θ' , the trivializing cocycle γ , and the two distinguished points $P.\text{pt}_i, P.\text{pt}_j$; its restriction to each $\mathcal{V}_x \cap D(\text{pairSection}(i, j))$ is the corrected witness value at x .

Theorem 851 (Diagonal triviality of the coherent witness). *The witness cochain θ of θ' is trivial along the diagonal $\Delta = \text{Spec}(\text{tensorMul})$: the Δ -pullback of θ is 1.*

Proof. Apply the diagonal-triviality transport (Theorem ??) with $\delta = \Delta$ and $\delta_r = q_1 \circ \Delta$; the required identity $\Delta \circ (q_1 \circ \Delta) = \Delta$ follows from $\Delta \circ q_1 = \text{id}$ (Lemma ??). Its hypothesis — triviality of the $(q_1 \circ \Delta)$ -pullback of θ — is Theorem ?? for the degeneracy σ against the three cofaces, whose section identities $\sigma \circ f_{23} = \text{id}$, $\sigma \circ f_{12} = q_1 \circ \Delta$, $\sigma \circ f_{13} = \text{id}$ are the degeneracy coincidences of Lemma ?? and whose coherence hypothesis is the Amitsur coherence of θ' . $\square$

Theorem 852 (The Amitsur cocycle identity of the glued components). *Over the triple basic open of $\text{Spec}(B \otimes_A (B \otimes_A B))$, the coface pullbacks of the glued witness components satisfy the Amitsur cocycle identity:*

$$f_{23}^\# \text{witnessBasicSection}(j, l) \cdot f_{12}^\# \text{witnessBasicSection}(i, j) = f_{13}^\# \text{witnessBasicSection}(i, l).$$

Proof. This is Theorem ?? applied along the cofaces f_{23}, f_{12}, f_{13} and coprojections q_1, q_2 , for the point pairs $(P.\text{pt}_j, P.\text{pt}_l)$, $(P.\text{pt}_i, P.\text{pt}_j)$, $(P.\text{pt}_i, P.\text{pt}_l)$ on the three double basic opens. Its simplicial coincidences $f_{12}q_1 = f_{13}q_1$, $f_{12}q_2 = f_{23}q_1$, $f_{13}q_2 = f_{23}q_2$ are the coface coincidences of Lemma ??, and its coherence input is the Amitsur coherence of θ' ; the containments of the triple basic open in the three coface preimages hold because tripleSection is divisible by each coface pullback of the relevant pairSection . $\square$

Theorem 853 (The diagonal collapse of the glued components). *The Δ -pullback of the glued witness component onto the Zariski overlap $D(P.r_i \cdot P.r_j) \subseteq \text{Spec } B$ is the Zariski cover cocycle value γ_{ij} of P :*

$$\Delta^\# \text{witnessBasicSection}(i, j) = \gamma_{ij}.$$

Proof. This is Theorem ?? applied with the common section $\delta = \Delta$ of q_1, q_2 , whose identities $\Delta \circ q_1 = \text{id} = \Delta \circ q_2$ are the degeneracy coincidences of Lemma ??. Its hypothesis of diagonal triviality of the witness is Theorem ??; the point pair is $(P.\text{pt}_i, P.\text{pt}_j)$, and the target open $D(P.r_i \cdot P.r_j)$ lands in $\mathcal{N}_{P.\text{pt}_i} \cap \mathcal{N}_{P.\text{pt}_j}$ and in the Δ -preimage of the double basic open. The resulting cocycle value $\gamma(P.\text{pt}_i, P.\text{pt}_j)$ restricted to the overlap is, by definition, the Zariski cover cocycle γ_{ij} of the basic refinement. $\square$

7.26 Section rings on basic opens as two-base localizations

To pass the three identities of Section ?? to the pure-algebra assembly of Section ??, the section rings on the basic opens of the tensor towers are identified with the algebraic tensor localizations. All section rings of $\text{Spec } R$ carry the canonical R -algebra structure whose structure map is the inverse of the global-sections isomorphism $\Gamma \text{Spec-iso}$ followed by restriction.

Theorem 854 (Section rings on basic opens are localizations of the ring). *For a commutative ring R and a global section $g \in \Gamma(\text{Spec } R, \top)$, the section ring on the basic open $D(g)$ is the localization of R itself away from the element $\Gamma \text{Spec-iso}(g) \in R$.*

Proof. The canonical $\Gamma(\text{Spec } R, \top)$ -algebra structure on $\Gamma(\text{Spec } R, D(g))$ exhibits it as the localization of $\Gamma(\text{Spec } R, \top)$ away from g . Transport this along the ring isomorphism $R \cong \Gamma(\text{Spec } R, \top)$ given by the global-sections isomorphism: the powers of g correspond to the powers of its image in R , and the two algebra structures agree, so $\Gamma(\text{Spec } R, D(g))$ is the localization of R away from that image. $\square$

Lemma 855 (The localization elements of the basic sections). *Write $b_i \in B$ for the element cutting out $D(P.r_i)$. The global-sections isomorphism carries the double basic section to the two-base localization element*

$$\text{pairSection}(i, j) \mapsto (b_i \otimes 1)(1 \otimes b_j) \quad \text{in } B \otimes_A B,$$

and the triple basic section to

$$\text{tripleSection}(i, j, l) \mapsto (b_i \otimes 1)(1 \otimes ((b_j \otimes 1)(1 \otimes b_l))) \quad \text{in } B \otimes_A (B \otimes_A B).$$

Proof. The global-sections isomorphism intertwines $q_1^\#$ with tensorInl and $q_2^\#$ with tensorInr (its naturality against $\text{Spec}(\text{tensorInl})$, $\text{Spec}(\text{tensorInr})$), so it sends $q_1^\#(P.r_i)$ to $b_i \otimes 1$ and $q_2^\#(P.r_j)$ to $1 \otimes b_j$; multiplying gives the double element. For the triple element, apply naturality once more against the cofaces f_{12}, f_{13} , computing the three insertions $f_{12}^\# q_1^\#(P.r_i)$, $f_{12}^\# q_2^\#(P.r_j)$, $f_{13}^\# q_2^\#(P.r_l)$ as $b_i \otimes (1 \otimes 1)$, $1 \otimes (b_j \otimes 1)$, $1 \otimes (1 \otimes b_l)$, and rearranging the product. $\square$

Definition 856 (The two-base localization identifications). Let $S_i := \Gamma(\text{Spec } B, D(P.r_i))$, a localization of B away from b_i (Theorem ??). The comparison of tensor localizations (Definition ??) gives the two-base identifications

$$\text{pairAwayEquiv}: S_i \otimes_A S_j \xrightarrow{\sim} \Gamma(\text{Spec}(B \otimes_A B), D(\text{pairSection}(i, j))),$$

$$\text{tripleAwayEquiv}: S_i \otimes_A (S_j \otimes_A S_l) \xrightarrow{\sim} \Gamma(\text{Spec}(B \otimes_A (B \otimes_A B)), D(\text{tripleSection}(i, j, l))),$$

the target section rings being the localizations of $B \otimes_A B$, resp. $B \otimes_A (B \otimes_A B)$, at the two-base elements of Lemma ?? . For the triple identification the inner factor $S_j \otimes_A S_l$ is first recognized as a localization of $B \otimes_A B$ (Theorem ??), so that $S_i \otimes_A (S_j \otimes_A S_l)$ is a two-base localization over $B \otimes_A (B \otimes_A B)$.

7.27 Transport of the simplicial maps through the two-base identifications

The restriction maps of the section rings along the diagonal Δ and the three cofaces correspond, under the identifications of Definition ??, to the index-wise collapse and faces of the pure-algebra assembly. Each is an instance of the uniqueness of maps out of a localization: both composites are computed on structure-map images and agree there, and extensionality is invoked over $B \otimes_A B$ — where the components are Away localizations — never over A .

Lemma 857 (The diagonal pullback is the index-wise collapse). *Under the identification pairAwayEquiv , the Δ -pullback $\Gamma(D(\text{pairSection}(i, j))) \rightarrow \Gamma(D(P.r_i \cdot P.r_j))$ is the index-wise collapse $\kappa_{ij}: S_i \otimes_A S_j \rightarrow T_{ij}$ onto the Zariski overlap ring $T_{ij} = \Gamma(\text{Spec } B, D(P.r_i \cdot P.r_j))$.*

Proof. Both maps are A -algebra homomorphisms out of the localization $S_i \otimes_A S_j$ of $B \otimes_A B$ (Theorem ??), so by uniqueness of maps out of a localization it suffices to compare their composites with the structure map $B \otimes_A B \rightarrow S_i \otimes_A S_j$. On a structure-map image the Δ -pullback computes, through the naturality of the global-sections isomorphism against $\text{Spec}(\text{tensorMul})$ and the diagonal coincidence (Lemma ??), as the multiplication $B \otimes_A B \rightarrow B$ followed by the structure map into T_{ij} ; and the index-wise collapse on a structure-map image is that same composite (Definition ??). The two agree, hence so do the maps. $\square$

Lemma 858 (The coface pullbacks are the index-wise faces). *Under the identifications pairAwayEquiv and tripleAwayEquiv , the three coface pullbacks $f_{23}^\#, f_{12}^\#, f_{13}^\#$ between the basic-open section rings are the index-wise faces $\text{face}_{23}, \text{face}_{12}, \text{face}_{13}$ of Definition ??.*

Proof. Fix the coface $\bullet \in \{12, 13, 23\}$. Both $\text{tripleAwayEquiv} \circ \text{face}_\bullet$ and $f_\bullet^\# \circ \text{pairAwayEquiv}$ are A -algebra homomorphisms out of the localization $S_j \otimes_A S_l$ (resp. $S_i \otimes_A S_j$, $S_i \otimes_A S_l$) of $B \otimes_A B$ (Theorem ??), so by uniqueness of maps out of a localization it suffices to compare their composites with the structure map $B \otimes_A B \rightarrow S_\bullet \otimes_A S_\bullet$. On a structure-map image the coface pullback computes, through the naturality of the global-sections isomorphism and the coface coincidences (Lemma ??), as the descent coface followed by the triple structure map; and the index-wise face intertwines the double and triple structure maps over that same descent coface (Lemma ??). The two composites agree. $\square$

7.28 The composite descent unit of the étale-separatedness assembly

Assembling the glued witness components into the composite A -descent unit yields the two halves of the deliverable and the headline class equation handed onward. Write $S_i := \Gamma(\text{Spec } B, D(P.r_i))$ and $P_\bullet := \prod_i S_i$.

Definition 859 (The witness component). The *witness component* $w_{ij} \in (S_i \otimes_A S_j)^\times$ is the glued witness component $\text{witnessBasicSection}(i, j)$ transported through the two-base identification pairAwayEquiv (Definition ??).

Definition 860 (The composite descent unit of the assembly). The *composite descent unit* $\text{assemblyUnit} \in (P_\bullet \otimes_A P_\bullet)^\times$ is the pi-assembly (Definition ??) of the family of witness components (w_{ij}) .

Theorem 861 (The witness components collapse onto the Zariski cover cocycle). *The index-wise collapse of the witness component is the Zariski cover cocycle value: $\kappa_{ij}(w_{ij}) = \gamma_{ij}$.*

Proof. By Lemma ?? the collapse κ_{ij} is, through pairAwayEquiv , the Δ -pullback; applied to the witness component it is the Δ -pullback of $\text{witnessBasicSection}(i, j)$, which is γ_{ij} by Theorem ??. $\square$

Theorem 862 (The witness components satisfy the index-wise cocycle identity). *The witness components satisfy the index-wise cocycle identity: $\text{face}_{23}(w_{jl}) \cdot \text{face}_{12}(w_{ij}) = \text{face}_{13}(w_{il})$.*

Proof. Apply the injective identification tripleAwayEquiv . By Lemma ?? it carries each index-wise face of a witness component to the corresponding coface pullback of the glued component $\text{witnessBasicSection}$; the transported identity is then the Amitsur cocycle identity of the glued components (Theorem ??). $\square$

Theorem 863 (The assembled witness unit is an A -descent 1-cocycle). *The composite descent unit assemblyUnit is an A -descent 1-cocycle relative to $A \rightarrow P_\bullet$.*

Proof. Apply the descent-cocycle criterion for a pi-assembly (Theorem ??). Its index-wise cocycle hypothesis is Theorem ?? . Its diagonal-normalization hypothesis is automatic: the witness components collapse onto the Zariski cover cocycle (Theorem ??), whose diagonal is 1 since the cover cocycle is a genuine Čech cocycle, so Theorem ?? gives $\text{diag}(w_{ii}) = 1$. This is precisely the descent construction of Kleiman's separatedness argument: the assembled unit $v'_{23}v'_{12} = v'_{13}$ descends the componentwise trivializations to a single datum on the base. $\square$

Theorem 864 (The collapse of the assembled unit is the descent unit of the cover cocycle). *The collapse of the composite descent unit along the tower $A \rightarrow \Gamma(\text{Spec } B, \top) \rightarrow P_\bullet$ is the descent unit of the Zariski cover cocycle of P :*

$$\pi(\text{assemblyUnit}) = \text{cocycleUnit}(\gamma_\bullet).$$

Proof. This is the collapse identity for a pi-assembly (Theorem ??), whose hypothesis — that the witness components collapse onto the cover cocycle, $\kappa_{ij}(w_{ij}) = \gamma_{ij}$ — is Theorem ??; the intermediate ring of the collapse tower is the scheme-side base ring $\Gamma(\text{Spec } B, \top)$. $\square$

Lemma 865 (Faithful flatness of the global sections of $\text{Spec } B$). *The global-sections isomorphism is an isomorphism of A -algebras $\Gamma(\text{Spec } B, \top) \cong B$ for the composed A -structure. Consequently, if $A \rightarrow B$ is faithfully flat, then $A \rightarrow \Gamma(\text{Spec } B, \top)$ is faithfully flat.*

Proof. The global-sections isomorphism is a ring isomorphism commuting with the two structure maps from A (both send a to the image of a , via restriction of Γ Spec-iso), hence an A -algebra isomorphism. Faithful flatness of a module transports along a linear isomorphism, so faithful flatness of $A \rightarrow B$ gives faithful flatness of $A \rightarrow \Gamma(\text{Spec } B, \top)$. $\square$

Theorem 866 (The class equation of the composite descent unit). *Assume $A \rightarrow B$ is faithfully flat. Then the Picard class of the composite descent unit maps, under the base-change homomorphism $\text{Pic}(A) \rightarrow \text{Pic}(\Gamma(\text{Spec } B, \top))$ along $A \rightarrow \Gamma(\text{Spec } B, \top)$, to the Picard class of the Zariski cover cocycle of P :*

$$\text{Pic}(A \rightarrow \Gamma(\text{Spec } B, \top))(\text{picClass}(\text{assemblyUnit})) = \text{pic}(\gamma),$$

which is $[N]$ when γ is the trivializing cocycle of an invertible module N .

Proof. Set $P_0 := \Gamma(\text{Spec } B, \top)$. The tower $A \rightarrow P_0 \rightarrow P_\bullet$ is faithfully flat: $A \rightarrow P_0$ by Lemma ??, and $P_0 \rightarrow P_\bullet$ because $P_\bullet = \prod_i S_i$ is the product of the localizations of P_0 at the covering family $P.r$ (Theorem ??); hence $A \rightarrow P_\bullet$ is faithfully flat. By naturality of the Picard class under descent in stages (Theorem ??) applied to assemblyUnit (Theorem ??), the base change to P_0 of $\text{picClass}(\text{assemblyUnit})$ is the Picard class over P_0 of the collapse $\pi(\text{assemblyUnit})$. By Theorem ?? that collapse is the descent unit of the Zariski cover cocycle, whose Picard class is $\text{pic}(\gamma)$ by definition (Definition ??); and when γ trivializes an invertible module N , $\text{pic}(\gamma) = [N]$ by Theorem ?. This is the invertible descended sheaf of Kleiman's separatedness argument, in its Čech incarnation. $\square$

7.29 Faithfully flat descent of sections along the base change of the curve product

Fix a field k , a curve C over k (proper, geometrically irreducible and geometrically reduced), and a tower $k \rightarrow A \rightarrow B$ with $A \rightarrow B$ faithfully flat. Write $X_A := (C \otimes \text{Spec } A)_{\text{left}}$, $X_B := (C \otimes \text{Spec } B)_{\text{left}}$, and $\text{cg} := (C \triangleleft \text{Spec}(A \rightarrow B))_{\text{left}}: X_B \rightarrow X_A$ for the base-change morphism of the curve product; the two coprojections $u_1, u_2: (C \otimes \text{Spec}(B \otimes_A B))_{\text{left}} \rightarrow X_B$ are the whiskered coprojections of $B \otimes_A B$. This section proves that sections of the structure sheaf, and of its units, descend along cg : the fppf sheaf property along the whiskered base inclusion.

Theorem 867 (The base-change square of the product is a pullback). *For C, T, T' over $\text{Spec } k$ and $g: T' \rightarrow T$, the square with horizontal legs $(C \triangleleft g)_{\text{left}}$ and g_{left} and vertical legs the two second projections is a pullback of schemes:*

$$\begin{array}{ccc} (C \otimes T')_{\text{left}} & \xrightarrow{(C \triangleleft g)_{\text{left}}} & (C \otimes T)_{\text{left}} \\ \downarrow \text{pr}' & & \downarrow \text{pr} \\ T'_{\text{left}} & \xrightarrow{g_{\text{left}}} & T_{\text{left}}. \end{array}$$

Proof. Paste the two cartesian squares expressing $C \otimes T$ and $C \otimes T'$ as products over $\text{Spec } k$. The first-projection triangle $(C \triangleleft g)_{\text{left}} \circ (\text{fst}_{C,T})_{\text{left}} = (\text{fst}_{C,T'})_{\text{left}}$ and the base identity $g_{\text{left}} \circ T.\text{hom} = T'.\text{hom}$ exhibit the outer rectangle over $C \times \text{Spec } k$ as the product square of $C \otimes T'$; together with the naturality square of the second projection (Lemma ??), the right-cancellation property of pullbacks gives that the second-projection square is itself a pullback. $\square$

Theorem 868 (Descent of sections along the base change of the curve product). *Let $U \subseteq X_A$ be an affine open. A section $t \in \Gamma(X_B, \text{cg}^{-1}U)$ whose two coprojection pullbacks to $C \otimes \text{Spec}(B \otimes_A B)$ agree, $u_1^*t = u_2^*t$, descends uniquely along cg : there is a unique $s \in \Gamma(X_A, U)$ with $\text{cg}^*s = t$.*

Proof. Over the affine open U , the base-change square (Theorem ??) identifies the sections of X_B over $\text{cg}^{-1}U$ with the base change $\Gamma(X_A, U) \otimes_A B$, and those of $C \otimes \text{Spec}(B \otimes_A B)$ with $\Gamma(X_A, U) \otimes_A (B \otimes_A B)$, the two coprojection pullbacks becoming the two faces $\text{id} \otimes (b \mapsto b \otimes 1)$ and $\text{id} \otimes (b \mapsto 1 \otimes b)$ (Definition ??). Under these identifications t corresponds to an element $x \in \Gamma(X_A, U) \otimes_A B$ with equal faces, so by the degree-zero Amitsur equalizer (Theorem ??) there is a unique $s \in \Gamma(X_A, U)$ with $s \otimes 1 = x$; this s is the unique section with $\text{cg}^*s = t$. $\square$

Theorem 869 (Injectivity of pullback along the base change). *For every open $W \subseteq X_A$, pullback of sections along cg is injective, $\Gamma(X_A, W) \rightarrow \Gamma(X_B, \text{cg}^{-1}W)$.*

Proof. First over an affine open V : by the base-change square (Theorem ??) the pullback is $s \mapsto s \otimes 1$ into $\Gamma(X_A, V) \otimes_A B$, which is injective because B is faithfully flat over A . For an arbitrary open W , if two sections have equal pullbacks, then their restrictions to each affine open $V \subseteq W$ have equal pullbacks (pullback commutes with restriction), hence agree by the affine case; as the affine opens contained in W cover W , the two sections agree by separation for the structure sheaf (Lemma ??). $\square$

Theorem 870 (Descent of unit sections along the base change). *Let $U \subseteq X_A$ be an affine open. A unit $t \in \Gamma(X_B, \text{cg}^{-1}U)^\times$ whose two coprojection pullbacks agree, $u_1^*t = u_2^*t$, descends to a unit: there is $s \in \Gamma(X_A, U)^\times$ with $\text{cg}^*s = t$.*

Proof. Both t and t^{-1} satisfy the equalizer hypothesis, so by Theorem ?? they descend to sections $a, b \in \Gamma(X_A, U)$ with $\text{cg}^*a = t$, $\text{cg}^*b = t^{-1}$. The pullback of ab is $tt^{-1} = 1 = \text{cg}^*1$, so $ab = 1$ by injectivity of the pullback over the affine open U (Theorem ??); likewise $ba = 1$. Hence a is a unit and $\text{cg}^*a = t$. $\square$

7.30 Saturated affine pieces of the base-change cover

Retain the field k , the curve C over k , and the tower $k \rightarrow A \rightarrow B$ of the previous section, with $X_A = (C \otimes \text{Spec } A)_{\text{left}}$, $X_B = (C \otimes \text{Spec } B)_{\text{left}}$, the projections $\text{pr}_A: X_A \rightarrow \text{Spec } A$, $\text{pr}_B: X_B \rightarrow \text{Spec } B$, and the base-change morphism $\text{cg}: X_B \rightarrow X_A$. This section refines the $\top$ -case identification $\Gamma(X_B, \top) \cong B$ (Lemma ??) to an arbitrary affine open of X_A : over such an open the section ring of the cover piece is the base change of the base section ring along $A \rightarrow B$. Nothing here uses flatness or properness (except where explicitly assumed); these are the geometric pieces on which a covering-and-overlaps effectivity argument runs.

Theorem 871 (The cover inclusion is an affine morphism). *The base-change morphism $\text{cg}: X_B \rightarrow X_A$ is affine. Consequently the preimage $\text{cg}^{-1}U$ of every affine open $U \subseteq X_A$ is affine.*

Proof. By the base-change square (Theorem ??, with $T = \text{Spec } A$, $T' = \text{Spec } B$ and $g = \text{Spec}(A \rightarrow B)$) the morphism cg is the base change of $\text{Spec}(A \rightarrow B): \text{Spec } B \rightarrow \text{Spec } A$ along the projection pr_A . The morphism $\text{Spec}(A \rightarrow B)$ is affine, being the spectrum of a ring homomorphism, and affine morphisms are stable under base change; hence cg is affine. The preimage of an affine open under an affine morphism is affine. $\square$

Theorem 872 (The section rings of a saturated affine piece form a pushout square). *Give $\Gamma(X_A, U)$ its A -algebra structure by pullback along the projection pr_A . For every affine open $U \subseteq X_A$ the square of commutative rings*

$$\begin{array}{ccc} A & \longrightarrow & \Gamma(X_A, U) \\ \downarrow & & \downarrow \text{cg}^\# \\ B & \longrightarrow & \Gamma(X_B, \text{cg}^{-1}U) \end{array}$$

— with top and left legs the two structure maps from A , right leg the pullback $\text{cg}^\sharp$, and bottom leg the pullback along pr_B — is a pushout (cocartesian). No flatness or properness is used.

Proof. The base-change square of schemes (Theorem ??) has one leg, $\text{Spec}(A \rightarrow B)$, affine, and U is affine; for such a fibre square the induced square of section rings over U — with the two $\top$ -corners $\Gamma(\text{Spec } A, \top)$ and $\Gamma(\text{Spec } B, \top)$ — is a pushout. Re-cornering along the canonical identifications $\Gamma(\text{Spec } A, \top) \cong A$ and $\Gamma(\text{Spec } B, \top) \cong B$ (which intertwine the structure maps, as in Lemma ??) replaces those two corners by A and B , yielding the displayed pushout. $\square$

Definition 873 (The section-ring identification). For every affine open $U \subseteq X_A$ there is a ring isomorphism

$$\Gamma(X_A, U) \otimes_A B \xrightarrow{\sim} \Gamma(X_B, \text{cg}^{-1}U), \quad s \otimes b \mapsto \text{cg}^\sharp(s) \cdot \text{pr}_B^\sharp(b),$$

the affine-open generalization of the $\top$ -case $\Gamma(X_B, \top) \cong B$ (Lemma ??). In particular $s \otimes 1 \mapsto \text{cg}^\sharp(s)$ and $1 \otimes b \mapsto \text{pr}_B^\sharp(b)$.

Proof. The base-change tensor product $\Gamma(X_A, U) \otimes_A B$ is the pushout of $\Gamma(X_A, U) \leftarrow A \rightarrow B$ in commutative rings. Comparing it with the pushout square of Theorem ??, which has the same span, yields a unique ring isomorphism between the two apices; its universal property sends the image of $\Gamma(X_A, U)$ to $\text{cg}^\sharp(s)$ and the image of B to $\text{pr}_B^\sharp(b)$, hence $s \otimes b$ to their product. $\square$

Definition 874 (The identification is an isomorphism of base-piece algebras). Give the cover-piece ring $\Gamma(X_B, \text{cg}^{-1}U)$ its $\Gamma(X_A, U)$ -algebra structure by pullback along cg . Then the identification of Definition ?? is an isomorphism of $\Gamma(X_A, U)$ -algebras

$$\Gamma(X_B, \text{cg}^{-1}U) \cong \Gamma(X_A, U) \otimes_A B.$$

Proof. The structure map of the $\Gamma(X_A, U)$ -algebra $\Gamma(X_B, \text{cg}^{-1}U)$ is $s \mapsto \text{cg}^\sharp(s)$, which the identification carries to $s \otimes 1$ (Definition ??); this is the structure map of the base-change algebra $\Gamma(X_A, U) \otimes_A B$. So the ring isomorphism intertwines the two structure maps from $\Gamma(X_A, U)$, hence is an isomorphism of $\Gamma(X_A, U)$ -algebras. $\square$

Theorem 875 (Faithful flatness of a saturated affine piece). Assume $A \rightarrow B$ is faithfully flat. Then for every affine open $U \subseteq X_A$ the structure map $\Gamma(X_A, U) \rightarrow \Gamma(X_B, \text{cg}^{-1}U)$ is faithfully flat.

Proof. Faithful flatness is stable under base change, so $\Gamma(X_A, U) \rightarrow \Gamma(X_A, U) \otimes_A B$ is faithfully flat. Transporting along the isomorphism of $\Gamma(X_A, U)$ -algebras of Definition ?? gives faithful flatness of $\Gamma(X_A, U) \rightarrow \Gamma(X_B, \text{cg}^{-1}U)$. $\square$

Theorem 876 (Restriction naturality of the identification). Let $W \leq V$ be affine opens of X_A , so $\text{cg}^{-1}W \leq \text{cg}^{-1}V$. The identification of Definition ?? commutes with restriction: for every $x \in \Gamma(X_A, V) \otimes_A B$,

$$(\text{cg}^{-1}W \leq \text{cg}^{-1}V)^*(\Phi_V(x)) = \Phi_W((\rho_W^V \otimes \text{id}_B)(x)),$$

where Φ_V, Φ_W are the identifications over V, W and $\rho_W^V: \Gamma(X_A, V) \rightarrow \Gamma(X_A, W)$ is restriction (an A -algebra homomorphism).

Proof. Both sides are A -algebra homomorphisms out of $\Gamma(X_A, V) \otimes_A B$, so it suffices to check them on pure tensors $s \otimes b$. There the left side restricts $\text{cg}^\sharp(s) \cdot \text{pr}_B^\sharp(b)$ and the right side is $\text{cg}^\sharp(\rho_W^V s) \cdot \text{pr}_B^\sharp(b)$; the two agree because restriction of sections commutes with the two pullbacks $\text{cg}^\sharp$ and $\text{pr}_B^\sharp$ (pullback of a section, then restrict, equals restrict, then pullback). $\square$

Theorem 877 (Saturated affine opens form a basis). *Every neighbourhood V of a point x of X_A contains an affine open W with $x \in W$ whose preimage $\text{cg}^{-1}W$ is again affine.*

Proof. The affine opens of X_A form a basis of its topology, so there is an affine open W with $x \in W \leq V$. Its preimage $\text{cg}^{-1}W$ is affine because cg is an affine morphism (Theorem ??). $\square$

Theorem 878 (The curve product over a proper curve is separated). *Assume C is proper over k . Then $X_A = (C \otimes \text{Spec } A)_{\text{left}}$ is a separated scheme.*

Proof. Properness of C makes its structure morphism separated, and separatedness is stable under base change, so the projection $\text{pr}_A : X_A \rightarrow \text{Spec } A$ is separated; composing with the affine (hence separated) morphisms $\text{Spec } A \rightarrow \text{Spec } k \rightarrow *$ shows the structure morphism of X_A is separated, i.e. X_A is a separated scheme. $\square$

Theorem 879 (The overlaps of the affine family are affine). *Assume C is proper over k . Then the affine opens of X_A are closed under intersection: if $V, W \subseteq X_A$ are affine open, so is $V \cap W$.*

Proof. A separated scheme (Theorem ??) has affine diagonal, and the intersection of two affine opens of a scheme with affine diagonal is affine. $\square$

7.31 The descent unit of a killed Čech class

Let $\gamma = \text{lam}$ be a unit Čech cocycle on a pointed cover $\mathcal{E}$ of X_A whose base-change class $\text{cg}^*[\mathcal{E}, \text{lam}]$ is trivial. Fixing a trivializing 0-cochain of the pulled-back cocycle, this section produces a single global unit of $\text{Spec}(B \otimes_A B)$ that records, on the curve product, the coprojection ratio of that 0-cochain and whose ring avatar is a descent 1-cocycle — the algebraic seed of the descended line bundle.

Theorem 880 (The descent unit of a killed class). *Let lam be a unit Čech cocycle on a pointed cover $\mathcal{E}$ of X_A , and let β be a unit 0-cochain on a pointed cover $\mathcal{A}$ of X_B refining the base-change pullback of $\mathcal{E}$, trivializing the pulled-back cocycle: on overlaps, $\beta_v \cdot (\text{cg}^\# \text{lam})_{vv'} = \beta_{v'}$. Then there is a global unit $w \in \Gamma(\text{Spec}(B \otimes_A B), \mathbb{T})^\times$ such that*

1. *on the common refinement of the two coprojection pullbacks of $\mathcal{A}$, the curve-product pullback of w is the ratio $u_1^* \beta / u_2^* \beta$ of the two coprojection pullbacks of β ; and*
2. *the ΓSpec -avatar of w in $(B \otimes_A B)^\times$ is a descent 1-cocycle relative to $A \rightarrow B$.*

Proof. The two coprojection pullbacks $u_1^* \beta, u_2^* \beta$ are units over the members of the common refinement $u_1^* \mathcal{A} \cap u_2^* \mathcal{A}$, and their ratio is compatible on overlaps: the coboundary hypothesis on β , pulled back through the coprojection compatibility $u_1 \circ \text{cg} = u_2 \circ \text{cg}$ of the whiskered faces, cancels the two lam -factors and leaves the ratios agreeing on double overlaps. By gluing of unit sections (Theorem ??) the ratio family glues to a global section, which by surjectivity of the global-section pullback (Theorem ??) is the curve-product pullback of a global unit w on $\text{Spec}(B \otimes_A B)$; this is (1).

For (2), work upstream on $\text{Spec}(B \otimes_A (B \otimes_A B))$. Applying the three Amitsur cofaces (Definition ??) to the local ratio form of w and using the simplicial coincidences among the coprojections, the β -telescope cancels exactly to give the cocycle identity $(f_{23}^\# w)(f_{12}^\# w) = f_{13}^\# w$ of the avatar. The diagonal normalization $\mu(w) = 1$ is the whiskered degeneracy of the tensor diagonal (Definition ??, Lemma ??, Lemma ??), under which the two coprojection pullbacks of β coincide and their ratio is 1. These are exactly the descent 1-cocycle conditions (Definition ??) for the avatar of w . $\square$

7.32 The Čech representative of a descended Picard class

The descent unit w of the previous section is the descent datum of an invertible module descended(w) over A . This section presents the Čech class of that module on $\text{Spec } A$, together with the trivializing units of its base change to $\text{Spec } B$, whose coprojection ratio is exactly w . Throughout, $\text{gS} := \text{Spec}(A \rightarrow B)_{\text{left}}: \text{Spec } B \rightarrow \text{Spec } A$ and $q_1, q_2: \text{Spec}(B \otimes_A B) \rightarrow \text{Spec } B$ are the two coprojections.

Definition 881 (The evaluation transform of a section-module tensor). Let N be an A -submodule of B and $V \subseteq \text{Spec } A$, $U \subseteq \text{Spec } B$ opens with $U \subseteq \text{gS}^{-1}V$. The *evaluation transform* is the A -linear map

$$\text{evalT}: \Gamma(\text{Spec } A, V) \otimes_A N \longrightarrow \Gamma(\text{Spec } B, U), \quad s \otimes m \longmapsto (\text{gS}^* s) \cdot (\Gamma \text{Spec}^{-1} m)|_U,$$

the product of the gS -pullback of the section s with the restriction to U of the section-avatar of $m \in B$. It is compatible with restriction in both opens.

Lemma 882 (The coprojection ratio of an evaluation). *Let $w \in \Gamma(\text{Spec}(B \otimes_A B), \top)^\times$ and suppose every $m \in N$ satisfies the descent equation of w : $1 \otimes m = \Gamma \text{Spec}(w) \cdot (m \otimes 1)$ in $B \otimes_A B$. Then, over any open O below both coprojection preimages of U , the two coprojection pullbacks of an evaluation differ by the restriction of w :*

$$q_2^*(\text{evalT}(x)) = w|_O \cdot q_1^*(\text{evalT}(x)), \quad x \in \Gamma(\text{Spec } A, V) \otimes_A N.$$

Proof. Both sides are additive in x , so it suffices to treat a pure tensor $s \otimes m$. The two coprojections send the section factor $\text{gS}^* s$ to the same value, the pullback along the common composite $q_1 \circ \text{gS} = q_2 \circ \text{gS}$; and they send the module factor to the two coprojection-avatars $\Gamma \text{Spec}^{-1}(m \otimes 1)$ and $\Gamma \text{Spec}^{-1}(1 \otimes m)$ (Definition ??). The descent equation of w applied to m turns the second avatar into w times the first, and the central unit w factors out, giving the stated ratio. $\square$

Definition 883 (The Čech representative of a descended class). Let w be a global unit of $\text{Spec}(B \otimes_A B)$ whose avatar is a descent 1-cocycle. A *Čech representative* of the descended class consists of a pointed cover $\mathcal{C}$ of $\text{Spec } A$, a unit Čech cocycle d on $\mathcal{C}$, and a family of trivializing units $\mu_a \in \Gamma(\text{Spec } B, \text{gS}^{-1}\mathcal{C}(a))^\times$ of the gS -pullback, subject to:

1. (*ratio*) the two coprojection pullbacks of each μ_a differ by w : $q_2^* \mu_a = w \cdot q_1^* \mu_a$ on any sub-open of the double preimage of $\mathcal{C}(a)$;
2. (*transition*) on overlaps the μ 's differ by the gS -pullback of the cocycle value: $\mu_a = \text{gS}^* d_{ab} \cdot \mu_b$ on $\text{gS}^{-1}(\mathcal{C}(a) \cap \mathcal{C}(b))$.

Theorem 884 (Existence of the Čech representative). *For every global unit w of $\text{Spec}(B \otimes_A B)$ whose avatar is a descent 1-cocycle, a Čech representative (Definition ??) of the descended class exists.*

Proof. Let $N := \text{descended}(w)$ be the invertible A -module descended by the avatar of w (Definition ??), and pass to its section-ring avatar $\Gamma(\text{Spec } A, \top) \otimes_A N$. Choose a trivializing family of that invertible module (Lemma ??): a pointed cover $\mathcal{C}$ of $\text{Spec } A$ by basic opens $D(g_a)$ with local trivializations, giving the cover $\mathcal{C}$ and its cover cocycle d . On the base change to $\text{Spec } B$, define μ_a as the evaluation of the collapsed generator of the a -trivialization through the surjective evaluation of the base-changed descent equivalence (Definition ??); this value is a unit by the unit-production principle (Theorem ??). The ratio condition is the coprojection ratio of an evaluation (Lemma ??), whose hypothesis is the descent equation of w satisfied by the elements of N . The transition condition follows because the trivializations are compatible across the cover and the transition units act on the restricted generators by the cover cocycle d . $\square$

7.33 The cobounding units on an affine chart

The trivializing 0-cochain β , the cocycle lam , and the trivializing units μ of the representative are assembled into a single unit on the base-change pullback of each chart of X_A , which then descends through cg . These descended units are the cobounding 0-cochain that exhibits $[\mathcal{E}, \text{lam}]$ as pulled back from $\text{Spec } A$.

Theorem 885 (The cobounding unit at an affine chart). *Keep the data $(\mathcal{E}, \text{lam}, \beta)$ of the descent unit w (Theorem ??) and a Čech representative $(\mathcal{C}, d, \mu)$ of the descended class (Definition ??). For every affine open $U \subseteq X_A$ contained in $\mathcal{E}(s)$ and in $\text{pr}_A^{-1}\mathcal{C}(\text{pr}_A(s))$, there is a unit $c \in \Gamma(X_A, U)^\times$ whose cg -pullback has, on every small enough sub-open, the glued local form*

$$(\text{cg}^*c) = \beta_x^{-1} \cdot (\text{cg}^\# \text{lam}_{\text{cg}(x),s})^{-1} \cdot (\text{pr}_B^\# \mu_{\text{pr}_A(s)})^{-1}.$$

Proof. On each member of the base-change pullback of $\mathcal{E}$ meeting $\text{cg}^{-1}U$, form the displayed product of the inverse trivialization β_x , the inverse cg -pullback of the cross-evaluation of lam , and the inverse pr_B -pullback of μ . These local pieces agree on overlaps: the coboundary relation of β against lam and the cocycle identity of lam telescope the β - and lam -factors, while the μ -factor is independent of x . By gluing of unit sections (Lemma ??) they glue to a unit G on $\text{cg}^{-1}U$. Its two coprojection pullbacks to $C \otimes \text{Spec}(B \otimes_A B)$ agree: the local ratio form of the descent unit w (Theorem ??) and the ratio field of the representative (Definition ??) exactly account for the two coprojection ratios of the three factors. Since U is affine, G therefore descends to a unit $c \in \Gamma(X_A, U)^\times$ with $\text{cg}^*c = G$ (Theorem ??), which is the claim. $\square$

7.34 The Čech kernel lemma and the close of étale separatedness

The pieces now compose. The kernel lemma isolates the genuinely fppf content — a Čech class killed by the cover pullback is pulled back from the base — and the class-level reduction and the plus unfold deliver, from it, the injectivity of the unit of the étale plus construction: Kleiman’s separatedness of the relative Picard functor.

Theorem 886 (The Čech kernel lemma). *Let C be proper, geometrically irreducible and geometrically reduced over k , and $A \rightarrow B$ faithfully flat. Then the kernel of the base-change pullback $\text{cg}^*: \text{Pic}(X_A) \rightarrow \text{Pic}(X_B)$ is contained in the image of the projection pullback $\text{pr}_A^*: \text{Pic}(\text{Spec } A) \rightarrow \text{Pic}(X_A)$: every class E on X_A with $\text{cg}^*E = 1$ is $\text{pr}_A^*M_1$ for some class M_1 on $\text{Spec } A$.*

Proof. Represent E by a unit cocycle lam on a pointed cover $\mathcal{E}$ of X_A . Since $\text{cg}^*E = 1$, the pulled-back cocycle is a coboundary: there is a trivializing 0-cochain β on a refinement of $\text{cg}^*\mathcal{E}$ with $\beta_v \cdot (\text{cg}^\# \text{lam})_{vv'} = \beta_{v'}$. Feed $(\mathcal{E}, \text{lam}, \beta)$ to the descent unit (Theorem ??), obtaining a global unit w whose avatar is a descent 1-cocycle; take a Čech representative $(\mathcal{C}, d, \mu)$ of its descended class (Theorem ??). Refine $\mathcal{E} \cap \text{pr}_A^{-1}\mathcal{C}$ to a cover $\mathcal{U}$ by affine opens (Lemma ??); on each chart $U = \mathcal{U}(s)$, the cobounding unit (Theorem ??) gives $c_s \in \Gamma(X_A, U)^\times$.

It remains to check that the c_s cobound lam , restricted to $\mathcal{U}$, against the pr_A -pullback of the representing cocycle d ; that is,

$$c_s|_{U_s \cap U_t} \cdot \text{pr}_A^* d_{\text{pr}_A s, \text{pr}_A t} = \text{lam}_{st}|_{U_s \cap U_t} \cdot c_t|_{U_s \cap U_t}.$$

Both sides pull back injectively along cg (Theorem ??); after pullback, the glued local forms of c_s, c_t , the μ -transition of the representative, and the cocycle identity of lam telescope to

the required identity. Hence lam and $\text{pr}_A^* d$ are cohomologous on $\mathcal{U}$, so $[\mathcal{U}, \text{lam}] = \text{pr}_A^* [\mathcal{C}, d]$ by refinement injectivity in degree one (Theorem ??, Corollary ??); that is, $E = \text{pr}_A^* M_1$ with $M_1 = [\mathcal{C}, d]$. This is the descent-datum step of Kleiman's separatedness argument in its Čech incarnation: the trivialization β descends the class from the faithfully flat cover to the base. $\square$

Theorem 887 (The class-level reduction). *Assume the kernel property of cg^* (Theorem ??). Let L be a Čech Picard class on X_A and N_0 a class on $\text{Spec } B$ with $\text{pr}_B^* N_0 = \text{cg}^* L$. Then L is pulled back from the base: $L = \text{pr}_A^* M$ for some class M on $\text{Spec } A$.*

Proof. From the hypothesis $\text{pr}_B^* N_0 = \text{cg}^* L$, the coherent Čech witness (Theorem ??) provides an Amitsur-coherent trivialization of the two coprojection pullbacks of N_0 , and the assembly of the previous chapter builds from it a composite A -descent unit whose avatar is a descent 1-cocycle (Theorem ??). Let M be the Čech class on $\text{Spec } A$ corresponding, through the Čech–Picard dictionary (Theorem ??), to the Picard class of the descended module (Definition ??) of that composite unit. By the class equation of the assembly (Theorem ??) and naturality of the dictionary in the affine scheme (Theorem ??), the base change $\text{gS}^* M$ has the same descent homomorphism image as N_0 ; since that homomorphism is injective (Theorem ??), $\text{gS}^* M = N_0$ on the nose. Then, using the projection base square (Lemma ??),

$$\text{cg}^*(\text{pr}_A^* M) = \text{pr}_B^*(\text{gS}^* M) = \text{pr}_B^* N_0 = \text{cg}^* L,$$

so $L \cdot (\text{pr}_A^* M)^{-1}$ is killed by cg^* . The kernel property yields M_1 with $\text{pr}_A^* M_1 = L \cdot (\text{pr}_A^* M)^{-1}$, whence $L = \text{pr}_A^*(M \cdot M_1)$. $\square$

Theorem 888 (The plus unfold of étale separatedness). *Fix A . Suppose the kernel property of the base-change pullback holds for every tower $A \rightarrow B$ with $A \rightarrow B$ faithfully flat. Then the unit $\eta_A: \text{relPic}(C, A) \rightarrow \text{PicEtAff}(C, A)$ of the plus construction (Definition ??) is injective.*

Proof. As η_A is a group homomorphism it suffices to show its kernel is trivial. Let $x = [\text{Spec } A, L] \in \text{relPic}(C, A)$ with $\eta_A(x) = 1$. Unwinding the plus-class equality $[\text{self}(A), \iota^* x] = 1$ through the presentation criterion (Theorem ??) and the essential uniqueness of A -algebra maps out of A , the datum reduces to a single presented étale cover $A \rightarrow B_G$: the base change $(\text{offd } A \text{ } B_G)^* L$ lies in the descent-trivial subgroup of $\text{relPic}(C, B_G)$ (Definition ??, Definition ??). Concretely there is a class N_0 on $\text{Spec } B_G$ with $\text{pr}_{B_G}^* N_0 = \text{cg}^* L$ — the hypothesis of the reduction. Since a presented étale cover is in particular faithfully flat, the reduction (Theorem ??) pulls L back from the base, $L = \text{pr}_A^* M$; that is exactly $x = 1$ in $\text{relPic}(C, A)$, using the normalization of the unit at the identity (Definition ??). $\square$

Theorem 889 (Étale separatedness of the relative Picard functor). *Let C be a curve over k that is proper, geometrically irreducible and geometrically reduced, and let A be any k -algebra. Then the unit $\eta_A: \text{relPic}(C, A) \rightarrow \text{PicEtAff}(C, A)$ of the one-step étale plus construction is injective. Equivalently, the relative Picard functor of C is separated for the étale topology — the separatedness half of Kleiman, The Picard Scheme, Theorem 2.5(1).*

Proof. The kernel property of the base-change pullback holds for every faithfully flat tower $A \rightarrow B$ by the Čech kernel lemma (Theorem ??). Feeding it to the plus unfold (Theorem ??) gives the injectivity of η_A . This is exactly Part 1 of Kleiman's comparison theorem in the étale case: the natural map $\text{Pic}_{X/S} \hookrightarrow \text{Pic}_{(X/S)_{\text{ét}}}$ is injective, so a single étale plus computes the sheafification and PicEtAff is separated on the étale site of affines. $\square$

7.35 Unit sections across an open immersion

Throughout this section $w: Z \rightarrow Y$ is an open immersion of schemes. Over an open $V \subseteq Y$ contained in the image of w , pullback of sections along w is bijective; this transports *unit* sections between Y and Z and is the substrate for decomposing Čech Picard classes along clopen partitions.

Lemma 890 (Units over an empty open form a trivial group). *For a scheme X and an open $O \subseteq X$ with $O \subseteq \emptyset$ (that is, $O = \emptyset$), the unit group $\Gamma(X, O)^\times$ is trivial (a subsingleton).*

Proof. Since $O = \emptyset$, the ring $\Gamma(X, O)$ is the zero ring, in which $0 = 1$ is the only element; hence its group of units has a single element. $\square$

Lemma 891 (Sections pull back isomorphically over an open inside the image). *Let $V \subseteq Y$ be an open with $V \subseteq \text{range}(w)$. Then pullback of sections along w is an isomorphism of rings $\Gamma(Y, V) \xrightarrow{\sim} \Gamma(Z, w^{-1}V)$.*

Proof. An open immersion w identifies Z with the open subscheme $\text{range}(w)$ of Y ; for an open V contained in that image the comparison map on sections is the restriction isomorphism of the structure sheaf between V and its homeomorphic preimage $w^{-1}V$. Hence pullback of sections over V is an isomorphism of rings. $\square$

Definition 892 (Transport of unit sections across an open immersion). For an open $V \subseteq Y$ with $V \subseteq \text{range}(w)$, the isomorphism of unit groups

$$\Gamma(Y, V)^\times \xrightarrow{\sim} \Gamma(Z, w^{-1}V)^\times$$

induced by pullback of sections along w .

Proof. Pullback of sections along w is an isomorphism of rings over V (Lemma ??); an isomorphism of rings restricts to an isomorphism of their groups of units. $\square$

7.36 Decomposition of the Čech Picard group along a clopen partition

A *clopen partition* of a scheme Y is a family of open immersions $w_i: Z_i \rightarrow Y$, $i \in \iota$, whose ranges are pairwise disjoint ($\text{range}(w_i) \cap \text{range}(w_j) = \emptyset$ for $i \neq j$) and cover Y ; each range is then open with open complement. This section proves that the Čech Picard group of Y decomposes along such a partition: separation along an arbitrary clopen partition, and gluing along a finite one. Throughout, $\text{Pic}(Y)$ is the definitional Čech Picard group recalled in Section ??, with pullback $w^*: \text{Pic}(Y) \rightarrow \text{Pic}(Z)$ along a morphism $w: Z \rightarrow Y$.

Theorem 893 (Separation of Čech Picard classes along a clopen partition). *Let $(w_i)_{i \in \iota}$ be a clopen partition of Y . Two Čech Picard classes $L, L' \in \text{Pic}(Y)$ with $w_i^*L = w_i^*L'$ for every i are equal.*

Proof. Choose for each point $y \in Y$ an index $p(y)$ with $y \in \text{range}(w_{p(y)})$. Represent $L = [\mathcal{U}_1, a_1]$ and $L' = [\mathcal{U}_2, a_2]$ and pass to the common adapted pointed cover $\mathcal{V}(y) := \mathcal{U}_1(y) \cap \mathcal{U}_2(y) \cap \text{range}(w_{p(y)})$, so that $\mathcal{V}(y) \subseteq \text{range}(w_{p(y)})$ for every y ; it suffices to show the ratio cocycle γ representing L/L' on $\mathcal{V}$ is a coboundary.

Triviality on each piece. Fix i . Since $w_i^*L = w_i^*L'$, the pullback classes $w_i^*[\mathcal{V}, a_1|_{\mathcal{V}}]$ and $w_i^*[\mathcal{V}, a_2|_{\mathcal{V}}]$ are represented on the pulled-back pointed cover $\mathcal{V}^{(i)} := w_i^{-1}\mathcal{V}$ by the pulled-back

cocycles, and they agree in $\text{Pic}(Z_i)$; by refinement injectivity (Corollary ??) already the pulled-back ratio cocycle $w_i^\# \gamma$ is a coboundary on $\mathcal{V}^{(i)}$: there are units $\alpha_z^{(i)} \in \mathcal{O}_{Z_i}(\mathcal{V}^{(i)}(z))^\times$ with $\alpha_z^{(i)} \cdot (w_i^\# \gamma)_{zz'} = \alpha_{z'}^{(i)}$ on the overlaps.

Gluing the piece coboundaries. For each $y \in Y$, let z_0 be the unique point of $Z_{p(y)}$ over y (an open immersion is injective on points), and set $\beta_y := \Phi^{-1}(\alpha_{z_0}^{(p(y))}|_{w_{p(y)}^{-1}\mathcal{V}(y)}) \in \Gamma(Y, \mathcal{V}(y))^\times$, where Φ is the unit transport of Definition ?? across $w_{p(y)}$ (applicable since $\mathcal{V}(y) \subseteq \text{range}(w_{p(y)})$). The choice of index is forced: if $y \in \text{range}(w_i)$ too, then $y \in \text{range}(w_i) \cap \text{range}(w_{p(y)})$, which is empty unless $i = p(y)$; and the choice of preimage point is forced by injectivity.

β cobounds γ . Fix y, y' . If $p(y') = p(y)$, both members lie in the single piece $Z_{p(y)}$; the coboundary relation for $\alpha^{(p(y))}$ pulls back injectively through $w_{p(y)}$ on the overlap $\mathcal{V}(y) \cap \mathcal{V}(y') \subseteq \text{range}(w_{p(y)})$ (pullback of unit sections is injective on opens inside the range, Definition ??), yielding $\beta_y \cdot \gamma_{yy'} = \beta_{y'}$ there. If $p(y') \neq p(y)$, then $\mathcal{V}(y) \cap \mathcal{V}(y') \subseteq \text{range}(w_{p(y)}) \cap \text{range}(w_{p(y')}) = \emptyset$, so the units over the overlap form a trivial group (Lemma ??) and the relation holds vacuously. Assembling over the cover of $\mathcal{V}(y) \cap \mathcal{V}(y')$ by these two cases and separating (Lemma ??; gluing across pieces uses Lemma ?? in the construction of β), $\gamma = \delta\beta$ is a coboundary, so $L/L' = 1$, i.e. $L = L'$. $\square$

Theorem 894 (The one-piece extension across a clopen decomposition). *Let $w: Z \rightarrow Y$ be an open immersion whose range is complemented by an open $\Omega' \subseteq Y$: $\text{range}(w) \cap \Omega' = \emptyset$ and $\text{range}(w) \cup \Omega' = Y$. For every $L_0 \in \text{Pic}(Z)$ there is a class $\tilde{L} \in \text{Pic}(Y)$ with*

$$w^* \tilde{L} = L_0, \quad v^* \tilde{L} = 1 \text{ for every } v: W \rightarrow Y \text{ with image contained in } \Omega'.$$

Proof. Represent $L_0 = [\mathcal{U}_0, \gamma_0]$ by a unit cocycle on a pointed cover of Z . Build the *extension cover* $\mathcal{E}$ of Y : at a point $y \in \text{range}(w)$ take the w -image of the member of $\mathcal{U}_0$ at the unique preimage of y ; at $y \notin \text{range}(w)$ take Ω' (a legitimate pointed cover, since $\text{range}(w) \cup \Omega' = Y$). Define its pair values: on range–range overlaps the transported cocycle value $\Phi^{-1}((\gamma_0)_\cdot)$ of Definition ?? across w , and 1 whenever either argument lies off the range. These satisfy the cocycle identity: on triple overlaps of range members it is the transported cocycle identity of γ_0 ; every other triple overlap meets a range member and a non-range member, hence lies in $\text{range}(w) \cap \Omega' = \emptyset$ and carries only the trivial unit (Lemma ??), so the identity degenerates. This produces $\tilde{L} := [\mathcal{E}, \text{extension cocycle}] \in \text{Pic}(Y)$.

Restriction on the piece. The given cover $\mathcal{U}_0$ refines the pullback $w^{-1}\mathcal{E}$ (the preimage of the image of a member is that member), and along this refinement the pulled-back extension cocycle restricts to γ_0 (its range–range values are the transported γ_0). Hence $w^* \tilde{L} = [\mathcal{U}_0, \gamma_0] = L_0$.

Restriction off the piece. For $v: W \rightarrow Y$ with image in Ω' , every point $v(x)$ lies off $\text{range}(w)$ (else it would lie in $\text{range}(w) \cap \Omega' = \emptyset$); so every pulled-back pair value is 1, the pulled-back cocycle is trivial, and $v^* \tilde{L} = 1$. $\square$

Theorem 895 (Gluing of Čech Picard classes along a finite clopen partition). *Let $(w_i)_{i \in \iota}$ be a finite clopen partition of Y . Every family $(L_i)_{i \in \iota}$ with $L_i \in \text{Pic}(Z_i)$ is the family of restrictions of a single class $M \in \text{Pic}(Y)$: $w_i^* M = L_i$ for all i .*

Proof. For each i , set $\Omega'_i := \bigcup_{j \neq i} \text{range}(w_j)$. By pairwise disjointness $\text{range}(w_i) \cap \Omega'_i = \emptyset$, and since the ranges cover Y , $\text{range}(w_i) \cup \Omega'_i = Y$. Let $M_i \in \text{Pic}(Y)$ be the one-piece extension of L_i across (w_i, Ω'_i) (Theorem ??). Put $M := \prod_i M_i$ (finite product). For each i , pullback along w_i is a homomorphism, so $w_i^* M = \prod_j w_i^* M_j$. The factor $j = i$ is $w_i^* M_i = L_i$; every factor $j \neq i$ is 1, because $\text{range}(w_i) \subseteq \Omega'_j$ means w_i has image in the complement of the j -th piece, so $w_i^* M_j = 1$ (Theorem ??). Hence $w_i^* M = L_i$. $\square$

7.37 The relative Picard group of a finite product of test algebras

Fix a field k and a curve C over k . For a finite family $(B_i)_{i \in \iota}$ of commutative k -algebras with product $P := \prod_i B_i$, the spectrum of P is the disjoint union of the spectra of the factors, clopen-partitioned by the basic opens of the coordinate idempotents; base-changing along the curve, the whiskered coordinate projections clopen-partition the curve-product. Through the clopen decomposition of the previous section this identifies the relative Picard group of the product test with the product of the relative Picard groups of the factors. Write $e_i \in P$ for the coordinate idempotent (1 in slot i , 0 elsewhere) and $\pi_i: P \rightarrow B_i$ for the projection.

Lemma 896 (Range of a whiskered open immersion). *Let $g: T' \rightarrow T$ be a morphism of test objects with g_{left} an open immersion. Then the whiskered morphism $(C \triangleleft g)_{\text{left}}$ is an open immersion, and its range is the preimage under the second projection $\text{pr}: (C \otimes T)_{\text{left}} \rightarrow T_{\text{left}}$ of the range of g_{left} :*

$$\text{range}((C \triangleleft g)_{\text{left}}) = \text{pr}^{-1}(\text{range}(g_{\text{left}})).$$

Proof. The whiskering square with horizontal legs $(C \triangleleft g)_{\text{left}}$, g_{left} and vertical legs the two second projections is a pullback of schemes (Theorem ??). Open immersions are stable under base change, so $(C \triangleleft g)_{\text{left}}$, being the base change of g_{left} along pr , is an open immersion; and the range of the base change of an open immersion along pr is the pr -preimage of its range. $\square$

Lemma 897 (A factor is the localization away from the coordinate idempotent). *For a finite product $P = \prod_j B_j$ and an index i , the projection $\pi_i: P \rightarrow B_i$ exhibits B_i as the localization of P away from the coordinate idempotent e_i .*

Proof. The element e_i is idempotent, $e_i^2 = e_i$. Two elements $x, y \in P$ have the same image in B_i iff they agree in slot i , iff $e_i x = e_i y$ (multiplication by e_i zeroes every other slot). And every $b \in B_i$ is the image of $\text{single}_i(b) \in P$. By the criterion for a localization away from an idempotent, $B_i = P[e_i^{-1}]$. $\square$

Lemma 898 (The coordinate projections clopen-partition the spectrum of a product). *The morphisms $\text{Spec } \pi_i: \text{Spec } B_i \rightarrow \text{Spec } P$ (for the finite family $(B_i)_{i \in \iota}$) are open immersions with pairwise disjoint ranges covering $\text{Spec } P$: the range of $\text{Spec } \pi_i$ is the basic open $D(e_i)$, and $D(e_i) \cap D(e_j) = \emptyset$ for $i \neq j$ while $\bigcup_i D(e_i) = \text{Spec } P$.*

Proof. By Lemma ??, π_i realizes B_i as a localization of P away from e_i , so $\text{Spec } \pi_i$ is an open immersion with range the basic open $D(e_i)$ (the spectrum of a localization away from an element is that element's basic open). For $i \neq j$, $e_i e_j = 0$, whence $D(e_i) \cap D(e_j) = D(e_i e_j) = D(0) = \emptyset$. Finally $\sum_i e_i = 1$; since ι is finite, no prime of P contains all the e_i (it would then contain 1), so every prime lies in some $D(e_i)$, i.e. the ranges cover. $\square$

Theorem 899 (Separation of relative Picard classes over a finite product). *For a finite family $(B_i)_{i \in \iota}$ with product P , two relative Picard classes $\zeta, \zeta' \in \text{relPic}(C, P)$ with $\pi_i^* \zeta = \pi_i^* \zeta'$ in $\text{relPic}(C, B_i)$ for every i are equal.*

Proof. Represent ζ, ζ' by Čech Picard classes L, L' on the curve-product $(C \otimes \text{Spec } P)_{\text{left}}$ (a relative Picard class is a class on the curve-product modulo those pulled back from the base along pr); restriction along π_i corresponds to Čech pullback of the underlying class along the whiskered projection $(C \triangleleft \text{Spec } \pi_i)_{\text{left}}$. The hypotheses give, for each i , a class N_i on $\text{Spec } B_i$ with $\text{pr}^* N_i = (C \triangleleft \text{Spec } \pi_i)_{\text{left}}^*(L/L')$ (the restriction of L/L' to the i -th piece is pulled back from

the base). Glue the N_i to a single class M on $\text{Spec } P$ by the clopen gluing (Theorem ??) for the coordinate partition of $\text{Spec } P$ (Lemma ??), so that $\text{Spec } \pi_i^* M = N_i$.

It remains to show $\text{pr}^* M = L/L'$ on the curve-product; then $\zeta = \zeta'$. The whiskered coordinate projections $(C \triangleleft \text{Spec } \pi_i)_{\text{left}}$ form a clopen partition of $(C \otimes \text{Spec } P)_{\text{left}}$: their ranges are the pr-preimages of the ranges of the coordinate projections (Lemma ??), which partition $\text{Spec } P$ (Lemma ??). By separation along this partition (Theorem ??) it suffices to check equal pullback along each whiskered projection. Using the base square of the projection (Lemma ??) and functoriality of Čech pullback,

$$(C \triangleleft \text{Spec } \pi_i)_{\text{left}}^* (\text{pr}^* M) = \text{pr}^* (\text{Spec } \pi_i^* M) = \text{pr}^* N_i = (C \triangleleft \text{Spec } \pi_i)_{\text{left}}^* (L/L'),$$

as required. $\square$

Theorem 900 (Gluing of relative Picard classes over a finite product). *For a finite family $(B_i)_{i \in I}$ with product P , every family $(\xi_i)_{i \in I}$ with $\xi_i \in \text{relPic}(C, B_i)$ is the family of restrictions of a class $\zeta \in \text{relPic}(C, P)$: $\pi_i^* \zeta = \xi_i$ for all i .*

Proof. Represent each ξ_i by a Čech Picard class L_i on the curve-product $(C \otimes \text{Spec } B_i)_{\text{left}}$. The whiskered coordinate projections $(C \triangleleft \text{Spec } \pi_i)_{\text{left}}$ form a clopen partition of $(C \otimes \text{Spec } P)_{\text{left}}$, their ranges being the pr-preimages (Lemma ??) of the coordinate ranges that partition $\text{Spec } P$ (Lemma ??). By the clopen gluing (Theorem ??) there is a class M on the curve-product with $(C \triangleleft \text{Spec } \pi_i)_{\text{left}}^* M = L_i$ for all i . Let ζ be the relative Picard class of M . Then $\pi_i^* \zeta$ is the relative Picard class of $(C \triangleleft \text{Spec } \pi_i)_{\text{left}}^* M = L_i$, which is ξ_i . $\square$

7.38 Zariski separation of the étale plus construction

Throughout, k is a field and C is a curve over k (proper, geometrically irreducible, geometrically reduced). This section proves that the affine étale plus $\text{PicEtAff}(C, -)$ is a separated presheaf for the Zariski topology of affine tests: two plus classes over A agreeing on a finite covering family of localizations agree. The argument is instance-free of étale separatedness; it consumes only the product decomposition of the relative Picard group.

Theorem 901 (Unfolding an equality of restricted plus classes). *Let $\varphi_1: A_1 \rightarrow R$ and $\varphi_2: A_2 \rightarrow R$ be homomorphisms of k -algebras, E_1 a presented étale cover of A_1 and E_2 of A_2 , and $\xi_1 \in \text{Desc}(C, E_1)$, $\xi_2 \in \text{Desc}(C, E_2)$. If $\varphi_1^*[E_1, \xi_1] = \varphi_2^*[E_2, \xi_2]$ in $\text{PicEtAff}(C, R)$, then there are a presented étale cover H of R and homomorphisms of k -algebras $m_1: B_{E_1} \rightarrow B_H$, $m_2: B_{E_2} \rightarrow B_H$, semilinear over φ_1, φ_2 ($m_1 \circ \iota_{E_1} = \iota_H \circ \varphi_1$ and $m_2 \circ \iota_{E_2} = \iota_H \circ \varphi_2$), with $m_1^* \xi_1 = m_2^* \xi_2$ in $\text{relPic}(C, B_H)$.*

Proof. Regard R as an A_1 - and A_2 -algebra through φ_1, φ_2 . Then $\varphi_i^*[E_i, \xi_i] = [(E_i)_R, \kappa_{E_i}^* \xi_i]$ is the restriction along the base extension $A_i \rightarrow R$ of the plus class (Definition ??), with $(E_i)_R$ the base change of E_i (Definition ??) and κ_{E_i} its canonical inclusion (Definition ??). The hypothesis $[(E_1)_R, \kappa_{E_1}^* \xi_1] = [(E_2)_R, \kappa_{E_2}^* \xi_2]$ unfolds by the presentation criterion (Theorem ??): there are a presented étale cover H of R and homomorphisms of R -algebras $r_i: B_{(E_i)_R} \rightarrow B_H$ with $(r_1)_*(\kappa_{E_1}^* \xi_1) = (r_2)_*(\kappa_{E_2}^* \xi_2)$. Set $m_i := r_i \circ \kappa_{E_i}: B_{E_i} \rightarrow B_H$. Since κ_{E_i} and r_i both cover the base inclusions, $m_i \circ \iota_{E_i} = \iota_H \circ \varphi_i$; and by functoriality of restriction (Lemma ??), $m_i^* \xi_i = r_i^*(\kappa_{E_i}^* \xi_i)$, so the refinement equality gives $m_1^* \xi_1 = m_2^* \xi_2$. $\square$

Theorem 902 (A finite covering family of localizations assembles into an étale cover). *Let A be a k -algebra and $(g_i)_{i \in I}$ a finite family with $\sum_i (g_i) = (1)$, and let S_i be the localization of A away from g_i . Given presented étale covers K_i of S_i , the product $\prod_i B_{K_i}$ is the carrier of a presented étale cover W of A , with an isomorphism of A -algebras $B_W \cong \prod_i B_{K_i}$.*

Proof. Each S_i , a localization of A away from g_i , is étale over A ; each B_{K_i} is étale over S_i , hence over A ; so the finite product $\prod_i B_{K_i}$ is étale over A . It remains that $\text{Spec}(\prod_i B_{K_i}) \rightarrow \text{Spec } A$ is surjective; then $\prod_i B_{K_i}$ is a presented étale cover of A (Definition ??, Lemma ??). Given a prime p of A : since $\sum_i (g_i) = (1)$, some $g_i \notin p$, so p lifts to $\text{Spec } S_i$ (the spectrum of the localization away from g_i is the basic open $D(g_i) \ni p$); as K_i is a cover of S_i it is surjective on spectra (Definition ??), so p lifts further to $\text{Spec } B_{K_i}$, and thence to $\text{Spec}(\prod_j B_{K_j})$ through the i -th coordinate. Hence the structure map is surjective. $\square$

Theorem 903 (Zariski separation of the étale plus construction). *Let A be a k -algebra, $(g_i)_{i \in \iota}$ a finite family with $\sum_i (g_i) = (1)$, and S_i the localization of A away from g_i . Two plus classes $x, y \in \text{PicEtAff}(C, A)$ with equal restrictions along every $A \rightarrow S_i$ are equal.*

Proof. Restriction is a homomorphism, so it suffices to show a class z with trivial restriction to every S_i is trivial. Represent $z = [G, \zeta]$ (Lemma ??). For each i , comparing the restriction of $[G, \zeta]$ with that of the trivial class $[\text{self}(A), 1]$ over S_i and unfolding (Theorem ??), there are a presented étale cover K_i of S_i and a homomorphism $m_i: B_G \rightarrow B_{K_i}$, semilinear over $A \rightarrow S_i$, with $m_i^* \zeta = 1$ in $\text{relPic}(C, B_{K_i})$. Assemble the product cover W of A from the K_i (Theorem ??), $B_W \cong \prod_i B_{K_i}$ over A . The joint map $\langle m_i \rangle: B_G \rightarrow \prod_i B_{K_i}$ pushes ζ to a class whose i -th factor is $\pi_i^*(\langle m_i \rangle^* \zeta) = m_i^* \zeta = 1$; by the product separation (Theorem ??) the pushed class is trivial in $\text{relPic}(C, \prod_i B_{K_i})$, hence in $\text{relPic}(C, B_W)$ after transport through the carrier isomorphism. As the joint map is an A -algebra refinement $B_G \rightarrow B_W$, $[G, \zeta] = [W, 1] = 1$ (Theorem ??). So $z = 1$. $\square$

7.39 Restriction of relative Picard classes to an étale cover is injective

This section extracts the usable corollary of the (C1) étale-separatedness theorem for the Layer-2 gluing arguments. Throughout C is a curve over k (proper, geometrically irreducible, geometrically reduced) and R is a commutative k -algebra.

Theorem 904 (The unit represented on an arbitrary cover). *For a presented étale cover E of R and $z \in \text{relPic}(C, R)$, the unit $\eta_R(z)$ is represented on E by the tautological descent datum of z :*

$$\eta_R(z) = [E, \iota_E^* z], \quad \iota_E^* z \in \text{Desc}(C, E),$$

where $\iota_E: R \rightarrow B_E$ is the structure map.

Proof. By definition (Definition ??) $\eta_R(z) = [\text{self}(R), \iota^* z]$ with $\iota^* z$ the tautological datum on the trivial cover (Lemma ??). Transport along the canonical map $\text{fromSelf}: B_{\text{self}(R)} \rightarrow B_E$ (Definition ??) carries the tautological datum on $\text{self}(R)$ to the tautological datum $\iota_E^* z$ on E : both are the restriction of z along the structure map, by functoriality of restriction, and the composite structure map $R \rightarrow B_{\text{self}(R)} \rightarrow B_E$ is ι_E (Definition ??). By Lemma ??, transporting a representative along a refinement map does not change the plus class, so $[E, \iota_E^* z] = [E, \text{fromSelf}_*(\iota^* z)] = [\text{self}(R), \iota^* z] = \eta_R(z)$. $\square$

Theorem 905 (Restriction to an étale cover is injective — the (C1) consumable). *For a presented étale cover E of R , restriction of relative Picard classes along the structure map $\iota_E: R \rightarrow B_E$ is injective: if $x, y \in \text{relPic}(C, R)$ satisfy $\iota_E^* x = \iota_E^* y$ in $\text{relPic}(C, B_E)$, then $x = y$. This is what licenses every gluing step of the Layer-2 functor.*

Proof. By the étale separatedness of the relative Picard functor (Theorem ??), the unit η_R is injective, so it suffices to show $\eta_R(x) = \eta_R(y)$. Represent both on E (Theorem ??): $\eta_R(x) = [E, \iota_E^* x]$ and $\eta_R(y) = [E, \iota_E^* y]$. Since $\iota_E^* x = \iota_E^* y$ by hypothesis, these classes coincide, so $\eta_R(x) = \eta_R(y)$ and hence $x = y$. $\square$

7.40 Zariski gluing of the étale plus construction

Throughout, C is a curve over k (proper, geometrically irreducible, geometrically reduced). This section proves the gluing half of the Zariski sheaf property of $\text{PicEtAff}(C, -)$ on affine tests. Its engine, the pair condition, is the single point where the Layer-2 functoriality consumes the (C1) étale separatedness.

Theorem 906 (The pair condition). *Let S_1, S_2 be localizations of a k -algebra A away from g_1, g_2 , and T_v a localization away from $g_1 g_2$, with A -algebra homomorphisms $\varphi_1: S_1 \rightarrow T_v$, $\varphi_2: S_2 \rightarrow T_v$. Let E_1 be a presented étale cover of S_1 , E_2 of S_2 , and $\xi_1 \in \text{Desc}(C, E_1)$, $\xi_2 \in \text{Desc}(C, E_2)$. If $\varphi_1^*[E_1, \xi_1] = \varphi_2^*[E_2, \xi_2]$ in $\text{PicEtAff}(C, T_v)$, then the two pullbacks of the underlying classes into the A -tensor product agree:*

$$\text{inl}^* \xi_1 = \text{inr}^* \xi_2 \quad \text{in } \text{relPic}(C, B_{E_1} \otimes_A B_{E_2}),$$

where $\text{inl}: B_{E_1} \rightarrow B_{E_1} \otimes_A B_{E_2}$ and $\text{inr}: B_{E_2} \rightarrow B_{E_1} \otimes_A B_{E_2}$ are the two coprojections.

Proof. Unfold the overlap compatibility (Theorem ??): there are a presented étale cover H of T_v and homomorphisms $m_1: B_{E_1} \rightarrow B_H$, $m_2: B_{E_2} \rightarrow B_H$, semilinear over φ_1, φ_2 , with $m_1^* \xi_1 = m_2^* \xi_2$ in $\text{relPic}(C, B_H)$.

In $B_{E_1} \otimes_A B_{E_2}$, the element $g_1 g_2$ is a unit: g_1 is a unit in B_{E_1} and g_2 in B_{E_2} (both localized-away), and their images multiply to $g_1 g_2$. Since T_v is the localization of A away from $g_1 g_2$, there is a unique homomorphism of A -algebras $\rho: T_v \rightarrow B_{E_1} \otimes_A B_{E_2}$. Base-change H along ρ to a presented étale cover $W := H_{B_{E_1} \otimes_A B_{E_2}}$ (Definition ??), with canonical inclusion $\theta: B_H \rightarrow B_W$. By the (C1) consumable (Theorem ??), restriction along the structure map $B_{E_1} \otimes_A B_{E_2} \rightarrow B_W$ is injective; it therefore suffices to show that the two sides become equal after transport into B_W .

Consider the two routes $B_{E_1} \rightarrow B_W$. The first is inl followed by the structure map into B_W ; the second is $\theta \circ m_1$. Both are homomorphisms of S_1 -algebras: for the second, m_1 is φ_1 -semilinear and $\rho \circ \varphi_1$ equals the S_1 -structure map $S_1 \rightarrow B_{E_1} \otimes_A B_{E_2}$ (two homomorphisms of A -algebras out of the localization S_1 , hence equal). By refinement independence (Theorem ??, applied to the two S_1 -algebra maps out of the cover E_1 of S_1 and the descent class ξ_1), the two routes transport ξ_1 to the same class in $\text{relPic}(C, B_W)$. Symmetrically the two routes $B_{E_2} \rightarrow B_W$ (namely inr into B_W and $\theta \circ m_2$) transport ξ_2 equally. Now

$$(\theta \circ m_1)^* \xi_1 = \theta^*(m_1^* \xi_1) = \theta^*(m_2^* \xi_2) = (\theta \circ m_2)^* \xi_2,$$

so, chaining the two refinement independences, the transports into B_W of $\text{inl}^* \xi_1$ and $\text{inr}^* \xi_2$ agree; by injectivity, $\text{inl}^* \xi_1 = \text{inr}^* \xi_2$. $\square$

Lemma 907 (Restriction along an algebra isomorphism is injective). *For an isomorphism of k -algebras $e: A \xrightarrow{\sim} B$, restriction of relative Picard classes along e , $e^*: \text{relPic}(C, A) \rightarrow \text{relPic}(C, B)$, is injective.*

Proof. Restriction along e^{-1} is a left inverse: $(e^{-1})^* \circ e^* = (e^{-1} \circ e)^* = (\text{id}_A)^* = \text{id}$, by functoriality (Lemma ??). A map with a left inverse is injective. $\square$

Lemma 908 (Computing a restricted plus class on a given cover). *Let $\varphi: A_1 \rightarrow R$ be a homomorphism of k -algebras, E_1 a presented étale cover of A_1 , F a presented étale cover of R , $\xi \in \text{Desc}(C, E_1)$ and $\eta \in \text{Desc}(C, F)$. Suppose there is a homomorphism of k -algebras $n: B_{E_1} \rightarrow B_F$, semilinear over φ ($n \circ \iota_{E_1} = \iota_F \circ \varphi$), with $n^*\xi = \eta$ in $\text{relPic}(C, B_F)$. Then $\varphi^*[E_1, \xi] = [F, \eta]$.*

Proof. Regard R as an A_1 -algebra through φ , so that $\varphi^*[E_1, \xi] = [(E_1)_R, \kappa_{E_1}^*\xi]$ (Definition ??, with $(E_1)_R$ the base change, Definition ??). The semilinear map n extends uniquely to an R -algebra refinement $n_R: B_{(E_1)_R} \rightarrow B_F$ with $n_R \circ \kappa_{E_1} = n$. By the presentation criterion (Theorem ??, with the common cover F , the map n_R out of $(E_1)_R$ and the identity out of F), $[(E_1)_R, \kappa_{E_1}^*\xi] = [F, (n_R)_*(\kappa_{E_1}^*\xi)]$; and $(n_R)_*(\kappa_{E_1}^*\xi) = n^*\xi = \eta$ by functoriality (Definition ??). Hence $\varphi^*[E_1, \xi] = [F, \eta]$. $\square$

Theorem 909 (Zariski gluing of the étale plus construction). *Let A be a k -algebra, $(g_i)_{i \in \iota}$ a finite family with $\sum_i (g_i) = (1)$, S_i the localization of A away from g_i , and T_{ij} the localization away from $g_i g_j$. A family $(x_i)_{i \in \iota}$ with $x_i \in \text{PicEtAff}(C, S_i)$, compatible on the overlaps (the restrictions of x_i and x_j to T_{ij} agree for all i, j), is the family of restrictions of a plus class $z \in \text{PicEtAff}(C, A)$: restriction of z to S_i is x_i . Together with Theorem ?? this is the Zariski sheaf property of $\text{PicEtAff}(C, -)$ on affine tests.*

Proof. Represent $x_i = [E_i, \xi_i]$ (Lemma ??). For each i, j , the overlap compatibility over T_{ij} yields, by the pair condition (Theorem ??), $\text{inl}^*\xi_i = \text{inr}^*\xi_j$ in $\text{relPic}(C, B_{E_i} \otimes_A B_{E_j})$. Lift the underlying classes to the product: by the product gluing (Theorem ??) there is $\zeta \in \text{relPic}(C, \prod_i B_{E_i})$ with $\pi_i^*\zeta = \xi_i$. Assemble the product cover W of A from the E_i (Theorem ??), $B_W \cong \prod_i B_{E_i}$ over A ; transport ζ to $\zeta_W \in \text{relPic}(C, B_W)$.

ζ_W is a descent class on W . The descent condition $\text{inl}^*\zeta_W = \text{inr}^*\zeta_W$ in $\text{relPic}(C, B_W \otimes_A B_W)$ is checked through the π -decomposition of the tensor square, $B_W \otimes_A B_W \cong \prod_{(i,j)} B_{E_i} \otimes_A B_{E_j}$: by injectivity of transport along this isomorphism (Lemma ??) and product separation over the index set $\iota \times \iota$ (Theorem ??), it suffices to compare components. The (i, j) -component of $\text{inl}^*\zeta_W$ is $\text{inl}^*(\pi_i^*\zeta) = \text{inl}^*\xi_i$ and of $\text{inr}^*\zeta_W$ is $\text{inr}^*(\pi_j^*\zeta) = \text{inr}^*\xi_j$; these agree by the pair conditions.

The glued class. Set $z := [W, \zeta_W]$. Its restriction to S_i is x_i : the i -th projection $B_W \cong \prod_j B_{E_j} \rightarrow B_{E_i}$ is a homomorphism semilinear over $A \rightarrow S_i$ carrying ζ_W onto $\pi_i^*\zeta = \xi_i$, so by Lemma ??, $(A \rightarrow S_i)^*[W, \zeta_W] = [E_i, \xi_i] = x_i$. $\square$

7.41 The étale-sheafified relative Picard functor: the affine-opens limit

The frozen group-object interface forces the relative Picard functor to exist on *all* of $\text{Over}(\text{Spec } k)$, not only on affine tests. Following the resolution of the Layer-2 vehicle question, the extension is the *bespoke affine-opens limit*: a section over a test object T is a compatible family of affine-level plus classes over the affine opens of T_{left} . The index poset is Type-small, which is the point of the vehicle. Throughout, k is a field and C is a curve over k .

Definition 910 (Transport of plus classes along an algebra isomorphism). For an isomorphism of k -algebras $e: A \xrightarrow{\sim} B$, restriction of plus classes along e is a group isomorphism $\text{PicEtAff}(C, A) \xrightarrow{\sim} \text{PicEtAff}(C, B)$, with inverse restriction along e^{-1} .

Proof. Restriction along e and along e^{-1} are homomorphisms of groups (Definition ??); their two composites are restriction along $e^{-1} \circ e = \text{id}_A$ and along $e \circ e^{-1} = \text{id}_B$, each the identity

by transitivity of restriction (Theorem ??) and its normalization at the identity (Theorem ??). Hence the two are mutually inverse. $\square$

Definition 911 (The affine identification of global sections). For a k -algebra A , the global sections of the affine test $\text{Spec } A$ over its top open are identified with A as a k -algebra: $\Gamma((\text{Spec } A)_{\text{left}}, \top) \xrightarrow{\sim} A$, through the canonical isomorphism $\Gamma(\text{Spec } A, \top) \cong A$ of the affine scheme.

Proof. The canonical isomorphism of rings $\Gamma(\text{Spec } A, \top) \cong A$ intertwines the two k -algebra structures: the structure map of the sections and that of A both come from the k -algebra structure of A through this same isomorphism. Hence it is an isomorphism of k -algebras. $\square$

Definition 912 (The étale-sheafified relative Picard functor at a test object). For a test object T of $\text{Over}(\text{Spec } k)$, $\text{picEt}(C, T)$ is the group of *compatible families*: an element assigns to each affine open $V \subseteq T_{\text{left}}$ a plus class $s_V \in \text{PicEtAff}(C, \Gamma(T, V))$ such that for affine opens $U \subseteq V$, the value s_U is the restriction of s_V along the section-restriction homomorphism $\Gamma(T, V) \rightarrow \Gamma(T, U)$ (Definition ??). Componentwise product makes $\text{picEt}(C, T)$ a commutative group, a subgroup of the product $\prod_V \text{PicEtAff}(C, \Gamma(T, V))$; evaluation at an affine open V is the homomorphism $s \mapsto s_V$. On affine tests this is the plus itself (the affine comparison, Definition ??); on general tests it is the canonical Zariski-continuous extension — the realization on all test objects of the étale sheafification $\text{Pic}_{(X/S)\text{ét}}$ of the relative Picard functor.

Proof. The compatibility condition is closed under the componentwise group operations: the constant family 1 is compatible (restriction preserves the identity); and if s, t are compatible, so are $s \cdot t$ and s^{-1} , because restriction of plus classes is a homomorphism of groups (Definition ??). Hence the compatible families form a subgroup of the product group, which is therefore a commutative group. $\square$

Definition 913 (The affine comparison isomorphism). On an affine test $\text{Spec } A$, evaluation at the top affine open $\top$ is a group isomorphism

$$\text{picEt}(C, \text{Spec } A) \xrightarrow{\sim} \text{PicEtAff}(C, A),$$

transported through the identification $\Gamma(\top) \cong A$ of Definition ??.

Proof. The affine-opens poset of $\text{Spec } A$ has a terminal element, the top open $\top$ (all of $\text{Spec } A$, an affine open containing every affine open), so a compatible family is determined by its value at $\top$. The forward map sends s to $s_{\top}$ carried through $\Gamma(\top) \cong A$ (Definition ??, Definition ??). The backward map sends $x \in \text{PicEtAff}(C, A)$ to the family whose value at V is x carried to $\Gamma(\top)$ and restricted to $\Gamma(V)$ (Definition ??); this family is compatible because restriction is functorial (restricting from $\top$ to V and then to U is restricting to U). The two are mutually inverse: forward-then-backward recovers a family s from its $\top$ -value, since s_V is the restriction of $s_{\top}$ by the compatibility of s at $V \subseteq \top$; backward-then-forward returns x , since restriction from $\top$ to $\top$ is the identity and $\Gamma(\top) \cong A$ cancels. Both maps are homomorphisms of groups, so this is a group isomorphism. $\square$

7.42 Functoriality by the unique-pullback-value characterization

The Layer-2 functor picEt is functorial along an *arbitrary* morphism $f: T' \rightarrow T$ of test objects. The value of the restricted family at an affine open $W \subseteq T'_{\text{left}}$ — whose f -image need not lie

in any affine open of T_{left} — is glued over a finite basic-open refinement of W subordinate to the preimages of affine opens, using the Zariski sheaf property of the affine étale Picard functor. Throughout, C is a curve over k .

Theorem 914 (Finite basic subcovers subordinate to a constraint). *Let W be an affine open of a scheme X and P a property of opens such that every point of W has an open neighbourhood inside W satisfying P . Then there is a finite family of sections $r \in \Gamma(X, W)$ spanning the unit ideal whose basic opens $D(r)$ each lie inside some open satisfying P .*

Proof. For each $w \in W$, choose an open $U_0 \ni w$ with $P(U_0)$; since the basic opens of the affine W form a basis of its topology, choose $r_w \in \Gamma(X, W)$ with $w \in D(r_w) \subseteq U_0$. The $D(r_w)$ cover W ; as W is quasi-compact (being affine), finitely many $D(r_{w_1}), \dots, D(r_{w_n})$ already cover W , which means the r_{w_j} span the unit ideal of $\Gamma(X, W)$. Each $D(r_{w_j}) \subseteq U_0$ lies inside an open with P . $\square$

Theorem 915 (Zariski separation of the plus over the section rings of a basic cover). *Let W be an affine open of a test object T and $(r_i)_{i \in S}$ a finite family in $\Gamma(T, W)$ spanning the unit ideal. Two plus classes $x, y \in \text{PicEtAff}(C, \Gamma(T, W))$ whose restrictions to $\Gamma(T, D(r_i))$ agree for every i are equal.*

Proof. The section ring $\Gamma(T, D(r_i))$ of a basic open of the affine open W is the localization of $\Gamma(T, W)$ away from r_i , and the r_i span the unit ideal. Apply the Zariski separation of the plus construction (Theorem ??) to this finite covering family of localizations of $\Gamma(T, W)$. $\square$

Definition 916 (The pullback-value property). Let $f: T' \rightarrow T$ be a morphism of test objects, $s \in \text{picEt}(C, T)$, and W an affine open of T'_{left} . A plus class $v \in \text{PicEtAff}(C, \Gamma(T', W))$ has the *pullback-value property* for (f, s, W) if, for every affine open $W_0 \subseteq W$ contained in the preimage $f^{-1}V$ of some affine open V of T_{left} , the restriction of v to $\Gamma(T', W_0)$ equals the pullback of s_V along the section-pullback homomorphism $\Gamma(T, V) \rightarrow \Gamma(T', W_0)$ (pullback of sections along f , a k -algebra map; Definition ??).

Theorem 917 (Choice independence). *Let $f: T' \rightarrow T$, $s \in \text{picEt}(C, T)$, W_0 an affine open of T'_{left} , and V, V' affine opens of T_{left} with $W_0 \subseteq f^{-1}V$ and $W_0 \subseteq f^{-1}V'$. Then the pullback of s_V and the pullback of $s_{V'}$ to $\Gamma(T', W_0)$ (along the two section-pullback maps) agree.*

Proof. Refine W_0 by basic opens: for each point $w \in W_0$, the image $f(w) \in V \cap V'$ lies in an affine open $V'' \subseteq V \cap V'$ of T_{left} ; choose a basic open $D(r) \ni w$ of W_0 with $f(D(r)) \subseteq V''$, i.e. $D(r) \subseteq f^{-1}V''$. By Theorem ?? a finite family of such r spans the unit ideal. On each $D(r)$, the pullback of s_V restricted to $\Gamma(T', D(r))$ factors as the pullback of $s_{V''}$: using the compatibility $s_{V''} = s_V|_{V''}$ (as $V'' \subseteq V$) and the functoriality of section pullback and restriction, it equals the pullback of $s_{V''}$ along f from V'' to $D(r)$; likewise for $s_{V'}$. Hence the two pullbacks agree on every $D(r)$, so by Theorem ?? they agree on $\Gamma(T', W_0)$. $\square$

Theorem 918 (Uniqueness of the pullback value). *For fixed (f, s, W) , a plus class with the pullback-value property is unique: if v, v' both have the property, then $v = v'$.*

Proof. Refine W by basic opens $D(r)$ each contained in the preimage of an affine open of T_{left} : every point of W maps into some affine open V of T_{left} , so has a basic-open neighbourhood in W inside $f^{-1}V$ by Theorem ?? a finite such family spans the unit ideal. On each $D(r) \subseteq f^{-1}V$, both v and v' restrict to the pullback of s_V (the defining property, Definition ??), hence agree. By Theorem ??, $v = v'$. $\square$

Theorem 919 (Existence of the pullback value). *For every $f: T' \rightarrow T$, $s \in \text{picEt}(C, T)$ and affine open W of T'_{left} , there exists a unique plus class $v \in \text{PicEtAff}(C, \Gamma(T', W))$ with the pullback-value property for (f, s, W) .*

Proof. Choose (Theorem ??) a finite family (r) in $\Gamma(T', W)$ spanning the unit ideal such that each $D(r)$ lies in the preimage $f^{-1}V_r$ of an affine open V_r of T_{left} (every point of W maps into some affine open). On $D(r)$ take the local value to be the pullback of s_{V_r} along f . On an overlap $D(r) \cap D(r') = D(rr')$, the two local values agree: both restrict, through the affine opens $V_r, V_{r'}$, to the same plus class by choice independence (Theorem ??). The section rings $\Gamma(T', D(r))$ are localizations of $\Gamma(T', W)$ away from the r , which span the unit ideal, and the overlaps are localizations away from rr' ; so by the Zariski gluing of the plus construction (Theorem ??) the local values glue to a plus class $v \in \text{PicEtAff}(C, \Gamma(T', W))$.

This v has the pullback-value property: for an affine $W_0 \subseteq W$ inside $f^{-1}V$, refine W_0 by basic opens $D(q)$ each inside some $D(r)$ (Theorem ??). On $D(q)$, v restricts through the glued datum on $D(r)$ to the pullback of s_{V_r} , which by choice independence (Theorem ??) equals the pullback of s_V ; by Theorem ??, the restriction of v to $\Gamma(T', W_0)$ equals the pullback of s_V . Uniqueness is Theorem ??.

Definition 920 (Restriction of picEt along an arbitrary morphism). For a morphism $f: T' \rightarrow T$ of test objects, the homomorphism of groups $f^*: \text{picEt}(C, T) \rightarrow \text{picEt}(C, T')$ whose value at an affine open W of T'_{left} is the unique plus class with the pullback-value property for (f, s, W) (Theorem ??).

Proof. The values form a compatible family: for affine opens $U \subseteq W$, the restriction of the value at W still has the pullback-value property at U (the property is stable under further restriction, by functoriality of section pullback and restriction), so by uniqueness it equals the value at U ; thus $(W \mapsto v_W)$ is a section of $\text{picEt}(C, T')$. It is a homomorphism: the pullback-value property is preserved by the componentwise group operations (pullback of s is a homomorphism), so by uniqueness the value at $s \cdot t$ is the product of the values and the value at 1 is 1.

Theorem 921 (The functor laws). *Restriction along the identity is the identity, $(\text{id}_T)^* = \text{id}$ on $\text{picEt}(C, T)$; and for $f: T' \rightarrow T$ and $g: T'' \rightarrow T'$, restriction along the composite is the composite of the restrictions, $(f \circ g)^* = g^* \circ f^*$.*

Proof. Identity. The value of $(\text{id}_T)^* s$ at an affine open W is the unique class with the pullback-value property; s_W itself has it, since for $W_0 \subseteq W$ inside $\text{id}^{-1}V = V$ the restriction of s_W is s_{W_0} , the pullback of s_V along the identity (by compatibility of s). By uniqueness (Theorem ??) the value is s_W , so $(\text{id}_T)^* s = s$.

Composite. It suffices to show the value $(g^*(f^*s))_W$ has the pullback-value property for $(f \circ g, s, W)$; uniqueness then identifies it with $((f \circ g)^*s)_W$. Fix an affine $W \subseteq T'_{\text{left}}$ and $W_0 \subseteq W$ inside $(f \circ g)^{-1}V$. Refine W_0 by basic opens $D(q)$ whose g -image lies in an affine open $V'' \subseteq f^{-1}V$ of T'_{left} (Theorem ??). On $D(q)$: the value of f^*s at V'' is the pullback of s_V (the pullback-value property of f^*s , as $V'' \subseteq f^{-1}V$), and the value of $g^*(f^*s)$ at $D(q)$ is the pullback of $(f^*s)_{V''}$ along g , which by functoriality of section pullback equals the pullback of s_V along $f \circ g$. By Theorem ??, the value of $g^*(f^*s)$ at W has the pullback-value property for $(f \circ g, s, W)$, as required.

Definition 922 (The étale-sheafified relative Picard functor). For a curve C over k (proper, geometrically irreducible, geometrically reduced), the contravariant functor

$$\text{picEt}(C, -): (\text{Over}(\text{Spec } k))^{\text{op}} \rightarrow \mathbf{CommGrp}$$

sending a test object T to the commutative group $\text{picEt}(C, T)$ (Definition ??) and a morphism f to the restriction homomorphism f^* (Definition ??).

Proof. The object and morphism assignments are given by Definition ?? and Definition ??; the identity and composition laws of a functor are exactly the two functor laws of restriction (Theorem ??).

Theorem 923 (Naturality of the affine comparison). *The affine comparison is natural in the test algebra: for a homomorphism of k -algebras $\varphi: A \rightarrow B$ and $s \in \text{picEt}(C, \text{Spec } A)$, the affine comparison of $(\text{Spec } \varphi)^*s$ equals the restriction along φ of the affine comparison of s . Under the comparison isomorphisms (Definition ??), restriction along $\text{Spec } \varphi$ corresponds to restriction of plus classes along φ .*

Proof. Both sides are computed at the top affine opens. Since $\top_B \subseteq (\text{Spec } \varphi)^{-1}\top_A$, the value of $(\text{Spec } \varphi)^*s$ at $\top_B$ is, by the pullback-value property (Theorem ??), the pullback of $s_{\top_A}$ along the section-pullback map $\Gamma(\text{Spec } A, \top) \rightarrow \Gamma(\text{Spec } B, \top)$. Under the identifications $\Gamma(\top) \cong A$ and $\Gamma(\top) \cong B$ (Definition ??), that section-pullback map corresponds to φ . Transporting through the comparison isomorphisms (Definition ??), the value corresponds to the restriction along φ (Definition ??) of the affine comparison of s . $\square$

Theorem 924 (Uniqueness of gluings for picEt). *Two sections $s, t \in \text{picEt}(C, T)$ agreeing on every affine open subordinate to an open cover of T_{left} are equal: if $(O_i)_i$ is a family of opens covering T_{left} and $s_U = t_U$ for every affine open U contained in some O_i , then $s = t$. This is the separation half of the sheaf property of the Layer-2 functor.*

Proof. Fix an affine open U of T_{left} . Refine U by basic opens $D(r)$ each contained in some O_i : every point of U lies in some O_i , so has a basic-open neighbourhood in U inside O_i ; by Theorem ?? a finite such family spans the unit ideal. On $D(r) \subseteq O_i$, the restriction of s_U equals $s_{D(r)} = t_{D(r)}$ (the hypothesis, as $D(r)$ is an affine open inside O_i), which equals the restriction of t_U (compatibility of s, t at $D(r) \subseteq U$). By Theorem ??, $s_U = t_U$. As U was an arbitrary affine open, $s = t$. $\square$

7.43 Rigidification along a section of the projection

Let C be an object of $\text{Over}(\text{Spec } k)$ (the curve). For a test object T in $\text{Over}(\text{Spec } k)$, a T -point $\sigma: T \rightarrow C$ determines a section of the projection $\text{pr} = \text{snd}: C \otimes T \rightarrow T$; rigidifying a Čech Picard class along that section is the cocycle recast of Kleiman’s g -rigidification. These are the inputs to the (C2) comparison of the next section.

Definition 925 (The section of the projection attached to a point). For a T -point $\sigma: T \rightarrow C$, the *section of the projection* $g_\sigma: T \rightarrow C \otimes T$ is the map $\langle \sigma, \text{id}_T \rangle$ into the product, characterised by

$$\text{fst} \circ g_\sigma = \sigma, \quad \text{snd} \circ g_\sigma = \text{id}_T.$$

It is natural in the test object: for $h: T' \rightarrow T$, $h \circ g_\sigma = g_{h \circ \sigma} \circ (C \triangleleft h)$. This is Kleiman’s g_T (*The Picard Scheme*, proof of Theorem 2.5).

Proof. By the universal property of the cartesian product, T -points of C and sections of the projection snd are the same datum: $\langle \sigma, \text{id}_T \rangle$ is the unique map with the two stated composites. Naturality is checked on the two projections, using $\text{fst} \circ (C \triangleleft h) = h \circ \text{fst}$ and $\text{snd} \circ (C \triangleleft h) = \text{snd}$. $\square$

Definition 926 (Rigidified Čech Picard class). A Čech Picard class L on the product $C \otimes T$ is *rigidified* along a point $\sigma: T \rightarrow C$ when its pullback along the section of the projection is trivial,

$$g_\sigma^* L = 1.$$

This is the cocycle form of Kleiman’s g -rigidification ($\text{df} : \text{rgd}$): on the cocycle model an isomorphism class carries no datum, so “a trivialisation $u: \mathcal{O}_T \xrightarrow{\sim} g_\sigma^* L$ exists” is exactly $g_\sigma^* L = 1$.

Theorem 927 (Rigidity of automorphisms of a rigidified pair). *Let the curve C be proper, geometrically irreducible and geometrically reduced. For every point $\sigma: \operatorname{Spec} A \rightarrow C$, pullback of global unit sections along the section of the projection,*

$$g_\sigma^\sharp: \Gamma(\operatorname{Spec} A, \mathbb{T})^\times \longrightarrow \Gamma(X_A, \mathbb{T})^\times,$$

is a bijection. In particular, a global unit of the curve X_A whose restriction along the section is 1 is itself 1. This is the cocycle form of Kleiman's automorphism rigidity: an automorphism of a rigidified pair is trivial.

Proof. Because C is proper, geometrically irreducible and geometrically reduced, the degree-zero cohomology of the relative curve is the test ring itself: pullback of global sections along the projection $\operatorname{pr} = \operatorname{snd}: X_A \rightarrow \operatorname{Spec} A$ is a ring isomorphism $\Gamma(\operatorname{Spec} A, \mathbb{T}) \xrightarrow{\sim} \Gamma(X_A, \mathbb{T})$ (Theorem ??, the case $H^0(X_A, \mathcal{O}) = A$; this is Kleiman's hypothesis $\mathcal{O}_S \xrightarrow{\sim} f_* \mathcal{O}_X$ holding universally in the test ring). The section retracts the projection, $\operatorname{pr} \circ g_\sigma = \operatorname{id}$ (Definition ??), and pullback of sections is contravariant, so on global sections $g_\sigma^\sharp \circ \operatorname{pr}^\sharp = (\operatorname{pr} \circ g_\sigma)^\sharp = \operatorname{id}$. As $\operatorname{pr}^\sharp$ is an isomorphism, $g_\sigma^\sharp$ is its inverse, hence an isomorphism, hence bijective on the unit groups. Kleiman's argument is exactly this: an automorphism v of a pair (L, u) rigidified along σ satisfies $g_\sigma^\sharp v = 1$, while v lies in $H^0(\mathcal{O}_{X_A}) = H^0(\mathcal{O}_{\operatorname{Spec} A})$; the displayed isomorphism forces $v = 1$. $\square$

Lemma 928 (The retraction identity). *For a point $\sigma: T \rightarrow C$, pullback of Čech Picard classes along the section of the projection retracts pullback along the projection: for every $N \in \operatorname{Pic}(T)$,*

$$g_\sigma^* \operatorname{snd}^* N = N.$$

This is $g_T^ f_T^* = 1$ in Kleiman's notation.*

Proof. Pullback of Čech classes is contravariantly functorial, and $\operatorname{snd} \circ g_\sigma = \operatorname{id}_T$ (Definition ??), so the composite pullback is the pullback along the identity, which is the identity. $\square$

Lemma 929 (Stability of rigidification under restriction). *The trivial class is rigidified along every point. Moreover, for $h: T' \rightarrow T$ and a class L on $C \otimes T$ rigidified along $\sigma: T \rightarrow C$, the pullback $(C \triangleleft h)^* L$ is rigidified along the restricted point $h \circ \sigma$.*

Proof. The first claim is $g_\sigma^* 1 = 1$. For the second, the naturality of the section in the test object (Definition ??) gives $h \circ g_\sigma = g_{h \circ \sigma} \circ (C \triangleleft h)$, so

$$g_{h \circ \sigma}^* (C \triangleleft h)^* L = ((C \triangleleft h) \circ g_{h \circ \sigma})^* L = (g_\sigma \circ h)^* L = h^* (g_\sigma^* L) = h^* 1 = 1. \quad \square$$

Theorem 930 (Kleiman's identification, injectivity: rigidified representatives are unique). *Two Čech Picard classes on $C \otimes T$, both rigidified along a point $\sigma: T \rightarrow C$, which represent the same relative Picard class, are equal on the nose. In particular a rigidified representative of the identity relative class is 1. This is the injectivity half of Kleiman's Lemma lm:iden (rigidified pairs are carried isomorphically onto $\operatorname{Pic}_{X/S}(T)$), in cocycle form.*

Proof. Let L, L' be the two classes. By the exact coset relation (Definition ??) the ratio is pulled back from the base: $L/L' = \operatorname{snd}^* N$ for some $N \in \operatorname{Pic}(T)$. Apply g_σ^* : by the retraction identity (Lemma ??),

$$N = g_\sigma^* \operatorname{snd}^* N = g_\sigma^* (L/L') = g_\sigma^* L / g_\sigma^* L' = 1/1 = 1,$$

using both rigidifications. Hence $L/L' = \operatorname{snd}^* 1 = 1$, i.e. $L = L'$. The special case takes $L' = 1$, which is rigidified. $\square$

Theorem 931 (Kleiman’s identification, surjectivity: existence of a rigidified representative). *Every relative Picard class on $C \otimes T$ has a representative rigidified along any given point $\sigma: T \rightarrow C$. This is the surjectivity half of Kleiman’s Lemma `lm:iden`, in cocycle form.*

Proof. Let M be any Čech representative of the class (the projection to the quotient is surjective, Definition ??). Kleiman’s identity-rigidification twist

$$L := M \cdot (\text{snd}^* g_\sigma^* M)^{-1}$$

$(\mathcal{L} := \mathcal{M} \otimes (f_T^* g_T^* \mathcal{M})^{-1})$ works. The twist factor is pulled back from the base, so L represents the same relative class. And by the retraction identity (Lemma ??), $g_\sigma^* L = g_\sigma^* M \cdot (g_\sigma^* M)^{-1} = 1$, so L is rigidified along σ . $\square$

Theorem 932 (The rigidified transport keystone). *Let A be a k -algebra, E a presented étale cover of A , R an A -algebra, $\sigma: \text{Spec } A \rightarrow C$ a point, and L a Čech Picard class on $C \otimes \text{Spec } B_E$ such that the relative class of L is a descent class on E and L is rigidified along the point of $\text{Spec } B_E$ induced by σ . Then for any two A -algebra maps $j_1, j_2: B_E \rightarrow R$, the two pullbacks of L agree on the nose:*

$$(C \triangleleft \text{Spec } j_1)^* L = (C \triangleleft \text{Spec } j_2)^* L \quad \text{in } \text{Pic}(C \otimes \text{Spec } R).$$

Proof. The plus-construction keystone `relPicAlgMap_congr` (Theorem ??) gives the equality of the two pullbacks as *relative* Picard classes, since the class of L is a descent class and the j_i are A -algebra maps. Both pullbacks are rigidified along the point of $\text{Spec } R$ induced by σ : the j_i are A -algebra maps, so the induced point of $\text{Spec } R$ restricts, along $\text{Spec } j_i$, the induced point of $\text{Spec } B_E$, and rigidification is stable under restriction (Lemma ??). Kleiman’s injectivity (Theorem ??) upgrades the coset equality of two rigidified classes to an equality of Čech classes. $\square$

Theorem 933 (The descent equation of a rigidified representative holds on the nose). *In the situation of Theorem ??, the two pullbacks of the rigidified representative L to the double cover $\text{Spec}(B_E \otimes_A B_E)$, along the two coprojections, are equal Čech Picard classes — not merely equal relative classes. This is Kleiman’s rigidified pair $(\mathcal{L}', u')$ having a canonical comparison on the double cover, the input to the descent-datum construction in the proof of his comparison theorem.*

Proof. The two coprojections $B_E \rightrightarrows B_E \otimes_A B_E$ (Definition ??) are A -algebra maps out of the cover carrier into the common A -algebra $R := B_E \otimes_A B_E$; apply the transport keystone (Theorem ??) to them. $\square$

7.44 The σ -normalized comparison cochain

We now fix a k -algebra A with a further A -algebra B (the carrier of an étale cover in the intended application) and a curve point $\sigma: \text{Spec } A \rightarrow C$. Write $X_R := (C \otimes \text{Spec } R)_{\text{left}}$ for the curve over $\text{Spec } R$. Let $q_1, q_2: \text{Spec}(B \otimes_A B) \rightarrow \text{Spec } B$ be the coprojections (Definition ??), $u_1, u_2: X_{B \otimes_A B} \rightarrow X_B$ their whiskerings $C \triangleleft q_i$, and $f_{12}, f_{13}, f_{23}, w_{ij} = C \triangleleft f_{ij}$ the cofaces of Definition ??. Write s_B, s_q for the sections of the projection (Definition ??) attached to the base changes of σ to B and $B \otimes_A B$, and $p_2: X_{B \otimes_A B} \rightarrow \text{Spec}(B \otimes_A B)$ for the projection. Throughout, $s^\sharp$ denotes pullback of unit cochains as in Section ??.

This section packages, and constructs with *no geometric hypotheses*, a cochain-level realisation of the descent equation $u_1^* L = u_2^* L$, normalized along σ .

Lemma 934 (The section squares). *The section of the projection is natural across the base-change coprojections and cofaces: as maps of curve products,*

$$s_q \circ u_1 = q_1 \circ s_B, \quad s_q \circ u_2 = q_2 \circ s_B, \quad s_{\text{cb}} \circ w_{ij} = f_{ij} \circ s_q \quad (ij \in \{12, 13, 23\}),$$

and s_q retracts the projection, $s_q \circ p_2 = \text{id}$.

Proof. Each square is the naturality of the section (Definition ??) applied to the relevant A -algebra map: the coprojections q_i and cofaces f_{ij} are Spec of A -algebra homomorphisms out of B respectively $B \otimes_A B$ (Definitions ??, ??), and restricting the base-changed point along such a map lands at the base-changed point over the target algebra; whiskering with C and passing to the section turns this point-compatibility into the stated square. The retraction is $\text{snd} \circ g_\sigma = \text{id}$ of Definition ??, whiskered to the curve product. $\square$

Lemma 935 (Pullback of a one-sided trivializing relation). *Let $\varphi: X \rightarrow Y$ and $r: Y \rightarrow Z$. Suppose a unit 0-cochain a on a pointed cover of Y trivializes the pullback along r of a pair-indexed unit family E : for all y, y' , on the overlap,*

$$a_y \cdot (r^\sharp E)_{yy'} = a_{y'}.$$

Then for every $x, x' \in X$ and every open $O \subseteq \varphi^{-1}(\dots)$ inside the relevant overlap pullback, the pulled-back cochain $\varphi^\sharp a$ trivializes the family pulled back along the composite:

$$\varphi_O^\sharp a_{\varphi x} \cdot ((r\varphi)_O^\sharp E)_{xx'} = \varphi_O^\sharp a_{\varphi x'}.$$

This is the one-sided companion of Lemma ??.

Proof. Apply the pullback homomorphism $\varphi_O^\sharp$ to the given relation at $\varphi x, \varphi x'$. Being a group homomorphism on units it distributes over the product; by functoriality $\varphi_O^\sharp(r^\sharp E) = (r\varphi)_O^\sharp E$ merges the pullback, while $\varphi_O^\sharp a$ is left unchanged. $\square$

Lemma 936 (Compatibility of a ratio cochain). *Let $\varphi_t: X \rightarrow Y_c$, $\varphi_1, \varphi_2: X \rightarrow Y_a$, $r_1, r_2: Y_c \rightarrow Z$, $s_1: Y_a \rightarrow Z$. Suppose a 0-cochain c on Y_c cobounds the comparison of a pair-indexed family E pulled back along r_1, r_2 (??), and a 0-cochain a on Y_a trivializes E pulled back along s_1 (??). If the pulled-back composites agree, $\varphi_t \circ r_1 = \varphi_1 \circ s_1$ and $\varphi_t \circ r_2 = \varphi_2 \circ s_1$, then the ratio cochain*

$$\varphi_t^\sharp c \cdot \frac{\varphi_2^\sharp a}{\varphi_1^\sharp a}$$

on X is Čech-compatible: its values at x and x' agree on overlaps.

Proof. Pull the coboundary relation for c back along φ_t (??) and the trivializing relation for a back along φ_1 and φ_2 (??). The hypotheses $\varphi_t r_i = \varphi_i s_1$ identify the two families through Lemma ??; combining the pulled-back relations, the comparison of the ratio cochain at x and x' reduces to an identity of units, giving compatibility. $\square$

Definition 937 (The normalization cover). For a pointed cover $\mathcal{N}$ of the curve X_B and a pointed cover $\mathcal{W}$ of the curve $X_{B \otimes_A B}$, the *normalization cover* of $\text{Spec}(B \otimes_A B)$ is

$\mathcal{W}$ pulled back along s_q , intersected with the two coprojection pullbacks q_1^*, q_2^* of $\mathcal{N}$ pulled back along s_B .

It is the cover on which the σ -normalization below is stated; it lies below $s_q^{-1}\mathcal{W}$ and below each $q_i^{-1}(s_B^{-1}\mathcal{N})$.

Definition 938 (The σ -normalized comparison datum). Fix a unit Čech cocycle γ on a pointed cover $\mathcal{N}$ of X_B , representing a class $L = [\gamma]$. A σ -normalized comparison of γ packages:

- a trivializing 0-cochain ρ of the section-pullback $s_B^\# \gamma$ on the cover $s_B^{-1} \mathcal{N}$ of $\text{Spec } B$: $\rho_b \cdot (s_B^\# \gamma)_{bb'} = \rho_{b'}$ on overlaps — the cochain shadow of rigidification (??) along the section over B ;
- a pointed cover $\mathcal{W}$ of $X_{B \otimes_A B}$ refining both coprojection pullbacks of $\mathcal{N}$, and a comparison 0-cochain θ on it, with
- (N1) $\theta_x \cdot (u_1^\# \gamma)_{xy} = (u_2^\# \gamma)_{xy} \cdot \theta_y$ on overlaps — θ cobounds the two coprojection pullbacks of γ , realising $u_1^* L = u_2^* L$ at the cochain level; and
- (N2) $(s_q^\# \theta)_y \cdot (q_2^\# \rho)_y = (q_1^\# \rho)_y$ on the normalization cover (??) — the σ -restriction of θ is the ρ -comparison $q_1^\# \rho / q_2^\# \rho$.

The normalization is ρ -relative: rigidification makes $s_B^\# \gamma$ trivial only as a class, giving the trivializing cochain ρ , so “ σ -restriction = 1” is replaced by (N2). Every step is stated pointwise via unit pullback and is blind to disconnectedness of $\text{Spec}(B \otimes_A B)$.

Theorem 939 (The σ -defect glues to a global unit). *Let γ be a unit cocycle on $\mathcal{N}$, let θ_0 be a comparison cochain on a refinement $\mathcal{W}$ satisfying (N1), and let ρ be a trivializing cochain of $s_B^\# \gamma$. The σ -defect cochain*

$$s_q^\# \theta_0 \cdot \frac{q_2^\# \rho}{q_1^\# \rho}$$

on the normalization cover is Čech-compatible, hence glues to a global unit $\delta \in \Gamma(\text{Spec}(B \otimes_A B), \mathbb{T})^\times$.

Proof. Compatibility is the ratio-compatibility Lemma ?? applied with $\varphi_t = s_q$, $\varphi_i = q_i$, $r_i = u_i$, $s_1 = s_B$: (N1) is the coboundary relation for θ_0 , the relation for ρ is the one-sided trivialization, and the composite coincidences $s_q \circ u_i = q_i \circ s_B$ are the section squares (??). A compatible unit family on a pointed cover glues to a global unit by the sheaf axiom (??). $\square$

Theorem 940 (Existence of the σ -normalized comparison). *Let γ be a unit cocycle on a pointed cover $\mathcal{N}$ of X_B whose class $L = [\gamma]$ is rigidified along the section over B (??) and satisfies the descent equation $u_1^* L = u_2^* L$ — as holds for any rigidified representative of a descent class, by Theorem ?? . Then a σ -normalized comparison of γ exists. No properness, geometric irreducibility or reducedness, or faithful flatness is required.*

Proof. Rigidification gives $g_{\sigma_B}^* L = 1$; on cocycles (??) this means the section-pullback $s_B^\# \gamma$ is a coboundary, so a trivializing cochain ρ exists on $s_B^{-1} \mathcal{N}$ itself (no refinement). The descent equation $u_1^* L = u_2^* L$ means the two coprojection pullbacks of γ cobound on a common refinement $\mathcal{W}$, yielding a comparison cochain θ_0 with (N1). By Theorem ?? the σ -defect of θ_0 glues to a global unit δ of $\text{Spec}(B \otimes_A B)$. Correct θ_0 by dividing out the projection pullback $p_2^\# \delta$: this preserves (N1) (a global factor cancels from both sides of the coboundary relation), and, because s_q retracts p_2 (??), it makes the σ -restriction of the corrected cochain equal to $q_1^\# \rho / q_2^\# \rho$ on the nose — that is (N2). The corrected data is the required datum. $\square$

7.45 Coherence and uniqueness of the σ -normalized comparison

Fix a unit Čech cocycle γ on a pointed cover $\mathcal{N}$ of X_B and a σ -normalized comparison of γ (Definition ??), with cover $\mathcal{W}$ of $X_{B \otimes_A B}$ and comparison cochain θ . We show that θ satisfies the Amitsur cocycle identity on the triple base *on the nose*, is unique, and is trivial on the diagonal — the three inputs the per-piece descent of the next section runs on. All three are the cochain form of Kleiman’s automorphism-rigidity argument (Lemma ??): glue a defect to a global unit, kill its σ -restriction by the normalization, and apply rigidity at the relevant affine test.

Write $X_{B \otimes_A (B \otimes_A B)}$ for the curve over the triple base. Its three tensor-slot *insertions* $v_1, v_2, v_3 : X_{B \otimes_A (B \otimes_A B)} \rightarrow X_B$ are the whiskerings $C \triangleleft (\text{Spec of})$ of the A -algebra maps $x \mapsto x \otimes 1 \otimes 1$, $x \mapsto 1 \otimes x \otimes 1$, $x \mapsto 1 \otimes 1 \otimes x$.

Definition 941 (The Amitsur product cover and its insertions). For a pointed cover $\mathcal{W}$ of the curve $X_{B \otimes_A B}$, the *Amitsur product cover* of $X_{B \otimes_A (B \otimes_A B)}$ is the common refinement of the three coface pullbacks,

$$w_{23}^{-1} \mathcal{W} \sqcap w_{12}^{-1} \mathcal{W} \sqcap w_{13}^{-1} \mathcal{W},$$

the curve-product mirror of the Amitsur cover (Definition ??). The three insertions v_1, v_2, v_3 are the pairwise-coincident values of the six coface–coprojection composites $w_{ij} \circ u_k : X_{B \otimes_A (B \otimes_A B)} \rightarrow X_B$:

$$v_1 = w_{12} \circ u_1 = w_{13} \circ u_1, \quad v_2 = w_{23} \circ u_1 = w_{12} \circ u_2, \quad v_3 = w_{23} \circ u_2 = w_{13} \circ u_2.$$

Proof. The cover is a finite intersection of pullbacks of the pointed cover $\mathcal{W}$, hence a pointed cover. Each of the three displayed pairs of composites coincides by the simplicial identities of the tensor tower relating the cofaces f_{ij} (Definition ??) to the coprojections $x \mapsto x \otimes 1$, $x \mapsto 1 \otimes x$ (Definition ??): whiskering to the curve preserves those equalities, and each v_i is the whiskered Spec of one tensor-slot insertion. $\square$

Theorem 942 (The Amitsur defect glues to a global unit). *Let γ be a unit cocycle on $\mathcal{N}$ and θ_0 a comparison cochain on a refinement $\mathcal{W}$ satisfying (N1), i.e. θ_0 cobounds the two coprojection pullbacks of γ . The Amitsur defect*

$$\varepsilon = \frac{(w_{23}^\# \theta_0)(w_{12}^\# \theta_0)}{w_{13}^\# \theta_0}$$

— a unit 0-cochain on the Amitsur product cover — is Čech-compatible, hence glues to a global unit $\bar{\varepsilon} \in \Gamma(X_{B \otimes_A (B \otimes_A B)}, \mathbb{T})^\times$. No geometric hypotheses are used.

Proof. Relation (N1) reads $\theta_{0,x} \cdot (u_1^\# \gamma)_{xy} = (u_2^\# \gamma)_{xy} \cdot \theta_{0,y}$ on overlaps. Pull it back along each of the three cofaces w_{23}, w_{12}, w_{13} ; by Definition ?? the coface–coprojection composites appearing are exactly the three insertions, so the pulled-back relations express the overlap discrepancy of each factor $w_{ij}^\# \theta_0$ through the cocycle γ evaluated at the insertions v_1, v_2, v_3 . The three insertion values are common to the numerator and denominator of ε ; multiplying the three relations telescopically, the γ -terms cancel and the value of ε at z agrees with its value at z' on the overlap. A compatible unit family on a pointed cover glues to a global unit by the sheaf axiom (Theorem ??). $\square$

Theorem 943 (Coherence of the σ -normalized comparison). *Let the curve C be proper, geometrically irreducible and geometrically reduced. Then the comparison cochain θ of a σ -normalized comparison of γ satisfies the Amitsur cocycle identity on the nose:*

$$(w_{23}^\# \theta)(w_{12}^\# \theta) = w_{13}^\# \theta,$$

in the face convention of the descent 1-cocycle (Definition ??). No faithful flatness is required.

Proof. By (N1) the Amitsur defect ε of θ glues to a global unit $\bar{\varepsilon}$ of $X_{B \otimes_A (B \otimes_A B)}$ (Theorem ??). We claim its σ -restriction is 1. Pull the glued identity back along the section s_{cb} over the triple base: by the section squares $s_{cb} \circ w_{ij} = f_{ij} \circ s_q$ (Lemma ??) the three coface pullbacks of θ become the coface pullbacks $f_{ij}^\#$ of the σ -restriction of θ . Now (N2) rewrites the σ -restriction of θ as the ρ -comparison $q_1^\# \rho / q_2^\# \rho$; substituting, the three coface pullbacks become three ratios of ρ -terms whose endpoints match up through the simplicial coincidences $f_{12} \circ q_2 = f_{23} \circ q_1$, $f_{13} \circ q_1 = f_{12} \circ q_1$, $f_{13} \circ q_2 = f_{23} \circ q_2$. These telescope to 1; hence the σ -restriction of $\bar{\varepsilon}$ is 1. Finally, at the affine test $B \otimes_A (B \otimes_A B)$, rigidity (Theorem ??) forces a global unit with trivial σ -restriction to be 1, so $\bar{\varepsilon} = 1$, i.e. $\varepsilon = 1$, which is the claimed identity. This is Kleiman's descent-cocycle step: the automorphism $v_{13}^{-1} v_{23} v_{12}$ of the first-projection pullback of the rigidified pair is trivial. $\square$

Theorem 944 (Uniqueness of the σ -normalized comparison). *Let the curve C be proper, geometrically irreducible and geometrically reduced. Two comparison cochains θ, θ' on the same cover $\mathcal{W}$, each satisfying (N1) against the same cocycle γ and (N2) against the same trivializing cochain ρ , are equal.*

Proof. The ratio θ/θ' has trivial coboundary: since θ and θ' cobound the same comparison of γ (both satisfy (N1)), the overlap discrepancies of θ and θ' coincide and cancel in the ratio. Hence θ/θ' is Čech-compatible and glues to a global unit χ of $X_{B \otimes_A B}$ (Theorem ??). Its σ -restriction is 1: by (N2) both $s_q^\# \theta$ and $s_q^\# \theta'$ equal the same ρ -comparison $q_1^\# \rho / q_2^\# \rho$ on the normalization cover, so their ratio restricts to 1. Rigidity (Theorem ??) at the affine test $B \otimes_A B$ forces $\chi = 1$; reading this pointwise gives $\theta = \theta'$. $\square$

Theorem 945 (The diagonal pullback of a witness glues to a global unit). *The multiplication $\nabla: B \otimes_A B \rightarrow B$ retracts both coprojections, $\nabla \circ q_1 = \text{id} = \nabla \circ q_2$; whiskered to the curve this is $u_1 \circ \Delta = \text{id} = u_2 \circ \Delta$, where $\Delta: X_B \rightarrow X_{B \otimes_A B}$ is the diagonal (the whiskered Spec of ∇). Consequently, for a comparison cochain θ_0 satisfying (N1), the diagonal pullback $\Delta^\# \theta_0$ is Čech-compatible and glues to a global unit $\bar{d} \in \Gamma(X_B, \mathbb{T})^\times$. No geometric hypotheses.*

Proof. The retractions are the identities $\nabla(b \otimes 1) = b$ and $\nabla(1 \otimes b) = b$; whiskering the resulting equalities of A -algebra maps to the curve gives $u_1 \circ \Delta = \text{id} = u_2 \circ \Delta$. Pull (N1) back along Δ : both insertion terms of the pulled relation are γ evaluated along $u_1 \circ \Delta = u_2 \circ \Delta$, hence coincide and cancel, leaving $\Delta^\# \theta_0$ with matching overlap values. Gluing by the sheaf axiom (Theorem ??) yields $\bar{d}$. $\square$

Theorem 946 (Triviality of the comparison on the diagonal). *Let the curve C be proper, geometrically irreducible and geometrically reduced. The pullback of the comparison cochain θ of a σ -normalized comparison along the diagonal Δ of $X_{B \otimes_A B}$ is trivial: $\Delta^\# \theta = 1$. This is the cochain source of the diagonal normalization $\mu(u) = 1$ of the descent 1-cocycle.*

Proof. Glue the diagonal seed of θ to a global unit $\bar{d}$ of X_B (Theorem ??). Its σ -restriction is 1: the section square relating s_B , Δ , the Spec of ∇ and s_q (Lemma ??) turns it into the ∇ -pullback of the σ -restriction of θ , which (N2) makes the ∇ -pullback of the ρ -comparison $q_1^\# \rho / q_2^\# \rho$; the two legs

coincide on the diagonal because the multiplication retracts both coprojections (Theorem ??), so the comparison collapses to 1. Rigidity (Theorem ??) at the affine test B forces $\bar{d} = 1$; pointwise this is $\Delta^\sharp\theta = 1$. $\square$

7.46 The per-piece descent datum

The σ -normalized comparison is descended piece by piece over a cg-saturated affine cover of the base curve X_A . On a piece V the cover-piece ring is $M := S \otimes_A B$ with $S := \Gamma(X_A, U)$ the base-piece ring (Definition ??), and the descent along $S \rightarrow M$ is governed by the piece restriction of the comparison. This section records the pure-algebra bridge: the identification of the descent-cocycle ring with the geometric double-base ring, and the extraction of a descent 1-cocycle from a comparison unit in geometric coordinates. Throughout, A is a commutative ring, S and B are A -algebras, and $M = S \otimes_A B$.

Definition 947 (The descent-cocycle ring identification). The ring $M \otimes_S M$ in which a descent 1-cocycle relative to $S \rightarrow M$ (Definition ??) lives is the base change to S of the whole-cover descent-cocycle ring $B \otimes_A B$: there is an S -algebra isomorphism

$$M \otimes_S M \xrightarrow{\sim} S \otimes_A (B \otimes_A B), \quad (s_1 \otimes b_1) \otimes (s_2 \otimes b_2) \mapsto (s_2 s_1) \otimes (b_1 \otimes b_2),$$

together with its triple analogue $M \otimes_S (M \otimes_S M) \xrightarrow{\sim} S \otimes_A (B \otimes_A (B \otimes_A B))$.

Proof. The map is the composite of the cancellation of the middle base ring, $(S \otimes_A B) \otimes_S (S \otimes_A B) \cong S \otimes_A (S \otimes_A B) \cong (S \otimes_A S) \otimes_A B$ contracted along $S \otimes_A S \rightarrow S$, with the associativity isomorphism of the iterated tensor product; the displayed value on pure tensors is read off from these two standard isomorphisms. The triple form is obtained by applying the same identification in the inner two factors and then cancelling and reassociating. $\square$

Lemma 948 (Compatibility with the multiplication and the cofaces). *Under the identification of Definition ??, the multiplication $\mu: M \otimes_S M \rightarrow M$ corresponds to $\text{id}_S \otimes \mu$ on $S \otimes_A (B \otimes_A B)$, and the three cofaces $\partial_{23}, \partial_{12}, \partial_{13}$ of $M \otimes_S M$ correspond to $\text{id}_S \otimes \partial_{23}, \text{id}_S \otimes \partial_{12}, \text{id}_S \otimes \partial_{13}$ on $S \otimes_A (B \otimes_A (B \otimes_A B))$.*

Proof. Each is an identity between two S -algebra maps out of $M \otimes_S M$; it therefore suffices to check it on pure tensors, where both sides evaluate to the same element by the computation of Definition ??, and to extend by bilinearity. $\square$

Theorem 949 (A coherent comparison unit is a descent 1-cocycle). *Let $v \in (S \otimes_A (B \otimes_A B))^\times$ be a comparison unit that is diagonal-normalized, $(\text{id}_S \otimes \mu)(v) = 1$, and satisfies the Amitsur identity through the three cofaces,*

$$(\text{id}_S \otimes \partial_{23})(v) \cdot (\text{id}_S \otimes \partial_{12})(v) = (\text{id}_S \otimes \partial_{13})(v).$$

Then the unit of $M \otimes_S M$ that corresponds to v under Definition ?? — the comparison descent unit — is a descent 1-cocycle relative to $S \rightarrow M$ (Definition ??).

Proof. Transport the two hypotheses through the identification. By Lemma ?? the normalization μ of the comparison descent unit is carried to $(\text{id}_S \otimes \mu)(v) = 1$, which is the normalization condition; and the cocycle product $\partial_{23} \cdot \partial_{12}$ of the comparison descent unit is carried, through the triple identification, to $(\text{id}_S \otimes \partial_{23})(v) \cdot (\text{id}_S \otimes \partial_{12})(v) = (\text{id}_S \otimes \partial_{13})(v)$, the image of ∂_{13} . As the triple identification is injective, the cocycle identity holds. $\square$

Definition 950 (The descended per-piece class and its cover pullback). Assume $A \rightarrow B$ is faithfully flat, and let v be a coherent comparison unit as in Theorem ???. The *descended per-piece class* $M_V \in \text{Pic}(S)$ is the Picard class (Definition ??) of the descent 1-cocycle of the comparison descent unit. Its image under $\text{Pic}(S) \rightarrow \text{Pic}(M)$ is trivial.

Proof. Faithful flatness passes from $A \rightarrow B$ to $S \rightarrow M = S \otimes_A B$ by base change, so the descent 1-cocycle of the comparison descent unit has a well-defined Picard class (Definition ??). Base change to M sends the descended module to M itself: the descended module of a cocycle is, by construction, the submodule of M on which the two coactions agree (Definition ??), and tensoring up to M recovers M as a free rank-one module through the descent isomorphism. Hence the class of M_V in $\text{Pic}(M)$ is that of a free module, namely 1. $\square$

Definition 951 (The geometric descent-cocycle ring identification). For a cg-saturated affine piece $U \subseteq X_A$, with base-piece ring $\Gamma(X_A, U)$ and cover-piece ring $\Gamma(X_B, \text{cg}^{-1}U)$, the descent-cocycle ring $\Gamma(X_B, \text{cg}^{-1}U) \otimes_{\Gamma(X_A, U)} \Gamma(X_B, \text{cg}^{-1}U)$ is identified, as a $\Gamma(X_A, U)$ -algebra, with $\Gamma(X_A, U) \otimes_A (B \otimes_A B)$, and hence with the section ring $\Gamma(X_{B \otimes_A B}, \text{cgq}^{-1}U)$ of the curve over the descent-cocycle base — the ring on whose spectrum the piece restriction of the comparison lives.

Proof. Apply the piece identification $\Gamma(X_B, \text{cg}^{-1}U) \cong \Gamma(X_A, U) \otimes_A B$ (Definition ??) to each tensor factor, then the algebraic descent-cocycle ring identification (Definition ??) with $S = \Gamma(X_A, U)$; this gives the first identification. For the second, apply the piece identification at the test algebra $B \otimes_A B$, giving $\Gamma(X_A, U) \otimes_A (B \otimes_A B) \cong \Gamma(X_{B \otimes_A B}, \text{cgq}^{-1}U)$. $\square$

7.47 The per-piece comparison unit and its descended class

We continue with the (C2) data: a commutative ring A , an A -algebra B , the curve C over $\text{Spec } k$, and the cover inclusion $\text{cg}: X_B \rightarrow X_A$ (the whiskering $C \triangleleft \text{Spec}(A \rightarrow B)$), with $\text{cgq}: X_{B \otimes_A B} \rightarrow X_A$ its double, and $u_1, u_2: X_{B \otimes_A B} \rightarrow X_B$ the coprojections. A rigidified class $L = [\gamma]$ on X_B is descended *piece by piece* over cg-saturated affine opens V of X_A . This section manufactures, on each such piece equipped with a trivialization of L , the comparison unit that the per-piece descent extraction (brick E2) consumes, and its descended class.

Definition 952 (A piece trivialization). A *piece trivialization* of a representing cocycle γ on a pointed cover $\mathcal{N}$ of X_B , over a piece $V \subseteq X_A$, is a trivializing 0-cochain t of γ on the $\text{cg}^{-1}V$ -trimmed opens: units $t_b \in \Gamma(X_B, \mathcal{N}(b) \cap \text{cg}^{-1}V)^\times$ with $t_b \cdot \gamma_{bb'} = t_{b'}$ on the trimmed pairwise overlaps. It is the cochain witness that the class $L = [\gamma]$ is trivial on the cover piece $\text{cg}^{-1}V$ (?); its existence over a covering family of pieces is the piece-selection input consumed by the final splice. The two coprojections both lie over the cover inclusions, $u_1 \circ \text{cg} = \text{cgq} = u_2 \circ \text{cg}$ as maps to X_A , so the double piece is bounded by each coprojection preimage of the piece, $\text{cgq}^{-1}V \leq u_i^{-1}(\text{cg}^{-1}V)$.

Proof. The two coprojection identities are the whiskerings of $q_i \circ (A \rightarrow B) = (A \rightarrow B \otimes_A B)$ as A -algebra maps (Definition ??); whiskering to the curve and passing to preimages gives $\text{cgq}^{-1}V \leq u_i^{-1}(\text{cg}^{-1}V)$. $\square$

Theorem 953 (The trivialization-twisted witness glues to a unit). *Over abstract schemes, let $r_1, r_2: Y \rightarrow Z$, let $\mathcal{C}$ be a pointed cover of Y and $\mathcal{N}$ one of Z , let c be a comparison 0-cochain on $\mathcal{C}$ and E a pair-indexed unit family on $\mathcal{N}$, and let t trivialize E on the DZ -trimmed opens of $\mathcal{N}$, for opens $DZ \subseteq Z$, $DY \subseteq Y$ with $DY \leq r_i^{-1}DZ$. Define the twisted value at y , on $\mathcal{C}(y) \cap DY$,*

$$(c_y) \cdot (r_2^\# t)_y \cdot (r_1^\# t)_y^{-1}.$$

If c cobounds the two pullbacks of E (the relation $c_x \cdot (r_1^\sharp E)_{xy} = (r_2^\sharp E)_{xy} \cdot c_y$) and t satisfies the trivializing relation, then the twisted values are Čech-compatible on the trimmed cover $\{\mathcal{C}(y) \cap DY\}$ of DY , hence glue to a global unit $v \in \Gamma(Y, DY)^\times$; and for a further map φ into Y with $\varphi \circ r_i = m_i$, the φ -pullback of v expands into the pulled witness value $\varphi^\sharp c$ times the two m_i -pullbacks of t .

Proof. Compatibility is a telescoping identity: on a pairwise overlap the coboundary relation for c and the two pullbacks of the trivializing relation for t (along r_1 and r_2) combine so that the E -terms cancel and the twisted value at y matches that at y' . A compatible unit family on a pointed cover glues to a global unit by the sheaf axiom (Theorem ??). The pullback expansion is the defining gluing relation pulled back along φ , with the composites $\varphi \circ r_i = m_i$ merging the two pullbacks. $\square$

Theorem 954 (The per-piece comparison unit). *Let W be a σ -normalized comparison of γ (Definition ??) with comparison cochain θ , and T a piece trivialization of γ on V . The per-piece comparison unit*

$$v_V \in \Gamma(X_{B \otimes_A B}, \text{cgq}^{-1}V)^\times$$

is the global unit glued from the values $\theta_x \cdot (u_2^\sharp T)(x) / (u_1^\sharp T)(x)$ on the trimmed comparison cover: on each member it restricts to that twisted witness value. It is the unique unit of the double piece with this restriction property.

Proof. Apply Theorem ?? with $r_1, r_2 = u_1, u_2$, comparison cover $\mathcal{C} = W.\text{cover}$, representing cover $\mathcal{N}$, comparison cochain $c = \theta$, pair family $E = \gamma$, trimming opens $DZ = \text{cg}^{-1}V$, $DY = \text{cgq}^{-1}V$, and trivialization $t = T$: the coboundary relation for θ is the (N1) witness relation of the normalized comparison, and the trivializing relation is that of the piece trivialization (Definition ??). Gluing yields v_V and its defining restriction property. Uniqueness is the separation half of the sheaf axiom: two units of $\text{cgq}^{-1}V$ with the same restrictions to a cover agree. $\square$

Theorem 955 (Naturality of the section-ring identification in the algebra). *The section-ring identification $\Gamma(X_R, \text{cg}_R^{-1}V) \cong \Gamma(X_A, V) \otimes_A R$ (Definition ??, R -uniform) is natural in the base algebra: for an A -algebra map $j: R \rightarrow R'$, the whiskered map $W_j = (C \triangleleft \text{Spec } j): X_{R'} \rightarrow X_R$ lies over the cover structure maps ($W_j \circ \text{cg}_R = \text{cg}_{R'}$), and pulling identified sections back along W_j is $\text{id}_{\Gamma(X_A, V)} \otimes j$ through the identifications of the two pieces:*

$$W_j^\sharp(\Phi_V^R(x)) = \Phi_V^{R'}((\text{id} \otimes j)(x)).$$

Proof. Both sides are A -algebra homomorphisms out of $\Gamma(X_A, V) \otimes_A R$, so it suffices to check them on a pure tensor $s \otimes r$, then extend by bilinearity. There the identification sends $s \otimes r$ to $\text{cg}_R^\sharp(s) \cdot \text{pr}_R^\sharp(r)$ (Definition ??). The cg -factor pulls back along W_j by the composite $W_j \circ \text{cg}_R = \text{cg}_{R'}$ (as A -algebra maps, $j \circ (A \rightarrow R) = (A \rightarrow R')$), giving $\text{cg}_{R'}^\sharp(s)$. The projection factor pulls back along the square $W_j \circ \text{pr}_R = \text{pr}_{R'} \circ \text{Spec } j$, and the Spec -side factor is the naturality of $\Gamma\text{-of-Spec}$ applied to j , giving $\text{pr}_{R'}^\sharp(jr)$. Their product is $\Phi_V^{R'}(s \otimes jr) = \Phi_V^{R'}((\text{id} \otimes j)(s \otimes r))$. This is the algebra-variable mirror of the open-variable restriction seam (Theorem ??). $\square$

Theorem 956 (Diagonal triviality of the comparison unit). *Let C be proper, geometrically irreducible and geometrically reduced. The pullback of the per-piece comparison unit v_V along the diagonal $\Delta = (C \triangleleft \text{Spec } \nabla): X_B \rightarrow X_{B \otimes_A B}$ (∇ the multiplication of $B \otimes_A B$) is 1.*

Proof. Work locally on the diagonal pullback of the comparison cover. The multiplication retracts both coprojections, $\Delta \circ u_1 = \text{id} = \Delta \circ u_2$ whiskered (Theorem ??), so the pullback expansion of v_V (Theorem ??, with $\varphi = \Delta$ and $m_1 = m_2 = \text{id}$) gives the diagonal witness value $\Delta^\sharp \theta$ times two trivialization factors along the *same* insertion, which cancel. The remaining witness factor $\Delta^\sharp \theta = 1$ is the landed diagonal triviality of the σ -normalized comparison (Theorem ??). So the local value is 1; by sheaf separation the diagonal pullback of v_V is 1. This is the cochain form of the diagonal normalization in Kleiman's descent-cocycle step. $\square$

Theorem 957 (Amitsur coherence of the comparison unit). *Let C be proper, geometrically irreducible and geometrically reduced. The three coface pullbacks of v_V to the triple piece $\Gamma(X_{B \otimes_A (B \otimes_A B)}, \text{cgcb}^{-1}V)^\times$ satisfy the Amitsur cocycle identity on the nose:*

$$(w_{23}^\sharp v_V)(w_{12}^\sharp v_V) = w_{13}^\sharp v_V.$$

Proof. Work locally on the trimmed Amitsur product cover of $X_{B \otimes_A (B \otimes_A B)}$ (Definition ??). By the pullback expansion (Theorem ??) each coface pullback $w_{ij}^\sharp v_V$ is the coface pullback of the witness θ times two insertion pullbacks of the trivialization; the three tensor-slot insertions v_1, v_2, v_3 are common to numerator and denominator, so the trivialization factors telescope, $(T_{23} t_c t_b^{-1})(T_{12} t_b t_a^{-1}) = T_{23} T_{12} t_c t_a^{-1}$. The landed coherence (Theorem ??) supplies $(w_{23}^\sharp \theta)(w_{12}^\sharp \theta) = w_{13}^\sharp \theta$, so the two sides agree locally; by sheaf separation the identity holds globally. This is Kleiman's descent-cocycle identity $v_{13}^{-1} v_{23} v_{12} = 1$. $\square$

Theorem 958 (The transported comparison unit and its descent hypotheses). *Let C be proper, geometrically irreducible and geometrically reduced. Through the section-ring identification $\Gamma(X_{B \otimes_A B}, \text{cgq}^{-1}V) \cong \Gamma(X_A, V) \otimes_A (B \otimes_A B)$ (Definition ??), the comparison unit v_V is a unit $\tilde{v}_V \in (\Gamma(X_A, V) \otimes_A (B \otimes_A B))^\times$ of the descent-cocycle ring. It is diagonal-normalized and Amitsur-coherent through the maps of the descent 1-cocycle:*

$$(\text{id} \otimes \mu)(\tilde{v}_V) = 1, \quad (\text{id} \otimes \partial_{23})(\tilde{v}_V)(\text{id} \otimes \partial_{12})(\tilde{v}_V) = (\text{id} \otimes \partial_{13})(\tilde{v}_V).$$

Proof. The whiskered maps Δ and w_{ij} are the whiskerings of Spec of μ and the three cofaces ∂_{ij} ; so by the algebra-naturality of the identification (Theorem ??) the map $\text{id} \otimes \mu$ applied to $\tilde{v}_V$ equals the transport of the diagonal pullback $\Delta^\sharp v_V = 1$ (Theorem ??), and likewise the three $\text{id} \otimes \partial_{ij}$ applied to $\tilde{v}_V$ are the transports of the three coface pullbacks of v_V . The cocycle identity of the coface pullbacks (Theorem ??) therefore transports to the stated identity of the three $\text{id} \otimes \partial_{ij}$ images (Definition ??). $\square$

Theorem 959 (The descended per-piece class and its cover pullback). *Assume in addition $A \rightarrow B$ is faithfully flat. The descended per-piece class*

$$M_V \in \text{Pic}(\Gamma(X_A, V))$$

is the Picard class of the descent 1-cocycle of the transported comparison unit $\tilde{v}_V$, whose diagonal-normalization and coherence are those of Theorem ??. *Its pullback to the cover-piece ring $\Gamma(X_A, V) \otimes_A B \cong \Gamma(X_B, \text{cg}^{-1}V)$ is trivial.*

Proof. Faithful flatness of $A \rightarrow B$ passes to $\Gamma(X_A, V) \rightarrow \Gamma(X_A, V) \otimes_A B$ by base change, so the coherent comparison unit $\tilde{v}_V$ (Theorem ??) is a descent 1-cocycle (Theorem ??) with a well-defined Picard class M_V (Definition ??); the same definition records that its base change to the cover-piece ring is the trivial class. $\square$

Theorem 960 (The piece restriction of the descending class is trivial). *The class $L = [\gamma]$ of the (C2) setting, restricted to the open subscheme $\text{cg}^{-1}V$, is trivial: $\iota^*L = 1$ for the open immersion $\iota: \text{cg}^{-1}V \hookrightarrow X_B$. Together with the cover-pullback triviality of M_V (Theorem ??), this is the recovery identity $\text{cg}^*M_V = [L|_{\text{cg}^{-1}V}]$, both sides trivial by construction.*

Proof. Over abstract schemes, if $f^{-1}(DZ) = \top$ and a 0-cochain t trivializes the pair values of γ on the DZ -trimmed opens, then the f -pullbacks of t cobound the pulled cocycle $f^\# \gamma$ against 1, so $f^\# \gamma$ is cohomologous to 1. Apply this to the open immersion ι of the piece $\text{cg}^{-1}V$, whose self-preimage is $\top$, with $t = T.\text{triv}$ the piece trivialization (Definition ??): the pulled cocycle is cohomologous to 1, hence its class is trivial (??). $\square$

Theorem 961 (Base-change naturality of the descent extraction). *Let $S \rightarrow S'$ be a map of A -algebras and v a coherent comparison unit over S . The class descended over S' from the base-changed comparison $(\text{algebraMap} \otimes \text{id})(v)$ is the Pic-image of the class descended over S :*

$$\text{comparisonDescentClass}_{S'}((\text{algebraMap} \otimes \text{id})(v)) = \text{Pic.mapAlgebra}_{S \rightarrow S'}(\text{comparisonDescentClass}_S(v)).$$

Proof. First, the comparison descent unit of the base-changed comparison is the image of the descent unit of v under the tensor-square base change along $S \rightarrow S'$, identified through the base-change cancellation on each tensor factor: this is an elementwise computation on pure tensors, using the descent-cocycle ring identification (Definition ??) and the tensor-square base change (Definition ??). Now chain the descent extraction (Theorem ??): the S' -class is the class of the base-changed descent cocycle further mapped by the cancellation (picClass is invariant under the mapping and under replacing a cocycle by a cohomologous one), which equals the class of the base-changed cocycle (Theorem ??), which in turn is the mapAlgebra -image of the S -class (Theorem ??). $\square$

Theorem 962 (Restriction canonicity of the descended per-piece classes). *Assume $A \rightarrow B$ faithfully flat and C proper, geometrically irreducible and geometrically reduced. For affine pieces $V' \leq V$ of X_A and a piece trivialization T on V : the trivialization restricts to one on V' , the glued comparison unit of the restricted datum is the restriction of the glued unit, and the descended classes are natural in the piece,*

$$M_{V'} = \text{Pic.mapAlgebra}_{\Gamma(X_A, V) \rightarrow \Gamma(X_A, V')}(M_V).$$

Proof. The trivialization restricts by restricting each trimmed cochain value along $\text{cg}^{-1}V' \leq \text{cg}^{-1}V$, preserving the trivializing relation. Its comparison unit is the restriction of v_V by uniqueness of gluing (Theorem ??): the restriction of v_V has the defining restriction property of the restricted datum. Through the section-ring identifications, this restriction is $\text{algebraMap} \otimes \text{id}$ on the descent-cocycle ring, by the open-variable restriction seam (Theorem ??) applied to the transported unit (Theorem ??). Hence $\tilde{v}_{V'}$ is the base change of $\tilde{v}_V$ along $\Gamma(X_A, V) \rightarrow \Gamma(X_A, V')$, and $M_{V'}$ is the class of the base-changed unit, which by the base-change naturality of the extraction (Theorem ??) is $\text{mapAlgebra}(M_V)$. On an overlap of two pieces both descend from the same global σ -normalized comparison, so the descended classes agree through their common restriction: this is what the refinement-splice consumes. $\square$

7.48 Trivializing an invertible module away from finitely many primes

The pure-algebra core of the (C2) piece-selection moving lemma. An invertible module over a commutative ring becomes free after inverting a single element chosen to avoid any prescribed

finite set of primes; and, for an integral extension, an element of the extension avoiding every prime over a base prime divides the image of a base element avoiding that prime — the *algebraic tube*. These are the two bricks that convert triviality of a class on a basic open of the cover into a basic open of the base.

Lemma 963 (A cyclic invertible module is free). *Let R be a commutative ring and N an invertible R -module generated by a single element m_0 , i.e. $R \cdot m_0 = N$. Then the scalar map $r \mapsto r \cdot m_0$ is an isomorphism $R \xrightarrow{\sim} N$; in particular N is free of rank one.*

Proof. Surjectivity is the hypothesis $R \cdot m_0 = N$. For injectivity, suppose $r \cdot m_0 = 0$. Since m_0 generates N , r annihilates every element of N , hence $r \cdot \text{id}_N = 0$ as an endomorphism, and therefore r annihilates every element of $N^\vee \otimes_R N$ (the endomorphism acts through the right tensor factor). Invertibility of N means the contraction pairing $N^\vee \otimes_R N \rightarrow R$ is surjective; choose w with $\text{contract}(w) = 1$. Then $r = r \cdot \text{contract}(w) = \text{contract}(r \cdot w) = \text{contract}(0) = 0$. So the scalar map is bijective, whence $N \cong R$ is free. $\square$

Theorem 964 (The avoidance brick). *Let R be a commutative ring, N an invertible R -module, and $(\mathfrak{q}_i)_{i \in I}$ a finite family of prime ideals of R . Then there is an element $f \in R$ with $f \notin \mathfrak{q}_i$ for every i such that, for every R -algebra S in which f becomes a unit, the base change $S \otimes_R N$ is a free S -module.*

Proof. If $I = \emptyset$ there are no primes to avoid and the claim holds trivially; assume I is nonempty. Localize R at the multiplicative set $S_0 := \bigcap_i (R \setminus \mathfrak{q}_i)$, the complement of $\bigcup_i \mathfrak{q}_i$. Every maximal ideal of R_{S_0} is the localization of one of the $\mathfrak{q}_i$: its contraction to R is a prime disjoint from S_0 , hence contained in $\bigcup_i \mathfrak{q}_i$, hence — by prime avoidance — inside some $\mathfrak{q}_i$, and maximality forces equality. Thus R_{S_0} is semilocal with finitely many maximal ideals, so its Picard group is trivial, and therefore $S_0^{-1}N$ is a *cyclic* R_{S_0} -module, generated by a single element z .

Fix finitely many generators $g_1, \dots, g_n$ of N over R , and write $z = m_0/s_0$ with $m_0 \in N$, $s_0 \in S_0$. Clearing denominators in the relations expressing each g_j as an R_{S_0} -multiple of z yields elements $d_j \in S_0$ and $r_j \in R$ with $d_j \cdot g_j = r_j \cdot m_0$ in N . Set $f := \prod_j d_j$; each $d_j \in S_0$ avoids every $\mathfrak{q}_i$, and a prime containing a product contains a factor, so f avoids every $\mathfrak{q}_i$.

Now let S be an R -algebra in which f is a unit. In $S \otimes_R N$ each $1 \otimes g_j$ lies in the S -span of $1 \otimes m_0$: from $(\prod_l d_l) \cdot g_j = (\prod_{l \neq j} d_l) r_j \cdot m_0$, inverting the unit $f = \prod_l d_l$ expresses $1 \otimes g_j$ as an S -multiple of $1 \otimes m_0$. Since the g_j generate N , every pure tensor $1 \otimes x$ — and hence all of $S \otimes_R N$ — lies in the S -span of $1 \otimes m_0$. So $S \otimes_R N$ is cyclic, and invertible because invertibility is preserved by base change; by Lemma ?? it is free. $\square$

Lemma 965 (The algebraic tube). *Let $S \rightarrow R$ be an integral extension of commutative rings, $\mathfrak{p}$ a prime of S , and $f \in R$ an element avoiding every prime of R lying over $\mathfrak{p}$ (that is, $f \notin Q$ whenever Q is prime with $Q \cap S = \mathfrak{p}$). Then there are $g \in S \setminus \mathfrak{p}$ and $h \in R$ with $g = f \cdot h$ in R .*

Proof. Consider the contracted ideal $(fR) \cap S$ of S . Were it contained in $\mathfrak{p}$, going-up for the integral extension $S \rightarrow R$ would produce a prime $Q \supseteq fR$ of R with $Q \cap S = \mathfrak{p}$; but then $f \in Q$ is a prime over $\mathfrak{p}$ containing f , contradicting the hypothesis. Hence $(fR) \cap S \not\subseteq \mathfrak{p}$: some $g \in (fR) \cap S$ lies outside $\mathfrak{p}$. By construction $g \in fR$, so $g = f \cdot h$ for some $h \in R$. $\square$

7.49 The moving lemma: pieces on which the class trivializes

We now select, for a *module-finite* faithfully flat cover $A \rightarrow B$, the affine pieces of the base curve X_A on which a prescribed Čech Picard class of the cover curve X_B becomes trivial after pullback

to the cover piece. The route is pure commutative algebra — no integrality of the curve, no divisors — through the affine-open class and the two bricks of the previous section. Throughout, $\text{cg}: X_B \rightarrow X_A$ is the cover inclusion.

Definition 966 (The affine-open Čech–Picard class). Let Z be a scheme and $O \subseteq Z$ an affine open. Pullback along the open immersion $O \hookrightarrow Z$ identifies the ambient sections $\Gamma(Z, O)$ with the global sections $\Gamma(O, \mathbb{T})$ of the open subscheme; write $\iota_O^\top$ for this ring isomorphism. The *affine-open class* of a Čech Picard class L on Z is

$$L\langle O \rangle := (\iota_O^\top)^{-1} \text{toPic}_O(\iota^* L) \in \text{Pic}(\Gamma(Z, O)),$$

the invertible $\Gamma(Z, O)$ -module obtained by transporting, through the affine Čech–Picard dictionary toPic (Definition ??), the restriction of L to O . It is natural in the open: for affine opens $O' \leq O$ the class $L\langle O' \rangle$ is the image of $L\langle O \rangle$ under the restriction homomorphism $\text{Pic}(\Gamma(Z, O)) \rightarrow \text{Pic}(\Gamma(Z, O'))$.

Proof. The section identification $\iota_O^\top$ is an isomorphism because $O \hookrightarrow Z$ is an open immersion with self-preimage O , so its pullback on sections over O is invertible; transporting the affine class $\text{toPic}_O(\iota^* L)$ back along it gives a well-defined element of $\text{Pic}(\Gamma(Z, O))$. Naturality is the commuting square of the two open immersions $O' \leq O \leq Z$: the restriction map of ambient sections and the two section identifications intertwine the toPic -images by naturality of the dictionary in the scheme (Theorem ??), so the class formed on O' is the restriction of the class formed on O . $\square$

Theorem 967 (Triviality detection for the affine-open class). *Let Z be a scheme and L a Čech Picard class on Z .*

1. *If the affine-open class $L\langle O \rangle$ is trivial for an affine open O , then the restriction $\iota^* L$ of L to O is trivial.*
2. *Restriction triviality is monotone: if $\iota_O^* L = 1$ and $O' \leq O$, then $\iota_{O'}^* L = 1$.*
3. *If, for an affine open O and $f \in \Gamma(Z, O)$, the invertible module $L\langle O \rangle$ becomes free over every $\Gamma(Z, O)$ -algebra in which f is a unit, then the affine-open class on the basic open $D(f)$ is trivial, $L\langle D(f) \rangle = 1$; hence L restricts trivially to $D(f)$.*

Proof. (1) The dictionary toPic is injective (Theorem ??), so $\iota^* L = 1$ follows from $\text{toPic}_O(\iota^* L) = 1$; the latter is the transport of $L\langle O \rangle = 1$ through the section identification, which reflects triviality because it has a two-sided inverse. (2) Restriction along $O' \leq O$ factors the restriction to O' through the restriction to O , and a group homomorphism sends 1 to 1. (3) The restriction homomorphism $\Gamma(Z, O) \rightarrow \Gamma(Z, D(f))$ inverts f , because f is a unit on its own basic open; so by hypothesis $L\langle O \rangle$ base-changes to a free, hence trivial, class over $\Gamma(Z, D(f))$, and by naturality of the affine-open class (Definition ??) this base change is $L\langle D(f) \rangle$, which is therefore 1. Triviality of L restricted to $D(f)$ then follows from (1). $\square$

Lemma 968 (Module-finiteness of the cover piece over the base piece). *Let $A \rightarrow B$ be a module-finite homomorphism of k -algebras and $V \subseteq X_A$ an affine open. Then the cover-piece ring $\Gamma(X_B, \text{cg}^{-1}V)$ is a module-finite $\Gamma(X_A, V)$ -algebra.*

Proof. By the piece identification (Definition ??) the cover-piece ring is $\Gamma(X_A, V) \otimes_A B$ as a $\Gamma(X_A, V)$ -algebra. Base change preserves module-finiteness: as B is module-finite over A , the base change $\Gamma(X_A, V) \otimes_A B$ is module-finite over $\Gamma(X_A, V)$. $\square$

Theorem 969 (The moving lemma). *Let $A \rightarrow B$ be module-finite. For every Čech Picard class L on the cover curve X_B and every point $x \in X_A$, there is an affine open $V \ni x$ of the base curve such that L restricts trivially to the open subscheme $\text{cg}^{-1}V$ of X_B .*

Proof. Choose a cg -saturated affine open $V_0 \ni x$ of X_A (Theorem ??); the cover piece $\text{cg}^{-1}V_0$ is then affine with ring $N := \Gamma(X_B, \text{cg}^{-1}V_0)$, and $L\langle \text{cg}^{-1}V_0 \rangle$ (Definition ??) is an invertible N -module. By module-finiteness (Lemma ??) the piece map $\Gamma(X_A, V_0) \rightarrow N$ is module-finite, hence integral and quasi-finite; therefore the fibre of $\text{Spec } N \rightarrow \text{Spec } \Gamma(X_A, V_0)$ over the germ prime $\mathfrak{p}_x$ of x is a *finite* set of primes $\mathfrak{q}_1, \dots, \mathfrak{q}_m$ of N .

Apply the avoidance brick (Theorem ??) to the invertible module $L\langle \text{cg}^{-1}V_0 \rangle$ and the finite family $(\mathfrak{q}_j)$: there is $f \in N$ with $f \notin \mathfrak{q}_j$ for all j such that $L\langle \text{cg}^{-1}V_0 \rangle$ becomes free wherever f is inverted. Since f avoids every prime of N over $\mathfrak{p}_x$, the algebraic tube (Lemma ??) for the integral extension $\Gamma(X_A, V_0) \rightarrow N$ produces $g \notin \mathfrak{p}_x$ and $h \in N$ with $\text{cg}^\sharp g = f \cdot h$. Set $V := D(g) \subseteq X_A$. Then $x \in V$, because $g \notin \mathfrak{p}_x$ means the germ of g at x is a unit, and

$$\text{cg}^{-1}V = D(\text{cg}^\sharp g) = D(f \cdot h) \leq D(f).$$

Freeness of $L\langle \text{cg}^{-1}V_0 \rangle$ after inverting f trivializes the class on $D(f)$ (Theorem ??(3)), and restriction triviality is monotone (Theorem ??(2)), so L restricts trivially to $\text{cg}^{-1}V \leq D(f)$.

Module-finiteness is necessary. For a merely quasi-finite (for instance étale) non-finite cover the conclusion fails. There the fibre of the piece map over $\mathfrak{p}_x$ can be infinite, and the bare restricted class need not become trivial on any cg -saturated neighbourhood of the fibre: take $B = A_u \times A_v$, a product of two localizations of a two-dimensional normal local ring A with non-trivial class group, and a class whose restrictions to the two factors are non-trivial classes of the two localizations; no single saturated piece trivializes it. The general faithfully-flat effectivity therefore genuinely needs the per-piece descent datum of Section ?? rather than bare piece-triviality, and survives with piece-selection taken as a hypothesis (Theorem ??). $\square$

7.50 From subscheme triviality to a trimmed trivializing cochain

The moving lemma delivers triviality of the class *as a class* on the cover piece. The per-piece descent of Section ?? instead consumes a trivializing *cochain* of the representing cocycle, trimmed to the piece and indexed by all ambient points. This section converts the one to the other.

Theorem 970 (The trimmed trivializing cochain of a class trivial on an open subscheme). *Let Z be a scheme, $\mathcal{N}$ a pointed cover of Z , γ a unit 1-cocycle on $\mathcal{N}$, and $D \subseteq Z$ an open subscheme. If the class $[\gamma]$ restricts trivially to D (that is $\iota_D^*[\gamma] = 1$), then there is a family of units*

$$t_b \in \Gamma(Z, \mathcal{N}(b) \cap D)^\times, \quad b \in Z,$$

indexed by all ambient points, satisfying the one-sided trivializing relation $t_b \cdot \gamma_{bb'} = t_{b'}$ on the D -trimmed pairwise overlaps.

Proof. Triviality of the restricted class means the pulled-back cocycle $\iota_D^\sharp \gamma$ on the cover $\iota_D^{-1}\mathcal{N}$ of D is cohomologous to 1 (Corollary ??); as restriction to an open subscheme needs no further refinement, this yields a trivializing 0-cochain α , indexed by points p of D , with $\alpha_p \cdot (\iota_D^\sharp \gamma)_{pp'} = \alpha_{p'}$. Transport each α_p across the open immersion ι_D : on the preimage of the trimmed member $\mathcal{N}(\iota_D p) \cap D$ the unit α_p is the image of an ambient unit (Definition ??, unit sections across an open immersion), giving units t'_p on $\mathcal{N}(\iota_D p) \cap D$ that satisfy the ambient trivializing relation

on subscheme pairs. Finally re-index from points of D to points of Z : for each ambient b the trimmed member $\mathcal{N}(b) \cap D$ is covered by the domains of the t'_p , the values $t'_p \cdot \gamma_{(\iota_{Dp})b}$ agree on overlaps by the cocycle identity, so they glue (Lemma ??) to a unit t_b on $\mathcal{N}(b) \cap D$. The glued family satisfies $t_b \cdot \gamma_{bb'} = t_{b'}$, again by the cocycle identity together with the local relations. $\square$

Theorem 971 (A piece trivialization from restriction triviality). *Let γ be a representing unit cocycle on a pointed cover $\mathcal{N}$ of X_B and $V \subseteq X_A$ a piece. If the class $[\gamma]$ restricts trivially to the cover piece $\text{cg}^{-1}V$, then V carries a piece trivialization of γ (Definition ??).*

Proof. Instantiate Theorem ?? at the open subscheme $D = \text{cg}^{-1}V$: the trimmed trivializing cochain it produces is precisely the field data of a piece trivialization (Definition ??). $\square$

7.51 Twisting a piece trivialization

Two piece trivializations of the same representing cocycle on the same piece differ, member by member, by the restrictions of a single global unit of the cover piece (this is proved in the next section). Here we record how the per-piece comparison unit responds to such a twist, and the algebraic and geometric seams that identify the twist with a descent coboundary — the mechanism behind both the trivialization-independence of the descended class and the flattening of a trivialization.

Theorem 972 (Twisting a piece trivialization by a global unit). *Let T be a piece trivialization of γ on V and $e \in \Gamma(X_B, \text{cg}^{-1}V)^\times$ a global unit of the cover piece. The twist $T \cdot e$ is the piece trivialization with members $t_b \cdot e|_{\mathcal{N}(b) \cap \text{cg}^{-1}V}$. Write the geometric descent coboundary of e ,*

$$\partial e := \frac{u_2^\sharp e}{u_1^\sharp e} \in \Gamma(X_{B \otimes_A B}, \text{cg}^{-1}V)^\times,$$

the ratio of its two coprojection pullbacks to the double piece. If two piece trivializations T, T' differ member by member by e , i.e. $T' = T \cdot e$, then for any σ -normalized comparison W their per-piece comparison units (Theorem ??) differ by ∂e :

$$v_V(T') = v_V(T) \cdot \partial e.$$

Proof. Multiplying every member of T by the restrictions of the single global unit e preserves the trivializing relation $t_b \cdot \gamma_{bb'} = t_{b'}$: the two e -factors coincide on the overlap. For the comparison units, the per-piece comparison unit is characterized by its restriction to the trimmed comparison cover (Theorem ??), where it restricts to $\theta_x \cdot (u_2^\sharp T)(x) / (u_1^\sharp T)(x)$. Replacing T by $T \cdot e$ multiplies this local value by $(u_2^\sharp e) / (u_1^\sharp e)$ at every point; by uniqueness of the glued unit this multiplies $v_V(T)$ by the global unit ∂e with that restriction. $\square$

Theorem 973 (The coboundary seams of the identification). *Fix an affine piece V and $e \in \Gamma(X_B, \text{cg}^{-1}V)^\times$; let $\beta \in (\Gamma(X_A, V) \otimes_A B)^\times$ be its image under the piece identification (Definition ??). Then:*

1. (algebra seam) *the comparison descent unit of the base-changed coboundary of β — the ratio of the two inclusions $\beta \mapsto \beta \otimes 1$, $\beta \mapsto 1 \otimes \beta$ of $B \rightarrow B \otimes_A B$, base changed to $\Gamma(X_A, V)$ — equals the descent coboundary of β relative to $\Gamma(X_A, V) \rightarrow \Gamma(X_A, V) \otimes_A B$;*
2. (geometric seam) *through the piece identification, the geometric descent coboundary ∂e (Theorem ??) is carried to the base-changed coboundary of β .*

Proof. (1) Transport both sides through the descent-cocycle ring identification (Definition ??) to $(\Gamma(X_A, V) \otimes_A (B \otimes_A B))^\times$: there the identification carries a left pure tensor to the base change of the first inclusion and a right pure tensor to that of the second (checked on pure tensors, extended by bilinearity), so the comparison descent unit of the inclusion ratio (Theorem ??) is the descent coboundary of β . (2) The two coprojections u_1, u_2 are the whiskerings of the two inclusions $B \rightarrow B \otimes_A B$, so by the algebra-naturality of the section-ring identification (Theorem ??) each pullback $u_i^* e$ is carried to the corresponding inclusion applied to β , and their ratio is the base-changed coboundary of β . $\square$

7.52 Trivialization independence and flat trivializations

We deduce the (B) deliverable of the (C2) close — the descended per-piece class is independent of the chosen trivialization — and its constructive upgrade: on a piece whose descended class is trivial some trivialization is *flat*, with per-piece comparison unit 1 on the nose. The flat pieces produced here are what the refinement splice glues.

Theorem 974 (Two piece trivializations differ by a glued global unit). *Any two piece trivializations T, T' of the same cocycle γ on the same piece V differ by a single global unit of the cover piece: there is $e \in \Gamma(X_B, \text{cg}^{-1}V)^\times$ with $T' = T \cdot e$, i.e. $t'_b = t_b \cdot e|_{\mathcal{N}(b) \cap \text{cg}^{-1}V}$ for every b .*

Proof. The member-wise ratio t'_b/t_b , a unit on the trimmed member $\mathcal{N}(b) \cap \text{cg}^{-1}V$, is Čech-compatible: the two trivializing relations $t_b \cdot \gamma_{bb'} = t_{b'}$ and $t'_b \cdot \gamma_{bb'} = t'_{b'}$ cancel the cocycle γ , so t'_b/t_b and $t'_{b'}/t_{b'}$ agree on the overlap. The trimmed members cover $\text{cg}^{-1}V$, so the ratios glue (Lemma ??) to a unit e with the stated restrictions. $\square$

Theorem 975 (Trivialization independence of the descended per-piece class). *Let C be proper, geometrically irreducible and geometrically reduced, and $A \rightarrow B$ faithfully flat. For a fixed σ -normalized comparison W and an affine piece V , the descended per-piece class M_V (Theorem ??) does not depend on the chosen piece trivialization.*

Proof. Let T, T' be two piece trivializations; by Theorem ?? they differ by a global unit e , $T' = T \cdot e$. Transported to the descent-cocycle ring, the comparison unit of T' is that of T multiplied by the descent coboundary of β , the piece identification of e : this is the twist lemma (Theorem ??) combined with the two coboundary seams (Theorem ??). The Picard class of a descent 1-cocycle is unchanged when the cocycle is multiplied by a descent coboundary, since a coboundary alters the descent datum by an isomorphism and not the isomorphism class of the descended module (Definition ??). Hence $M_V(T') = M_V(T)$. $\square$

Theorem 976 (Flattening a trivialization along a trivial descended class). *Let C be proper, geometrically irreducible and geometrically reduced, $A \rightarrow B$ faithfully flat, W a σ -normalized comparison, and T a piece trivialization on V whose descended class $M_V(T)$ is trivial. Then V carries a piece trivialization whose per-piece comparison unit is 1 on the nose.*

Proof. Triviality of $M_V(T)$ means the descent 1-cocycle of the transported comparison unit cobounds: there is $\beta \in (\Gamma(X_A, V) \otimes_A B)^\times$ whose descent coboundary is that cocycle (Definition ??: the class is 1 exactly when the descended module is free). Pull β back through the piece identification to a global unit $e \in \Gamma(X_B, \text{cg}^{-1}V)^\times$. By the coboundary seams (Theorem ??) the per-piece comparison unit of T is the geometric descent coboundary ∂e . Twisting T by e^{-1} multiplies the comparison unit by $\partial(e^{-1}) = (\partial e)^{-1}$ (Theorem ??), giving comparison unit $v_V(T) \cdot (\partial e)^{-1} = 1$. $\square$

Theorem 977 (Localize and flatten). *Let C be proper, geometrically irreducible and geometrically reduced, $A \rightarrow B$ faithfully flat, and W a σ -normalized comparison. Every point x of a trivialized affine piece (V_1, T_1) lies in a basic affine subpiece $V \leq V_1$ carrying a flat piece trivialization — one whose per-piece comparison unit is 1.*

Proof. Apply the avoidance brick (Theorem ??) to the invertible module $M_{V_1}(T_1)$ and the single germ prime $\mathfrak{p}_x$ of x on $\Gamma(X_A, V_1)$: it yields $f \notin \mathfrak{p}_x$ such that $M_{V_1}(T_1)$ becomes free wherever f is inverted. Set $V := D(f) \subseteq X_A$; then $x \in V$ (the germ of f at x is a unit) and $V \leq V_1$. Restrict T_1 to V ; by restriction canonicity of the descended classes (Theorem ??) the restricted descended class is the base change of $M_{V_1}(T_1)$ along $\Gamma(X_A, V_1) \rightarrow \Gamma(X_A, V)$, which inverts f and so is free, hence trivial. Flattening (Theorem ??) then produces a piece trivialization on V with comparison unit 1. $\square$

7.53 The refinement splice

The assembly of the descended class from a covering family of flat pieces, run in the constructive direction of the (C1) kernel lemma's final assembly.

Theorem 978 (The refinement splice). *Let C be proper, geometrically irreducible and geometrically reduced, and $A \rightarrow B$ faithfully flat. Suppose every point of X_A lies in an affine piece carrying a flat piece trivialization of γ (per-piece comparison unit 1). Then the class $L = [\gamma]$ descends along the cover inclusion: there is a Čech Picard class M on X_A with $\text{cg}^*M = L$.*

Proof. Fix a covering family (V_x, T_x) of flat trivialized pieces. On the overlap $V_x \cap V_y$ the two restricted trivializations differ by a glued unit c_{xy} of the overlap cover piece (Theorem ??); both restrictions are still flat, so the twist lemma (Theorem ??) forces the geometric descent coboundary of c_{xy} to be 1 — the two coprojection pullbacks of c_{xy} to the double piece agree on the nose. Faithfully flat descent of unit sections along cg (Theorem ??) therefore descends each c_{xy} to a unit m_{xy} of the affine overlap $V_x \cap V_y$ downstairs, the overlaps being affine by separatedness of the curve product (Theorem ??). The m_{xy} satisfy the Čech cocycle identity: their cg -pullbacks telescope through the trivializations, and cg -pullback is injective on sections (Theorem ??). Let M be the class of the resulting unit cocycle m on the pointed cover (V_x) . Then $\text{cg}^*M = L$ by mk-calculus over the common refinement of the cg -pullback of (V_x) and $\mathcal{N}$: the trivializations T_x themselves cobound the two pulled cocycles (Corollary ??). $\square$

7.54 The (C2) effectivity theorem and the surjectivity close

The campaign close: the effectivity theorem in both forms, and the resulting bijectivity of the unit of the étale plus construction over a section-admitting field test.

Theorem 979 (Effectivity from a covering family of trivialized pieces). *Let C be proper, geometrically irreducible and geometrically reduced, and $A \rightarrow B$ faithfully flat. Let γ be a representing cocycle on X_B whose class $L = [\gamma]$ is rigidified along the base-changed section (Definition ??) and satisfies the on-the-nose descent equation $u_1^*L = u_2^*L$. If every point of X_A lies in an affine piece on which γ admits a piece trivialization, then L descends along the cover inclusion: $\exists M, \text{cg}^*M = L$.*

Proof. The descent equation together with rigidification produces a σ -normalized comparison W of γ , with no further geometric hypotheses (Theorem ??). At each point, localize-and-flatten (Theorem ??) turns the given trivialized piece into a flat one through the point. The resulting covering family of flat pieces feeds the refinement splice (Theorem ??), which descends L . $\square$

Theorem 980 (The (C2) effectivity theorem). *Let C be proper, geometrically irreducible and geometrically reduced, and $A \rightarrow B$ a module-finite faithfully flat cover. Every Čech Picard class L on the cover curve X_B that is rigidified along the base-changed section and satisfies the on-the-nose descent equation $u_1^*L = u_2^*L$ descends along the cover inclusion: $\exists M, \text{cg}^*M = L$.*

Proof. Represent $L = [\gamma]$. Because the cover is module-finite, the moving lemma (Theorem ??) applies: every point of X_A has an affine neighbourhood on which L restricts trivially to the cover piece, and such a restriction triviality furnishes a piece trivialization (Theorem ??). The covering family of trivialized pieces feeds the general effectivity Theorem ?. This is the cocycle recast of the descent step in Kleiman's Comparison Theorem, Part 2: the two projection pullbacks of the rigidified pair agree, so the automorphism $v_{13}^{-1}v_{23}v_{12}$ of its first-projection pullback is trivial and the pair descends along the cover. $\square$

Theorem 981 (Surjectivity of the unit over a section-admitting field test). *Let C be a curve over k that is proper, geometrically irreducible and geometrically reduced, and let K be a field extension of k admitting a K -point $\sigma: \text{Spec } K \rightarrow C$. Then the unit $\eta_K: \text{relPic}(C, \text{Spec } K) \rightarrow \text{PicEtAff}(C, K)$ of the étale plus construction is surjective.*

Proof. A plus class is represented on an étale cover E of $\text{Spec } K$ (Lemma ??). Étale covers of a field are cofinal in finite separable field extensions (Theorem ??): refine E to a finite separable field extension L/K , giving a module-finite — hence faithfully flat — cover $F = \text{Spec } L$ and a descent class y over F representing the same plus class (Lemma ??). The rigidification layer produces a rigidified Čech representative L_c of y (Definition ??) whose two double-cover pullbacks agree on the nose. As $K \rightarrow L$ is module-finite faithfully flat, the effectivity theorem (Theorem ??) descends L_c along the cover inclusion to a class M on X_K with $\text{cg}^*M = L_c$. The relative Picard class $[M]$ maps under η_K to the original plus class, by the identification of the unit with the tautological mk (Theorem ??) and mk -calculus for the algebra map $K \rightarrow L$ (Definition ??, Definition ??). Hence η_K is surjective. This is the surjectivity direction of Kleiman's Comparison Theorem Part 2 for a base admitting a section — over a field test the étale-local section is an actual point. $\square$

Theorem 982 (The unit is an isomorphism over a section-admitting field test). *Under the hypotheses of Theorem ??, the unit is a group isomorphism*

$$\eta_K: \text{relPic}(C, \text{Spec } K) \xrightarrow{\sim} \text{PicEtAff}(C, K).$$

Proof. The unit is injective by étale separatedness (the (C1) close, Theorem ??) and surjective by Theorem ??; a bijective group homomorphism is an isomorphism. $\square$

7.55 Divisor classes from local equations

An effective Cartier divisor D on a scheme X is cut out, near each point, by a single regular equation. Adapted to the pointed covers of the Čech Picard group $\text{Pic}(X)$ recalled earlier in this chapter, this data — one regular section f_x on each member of a pointed cover, the sections agreeing up to a unit on overlaps — is a trivialization of the invertible sheaf $\mathcal{O}_X(D)$: on the member $\mathcal{U}(x)$ the sheaf is trivialized by multiplication by f_x . The transition units $g_{xy} = f_x/f_y \in \mathcal{O}_X(\mathcal{U}(x) \cap \mathcal{U}(y))^\times$ form a unit Čech 1-cocycle, and its class in $\text{Pic}(X)$ is $\mathcal{O}_X(D)$. This section records that construction and its independence of the chosen equations up to units; the next builds from it the class of a Weil divisor on a curve.

Definition 983 (Local equations for an effective divisor). A system of *local equations* on X , adapted to a pointed cover $\mathcal{U}$, consists of a section $f_x \in \mathcal{O}_X(\mathcal{U}(x))$ on each member, subject to two conditions:

- (regularity) the germ of f_x at every point $y \in \mathcal{U}(x)$ is a nonzerodivisor in the stalk $\mathcal{O}_{X,y}$;
- (overlap) for every pair x, y the restrictions of f_x and f_y to the overlap $\mathcal{U}(x) \cap \mathcal{U}(y)$ differ by a unit factor: there is a unit $u \in \mathcal{O}_X(\mathcal{U}(x) \cap \mathcal{U}(y))^\times$ with $f_x|_\cap = u \cdot f_y|_\cap$.

Lemma 984 (The transition units). *For a local-equation system and points x, y there is a unique unit $g_{xy} \in \mathcal{O}_X(\mathcal{U}(x) \cap \mathcal{U}(y))^\times$ with $f_x|_\cap = g_{xy} \cdot f_y|_\cap$ on the overlap; write $g_{xy} = f_x/f_y$ for the transition unit.*

Proof. Existence is the overlap condition of Definition ???. For uniqueness, the restriction of f_y to the overlap is again a nonzerodivisor: a section of a sheaf is determined by its germs, so it is a nonzerodivisor as soon as each of its germs is, and the germs of f_y are nonzerodivisors by regularity. Hence in the equation $f_x|_\cap = u \cdot f_y|_\cap$ the right factor $f_y|_\cap$ can be cancelled on the right, determining the unit u uniquely. $\square$

Lemma 985 (The cocycle identity of the transition units). *On the triple overlap $\mathcal{U}(x) \cap \mathcal{U}(y) \cap \mathcal{U}(z)$ the transition units satisfy $g_{xy} \cdot g_{yz} = g_{xz}$.*

Proof. Restrict everything to the triple overlap and multiply through by f_z . By the defining equation (Lemma ??) applied to the pairs (y, z) and (x, y) ,

$$g_{xy} g_{yz} f_z = g_{xy} f_y = f_x,$$

while applied to (x, z) it gives $g_{xz} f_z = f_x$. The restriction of f_z to the triple overlap is a nonzerodivisor (regularity, as in Lemma ??), so cancelling it yields $g_{xy} g_{yz} = g_{xz}$. $\square$

Definition 986 (The Picard class of a local-equation system). The transition units g_{xy} of a local-equation system d form a unit Čech 1-cocycle on its cover $\mathcal{U}$; its class

$$\text{picClass}(d) = [\mathcal{U}, g] \in \text{Pic}(X)$$

is the *Picard class* $\mathcal{O}_X(D)$ of the effective divisor D cut out by d .

Proof. Each g_{xy} is a unit on the double overlap $\mathcal{U}(x) \cap \mathcal{U}(y)$, and the cocycle identity $g_{xy} g_{yz} = g_{xz}$ holds on triple overlaps (Lemma ??); this is exactly the data of a unit Čech 1-cocycle on $\mathcal{U}$. Take its class in the colimit $\text{Pic}(X)$. $\square$

Lemma 987 (Independence of the class under unit rescaling). *Let d be a local-equation system on $\mathcal{U}$ and $(v_x)_x$ a family of units $v_x \in \mathcal{O}_X(\mathcal{U}(x))^\times$. Rescaling each equation, $f'_x = v_x \cdot f_x$, is again a local-equation system on $\mathcal{U}$, and it has the same Picard class: $\text{picClass}(d \text{ rescaled by } v) = \text{picClass}(d)$. In particular $\mathcal{O}_X(D)$ depends only on the divisor, not on the chosen equations.*

Proof. Each $v_x \cdot f_x$ is regular — at every point its germ is the product of a unit germ and a nonzerodivisor germ, hence a nonzerodivisor — and its overlap ratio is the conjugated unit

$$g'_{xy} = (v_x|_\cap) \cdot g_{xy} \cdot (v_y|_\cap)^{-1},$$

so the rescaled data is a local-equation system on the same cover $\mathcal{U}$ with transition units g' . The rings $\mathcal{O}_X(U)$ are commutative, so their units form an abelian group; hence $g'_{xy} = (v_x|_\cap)(v_y|_\cap)^{-1} g_{xy}$, which exhibits g' as the product of g with the coboundary of the 0-cochain $(v_x)_x$. Cohomologous cocycles on the same cover have the same class in $\text{Pic}(X)$, so picClass is unchanged. $\square$

7.56 Pullback of a local-equation system

A morphism $f: Y \rightarrow X$ pulls a local-equation system on X back to one on Y , provided the pulled-back equations remain regular; on Picard classes this realizes the pullback of the associated invertible sheaf, $f^*\mathcal{O}_X(D) = \mathcal{O}_Y(f^*D)$.

Definition 988 (Pullback of a local-equation system). Let $f: Y \rightarrow X$ be a morphism of schemes and E a local-equation system on X (??), adapted to a pointed cover $\mathcal{U}$. Pull the cover back pointwise: the pulled-back pointed cover $f^*\mathcal{U}$ assigns to $y \in Y$ the open $f^{-1}\mathcal{U}(f(y))$. The *pulled-back equations* are the f -pullbacks $f^*(f_{f(y)}) \in \mathcal{O}_Y(f^{-1}\mathcal{U}(f(y)))$ of the equations of E .

These form a local-equation system f^*E on $f^*\mathcal{U}$, under the additional hypothesis that the pulled-back equations are regular: that the germ of $f^*(f_{f(y)})$ at every point of its member is a nonzerodivisor. This regularity is not automatic — pulling a nonzerodivisor back along a non-flat morphism can produce a zerodivisor, so the naive claim “pull the equations back and stay a local-equation system” is false in general, and the hypothesis is genuinely required. (Flatness of f would be a stronger sufficient condition, since flat pullback preserves nonzerodivisors; it is not assumed here.)

Proof. Regularity of the pulled-back equations is the added hypothesis. For the overlap condition, fix y, y' and restrict the defining equation $f_{f(y)}|_{\cap} = g_{f(y)f(y')} \cdot f_{f(y')}|_{\cap}$ of E (??) along f . Since pullback of sections is a ring homomorphism commuting with restriction, the two pulled-back equations differ on the overlap $f^{-1}\mathcal{U}(f(y)) \cap f^{-1}\mathcal{U}(f(y'))$ by the factor $f^*(g_{f(y)f(y')})$, the f -pullback of the transition unit of E . The pullback of a unit section is again a unit, so this factor is a unit and the overlap condition holds. $\square$

Lemma 989 (Transition data of a pullback). *The transition units of the pulled-back system f^*E are the f -pullbacks of those of E : for all y, y' ,*

$$g_{yy'}^{f^*E} = f^*(g_{f(y)f(y')}).$$

*Consequently the unit Čech 1-cocycle of f^*E is the f -pullback of the cocycle of E .*

Proof. The overlap ratio arising in the construction of f^*E (??) is $f^*(g_{f(y)f(y')})$, and the transition unit of f^*E is the unique unit realizing that ratio (??); the two therefore agree. Assembling over all pairs, the cocycle of f^*E (??) is the entry-by-entry f -pullback of the cocycle of E . $\square$

Theorem 990 (The Picard class of a pullback is the pullback of the class). *For a morphism $f: Y \rightarrow X$ and a local-equation system E on X with regular pullback, the Picard class of f^*E is the pullback of the class of E :*

$$\text{picClass}(f^*E) = f^* \text{picClass}(E) \quad \text{in } \text{Pic}(Y),$$

*that is, $\mathcal{O}_Y(f^*D) = f^*\mathcal{O}_X(D)$.*

Proof. Both sides are represented on the pulled-back cover $f^*\mathcal{U}$. The class of f^*E is the class of its cocycle (??), and that cocycle is the f -pullback of the cocycle of E (??); pullback of Picard classes is, by definition, induced by pulling cocycles back along f . Hence the class of the pulled-back cocycle is the pullback f^* of $\text{picClass}(E)$. $\square$

7.57 The point divisor and the class of a Weil divisor

We return to the curve bundle X of Chapter ??: integral, smooth of relative dimension one and quasi-compact over the field K , with generic point η . To each closed point x we attach the local equations of the effective divisor $1 \cdot x$, whose only zero is a simple one at x ; passing to Picard classes and taking products to the multiplicities sends each Weil divisor $D \in \text{Div}(X)$ to its class $\mathcal{O}_X(D) \in \text{Pic}(X)$, a homomorphism extending the point classes.

Lemma 991 (A uniformizer isolated to a single zero). *For a closed point x of the curve there are an open neighbourhood $V \ni x$ and a section $s \in \mathcal{O}_X(V)$ whose germ at the generic point η is the chosen uniformizer t at x , whose germ at every point $y \in V$ is a nonzerodivisor of $\mathcal{O}_{X,y}$, and is moreover a unit whenever $y \neq x$. Thus s is a uniformizer at x whose only zero on V is the point x itself, and whose realization as a rational function is recorded to be t .*

Proof. Let $t \in K(X)$ be a uniformizer at x , so $\text{ord}_x t = 1$ (Lemma ??); it is a nonzero rational function. Having nonnegative order at x , it is integral there: t is the image of an element of the stalk $\mathcal{O}_{X,x}$ (Lemma ??). Represent that stalk element by a section s_0 over some open neighbourhood $W_0 \ni x$. Its germ at the generic point is the image of its germ at x under $\mathcal{O}_{X,x} \rightarrow K(X)$ (Lemma ??), so $\text{germ}_\eta(s_0) = t$.

Now t , being a nonzero rational function, has nonzero order at only finitely many closed points (Theorem ??); let bad be the finite set of those points *other than* x . Each closed point is a closed singleton (Theorem ??), so bad is a finite closed set; removing it from W_0 leaves an open set

$$V := W_0 \setminus \text{bad}, \quad x \in V$$

(indeed $x \notin \text{bad}$). Restrict s_0 to V and call the result s ; its germ at $y \in V$ still maps to t under $\mathcal{O}_{X,y} \rightarrow K(X)$. In particular at the generic point the germ is exactly $\text{germ}_\eta(s) = t$, the tracked realization.

Regularity. For $y \in V$, the germ $\text{germ}_y(s)$ maps to $t \neq 0$ in the function field under the localization $\mathcal{O}_{X,y} \rightarrow K(X)$; since that map is injective on the domain stalk, $\text{germ}_y(s) \neq 0$, i.e. it is a nonzerodivisor.

Unit away from x . Let $y \in V$ with $y \neq x$. If $y = \eta$, then $\text{germ}_\eta(s) = t \neq 0$ is a unit of the field $K(X)$. Otherwise y is a closed point; suppose $\text{germ}_y(s)$ were not a unit of $\mathcal{O}_{X,y}$. A section invertible at a closed point has order of vanishing 0 there (Lemma ??); contrapositively, non-invertibility of s at y forces $\text{ord}_y t \neq 0$ (recall $\text{germ}_\eta(s) = t$, and ord_y is the order homomorphism ??). Then y is a closed point where t has nonzero order, distinct from x , so $y \in \text{bad}$ — contradicting $y \in V = W_0 \setminus \text{bad}$. Hence $\text{germ}_y(s)$ is a unit. $\square$

Definition 992 (The point divisor $1 \cdot x$). For a closed point x , the *point divisor* $1 \cdot x$ is the local-equation system on the pointed cover

$$\mathcal{U}(y) = \begin{cases} V & y = x, \\ X \setminus \{x\} & y \neq x, \end{cases}$$

with V the neighbourhood of Lemma ??, whose equation on the distinguished member V is the uniformizer-section s and whose equation on every other member $X \setminus \{x\}$ is the constant 1. Its defining data is fully accessible: in particular the germ at the generic point η of the equation on the member at x is the chosen uniformizer t , which is what pins the Weil divisor these equations cut out.

Proof. The complement $X \setminus \{x\}$ is open because $\{x\}$ is closed (Theorem ??), and it contains every $y \neq x$, so $\mathcal{U}$ is a pointed cover. Regularity holds member by member: on V the germs of s are nonzerodivisors (Lemma ??), and on $X \setminus \{x\}$ the germs of the constant 1 are units, hence nonzerodivisors. For the overlap ratios, consider a pair (a, b) . If neither or both of a, b equal x , the two equations on the overlap are equal (both 1, or both restrictions of s on $V \cap V = V$), so the ratio is the unit 1. If exactly one of a, b equals x , the overlap is contained in $V \cap (X \setminus \{x\}) = V \setminus \{x\}$, where s is a unit (Lemma ??); the ratio of the two equations s and 1 is then s or s^{-1} , a unit there. Thus the data is a local-equation system. The equation on the member at x is the section s , whose germ at η is the tracked uniformizer t (Lemma ??). $\square$

Definition 993 (The class of a Weil divisor). For a Weil divisor $D \in \text{Div}(X)$ (??), its *Picard class* is the finite product

$$\mathcal{O}_X(D) = \prod_x \mathcal{O}_X(x)^{D_x} \in \text{Pic}(X),$$

over the finite support of D , where $\mathcal{O}_X(x) := \text{picClass}(1 \cdot x) \in \text{Pic}(X)$ is the class of the point divisor (??, ??). A negative coefficient D_x contributes the inverse class, so the exponent is an integer power in the abelian group $\text{Pic}(X)$; the assignment is defined on all of $\text{Div}(X)$, with no effective/anti-effective case split. The product is finite because D is finitely supported and $\mathcal{O}_X(x)^0 = 1$.

Theorem 994 (Additivity of the divisor class). *The class map $D \mapsto \mathcal{O}_X(D)$ is a homomorphism $\text{Div}(X) \rightarrow \text{Pic}(X)$: for all $D, D' \in \text{Div}(X)$,*

$$\mathcal{O}_X(D + D') = \mathcal{O}_X(D) \cdot \mathcal{O}_X(D').$$

Proof. The class is the finitely supported product $D \mapsto \prod_x \mathcal{O}_X(x)^{D_x}$. At each point x the exponentiation $n \mapsto \mathcal{O}_X(x)^n$ is a homomorphism $\mathbb{Z} \rightarrow \text{Pic}(X)$: it sends 0 to the identity and satisfies $\mathcal{O}_X(x)^{m+n} = \mathcal{O}_X(x)^m \cdot \mathcal{O}_X(x)^n$. Hence, taking the product over the pointwise sum $(D + D')_x = D_x + D'_x$ in the abelian group $\text{Pic}(X)$, the summands separate and the product over $D + D'$ factors as the product over D times the product over D' , giving $\mathcal{O}_X(D + D') = \mathcal{O}_X(D) \cdot \mathcal{O}_X(D')$. $\square$

Theorem 995 (Normalization at a one-point divisor). *For a closed point x , the class of the one-point divisor $1 \cdot x$ (??) is the point-divisor class:*

$$\mathcal{O}_X(1 \cdot x) = \text{picClass}(1 \cdot x) = \mathcal{O}_X(x).$$

Proof. The one-point divisor $1 \cdot x$ is supported at the single point x with coefficient 1 (??), so the defining product of $\mathcal{O}_X(1 \cdot x)$ (??) reduces to the single factor $\mathcal{O}_X(x)^1 = \mathcal{O}_X(x)$. $\square$

7.58 Meromorphic presentations of Čech Picard classes

The remainder of this chapter carries out, at the level of Čech cocycles, the classical identification of the group of Weil divisors modulo principal equivalence with the Picard group of an integral curve. The device that replaces every sheaf-theoretic gluing argument is the *generic-point embedding*: on an integral scheme every nonempty open contains the generic point η , and taking germs at η embeds the sections over such an open into the function field $K(X)$. A Čech Picard class can therefore be *presented meromorphically* — by a pointed cover, a unit cocycle on it, and one rational function per member — after which every cochain comparison downstream is a literal equality of elements of the field $K(X)$, never a gluing of sheaf sections.

Definition 996 (The generic-point embedding of unit sections). Let X be an integral scheme with generic point η and U an open with $\eta \in U$. Taking the germ at η is a homomorphism of groups

$$\text{germ}_\eta : \Gamma(X, U)^\times \longrightarrow K(X)^\times,$$

the *generic-point embedding* of unit sections into the units of the function field. It is restriction-invariant — for $\eta \in W \leq U$ it agrees with restriction $\Gamma(X, U)^\times \rightarrow \Gamma(X, W)^\times$ followed by the embedding on W — and, on an integral scheme, injective.

Proof. The germ at η is a ring homomorphism $\Gamma(X, U) \rightarrow K(X)$ carrying units to units, so it restricts to a group homomorphism on the unit groups. Restriction-invariance is the functoriality of germs along $W \leq U$. For injectivity, on an integral scheme a section over a nonempty open is determined by its germ at η (which lies in every nonempty open); hence two unit sections with the same germ at η are equal. $\square$

Lemma 997 (Pointed covers contain the generic point). *On an irreducible scheme X , every member $\mathcal{U}(x)$ of a pointed cover contains the generic point η , and hence so does every pairwise overlap $\mathcal{U}(x) \cap \mathcal{U}(y)$.*

Proof. A pointed cover assigns to each x an open $\mathcal{U}(x) \ni x$, in particular nonempty; on an irreducible space the generic point lies in every nonempty open, so $\eta \in \mathcal{U}(x)$. Intersecting two such memberships gives $\eta \in \mathcal{U}(x) \cap \mathcal{U}(y)$. $\square$

Definition 998 (Meromorphic presentation). A *meromorphic presentation* of a Čech Picard class on an integral scheme X consists of a pointed cover $\mathcal{U}$, a unit Čech 1-cocycle γ on $\mathcal{U}$, and a family $(f_x)_x$ of units $f_x \in K(X)^\times$, one per member, satisfying the *ratio law*

$$f_x \cdot f_y^{-1} = \text{germ}_\eta(\gamma_{xy}) \quad \text{for every pair } x, y,$$

where $\gamma_{xy} \in \Gamma(X, \mathcal{U}(x) \cap \mathcal{U}(y))^\times$ is the cocycle value. Its *Picard class* is

$$\text{picClass}(P) = [\mathcal{U}, \gamma] \in \text{Pic}(X).$$

Proof. Each overlap $\mathcal{U}(x) \cap \mathcal{U}(y)$ contains η (??), so the germ $\text{germ}_\eta(\gamma_{xy})$ (??) is defined and the ratio law is a condition in $K(X)^\times$. The Picard class is the class in the colimit $\text{Pic}(X)$ of the unit cocycle γ on $\mathcal{U}$. $\square$

Theorem 999 (Existence of meromorphic presentations). *Every Čech Picard class $L \in \text{Pic}(X)$ of an integral scheme is the class of a meromorphic presentation.*

Proof. Represent L by a unit 1-cocycle γ on a pointed cover $\mathcal{U}$. Use the generic point itself as base index: set

$$f_x := \text{germ}_\eta(\gamma_{x\eta}) \in K(X)^\times,$$

the germ at η of the cocycle value on the pair (x, η) , whose overlap contains η (??). The cocycle identity for the triple (x, y, η) reads $\gamma_{xy} \gamma_{y\eta} = \gamma_{x\eta}$ on the triple overlap, which contains η ; applying the generic-point embedding, whose value is unchanged by the restriction to the triple overlap (??), and rearranging gives

$$\text{germ}_\eta(\gamma_{xy}) = \text{germ}_\eta(\gamma_{x\eta}) \cdot \text{germ}_\eta(\gamma_{y\eta})^{-1} = f_x \cdot f_y^{-1},$$

the ratio law. Thus $(\mathcal{U}, \gamma, f)$ is a meromorphic presentation (??) with class $[\mathcal{U}, \gamma] = L$. $\square$

Lemma 1000 (The presentation of a local-equation system). *Let E be a local-equation system on X (??). The germs at η of its regular equations, $f_x := \text{germ}_\eta(E.\text{eqn}_x) \in K(X)^\times$, form a meromorphic presentation whose cocycle is the transition-unit cocycle of E and whose class is the Picard class $\text{picClass}(E)$ of E (??).*

Proof. Each equation $E.\text{eqn}_x$ is regular, so its germ at η is a nonzerodivisor of the field $K(X)$, that is, a nonzero element, hence a unit $f_x \in K(X)^\times$. The defining equation $E.\text{eqn}_x = g_{xy} \cdot E.\text{eqn}_y$ on the overlap (??), pushed to the generic point, gives $f_x = \text{germ}_\eta(g_{xy}) \cdot f_y$, i.e. the ratio law $f_x \cdot f_y^{-1} = \text{germ}_\eta(g_{xy})$ with the transition-unit cocycle g of E . The presented class is $[\mathcal{U}, g] = \text{picClass}(E)$ (??). $\square$

7.59 The divisor of a meromorphic presentation

We now specialize to the curve bundle X of Chapter ??: integral, smooth of relative dimension one and quasi-compact over the field K , with generic point η . The trivializing elements of a meromorphic presentation cut out a Weil divisor, read off point by point from their orders of vanishing.

Theorem 1001 (Unit sections have trivial order). *For an open $U \ni \eta$, a unit section $u \in \Gamma(X, U)^\times$, and a closed point $x \in U$, the germ at η of u has trivial order at x :*

$$\text{ord}_x(\text{germ}_\eta(u)) = 0.$$

Proof. The section u is invertible at x , so x lies in the basic open set where the underlying section is invertible. A rational function that is the germ at η of a section invertible at x has order 0 there (??). $\square$

Theorem 1002 (Piece-independence of the order). *Let P be a meromorphic presentation and x a closed point lying in a member $\mathcal{U}(y)$ of its cover. Then the trivializing element f_y of that piece has the same order at x as the element f_x indexed by x itself:*

$$\text{ord}_x(f_y) = \text{ord}_x(f_x).$$

Proof. By the ratio law, $f_x \cdot f_y^{-1} = \text{germ}_\eta(\gamma_{xy})$ with γ_{xy} a unit section over $\mathcal{U}(x) \cap \mathcal{U}(y)$, an overlap containing the closed point x (which lies in $\mathcal{U}(x)$ and, by hypothesis, in $\mathcal{U}(y)$). Its germ at η has order 0 at x (??). Since ord_x is a homomorphism (??), $\text{ord}_x(f_x) - \text{ord}_x(f_y) = \text{ord}_x(f_x \cdot f_y^{-1}) = 0$. $\square$

Theorem 1003 (Finite support of the orders). *For a meromorphic presentation P on the quasi-compact curve, the set of closed points x at which the piece-indexed element f_x has nonzero order is finite.*

Proof. The curve is quasi-compact, so the members $\mathcal{U}(i)$ of the cover admit a finite subcover indexed by a finite set T . If a closed point x has $\text{ord}_x(f_x) \neq 0$, choose $i \in T$ with $x \in \mathcal{U}(i)$; by piece-independence (??), $\text{ord}_x(f_x) = \text{ord}_x(f_i) \neq 0$, so x lies in the order support of the single rational function f_i , which is finite (??). Hence the whole support is contained in the finite union over $i \in T$ of these finite sets. $\square$

Definition 1004 (The divisor of a presentation). For a meromorphic presentation P on the curve, its *divisor* is the Weil divisor

$$\text{presentationDivisor}(P) = \sum_x \text{ord}_x(f_x) x \in \text{Div}(X),$$

whose coefficient at each closed point x is the order at x of the piece-indexed trivializing element f_x .

Proof. The assignment $x \mapsto \text{ord}_x(f_x)$ has finite support (??), so it defines an element of $\text{Div}(X)$ (??); by piece-independence (??) the coefficient is the order of the local trivialization near x , whichever piece is used to read it off. $\square$

7.60 The calculus of meromorphic presentations

Meromorphic presentations multiply, invert, and have a unit — pointwise on the trivializing elements, and, on the cocycles, by restriction to the common refinement. No group laws are claimed (the covers of $(P \cdot Q) \cdot R$ and $P \cdot (Q \cdot R)$ differ by associativity of intersection), and none are needed: every identification is made at the level of the Picard class and of the divisor.

Definition 1005 (The principal presentation). For $g \in K(X)^\times$, the *principal presentation* of g is the trivial cocycle on the one-member cover $\{X\}$, trivialized by the constant g (so $f_x = g$ for all x).

Proof. The cocycle is trivial, and the ratio law holds because $f_x \cdot f_y^{-1} = g \cdot g^{-1} = 1 = \text{germ}_\eta(1)$. $\square$

Theorem 1006 (Multiplicativity of the presented class). Write 1 for the unit presentation (the trivial cocycle on $\{X\}$ with all elements 1), $P \cdot Q$ for the product on the common refinement $\mathcal{U}_P \cap \mathcal{U}_Q$ (trivializing elements multiplied pointwise, cocycles restricted and multiplied), and P^{-1} for the inverse (trivializing elements and cocycle inverted). Then the presented class is multiplicative:

$$\text{picClass}(1) = 1, \quad \text{picClass}(P \cdot Q) = \text{picClass}(P) \cdot \text{picClass}(Q), \quad \text{picClass}(P^{-1}) = \text{picClass}(P)^{-1},$$

and $\text{picClass}(\text{principal } g) = 1$.

Proof. The class of a presentation is the class of its cocycle. The trivial cocycle has trivial class, giving $\text{picClass}(1) = 1$; the principal presentation (??) also carries the trivial cocycle, so $\text{picClass}(\text{principal } g) = 1$. Inverting a cocycle inverts its class, giving $\text{picClass}(P^{-1}) = \text{picClass}(P)^{-1}$. For the product, restricting each cocycle to the common refinement leaves its class unchanged, and the class of a product of cocycles is the product of the classes, so $\text{picClass}(P \cdot Q) = \text{picClass}(P) \cdot \text{picClass}(Q)$. $\square$

Theorem 1007 (Additivity of the presentation divisor). With the operations of ??, the divisor is additive:

$$\text{presentationDivisor}(1) = 0, \quad \text{presentationDivisor}(P \cdot Q) = \text{presentationDivisor}(P) + \text{presentationDivisor}(Q),$$

$\text{presentationDivisor}(P^{-1}) = -\text{presentationDivisor}(P)$, and the divisor of the principal presentation of g is the principal divisor:

$$\text{presentationDivisor}(\text{principal } g) = \text{div}(g).$$

Proof. Compare coefficients at each closed point x (??). The element of 1 at x is 1, of order 0, so $\text{presentationDivisor}(1) = 0$. The element of $P \cdot Q$ at x is $f_x \cdot f'_x$, and ord_x is a homomorphism (??), giving additivity; inversion negates the order. For the principal presentation the element at x is g , so the coefficient is $\text{ord}_x(g)$, the coefficient of $\text{div}(g)$ (??). $\square$

7.61 The class law: equal divisors give equal classes

The heart of the meromorphic bridge is that the Čech Picard class of a presentation depends only on its divisor. Its local input is that a rational function of trivial order is the germ of a unit section; its global mechanism is that the cohomology relating two presentations of equal divisor can be verified entirely inside $K(X)^\times$, through the injective generic-point embedding — on an integral scheme no gluing of sheaf sections is ever required.

Lemma 1008 (Order matching produces a unit section). *Let V be a nonempty open ($\eta \in V$). A rational function $q \in K(X)^\times$ whose order vanishes at every closed point of V is the germ at η of a unit section over V : there is $u \in \Gamma(X, V)^\times$ with $\text{germ}_\eta(u) = q$. Consequently two rational functions $g, h \in K(X)^\times$ with equal order at every closed point of V differ by the germ of a unit section: $\text{germ}_\eta(u) = g \cdot h^{-1}$ for some $u \in \Gamma(X, V)^\times$.*

Proof. Since $\text{ord}_x(q) = 0$ at every closed point $x \in V$, both q and q^{-1} have poles bounded by the zero divisor on V , so both lie in $\mathcal{O}(0)(V)$ (??). Each is therefore the germ at η of a section over V (??); let s, s' be sections with $\text{germ}_\eta(s) = q$ and $\text{germ}_\eta(s') = q^{-1}$. Their product has $\text{germ}_\eta(ss') = qq^{-1} = 1$, and on the integral scheme a section is determined by its germ at η , so $ss' = 1$; thus s is a unit section with $\text{germ}_\eta(s) = q$. The two-element form is this applied to $q = g \cdot h^{-1}$, whose order at each closed point of V is $\text{ord}_x(g) - \text{ord}_x(h) = 0$. $\square$

Theorem 1009 (The class law). *Two meromorphic presentations with equal divisors present equal Čech Picard classes: if $\text{presentationDivisor}(P) = \text{presentationDivisor}(Q)$ then $\text{picClass}(P) = \text{picClass}(Q)$.*

Proof. Equal divisors mean $\text{ord}_z(f_z) = \text{ord}_z(f'_z)$ at every closed point z , comparing coefficients (??). Fix a member indexed by x of the common refinement $\mathcal{W} := \mathcal{U}_P \cap \mathcal{U}_Q$ and a closed point $z \in \mathcal{W}(x)$. Piece-independence (??) gives $\text{ord}_z(f_x) = \text{ord}_z(f_z)$ and $\text{ord}_z(f'_x) = \text{ord}_z(f'_z)$, so $\text{ord}_z(f_x) = \text{ord}_z(f'_x)$: the elements f_x and f'_x have equal order at every closed point of $\mathcal{W}(x)$. By order matching (??) there is a unit section $u_x \in \Gamma(X, \mathcal{W}(x))^\times$ with $\text{germ}_\eta(u_x) = f_x \cdot (f'_x)^{-1}$.

The 0-cochain $(u_x^{-1})_x$ exhibits the restricted cocycles $\gamma|_{\mathcal{W}}$ and $\gamma'|_{\mathcal{W}}$ as cohomologous. The required coboundary identity is, on each overlap $\mathcal{W}(x) \cap \mathcal{W}(y)$, an equality of unit sections; applying the injective generic-point embedding (??) turns it into the identity

$$\text{germ}_\eta(u_x)^{-1} \cdot (f_x f_y^{-1}) = (f'_x (f'_y)^{-1}) \cdot \text{germ}_\eta(u_y)^{-1}$$

in the abelian group $K(X)^\times$, where $f_x f_y^{-1}$ and $f'_x (f'_y)^{-1}$ are the germs of the restricted cocycle values by the ratio law; substituting $\text{germ}_\eta(u_x) = f_x (f'_x)^{-1}$ makes both sides equal. Cohomologous cocycles on $\mathcal{W}$ have the same class, and restricting a cocycle to a refinement preserves its Čech class, so $\text{picClass}(P) = [\mathcal{W}, \gamma|_{\mathcal{W}}] = [\mathcal{W}, \gamma'|_{\mathcal{W}}] = \text{picClass}(Q)$. $\square$

7.62 Extraction: equal classes differ by a principal divisor

Theorem 1010 (Extraction and the meromorphic characterization). *Two meromorphic presentations P, Q present the same Čech Picard class if and only if their divisors differ by a principal divisor:*

$$\text{picClass}(P) = \text{picClass}(Q) \iff \text{presentationDivisor}(P) - \text{presentationDivisor}(Q) = \text{div}(g) \text{ for some } g \in K(X)^\times.$$

Proof. ($\Rightarrow$) Equal classes are, on a common refinement $\mathcal{W}$, cohomologous cocycles: there is a 0-cochain (α_x) of unit sections $\alpha_x \in \Gamma(X, \mathcal{W}(x))^\times$ with $\alpha_x \cdot \gamma_{xy}|_{\mathcal{W}} = \gamma'_{xy}|_{\mathcal{W}} \cdot \alpha_y$ on each overlap. Put

$$c_x := f_x \cdot (f'_x)^{-1} \cdot \text{germ}_\eta(\alpha_x) \in K(X)^\times.$$

Applying the generic-point embedding to the coboundary identity and rearranging in the abelian group $K(X)^\times$ shows $c_x = c_y$ for all x, y ; as the members cover X , the c_x are a single rational function g . At a closed point z , choosing x with $z \in \mathcal{W}(x)$, the germ $\text{germ}_\eta(\alpha_x)$ has order 0 at z (??), so

$$\text{ord}_z(g) = \text{ord}_z(f_z) - \text{ord}_z(f'_z) = \text{presentationDivisor}(P)_z - \text{presentationDivisor}(Q)_z.$$

Hence $\text{div}(g) = \text{presentationDivisor}(P) - \text{presentationDivisor}(Q)$.

($\Leftarrow$) If $\text{presentationDivisor}(P) - \text{presentationDivisor}(Q) = \text{div}(g)$, then by additivity (??) the presentation $Q \cdot \text{principal } g$ has divisor $\text{presentationDivisor}(Q) + \text{div}(g) = \text{presentationDivisor}(P)$; the class law (??) gives $\text{picClass}(Q \cdot \text{principal } g) = \text{picClass}(P)$, while $\text{picClass}(Q \cdot \text{principal } g) = \text{picClass}(Q)$ by multiplicativity and $\text{picClass}(\text{principal } g) = 1$ (??). Hence $\text{picClass}(P) = \text{picClass}(Q)$. $\square$

7.63 The point presentation

To feed the class map with concrete divisors we realize each one-point divisor $1 \cdot x$ by a meromorphic presentation whose defining data is tracked: its equation at x is a uniformizer whose germ at η is recorded.

Lemma 1011 (A tracked point-uniformizer). *For a closed point x of the curve, a tracked point-uniformizer at x is an open neighbourhood $V \ni x$ together with a section $s \in \Gamma(X, V)$ that is regular (its germs are nonzerodivisors), whose germ at η is a chosen uniformizer t at x ($\text{ord}_x t = 1$), and which is a unit at every point of V other than x . Such data exists.*

Proof. Choose a uniformizer $t \in K(X)$ at x (??), a nonzero rational function with $\text{ord}_x t = 1$. Having nonnegative order at x , t is integral there, so it is the germ at η of a section s_0 over a neighbourhood $W_0 \ni x$ (??, ??). Remove from W_0 the finitely many closed points other than x at which t has nonzero order (finite by ??, and a finite union of closed points is closed by ??); the result $V := W_0 \setminus \text{bad}$ is an open neighbourhood of x , and the restriction s of s_0 to V still has germ t at η .

Regularity. For $y \in V$, the germ of s maps to $t \neq 0$ under the injective localization $\mathcal{O}_{X,y} \rightarrow K(X)$, hence is a nonzerodivisor.

Unit away from x . Let $y \in V$, $y \neq x$. If $y = \eta$, then $\text{germ}_\eta(s) = t$ is a unit of the field. If y is a closed point and the germ of s at y were not a unit, then t would have nonzero order at y (a section invertible at a closed point has order 0 there, ??, contrapositively, reading order through ??); but then y is a removed point, contradicting $y \in V$. So s is a unit at y . $\square$

Definition 1012 (The local equations of the point divisor). From a tracked point-uniformizer d at x (??), the *point equations* of $1 \cdot x$ are the local-equation system on the pointed cover

$$\mathcal{U}(z) = \begin{cases} V & z = x, \\ X \setminus \{x\} & z \neq x, \end{cases}$$

whose equation on the member at x is the tracked section s and whose equation on every other member is the constant 1; the germ at η of the equation at x is the chosen uniformizer.

Proof. The complement $X \setminus \{x\}$ is open (??) and contains every $z \neq x$, so $\mathcal{U}$ is a pointed cover. Regularity holds member by member: on V the germs of s are nonzerodivisors (??), and the constant 1 is a unit. For overlap ratios, a pair both at x or both away from x has equal equations, ratio 1; a mixed pair has overlap inside $V \setminus \{x\}$, where s is a unit (??), so the ratio of s and 1 is a unit. Thus the data is a local-equation system (??), and the germ at η of the equation at x is $\text{germ}_\eta(s) = t$. $\square$

Theorem 1013 (The divisor of the point presentation). *The divisor of the meromorphic presentation of the point equations at a closed point x is the one-point divisor:*

$$\text{presentationDivisor}((\text{point equations at } x)\text{'s presentation}) = 1 \cdot x.$$

Proof. The presentation of the point equations (??) has, at each closed point z , trivializing element the germ at η of the equation at z . At $z = x$ this is the uniformizer t , of order $\text{ord}_x t = 1$ (??), so the coefficient at x is 1; at $z \neq x$ the equation is the constant 1, of order 0, so the coefficient vanishes. Hence the divisor is $1 \cdot x$ (??, ??). $\square$

7.64 The Čech Picard class of a Weil divisor

Assembling the realization of divisors, the class law, and extraction yields the total class map $D \mapsto \mathcal{O}_X(D)$ from Weil divisors to Čech Picard classes, and its kernel: the class map descends to an isomorphism of $\text{Div}(X)$ modulo principal divisors onto $\text{Pic}(X)$.

Theorem 1014 (Realization of Weil divisors). *Every Weil divisor $D \in \text{Div}(X)$ on the curve is the divisor of a meromorphic presentation.*

Proof. First realize the one-point divisors $n \cdot x$. For $n \geq 0$, induct on n : the divisor $0 \cdot x = 0$ is realized by the unit presentation ($\text{presentationDivisor}(1) = 0$, ??), and $(n+1) \cdot x = n \cdot x + 1 \cdot x$ by the product of a realization of $n \cdot x$ with the point presentation at x , using additivity (??), the point-divisor anchor (??, ??), and the one-point arithmetic (??). Negative multiplicities use the inverse presentation, whose divisor is negated (??). A general D is a finite sum of one-point divisors (??), realized by the product of their presentations, again by additivity. $\square$

Definition 1015 (The class map and its surjectivity). The Čech Picard class $\mathcal{O}_X(D) = \text{picClass}(D) \in \text{Pic}(X)$ of a Weil divisor D is the class of any meromorphic presentation whose divisor is D . It is well-defined, satisfies $\text{picClass}(\text{presentationDivisor}(P)) = \text{picClass}(P)$ for every presentation P , and the map $D \mapsto \mathcal{O}_X(D)$ is surjective onto $\text{Pic}(X)$.

Proof. A realizing presentation exists (??). If P, P' both have divisor D , the class law (??) gives $\text{picClass}(P) = \text{picClass}(P')$, so $\text{picClass}(D)$ is well-defined; taking $D = \text{presentationDivisor}(P)$ with the realization P itself gives $\text{picClass}(\text{presentationDivisor}(P)) = \text{picClass}(P)$. For surjectivity, every class L is presented by some P (??), and then $\mathcal{O}_X(\text{presentationDivisor}(P)) = \text{picClass}(P) = L$. $\square$

Theorem 1016 (The class map is a homomorphism trivial on principal divisors). *The class map is a group homomorphism $\text{Div}(X) \rightarrow \text{Pic}(X)$, trivial on principal divisors:*

$$\mathcal{O}_X(0) = 1, \quad \mathcal{O}_X(D + D') = \mathcal{O}_X(D) \cdot \mathcal{O}_X(D'), \quad \mathcal{O}_X(\text{div } g) = 1 \quad (g \in K(X)^\times).$$

Proof. Realizing divisors by presentations and using $\text{picClass}(\text{presentationDivisor}(P)) = \text{picClass}(P)$ (??): $\mathcal{O}_X(0) = \text{picClass}(1) = 1$ since $\text{presentationDivisor}(1) = 0$ (??, ??). If P, P' realize

D, D' , then $P \cdot P'$ realizes $D + D'$ (??), so $\mathcal{O}_X(D + D') = \text{picClass}(P \cdot P') = \text{picClass}(P) \cdot \text{picClass}(P') = \mathcal{O}_X(D) \cdot \mathcal{O}_X(D')$ (??). Finally $\text{div } g = \text{presentationDivisor}(\text{principal } g)$ (??), so $\mathcal{O}_X(\text{div } g) = \text{picClass}(\text{principal } g) = 1$ (??). $\square$

Theorem 1017 (The kernel is the principal divisors). *Two Weil divisors have the same Čech Picard class if and only if they differ by a principal divisor:*

$$\mathcal{O}_X(D) = \mathcal{O}_X(D') \iff D - D' = \text{div}(g) \text{ for some } g \in K(X)^\times.$$

Together with surjectivity (??) this identifies $\text{Pic}(X)$ with the group of Weil divisors modulo principal divisors, the classical isomorphism $\text{CaCl}(X) = \text{Pic}(X)$ on an integral curve.

Proof. $(\Rightarrow)$ Realize D, D' by presentations P, P' , so $\text{picClass}(P) = \mathcal{O}_X(D) = \mathcal{O}_X(D') = \text{picClass}(P')$; extraction (??) yields g with $\text{presentationDivisor}(P) - \text{presentationDivisor}(P') = \text{div}(g)$, that is $D - D' = \text{div}(g)$. $(\Leftarrow)$ If $D - D' = \text{div}(g)$, then $D = D' + \text{div}(g)$, so $\mathcal{O}_X(D) = \mathcal{O}_X(D') \cdot \mathcal{O}_X(\text{div } g) = \mathcal{O}_X(D')$ by additivity and triviality on principal divisors (??). Since the class map is onto $\text{Pic}(X)$ with fibres the cosets of the subgroup of principal divisors, it descends to an isomorphism of $\text{Div}(X)$ modulo principal divisors onto $\text{Pic}(X)$. $\square$

Theorem 1018 (Normalization at a one-point divisor). *For a closed point x , the class of the one-point divisor $1 \cdot x$ is the class of the tracked point equations at x :*

$$\mathcal{O}_X(1 \cdot x) = \text{picClass}(\text{point equations at } x).$$

Proof. The presentation of the point equations at x has divisor $1 \cdot x$ (??), so $\mathcal{O}_X(1 \cdot x)$ is its class (??), which is the class of the point equations (??, ??). $\square$

7.65 The point-divisor class map and the meromorphic class map coincide

The chapter carries two constructions of the divisor-to-class assignment $D \mapsto \mathcal{O}_X(D)$. The first, divisorClass (??), is the finitely supported product of the point-divisor classes $\mathcal{O}_X(x) = \text{picClass}(1 \cdot x)$ read off the point local equations (??); the second, the anchored picClass (??), is the class of any meromorphic presentation realizing D . This closing section identifies them. The bridge is that the point local equations, whose equation at x has generic-point germ the uniformizer (??), present the one-point divisor $1 \cdot x$; their two classes agree, and the agreement propagates from one-point divisors to all divisors by homomorphism extensionality.

Theorem 1019 (The divisor of the point-divisor presentation). *The divisor of the meromorphic presentation of the point-divisor local equations at a closed point x is the one-point divisor:*

$$\text{presentationDivisor}((1 \cdot x)\text{'s presentation}) = 1 \cdot x.$$

Proof. Compare coefficients at each closed point z (??). The presentation of the point-divisor equations (??) has, at z , trivializing element the germ at η of the equation at z . At $z = x$ that germ is the chosen uniformizer t (??), of order $\text{ord}_x t = 1$ (??), so the coefficient at x is 1; at $z \neq x$ the equation is the constant 1, of order 0, so the coefficient vanishes. Hence the divisor is $1 \cdot x$ (??). $\square$

Theorem 1020 (The point-divisor class is the anchored one-point class). *For a closed point x , the Picard class of the point-divisor local equations at x equals the anchored class of the one-point divisor $1 \cdot x$:*

$$\text{picClass}(1 \cdot x) = \mathcal{O}_X(1 \cdot x).$$

Proof. The presentation of the point-divisor equations has divisor $1 \cdot x$ (??), so the anchored class $\mathcal{O}_X(1 \cdot x)$ is the class of that presentation (??); and the class of the presentation of a local-equation system is the Picard class of the system itself (??), which is $\text{picClass}(1 \cdot x)$ (??). Chaining the two equalities gives $\text{picClass}(1 \cdot x) = \mathcal{O}_X(1 \cdot x)$. $\square$

Theorem 1021 (The two divisor-class maps coincide). *The point-product class map (??) and the anchored meromorphic class map (??) agree on every Weil divisor:*

$$\text{divisorClass}(D) = \mathcal{O}_X(D), \quad D \in \text{Div}(X).$$

Proof. Both assignments are homomorphisms $\text{Div}(X) \rightarrow \text{Pic}(X)$ from the additive group of divisors to the multiplicative Picard group: divisorClass by additivity (??) and the anchored class map by Theorem ???. Since $\text{Div}(X)$ is freely generated by the one-point divisors (??), it suffices to check agreement on each $n \cdot x$. On the point-product side the defining product (??) collapses to the single factor $\text{divisorClass}(n \cdot x) = \text{picClass}(1 \cdot x)^n$. On the anchored side, additivity (??) gives $\mathcal{O}_X(n \cdot x) = \mathcal{O}_X(1 \cdot x)^n$. The two base classes are equal, $\text{picClass}(1 \cdot x) = \mathcal{O}_X(1 \cdot x)$ (??), so their n -th powers agree and $\text{divisorClass}(n \cdot x) = \mathcal{O}_X(n \cdot x)$. By homomorphism extensionality on the generators, the two maps coincide on all of $\text{Div}(X)$. $\square$

7.66 The unit of the étale sheafification

The honest relative Picard functor maps into its étale-sheafified Layer-2 vehicle picEt by a natural transformation — the unit of the sheafification. On an affine test it is the unit of the plus construction; on a general test it is assembled over the affine opens.

Definition 1022 (Restriction of plus classes along an explicit algebra map). For an explicit homomorphism of k -algebras $\varphi: A \rightarrow A'$, restriction of plus classes $\varphi^*: \text{PicEtAff}(C, A) \rightarrow \text{PicEtAff}(C, A')$: regard A' as an A -algebra through φ and apply the base-extension map of Definition ???. It satisfies $(\text{id}_A)^* = \text{id}$ and $(\psi \circ \varphi)^* = \psi^* \circ \varphi^*$, and it is natural on units of the plus construction:

$$\varphi^*(\text{unit}(x)) = \text{unit}(\varphi^*x), \quad x \in \text{relPic}(C, A).$$

Proof. The scalar tower $k \rightarrow A \rightarrow A'$ determined by φ makes Definition ?? applicable; the two functor laws are Theorems ?? and ?? read through this tower, and naturality on units is Theorem ???. $\square$

Definition 1023 (The affine-open test objects of a test object). For a test object T and an affine open $U \subseteq T_{\text{left}}$, the spectrum of the section ring $\text{Spec } \Gamma(T, U)$ is a test object over T , via the canonical morphism $\text{Spec } \Gamma(T, U) \rightarrow T$ of the affine open. This assignment is coherent:

1. (restriction) for $U \leq V$, the map of U factors through that of V along Spec of the section restriction $\Gamma(T, V) \rightarrow \Gamma(T, U)$;
2. (naturality) for a morphism of tests $f: T' \rightarrow T$ and affine opens $W \leq f^{-1}V$, the square formed by Spec of the section pullback $\Gamma(T, V) \rightarrow \Gamma(T', W)$ and the two affine-open test maps commutes;
3. (top of an affine test) for $T = \text{Spec } A$, the affine-open test object of the top open, composed with Spec of the identification $\Gamma(\text{Spec } A, \top) \cong A$ (Definition ??), is the identity of $\text{Spec } A$.

Proof. The structure morphism to $\mathrm{Spec} k$ is Spec of the k -algebra structure of the section ring, so the canonical morphism of the affine open lies over $\mathrm{Spec} k$; this is the compatibility of the affine-open inclusion with the structure morphisms. Coherences (i) and (ii) are both instances of the standard identity $\mathrm{Spec}(f_{V,W}^\sharp) \circ \iota_W = \iota_V \circ f$ for the canonical maps of affine opens, applied to the identity of T and to f respectively; (iii) is the fact that the canonical map of the top affine open of an affine scheme is the isomorphism $\mathrm{Spec} \Gamma(\mathrm{Spec} A, \top) \cong \mathrm{Spec} A$, inverse to Spec of the section identification. $\square$

Definition 1024 (The component of the unit at a test object). For a test object T , the homomorphism $\mathrm{relPic}(C, T) \rightarrow \mathrm{picEt}(C, T)$ sending a relative Picard class z to the family whose value at an affine open $U \subseteq T_{\mathrm{left}}$ is the plus unit of the restriction of z along the affine-open test object of U :

$$z \mapsto \left(U \mapsto \mathrm{unit}(\iota_U^* z) \right)_U.$$

Proof. The family is compatible under restriction of affine opens, as required by Definition ??; for $U \leq V$, the restriction coherence of the affine-open test objects (Definition ??) gives $\iota_U = \iota_V \circ \mathrm{Spec}(\mathrm{res}_{V,U})$, so functoriality of restriction (Definition ??) and naturality of the unit on explicit algebra maps (Definition ??) identify $\mathrm{res}_{V,U}^* \mathrm{unit}(\iota_V^* z) = \mathrm{unit}(\iota_U^* z)$. The assignment is a homomorphism because each constituent map is. $\square$

Theorem 1025 (Affine consistency of the unit). *On an affine test $\mathrm{Spec} A$, the component of the unit collapses through the affine comparison $\mathrm{picEt}(C, \mathrm{Spec} A) \cong \mathrm{PicEtAff}(C, A)$ (Definition ??) to the unit of the plus construction:*

$$\mathrm{picEtAffineEquiv}(\mathrm{relPicToPicEt}(z)) = \mathrm{unit}(z), \quad z \in \mathrm{relPic}(C, A).$$

Proof. The affine comparison evaluates the family at the top affine open and transports along the section identification $\Gamma(\mathrm{Spec} A, \top) \cong A$. The value there is $\mathrm{unit}(\iota_\top^* z)$; transporting along the identification and using naturality of the unit (Definition ??) yields the unit of z restricted along $\mathrm{Spec}(\Gamma(\mathrm{Spec} A, \top) \cong A) \circ \iota_\top$, which is the identity of $\mathrm{Spec} A$ by coherence (iii) of Definition ??; so the value is $\mathrm{unit}(z)$ by functoriality (Definition ??). $\square$

Theorem 1026 (Naturality of the unit). *Let the curve C be proper, geometrically irreducible and geometrically reduced. For a morphism of test objects $f: T' \rightarrow T$ and $z \in \mathrm{relPic}(C, T)$,*

$$f^*(\mathrm{relPicToPicEt}(z)) = \mathrm{relPicToPicEt}(f^* z) \quad \text{in } \mathrm{picEt}(C, T').$$

Proof. The restriction map of picEt is characterised value-by-value by the $\exists!$ -property of Definition ?? together with its uniqueness (Theorem ??); it therefore suffices to verify that the right-hand family has the characterising property of the pullback of the left-hand family. Unwinding, this is the statement that for affine opens $W_0 \leq W \subseteq T'_{\mathrm{left}}$ with $W_0 \leq f^{-1}V$,

$$\mathrm{res}^* \mathrm{unit}(\iota_W^* f^* z) = (f_{V,W_0}^\sharp)^* \mathrm{unit}(\iota_V^* z).$$

Both sides are units of restrictions of z (naturality of the unit, Definition ??), and the two restrictions agree because the two composites of affine-open test maps agree: $\iota_{W_0} \circ (\mathrm{res}) = \iota_W$ and $f \circ \iota_{W_0} = \iota_V \circ \mathrm{Spec}(f_{V,W_0}^\sharp)$, the two coherences of Definition ??). $\square$

Definition 1027 (The unit of the étale sheafification). Let the curve C be proper, geometrically irreducible and geometrically reduced. The components of Definition ?? assemble into a natural transformation

$$\mathrm{relPicFunctor}(C) \longrightarrow \mathrm{picEtFunctor}(C)$$

of presheaves of commutative groups on $\text{Over}(\text{Spec } k)$, whose affine components collapse to the unit of the plus construction.

Proof. Naturality of the components is Theorem ??; the affine collapse is Theorem ?. $\square$

7.67 The degree of a plus class

The relative degree relPicDeg of Definition ?? reads an honest integer off a relative Picard class over a *field*. This section descends it to the plus construction: a plus class over K is represented on an étale cover, every cover is refined by a single finite separable field extension (cofinality, Theorem ??), and the *zigzag* shows the field reading is independent of every choice.

Lemma 1028 (The relative degree is additive). *For a field extension K of k and relative Picard classes $x, y \in \text{relPic}(C, K)$,*

$$\text{relPicDeg}_K(x \cdot y) = \text{relPicDeg}_K(x) + \text{relPicDeg}_K(y), \quad \text{relPicDeg}_K(1) = 0.$$

Proof. relPicDeg_K is a homomorphism from the multiplicative relative Picard group to $\mathbb{Z}$ by construction (Definition ??); these are its two homomorphism laws. $\square$

Theorem 1029 (The zigzag, leg (i): one degree through any two field readings). *Let K be a field extension of k , E a presented étale cover of K , and $x \in \text{relPic}(C, B_E)$ a descent class on E . Let L_1, L_2 be finite separable field extensions of K and $j_1: B_E \rightarrow L_1$, $j_2: B_E \rightarrow L_2$ two K -algebra maps. Then*

$$\text{relPicDeg}_{L_1}(j_1^*x) = \text{relPicDeg}_{L_2}(j_2^*x).$$

Proof. Refine the product of the two field covers $\text{offield}(L_1) \times \text{offield}(L_2)$ (Definitions ??, ??) to a single finite separable field extension M of K (cofinality, Theorem ??); composing with the coprojections (Definition ??) yields K -algebra maps $\theta_1: L_1 \rightarrow M$ and $\theta_2: L_2 \rightarrow M$. Then

$$\text{relPicDeg}_{L_1}(j_1^*x) = \text{relPicDeg}_M(\theta_1^*j_1^*x) = \text{relPicDeg}_M((\theta_1 j_1)^*x),$$

using invariance of the relative degree under base-field transitions (E-iv-alg descended, Theorem ??) and functoriality (Lemma ??); likewise for the second leg. Now $\theta_1 j_1$ and $\theta_2 j_2$ are two K -algebra maps $B_E \rightarrow M$ out of the cover carrier, and x is a descent class, so the transport-independence keystone of the plus construction (Theorem ??) gives $(\theta_1 j_1)^*x = (\theta_2 j_2)^*x$. The two degrees over M therefore agree, hence so do the original readings. $\square$

Theorem 1030 (The zigzag: well-definedness of the plus-class degree). *Let E_1, E_2, G be presented étale covers of K , x_1, x_2 descent classes on E_1, E_2 , and $f: B_{E_1} \rightarrow B_G$, $g: B_{E_2} \rightarrow B_G$ K -algebra maps with $f_*x_1 = g_*x_2$ as descent classes on G (the identification of the plus-construction setoid). Then for any two field readings — finite separable field extensions L_1, L_2 of K with K -algebra maps $j_1: B_{E_1} \rightarrow L_1$ and $j_2: B_{E_2} \rightarrow L_2$ —*

$$\text{relPicDeg}_{L_1}(j_1^*x_1) = \text{relPicDeg}_{L_2}(j_2^*x_2).$$

No Galois theory appears: Galois twisting of the transport is a special case of the transport-independence keystone.

Proof. Refine G to a single finite separable field extension N of K by cofinality (Theorem ??), with refinement map $\ell: B_G \rightarrow N$. By leg (i) of the zigzag (Theorem ??) applied to the descent class x_1 on E_1 and the two readings j_1 and ℓf ,

$$\text{relPicDeg}_{L_1}(j_1^* x_1) = \text{relPicDeg}_N((\ell f)^* x_1) = \text{relPicDeg}_N(\ell^* f^* x_1)$$

(functoriality, Lemma ??). Cross over along the identification $f^* x_1 = g^* x_2$ (the underlying relative classes of $f_* x_1 = g_* x_2$, Definition ??) and run the same two steps backwards on the second leg: $\text{relPicDeg}_N(\ell^* g^* x_2) = \text{relPicDeg}_N((\ell g)^* x_2) = \text{relPicDeg}_{L_2}(j_2^* x_2)$, again by leg (i). $\square$

Definition 1031 (The degree of a plus class). The *degree* $\deg_K^+: \text{PicEtAff}(C, K) \rightarrow \mathbb{Z}$ of an étale-locally trivial relative Picard class: represent the class by a descent class x on a cover E , refine E to a finite separable field extension L/K by cofinality (Theorem ??), and read relPicDeg_L of the transported class. The value is independent of the representative and of the refinement, and it is pinned by the *master equality*: for every descent-class representative (E, x) , every finite separable field extension L/K and every K -algebra map $j: B_E \rightarrow L$,

$$\deg_K^+[E, x] = \text{relPicDeg}_L(j^* x).$$

In particular, on a field-cover representative $(\text{ofField}(L), y)$ the degree is relPicDeg_L of y transported through the carrier identification.

Proof. Existence of a field refinement of each cover is cofinality (Theorem ??). Independence of the representative is exactly the zigzag (Theorem ??): two representatives of one plus class are identified after transport to a common refinement (Definition ??), and the zigzag equates any two field readings of such a pair — in particular the readings through the chosen refinements. The master equality is leg (i) of the zigzag (Theorem ??), comparing the chosen refinement map with the given j ; the field-cover normalization is the master equality at the carrier identification of $\text{ofField}(L)$ (Definition ??). $\square$

Theorem 1032 (Additivity of the plus-class degree). $\deg_K^+$ is a homomorphism: $\deg_K^+(1) = 0$, $\deg_K^+(ab) = \deg_K^+(a) + \deg_K^+(b)$ and $\deg_K^+(a^{-1}) = -\deg_K^+(a)$.

Proof. Unit. The trivial plus class is represented by the trivial descent class on the trivial cover (Definitions ??, ??); its field reading is $\text{relPicDeg}_K(1) = 0$ (Lemma ??), and the master equality (Definition ??) transfers this to $\deg_K^+$.

Additivity. Represent $a = [E, x]$ and $b = [F, y]$ (Lemma ??); the product is represented on the product cover $E \times F$ by the product of the transported classes (Theorem ??). Refine $E \times F$ to a single finite separable field L (Theorem ??) with refinement map ℓ , and read all three degrees there through the master equality: the reading of ab is relPicDeg_L of the product of the two transported classes (transport is a homomorphism, Lemma ??), which splits as the sum by Lemma ??; and the two summands are the readings of a and b through ℓ composed with the coprojections (Definition ??).

Inverses. From $\deg_K^+(a^{-1}) + \deg_K^+(a) = \deg_K^+(1) = 0$. $\square$

Theorem 1033 (The unit normalization of the plus-class degree). *On a class in the image of the unit of the plus construction, the degree is the relative degree over the base field itself:*

$$\deg_K^+(\text{unit}(z)) = \text{relPicDeg}_K(z), \quad z \in \text{relPic}(C, K).$$

Proof. The unit represents its value by the tautological descent class of z on the trivial cover (Definition ??, Lemma ??), and the trivial cover is refined by the field cover of K itself. The master equality (Definition ??) reads relPicDeg_K of z transported along the composite $K \rightarrow B_{\text{self}} \rightarrow K$, which is the identity (Lemma ??). $\square$

7.68 The degree at field points and the degree-zero Picard functor

Definition 1034 (The degree of an étale Picard class at a field point). Let T be a test object, $\lambda \in \text{picEt}(C, T)$, K a field extension of k and $t: \text{Spec } K \rightarrow T$ a *field point* of T . The *degree of λ at t* is the plus-class degree of the restriction:

$$\deg_t(\lambda) := \deg_K^+(\text{picEtAffineEquiv}(t^*\lambda)) \in \mathbb{Z},$$

the restriction along t (Definition ??) collapsed to a plus class by the affine comparison (Definition ??).

Lemma 1035 (The degree at a field point is additive). *For classes $\lambda, \mu \in \text{picEt}(C, T)$ and a field point t ,*

$$\deg_t(1) = 0, \quad \deg_t(\lambda\mu) = \deg_t(\lambda) + \deg_t(\mu), \quad \deg_t(\lambda^{-1}) = -\deg_t(\lambda).$$

Proof. Restriction and the affine comparison are homomorphisms, and $\deg^+$ is additive (Theorem ??). $\square$

Theorem 1036 (Functoriality of the degree at field points). *For $f: T' \rightarrow T$, $\lambda \in \text{picEt}(C, T)$ and a field point $t': \text{Spec } K \rightarrow T'$,*

$$\deg_{t'}(f^*\lambda) = \deg_{f \circ t'}(\lambda).$$

Proof. Both sides collapse the restriction of λ along the same composite field point: $t'^*(f^*\lambda) = (f \circ t')^*\lambda$ by functoriality of restriction (Theorem ??). $\square$

Definition 1037 (The degree-zero subgroup). For a test object T , the subgroup

$$\text{pic}^0(C, T) := \{ \lambda \in \text{picEt}(C, T) : \deg_t(\lambda) = 0 \text{ for every field point } t \text{ of } T \}$$

of classes of degree zero at *every* field point of T .

Proof. Closure under unit, product and inverse is Lemma ??, pointwise in the field point. $\square$

Definition 1038 (The degree-zero relative Picard functor). The degree-zero condition is stable under restriction: for $f: T' \rightarrow T$, the restriction map of picEt carries $\text{pic}^0(C, T)$ into $\text{pic}^0(C, T')$. The functor

$$\text{Pic}_{C/k}^0: \text{Over}(\text{Spec } k)^{\text{op}} \longrightarrow \text{CommGrp}, \quad T \mapsto \text{pic}^0(C, T),$$

with these restrictions, is the *degree-zero relative Picard functor* — a subgroup subfunctor of $\text{picEtFunctor}(C)$, with inclusion natural transformation $\text{Pic}_{C/k}^0 \rightarrow \text{picEtFunctor}(C)$. This is the carrier of the representability statement: the Jacobian represents *this* functor.

Proof. Stability: for λ of degree zero at every field point of T and t' a field point of T' , functoriality (Theorem ??) gives $\deg_{t'}(f^*\lambda) = \deg_{f \circ t'}(\lambda) = 0$. The functor laws are inherited from those of picEtFunctor (Theorem ??) on underlying classes, and the inclusion is natural because the restriction of a degree-zero class *is* the restriction of its underlying class. $\square$

7.69 Functoriality of the étale-sheafified Picard functor in the curve

Throughout this section k is a field and D, E, F are objects of $\text{Over}(\text{Spec } k)$; for the statements involving restriction along arbitrary morphisms of test objects the curves are in addition proper, geometrically irreducible and geometrically reduced over k .

Definition 1039 (Curve-direction transport of the limit). Let $g: D \rightarrow E$ be a morphism of $\text{Over}(\text{Spec } k)$ and T a test object. For a compatible family $s = (s_U)_U \in \text{picEt}(E, T)$ over the affine opens of T , the componentwise transported family

$$g^*s := (g^*(s_U))_U,$$

each component transported by the affine curve transport $g^*: \text{PicEtAff}(E, \Gamma(T, U)) \rightarrow \text{PicEtAff}(D, \Gamma(T, U))$ of Definition ??, is again compatible, and the assignment $g^*: \text{picEt}(E, T) \rightarrow \text{picEt}(D, T)$ is a group homomorphism.

Proof. Compatibility: for affine opens $U \subseteq V$ of T , the restriction of the transported value is $\text{res}_U^V(g^*(s_V)) = g^*(\text{res}_U^V(s_V)) = g^*(s_U)$, by the bilateral restriction square of Theorem ?? applied to the k -algebra map res_U^V . The homomorphism property holds componentwise. $\square$

Theorem 1040 (Functor laws and affine compatibility). *The curve-direction transport of the limit is contravariantly functorial in the curve: $\text{id}_E^* = \text{id}$, and $(g' \circ g)^* = g^* \circ g'^*$ for $g: D \rightarrow E$ and $g': E \rightarrow F$. At an affine test $\text{Spec } A$ it agrees with the affine curve transport under the affine comparison of Definition ??:*

$$\text{picEtAffineEquiv}_{D,A} \circ g^* = g^* \circ \text{picEtAffineEquiv}_{E,A},$$

and likewise with the inverse comparisons.

Proof. The functor laws hold componentwise by Theorem ??. For the affine compatibility: both sides evaluate the family at the top affine open and transport, and the Γ -Spec identification of the test algebra is a k -algebra map, so it commutes with the curve transport by the bilateral restriction square. The inverse form follows because both comparisons are isomorphisms. $\square$

Theorem 1041 (Naturality of the curve transport in the test object). *Let D, E be proper, geometrically irreducible and geometrically reduced over k , let $g: D \rightarrow E$, and let $f: T' \rightarrow T$ be an arbitrary morphism of test objects. Then restriction commutes with the curve transport:*

$$f^* \circ g^* = g^* \circ f^*: \text{picEt}(E, T) \longrightarrow \text{picEt}(D, T').$$

Proof. The value of the restricted family f^* at each affine open of T' is characterized uniquely by its pullback property (Theorem ??), so it suffices to show that the curve transport of the glued value for s has the pullback property for the transported family g^*s . On every affine sub-open lying in the preimage of an affine open V of T , the required identity is obtained from the pullback property of the glued value for s by applying g^* and commuting it past the two restriction k -algebra maps by the bilateral square of Theorem ??. $\square$

Definition 1042 (The bundled curve-direction natural transformation). For D, E proper, geometrically irreducible and geometrically reduced over k and $g: D \rightarrow E$, the componentwise transports assemble into a natural transformation

$$g^*: \text{picEtFunctor}(E) \longrightarrow \text{picEtFunctor}(D)$$

of CommGrp -valued functors on the test category, with id_E^* the identity transformation and $(g' \circ g)^* = g^* \circ g'^*$.

Proof. Naturality is Theorem ??; the identity and composition laws are the pointwise laws of Theorem ?? at each test object. $\square$

Theorem 1043 (Degree of a pulled-back class at a field point). *Let D be smooth of relative dimension 1, proper and geometrically irreducible over k , let E be proper, geometrically irreducible and geometrically reduced over k , let $g: D \rightarrow E$, let $\lambda \in \text{picEt}(E, T)$, and let $t: \text{Spec } K \rightarrow T$ be a field point of the test object. Then the degree of the pulled-back class at t is the degree of the affine curve transport of the collapsed restriction of λ :*

$$\deg_t(g^*\lambda) = \deg\left(g^*(\text{picEtAffineEquiv}_{E,K}(t^*\lambda))\right).$$

Proof. By definition, $\deg_t(g^*\lambda)$ is the degree of the plus class obtained by restricting $g^*\lambda$ along t and collapsing by the affine comparison. Restriction along t commutes with the curve transport (Theorem ??), and at the affine test $\text{Spec } K$ the transport is the affine curve transport under the comparison (Theorem ??). $\square$

7.70 Degree-zero preservation under the curve transport

Throughout this section k is a field, D and E (and, where a composite appears, F) are curve bundles over k — objects of $\text{Over}(\text{Spec } k)$, proper, smooth of relative dimension one and geometrically irreducible over k — and $g: D \rightarrow E$ is an arbitrary morphism of $\text{Over}(\text{Spec } k)$. Pullback along g preserves degree zero. The degree control is obtained *over each field independently*: over a field extension L of k the base-changed morphism is finite surjective or constant, and in either case a degree-zero class pulls back to a degree-zero class — the multiplier of the finite case is never compared across different fields.

Theorem 1044 (Pullback preserves degree zero on every fibre). *Let L be a field extension of k and let Λ be a Čech Picard class on the fibre $(E \otimes \text{Spec } L)_{\text{left}}$ with $\deg_{\text{Pic}}(\Lambda) = 0$. Then the pullback of Λ along the whisker $(g \triangleright \text{Spec } L): (D \otimes \text{Spec } L)_{\text{left}} \rightarrow (E \otimes \text{Spec } L)_{\text{left}}$ satisfies $\deg_{\text{Pic}}((g \triangleright \text{Spec } L)^*\Lambda) = 0$.*

Proof. Both fibres $D_L := (D \otimes \text{Spec } L)_{\text{left}}$ and E_L are proper, smooth of relative dimension one and geometrically irreducible over L via their second projections (Theorem ??), and the whisker commutes with the second projections, so it is a morphism of curve bundles over L . By the dichotomy (Theorem ??) it is either finite and surjective, or constant with a single closed point as set-range. In the finite surjective case the generic points match (Lemma ??) and degree multiplicativity (Theorem ??) gives $\deg_{\text{Pic}}((g \triangleright \text{Spec } L)^*\Lambda) = [L(D_L) : L(E_L)] \cdot \deg_{\text{Pic}}(\Lambda) = 0$. In the constant case the whisker is proper (Theorem ??), in particular quasi-compact, so the pulled-back class is already trivial (Corollary ??), of degree zero. $\square$

Theorem 1045 (Relative-degree-zero preservation). *Let L be a field extension of k and $y \in \text{relPic}(E, \text{Spec } L)$ a relative Picard class with $\text{relPicDeg}_L(y) = 0$. Then the curve transport satisfies $\text{relPicDeg}_L(g^*y) = 0$.*

Proof. Choose a representative Čech class Λ of y ; then $(g \triangleright \text{Spec } L)^*\Lambda$ represents g^*y (Lemma ??), and the relative degree reads the class degree of any representative (Definition ??). Conclude by Theorem ??). $\square$

Theorem 1046 (The affine curve transport preserves degree zero). *Let K be a field extension of k and $a \in \text{PicEtAff}(E, K)$ a plus class with $\deg_K^+(a) = 0$. Then $\deg_K^+(g^*a) = 0$.*

Proof. Write a as the class of a descent-class representative x on an étale cover U of K , so that g^*a is the class of the cover-wise transport g^*x on the same cover (Theorem ??). Choose a finite separable field refinement $j: B_U \rightarrow L$ of the cover (Theorem ??). By the master equality of the plus degree (Definition ??), $\deg_K^+(a) = \text{relPicDeg}_L(j^*x)$ and $\deg_K^+(g^*a) = \text{relPicDeg}_L(j^*(g^*x))$. The transports along j and along the curve morphism commute (Lemma ??), so the latter equals $\text{relPicDeg}_L(g^*(j^*x))$, which vanishes by Theorem ?? applied over L to the class j^*x , of relative degree $\deg_K^+(a) = 0$. $\square$

Definition 1047 (Curve transport of the degree-zero functor). For a test object T , the curve transport of Definition ?? restricts to the degree-zero subgroups: a group homomorphism $g^*: \text{pic}^0(E, T) \rightarrow \text{pic}^0(D, T)$ whose underlying map of étale Picard classes is the curve transport.

Proof. Let $\lambda \in \text{pic}^0(E, T)$ and let $t: \text{Spec } K \rightarrow T$ be a field point of the test object. The degree of $g^*\lambda$ at t is the plus degree of the affine curve transport of the collapsed restriction $t^*\lambda$ (Theorem ??); the collapsed restriction has plus degree $\deg_t(\lambda) = 0$ (Definition ??), so the transport has degree zero (Theorem ??, applied over the field K of the point itself). Hence $g^*\lambda$ has degree zero at every field point of T . $\square$

Theorem 1048 (Naturality and functor laws of the degree-zero transport). *The degree-zero curve transport commutes with restriction along an arbitrary morphism of test objects $f: T' \rightarrow T$, $f^* \circ g^* = g^* \circ f^*$; it is the identity for $g = \text{id}_E$; and $(g' \circ g)^* = g^* \circ g'^*$ for composable curve morphisms $g: D \rightarrow E$, $g': E \rightarrow F$.*

Proof. Membership in the degree-zero subgroup is a property of the underlying class, and all three equalities hold on underlying classes: naturality is Theorem ??, and the functor laws are Theorem ??. $\square$

Definition 1049 (The bundled degree-zero curve transport). The degree-zero curve transports assemble into a natural transformation

$$g^*: \text{Pic}_{E/k}^0 \longrightarrow \text{Pic}_{D/k}^0$$

of CommGrp -valued functors on the test category, with id_E^* the identity transformation and $(g' \circ g)^* = g^* \circ g'^*$, compatibly with the inclusions into the étale-sheafified relative Picard functors and their curve transport (Definition ??).

Proof. Naturality and the two functor laws are Theorem ?? at each test object; the inclusion square commutes because the transport of a degree-zero class is, by construction, the transport of its underlying class. $\square$

7.71 The degree seam at field points

For a *cocycle-presented* plus class on an affine test — one which is the unit image of an honest Čech Picard class — the degree at a field point collapses to the curve-level class degree $\deg_{\text{Pic}}$ of the fibre class. This seam is where the campaign's coverage and stratification arguments spend degree-zero-ness.

Theorem 1050 (The degree seam, unit form). *Let A be a k -algebra, $\lambda \in \text{picEt}(C, \text{Spec } A)$ a class whose affine collapse is the plus unit of a relative Picard class $z \in \text{relPic}(C, A)$, and $\varphi: A \rightarrow K$ a k -algebra map into a field. Then the degree of λ at the field point $\text{Spec } \varphi$ is the relative degree of the fibre class:*

$$\deg_{\text{Spec } \varphi}(\lambda) = \text{relPicDeg}_K(\varphi^*z).$$

Proof. The affine comparison intertwines restriction along $\text{Spec } \varphi$ with the explicit transport φ^* of plus classes (Theorem ??), which is natural on units (Definition ??); so the collapse of $t^*\lambda$ is $\text{unit}(\varphi^*z)$, whose degree is $\text{relPicDeg}_K(\varphi^*z)$ by the unit normalization (Theorem ??). $\square$

Theorem 1051 (The degree seam: cocycle-presented classes). *In the situation of Theorem ??, suppose $z = \pi_{\text{Spec } A}(L)$ for an honest Čech Picard class L on $C \otimes \text{Spec } A$. Then the degree at the field point is the class degree of the fibre Čech class:*

$$\deg_{\text{Spec } \varphi}(\lambda) = \deg_{\text{Pic}}((C \triangleleft \text{Spec } \varphi)^*L),$$

the pullback of L along the point.

Proof. By Theorem ?? the degree is $\text{relPicDeg}_K(\varphi^*\pi_{\text{Spec } A}(L))$; the restriction of a projected class is the projection of the pullback (Definition ??), and on projected classes over a field the relative degree reads the class degree of the representative (Definition ??). $\square$

Theorem 1052 (Base-field constancy of the degree). *In the situation of Theorem ??, suppose z is pulled back from the base: $z = \iota_A^*w$ for a class $w \in \text{relPic}(C, k)$, with $\iota_A: k \rightarrow A$ the structure map. Then the degree of λ is $\text{relPicDeg}_k(w)$ at every field point of $\text{Spec } A$ — independent of the point. If moreover $w = \pi(L_0)$ for a Čech class L_0 on $C \otimes \text{Spec } k$, the constant is $\deg_{\text{Pic}}(L_0)$.*

Proof. At a field point $\text{Spec } \varphi$, Theorem ?? reads $\text{relPicDeg}_K(\varphi^*\iota_A^*w) = \text{relPicDeg}_K((\varphi\iota_A)^*w)$ (functoriality, Lemma ??); invariance of the relative degree under base-field transitions (Theorem ??) collapses this to $\text{relPicDeg}_k(w)$, with no dependence on φ . The Čech form reads the representative through Definition ??). $\square$

Theorem 1053 (The fibre degree of a degree-zero twist). *For any class $\theta \in \text{picEt}(C, T)$, $m \in \mathbb{N}$, and a field point t , $\deg_t(\theta^m) = m \cdot \deg_t(\theta)$; and for $\lambda \in \text{pic}^0(C, T)$,*

$$\deg_t(\lambda \cdot \theta^m) = m \cdot \deg_t(\theta)$$

on the nose — the degree-zero factor contributes nothing.

Proof. The power law is additivity (Lemma ??) by induction on m ; the twist law adds $\deg_t(\lambda) = 0$ (Definition ??). $\square$

7.72 Field points across a base-field extension

Let $k \rightarrow L$ be a field extension, $\sigma: \text{Spec } L \rightarrow \text{Spec } k$ the induced morphism, and T a test object of $\text{Over}(\text{Spec } L)$. Write $\sigma_1 T$ for T regarded as a test object of $\text{Over}(\text{Spec } k)$ through σ : the same underlying scheme, with structure morphism the structure morphism of T followed by σ ; and write $\text{Spec } K / \text{Spec } F$ for the affine test of an F -algebra K , as in Chapter ??). This section matches the k -field-points of $\sigma_1 T$ with the L -field-points of T : a k -morphism $\text{Spec } K / \text{Spec } k \rightarrow \sigma_1 T$ from the affine test of a field K is the same thing as an L -algebra structure on K completing the

tower $k \rightarrow L \rightarrow K$ together with an L -morphism $\text{Spec } K / \text{Spec } L \rightarrow T$, both directions preserving the underlying morphism of schemes. The matching then exchanges the quantifier defining the degree-zero subgroup on $\sigma_1 T$ for a quantifier over L -field-points of T . For the last two statements, C is a curve bundle over k — proper, smooth of relative dimension one, geometrically irreducible — with base change $C_L := \sigma^* C$ over L (??).

Definition 1054 (Pushing a field point forward). Let K be a field in a scalar tower $k \rightarrow L \rightarrow K$ and $s : \text{Spec } K / \text{Spec } L \rightarrow T$ an L -field-point of T . The *pushed field point* is the k -field-point

$$\text{push}(s) : \text{Spec } K / \text{Spec } k \longrightarrow \sigma_1 T$$

obtained by composing the inverse of the affine matching $\sigma_1(\text{Spec } K / \text{Spec } L) \cong \text{Spec } K / \text{Spec } k$ (??) with s pushed forward along σ . Its underlying morphism of schemes is that of s .

Proof. The affine matching has underlying morphism the identity of $\text{Spec } K$ (??), and pushing forward along σ leaves underlying morphisms unchanged; so the composite is a morphism of $\text{Over}(\text{Spec } k)$ whose underlying morphism is that of s . $\square$

Definition 1055 (The algebra structure unpacked from a field point). Let K be a field extension of k and $t : \text{Spec } K / \text{Spec } k \rightarrow \sigma_1 T$ a k -field-point of the pushed test. The composite of the underlying morphism of t with the structure morphism of T is a morphism of schemes $\text{Spec } K \rightarrow \text{Spec } L$, hence Spec of a unique ring homomorphism $L \rightarrow K$, by full faithfulness of Spec . This homomorphism makes K an L -algebra: the *field-point algebra structure* of t , whose algebra map is that unique homomorphism.

Proof. Spec is a fully faithful contravariant functor from commutative rings to schemes, so every morphism $\text{Spec } K \rightarrow \text{Spec } L$ is $\text{Spec } \varphi$ for a unique ring homomorphism $\varphi : L \rightarrow K$, and a ring homomorphism into K equips K with the structure of an L -algebra. $\square$

Lemma 1056 (The unpacked structure completes the tower). *In the situation of Definition ??, write $\varphi : L \rightarrow K$ for the unpacked algebra map. Then $\varphi \circ (k \rightarrow L) = (k \rightarrow K)$: the field-point algebra structure, the given k -structure of L and the given k -structure of K form a scalar tower $k \rightarrow L \rightarrow K$.*

Proof. The point t lies over $\text{Spec } k$: the underlying morphism of t followed by the structure morphism of $\sigma_1 T$ is the structure morphism of the affine test $\text{Spec } K / \text{Spec } k$, which is Spec of $k \rightarrow K$. The structure morphism of $\sigma_1 T$ is that of T followed by σ , so the left-hand side is $\text{Spec } \varphi$ followed by $\text{Spec}(k \rightarrow L)$, which by contravariant functoriality is $\text{Spec}(\varphi \circ (k \rightarrow L))$. Faithfulness of Spec turns $\text{Spec}(\varphi \circ (k \rightarrow L)) = \text{Spec}(k \rightarrow K)$ into the tower identity. $\square$

Definition 1057 (Lifting a field point of the pushed test). In the situation of Definition ??, endow K with the field-point algebra structure of t . Then the underlying morphism of t is an L -field-point

$$\text{lift}(t) : \text{Spec } K / \text{Spec } L \longrightarrow T$$

with the same underlying morphism of schemes as t .

Proof. The required triangle over $\text{Spec } L$ — the underlying morphism of t followed by the structure morphism of T equals Spec of the algebra map $L \rightarrow K$ — holds by the very definition of the field-point algebra structure (??). $\square$

Theorem 1058 (The field-point matching round trips). *The pack and unpack constructions are mutually inverse: (i) for a k -field-point t of $\sigma_1 T$, pushing $\text{lift}(t)$ forward — over the field-point algebra structure of t and its tower — recovers t : $\text{push}(\text{lift}(t)) = t$; (ii) for a field K in a scalar tower $k \rightarrow L \rightarrow K$ and an L -field-point s of T , the field-point algebra structure of $\text{push}(s)$ is the given L -structure of K ; (iii) and $\text{lift}(\text{push}(s))$ has the same underlying morphism of schemes as s .*

Proof. (i) and (iii): push and lift both preserve the underlying morphism of schemes (??, ??), and a morphism of a slice category is determined by its underlying morphism; the tower of Lemma ?? guarantees that source and target objects agree on both sides. (ii): the underlying morphism of $\text{push}(s)$ followed by the structure morphism of T is the underlying morphism of s so followed, which is Spec of the given algebra map $L \rightarrow K$ — the defining triangle of s . By uniqueness in Definition ??, the unpacked algebra map is the given one, so the two L -algebra structures coincide. $\square$

Theorem 1059 (The quantifier exchange for degree zero on a pushed test). *For a class $\lambda \in \text{picEt}(C, \sigma_1 T)$, the following are equivalent: $\lambda \in \text{pic}^0(C, \sigma_1 T)$; and for every field K in a scalar tower $k \rightarrow L \rightarrow K$ and every L -field-point s of T ,*

$$\deg_{\text{push}(s)}(\lambda) = 0.$$

Proof. Membership in $\text{pic}^0(C, \sigma_1 T)$ is vanishing of the degree at every k -field-point of $\sigma_1 T$ (??). Forward: each pushed point $\text{push}(s)$ is such a field point (??). Backward: let $t : \text{Spec } K \rightarrow \sigma_1 T$ be any k -field-point. Endow K with the field-point algebra structure of t ; it completes the tower $k \rightarrow L \rightarrow K$ (??, ??), and over it $t = \text{push}(\text{lift}(t))$ (??, ??). The hypothesis at this tower and at the point $\text{lift}(t)$ gives $\deg_t(\lambda) = \deg_{\text{push}(\text{lift}(t))}(\lambda) = 0$. $\square$

Theorem 1060 (Classes matching in degree at pushed points are simultaneously degree zero). *Let $\lambda \in \text{picEt}(C, \sigma_1 T)$ and $\lambda_L \in \text{picEt}(C_L, T)$ — the étale Picard group of the base-changed curve C_L at the test T , formed over the base field L exactly as in Definition ???. Suppose the two classes match in degree at every pushed field-point pair: for every field K in a scalar tower $k \rightarrow L \rightarrow K$ and every L -field-point s of T ,*

$$\deg_{\text{push}(s)}(\lambda) = \deg_s(\lambda_L).$$

Then $\lambda \in \text{pic}^0(C, \sigma_1 T)$ if and only if $\lambda_L \in \text{pic}^0(C_L, T)$.

Proof. By the quantifier exchange (??), membership of λ is equivalent to the vanishing of $\deg_{\text{push}(s)}(\lambda)$ over all towers K and L -field-points s of T .

Suppose this vanishing holds, and let $s : \text{Spec } K \rightarrow T$ be an L -field-point with K a field extension of L . Restricting the L -structure of K through $k \rightarrow L$ makes K a field in a scalar tower $k \rightarrow L \rightarrow K$, so the matching hypothesis applies: $\deg_s(\lambda_L) = \deg_{\text{push}(s)}(\lambda) = 0$. As s was arbitrary, $\lambda_L \in \text{pic}^0(C_L, T)$ (??, over the base field L).

Conversely, if λ_L is of degree zero at every L -field-point of T , then for every tower K and every L -field-point s , $\deg_{\text{push}(s)}(\lambda) = \deg_s(\lambda_L) = 0$; conclude by the exchange. $\square$

7.73 The base-field comparison of the degree-zero Picard functor

Fix a field embedding $k \rightarrow L$, write $\sigma : \text{Spec } L \rightarrow \text{Spec } k$ for the induced morphism and $\sigma_!$ for the postcomposition functor $\text{Over}(\text{Spec } L) \rightarrow \text{Over}(\text{Spec } k)$, and let C_L denote the base change of the

curve C along $k \rightarrow L$. The base-field shuffle of the plus construction (Definition ??) extends from affine tests to all test objects, respects the degree, and assembles into a natural isomorphism θ between the degree-zero Picard functor of C_L and the pullback of that of C along $\sigma_!$.

Lemma 1061 (The section-ring scalar tower of a pushed test). *Let T be a test object of $\text{Over}(\text{Spec } L)$ and $U \subseteq T_{\text{left}}$ an open. The section ring $\Gamma(T, U)$ carries an L -algebra structure through the structure morphism of T , and a k -algebra structure through the structure morphism of the pushed test $\sigma_!T$ (the same scheme, viewed over $\text{Spec } k$ through σ). These form a scalar tower: the k -structure map is the composite of $k \rightarrow L$ with the L -structure map.*

Proof. The structure morphism of $\sigma_!T$ is the composite of that of T with σ , so its action on sections over U factors as $\Gamma(\text{Spec } k, \top) \rightarrow \Gamma(\text{Spec } L, \top) \rightarrow \Gamma(T, U)$. Under the canonical identifications $\Gamma(\text{Spec } k, \top) \cong k$ and $\Gamma(\text{Spec } L, \top) \cong L$, the first map is the embedding $k \rightarrow L$ and the second is the L -structure map of $\Gamma(T, U)$. $\square$

Definition 1062 (The componentwise lift of the base-field shuffle). Let T be a test object of $\text{Over}(\text{Spec } L)$. The underlying schemes of T and $\sigma_!T$ coincide, so their affine opens and section rings are literally shared, and by the scalar tower of Lemma ?? each section ring $\Gamma(T, U)$ is a legal test algebra for the base-field shuffle. Applying the shuffle $\text{Sh}_{\Gamma(T, U)}$ of Definition ?? componentwise to a compatible family defines a group isomorphism

$$\text{picEt}(C, \sigma_!T) \cong \text{picEt}(C_L, T),$$

the *componentwise lift* of the shuffle.

Proof. A family transported componentwise remains compatible: the shuffle commutes with restriction of sections, since restriction along $U \subseteq V$ is the same ring homomorphism $\Gamma(T, V) \rightarrow \Gamma(T, U)$ for the two base structures, and the shuffle satisfies the restriction square of Theorem ?? at it. Componentwise application of a group isomorphism preserves products, and the two directions are componentwise inverse, so the lift is a group isomorphism. $\square$

Theorem 1063 (Naturality of the componentwise lift in the test object). *For a morphism $f: T' \rightarrow T$ of test objects of $\text{Over}(\text{Spec } L)$, restriction along f commutes with the componentwise lift: restricting a class of $\text{picEt}(C, \sigma_!T)$ along $\sigma_!f$ and lifting equals lifting and restricting along f .*

Proof. The value at an affine open $W \subseteq T'$ of a restricted family is characterized by the pullback property of Definition ??: on every affine sub-open of W contained in the preimage of an affine open V of the target, it restricts to the pullback of the family value at V . The lift of the $\sigma_!f$ -restriction satisfies this property for f : pullback of sections along f and along $\sigma_!f$ are the same ring homomorphism, and the shuffle commutes with both restriction and pullback of sections (Theorem ?? at the section rings, through the tower of Lemma ??). Uniqueness of the characterized value concludes. $\square$

Theorem 1064 (The affine face of the componentwise lift). *Let A be an algebra in a scalar tower $k \rightarrow L \rightarrow A$. Under the affine comparisons of Definition ?? on the two sides, the componentwise lift at the affine test $\text{Spec } A$ over L is the base-field shuffle Sh_A itself, the pushed side being read through the affine matching $\sigma_!(\text{Spec } A/L) \cong (\text{Spec } A/k)$ of Definition ??.*

Proof. Evaluate both sides at the top affine open. On the lifted side the value is the shuffle of the value of the original family at the top open, transported into A by the L -comparison $\Gamma(\text{Spec } A, \top) \cong A$; by the restriction square of the shuffle this is Sh_A applied to the transport of

the original value along the k -restriction of that comparison. On the pushed side, the restriction along the affine matching has value the pullback of the original value along the identity carrier, transported by the k -comparison; the two composite transports $\Gamma((\mathrm{Spec} A/L)_{\mathrm{left}}, \top) \rightarrow A$ agree, both being the canonical identification. $\square$

Theorem 1065 (Degree invariance of the base-field shuffle at a field). *Let K be a field in a scalar tower $k \rightarrow L \rightarrow K$. For every plus class $x \in \mathrm{PicEtAff}(C, K)$,*

$$\deg_K(\mathrm{Sh}_K(x)) = \deg_K(x),$$

where the left degree is that of the base-changed curve C_L over the base field L and the right one that of C over k .

Proof. Represent x by a descent class y on an étale cover E of $\mathrm{Spec} K$; then $\mathrm{Sh}_K(x)$ is represented on the same cover by the relative Picard shuffle of y (Theorem ??). Choose a finite separable field extension M/K refining E through some $j: E \rightarrow M$, and endow M with the composite k - and L -structures of the tower $k \rightarrow L \rightarrow K \rightarrow M$. By the master equality of the plus-class degree (Definition ??) both sides are read at (M, j) : the left side is $\mathrm{relPicDeg}_M$ of the transport along j over L of the shuffled class, which by the naturality square of the relative Picard shuffle (Theorem ??) is the shuffle at M of the transport along j over k of y ; by the degree invariance of the relative Picard shuffle at the field M (Theorem ??) this reads the same integer as the transport itself, which is the right side. $\square$

Theorem 1066 (The degree matching at pushed field points). *Let T be a test object of $\mathrm{Over}(\mathrm{Spec} L)$ and $\lambda \in \mathrm{picEt}(C, \sigma_1 T)$. For every field K in a scalar tower $k \rightarrow L \rightarrow K$ and every L -field-point s of T ,*

$$\deg_{\mathrm{push}(s)}(\lambda) = \deg_s(\mathrm{lift}(\lambda)),$$

where $\mathrm{lift}(\lambda) \in \mathrm{picEt}(C_L, T)$ is the componentwise lift of λ .

Proof. The pushed point factors as the affine matching followed by $\sigma_1 s$, so the left side is the degree at K of the restriction of λ along $\sigma_1 s$, read through the affine matching and the k -side affine comparison. By naturality of the lift (Theorem ??) the restriction of $\mathrm{lift}(\lambda)$ along s is the lift of the restriction of λ along $\sigma_1 s$; by the affine face (Theorem ??) its reading under the L -side affine comparison is the shuffle Sh_K of the k -side reading. The shuffle preserves the degree at the field K (Theorem ??), so the two sides agree. $\square$

Theorem 1067 (The componentwise lift restricts to the degree-zero subgroups). *A class $\lambda \in \mathrm{picEt}(C, \sigma_1 T)$ is of degree zero at every field point if and only if its componentwise lift $\mathrm{lift}(\lambda) \in \mathrm{picEt}(C_L, T)$ is.*

Proof. The pair $(\lambda, \mathrm{lift}(\lambda))$ matches in degree at every pushed field-point pair (Theorem ??), so the simultaneous-vanishing criterion (Theorem ??) applies. $\square$

Definition 1068 (θ : the base-field comparison of the degree-zero Picard functor). The componentwise lift restricts to the degree-zero subgroups (Theorem ??) and so induces, for each test object T of $\mathrm{Over}(\mathrm{Spec} L)$, a group isomorphism

$$\theta_T: \mathrm{pic}^0(C_L, T) \cong \mathrm{pic}^0(C, \sigma_1 T).$$

These are the components of a natural isomorphism of commutative-group-valued functors

$$\theta: \mathrm{pic}_{C/L}^0 \cong \sigma_!^{\mathrm{op}} \circ \mathrm{pic}_{C/k}^0$$

on $\mathrm{Over}(\mathrm{Spec} L)^{\mathrm{op}}$.

Proof. Naturality in T is the naturality of the componentwise lift in the test object (Theorem ??), restricted to the degree-zero subgroups, whose restriction maps are those of the ambient étale Picard groups. $\square$

Definition 1069 (The set-valued form of θ). Composing θ with the forgetful functor from commutative groups to sets yields the natural isomorphism $\text{pic}_{\text{Set}}^0(C_L) \cong \sigma_1^{\text{op}} \circ \text{pic}_{\text{Set}}^0(C)$ of the set-valued degree-zero functors of Definition ??, with the same components as θ .

7.74 The θ -family and the multiplication shift

Fix a Čech Picard class L_0 on the base-changed curve $C_k := (C \otimes \text{Spec } k)_{\text{left}}$. Pulling its plus class back from the base test along the canonical structure maps produces a family θ_T of étale Picard classes, natural in T and of constant degree $d_1 = \deg_{\text{Pic}}(L_0)$ at every field point; multiplication by θ_T^m then shifts the degree-zero functor onto the degree- md_1 layer.

Definition 1070 (The canonical map to the base test). The base test $\text{Spec } k$ (the affine test of the k -algebra k) is terminal in $\text{Over}(\text{Spec } k)$ up to the identification of its structure morphism with the identity: every test object T has a canonical morphism $T \rightarrow \text{Spec } k$ given by its structure morphism, natural in T ($f \circ (\text{canonical}) = \text{canonical}$); on an affine test $\text{Spec } A$ it is Spec of the structure map $k \rightarrow A$.

Proof. The structure morphism of the base test is Spec of id_k , the identity; hence the structure morphism of T is a morphism of $\text{Over}(\text{Spec } k)$ into it. Naturality is the defining triangle of morphisms of $\text{Over}(\text{Spec } k)$; the affine case is the compatibility of Spec with structure maps (Definition ??). $\square$

Definition 1071 (The θ -family). For a Čech class L_0 on C_k , let $\theta_{\text{base}} \in \text{picEt}(C, \text{Spec } k)$ be the affine-comparison preimage of the plus unit of $\pi(L_0)$, and for every test T set

$$\theta_T := (\text{canonical } T \rightarrow \text{Spec } k)^* \theta_{\text{base}} \in \text{picEt}(C, T).$$

The family is natural: $f^* \theta_T = \theta_{T'}$ for every $f: T' \rightarrow T$ — θ pulls back to θ , so twisting by θ_T^m is uniform in the test.

Proof. Naturality: $f^* \theta_T$ is the pullback of θ_{base} along the composite $f \circ (\text{canonical})$, which is the canonical map of T' (Definition ??); conclude by functoriality of restriction (Theorem ??). $\square$

Theorem 1072 (The degree of the θ -family at every field point). *For every test T , every field point $t: \text{Spec } K \rightarrow T$ and every $m \in \mathbb{N}$,*

$$\deg_t(\theta_T) = \deg_{\text{Pic}}(L_0) =: d_1, \quad \deg_t(\theta_T^m) = m d_1,$$

a base-field-invariant constant.

Proof. On an affine test $\text{Spec } A$, the affine collapse of $\theta_{\text{Spec } A}$ is the plus unit of the base-changed class $\iota_A^* \pi(L_0)$: the canonical map is $\text{Spec } \iota_A$ (Definition ??), the affine comparison intertwines restriction with the explicit transport (Theorem ??), and the transport is natural on units (Definition ??). Base-field constancy (Theorem ??) then reads $\deg = \deg_{\text{Pic}}(L_0)$ at every field point of $\text{Spec } A$. For a general test T and field point t , naturality of the family (Definition ??) plus functoriality of the degree (Theorem ??) reduce $\deg_t(\theta_T)$ to the degree of $\theta_{\text{Spec } K}$ at the identity point of $\text{Spec } K$, covered by the affine case. The power law is Theorem ??). $\square$

Definition 1073 (The degree- d layer functor). For $d \in \mathbb{Z}$, the *degree- d layer*

$$\mathrm{pic}^d(C, T) := \{ \lambda \in \mathrm{picEt}(C, T) : \deg_t(\lambda) = d \text{ for every field point } t \},$$

a *set-valued* subfunctor $\mathrm{Over}(\mathrm{Spec} k)^{\mathrm{op}} \rightarrow \mathrm{Set}$ of the étale Picard functor. For $d \neq 0$ the layer is a coset of the degree-zero subgroup, not a subgroup: it carries no group structure.

Proof. Stability of the degree- d condition under restriction is functoriality of the degree (Theorem ??); the functor laws are inherited from picEt Functor (Theorem ??). $\square$

Theorem 1074 (The multiplication-by- θ^m natural isomorphism and the shift of a representation). For every $m \in \mathbb{N}$, multiplication by θ_T^m is a bijection

$$\mathrm{pic}^0(C, T) \xrightarrow{\sim} \mathrm{pic}^{md_1}(C, T), \quad \lambda \mapsto \lambda \cdot \theta_T^m,$$

natural in T : an isomorphism of set-valued functors from (the underlying-set functor of) $\mathrm{Pic}_{C/k}^0$ onto the degree- md_1 layer functor. Consequently a representation of the shifted layer functor by an object J transports to a representation of $\mathrm{Pic}_{C/k}^0$ (as a set-valued functor) by J .

Proof. Forward: $\deg_t(\lambda \theta_T^m) = m \deg_t(\theta_T) = md_1$ at every field point (Theorem ??, Theorem ??). Backward, $\mu \mapsto \mu \cdot (\theta_T^m)^{-1}$ lands in degree zero by additivity (Lemma ??); the two maps are mutually inverse since the ambient group is commutative. Naturality is naturality of the θ -family (Definition ??): restriction is a homomorphism carrying θ_T^m to $\theta_{T'}^m$. Transport of a representation along a natural isomorphism of set-valued functors is formal. $\square$

Definition 1075 (The pinned Θ -class). The base-changed curve $C_k = (C \otimes \mathrm{Spec} k)_{\mathrm{left}}$ is again a curve over k through the second projection (Theorem ??), so it carries a finite dominant morphism $\pi: C_k \rightarrow \mathbb{P}^1$ (Theorem ??). The *pinned Θ -class* is the unit fiber twist

$$\Theta := \Theta_1(\pi) = \mathrm{fiberTwist}(\pi, 1) \in \mathrm{Pic}(C_k),$$

and it satisfies $1 \leq \deg_{\mathrm{Pic}}(\Theta)$. It is the canonical choice of L_0 for the θ -family; its degree $d_1 := \deg_{\mathrm{Pic}}(\Theta) \geq 1$ is the shift constant.

Proof. Positivity is the Θ -positivity theorem (Theorem ??) applied to the finite dominant π on the curve C_k . $\square$

7.75 The Zariski sheaf property of the degree-zero functor on the slice

The functor picEt satisfies the Zariski sheaf axioms for arbitrary open covers of arbitrary tests — separation was established as Theorem ??; here we add gluing, and the S-lemma making the degree-zero subfunctor a subsheaf.

Definition 1076 (Compatible local data on an open cover). Let T be a test object and $(O_i)_{i \in \iota}$ an open cover of T_{left} . *Compatible local data* on the cover is a family assigning to every index i and every affine open $W \leq O_i$ a plus class $v_{i,W} \in \mathrm{PicEtAff}(C, \Gamma(T, W))$, such that (*res*) within a member, $v_{i,W}$ restricts to $v_{i,W'}$ for $W' \leq W \leq O_i$, and (*glue*) across members, $v_{i,W} = v_{j,W}$ whenever $W \leq O_i$ and $W \leq O_j$. A plus class z at an affine open W is a *glue value* of the data when its restriction to every affine sub-open $W_0 \leq W$ contained in some member O_i equals v_{i,W_0} .

Theorem 1077 (Uniqueness of glue values). *If the O_i cover T_{left} , then any two glue values of the same data at the same affine open agree.*

Proof. Choose a finite family of basic opens of W covering W , each contained in some member (Theorem ??, with the subordination predicate “contained in some O_i ”). Two glue values restrict, on each such basic open, to the same local datum; classes agreeing on a spanning family of basic opens agree (Theorem ??). $\square$

Theorem 1078 (Existence of glue values). *Compatible local data on an open cover of T_{left} has a glue value at every affine open of T_{left} .*

Proof. Fix an affine open W and choose a finite basic-open refinement $(D(r))_{r \in s}$ of W subordinate to the cover, with $\sum_{r \in s} (r) = (1)$ (Theorem ??). The local data restricted to the $D(r)$ is a compatible family in the sense of the affine Zariski gluing: on an overlap $D(rr') = D(r) \cap D(r')$ the two restrictions agree by *(res)* within each chosen member followed by *(glue)* across the two members. By the affine gluing theorem (Theorem ??) there is a plus class z at W restricting to the data on each $D(r)$.

z is a glue value: given an affine sub-open $W_0 \leq W$ with $W_0 \leq O_i$, refine W_0 by basic opens of W_0 each of the form $D(q) \subseteq D(r)$ for some $r \in s$ (Theorem ?? again). On each such $D(q)$, the restriction of z computes through $D(r)$ as the local datum at $D(q)$ by *(res)*, and the local datum at $D(q)$ read through member i agrees with that read through the member of r by *(glue)*. So the restrictions of z and of v_{i,W_0} to a spanning family of basic opens of W_0 agree, and Theorem ?? concludes. $\square$

Theorem 1079 (The gluing half of the Zariski sheaf property of picEt). *Compatible local data on an open cover of T_{left} glues to a unique section $s \in \text{picEt}(C, T)$ whose value at every affine open subordinate to a member is the local datum there.*

Proof. Take s to be the family of glue values (existence: Theorem ??; they are unique by Theorem ??). The family is compatible under restriction of affine opens — the restriction of the glue value at V to $U \leq V$ is itself a glue value at U , since glue-value restrictions compose — so s is a section of $\text{picEt}(C, T)$ (Definition ??). At an affine open $W \leq O_i$, the glue value restricted along $W \leq W$ is the local datum, so s extends the data. Uniqueness: two such sections agree at every affine open subordinate to a member, hence agree by separation (Theorem ??). $\square$

Theorem 1080 (Separation of the degree-zero subfunctor). *Two degree-zero classes on T whose values agree at every affine open subordinate to an open cover of T_{left} are equal.*

Proof. The underlying picEt classes agree by separation (Theorem ??); membership in the subgroup carries no further data. $\square$

Theorem 1081 (The S-lemma: degree-zero membership is Zariski-local). *Let $(f_i : T_i \rightarrow T)$ be a family of test morphisms whose left legs are open immersions with jointly surjective ranges, and $\lambda \in \text{picEt}(C, T)$ a class whose restriction $f_i^* \lambda$ is degree-zero for every i . Then λ is degree-zero. This is what makes the degree-zero subfunctor a subsheaf, not merely a subfunctor, of the Zariski sheaf picEt .*

Proof. Let $t : \text{Spec } K \rightarrow T$ be a field point. Its underlying space is a single point, so its topological image lies in the range of some f_i ; as that left leg is an open immersion, t factors through it: $t = f_i \circ t'$ for a field point t' of T_i (lift along an open immersion containing the image). Then

$$\deg_t(\lambda) = \deg_{t'}(f_i^* \lambda) = 0$$

by functoriality of the degree (Theorem ??) and the hypothesis. As t was arbitrary, $\lambda \in \text{pic}^0(C, T)$. $\square$

7.76 The morphism-form cover bridge

The sheaf statements of the previous section speak *topological* covers — opens of T_{left} and data at subordinate affine opens. The big-site packaging of the next section speaks *morphism-form* covers: families $f_i: T_i \rightarrow T$ of test morphisms with open-immersion left legs carrying sections. This section is the bridge.

Lemma 1082 (The section-ring isomorphism of an open immersion of tests). *Let $f: T' \rightarrow T$ be a test morphism whose left leg is an open immersion, and $V \subseteq T_{\text{left}}$ an open contained in its range. Then pullback of sections $\Gamma(T, V) \rightarrow \Gamma(T', f^{-1}V)$ is a k -algebra isomorphism. Moreover transport of plus classes along any bijective k -algebra map is injective.*

Proof. An open immersion restricts to an isomorphism from $f^{-1}V$ onto V when V lies in the range, and an isomorphism of schemes induces an isomorphism on sections; the algebra structures match because f lies over $\text{Spec } k$. For the second claim, transport along the inverse algebra isomorphism is a two-sided inverse up to the functor laws of Definition ??, so the transport is injective. $\square$

Definition 1083 (The evaluation of a morphism-form family into local data). Let $(f_i: T_i \rightarrow T)$ have open-immersion left legs and let $x_i \in \text{picEt}(C, T_i)$ be a family of sections, *compatible* in the morphism form: for every pair i, j , every test Z and every pair of morphisms $g_i: Z \rightarrow T_i$, $g_j: Z \rightarrow T_j$ with $f_i g_i = f_j g_j$, the two restrictions of the family to Z agree. The *cover value* of the family at an affine open W contained in the range of f_i is the value of x_i at the preimage affine open $f_i^{-1}W$, transported back along the section-ring isomorphism (Lemma ??). The cover values are compatible local data (Definition ??) on the ranges $O_i := \text{range}(f_i)$.

Proof. (res) Restriction within a member commutes with the section-ring transports and with the internal compatibility of the section x_i , by the restriction calculus of the vehicle (Theorem ??) and injectivity of transport along the bijective pullback (Lemma ??).

(glue) Fix a shared affine open $W \leq O_i \cap O_j$ and form the mediating test: the open subscheme of T at W , over $\text{Spec } k$ through the inclusion. The inclusion lifts through both members (an open immersion whose range contains W admits a lift of the inclusion of W), producing g_i, g_j with $f_i g_i = f_j g_j = (\text{inclusion of } W)$. The compatibility hypothesis identifies the two restricted sections on the mediating test; evaluating both at its top affine open and cancelling the common (bijective) evaluation map (Lemma ??) identifies the two cover values. $\square$

Theorem 1084 (The sheaf property of picEt in morphism-cover form). *Let $(f_i: T_i \rightarrow T)$ have open-immersion left legs with jointly surjective ranges. Then two sections of $\text{picEt}(C, T)$ agreeing after restriction to every member agree; and every compatible family (x_i) as in Definition ?? glues to a unique section s with $f_i^* s = x_i$ for all i .*

Proof. Separation. At an affine open subordinate to a member the values of the two sections compute through the member's restriction, which agree; conclude by topological separation (Theorem ??) after cancelling the bijective transports (Lemma ??).

Gluing. The cover values of the family are compatible local data on the ranges (Definition ??), which glue to a unique section s (Theorem ??). That $f_i^* s = x_i$ is checked at every affine open U' of T_i : the value of $f_i^* s$ there computes from the value of s at the image affine open $f_i(U')$, which is the cover value, i.e. the value of x_i at $f_i^{-1}f_i(U') \supseteq U'$, and the internal compatibility of x_i restricts it to U' . Uniqueness follows from separation. $\square$

Theorem 1085 (The sheaf property of the degree-zero functor in morphism-cover form). *In the situation of Theorem ??, a compatible family of degree-zero classes $x_i \in \text{pic}^0(C, T_i)$ glues to a unique degree-zero class $s \in \text{pic}^0(C, T)$ with $f_i^* s = x_i$; and two degree-zero classes agreeing on every member agree.*

Proof. Glue the underlying picEt classes (Theorem ??); the glued class is degree-zero by the S-lemma (Theorem ??), since each restriction is. Both uniqueness claims reduce to the underlying classes. $\square$

7.77 The Σ -extension and the big-site sheaf

A presheaf on the slice $\text{Over}(S)$ extends to a presheaf on the ambient category by summing over structure morphisms. For $S = \text{Spec } k$ and the degree-zero Picard functor this *slice trick* produces the big-site carrier consumed by the local-representability engine, and a representation of the extension descends to one of the original functor.

Definition 1086 (The canonical slice morphism of an equality). Let $\mathcal{C}$ be a category, S an object, and $a = b$ two equal morphisms $T \rightarrow S$. The identity of T is a morphism of slice objects $(T, a) \rightarrow (T, b)$ in $\text{Over}(S)$, the *canonical morphism of the equality*. It satisfies: the canonical morphism of a reflexive equality is the identity; canonical morphisms compose to the canonical morphism of the transitive equality; and every slice morphism $g: X \rightarrow (T, b)$ factors as the canonical morphism of its own compatibility equation followed by the slice morphism induced by its underlying map.

Proof. The identity of T satisfies $\text{id}_T \circ b = a$ precisely because $a = b$; the three laws are immediate from composition of identities. $\square$

Definition 1087 (The Σ -extension of a slice presheaf). For a presheaf F on the slice $\text{Over}(S)$, the Σ -extension $\tilde{F}$ is the presheaf on the ambient category with

$$\tilde{F}(T) := \coprod_{a: T \rightarrow S} F(T, a),$$

an element being a pair $\langle a, \xi \rangle$ of a structure morphism and a fibre element. Restriction along $g: T' \rightarrow T$ composes on the structure component and applies F of the induced slice morphism on the fibre: $g^* \langle a, \xi \rangle = \langle g \circ a, F(g)^* \xi \rangle$. Two Σ -elements over equal structure morphisms are equal iff their fibres correspond under F of the canonical morphism of the equality (Definition ??).

Proof. Functoriality: restriction along the identity composes with a to a , and the induced slice morphism is the identity, so F fixes the fibre. For a composite, the two structure components agree by associativity, and the two fibre components correspond under F of the canonical morphism, since the induced slice morphisms compose accordingly — the transport calculus of Definition ?? mediates without any cast. The final characterisation of equality of Σ -elements is the elimination rule of the coproduct. $\square$

Lemma 1088 (The amalgamation seam and the compatibility descent). *Let F be a presheaf on $\text{Over}(S)$ and $\tilde{F}$ its Σ -extension.*

1. (Amalgamation seam) *For $g: T' \rightarrow T$, $a: T \rightarrow S$, $b: T' \rightarrow S$ with $g \circ a = b$, and elements $s \in F(T, a)$, $y \in F(T', b)$: $g^* \langle a, s \rangle = \langle b, y \rangle$ in $\tilde{F}(T')$ iff s restricts to y along the slice morphism $(T', b) \rightarrow (T, a)$ induced by g .*
2. (Left-leg rule) *For a slice morphism $g: Z \rightarrow (T, a)$ and $x \in F(T, a)$: restriction of $\langle a, x \rangle$ along the underlying map of g is $\langle Z_{\text{struct}}, F(g)^* x \rangle$, the honest slice restriction on the fibre.*
3. (Compatibility descent) *If two Σ -elements $\langle a, x \rangle, \langle b, y \rangle$ have equal restrictions along the underlying maps of a pair of slice morphisms $g_a: Z \rightarrow (T, a)$, $g_b: Z' = Z \rightarrow (T', b)$, then $F(g_a)^* x = F(g_b)^* y$.*

Proof. (i) After substituting the equality $g \circ a = b$, both sides live over the same structure morphism and the characterisation of equal Σ -elements (Definition ??) reduces the claim to the fibres, where the canonical morphism of the reflexive equality acts as the identity. (ii) is (i) applied to the compatibility equation of g , using that the induced slice morphism of the left leg is g (Definition ??, factorisation law). (iii) Combine (ii) on both sides: both restrictions are Σ -elements over the structure morphism of Z , and equality of Σ -elements over one structure morphism gives equality of fibres. $\square$

Theorem 1089 (The Σ -descent of a representation). *Let F be a presheaf on $\text{Over}(S)$ and suppose the Σ -extension $\tilde{F}$ is representable by an object J of the ambient category, with universal element $\langle u_1, u_2 \rangle \in \tilde{F}(J)$ (the image of id_J). Then F is representable on the slice by (J, u_1) , with universal correspondence sending a slice morphism $g: X \rightarrow (J, u_1)$ to the slice restriction $F(g)^*u_2$.*

Proof. Injectivity. If $F(g)^*u_2 = F(g')^*u_2$ then, by the left-leg rule (Lemma ??(ii)), the restrictions of the universal element along the underlying maps of g and g' are the same Σ -element of $\tilde{F}(X_{\text{left}})$; by representability of $\tilde{F}$ the underlying maps agree, hence $g = g'$ (slice morphisms are determined by their underlying maps).

Surjectivity. Given $\xi \in F(X)$, the Σ -element $\langle X_{\text{struct}}, \xi \rangle$ corresponds under the representation to a map $h: X_{\text{left}} \rightarrow J$ with $h^*\langle u_1, u_2 \rangle = \langle X_{\text{struct}}, \xi \rangle$. Comparing structure components, $h \circ u_1 = X_{\text{struct}}$, so h underlies a slice morphism $g: X \rightarrow (J, u_1)$; comparing fibres through the amalgamation seam (Lemma ??(i)) gives $F(g)^*u_2 = \xi$.

Compatibility with composition is functoriality of F . $\square$

Definition 1090 (The big-site carrier of the degree-zero functor). Write $\text{pic}_{\text{Set}}^0(C)$ for the set-valued functor underlying the degree-zero relative Picard functor (Definition ??). Its Σ -extension along the slice $\text{Over}(\text{Spec } k)$,

$$T \longmapsto \coprod_{a: T \rightarrow \text{Spec } k} \text{pic}^0(C, (T, a)), \quad T \in \text{Sch},$$

is a presheaf of sets on the category of schemes — the big-site carrier of the slice trick.

Theorem 1091 (The big-site Zariski sheaf certificate). *The Σ -extension of the degree-zero Picard functor is a sheaf for the big Zariski topology on schemes.*

Proof. The Zariski pretopology is generated by open covers and is stable under base change, so the sheaf condition reduces to elementwise unique amalgamation over an honest open cover $(\iota_i: U_i \rightarrow Y)$ carrying a compatible family $x_i = \langle a_i, \xi_i \rangle$ of Σ -elements.

Structure components. Compatibility on pullbacks forces the a_i to agree on overlaps, so they glue to a unique $a_Y: Y \rightarrow \text{Spec } k$ with $\iota_i \circ a_Y = a_i$ (gluing of scheme morphisms along an open cover).

Fibre components. The cover members induce slice morphisms $(U_i, a_i) \rightarrow (Y, a_Y)$ with open-immersion left legs and jointly surjective ranges. The fibres ξ_i are a compatible family on this slice cover: by the compatibility descent (Lemma ??(iii)), the Σ -level compatibility of the x_i descends to slice-level compatibility of the ξ_i . By the morphism-form sheaf property of the degree-zero functor (Theorem ??) the ξ_i glue to a unique degree-zero class s on (Y, a_Y) restricting to each.

Stitching. The Σ -element $\langle a_Y, s \rangle$ restricts to each x_i by the amalgamation seam (Lemma ??(i)). Uniqueness: the structure component of any amalgamation is determined by gluing of morphisms, and its fibre by the separation half of Theorem ??, again through the seam. $\square$

Definition 1092 (Local representability descends to the slice). Suppose given a family of schemes X_i with maps $f_i: \text{yoneda}(X_i) \rightarrow \widetilde{\text{pic}}^0(C)$ of big-site sheaves which are relatively representable open immersions, jointly Zariski-locally surjective. Then the local-representability theorem glues the X_i into a scheme J representing the Σ -extension, and by Σ -descent (Theorem ??) the set-valued degree-zero Picard functor is representable on the slice by J placed over $\text{Spec } k$ via the structure component of the universal element.

Proof. The local-representability theorem for Zariski sheaves on schemes (mathlib’s 01JJ engine) applies to the sheaf of Theorem ?? and the given chart family, producing a representation of the Σ -extension by the glued scheme; compose with Theorem ??. $\square$

This remains a valid conditional local-representability theorem. Spelled out, its hypothesis asks for a family of schemes X_i together with maps $\text{yoneda}(X_i) \rightarrow \widetilde{\text{pic}}^0(C)$ such that

- each f_i is *relatively representable by an open immersion*: for every scheme T and every Σ -element $\xi \in \widetilde{\text{pic}}^0(C)(T)$, the fibre product $\text{yoneda}(X_i) \times_{\widetilde{\text{pic}}^0(C)} T$ is representable by an open subscheme of T — so each f_i exhibits X_i as an *open piece* of the functor, with the openness verified at every test, not merely at points;
- the family is *jointly Zariski-locally surjective*: every element of $\widetilde{\text{pic}}^0(C)(T)$ is, after restriction to the members of some open cover of T , in the image of some f_i .

These hypotheses are not supplied by the raw admissible Abel map. Its fibres are complete linear systems and can have positive dimension, so that map is neither an open immersion nor an open atlas. Section ?? instead restricts the degree- $g(C)$ Abel map to the locus where H^1 vanishes and the pushforward has rank one. The canonical evaluation divisor identifies that divisor open with the corresponding Picard open. Over a separably closed field, its translates cover every field-valued Picard class and therefore supply the chart family above; finite Galois descent then returns the arbitrary-field representer together with its representation certificate. The larger admissible divisor source remains available for arbitrary-degree divisor geometry, but its positive-dimensional linear systems do not enter this open-atlas construction.

7.78 The Abel element

For a rational point P of the curve, the class of the diagonal minus the constant graph is a degree-zero family on the test $T = C$ — the element whose image under the universal correspondence of the representation is the Abel map.

Theorem 1093 (The unit seam at arbitrary tests). *For a test object T , a relative Picard class $z \in \text{relPic}(C, T)$ and a field point $t: \text{Spec } K \rightarrow T$, the degree at t of the unit image of z is the relative degree of the restriction:*

$$\deg_t(\text{relPicToPicEt}(z)) = \text{relPicDeg}_K(t^*z).$$

Proof. Naturality of the unit (Theorem ??) collapses the restriction along t to the unit image of t^*z over $\text{Spec } K$; affine consistency (Theorem ??) identifies its affine collapse with the plus unit of t^*z ; the unit normalization of the degree (Theorem ??) reads relPicDeg_K . $\square$

Definition 1094 (The Abel Čech class). Let $P: \mathbf{1} \rightarrow C$ be a rational point of the curve (a morphism from the unit test). The *Abel Čech class* on the test $T = C$ is

$$\mathcal{A}_P := \mathcal{O}_{C \otimes C}(\Delta) \cdot \mathcal{O}_{C \otimes C}(P \times C)^{-1} \in \text{Pic}(C \otimes C),$$

the graph class of the identity id_C (the diagonal) times the inverse graph class of the constant point $C \rightarrow \mathbf{1} \xrightarrow{P} C$ — both factors through the graph divisor of Definition ??, so one degree certificate serves both. The *Abel class* $\alpha_P \in \text{picEt}(C, C)$ is the unit image of its relative class. Milne’s cited section assumes $g(C) > 0$; this graph-class construction uses only the rational point P and therefore also applies in genus zero.

Theorem 1095 (The functor-point formula, Čech level). *For any test T and any point $f: T \rightarrow C$, the pullback of the Abel Čech class along $C \triangleleft f$ is the graph-minus-constant-graph class of f :*

$$(C \triangleleft f)^* \mathcal{A}_P = \mathcal{O}_{C \otimes T}(\Gamma_f) \cdot \mathcal{O}_{C \otimes T}(\Gamma_{P_T})^{-1},$$

where $P_T: T \rightarrow \mathbf{1} \xrightarrow{P} C$ is the constant point of T .

Proof. Base change of the graph class (Theorem ??) sends $\mathcal{O}(\Gamma_{\text{id}_C})$ to $\mathcal{O}(\Gamma_{f \circ \text{id}}) = \mathcal{O}(\Gamma_f)$ and $\mathcal{O}(\Gamma_{P_C})$ to $\mathcal{O}(\Gamma_{f \circ P_C})$; and $f \circ P_C = P_T$ by terminality of the unit test (both factor through $\mathbf{1}$). Pullback is a homomorphism, so the product formula follows. $\square$

Theorem 1096 (The degree-zero certificate of the Abel class). *At every field point $t: \text{Spec } K \rightarrow C$, the Abel class has degree zero: $\deg_t(\alpha_P) = 0$. The certificate is a proved equality at every point — no choice is involved.*

Proof. By the unit seam (Theorem ??) the degree is relPicDeg_K of the restriction of the relative Abel class along t , which the relative degree reads as $\deg_{\text{pic}}$ of the fibre Čech class (Definition ?? on the projected representative, Definition ??). By the functor-point formula (Theorem ??) at $f := t$, the fibre class is $\mathcal{O}(\Gamma_t) \cdot \mathcal{O}(\Gamma_{P_{\text{Spec } K}})^{-1}$ — the ratio of two graph classes of K -points. Each has degree one by the degree-one certificate (Theorem ??), so the degree is $1 - 1 = 0$. $\square$

Definition 1097 (The Abel element). The *Abel element* is the Abel class as an element of the degree-zero subgroup,

$$\alpha_P \in \text{pic}^0(C, C),$$

membership being the certificate of Theorem ??. On points it is the family $t \mapsto [\Gamma_t - \Gamma_P]$, the divisor-class form $t \mapsto [t - P]$ of the Abel map with base point P . As above, Milne states this formula under the standing hypothesis $g(C) > 0$, whereas the formal element is defined in genus zero as well.

Theorem 1098 (The functor-point formula of the Abel element). *For any point $f: T \rightarrow C$, the restriction of the Abel element along f is the unit image of the graph-minus-constant-graph class of f :*

$$f^* \alpha_P = \text{relPicToPicEt} \left(\pi(\mathcal{O}_{C \otimes T}(\Gamma_f) \cdot \mathcal{O}_{C \otimes T}(\Gamma_{P_T})^{-1}) \right) \quad \text{in } \text{pic}^0(C, T).$$

Proof. Naturality of the unit (Theorem ??) commutes the restriction into the relative class, where it is the projection of the pullback of the Abel Čech class (Definition ??); conclude by the Čech-level functor-point formula (Theorem ??). $\square$

Theorem 1099 (The pointing law). *Restriction of the Abel element along the rational point P itself is trivial:*

$$P^* \alpha_P = 1 \quad \text{in } \text{pic}^0(C, \mathbf{1}).$$

Proof. By the functor-point formula (Theorem ??) at $f := P$, the restricted class is the unit image of $\mathcal{O}(\Gamma_P) \cdot \mathcal{O}(\Gamma_{P_1})^{-1}$ on $C \otimes \mathbf{1}$. The constant point of the unit test is P itself (terminality: the canonical map $\mathbf{1} \rightarrow \mathbf{1}$ is the identity), so the two graph factors coincide and the Čech class is already 1; its unit image is 1. $\square$

Chapter 8

The divisor scheme and the representability of Pic^0

Grothendieck's construction of the Picard scheme does not represent Pic directly. It represents first the functor of relative effective divisors, and then presents Pic as a quotient of that divisor scheme along the Abel map: the Abel map is a surjection of sheaves whose fibres are linear systems, and a scheme representing Pic is obtained by descending along it (??, and Kleiman's Theorem 4.8 and its proof).

This chapter builds the divisor half of that argument for the curve C/k of Chapter ??, in the form in which the representability seam ?? can consume it. The three steps are:

1. a functor $\mathrm{Div}_{C/k}^n$ of degree- n relative divisors, defined on affine test algebras by *local equations plus a finite chart adaptation carrying a colength certificate*, and its Zariski-localised form (Sections ??–??);
2. an embedding ε of that functor into a product of two Grassmannians, sending a divisor D to the pair of window submodules $H^0(\mathcal{O}(aF - D)) \subseteq H^0(\mathcal{O}(aF))$ at two pinned exponents a , and a closed subscheme **Div** of that Grassmannian pair carved out by the multiplication condition which the image satisfies (Section ??); this is Grothendieck's embedding of a Quot-type functor into a Grassmannian by a twisted pushforward;
3. the *backward* half of representability — every divisor class over an affine test induces a unique morphism to **Div**, and the class is recoverable from that morphism (Section ??).

The forward half — a universal divisor family on **Div** — is not built here. Section ?? states a conditional package for it and isolates the unproved statement (??). It does not by itself produce the open-immersion certificates or the local-surjectivity instance consumed by ??; those are separate hypotheses at the representability seam.

The finite two-chart presentation in this chapter is an auxiliary window engine. The divisor functor used in the representability argument instead permits certificates on arbitrary affine-open covers of the relative curve. Thus the fixed chart cover data below is not the definition of that widened functor.

Throughout, k is a field, C is a smooth proper geometrically irreducible curve over k of genus g , $\pi: C \rightarrow \mathbb{P}_k^1$ is a finite dominant morphism, and F is its fibre Weil divisor (??). For a k -algebra R we write C_R for the relative curve (??) and V_0^R, V_1^R for the two pinned charts of C_R obtained by pulling back the two standard charts of $\mathbb{P}^1$ along π (??, ??). We abbreviate

$H_a := H^0(C, \mathcal{O}(aF))$ (??). These pinned charts occur only in the auxiliary window calculations. Kleiman's Main theorem, which is cited below, asks that the morphism be projective (Zariski locally on the base), flat, and have integral geometric fibres (??); those hypotheses are not synonyms for smooth, proper, and geometrically irreducible, and require an explicit bridge.

8.1 Divisor equality of local-equation systems

A local-equation system (??) is a presentation of an effective Cartier divisor, not the divisor itself: the same divisor is presented by many systems, differing by refinement of the cover and by unit rescaling of the equations. The relation below identifies exactly those, in one move.

Lemma 1100 (Restriction to a refinement). *Let d be a local-equation system on X adapted to a pointed cover $\mathcal{U}$, and let $\mathcal{V} \leq \mathcal{U}$ be a refinement ($\mathcal{V}(x) \subseteq \mathcal{U}(x)$ for every x). Restricting each equation, $f'_x := f_x|_{\mathcal{V}(x)}$, is again a local-equation system, and it has the same Picard class: $\text{picClass}(d|_{\mathcal{V}}) = \text{picClass}(d)$.*

Proof. Restriction of sections is a ring homomorphism and germs are unchanged along it, so each f'_x is again regular at every point of $\mathcal{V}(x)$; restricting the overlap equation $f_x|_{\cap} = g_{xy}f_y|_{\cap}$ of d (??) to $\mathcal{V}(x) \cap \mathcal{V}(y)$ shows the overlap condition, with transition unit the restriction $g_{xy}|_{\cap}$. Thus the unit cocycle of $d|_{\mathcal{V}}$ is the restriction of the cocycle of d along the refinement $\mathcal{V} \leq \mathcal{U}$. In the colimit defining $\text{Pic}(X)$ (??) a cocycle and its restriction along a refinement have the same class, by the definition of the colimit. $\square$

Lemma 1101 (Divisor equality). *Call two local-equation systems d_1, d_2 on X divisor-equal, written $d_1 \sim d_2$, if there is a pointed cover $\mathcal{W}$ refining both covers such that for every $x \in X$ the two restricted equations differ by a unit on $\mathcal{W}(x)$:*

$$f_x^{(1)}|_{\mathcal{W}(x)} = u_x \cdot f_x^{(2)}|_{\mathcal{W}(x)}, \quad u_x \in \mathcal{O}_X(\mathcal{W}(x))^\times.$$

This is an equivalence relation, and picClass is constant on its classes: $d_1 \sim d_2$ implies $\mathcal{O}_X(D_1) = \mathcal{O}_X(D_2)$ in $\text{Pic}(X)$.

Proof. *Reflexivity:* take $\mathcal{W} = \mathcal{U}$ and $u_x = 1$. *Symmetry:* the same $\mathcal{W}$ with u_x^{-1} . *Transitivity:* given $d_1 \sim d_2$ on $\mathcal{W}$ with units u , and $d_2 \sim d_3$ on $\mathcal{V}$ with units v , take the pointwise intersection $\mathcal{W} \wedge \mathcal{V}$, which refines all three covers. Restricting the two equations to it and multiplying,

$$f_x^{(1)} = (u_x|) f_x^{(2)} = (u_x|)(v_x|) f_x^{(3)},$$

and $(u_x|)(v_x|)$ is a unit, being a product of restrictions of units.

Invariance of the class: restrict both systems to the witnessing cover $\mathcal{W}$; by Lemma ?? neither class moves. On $\mathcal{W}$ the first restricted system is the second one rescaled by the unit family $(u_x)_x$, so the two classes agree by Lemma ??. $\square$

8.2 The relative divisor functor of the curve

Fix a k -algebra R . A relative effective divisor on C_R/R is, locally on C_R , cut out by one equation which is regular on the fibres; a local-equation system on C_R records exactly that, but on an *arbitrary* pointed cover, and with regularity asked in the stalks of C_R rather than on the fibres. For the auxiliary pinned-window presentation, a refinement by finitely many basic opens of the two pinned charts is paired with a certificate asserting that the resulting colength

modules are finite projective over R of constant rank n . The widened divisor carrier used by the representability route is a separate quotient in which each local certificate is carried by an arbitrary affine-open cover of the relative curve. The $\text{Div}_{\text{Zar}}^n$ carrier defined below retains the pinned-window certificate and is used only for the auxiliary comparison. In either presentation the certificate is the proposed formal analogue of R -flatness with fibres of degree n in Kleiman's Definition of $\text{Div}_{X/S}^n$.

Definition 1102 (Auxiliary finite chart cover data). This auxiliary pinned-window datum on C_R consists of finitely many sections $h_1^0, \dots, h_{m_0}^0 \in \mathcal{O}(V_0^R)$ and $h_1^1, \dots, h_{m_1}^1 \in \mathcal{O}(V_1^R)$, together with coefficients a_j^t in the same rings satisfying the two partition identities

$$\sum_j a_j^0 h_j^0 = 1 \text{ in } \mathcal{O}(V_0^R), \quad \sum_j a_j^1 h_j^1 = 1 \text{ in } \mathcal{O}(V_1^R).$$

Its *pieces* are the basic opens $P_j := D(h_j^t) \subseteq V_t^R$, indexed by the disjoint union of the two index sets. Each piece is affine, and the pieces of each chart cover that chart: $V_0^R = \bigcup_j D(h_j^0)$ and $V_1^R = \bigcup_j D(h_j^1)$.

Proof. A basic open of an affine open is affine, and the pinned charts are affine over the base, so each P_j is affine.

For the covering claim on V_0^R : let $z \in V_0^R$ and suppose $z \notin D(h_j^0)$ for every j , that is, the germ of every h_j^0 at z lies in the maximal ideal of $\mathcal{O}_{C_R, z}$. Taking germs at z in the partition identity gives $1 = \sum_j (a_j^0)(h_j^0)_z$, which exhibits 1 as an element of the maximal ideal — impossible in a local ring. Hence some h_j^0 is a unit at z , i.e. $z \in D(h_j^0)$. The same argument applies on V_1^R . $\square$

Definition 1103 (Finite chart adaptation of a local-equation system). Let d be a local-equation system on C_R , adapted to a pointed cover $\mathcal{U}$. A *finite chart adaptation* A of d consists of the auxiliary finite chart cover data (??) with pieces P_j , together with sections $f_j \in \mathcal{O}(P_j)$ subject to the *refinement clause*: for every piece index j and every point $y \in C_R$ there is a unit $u \in \mathcal{O}(P_j \cap \mathcal{U}(y))^\times$ with

$$f_j|_{P_j \cap \mathcal{U}(y)} = u \cdot d_y|_{P_j \cap \mathcal{U}(y)}.$$

(The quantifier ranges over *all* y ; over an empty overlap the section ring is the zero ring and the clause is vacuous.)

In particular, anchored data suffice to build an adaptation: if for each j one is given a point p_j with $P_j \subseteq \mathcal{U}(p_j)$ and a unit w_j on P_j with $f_j = w_j \cdot d_{p_j}|_{P_j}$, then the refinement clause holds for every y , with unit the product of $w_j|$ and the restriction of the transition unit $g_{p_j y}$ of d (??).

Proof. Only the last assertion needs an argument. Fix j and y , and restrict everything to $W := P_j \cap \mathcal{U}(y)$. Restricting $f_j = w_j d_{p_j}|_{P_j}$ to W gives $f_j|_W = (w_j|_W) d_{p_j}|_W$. Restricting the defining equation of the transition unit, $d_{p_j}|_{\mathcal{U}(p_j) \cap \mathcal{U}(y)} = g_{p_j y} d_y|_{\mathcal{U}(p_j) \cap \mathcal{U}(y)}$, along $W \subseteq \mathcal{U}(p_j) \cap \mathcal{U}(y)$ gives $d_{p_j}|_W = (g_{p_j y}|_W) d_y|_W$. Substituting, $f_j|_W = (w_j|_W)(g_{p_j y}|_W) d_y|_W$, and the left factor is a unit on W , a product of restrictions of units. $\square$

Lemma 1104 (The adapted equations are regular). *Let A be a finite chart adaptation of d . For every piece index j and every point $z \in P_j$, the germ of f_j at z is a nonzerodivisor in $\mathcal{O}_{C_R, z}$.*

Proof. Apply the refinement clause of ?? at the piece j and at the point $y := z$ itself. The point z lies in P_j and in $\mathcal{U}(z)$, hence in the overlap $W := P_j \cap \mathcal{U}(z)$, so germs at z may be taken in the identity $f_j|_W = u d_z|_W$; germs commute with restriction, so

$$(f_j)_z = u_z \cdot (d_z)_z \text{ in } \mathcal{O}_{C_R, z}.$$

The germ u_z of a unit section is a unit, hence a nonzerodivisor; and $(d_z)_z$ is a nonzerodivisor by the regularity clause of d (??). A product of nonzerodivisors is a nonzerodivisor. $\square$

Definition 1105 (The chart colength modules and their glued equalizer). Let A be a finite chart adaptation of d , with pieces P_j and equations f_j . Its *chart colength modules* are the R -algebras

$$Q_j := \mathcal{O}(P_j)/(f_j),$$

the structure sheaves of the pieces of the divisor cut out by d . On an overlap put $Q_{ij} := \mathcal{O}(P_i \cap P_j)/(f_i|, f_j|)$, the quotient by the ideal generated by *both* restricted equations, and let $\delta^-, \delta^+ : \prod_j Q_j \rightarrow \prod_{(i,j)} Q_{ij}$ be the two restriction maps, δ^- restricting the i -component and δ^+ the j -component. The *glued colength module* is their equalizer

$$W(A) := \ker(\delta^- - \delta^+) \subseteq \prod_j Q_j,$$

an R -submodule; concretely $s \in W(A)$ iff s_i and s_j have the same image in Q_{ij} for every pair (i, j) . It is the module of global sections of the structure sheaf of the divisor over the union of the pieces.

Definition 1106 (The colength certificate of degree n). A finite chart adaptation A is *certified of degree n* if

- (c1) every chart colength module Q_j is finite and projective over R ;
- (c2) the glued colength module $W(A)$ is finite and projective over R , and has constant rank n at every prime: $\text{rank}_{\mathfrak{p}} W(A) = n$ for all $\mathfrak{p} \in \text{Spec } R$;
- (c3) the cokernel of the inclusion $W(A) \hookrightarrow \prod_j Q_j$ is flat over R ;
- (c4) the cokernel of $\delta^- - \delta^+$ is flat over R .

Clauses (c1) and (c2) say that the divisor is finite flat of degree n over the base — Kleiman's condition that a relative effective divisor of degree n have S -flat structure sheaf with fibres of length n . Clauses (c3) and (c4) are what make the certificate stable under an *arbitrary* base change $R \rightarrow R'$: they are the flatness hypotheses under which tensoring the defining left-exact sequence $0 \rightarrow W(A) \rightarrow \prod_j Q_j \rightarrow \prod_{i,j} Q_{ij}$ with R' stays exact, so that the base-changed glued module is again the equalizer of the base-changed arrows. Over a field they are automatic.

Definition 1107 (Certified divisor families and the divisor functor value). A *certified divisor family of degree n* over R is a triple (d, A, c) : a local-equation system d on C_R , a finite chart adaptation A of d , and a certificate c of degree n for A . Divisor equality of the underlying systems (??) is an equivalence relation on such triples — the adaptation and the certificate are presentation data and do not enter the relation — and

$$\text{Div}^n(R) := \{ \text{certified divisor families of degree } n \text{ over } R \} / \sim$$

is the value of this certified subfunctor at R . The Picard class descends to it: $\mathcal{O}(D) \in \text{Pic}(C_R)$ is well defined on $\text{Div}^n(R)$. On a certified triple this is the Abel-class map; it agrees with Kleiman's Abel construction only after the triple is shown to represent a flat effective divisor of degree n . No theorem here says that every effective flat Cartier divisor admits this pinned certificate.

Proof. The relation is an equivalence relation because divisor equality is (??), and it ignores the last two components by definition. The Picard class of a certified family depends only on d , and is constant on divisor-equality classes by the invariance half of ??; hence it factors through the quotient. $\square$

8.3 The Zariski-local certificate

The certificate of ?? is a condition on *global* modules over R , and it is genuinely restrictive: it forces the divisor to sit, chart-trace by chart-trace, in a closed way inside the two pinned charts. The following is the obstruction for this auxiliary pinned certificate. It is not a no-go theorem for the widened functor, whose pieces may be arbitrary affine opens, and it is the reason that carrier is kept separate from this window engine.

Theorem 1108 (The chart obstruction to a global certificate). *Write $\text{Supp}(d) \subseteq C_R$ for the support of d , the closed set of points at which the equation of d is not a unit.*

1. *At a single piece P_j , the chart colength module Q_j is a finite R -module if and only if $\text{Supp}(d) \cap P_j$ is closed in C_R — that is, iff the support does not leak out of the piece.*
2. *Consequently, if A is a certified adaptation of d in any degree, then both pinned-chart traces $\text{Supp}(d) \cap V_0^R$ and $\text{Supp}(d) \cap V_1^R$ are closed in C_R .*
3. *In particular a divisor one of whose chart traces is not closed admits no certified adaptation in any degree; and if some representative of a divisor class has preconnected support containing a point outside V_0^R and a point outside V_1^R , then no representative of that class admits one.*

Proof. Say the *leak* of the piece P_j is $\overline{\text{Supp}(d) \cap P_j} \setminus P_j$; it is empty exactly when $\text{Supp}(d) \cap P_j$ is closed in C_R (a set is closed iff it contains its closure, and the trace is closed in the open P_j always).

(i) The piece P_j is affine (??), and $\text{Supp}(d) \cap P_j$ is the closed subscheme $Z_j := V(f_j) \subseteq P_j$ with coordinate ring Q_j — a point of P_j is a support point exactly when f_j is not a unit there, i.e. when f_j lies in the corresponding prime.

If Q_j is a finite R -module, then $Z_j \rightarrow \text{Spec } R$ is a finite, hence proper, morphism. The structure morphism $C_R \rightarrow \text{Spec } R$ is separated (it is a base change of the proper $C \rightarrow \text{Spec } k$), so the immersion $Z_j \rightarrow C_R$ over $\text{Spec } R$ is proper, and the image of a proper morphism is closed. Hence $\text{Supp}(d) \cap P_j$ is closed in C_R , i.e. the leak is empty.

Conversely, if $\text{Supp}(d) \cap P_j$ is closed in C_R , then Z_j is a closed subscheme of C_R contained in the affine open P_j ; $C_R \rightarrow \text{Spec } R$ is proper, so $Z_j \rightarrow \text{Spec } R$ is proper, and a proper affine morphism is finite. Hence Q_j is a finite R -module.

(ii) Clause (c1) of the certificate supplies finiteness of Q_j at every piece, hence by (i) an empty leak at every piece. The pieces indexed by the first chart cover V_0^R (??), so $\text{Supp}(d) \cap V_0^R$ is the finite union $\bigcup_j (\text{Supp}(d) \cap P_j)$ of sets closed in C_R , hence closed in C_R . Likewise for V_1^R .

(iii) The first assertion is the contrapositive of (ii). For the second, first note that divisor equality does not move the support: on a common refinement the two equations differ by a unit at every point, and being a unit is a germ-local condition, so the two systems have the same non-unit locus. Hence the connectivity hypothesis and the two witnesses transport from the given representative to any other, and it suffices to rule out a certified adaptation of a system d whose own support is preconnected and meets the complements of both charts.

Suppose it had one. By (ii) both traces $\text{Supp}(d) \cap V_\ell^R$ are closed in C_R , hence closed in $\text{Supp}(d)$; each is also open in $\text{Supp}(d)$, being the intersection with an open. So each trace is clopen in the preconnected space $\text{Supp}(d)$, hence is empty or everything. If $\text{Supp}(d) \cap V_0^R = \text{Supp}(d)$ then $\text{Supp}(d) \subseteq V_0^R$, contradicting the witness outside V_0^R ; otherwise $\text{Supp}(d) \cap V_0^R = \emptyset$, and since $V_0^R \cup V_1^R = C_R$ this forces $\text{Supp}(d) \subseteq V_1^R$, contradicting the witness outside V_1^R . $\square$

Definition 1109 (The Zariski-local certificate). A local-equation system d on C_R is *locally certified of degree n* if there are finitely many $g_1, \dots, g_m \in R$ generating the unit ideal such that for each i there is a certified divisor family of degree n over the localisation R_{g_i} whose local-equation system is divisor-equal to the pullback of d along $C_{R_{g_i}} \rightarrow C_R$.

Being locally certified is invariant under divisor equality, and every certified family is locally certified.

Proof. The comparison $C_{R_g} \rightarrow C_R$ of an away localisation is an open immersion, so pullback of a local-equation system along it is defined without further hypothesis: the regularity required by ?? holds because an open immersion induces isomorphisms on stalks, and an isomorphism carries nonzerodivisors to nonzerodivisors.

Invariance: if $d \sim d'$ then their pullbacks along each $C_{R_{g_i}} \rightarrow C_R$ are divisor-equal (restrict the witnessing cover and pull the comparison units back; pullback of a unit is a unit), so the recorded divisor equalities for d compose with these to give the ones needed for d' (??, transitivity).

Global implies local: take $m = 1$ and $g_1 = 1$, whose span is the unit ideal. Over R_1 take the base change of the given certified family; the certificate base-changes by clauses (c3)/(c4) of ??, and its local-equation system is the pullback of d on the nose, so the required divisor equality is reflexivity. $\square$

Theorem 1110 (Regularity of the pulled equations of a locally certified system). *Let d be locally certified of degree n over R , and let $R \rightarrow R'$ be an arbitrary morphism of k -algebras. Then the equations of d pulled back along $C_{R'} \rightarrow C_R$ are regular; hence the pullback $d|_{R'}$ is again a local-equation system (??), with no flatness hypothesis on $R \rightarrow R'$.*

Proof. Regularity of a section is a condition on its germ at each point, hence may be checked on any open cover of $C_{R'}$. Let $g_1, \dots, g_m$ be the unit-ideal family certifying d . Their images $\varphi(g_i) \in R'$ again generate the unit ideal, so the open immersions $C_{R'_{\varphi(g_i)}} \rightarrow C_{R'}$ cover $C_{R'}$; it suffices to prove regularity of the pulled germs at points of each such chart.

Fix i . The localisation map factors: $R \rightarrow R_{g_i} \rightarrow R'_{\varphi(g_i)}$, the second arrow being the induced map of away localisations. Correspondingly

$$C_{R'_{\varphi(g_i)}} \rightarrow C_{R_{g_i}} \rightarrow C_R$$

composes to the restriction of $C_{R'} \rightarrow C_R$, so the pulled equations over $R'_{\varphi(g_i)}$ are the equations of d pulled first to $C_{R_{g_i}}$ and then along the second arrow. On $C_{R_{g_i}}$ the pulled system is divisor-equal to the local-equation system of a *certified* family G_i by hypothesis. For a certified family the pulled equations along an arbitrary base change are regular: clause (c1) of the certificate makes each chart colength module projective, so the multiplication-by- f_j sequence $0 \rightarrow \mathcal{O}(P_j) \rightarrow \mathcal{O}(P_j) \rightarrow Q_j \rightarrow 0$ is split as R -modules and stays exact after $-\otimes_R R'$, which is precisely the statement that the pulled equation is again a nonzerodivisor. Finally, divisor equality is preserved by pullback and units pull back to units, so regularity transports from the pulled equations of G_i to those of d . $\square$

Definition 1111 (The auxiliary locally certified divisor functor $\text{Div}_{\text{Zar}}^n$). The value at R of the *locally certified* divisor functor is

$$\text{Div}_{\text{Zar}}^n(R) := \{ d \text{ locally certified of degree } n \text{ on } C_R \} / \sim,$$

the quotient by divisor equality (well posed by ??). The Picard class descends to it, and passing from a global to a local certificate gives a natural comparison $\text{Div}^n(R) \rightarrow \text{Div}_{\text{Zar}}^n(R)$ compatible with the Picard class.

This is the auxiliary carrier used by the statements in this section. The widened representability route uses the separate arbitrary-affine carrier described above. The reason for retaining this auxiliary carrier is Theorem ??: the condition “carries a global certificate” is not local on $\text{Spec } R$, so Div^n cannot be a Zariski sheaf on affine test algebras, whereas $\text{Div}_{\text{Zar}}^n$ is local on the base by construction.

Proof. Divisor equality is an equivalence relation on all local-equation systems (??) and preserves the property of being locally certified (??), so it restricts to an equivalence relation on the locally certified ones. The Picard class is constant on divisor-equality classes (??), so it factors through the quotient. The comparison sends the class of a certified family (d, A, c) to the class of d , which is locally certified by the last part of ??; it is well defined because both sides are quotients by the same relation, and it commutes with the Picard class because both are computed from d . $\square$

Definition 1112 (The widened arbitrary-affine divisor functor). For an affine test algebra R , the $R2$ carrier is the quotient

$$\text{Div}_{\text{Aff}}^n(R) := \{d \text{ on } C_R \text{ with an arbitrary affine-open cover carrying a degree-}n \text{ certificate}\} / \sim,$$

where the equivalence relation is equality of local divisors. The cover is part of the presentation and may vary with the divisor; no pair of pinned $\mathbb{P}^1$ -charts is required. Base change makes these values into the presheaf $\text{divFunctorAff}(C, n)$. Its representability target is therefore

$$\text{divFunctorAff}(C, n) \text{ is represented by } D,$$

not a representation of the auxiliary $\text{Div}_{\text{Zar}}^n$ carrier above. The carrier construction alone does not supply a representation witness. The unconditional witnesses at degree $g(C)$ and at the admissible parameter are recorded in Definitions ?? and ??.

8.4 The window embedding and the divisor scheme

Grothendieck’s route from a Quot-type functor to a scheme is: twist high enough that the relevant H^1 vanishes, push forward, and read the result as a point of a Grassmannian; then cut out the image by the condition that the submodule be closed under multiplication by sections of a further twist. The window engine in this section is deliberately the auxiliary pinned-chart calculation for $\text{Div}_{\text{Zar}}^n$; it is not an identification with the widened carrier of Definition ??.

Fix two exponents a with $H^1(C_R, \mathcal{O}(aF)) = 0$ universally — concretely the two *pinned window exponents* M and $M + s$ attached to π and the genus g , which are chosen past the bound at which the fibre twists have vanishing relative H^1 and the multiplication maps are surjective.

Definition 1113 (The affine Grassmannian functor). For a finite-dimensional k -vector space H and $n \in \mathbb{N}$, the *Grassmannian functor* $\text{Gr}(H, n)$ assigns to a k -algebra R the set of R -submodules $K \subseteq R \otimes_k H$ whose quotient $(R \otimes_k H)/K$ is finite projective over R of constant rank n at every prime. A k -algebra map $R \rightarrow R'$ acts by $K \mapsto \text{im}(R' \otimes_R K \rightarrow R' \otimes_k H)$. Corank rather than rank is the right index because it is what base-changes: a projective quotient of constant rank stays one.

Definition 1114 (The window submodule and the embedding ε). Let a be an exponent at which the relative twist $\mathcal{O}(aF)$ on C_R has vanishing H^1 , so that base change identifies

$$R \otimes_k H_a \xrightarrow{\sim} H^0(C_R, \mathcal{O}(aF)).$$

For a local-equation system d on C_R , the *window submodule*

$$K_a(d) \subseteq R \otimes_k H_a$$

is the preimage under this identification of the submodule of sections of $\mathcal{O}(aF)$ vanishing along d — i.e. of $H^0(C_R, \mathcal{O}(aF - D))$, read inside the free window. It depends only on d , not on any chart adaptation, and it is unchanged when d is replaced by a divisor-equal system; hence it descends to $\text{Div}^n(R)$ and to $\text{Div}_{\text{Zar}}^n(R)$.

Taking the two pinned exponents gives the *window pair*

$$\varepsilon(D) = (K_M(D), K_{M+s}(D)) \in \left(R \otimes_k H_M \text{ submodules} \right) \times \left(R \otimes_k H_{M+s} \text{ submodules} \right).$$

Proof. Vanishing along a divisor is a condition on the section, not on its presentation: a section σ of $\mathcal{O}(aF)$ lies in $H^0(\mathcal{O}(aF - D))$ iff on each member of the cover of d the local expression of σ is divisible by the equation of d . If $d \sim d'$, then on a common refinement the two equations differ by a unit at every point, and divisibility by an element is unchanged when the element is multiplied by a unit; divisibility is local on the cover, so the two vanishing submodules agree. Hence $K_a(-)$ is constant on divisor-equality classes and factors through the quotients of ?? and ??. $\square$

Lemma 1115 (Equal rank at every stalk forces equality). *Let R be a commutative ring.*

1. *A surjection $\varphi: M \twoheadrightarrow N$ of R -modules with M finite projective and N projective, whose ranks at every prime agree, is an isomorphism.*
2. *Consequently, if $K_1 \subseteq K_2$ are submodules of M with M/K_1 finite projective, M/K_2 projective, and $\text{rank}_{\mathfrak{p}}(M/K_1) = \text{rank}_{\mathfrak{p}}(M/K_2)$ for every prime $\mathfrak{p}$, then $K_1 = K_2$.*

Proof. (i) Since N is projective, φ splits: there is $\ell: N \rightarrow M$ with $\varphi\ell = \text{id}_N$, and $M \cong \ker \varphi \oplus N$. Hence $\ker \varphi$ is a direct summand of the finite projective module M , so it is itself finite projective, and rank is additive over direct sums: $\text{rank}_{\mathfrak{p}} \ker \varphi + \text{rank}_{\mathfrak{p}} N = \text{rank}_{\mathfrak{p}} M = \text{rank}_{\mathfrak{p}} N$ for every $\mathfrak{p}$, so $\text{rank}_{\mathfrak{p}} \ker \varphi = 0$ identically. A finite projective module of rank 0 at every prime is trivial (it is trivial after localisation at every prime, and triviality of a module is local). Hence $\ker \varphi = 0$ and φ is an isomorphism.

(ii) The canonical map $M/K_1 \rightarrow M/K_2$ is surjective, and (i) applies to it, so it is injective. If $x \in K_2$, its class in M/K_1 maps to 0, hence is 0, i.e. $x \in K_1$. Thus $K_2 \subseteq K_1$, and the reverse inclusion is the hypothesis. $\square$

Theorem 1116 (The window identity). *Let A be a finite chart adaptation of d over R , certified of degree n , let a be an exponent as in ??, and assume the evaluation map*

$$R \otimes_k H_a \longrightarrow W(A)^{(a)}$$

onto the $\mathcal{O}(aF)$ -twisted glued colength module is surjective. Then:

1. *$(R \otimes_k H_a)/K_a(d) \cong W(A)^{(a)}$; in particular $K_a(d)$ is a point of $\text{Gr}(H_a, n)$ over R ;*
2. *any point x of $\text{Gr}(H_a, n)$ over R with $x \subseteq K_a(d)$ equals $K_a(d)$.*

Consequently, for a certified family whose two window submodules contain a given pair of Grassmannian points at the pinned exponents, ε of the family is that pair.

Proof. (i) By definition $K_a(d)$ is the kernel of the evaluation map: a section of $\mathcal{O}(aF)$ vanishes along d exactly when its image in each chart colength module — hence in their glued equalizer — is zero. The evaluation is surjective by hypothesis, so the first isomorphism theorem gives the displayed isomorphism. The certificate makes $W(A)^{(a)}$ finite projective of constant rank n at every prime: this is clause (c2) transported across the twist by $\mathcal{O}(aF)$, which is an invertible sheaf and hence does not change finiteness, projectivity or rank of the glued colength module. So the quotient of $R \otimes_k H_a$ by $K_a(d)$ is finite projective of constant rank n , which is exactly membership in $\text{Gr}(H_a, n)(R)$ (??).

(ii) Both $(R \otimes_k H_a)/x$ and $(R \otimes_k H_a)/K_a(d)$ are finite projective of constant rank n — the first because x is a Grassmannian point, the second by (i) — so they have equal rank at every prime, and Lemma ??(ii) applied to the inclusion $x \subseteq K_a(d)$ gives $x = K_a(d)$.

The last sentence is (ii) applied at each of the two pinned exponents separately, the certificate slots at both exponents being supplied by the family's own certificate. $\square$

Theorem 1117 (ε is injective on divisor classes). *Assume the base-field normalisations $h^0(\mathcal{O}_C) = 1$ and $\chi(\mathcal{O}_C) = 1 - g$. Let G, G' be certified divisor families of degree g over an arbitrary k -algebra R — no Noetherian hypothesis — with $\varepsilon(G) = \varepsilon(G')$. Then G and G' are divisor-equal; that is, ε is injective on $\text{Div}^g(R)$.*

Proof. *Step 1: stalk ideals determine the class.* Two local-equation systems with the same ideal $(d_z)\mathcal{O}_{C_R, z}$ in every stalk are divisor-equal. Indeed, fix a point z ; the two equations generate the same stalk ideal, so each is a unit multiple of the other in the stalk, and a germ-level unit factor lifts to a unit on some neighbourhood of z contained in both members. Letting $\mathcal{W}(z)$ be such a neighbourhood defines a pointed cover refining both, on which the equations differ by a unit at every point — which is divisor equality (??). Descending through the quotient of ?? gives equality in $\text{Div}^g(R)$.

Step 2: the window generates. For a certified family the first window generates the divisor: at every point z of C_R the stalk ideal of d is generated by the germs at z of the sections in $K_M(d)$, divided by a local generator of $\mathcal{O}(MF)$. This is where the exponent M is used: it is past the bound at which $\mathcal{O}(MF - D)$ is relatively globally generated for every degree- g certified family, and global generation of $\mathcal{O}(MF - D)$ says exactly that the image of $H^0(\mathcal{O}(MF - D)) \otimes \mathcal{O} \rightarrow \mathcal{O}(MF)$ is the subsheaf $\mathcal{O}(MF - D)$, whose local generator is (a local generator of $\mathcal{O}(MF)$ times) the equation of D . The normalisations $h^0 = 1$, $\chi = 1 - g$ are what pin the numerical bound.

Conclusion. If $\varepsilon(G) = \varepsilon(G')$ then in particular $K_M(G) = K_M(G')$, so by Step 2 the two systems have equal stalk ideals at every point, and by Step 1 they are divisor-equal. $\square$

Definition 1118 (The divisor scheme **Div**). Choose k -bases of H_M and H_{M+s} , giving the Grassmannian functors $\text{Gr}(H_M, g)$ and $\text{Gr}(H_{M+s}, g)$ their standard affine charts indexed by coordinate subsets, and let Gr-pair be the product of the two chart-glued Grassmannian schemes. For $\sigma \in H_s = H^0(C, \mathcal{O}(sF))$, multiplication by σ is a k -linear map $\mu_\sigma: H_M \rightarrow H_{M+s}$.

The *divisor scheme* **Div** is the closed subscheme of Gr-pair cut out, chart by chart, by the ideal generated by the matrix entries of the composites

$$K_1 \hookrightarrow R \otimes_k H_M \xrightarrow{\text{id} \otimes \mu_\sigma} R \otimes_k H_{M+s} \twoheadrightarrow (R \otimes_k H_{M+s})/K_2, \quad \sigma \in H_s,$$

where (K_1, K_2) is the tautological pair on the chart. Its R -points are therefore the pairs (K_1, K_2) of Grassmannian points of corank g satisfying $\mu_\sigma(K_1) \subseteq K_2$ for every $\sigma \in H_s$. It comes with a closed immersion **Div** $\hookrightarrow$ Gr-pair ; in particular morphisms into **Div** are determined by their composites with that immersion. Placed over $\text{Spec } k$ by its structure morphism, it is an object $\mathbf{Div}_{/k}$ of $\text{Over}(\text{Spec } k)$.

Proof. The description of the R -points is the standard reading of a determinantal-type curve on a Grassmannian chart: on the chart the tautological submodules are free with an explicit basis, the composite above is an R -linear map between free modules, and a map between free modules vanishes exactly when all its matrix entries do. Hence a k -algebra map from the chart ring kills the curve ideal iff the base-changed composite is zero for every σ , i.e. iff $\mu_\sigma(K_1) \subseteq K_2$. The curve ideals on overlapping charts agree because the condition they express is chart-independent, so they glue to a quasi-coherent ideal sheaf and the closed subschemes glue. Determination of morphisms by their composite with the immersion holds because a closed immersion is a monomorphism. $\square$

The relation of ?? to ?? is the reason for the curve: if D is an effective divisor then $\sigma \cdot H^0(\mathcal{O}(MF - D)) \subseteq H^0(\mathcal{O}((M + s)F - D))$ for every $\sigma \in H^0(\mathcal{O}(sF))$, so $\varepsilon(D)$ is an R -point of **Div**. The content of the construction is the converse direction, that a point of **Div** determines a divisor; that is the forward map of Section ??, which is not built.

8.5 The classifying morphism

This section is the backward half of representability: a divisor class over an affine test determines a morphism to **Div**, uniquely, and the class is recoverable from the morphism. Both directions are proved.

Definition 1119 (The characterizing clause of a classifying morphism). Let S be a k -algebra, $D \in \text{Div}_{\text{Zar}}^g(S)$, and $v: \text{Spec } S \rightarrow \mathbf{Div}$ a morphism. Say v *classifies* D if for every k -algebra T under S , every certified family G over T whose class is the restriction of D to T , and every presentation of $\varepsilon(G)$ as a point of a chart of Gr-pair — that is, a k -algebra map w from the chart ring to T carrying the tautological pair to $\varepsilon(G)$ — the square

$$\text{Spec } T \longrightarrow \text{Spec } S \xrightarrow{v} \mathbf{Div} \hookrightarrow \text{Gr-pair}$$

agrees with the chart morphism $\text{Spec } T \xrightarrow{w} \text{Gr-pair}$.

Quantifying over all tests T , all certified representatives and all chart presentations makes the clause stable under refinement of covers and change of representative by construction.

Theorem 1120 (Existence and uniqueness of the classifying morphism). *For every k -algebra S and every $D \in \text{Div}_{\text{Zar}}^g(S)$ there is a unique morphism $\text{Spec } S \rightarrow \mathbf{Div}$ classifying D in the sense of ??. It lifts canonically to a morphism*

$$\gamma_S(D): \text{Spec } S \longrightarrow \mathbf{Div}/_k$$

in $\text{Over}(\text{Spec } k)$, and any morphism of $\text{Over}(\text{Spec } k)$ whose underlying scheme morphism classifies D equals $\gamma_S(D)$.

Proof. Existence. By ?? there is a finite family $r_1, \dots, r_m \in S$ generating the unit ideal such that over each S_{r_p} the restriction of D has a certified representative G_p ; refining that family further we may also assume that $\varepsilon(G_p)$ admits a chart presentation w_p over S_{r_p} , since the Grassmannian charts cover Gr-pair and a point of a scheme covered by opens lies in one of them after a further localisation of the base. Over each S_{r_p} the chart presentation kills the curve ideal — by the multiplication inclusion recorded after ?? — so it factors through the closed immersion of ??, giving $v_p: \text{Spec } S_{r_p} \rightarrow \mathbf{Div}$ satisfying the clause for G_p .

The v_p agree on overlaps: on $\text{Spec } S_{r_p r_q}$ both restrictions satisfy the clause for the restriction of D , and the uniqueness argument below (applied over the ring $S_{r_p r_q}$) identifies them. Gluing

along the affine open cover $\{\mathrm{Spec} S_{r_p}\}$ of $\mathrm{Spec} S$ yields v , and the clause for v follows from the clause for each v_p because it may be tested after pulling back to the members of a cover.

Uniqueness. Let v, v' both classify D . Take a family r_p and data G_p, w_p as above. Restricting to $\mathrm{Spec} S_{r_p}$, the clause applied to v and to v' with the same $(T, G, w) = (S_{r_p}, G_p, w_p)$ gives that both composites $\mathrm{Spec} S_{r_p} \rightarrow \mathbf{Div} \hookrightarrow \mathbf{Gr}\text{-pair}$ equal the same chart morphism. Since $\mathbf{Div} \hookrightarrow \mathbf{Gr}\text{-pair}$ is a monomorphism (??), the two restrictions to $\mathrm{Spec} S_{r_p}$ agree; as the $\mathrm{Spec} S_{r_p}$ cover $\mathrm{Spec} S$, $v = v'$.

The lift. The composite of v with the immersion is, over each member of the cover, a chart morphism, and every chart of $\mathbf{Gr}\text{-pair}$ lies over $\mathrm{Spec} k$ compatibly; so v commutes with the structure morphisms to $\mathrm{Spec} k$ and underlies a morphism $\gamma_S(D)$ of $\mathrm{Over}(\mathrm{Spec} k)$. A morphism of $\mathrm{Over}(\mathrm{Spec} k)$ is determined by its underlying scheme morphism, so the last assertion is uniqueness again. $\square$

Theorem 1121 (Separation of divisor classes). *If $D_0, D_1 \in \mathrm{Div}_{\mathrm{Zar}}^g(S)$ are classified by the same morphism $v: \mathrm{Spec} S \rightarrow \mathbf{Div}$, then $D_0 = D_1$. Equivalently, $D \mapsto \gamma_S(D)$ is injective.*

Proof. Choose, as in ??, unit-ideal families $c^{(0)}$ and $c^{(1)}$ in S over whose localisations D_0 , respectively D_1 , have certified representatives with chart-presented ε -pairs. The products $c_p^{(0)} c_q^{(1)}$ again generate the unit ideal (a product of two unit ideals is the unit ideal), so it suffices to prove that D_0 and D_1 have the same restriction to each $S_{c_p^{(0)} c_q^{(1)}}$; membership in $\mathrm{Div}_{\mathrm{Zar}}^g$ is a quotient of local-equation systems and equality of classes is local on $\mathrm{Spec} S$.

Fix such a common localisation B . Over B we have certified representatives G_0 of $D_0|_B$ and G_1 of $D_1|_B$, together with chart presentations w_0 of $\varepsilon(G_0)$ and w_1 of $\varepsilon(G_1)$. Applying the clause of ?? to the single morphism v , once with (G_0, w_0) and once with (G_1, w_1) , shows that w_0 and w_1 present the *same* morphism $\mathrm{Spec} B \rightarrow \mathbf{Gr}\text{-pair}$; hence the two tautological images agree, i.e.

$$\varepsilon(G_0) = \varepsilon(G_1)$$

as pairs of submodules of the free windows over B . By the mono theorem (??) two certified families of degree g over B with equal ε -pairs are divisor-equal, so G_0 and G_1 define the same class; hence $D_0|_B = D_1|_B$.

Note that this argument consumes neither a certificate over the chart ring of $\mathbf{Div}$ nor any universal family: the only certified objects are the local representatives supplied by ??. $\square$

8.6 Affine representability, and what remains

Everything above is the backward direction: from divisor classes to morphisms. Turning $\mathbf{Div}$ into a representing object requires the forward direction — a universal divisor family on $\mathbf{Div}$, from which a divisor class can be pulled back along any test morphism. This section states exactly what that datum is, shows that it suffices, and records that it has not been constructed.

Definition 1122 (The affine pullback package). An *affine pullback package* for $\mathbf{Div}$ consists of, for every k -algebra S , a map

$$\rho_S: \mathrm{Hom}_{\mathrm{Over}(\mathrm{Spec} k)}(\mathrm{Spec} S, \mathbf{Div}_{/k}) \longrightarrow \mathrm{Div}_{\mathrm{Zar}}^g(S),$$

subject to two conditions: $\rho_S(v)$ is classified by v in the sense of ??, for every v ; and ρ is natural, $\rho_B(\varphi \circ v) = \varphi_* \rho_A(v)$ for every k -algebra map $\varphi: A \rightarrow B$.

The canonical way to produce such a ρ is from a *chart family*: a certified divisor family U_{ij} over the coordinate ring of each affine chart of $\mathbf{Div}$, pulled back along a chart presentation of

the given v ; this is well defined precisely when the family is *compatible*, meaning that two chart presentations inducing the same morphism to $\mathbf{Div}$ pull U back to the same class in $\mathrm{Div}_{\mathrm{Zar}}^g$.

Theorem 1123 (The affine equivalence). *An affine pullback package ρ makes $\mathbf{Div}_{/k}$ represent $\mathrm{Div}_{\mathrm{Zar}}^g$ on affine tests: for every k -algebra S the maps ρ_S and γ_S (??) are mutually inverse bijections*

$$\mathrm{Hom}_{\mathrm{Over}(\mathrm{Spec} k)}(\mathrm{Spec} S, \mathbf{Div}_{/k}) \simeq \mathrm{Div}_{\mathrm{Zar}}^g(S),$$

natural in S .

Proof. $\gamma_S \circ \rho_S = \mathrm{id}$: given v , the class $\rho_S(v)$ is classified by v by hypothesis, and by the uniqueness clause of ?? any morphism classifying a class is the classifying morphism of that class; hence $\gamma_S(\rho_S(v)) = v$.

$\rho_S \circ \gamma_S = \mathrm{id}$: given D , both $\rho_S(\gamma_S(D))$ and D are classified by the single morphism $\gamma_S(D)$ — the first by the hypothesis on ρ , the second by the characterizing property of γ_S (??). By separation (??) two classes classified by the same morphism are equal, so $\rho_S(\gamma_S(D)) = D$. (In particular this round-trip law need not be assumed in ??: it is a consequence of the clause law and separation.)

Naturality of the bijection in S is the naturality assumed of ρ . $\square$

Theorem 1124 (The auxiliary fixed-window universal family — unproved). *There exists an affine pullback package for $\mathbf{Div}$ in the sense of ??; equivalently, there is a compatible chart family — a certified divisor family U_{ij} of degree g over the coordinate ring of each affine chart of $\mathbf{Div}$, whose window pair is the tautological pair of that chart, and whose pullbacks along two presentations of one morphism agree in $\mathrm{Div}_{\mathrm{Zar}}^g$.*

This statement is not proved. It is the forward direction of representability: given a point (K_1, K_2) of $\mathbf{Div}$ over R , one must produce the divisor whose windows are K_1 and K_2 . The intended construction takes a generating family of K_1 inside $R \otimes_k H_M$ and reads its common vanishing locus as an effective divisor, with local equations extracted chart by chart; the two obstacles are that this produces the local equations only after a Zariski localisation of R — which is why the target is $\mathrm{Div}_{\mathrm{Zar}}^g$ and not Div^g , and why the global certificate of ?? is unavailable (??) — and that the resulting family's second window must be shown to be exactly K_2 , for which the containment $K_2 \subseteq K_{M+s}$ has to be upgraded from the fibrewise statement to a relative one, an upgrade that fails over nonreduced base rings if attempted fibrewise.

Everything in Sections ??–?? is independent of this statement. This is a conditional result for the pinned-window auxiliary carrier, not the current R2 representability target $\mathrm{div}\mathrm{FunctorAff}(C, n)$.

8.7 The rank-one Abel chart and descent

The Abel fibre over a line-bundle class is its complete linear system. At degree $g(C)$, the locus with $H^1 = 0$ has one-dimensional space of sections, so the fibre is a point and the Abel map can be inverted without choosing a generator. At a larger degree the same fibre generally has positive dimension. Thus the represented degree- $g(C)$ divisor functor supplies the chart, while the arbitrary-degree construction remains auxiliary.

8.7.1 The represented sources and the fibre obstruction

Definition 1125 (The represented genus-degree divisor source). There is a scheme D_g over $\mathrm{Spec} k$ representing $\mathrm{div}\mathrm{FunctorAff}(C, g(C))$. In particular, a morphism $T \rightarrow D_g$ is naturally the same as a relative effective divisor of degree $g(C)$ on C_T/T , presented on an arbitrary affine-open cover.

Definition 1126 (The represented admissible divisor source). There is a parameter n_{adm} , determined by the curve data, and a scheme D_{adm} over $\text{Spec } k$ representing $\text{divFuncAff}(C, n_{\text{adm}})$, without a rational-point or further representability hypothesis.

Definition 1127 (The represented genus-degree Abel map). Sending a relative effective divisor to the class of its associated invertible sheaf gives a natural transformation

$$a_g: h_{D_g} \longrightarrow \text{pic}^{g(C)}(C).$$

More generally the same construction exists in every degree and is compatible with the Sigma-extension from the slice over $\text{Spec } k$ to the big site.

Theorem 1128 (The higher-degree Abel map is not an open chart). *Let $n > g(C)$, fix a representation of the degree- n divisor functor, and fix any degree-calibrated translation of its Abel map. If that divisor functor has a point after a field extension, then the corresponding unrestricted Abel map is not an open immersion of presheaves.*

Proof. Let D be such a divisor. Riemann–Roch gives $h^0(\mathcal{O}(D)) \geq n + 1 - g(C) \geq 2$. Two independent sections determine two distinct effective divisors in the complete linear system of D : if their divisors were equal, their ratio would be a global unit on the proper geometrically integral curve and hence a scalar, contrary to independence. The two divisors therefore have the same Abel class but are unequal. Thus the Abel map is not a monomorphism, whereas every open immersion is a monomorphism. $\square$

Admissible-degree Abel properties

The admissible-degree Abel map remains useful for divisor geometry and for detecting the relative-linear-equivalence relation. The properties below do not by themselves assert the existence of a scheme quotient.

Definition 1129 (The concrete admissible Abel map). Choosing the admissible chart index and reference divisor internally gives a natural transformation from the represented admissible divisor functor to $\text{pic}^0(C)$. Its Sigma-extension is the corresponding map on the big site.

Theorem 1130 (The admissible source is project-local). *The source D_{adm} has a closed immersion into a product of two Grassmannians. Consequently it is quasi-compact, locally of finite type, and separated, and it has a finite affine cover obtained by pulling back standard affine Grassmannian charts.*

Proof. The two universal subbundles associated to an admissible divisor give the classifying map to the product of Grassmannians. Equality of the classified subbundles recovers the divisor datum, so the map is a closed immersion. Each Grassmannian is projective and has a finite standard affine cover. A closed subscheme of their product is separated, locally of finite type, and quasi-compact, and the inverse images of the finitely many product charts give the asserted affine cover. $\square$

Theorem 1131 (Kernel of the admissible Abel map). *Two admissible divisor families have the same Abel value exactly when, on every affine open of the test scheme, the ratio of their Picard classes is pulled back from that affine open. Thus its kernel is relative linear equivalence, including the quotient by line bundles from the base.*

Proof. The two Abel values are obtained from the divisor classes by multiplying by the same fixed reference class, so equality is unchanged after cancelling that class. Equality in the relative Picard functor means precisely that, locally on the test, the two representing invertible sheaves differ by the pullback of an invertible sheaf on the test. Restricting to an affine-open cover yields the stated condition. Conversely those local pullback classes identify the two relative classes and hence the two Abel values. $\square$

Theorem 1132 (Fibre dimension at the admissible degree). *On every field fibre, a divisor of exact admissible degree has vanishing H^1 , section rank $n_{\text{adm}} + 1 - g(C)$, and complete linear system of dimension $n_{\text{adm}} - g(C)$.*

Proof. The admissible bound gives $H^1(\mathcal{O}(D)) = 0$. Riemann–Roch therefore gives $h^0(\mathcal{O}(D)) = \deg D + 1 - g(C) = n_{\text{adm}} + 1 - g(C)$. The complete linear system is the projective space of nonzero sections modulo scalars, so its dimension is one less, namely $n_{\text{adm}} - g(C)$. $\square$

Definition 1133 (The admissible Abel map after big-etale sheafification). Subcanonicity identifies the sheafification of the represented source with the big-etale Yoneda sheaf of D_{adm} . Sheafifying the admissible Abel transformation gives a morphism from this Yoneda sheaf to the big-etale sheafification of the Sigma Picard presheaf.

Theorem 1134 (Kernel after big-etale sheafification). *Two sections have equal images under the sheafified Abel map exactly when the sieve on which their pulled-back divisor classes differ by a line bundle from the base is covering.*

Proof. Equality in a sheafification holds precisely when the equalizer sieve of the two raw sections is covering. By Theorem ??, an arrow belongs to that equalizer exactly when, on every affine open after pullback, the ratio of the two divisor classes comes from the base. Substituting this description into the equalizer criterion gives the assertion. $\square$

Theorem 1135 (Etale-local surjectivity of the admissible Abel map). *Every degree-zero Picard class lifts to an admissible divisor family after an etale cover of the test scheme.*

Proof. After an etale cover, represent the Picard class by an invertible sheaf and apply the fixed admissible translation, obtaining an invertible sheaf of degree n_{adm} . Its first cohomology vanishes by Theorem ??, so its pushforward is locally free, commutes with base change, and has positive rank. The projective bundle of rank-one quotients of the dual pushforward is smooth and surjective over the test. It therefore has a section after a further etale cover. The corresponding nowhere-zero fibrewise section of the invertible sheaf cuts out a relative effective divisor whose translated Abel class is the original class. $\square$

Theorem 1136 (The sheafified Abel map is the coequalizer of its kernel pair). *The sheafified admissible Abel map is the coequalizer of its pullback kernel pair in the category of big-etale sheaves.*

Proof. Etale-local surjectivity says exactly that the sheafified map is an epimorphism. A category of sheaves of sets is a topos, and every epimorphism in a topos is effective: it is the coequalizer of the two projections from its pullback kernel pair. Apply this fact to the sheafified Abel map. $\square$

Theorem 1137 (Geometry of the genus-degree divisor source). *The scheme D_g is projective and geometrically irreducible over k .*

Proof. The degree- $g(C)$ relative effective-divisor functor is represented by the symmetric power $C^{(g(C))}$: a tuple of points is sent to the sum of its graph divisors, and the universal property of the symmetric quotient identifies this with every relative effective divisor after base change. The quotient map $C^{g(C)} \rightarrow C^{(g(C))}$ is finite and surjective. Since $C^{g(C)}$ is projective and geometrically irreducible, its finite symmetric quotient is projective, and its underlying space remains geometrically irreducible. Uniqueness of a representing object identifies this symmetric power with D_g . $\square$

The admissible representer in Definition ?? provides arbitrary-degree divisor geometry, but Theorem ?? precludes using its raw Abel map as the chart below. A quotient of that map would require, as additional mathematics, a represented kernel relation, an effective scheme quotient compatible with base change, and a proof that its Yoneda sheaf is the Picard sheaf. Those assertions do not follow from the admissible-degree properties above.

8.7.2 The intrinsic rank-one locus

Definition 1138 (A rank-one local presentation of a Picard class). Let T be a k -scheme, let $f_T: C_T \rightarrow T$, and let $\lambda \in \text{pic}^{g(C)}(C, T)$. A *rank-one local presentation* of λ consists of an étale cover $p_i: T_i \rightarrow T$ and invertible sheaves $\mathcal{L}_i$ on C_{T_i} whose classes are exactly $p_i^*\lambda$, such that

$$R^1 f_{i*} \mathcal{L}_i = 0, \quad \mathcal{N}_i := f_{i*} \mathcal{L}_i \text{ is invertible,}$$

and the base-change map $q^* \mathcal{N}_i \rightarrow f'_* q_C^* \mathcal{L}_i$ is an isomorphism for every $q: T' \rightarrow T_i$. The equation $[\mathcal{L}_i] = p_i^* \lambda$ is part of the data; a line bundle unrelated to λ is not a presentation of it. The section used below is fixed to be the counit $f_i^* \mathcal{N}_i \rightarrow \mathcal{L}_i$; no local generator of $\mathcal{N}_i$ is included in the presentation.

Lemma 1139 (Changing a local representative). *Suppose $\mathcal{L}$ and $\mathcal{L}'$ represent the same relative Picard class over a test T . After an étale refinement, write $\mathcal{L}' \simeq \mathcal{L} \otimes f_T^* \mathcal{M}$ for an invertible sheaf $\mathcal{M}$ on T . Then $\mathcal{L}$ satisfies the rank-one conditions of Definition ?? if and only if $\mathcal{L}'$ does. Under this equivalence their evaluation morphisms determine the same closed subscheme of C_T .*

Proof. The projection formula gives functorial isomorphisms

$$f_{T*} \mathcal{L}' \simeq f_{T*} \mathcal{L} \otimes \mathcal{M}, \quad R^1 f_{T*} \mathcal{L}' \simeq R^1 f_{T*} \mathcal{L} \otimes \mathcal{M}.$$

Tensoring by the invertible sheaf $\mathcal{M}$ preserves vanishing, invertibility, and all base-change isomorphisms. Under the first displayed isomorphism, the counit $f_T^* f_{T*} \mathcal{L}' \rightarrow \mathcal{L}'$ is the counit for $\mathcal{L}$ tensored by $f_T^* \mathcal{M}$. After tensoring source and target by the inverse line bundle, the two counits therefore have the same vanishing ideal, so their zero schemes agree. $\square$

Definition 1140 (The rank-one Picard subfunctor). Define $\text{PicRankOneOpen}(C, T)$ to be the set of $\lambda \in \text{pic}^{g(C)}(C, T)$ admitting a rank-one local presentation. Pullback of the presenting cover and line bundles makes these sets a subfunctor

$$\text{PicRankOneOpen}(C) \subseteq \text{pic}^{g(C)}(C).$$

Theorem 1141 (Openness of the rank-one Picard locus). *The inclusion $\text{PicRankOneOpen}(C) \hookrightarrow \text{pic}^{g(C)}(C)$ is relatively representable by open immersions, and its formation commutes with arbitrary base change.*

Proof. Pull a test class back to an étale cover on which it is represented by an invertible sheaf $\mathcal{L}$. Upper semicontinuity makes the locus where $H^1(C_t, \mathcal{L}_t) = 0$ open. On that locus, cohomology and base change identifies $f_*\mathcal{L} \otimes k(t)$ with $H^0(C_t, \mathcal{L}_t)$, and Riemann–Roch at degree $g(C)$ gives

$$\dim H^0(C_t, \mathcal{L}_t) = g(C) + 1 - g(C) = 1.$$

Hence $f_*\mathcal{L}$ is invertible there and its formation commutes with every base change. Conversely the rank-one conditions include the same H^1 -vanishing, so this open is exactly the desired locus. Lemma ?? shows that the opens obtained from different representatives agree on a common étale refinement; open subsets descend uniquely along an étale cover. The construction by fibrewise vanishing and base change also shows compatibility with an arbitrary pullback of the test. $\square$

8.7.3 The canonical divisor and the Abel isomorphism

Definition 1142 (The divisor rank-one open). Let $W \subseteq D_g$ be the open subscheme representing the inverse image of $\text{PicRankOneOpen}(C)$ under a_g . Thus a T -point of W is precisely a degree- $g(C)$ relative effective divisor whose associated Picard class belongs to the rank-one locus. Write $\text{DivRankOneOpen}(C) = W$.

Proof. The inverse image of a relatively representable open subfunctor along the represented Abel map is represented by the corresponding open subscheme of D_g . The pointwise description is the defining property of this fibre product. $\square$

Definition 1143 (The restricted rank-one Abel map). Restriction of a_g to W gives a natural transformation

$$\text{rankOneAbel}: h_W \longrightarrow \text{PicRankOneOpen}(C).$$

Definition 1144 (The canonical divisor of a rank-one class). There is a natural transformation

$$\text{divisorOfRankOne}: \text{PicRankOneOpen}(C) \longrightarrow h_W$$

defined by the zero scheme of the canonical evaluation morphism. Its value at λ is finite locally free of degree $g(C)$, commutes with arbitrary base change, and has relative Picard class λ .

Proof. On a member T_i of a rank-one presentation, set $\mathcal{N}_i = f_{i*}\mathcal{L}_i$. The counit $f_i^*\mathcal{N}_i \rightarrow \mathcal{L}_i$ is equivalently a canonical section

$$s_i \in H^0(C_{T_i}, \mathcal{L}_i \otimes f_i^*\mathcal{N}_i^{-1}).$$

On every geometric fibre, base change identifies s_i with the nonzero generator of the one-dimensional vector space $H^0(C_t, \mathcal{L}_{i,t})$. Since the fibre curve is integral, this section is regular. The fibrewise criterion for a relative effective Cartier divisor therefore shows that its zero scheme E_i is a relative effective Cartier divisor. Its fibre degree is the degree of $\mathcal{L}_{i,t}$, namely $g(C)$; hence $E_i \rightarrow T_i$ is proper and quasi-finite, therefore finite, and its flat fibres have constant length $g(C)$, so it is finite locally free of that rank.

On a common refinement, two representatives differ by a pullback from the base. The projection-formula comparison in Lemma ?? identifies their evaluation sections and hence their zero schemes. Effective descent for closed subschemes glues the E_i uniquely to a divisor $E \subset C_T$. The same comparison after an arbitrary map $T' \rightarrow T$ proves that E commutes with base change. Finally,

$$\mathcal{O}_{C_T}(E)|_{C_{T_i}} \simeq \mathcal{L}_i \otimes f_i^*\mathcal{N}_i^{-1},$$

so $\mathcal{O}(E)$ and $\mathcal{L}_i$ differ only by a line bundle pulled back from the base. Their relative Picard classes are equal to λ , which also shows that E is a T -point of W . These constructions are canonical, and therefore assemble into the asserted natural transformation. $\square$

Theorem 1145 (The two rank-one constructions are inverse). *The natural transformations rankOneAbel and divisorOfRankOne are mutually inverse.*

Proof. Starting with a rank-one class λ , the construction of Definition ?? gives a divisor E with $[\mathcal{O}(E)] = \lambda$ in the relative Picard functor. Applying the Abel map therefore returns λ .

Conversely, start with a divisor $E \in W(T)$ and put $\mathcal{L} = \mathcal{O}(E)$. The canonical divisor section gives a morphism $\mathcal{O}_T \rightarrow f_{T*}\mathcal{L}$. Both sides are invertible, and on each geometric fibre this morphism is the nonzero element represented by the divisor E_t ; it is therefore an isomorphism. Under this isomorphism the evaluation section is exactly the canonical section of $\mathcal{O}(E)$, whose zero scheme is E . Thus applying divisorOfRankOne after the Abel map returns E . Both equalities are compatible with pullback, so they are equalities of natural transformations. $\square$

Definition 1146 (The rank-one Abel isomorphism). The mutually inverse transformations form a natural isomorphism

$$\text{rankOneAbelIso}: h_W \xrightarrow{\sim} \text{PicRankOneOpen}(C).$$

Proof. Use rankOneAbel as the forward map and divisorOfRankOne as the inverse; the two triangle identities are Theorem ??. $\square$

Theorem 1147 (The rank-one Abel map is an open immersion). *The composite*

$$h_W \xrightarrow{\text{rankOneAbelIso}} \text{PicRankOneOpen}(C) \hookrightarrow \text{pic}^{g(C)}(C)$$

is relatively representable by open immersions.

Proof. A natural isomorphism is relatively representable by isomorphisms, and the second arrow is relatively representable by open immersions by Theorem ??. The composite of an isomorphism with an open immersion is an open immersion, and the same argument holds after every base change. $\square$

8.7.4 The separably closed cover

Definition 1148 (Translated rank-one opens in degree zero). Let $\mathcal{M}$ be an invertible sheaf of degree $g(C)$ on C . Define

$$U_{\mathcal{M}}(T) = \{\lambda \in \text{pic}^0(C, T) : \lambda \otimes \mathcal{M}_T \in \text{PicRankOneOpen}(C, T)\}.$$

Translation by $\mathcal{M}$ identifies $U_{\mathcal{M}}$ with $\text{PicRankOneOpen}(C)$.

Theorem 1149 (Representability of each translated open). *Each $U_{\mathcal{M}}$ is represented by a scheme, and its inclusion in $\text{pic}^0(C)$ is relatively representable by open immersions.*

Proof. Tensoring by $\mathcal{M}_T$ is a natural isomorphism from the degree-zero layer to the degree- $g(C)$ layer, with inverse tensoring by $\mathcal{M}_T^{-1}$. It carries $U_{\mathcal{M}}$ isomorphically onto $\text{PicRankOneOpen}(C)$, which is represented by W through Definition ??. Transporting that representation and the open-immersion certificate through the translation proves both assertions. $\square$

Theorem 1150 (Translated rank-one opens cover over a separably closed field). *Suppose k is separably closed. For every field extension K/k and every $\lambda \in \text{pic}^0(C, \text{Spec } K)$, there is an invertible sheaf $\mathcal{M}$ of degree $g(C)$ on C such that $\lambda \in U_{\mathcal{M}}(\text{Spec } K)$.*

Proof. Membership in the rank-one locus is étale local, so after a finite separable extension of K we may represent λ by an invertible sheaf $\mathcal{L}$ on the base-changed curve. Fix a positive invertible sheaf $\mathcal{A}$ on C . Serre vanishing gives an n such that $H^1(\mathcal{L} \otimes \mathcal{A}_K^n) = 0$, and increasing n if necessary makes its space of sections nonzero. Write this sheaf as $\mathcal{L}_0$.

If $h^0(\mathcal{L}_j) > 1$, choose a nonzero section s_j . Its zero scheme has finite support. The k -rational points of C are infinite and dense by Theorem ??, so there is a point $P_j \in C(k)$ outside that support. Evaluation at P_j is then onto, and the exact sequence

$$0 \longrightarrow \mathcal{L}_j(-P_j) \longrightarrow \mathcal{L}_j \longrightarrow \mathcal{L}_j|_{P_j} \longrightarrow 0$$

shows that h^0 drops by one while H^1 remains zero. Repeating this finite process gives

$$\mathcal{L}_r = \mathcal{L} \otimes (\mathcal{A}^n(-P_0 - \cdots - P_{r-1}))_K$$

with $h^0(\mathcal{L}_r) = 1$ and $H^1(\mathcal{L}_r) = 0$. Put $\mathcal{M} = \mathcal{A}^n(-P_0 - \cdots - P_{r-1})$, which is defined over k . Riemann–Roch gives $1 = \deg(\mathcal{L}_r) + 1 - g(C)$. Since λ has degree zero, this says $\deg \mathcal{M} = g(C)$. Cohomology and base change now gives a rank-one local presentation of $\lambda \otimes \mathcal{M}_K$. The construction was made after an étale extension of K , so étale descent of membership places the original class in $U_{\mathcal{M}}(\text{Spec } K)$. $\square$

Definition 1151 (The separably closed degree-zero Picard representer). If k is separably closed, there is a scheme J_s over k and a representation

$$h_{J_s} \xrightarrow{\sim} \text{pic}^0(C).$$

Proof. Use the represented opens $U_{\mathcal{M}}$, indexed by the degree- $g(C)$ invertible sheaves on C . Let T be a test scheme, $\lambda \in \text{pic}^0(C, T)$, and $t \in T$. Apply Theorem ?? to the residue field $k(t)$ and the restriction λ_t . It supplies an $\mathcal{M}$ for which t belongs to the open inverse image of $U_{\mathcal{M}}$ in T . As t varies, these inverse images form an open cover of T . Hence the represented open immersions are jointly Zariski-locally surjective. The local representability theorem (Definition ??) glues them and gives the asserted representation. $\square$

8.7.5 Finite Galois descent and the Jacobian datum

Definition 1152 (The separably closed descent package). A separably closed descent package consists of a scheme J_s representing $\text{pic}^0(C)$, its universal Picard class, and certificates that J_s is of finite type, proper, geometrically irreducible, and a group scheme. The representation and the universal class in the package are the same datum, related by Yoneda, rather than two independently chosen witnesses.

Theorem 1153 (The rank-one representer supplies the descent package). *Over a separably closed field, the representation of Definition ?? extends to a separably closed descent package.*

Proof. The commutative-group law on $\text{pic}^0(C)$ transfers uniquely across the representation to a group-scheme law on J_s , and the image of the identity morphism is its universal class. The translated rank-one charts show that J_s is locally of finite type. Its identity point is closed because it is a rational point of a scheme locally of finite type over a field. The diagonal is the inverse image of that point under the difference morphism $J_s \times J_s \rightarrow J_s$, so the group scheme is separated.

Choose a point of C over the separably closed base and use its $g(C)$ -fold divisor to translate the degree- $g(C)$ Abel map to $\text{pic}^0(C)$. After any algebraically closed extension, Riemann–Roch gives

a section of every degree- $g(C)$ line bundle, so this translated Abel map $D_g \rightarrow J_s$ is surjective. It is proper because D_g is projective and J_s is separated. A surjective image of the quasi-compact scheme D_g makes J_s quasi-compact; combined with local finite type, this gives finite type. Properness descends through the surjective proper map: after every base change, the image in the base of a closed subset of J_s is the image of its closed inverse image in D_g , hence is closed. Finally, the geometric fibres of D_g are irreducible by Theorem ??, and their surjective images are irreducible. These are exactly the certificates required by the package. $\square$

Theorem 1154 (Limit and faithfully flat compatibility of pic^0). *The degree-zero etale Picard functor commutes with filtered unions of field extensions and satisfies effective descent along faithfully flat field extensions, compatibly with its universal classes.*

Proof. An invertible sheaf is given on a finite affine cover by finitely many finitely presented rank-one modules, transition isomorphisms, and finitely many cocycle equations. All this data, as well as an isomorphism between two such data, descends to a finite stage of a filtered union. The degree-zero condition is preserved and reflected after field extension because degree is invariant under scalar extension. This proves filtered compatibility. Faithfully flat descent for finitely presented modules descends the local line bundles and their transition maps, while the cocycle equation gives effectivity and uniqueness. Passing to the quotient by line bundles from the base and then to the etale sheafification preserves this descent statement, since both equivalence and effectivity may be checked after a common faithfully flat refinement. $\square$

Theorem 1155 (A finite Galois stage for the Picard representer). *Let k^s be a separable closure of k . The separably closed descent package descends to a finite Galois extension L/k : there are a finite-type proper group scheme J_L representing $\text{pic}^0(C_L)$, its universal class, and a semilinear action of $\text{Gal}(L/k)$ preserving that class and satisfying the cocycle condition. Every orbit of this action is contained in an affine open of J_L .*

Proof. Because J_{k^s} is of finite presentation, it descends to a finite subextension L_0/k . The universal class descends after enlarging L_0 , by Theorem ??. The natural transformation defined by the descended class becomes the given representation after extension to k^s ; faithful flat descent for both the Yoneda functor and the Picard sheaf shows that it is already an isomorphism over L_0 . Properness, the group operations, and their finitely many axioms also descend after one further finite extension. Enlarge L_0 to a finite Galois extension L/k .

For $\sigma \in \text{Gal}(L/k)$, both σ^*J_L and J_L , equipped with their transported universal classes, represent the same functor. Yoneda uniqueness gives a unique semilinear isomorphism between them. The two composites associated to $\sigma\tau$ carry the universal class to the same element, so uniqueness proves the cocycle equation. A proper finite-type group scheme over a field is projective. Given a finite Galois orbit, choose a closed specialization of each of its finitely many points. For a sufficiently high power of an ample line bundle, restriction of global sections to the resulting finite closed subscheme is surjective. Lift the section that is 1 at every member of that subscheme. Its nonvanishing locus contains the chosen closed specializations and therefore the original orbit; as the complement of an ample effective divisor, this locus is affine. This proves the final certificate. $\square$

Lemma 1156 (Stable affine neighbourhoods for a finite action). *Let a finite group G act on a separated scheme X . If every G -orbit is contained in an affine open, then X has a G -stable affine open cover.*

Proof. Fix $x \in X$ and choose an affine open V containing its whole orbit. Every translate gV also contains that orbit. Hence $U_x = \bigcap_{g \in G} gV$ is a G -stable open containing x . A finite intersection

of affine opens in a separated scheme is affine: the intersection of two is the inverse image of the affine $V \times V'$ under the affine diagonal, and induction handles the finite intersection. The opens U_x , as x varies, form the required cover. $\square$

Theorem 1157 (Finite Galois descent of a scheme). *Let L/k be finite Galois and let an L -scheme X_L carry a semilinear Galois action satisfying the cocycle condition. If every orbit lies in an affine open, then there are a k -scheme X and an isomorphism $X \times_k L \simeq X_L$ inducing the given action.*

Proof. By Lemma ??, cover X_L by stable affine opens $U_i = \text{Spec } A_i$. Semilinear Galois descent for algebras gives $B_i = A_i^G$ and an isomorphism $L \otimes_k B_i \simeq A_i$. Since X_L is separated, every overlap $U_i \cap U_j$ and every triple overlap is affine and stable. Applying the same invariant construction to these overlaps descends both open immersions $U_i \cap U_j \rightrightarrows U_i, U_j$; faithful flatness of L/k reflects the open immersions and their cocycle identities. The affine schemes $\text{Spec } B_i$ therefore glue along the descended overlaps to a k -scheme X . The local isomorphisms $L \otimes_k B_i \simeq A_i$ agree on overlaps and glue to $X \times_k L \simeq X_L$. Their defining invariant maps identify the canonical descent action with the prescribed semilinear action. $\square$

Theorem 1158 (Finite Galois descent of a representation). *Let F be a set-valued sheaf on k -schemes satisfying faithfully flat descent, and let L/k be finite Galois. Suppose F_L is represented by X_L , and the universal element is preserved by a semilinear Galois action on X_L satisfying the cocycle condition and the orbit-in-affine hypothesis. Then there are a k -scheme X and a representation $h_X \simeq F$; the pair $(X, h_X \simeq F)$ is returned together, and its base change is the given pair over L .*

Proof. Descend X_L to X by Theorem ??. The invariance of the universal element is precisely a descent datum on its two pullbacks to $X_L \times_X X_L$. The sheaf condition gives a unique element $u \in F(X)$ descending it. Yoneda turns u into a natural map $h_X \rightarrow F$. After base change to L this is the given representing isomorphism. For any test T , apply the sheaf equalizer diagrams for the finite faithfully flat cover $T_L \rightarrow T$ to both h_X and F . The maps on T_L and $T_L \times_T T_L$ are bijections and commute with the two restriction maps; hence the map on their equalizers is a bijection. Thus u represents F . The construction keeps X and this certificate in the same pair, and the glued base-change isomorphism identifies it with the original represented pair over L . $\square$

Definition 1159 (The arbitrary-field degree-zero Picard representer). For every field k , there is a distinguished pair

$$(J, h_J \xrightarrow{\sim} \text{pic}^0(C))$$

obtained by finite Galois descent from the separably closed rank-one construction.

Proof. Apply Theorem ?? to obtain the represented finite Galois model, its action, cocycle, invariant universal class, and orbit-in-affine certificate. The Picard functor has the descent property of Theorem ??, so Theorem ?? returns the required scheme and representation certificate as one pair. This is the asserted distinguished datum. $\square$

Definition 1160 (The Jacobian datum from the chosen representation). Let (J, r) be the pair of Definition ??. The *rank-one descent Jacobian datum* is the Jacobian datum whose scheme is J , whose representation is exactly r , and whose finite-type and quasi-compact certificates are the descents of the corresponding certificates at the finite Galois stage.

Proof. Finite type and properness descend and are reflected by the finite faithfully flat extension L/k in Theorem ??; in particular $J \rightarrow \operatorname{Spec} k$ is locally of finite type and quasi-compact. Insert J , the same certificate r , and these two properties into Definition ?. No second representing object or universal element is chosen: the group law and universal class are transported from the same r by Yoneda. $\square$

Chapter 9

The tangent space of the Picard functor: the truncated exponential

Kleiman's computation of the tangent space of the Picard scheme at the identity, $T_0 \text{Pic}_{X/k} = H^1(X, \mathcal{O}_X)$, rests on the *truncated exponential sequence*: over the dual numbers $R[\varepsilon] = R \oplus R\varepsilon$ ($\varepsilon^2 = 0$), the units $1 + b\varepsilon$ form a copy of the additive group of R inside $R[\varepsilon]^\times$, split off by reduction mod ε . This chapter develops that layer in pure commutative algebra — the unit group of the dual numbers, its functoriality, the $\varepsilon \mapsto a\varepsilon$ scaling, dual numbers under base change, and the resulting two-chart Čech computation

$$H^1(X, \mathcal{O}_X) \cong \ker \left(\check{H}^{1 \times}(\Gamma(U_0 \cap U_1)[\varepsilon]) \longrightarrow \check{H}^{1 \times}(\Gamma(U_0 \cap U_1)) \right)$$

for a scheme covered by two affine opens — the two-cover Čech form of $H^1(C, \mathcal{O}_C) \cong \ker(\text{Pic}(C_\varepsilon) \rightarrow \text{Pic}(C))$.

(Here and below we write $R[\varepsilon]$ for the dual numbers of a commutative ring R , and, for an element $x = a + b\varepsilon$, $\text{fst}(x) = a$ and $\text{snd}(x) = b$ for its two components.)

9.1 Units of the dual numbers and the truncated exponential

Definition 1161 (The truncated exponential unit). Let R be a commutative ring and $b \in R$. Since $(b\varepsilon)(c\varepsilon) = 0$ in $R[\varepsilon]$, the element $1 + b\varepsilon$ is a unit of $R[\varepsilon]$, with inverse $1 - b\varepsilon$. We write $\text{tExp}(b) := 1 + b\varepsilon \in R[\varepsilon]^\times$; its components are $\text{fst} = 1$ and $\text{snd} = b$, and $\text{tExp}(0) = 1$.

Lemma 1162 (The truncated exponential is an injective homomorphism). *The map $\text{tExp} : (R, +) \rightarrow R[\varepsilon]^\times$ is a group homomorphism,*

$$\text{tExp}(b + c) = \text{tExp}(b) \cdot \text{tExp}(c),$$

and it is injective.

Proof. $(1 + b\varepsilon)(1 + c\varepsilon) = 1 + (b + c)\varepsilon + bc\varepsilon^2 = 1 + (b + c)\varepsilon$. Injectivity: the second component of $\text{tExp}(b)$ is b . $\square$

Definition 1163 (Reduction mod ε on units, and its section). The component projection $\text{fst} : R[\varepsilon] \rightarrow R$, $a + b\varepsilon \mapsto a$, and the constant inclusion $\text{inl} : R \rightarrow R[\varepsilon]$, $a \mapsto a + 0\varepsilon$, are ring homomorphisms. They induce group homomorphisms on unit groups,

$$\text{fst}^\times : R[\varepsilon]^\times \rightarrow R^\times \quad \text{and} \quad \text{inl}^\times : R^\times \rightarrow R[\varepsilon]^\times.$$

Lemma 1164 (Splitting identities). $\text{fst}^\times \circ \text{inl}^\times = \text{id}$ on $R^\times$; $\text{fst}^\times(\text{tExp}(b)) = 1$ for every $b \in R$; and for any unit $u \in R[\varepsilon]^\times$, $\text{fst}(u^{-1}) \cdot \text{fst}(u) = 1 = \text{fst}(u) \cdot \text{fst}(u^{-1})$.

Proof. The first two are componentwise computations ($\text{fst}(\text{inl}(a)) = a$, $\text{fst}(1 + b\varepsilon) = 1$); the third applies the ring homomorphism fst to $u^{-1}u = 1 = uu^{-1}$. $\square$

Theorem 1165 (Exactness in the middle). *The range of the truncated exponential is exactly the kernel of reduction mod ε on units: a unit $u \in R[\varepsilon]^\times$ satisfies $\text{fst}^\times(u) = 1$ if and only if $u = 1 + b\varepsilon$ for a (unique) $b \in R$. Together with ?? this exhibits the split short exact sequence*

$$1 \longrightarrow (R, +) \xrightarrow{\text{tExp}} R[\varepsilon]^\times \xrightarrow{\text{fst}^\times} R^\times \longrightarrow 1.$$

Proof. One inclusion is ??. Conversely, if $\text{fst}^\times(u) = 1$, then $u = 1 + b\varepsilon$ for $b = \text{snd}(u)$, and $u = \text{tExp}(b)$; uniqueness is the injectivity of ??. $\square$

Lemma 1166 (The unit decomposition of the dual numbers). *Every unit $u \in R[\varepsilon]^\times$ factors as a constant unit times a truncated exponential:*

$$u = \text{inl}^\times(\text{fst}^\times(u)) \cdot \text{tExp}(\text{fst}(u^{-1}) \cdot \text{snd}(u)).$$

Proof. Compare components. First components: $\text{fst}(u) \cdot 1 = \text{fst}(u)$. Second components: the product's second component is $\text{fst}(u) \cdot \text{fst}(u^{-1}) \cdot \text{snd}(u) = \text{snd}(u)$ by ??. $\square$

Definition 1167 (Functoriality of the dual numbers). A ring homomorphism $f : R \rightarrow S$ induces the ring homomorphism

$$f[\varepsilon] : R[\varepsilon] \rightarrow S[\varepsilon], \quad a + b\varepsilon \mapsto f(a) + f(b)\varepsilon,$$

applying f to both components.

Lemma 1168 (Functoriality identities). $\text{id}[\varepsilon] = \text{id}$; $(g \circ f)[\varepsilon] = g[\varepsilon] \circ f[\varepsilon]$; $f[\varepsilon] \circ \text{inl} = \text{inl} \circ f$; and on unit groups $f[\varepsilon]^\times \circ \text{inl}^\times = \text{inl}^\times \circ f^\times$.

Proof. All four are componentwise computations from Definition ??. $\square$

Lemma 1169 (Naturality of the truncated exponential and of the reduction). *For a ring homomorphism $f : R \rightarrow S$: $f[\varepsilon]^\times(\text{tExp}(b)) = \text{tExp}(f(b))$ for all $b \in R$, and $\text{fst}^\times \circ f[\varepsilon]^\times = f^\times \circ \text{fst}^\times$ on $R[\varepsilon]^\times$.*

Proof. Componentwise: $f[\varepsilon](1 + b\varepsilon) = 1 + f(b)\varepsilon$, and $\text{fst}(f[\varepsilon](x)) = f(\text{fst}(x))$. $\square$

Definition 1170 (The $\varepsilon \mapsto a\varepsilon$ scaling). For $a \in R$, the scaling $\mu_a : R[\varepsilon] \rightarrow R[\varepsilon]$ is the ring homomorphism $r + m\varepsilon \mapsto r + (am)\varepsilon$ (the identity on constants, multiplication by a on the infinitesimal part). Scalar multiplication on the tangent space of a functor at a rational point is functoriality along μ_a .

Lemma 1171 (Scaling identities). $\mu_a \circ \mu_b = \mu_{ab}$; $\mu_1 = \text{id}$; on truncated exponentials $\mu_a^\times(\text{tExp}(b)) = \text{tExp}(ab)$; and for a ring homomorphism $\rho : R \rightarrow S$ and $s \in R$, $\rho[\varepsilon] \circ \mu_s = \mu_{\rho(s)} \circ \rho[\varepsilon]$.

Proof. All componentwise computations. $\square$

9.2 Dual numbers under base change

Definition 1172 (The dual-number map over an algebra map). For a k -algebra A , the functorial map (algebra map) $[\varepsilon] : k[\varepsilon] \rightarrow A[\varepsilon]$ of Definition ?? is a k -algebra homomorphism (the compatibility with the structure constants holds componentwise).

Theorem 1173 (Dual numbers under base change). *Let k be a commutative ring and A a k -algebra. Extension of scalars of the dual numbers is the dual numbers of A : there is an A -algebra isomorphism*

$$A \otimes_k k[\varepsilon] \cong A[\varepsilon], \quad a \otimes (c + d\varepsilon) \mapsto a \cdot (c + d\varepsilon) = ac + ad\varepsilon,$$

with inverse $x \mapsto \text{fst}(x) \otimes 1 + \text{snd}(x) \otimes \varepsilon$.

Proof. The displayed map is the A -algebra extension of the k -algebra map of ?? along the scalar extension $k \rightarrow A$; it is a well-defined A -algebra homomorphism. The candidate inverse $g(x) = \text{fst}(x) \otimes 1 + \text{snd}(x) \otimes \varepsilon$ is additive. Composing one way, $g(ac + ad\varepsilon) = ac \otimes 1 + ad \otimes \varepsilon = a \otimes (c + d\varepsilon)$ after moving scalars across the tensor ($ac \otimes 1 = a \otimes c \cdot 1$ etc.), and every element of $A \otimes_k k[\varepsilon]$ is a sum of pure tensors, on which this computation applies; composing the other way, $\text{fst}(x) \cdot 1 + \text{snd}(x) \cdot \varepsilon = x$. So the map is bijective. (The dual numbers are finite free over k , so no flatness input is needed — the inverse is written down explicitly.) $\square$

9.3 The two-chart Čech unit-cocycle engine

A *two-chart datum* is a pair of ring homomorphisms $\rho_1 : A_1 \rightarrow B$, $\rho_2 : A_2 \rightarrow B$ — for the section rings of a two-affine cover of a scheme, $A_i = \Gamma(U_i)$, $B = \Gamma(U_0 \cap U_1)$ and the ρ_i the restrictions. Everything in this section is elementary commutative algebra over such a datum.

Definition 1174 (The multiplicative Čech coboundaries). For a two-chart datum $\rho_1 : A_1 \rightarrow B$, $\rho_2 : A_2 \rightarrow B$, the *coboundary subgroup* $\mathcal{B}^\times(\rho_1, \rho_2) \leq B^\times$ is the subgroup generated by the images of $\rho_1^\times$ and $\rho_2^\times$; its elements are exactly the products $\rho_1^\times(v_1) \rho_2^\times(v_2)$ with $v_i \in A_i^\times$ (the two images are subgroups, so the products already form a subgroup). The quotient $B^\times / \mathcal{B}^\times(\rho_1, \rho_2)$ is the two-cover Čech $\check{H}^1$ of units — for a two-affine cover, the Čech Picard group of the cover.

Definition 1175 (The additive Čech coboundaries). For a two-chart datum as above, the *additive coboundary subgroup* $\mathcal{B}^+(\rho_1, \rho_2) \leq (B, +)$ is $\rho_1(A_1) + \rho_2(A_2)$; its elements are exactly the sums $\rho_1(a_1) + \rho_2(a_2)$, equivalently the differences $\rho_1(a_1) - \rho_2(a_2)$ (replace a_2 by $-a_2$). The quotient $B / \mathcal{B}^+(\rho_1, \rho_2)$ is the two-cover Čech cokernel — for the section rings of a two-affine cover, the carrier of ??.

Theorem 1176 (The truncated exponential detects the additive coboundaries). *Let ρ_1, ρ_2 be a two-chart datum and $b \in B$. The truncated exponential $\text{tExp}(b) \in B[\varepsilon]^\times$ is a coboundary of the thickened datum $\rho_1[\varepsilon], \rho_2[\varepsilon]$ if and only if b is an additive coboundary:*

$$\text{tExp}(b) \in \mathcal{B}^\times(\rho_1[\varepsilon], \rho_2[\varepsilon]) \iff b \in \mathcal{B}^+(\rho_1, \rho_2).$$

Proof. ($\Leftarrow$) If $b = \rho_1(a_1) + \rho_2(a_2)$, then by naturality (??) and additivity (??), $\text{tExp}(b) = \text{tExp}(\rho_1(a_1)) \cdot \text{tExp}(\rho_2(a_2)) = \rho_1[\varepsilon]^\times(\text{tExp}(a_1)) \cdot \rho_2[\varepsilon]^\times(\text{tExp}(a_2))$, a coboundary.

($\Rightarrow$) Suppose $\text{tExp}(b) = \rho_1[\varepsilon]^\times(w_1) \cdot \rho_2[\varepsilon]^\times(w_2)$ with $w_i \in A_i[\varepsilon]^\times$. Decompose each chart unit (??) as $w_i = \text{inl}^\times(v_i) \cdot \text{tExp}(c_i)$ with $v_i \in A_i^\times$, $c_i \in A_i$. Pushing through $\rho_i[\varepsilon]$ (??, ??) and

collecting constant and exponential factors (truncated exponentials commute with everything and multiply additively),

$$\mathrm{tExp}(b) = \mathrm{inl}^\times(\rho_1^\times(v_1)\rho_2^\times(v_2)) \cdot \mathrm{tExp}(\rho_1(c_1) + \rho_2(c_2)).$$

Applying $\mathrm{fst}^\times$ (??, ??) kills both exponentials and gives $1 = \rho_1^\times(v_1)\rho_2^\times(v_2)$; substituting back and using injectivity of tExp (??) yields $b = \rho_1(c_1) + \rho_2(c_2) \in \mathcal{B}^+(\rho_1, \rho_2)$. $\square$

Definition 1177 (The Čech units reduction). Reduction mod ε on units carries thickened coboundaries to coboundaries: by naturality (??, ??), $\mathrm{fst}^\times(\rho_1[\varepsilon]^\times(w_1)\rho_2[\varepsilon]^\times(w_2)) = \rho_1^\times(\mathrm{fst}^\times w_1)\rho_2^\times(\mathrm{fst}^\times w_2)$. Hence $\mathrm{fst}^\times$ descends to a group homomorphism of Čech $\check{H}^1$ -of-units groups,

$$\pi : B[\varepsilon]^\times / \mathcal{B}^\times(\rho_1[\varepsilon], \rho_2[\varepsilon]) \longrightarrow B^\times / \mathcal{B}^\times(\rho_1, \rho_2),$$

the Čech-cocycle incarnation of the restriction $\mathrm{Pic}(X \times_k \mathrm{Spec} k[\varepsilon]) \rightarrow \mathrm{Pic}(X)$ along $\varepsilon \mapsto 0$.

Lemma 1178 (Truncated-exponential classes lie in the kernel, and vanish exactly on coboundaries). *For $b \in B$, the class $[\mathrm{tExp}(b)]$ lies in $\ker \pi$; and $[\mathrm{tExp}(b)] = 1$ in the thickened Čech group if and only if $b \in \mathcal{B}^+(\rho_1, \rho_2)$.*

Proof. $\pi[\mathrm{tExp}(b)] = [\mathrm{fst}^\times(\mathrm{tExp}(b))] = [1]$ (??). Vanishing of the class means membership of $\mathrm{tExp}(b)$ in the thickened coboundary subgroup, which is ?? $\square$

Theorem 1179 (Every kernel class is a truncated exponential). *Every class x of $B[\varepsilon]^\times / \mathcal{B}^\times(\rho_1[\varepsilon], \rho_2[\varepsilon])$ with $\pi(x) = 1$ is represented by a truncated exponential: $x = [\mathrm{tExp}(b)]$ for some $b \in B$.*

Proof. Let $u \in B[\varepsilon]^\times$ represent x . Since $\pi(x) = 1$, the reduction $\mathrm{fst}^\times(u)$ is a coboundary: $\mathrm{fst}^\times(u) = \rho_1^\times(v_1)\rho_2^\times(v_2)$ with $v_i \in A_i^\times$. Correct u by the constant lifts of these chart units:

$$u' := \left(\rho_1[\varepsilon]^\times(\mathrm{inl}^\times v_1)\right)^{-1} \cdot u \cdot \left(\rho_2[\varepsilon]^\times(\mathrm{inl}^\times v_2)\right)^{-1}.$$

The two correction factors are thickened coboundaries (??), so $[u'] = [u] = x$; and by ?? the reduction of u' is $\rho_1^\times(v_1)^{-1} \cdot \mathrm{fst}^\times(u) \cdot \rho_2^\times(v_2)^{-1} = 1$. By exactness (??), $u' = \mathrm{tExp}(b)$ for some $b \in B$. $\square$

Theorem 1180 (The truncated-exponential kernel computation). *For a two-chart datum $\rho_1 : A_1 \rightarrow B$, $\rho_2 : A_2 \rightarrow B$, the truncated exponential $b \mapsto [\mathrm{tExp}(b)]$ induces an isomorphism of abelian groups*

$$B/(\rho_1(A_1) + \rho_2(A_2)) \cong \ker\left(\pi : B[\varepsilon]^\times / \mathcal{B}^\times(\rho_1[\varepsilon], \rho_2[\varepsilon]) \rightarrow B^\times / \mathcal{B}^\times(\rho_1, \rho_2)\right),$$

sending the residue class of b to the class of $1 + b\varepsilon$.

Proof. The assignment $b \mapsto [\mathrm{tExp}(b)]$ is a homomorphism from $(B, +)$ into $\ker \pi$ (?? for additivity, ?? for landing in the kernel). It kills exactly $\mathcal{B}^+(\rho_1, \rho_2)$ (??), so it descends to an injective homomorphism on the quotient $B/(\rho_1(A_1) + \rho_2(A_2))$. It is surjective onto $\ker \pi$ by ?? $\square$

Lemma 1181 (Scaling equivariance of the engine). *Let ρ_1, ρ_2 be a two-chart datum and let $t \in B$ be chart-compatible: $t = \rho_1(s_1) = \rho_2(s_2)$ for some $s_i \in A_i$. Then the scaling $\mu_t^\times$ of $B[\varepsilon]^\times$ preserves the thickened coboundary subgroup $\mathcal{B}^\times(\rho_1[\varepsilon], \rho_2[\varepsilon])$, hence descends to the thickened Čech group; and on truncated-exponential classes it acts by $[\mathrm{tExp}(b)] \mapsto [\mathrm{tExp}(t \cdot b)]$.*

Proof. For a coboundary $\rho_1[\varepsilon]^\times(w_1)\rho_2[\varepsilon]^\times(w_2)$, the compatibility $\rho_i[\varepsilon] \circ \mu_{s_i} = \mu_{\rho_i(s_i)} \circ \rho_i[\varepsilon] = \mu_t \circ \rho_i[\varepsilon]$ (??) rewrites its image under $\mu_t^\times$ as $\rho_1[\varepsilon]^\times(\mu_{s_1}^\times w_1)\rho_2[\varepsilon]^\times(\mu_{s_2}^\times w_2)$, again a coboundary. The action on truncated exponentials is $\mu_t^\times(\mathrm{tExp}(b)) = \mathrm{tExp}(tb)$ (??), passed to classes. $\square$

9.4 H^1 as the ε -kernel on a two-cover

We now assemble the engine on the pinned two-cover section rings of a scheme X over $\text{Spec } k$ with two affine opens $U_0 \cup U_1 = X$: the two-chart datum is $B = \Gamma(X, U_0 \cap U_1)$ with the two restrictions $\rho_i : \Gamma(X, U_i) \rightarrow B$.

Lemma 1182 (Module quotients are additive-subgroup quotients). *For a module M over a ring R and a submodule $p \subseteq M$, the quotient module M/p coincides, as an abelian group, with the quotient of $(M, +)$ by the underlying additive subgroup of p , compatibly with the two quotient maps.*

Proof. The two quotients are defined by the same equivalence relation $(x \sim y \iff x - y \in p)$, so the identity map is the required isomorphism. $\square$

Lemma 1183 (Restriction intertwines the structure constants). *For a scheme X over $\text{Spec } k$, opens U_0, U_1 and $a \in k$, restricting the structure constant of a chart to the overlap gives the structure constant of the overlap: $\rho_i(a \cdot 1_{\Gamma(U_i)}) = a \cdot 1_{\Gamma(U_0 \cap U_1)}$ for $i = 0, 1$, where $a \cdot 1$ denotes the image of a under the structure map of $??$.*

Proof. The structure map of $??$ is compatible with restriction: it is defined by pulling back the global section a of the base, and pullback commutes with restriction to a smaller open. $\square$

Definition 1184 (The two-cover units reduction). The Čech units reduction π of $??$, instantiated at the two-chart datum of the cover: $B = \Gamma(X, U_0 \cap U_1)$, ρ_i the restrictions from $\Gamma(X, U_i)$.

Definition 1185 (Truncated-exponential kernel classes of overlap sections). For an overlap section $b \in \Gamma(X, U_0 \cap U_1)$, the class $\text{tCl}(b) := [\text{tExp}(b)] \in \ker \pi$ ($??$), regarded in the additive notation of the kernel; $\ker \text{Val}$ denotes the underlying thickened Čech class of an element of $\ker \pi$, so $\ker \text{Val}(\text{tCl}(b)) = [\text{tExp}(b)]$. Under the identification of $??$, $\text{tCl}(b)$ is the image of the residue class of b .

Lemma 1186 (Properties of the kernel classes). $\text{tCl}(b + c) = \text{tCl}(b) + \text{tCl}(c)$; $\text{tCl}(b) = 0$ iff b is an additive coboundary $\rho_1(a_1) + \rho_2(a_2)$ of the two charts; and every element of $\ker \pi$ is $\text{tCl}(b)$ for some overlap section b .

Proof. All three restate $??$ through the identification of $??$: additivity and the vanishing criterion are the well-definedness and injectivity of the induced map, and surjectivity is its surjectivity. $\square$

Lemma 1187 (The range of the difference map is the additive coboundary subgroup). *The range of the restriction-difference map diff of $??$, viewed as an additive subgroup of $\Gamma(X, U_0 \cap U_1)$, is exactly the additive coboundary subgroup $\mathcal{B}^+(\rho_1, \rho_2)$ of the restriction datum ($??$). Consequently the two-cover cokernel H1Cok ($??$) is, as an abelian group, the Čech quotient $\Gamma(X, U_0 \cap U_1) / \mathcal{B}^+(\rho_1, \rho_2)$, compatibly with the quotient maps.*

Proof. A difference $\rho_1(a_1) - \rho_2(a_2)$ and a sum $\rho_1(a_1) + \rho_2(a_2)$ span the same additive subgroup ($??$), so the two subgroups coincide elementwise. The quotient statement combines this with $??$ applied to the submodule $\text{range}(\text{diff})$. $\square$

Theorem 1188 (H^1 is the ε -kernel of the two-cover). *Let X be a scheme over $\text{Spec } k$ and U_0, U_1 affine opens with $U_0 \cup U_1 = X$. Then the degree-one cohomology of the structure sheaf is, additively, the kernel of the dual-number Čech units reduction of the cover:*

$$H^1(X, \mathcal{O}_X) \cong \ker \left(\pi : \check{H}^{1 \times}(\Gamma(U_0 \cap U_1)[\varepsilon]) \rightarrow \check{H}^{1 \times}(\Gamma(U_0 \cap U_1)) \right),$$

the two-cover Čech form of $H^1(C, \mathcal{O}_C) \cong \ker(\text{Pic}(C_\varepsilon) \rightarrow \text{Pic}(C))$.

Proof. Compose three identifications: the two-affine-cover cokernel bridge $H^1(X, \mathcal{O}_X) \cong \Gamma(U_0 \cap U_1)/\text{range}(\text{diff})$ (??); the carrier translation $\Gamma(U_0 \cap U_1)/\text{range}(\text{diff}) \cong \Gamma(U_0 \cap U_1)/\mathcal{B}^+(\rho_1, \rho_2)$ (??); and the truncated-exponential kernel computation $\Gamma(U_0 \cap U_1)/\mathcal{B}^+(\rho_1, \rho_2) \cong \ker \pi$ (??). $\square$

Lemma 1189 (Generator formula). *The identification of ?? sends the connecting class $\text{delta}(s)$ of an overlap section s (??) to the truncated-exponential class $\text{tCl}(s)$.*

Proof. The cokernel bridge sends $\text{delta}(s)$ to the residue class of s (??); the carrier translation preserves residue classes; and the kernel computation sends the residue class of s to $\text{tCl}(s)$ (??). $\square$

Lemma 1190 (Scaled generators). *For $a \in k$ and an overlap section s , the k -action on H^1 hits the connecting classes as multiplication by the structure constant: $a \cdot \text{delta}(s) = \text{delta}((a \cdot 1)s)$; consequently the identification of ?? sends $a \cdot \text{delta}(s)$ to $\text{tCl}((a \cdot 1)s)$.*

Proof. The connecting map delta is k -linear (??), and the k -action on the overlap sections is multiplication by the structure constant $a \cdot 1$ (??). The second claim applies ?? $\square$

Definition 1191 (The Mumford scaling of the thickened Čech group). *For $a \in k$, the scaling $\mu_{a \cdot 1}$ of the overlap ring's dual numbers, with $a \cdot 1$ the structure constant of the overlap, descends to an endomorphism of the thickened Čech group of the two-cover: the structure constant is chart-compatible by ??, so ?? applies; on truncated-exponential classes it acts by $[\text{tExp}(b)] \mapsto [\text{tExp}((a \cdot 1)b)]$.*

Theorem 1192 (Mumford-scaling equivariance of the keystone). *Through the identification of ??, the k -scalar action on $H^1(X, \mathcal{O}_X)$ corresponds to the Mumford scaling of the thickened Čech group: for $a \in k$ and an overlap section s , the underlying thickened class of the image of $a \cdot \text{delta}(s)$ equals the Mumford scaling by a of the underlying thickened class of the image of $\text{delta}(s)$. Since the connecting classes generate H^1 (??) and both actions are additive, this determines the correspondence on all of H^1 .*

Proof. By ??, the image of $a \cdot \text{delta}(s)$ is $\text{tCl}((a \cdot 1)s)$, with underlying class $[\text{tExp}((a \cdot 1)s)]$; by ?? and ??, the Mumford scaling by a of the underlying class $[\text{tExp}(s)]$ of the image of $\text{delta}(s)$ is the same $[\text{tExp}((a \cdot 1)s)]$. $\square$

Chapter 10

The rigidity lemma

Throughout this chapter K is a field; the main results assume it algebraically closed, and each statement records the hypothesis where it is needed. We work in the category of schemes over K : its objects are schemes X equipped with a structure morphism $t_X : X \rightarrow \operatorname{Spec} K$, its morphisms commute with the structure morphisms, $\operatorname{Spec} K$ is a terminal object, and the categorical product of X and Y is the fibre product $X \times_K Y$ with projections $\operatorname{pr}_1, \operatorname{pr}_2$. A *point* of X is a morphism $x : \operatorname{Spec} K \rightarrow X$ over K , that is, a section of t_X ; for a point y of Y we write

$$\sigma_y = (\operatorname{id}_X, y \circ t_X) : X \longrightarrow X \times_K Y$$

for the *slice embedding* over y , and symmetrically $\tau_x = (x \circ t_Y, \operatorname{id}_Y) : Y \rightarrow X \times_K Y$ for a point x of X .

The subject is the *rigidity lemma* of Mumford and Milne: a morphism from a product $X \times_K Y$, with X proper, that collapses one slice $X \times \{y_0\}$ to a point factors through the second projection. Its engine application makes every pointed morphism from a proper geometrically integral monoid scheme to a group scheme a homomorphism. The chapter proves the lemma for schemes over an algebraically closed field, after three preparatory sections: slice bookkeeping valid in any category with finite products, a constancy theorem for morphisms out of schemes with only constant global functions, and the dictionary between closed points and K -points.

10.1 Slices in a category with finite products

In this section $\mathcal{C}$ is any category with a terminal object 1 and binary products. For an object T write $t_T : T \rightarrow 1$ for the unique morphism, and call a morphism $1 \rightarrow Y$ a *point* of Y ; the slice embedding $\sigma_y : X \rightarrow X \times Y$ of a point y of Y is the morphism with components id_X and $y \circ t_X$.

Lemma 1193 (Morphisms with constant second component factor through a slice). *Let T, X, Y be objects of $\mathcal{C}$, let y be a point of Y , and let $p : T \rightarrow X \times Y$ be a morphism whose second component is constant with value y , i.e. $\operatorname{pr}_2 \circ p = y \circ t_T$. Then*

$$p = \sigma_y \circ (\operatorname{pr}_1 \circ p).$$

Proof. Two morphisms into the product $X \times Y$ are equal as soon as their two components are. The first component of the right-hand side is $\operatorname{pr}_1 \circ \sigma_y \circ (\operatorname{pr}_1 \circ p) = \operatorname{id}_X \circ \operatorname{pr}_1 \circ p = \operatorname{pr}_1 \circ p$. Its second component is $\operatorname{pr}_2 \circ \sigma_y \circ (\operatorname{pr}_1 \circ p) = y \circ t_X \circ \operatorname{pr}_1 \circ p = y \circ t_T$, because $t_X \circ \operatorname{pr}_1 \circ p$ and t_T are both morphisms $T \rightarrow 1$ and 1 is terminal. By hypothesis these are the two components of p . $\square$

Lemma 1194 (Transfer of slice constancy). *In the situation of ??, let moreover $f : X \times Y \rightarrow Z$ be a morphism which is constant on the slice over y , with value a point c of Z : $f \circ \sigma_y = c \circ t_X$. Then*

$$f \circ p = c \circ t_T.$$

Proof. By ??, $f \circ p = f \circ \sigma_y \circ (\text{pr}_1 \circ p) = c \circ t_X \circ \text{pr}_1 \circ p = c \circ t_T$, the last step again by uniqueness of morphisms into the terminal object. $\square$

10.2 Constancy of morphisms into affine opens

Lemma 1195 (Constancy). *Let R be a commutative ring and $s_V : V \rightarrow \text{Spec } R$ a morphism of schemes inducing an isomorphism $\Gamma(\text{Spec } R, \mathcal{O}_{\text{Spec } R}) \rightarrow \Gamma(V, \mathcal{O}_V)$ on global sections. Let $q : V \rightarrow W$ be a morphism of schemes whose set-theoretic image is contained in an affine open $U \subseteq W$. Then q is constant: there is a morphism $z : \text{Spec } R \rightarrow W$ with $q = z \circ s_V$.*

Proof. Since the image of q lies in the open subscheme U , the morphism q factors as $q = \iota \circ q'$ with $q' : V \rightarrow U$ and $\iota : U \hookrightarrow W$ the open immersion. Write $A = \Gamma(U, \mathcal{O}_U)$, so $U \cong \text{Spec } A$ as U is affine, and recall the affine adjunction: for every scheme T there is a bijection $\Phi_T : \text{Hom}_{\text{Sch}}(T, U) \rightarrow \text{Hom}_{\text{Ring}}(A, \Gamma(T, \mathcal{O}_T))$, natural in T , sending a morphism to its action on global sections.

Let $\varphi = \Phi_V(q') : A \rightarrow \Gamma(V, \mathcal{O}_V)$, let $\gamma : \Gamma(\text{Spec } R, \mathcal{O}_{\text{Spec } R}) \rightarrow \Gamma(V, \mathcal{O}_V)$ be the isomorphism induced by s_V , and set $z' = \Phi_{\text{Spec } R}^{-1}(\gamma^{-1} \circ \varphi) : \text{Spec } R \rightarrow U$. Naturality of Φ applied to s_V gives

$$\Phi_V(z' \circ s_V) = \gamma \circ \Phi_{\text{Spec } R}(z') = \gamma \circ \gamma^{-1} \circ \varphi = \varphi = \Phi_V(q'),$$

and since Φ_V is a bijection, $q' = z' \circ s_V$. Take $z = \iota \circ z'$. $\square$

Lemma 1196 (Constancy, relative form). *Let R be a commutative ring and let V, W be schemes over $\text{Spec } R$, with structure morphisms v, w , such that v induces an isomorphism on global sections. Let $q : V \rightarrow W$ be a morphism over $\text{Spec } R$ whose set-theoretic image is contained in an affine open $U \subseteq W$. Then there is a point $c : \text{Spec } R \rightarrow W$ over $\text{Spec } R$ (a section of w) with $q = c \circ v$.*

Proof. By ?? there is $z : \text{Spec } R \rightarrow W$, a priori not over $\text{Spec } R$, with $q = z \circ v$. It remains to prove $w \circ z = \text{id}$. Since q is a morphism over $\text{Spec } R$,

$$(w \circ z) \circ v = w \circ q = v.$$

Applying the (contravariant) global-sections functor Γ , $\Gamma(v) \circ \Gamma(w \circ z) = \Gamma(v)$; as $\Gamma(v)$ is an isomorphism, in particular injective, $\Gamma(w \circ z) = \text{id}$. Finally, a morphism between affine schemes is determined by its action on global sections (the affine adjunction of ??), so $w \circ z = \text{id}_{\text{Spec } R}$. $\square$

10.3 Points and products

The first lemma of this section is where algebraic closure enters; the second holds over an arbitrary field.

Lemma 1197 (Points are determined by their underlying points). *Let T be a scheme locally of finite type over the algebraically closed field K , and let $p, q : \text{Spec } K \rightarrow T$ be two points of T over K . If p and q send the unique point $*$ of $\text{Spec } K$ to the same point of the topological space of T , then $p = q$.*

Proof. Let $t = p(*) = q(*)$. A morphism $\text{Spec } K \rightarrow T$ with set-theoretic image $\{t\}$ is the same datum as a local ring homomorphism $\mathcal{O}_{T,t} \rightarrow K$ out of the local ring of T at t , and the morphism lies over K exactly when this homomorphism is a K -algebra map. A local homomorphism into a field kills the maximal ideal, hence factors uniquely through the residue field $\kappa(t)$; so each of p, q is determined by a K -algebra homomorphism $r : \kappa(t) \rightarrow K$, where $\kappa(t)$ is a field extension of K via a canonical embedding $\iota : K \rightarrow \kappa(t)$, and being a K -algebra map says $r \circ \iota = \text{id}_K$.

Such an r is unique: it is injective, being a homomorphism of fields, and for any $x \in \kappa(t)$ we have $r(\iota(r(x))) = r(x)$, whence $\iota(r(x)) = x$ by injectivity. Thus ι is surjective, $\kappa(t) = K$, and $r = \iota^{-1}$ is forced. Hence p and q induce the same homomorphism $\mathcal{O}_{T,t} \rightarrow K$ and the same image point, so $p = q$. $\square$

Lemma 1198 (Products of geometrically integral schemes are integral). *Let X and Y be schemes over a field k , each geometrically integral and locally of finite type over k . Then the product $X \times_k Y$ is an integral scheme.*

Proof. Fix an algebraic closure $\bar{k}$ of k . Base change commutes with fibre products, so

$$(X \times_k Y)_{\bar{k}} \cong X_{\bar{k}} \times_{\bar{k}} Y_{\bar{k}},$$

and $X_{\bar{k}}, Y_{\bar{k}}$ are integral by geometric integrality. Over the algebraically closed field $\bar{k}$, the product of two integral schemes is integral: it is covered by affine opens $\text{Spec}(A \otimes_{\bar{k}} B)$ with A, B reduced $\bar{k}$ -algebras, and the tensor product over an algebraically closed field of two reduced algebras is reduced; and it is irreducible, because over a separably closed field every irreducible scheme is geometrically irreducible, and the product of an irreducible scheme with a geometrically irreducible scheme is irreducible.

It remains to descend integrality along $k \rightarrow \bar{k}$. The projection $(X \times_k Y)_{\bar{k}} \rightarrow X \times_k Y$ is surjective (a base field extension of schemes is surjective), and the continuous image of an irreducible space is irreducible; so $X \times_k Y$ is irreducible. On any affine open $\text{Spec } C \subseteq X \times_k Y$ the base-changed sections are $C \otimes_k \bar{k}$, and the canonical map $C \rightarrow C \otimes_k \bar{k}$ is injective because $\bar{k}$ is a faithfully flat k -module; a subring of a reduced ring is reduced, so $X \times_k Y$ is reduced. Reduced and irreducible is integral. $\square$

10.4 The rigidity lemma

Theorem 1199 (Rigidity). *Let K be an algebraically closed field and let X, Y, Z be schemes over K such that*

1. *X is proper and geometrically integral over K ;*
2. *Y is locally of finite type over K and the product $X \times_K Y$ is an integral scheme (automatic when Y is also geometrically integral, by ??);*
3. *Z is separated and locally of finite type over K .*

Let $f : X \times_K Y \rightarrow Z$ be a morphism over K , let y_0 be a point of Y and z_0 a point of Z , and suppose f contracts the slice over y_0 to z_0 :

$$f \circ \sigma_{y_0} = z_0 \circ t_X.$$

Then there is a unique morphism $g : Y \rightarrow Z$ over K with $f = g \circ \text{pr}_2$.

Proof. Step 0: closed points and K -points. The schemes X , Y , Z and $X \times_K Y$ are all locally of finite type over K (X because properness includes finite type; the product as a fibre product of such). The underlying space of a scheme locally of finite type over a field is Jacobson, so every nonempty locally closed subset contains a point that is closed in the ambient space. By the Nullstellensatz the residue field at a closed point of such a scheme is a finite extension of K , hence equals K as K is algebraically closed; therefore every closed point underlies a point over K , unique by ??, and conversely the underlying point of any point over K is closed (its residue field is K). We use this dictionary freely.

Step 1: uniqueness. Since X is integral it is nonempty, so by Step 0 it has a point x_0 over K . The slice $\tau_{x_0} : Y \rightarrow X \times_K Y$ satisfies $\text{pr}_2 \circ \tau_{x_0} = \text{id}_Y$. Hence if $f = g \circ \text{pr}_2$ then $g = g \circ \text{pr}_2 \circ \tau_{x_0} = f \circ \tau_{x_0}$ is determined by f . Define, accordingly,

$$g = f \circ \tau_{x_0} : Y \rightarrow Z.$$

Step 2: the bad locus. Let $z \in Z$ be the underlying point of z_0 and let $U \subseteq Z$ be an affine open neighbourhood of z (affine opens form a basis). Set

$$A = f^{-1}(Z \setminus U) \subseteq X \times_K Y, \quad W = \text{pr}_2(A) \subseteq Y.$$

A is closed. The projection $\text{pr}_2 : X \times_K Y \rightarrow Y$ is universally closed, being the base change along t_Y of the universally closed structure morphism t_X (X is proper); in particular it is a closed map, so W is closed in Y .

Step 3: slice constancy off the bad locus. Let y be any point of Y over K whose underlying point lies outside W . The composite $f \circ \sigma_y : X \rightarrow Z$ has set-theoretic image inside U : a point of the slice mapping outside U would lie in A , and its image under pr_2 — which is the underlying point of y , since $\text{pr}_2 \circ \sigma_y = y \circ t_X$ — would lie in W . The scheme X is proper and geometrically integral over K , so its structure morphism induces an isomorphism on global sections (??); by the relative constancy lemma ?? there is a point c of Z over K with

$$f \circ \sigma_y = c \circ t_X.$$

Step 4: the distinguished point avoids the bad locus. Let y_0^* be the underlying point of y_0 ; we claim $y_0^* \notin W$. Suppose otherwise. The point y_0^* is closed in Y (Step 0), so $A \cap \text{pr}_2^{-1}(\{y_0^*\})$ is a nonempty closed subset of $X \times_K Y$ and therefore contains a closed point p , which underlies a point $\hat{p}$ of $X \times_K Y$ over K (Step 0). The two points $\text{pr}_2 \circ \hat{p}$ and y_0 of Y have the same underlying point y_0^* , so $\text{pr}_2 \circ \hat{p} = y_0$ by ??; thus the second component of $\hat{p}$ is constant with value y_0 , and the transfer lemma ??, applied with the contraction hypothesis $f \circ \sigma_{y_0} = z_0 \circ t_X$, gives $f \circ \hat{p} = z_0$. Taking underlying points, $f(p) = z \in U$, contradicting $p \in A = f^{-1}(Z \setminus U)$.

Step 5: a dense open of agreement. Let $S = \text{pr}_2^{-1}(Y \setminus W)$, an open subset of $X \times_K Y$. It is nonempty: the point $(x_0, y_0) := \sigma_{y_0} \circ x_0$ of $X \times_K Y$ over K has second projection with underlying point $y_0^* \notin W$ (Step 4). Since $X \times_K Y$ is integral, its space is irreducible, and the nonempty open S is dense.

Step 6: f and $g \circ \text{pr}_2$ agree on closed points of S . Let $p \in S$ be a closed point, with associated point $\hat{p}$ over K , and let $y^* = \text{pr}_2(p)$. Then $\{y^*\} = \text{pr}_2(\{p\})$ is closed (pr_2 is a closed map), so y^* underlies a point $\hat{y}$ of Y over K , and $y^* \notin W$ because $p \in S$. Step 3 provides a point c of Z with $f \circ \sigma_{\hat{y}} = c \circ t_X$. The points $\text{pr}_2 \circ \hat{p}$ and $\hat{y}$ share the underlying point y^* , so $\text{pr}_2 \circ \hat{p} = \hat{y}$ by ??, and the transfer lemma ?? gives

$$f \circ \hat{p} = c.$$

On the other side, $(g \circ \text{pr}_2) \circ \hat{p} = f \circ \tau_{x_0} \circ \hat{y}$, and $\tau_{x_0} \circ \hat{y}$ is a morphism $\text{Spec } K \rightarrow X \times_K Y$ whose second component is $\text{pr}_2 \circ \tau_{x_0} \circ \hat{y} = \hat{y}$; the transfer lemma, with the same slice constancy over

$\hat{y}$, gives $f \circ (\tau_{x_0} \circ \hat{y}) = c$ as well. Hence $f \circ \hat{p} = (g \circ \text{pr}_2) \circ \hat{p}$ — the two composites are equal as morphisms $\text{Spec } K \rightarrow Z$.

Step 7: conclusion. Consider the morphism $(f, g \circ \text{pr}_2) : X \times_K Y \rightarrow Z \times_K Z$ and the diagonal $\Delta : Z \rightarrow Z \times_K Z$, a closed immersion because Z is separated over K . Let $E \subseteq X \times_K Y$ be the preimage of Δ , i.e. the fibre product of $(f, g \circ \text{pr}_2)$ and Δ ; it is a closed subscheme, and a morphism $T \rightarrow X \times_K Y$ factors through E exactly when its composites with f and with $g \circ \text{pr}_2$ coincide. By Step 6 every closed point of S lies in E : its associated K -point factors through E . In the Jacobson space $X \times_K Y$ the closed points of S are dense in S (every nonempty open subset of S is locally closed in $X \times_K Y$, hence contains a closed point), and S is dense (Step 5); so the closed set underlying E is all of $X \times_K Y$. A closed subscheme of a reduced scheme whose underlying space is everything is the whole scheme: its quasi-coherent ideal sheaf is contained in the nilradical, which vanishes, and $X \times_K Y$ is reduced because it is integral. Hence $E = X \times_K Y$ and $f = g \circ \text{pr}_2$. $\square$

10.5 Pointed morphisms of group schemes

A *monoid scheme* over K is a monoid object in the category of schemes over K : a scheme A over K together with a multiplication $\mu_A : A \times_K A \rightarrow A$ and a unit point $e_A : \text{Spec } K \rightarrow A$ satisfying the associativity and two-sided unit laws; a *group scheme* is a group object, i.e. a monoid object with an inversion morphism. For a scheme V and a group scheme B , the set of morphisms $V \rightarrow B$ over K is a group under the pointwise product $u \cdot v = \mu_B \circ (u, v)$, with unit the constant morphism $e_B \circ t_V$ and inverse given by postcomposition with the inversion of B ; precomposition with any morphism $V' \rightarrow V$ is a group homomorphism of such groups. A morphism of monoid schemes $g : A \rightarrow B$ is a *homomorphism* if it preserves the unit, $g \circ e_A = e_B$, and the multiplication, $g \circ \mu_A = \mu_B \circ (g \times g)$.

Corollary 1200 (Pointed morphisms are homomorphisms). *Let K be an algebraically closed field, let A be a monoid scheme over K which is proper and geometrically integral over K , and let B be a group scheme over K which is separated and locally of finite type over K . Then every morphism $g : A \rightarrow B$ over K with $g \circ e_A = e_B$ is a homomorphism of monoid schemes.*

Proof. Form the defect of g , computed in the group of morphisms $A \times_K A \rightarrow B$:

$$\varphi = (g \circ \mu_A) \cdot \left((g \circ \text{pr}_1) \cdot (g \circ \text{pr}_2) \right)^{-1}.$$

It suffices to prove $\varphi = 1$ (the constant morphism $e_B \circ t_{A \times_K A}$): this says exactly $g \circ \mu_A = (g \circ \text{pr}_1) \cdot (g \circ \text{pr}_2) = \mu_B \circ (g \times g)$, and g preserves units by hypothesis.

The slice over e_A collapses. Precomposition with the slice embedding $\sigma_{e_A} : A \rightarrow A \times_K A$ is a group homomorphism, so $\varphi \circ \sigma_{e_A} = (g \circ \mu_A \circ \sigma_{e_A}) \cdot ((g \circ \text{pr}_1 \circ \sigma_{e_A}) \cdot (g \circ \text{pr}_2 \circ \sigma_{e_A}))^{-1}$. The right unit law of the monoid A gives $\mu_A \circ \sigma_{e_A} = \text{id}_A$; moreover $\text{pr}_1 \circ \sigma_{e_A} = \text{id}_A$ and $\text{pr}_2 \circ \sigma_{e_A} = e_A \circ t_A$, so, using $g \circ e_A = e_B$,

$$\varphi \circ \sigma_{e_A} = g \cdot (g \cdot (e_B \circ t_A))^{-1} = g \cdot g^{-1} = 1 = e_B \circ t_A.$$

Rigidity applies. In Theorem ?? take $X = Y = A$, $Z = B$, $f = \varphi$, $y_0 = e_A$, $z_0 = e_B$: the scheme A is proper (hence of finite type) and geometrically integral over K , so the product $A \times_K A$ is integral by ??; and B is separated and locally of finite type. The displayed identity is the contraction hypothesis, so there is a (unique) morphism $\psi : A \rightarrow B$ with $\varphi = \psi \circ \text{pr}_2$.

The other slice identifies the factor. Precompose with $\tau_{e_A} = (e_A \circ t_A, \text{id}_A) : A \rightarrow A \times_K A$. Since $\text{pr}_2 \circ \tau_{e_A} = \text{id}_A$, we get $\psi = \varphi \circ \tau_{e_A}$. Computing as before, with the left unit law $\mu_A \circ \tau_{e_A} = \text{id}_A$, and $\text{pr}_1 \circ \tau_{e_A} = e_A \circ t_A$, $\text{pr}_2 \circ \tau_{e_A} = \text{id}_A$:

$$\psi = g \cdot ((e_B \circ t_A) \cdot g)^{-1} = g \cdot g^{-1} = 1.$$

Hence $\varphi = 1 \circ \text{pr}_2 = 1$, as required. $\square$

When A is itself a group scheme, a homomorphism of monoid schemes between group schemes automatically commutes with the inversions, so Corollary ?? says: a pointed morphism of group schemes from a proper geometrically integral group scheme to a separated, locally finite type group scheme over an algebraically closed field is a homomorphism of group schemes. The hypotheses are strictly weaker than in the classical statements of Mumford and Milne, which take both source and target to be abelian varieties: here the source needs no inversion and no smoothness, and the target no properness, connectedness or commutativity.

Chapter 11

Rational maps into group varieties

This chapter develops the theory of rational maps needed for Milne's extension theorem: *a rational map from a nonsingular variety to an abelian variety extends, uniquely, to a regular morphism*. The chapter proves the two halves that do not mention the group structure of the target in an essential way — the calculus of rational maps (precomposition, pairing, function fields, the difference map $\Phi(x, y) = f(x)f(y)^{-1}$), and the geometry of a smooth variety (purity of polar loci, regular stalks, discrete valuation rings at codimension-one points, Milne's codimension- ≥ 2 indeterminacy theorem) — and assembles them into the extension theorem, taking as an explicit hypothesis the one remaining ingredient: Milne's Lemma 3.3, that the indeterminacy locus of a rational map into a group variety is empty or of pure codimension one. That lemma's statement is isolated as a named property (??), and its principal substeps are proved here (??, ??, ??, ??, ??).

Throughout, a *partial map* (U, φ) from a scheme X to a scheme Y is a dense open subset $U \subseteq X$ together with a morphism $\varphi : U \rightarrow Y$; two partial maps are *equivalent* if they agree on a dense open subset contained in both domains. A *rational map* $f : X \dashrightarrow Y$ is an equivalence class of partial maps. A point $x \in X$ is a *point of definition* of f if some representative of f is defined at x ; the points of definition form the dense open *domain of definition* $\text{Dom}(f)$. For a base scheme S with X, Y over S , the rational map f is *over* S when some (equivalently, by shrinking, any small enough) representative is a morphism over S ; when the structure morphisms are written s_X, s_Y , this is the equation $s_Y \circ f = s_X$ of rational maps $X \dashrightarrow S$. Composition of a rational map with a (total) morphism $g : Y \rightarrow Z$ on the left of the arrow, written $g \circ f$, is represented by $(U, \varphi \circ g)$ for any representative (U, φ) of f .

11.1 Rational maps: precomposition, pairing, function fields

Lemma 1201 (Restriction square). *Let $p : W \rightarrow X$ be a morphism of schemes and $V \subseteq U$ open subsets of X . Write $p_U : p^{-1}U \rightarrow U$ and $p_V : p^{-1}V \rightarrow V$ for the restrictions of p . Then the square of restrictions commutes:*

$$p_U \circ \iota_{p^{-1}V \subseteq p^{-1}U} = \iota_{V \subseteq U} \circ p_V \quad \text{as morphisms } p^{-1}V \rightarrow U.$$

Proof. The open immersion $U \hookrightarrow X$ is a monomorphism, so it suffices to compare the two composites $p^{-1}V \rightarrow U \hookrightarrow X$. Composing the left-hand side with $U \hookrightarrow X$ gives the composite

$p^{-1}V \hookrightarrow p^{-1}U \hookrightarrow W \xrightarrow{p} X$, i.e. the restriction of p to $p^{-1}V$ followed by the inclusion into X ; composing the right-hand side with $U \hookrightarrow X$ gives $p^{-1}V \xrightarrow{p_V} V \hookrightarrow X$, which is the same morphism. Cancel the monomorphism. $\square$

Definition 1202 (Precomposition of a rational map with an open morphism). Let $f : X \dashrightarrow Y$ be a rational map and $p : W \rightarrow X$ a morphism whose underlying continuous map is *open*. The composite

$$f \circ p : W \dashrightarrow Y$$

is the rational map represented by $(p^{-1}U, \varphi \circ p_U)$ for any representative (U, φ) of f , where $p_U : p^{-1}U \rightarrow U$ is the restriction of p .

Proof. Two things need proof: that $(p^{-1}U, \varphi \circ p_U)$ is a partial map, and that its class does not depend on the representative.

Density. If $D \subseteq X$ is dense and p is an open map, then $p^{-1}D$ is dense in W : for any nonempty open $O \subseteq W$ the image $p(O)$ is a nonempty open subset of X , hence meets D , so there is $w \in O$ with $p(w) \in D$, i.e. O meets $p^{-1}D$. Applied to $D = U$, the preimage $p^{-1}U$ is a dense open of W .

Independence of the representative. Let (U_1, φ_1) and (U_2, φ_2) be equivalent representatives of f , agreeing on a dense open $V \subseteq U_1 \cap U_2$. Then $p^{-1}V$ is dense (as above) and contained in $p^{-1}U_1 \cap p^{-1}U_2$, and on $p^{-1}V$ the two pulled-back morphisms agree: by the restriction square ??, restricting $\varphi_i \circ p_{U_i}$ to $p^{-1}V$ gives $\varphi_i|_V \circ p_V$, and $\varphi_1|_V = \varphi_2|_V$. Hence the two pullbacks are equivalent partial maps. $\square$

Lemma 1203 (Precomposition preserves being over the base). *Let S be a scheme, let W, X, Y be schemes over S , let (U, φ) be a partial map from X to Y whose morphism φ is a morphism over S , and let $p : W \rightarrow X$ be a morphism over S with open underlying map. Then the pulled-back partial map $(p^{-1}U, \varphi \circ p_U)$ is again over S ; in particular $f \circ p$ is a rational map over S whenever f is.*

Proof. The restriction $p_U : p^{-1}U \rightarrow U$ is a morphism over S : composing it with the inclusion $U \hookrightarrow X$ and the structure morphism of X gives the structure morphism of $p^{-1}U$, because p is over S and open subschemes carry the restricted structure morphisms. A composite of morphisms over S is over S . $\square$

Lemma 1204 (Precomposition commutes with post-composition). *For a rational map $f : X \dashrightarrow Y$, a morphism $p : W \rightarrow X$ with open underlying map, and a morphism $g : Y \rightarrow Z$,*

$$g \circ (f \circ p) = (g \circ f) \circ p \quad \text{as rational maps } W \dashrightarrow Z.$$

Proof. Choose a representative (U, φ) of f . Both sides are represented by the partial map $(p^{-1}U, g \circ (\varphi \circ p_U)) = (p^{-1}U, (g \circ \varphi) \circ p_U)$, by associativity of composition of morphisms. $\square$

Lemma 1205 (Precomposing a morphism). *Let $g : X \rightarrow Y$ be a morphism, regarded as a rational map, and let $p : W \rightarrow X$ have open underlying map. Then the precomposition of the rational map g with p is the rational map of the composite morphism $g \circ p : W \rightarrow Y$.*

Proof. The morphism g is represented by the total partial map (X, g) ; its pullback along p is $(p^{-1}X, g \circ p) = (W, g \circ p)$, the total partial map of the composite. $\square$

Lemma 1206 (Precomposition enlarges the domain by the preimage). *In the situation of ??,*

$$p^{-1}(\text{Dom } f) \subseteq \text{Dom}(f \circ p).$$

Proof. Let $w \in W$ with $p(w) \in \text{Dom } f$. By definition of the domain there is a representative (U, φ) of f with $p(w) \in U$. Its pullback $(p^{-1}U, \varphi \circ p_U)$ represents $f \circ p$ and is defined at w . $\square$

Lemma 1207 (Post-composition only enlarges the domain). *For a rational map $F : X \dashrightarrow Y$ and a morphism $g : Y \rightarrow Z$,*

$$\text{Dom}(F) \subseteq \text{Dom}(g \circ F).$$

Proof. A representative (U, φ) of F defined at x yields the representative $(U, g \circ \varphi)$ of $g \circ F$, with the same domain, defined at x . $\square$

Lemma 1208 (Associativity of post-composition). *For a rational map $f : X \dashrightarrow Y$ and morphisms $g : Y \rightarrow Z$, $h : Z \rightarrow T$,*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Proof. Both sides are represented by $(U, h \circ g \circ \varphi)$ for any representative (U, φ) of f . $\square$

The pairing of rational maps is built through the function field. Let X be an *integral* scheme with generic point η and function field $K(X) = \mathcal{O}_{X, \eta}$. Every open $U \ni \eta$ receives a canonical morphism $\text{Spec } K(X) \rightarrow U$ (the morphism of the local ring at η), so each representative (U, φ) of a rational map $f : X \dashrightarrow Y$ restricts to a morphism $\text{Spec } K(X) \rightarrow Y$; two representatives agree on a dense open, which contains η , so this morphism is independent of the representative. We call it the *generic-point morphism* $f_\eta : \text{Spec } K(X) \rightarrow Y$ of f .

Theorem 1209 (The function-field correspondence). *Let X be an integral scheme. Two rational maps $X \dashrightarrow Y$ are equal if and only if their generic-point morphisms agree. Moreover, if X and Y lie over a base S and the structure morphism s_Y is locally of finite type, every S -morphism $\text{Spec } K(X) \rightarrow Y$ is the generic-point morphism of a unique rational map $X \dashrightarrow Y$ over S : the finitely many coordinates of the image point, in an affine chart of Y , are germs at η of functions defined on a common open of X , so the morphism spreads out to a partial map.*

Lemma 1210 (Generic-point morphism of a post-composition). *Let X be irreducible, $f : X \dashrightarrow Y$ a rational map and $g : Y \rightarrow Z$ a morphism. Then $(g \circ f)_\eta = g \circ f_\eta$.*

Proof. Choose a representative (U, φ) of f ; then $(U, g \circ \varphi)$ represents $g \circ f$, and restricting to the generic point gives $(g \circ \varphi)_\eta = g \circ \varphi_\eta$ by associativity. $\square$

Lemma 1211 (Generic-point morphism of a map over the base). *Let X be integral, let $s_X : X \rightarrow S$, $s_Y : Y \rightarrow S$ be structure morphisms and $a : X \dashrightarrow Y$ a rational map over S ($s_Y \circ a = s_X$ as rational maps). Then*

$$s_Y \circ a_\eta = s_X \circ c_X,$$

where $c_X : \text{Spec } K(X) \rightarrow X$ is the canonical morphism.

Proof. Apply ?? with $g = s_Y$: $(s_Y \circ a)_\eta = s_Y \circ a_\eta$. By hypothesis $s_Y \circ a$ is the rational map of the morphism s_X , whose generic-point morphism is the restriction of s_X to the generic point, i.e. $s_X \circ c_X$. $\square$

Definition 1212 (Pairing of rational maps into a fibre product). Let X be an integral scheme with structure morphism $s_X : X \rightarrow S$, and let $s_Y : Y \rightarrow S$, $s_Z : Z \rightarrow S$ be locally of finite type. Let $a : X \dashrightarrow Y$ and $b : X \dashrightarrow Z$ be rational maps over S . The *pairing*

$$(a, b) : X \dashrightarrow Y \times_S Z$$

is the unique rational map over S whose generic-point morphism is the morphism $(a_\eta, b_\eta) : \text{Spec } K(X) \rightarrow Y \times_S Z$ induced by the universal property of the fibre product.

Proof. The two morphisms a_η, b_η out of $\text{Spec } K(X)$ agree after composing with s_Y , resp. s_Z : both composites equal $s_X \circ c_X$ by ???. Hence they factor through the fibre product. The structure morphism $Y \times_S Z \rightarrow S$ is the composite of the projection to Y (a base change of s_Z , hence locally of finite type) with s_Y , so it is locally of finite type, and the function-field correspondence ?? produces the unique rational map over S with the prescribed generic-point morphism. $\square$

Lemma 1213 (The projections of the pairing recover the factors). *In the situation of ??, writing pr_1, pr_2 for the projections of $Y \times_S Z$,*

$$\text{pr}_1 \circ (a, b) = a \quad \text{and} \quad \text{pr}_2 \circ (a, b) = b.$$

Proof. Rational maps out of the integral X are determined by their generic-point morphisms (??). By ?? the generic-point morphism of $\text{pr}_1 \circ (a, b)$ is $\text{pr}_1 \circ (a_\eta, b_\eta) = a_\eta$, and symmetrically for the second projection. $\square$

Lemma 1214 (The pairing is over the base). *In the situation of ??, composing (a, b) with the structure morphism $s_Y \circ \text{pr}_1$ of $Y \times_S Z$ gives s_X , as rational maps $X \dashrightarrow S$.*

Proof. By associativity (??) and ??, $(s_Y \circ \text{pr}_1) \circ (a, b) = s_Y \circ (\text{pr}_1 \circ (a, b)) = s_Y \circ a = s_X$. $\square$

For a *reduced* scheme X and a *separated* scheme Y , every rational map $f : X \dashrightarrow Y$ has a largest representative: the representatives glue on overlaps (two morphisms into a separated scheme that agree on a dense open of a reduced scheme agree, cf. ??), producing a representative defined on all of $\text{Dom}(f)$.

Definition 1215 (The maximal representative). For X reduced and Y separated, the *maximal representative* of a rational map $f : X \dashrightarrow Y$ is the partial map with domain $\text{Dom}(f)$ obtained by gluing all representatives; its class is f .

Definition 1216 (The explicit pairing representative). Let X be reduced and let Y, Z, S be separated. Let $a : X \dashrightarrow Y, b : X \dashrightarrow Z$ be rational maps over S , with maximal representatives α on $\text{Dom}(a)$ and β on $\text{Dom}(b)$. The *pairing representative* is the partial map from X to $Y \times_S Z$ with domain $\text{Dom}(a) \cap \text{Dom}(b)$ and morphism (α, β) , the fibre-product pairing of the two restrictions.

Proof. The intersection of the dense open $\text{Dom}(a)$ with the dense open $\text{Dom}(b)$ is dense: a nonempty open O meets $\text{Dom}(a)$ in a nonempty open, which meets $\text{Dom}(b)$. The maximal representative of a rational map over the separated base S is a morphism over S (its restriction to a small enough dense open is, and the locus where the two composites to S agree is closed since S is separated, contains a dense open, and X is reduced); hence α and β agree after composing to S and pair into the fibre product. $\square$

Lemma 1217 (The pairing representative represents the pairing). *Let X be integral and reduced, Y, Z, S separated, s_Y, s_Z locally of finite type, and a, b as in ??. Then the class of the pairing representative equals (a, b) .*

Proof. Both rational maps have the same generic-point morphism: restricting the pairing representative to the generic point and composing with each projection of $Y \times_S Z$ gives the restriction of α , resp. β , to the generic point, i.e. a_η , resp. b_η ; and morphisms into a fibre product agree once their two projections do. Conclude by ??. $\square$

Lemma 1218 (The pairing is defined wherever both factors are). *In the situation of ??,*

$$\text{Dom}(a) \cap \text{Dom}(b) \subseteq \text{Dom}((a, b)).$$

Proof. The pairing representative is a representative of (a, b) (??) with domain $\text{Dom}(a) \cap \text{Dom}(b)$, and the domain of a rational map contains the domain of each of its representatives. $\square$

Definition 1219 (Germ pullback along a rational map). Let X be an integral scheme and $f : X \dashrightarrow Y$ a rational map. Let $y \in Y$ be the image under f_η of the unique point of $\text{Spec } K(X)$. The *germ pullback* of f is the local ring homomorphism

$$f^\# : \mathcal{O}_{Y,y} \longrightarrow K(X)$$

induced by f_η on the local ring at the image point.

Theorem 1220 (A dominant rational map hits the generic point). *Let X be integral, Y irreducible, and $f : X \dashrightarrow Y$ a dominant rational map (some, equivalently every, representative has dense image). Then the generic-point morphism f_η sends the unique point of $\text{Spec } K(X)$ to the generic point η_Y of Y .*

Proof. Choose a representative (U, φ) of f ; it is dominant. The open U is irreducible with generic point $\eta = \eta_X$ (a dense open of an irreducible space is irreducible, and its generic point is that of the ambient space, which lies in every nonempty open). The canonical morphism $\text{Spec } K(X) \rightarrow U$ sends the unique point to η , so f_η sends it to $\varphi(\eta)$.

It remains to see $\varphi(\eta) = \eta_Y$. Since $\{\eta\}$ is dense in U and φ is continuous, $\varphi(U) = \varphi(\overline{\{\eta\}} \cap U) \subseteq \overline{\{\varphi(\eta)\}}$; taking closures and using dominance, $\overline{\{\varphi(\eta)\}} \supseteq \overline{\varphi(U)} = Y$. So $\varphi(\eta)$ is a generic point of the irreducible sober space Y , hence equals η_Y by uniqueness of generic points. $\square$

Definition 1221 (Function-field pullback of a dominant rational map). Let X be integral, Y irreducible and $f : X \dashrightarrow Y$ dominant. By ?? the source of the germ pullback ?? is the local ring of Y at its generic point, i.e. the function field $K(Y)$; the resulting field homomorphism

$$f^* : K(Y) \longrightarrow K(X)$$

is the *function-field pullback* of f .

11.2 The difference map

Let $\mathcal{C}$ be a cartesian monoidal category (finite products, with $\otimes$ the product and 1 the terminal object) and G a group object of $\mathcal{C}$, with multiplication $\mu : G \otimes G \rightarrow G$, unit $e : 1 \rightarrow G$ and inversion $\iota : G \rightarrow G$. For every object T the set $\text{Hom}(T, G)$ is a group under $u \cdot v = \mu \circ \langle u, v \rangle$, with unit the constant $e \circ t_T$ and inverse $u^{-1} = \iota \circ u$; precomposition with any morphism $T' \rightarrow T$ is a group homomorphism. We use this *hom-group calculus* freely.

Definition 1222 (The difference morphism of a group object). For a group object G in a cartesian monoidal category, the *difference morphism* is

$$\delta = \text{pr}_1 \cdot \text{pr}_2^{-1} : G \otimes G \longrightarrow G,$$

the quotient, in the hom-group $\text{Hom}(G \otimes G, G)$, of the two projections; on T -points it is $(g, h) \mapsto gh^{-1}$.

Lemma 1223 (Reconstruction identity). *For T -points $a, b : T \rightarrow G$ of a group object G ,*

$$\mu \circ \langle \delta \circ \langle a, b \rangle, b \rangle = a.$$

In pointwise notation: $(ab^{-1})b = a$.

Proof. Precomposition with $\langle a, b \rangle$ is compatible with the hom-group operations, so $\delta \circ \langle a, b \rangle = (\text{pr}_1 \circ \langle a, b \rangle) \cdot (\text{pr}_2 \circ \langle a, b \rangle)^{-1} = a \cdot b^{-1}$ in $\text{Hom}(T, G)$. Then $\mu \circ \langle ab^{-1}, b \rangle = (ab^{-1}) \cdot b = a$ by the group axioms of the hom-group. $\square$

Now fix a field k . We work in the category of schemes over k ; the categorical product of X and Y over k is the fibre product $X \times_k Y$. A *group scheme* over k is a group object G in this category.

Definition 1224 (The difference and multiplication morphisms of a group scheme). For a group scheme G over k , write

$$\delta_G : G \times_k G \longrightarrow G, \quad m_G : G \times_k G \longrightarrow G$$

for the underlying scheme morphisms of the difference morphism ?? and of the multiplication. Both are morphisms over k : writing s_G for the structure morphism of G , one has $s_G \circ \delta_G = s_G \circ \text{pr}_1$, the structure morphism of $G \times_k G$.

Lemma 1225 (Scheme-level reconstruction identity). *Let G be a group scheme over k , T a scheme with a morphism $t : T \rightarrow \text{Spec } k$, and $A, B : T \rightarrow G$ two morphisms with $s_G \circ A = s_G \circ B = t$, where s_G is the structure morphism. Then*

$$m_G \circ \langle \delta_G \circ \langle A, B \rangle, B \rangle = A,$$

where the pairings are taken into the fibre product $G \times_k G$.

Proof. The morphisms A, B are morphisms over k from the object (T, t) , i.e. T -points of the group object G in the category of schemes over k ; the fibre-product pairings are the categorical pairings there. The identity is ?? for these points, read off on underlying scheme morphisms (the forgetful functor from schemes over k to schemes preserves the relevant composites and sends categorical pairings to fibre-product pairings). $\square$

Lemma 1226 (The projections of a smooth self-product are open). *Let X be a scheme smooth over k . Then both projections $\text{pr}_1, \text{pr}_2 : X \times_k X \rightarrow X$ are open maps.*

Proof. Each projection is a base change of the smooth structure morphism $X \rightarrow \text{Spec } k$, hence smooth; smooth morphisms are flat and locally of finite presentation, and such morphisms are open. $\square$

Definition 1227 (Milne's difference rational map). Let X be a scheme smooth over k whose self-product $X \times_k X$ is integral, and let G be a group scheme locally of finite type over k . For a rational map $f : X \dashrightarrow G$ over k , the *difference rational map*

$$\Phi : X \times_k X \dashrightarrow G, \quad \Phi(x, y) = f(x) f(y)^{-1},$$

is the composite $\delta_G \circ (f \circ \text{pr}_1, f \circ \text{pr}_2)$: precompose f with the two (open, by ??) projections, pair the results into $G \times_k G$ (??; both factors are over k by ??, and the pairing needs the integrality of $X \times_k X$), and post-compose with the difference morphism δ_G .

Lemma 1228 (The difference map is over the base). *In the situation of ??, Φ is a rational map over k : composing Φ with the structure morphism of G gives the structure morphism of $X \times_k X$, as rational maps.*

Proof. By associativity of post-composition (??) and since δ_G is a morphism over k (??), composing Φ with the structure morphism of G equals composing the pairing with the structure morphism of $G \times_k G$, which is the structure morphism of $X \times_k X$ by ??. $\square$

Definition 1229 (The explicit pairing representative of the difference map). Let X be smooth over k with X reduced, and G a separated group scheme over k . Let $f : X \dashrightarrow G$ be a rational map over k , with maximal representative φ on $U = \text{Dom}(f)$. The *pairing representative* is the partial map from $X \times_k X$ to $G \times_k G$ with domain $\text{pr}_1^{-1}(U) \cap \text{pr}_2^{-1}(U)$ and morphism the fibre-product pairing $(\varphi \circ \text{pr}_1, \varphi \circ \text{pr}_2)$ of the two pulled-back restrictions.

Proof. The two preimages are dense opens (preimages of the dense open U under the open projections, as in ??), and the intersection of two dense opens is dense. Both pulled-back morphisms are morphisms over k (??, using that both projections are morphisms over k), so they pair into the fibre product $G \times_k G$. $\square$

Lemma 1230 (The pairing representative represents the pairing of the pullbacks). *In the situation of ??, assume moreover that $X \times_k X$ is integral and G is locally of finite type over k . Then the class of the pairing representative equals the pairing $(f \circ \text{pr}_1, f \circ \text{pr}_2)$ of ??.*

Proof. As in ??: both sides are rational maps out of the integral scheme $X \times_k X$, and their generic-point morphisms agree component-wise — composing with each projection of $G \times_k G$ gives the generic-point morphism of $f \circ \text{pr}_i$, computed either from the pulled-back maximal representative or from the pairing. Conclude by ??. $\square$

Theorem 1231 (The difference map is defined on $\text{Dom}(f) \times \text{Dom}(f)$). *In the situation of ??, with X reduced and G separated,*

$$\text{pr}_1^{-1}(\text{Dom } f) \cap \text{pr}_2^{-1}(\text{Dom } f) \subseteq \text{Dom}(\Phi).$$

In particular Φ is defined at every point (x, y) with $x, y \in \text{Dom}(f)$.

Proof. Post-composition with δ_G only enlarges the domain (??), so it suffices to bound the domain of the pairing; and the pairing representative (??) is a representative of the pairing defined exactly on $\text{pr}_1^{-1}(\text{Dom } f) \cap \text{pr}_2^{-1}(\text{Dom } f)$. $\square$

Theorem 1232 (Reconstruction of $f \circ \text{pr}_1$ from the difference map). *In the situation of ??,*

$$m_G \circ (\Phi, f \circ \text{pr}_2) = f \circ \text{pr}_1 \quad \text{as rational maps } X \times_k X \dashrightarrow G,$$

where the pairing is formed as in ?? (both entries are over k , by ??). This is Milne's formula $f(x) = \Phi(x, u) \cdot f(u)$.

Proof. Both sides are rational maps out of the integral scheme $X \times_k X$; we compare generic-point morphisms (??). Write $A = (f \circ \text{pr}_1)_\eta$ and $B = (f \circ \text{pr}_2)_\eta$, two morphisms $\text{Spec } K(X \times_k X) \rightarrow G$ with equal composites to $\text{Spec } k$ (?). Unwinding ?? and ?? with ??, the generic-point morphism of the left-hand side is $m_G \circ \langle \delta_G \circ \langle A, B \rangle, B \rangle$, which equals A by the scheme-level reconstruction identity ??. $\square$

Theorem 1233 (Domain reflection through the difference map). *In the situation of ??, with X reduced and G separated,*

$$\text{Dom}(\Phi) \cap \text{pr}_2^{-1}(\text{Dom } f) \subseteq \text{Dom}(f \circ \text{pr}_1).$$

At a point where Φ is defined and whose second coordinate is a point of definition of f , the reconstruction $f(x) = \Phi(x, u) f(u)$ exhibits $f \circ \text{pr}_1$ as defined.

Proof. Rewrite $f \circ \text{pr}_1 = m_G \circ (\Phi, f \circ \text{pr}_2)$ by ??. The right-hand side is defined wherever the pairing is (??), the pairing is defined on $\text{Dom}(\Phi) \cap \text{Dom}(f \circ \text{pr}_2)$ (??), and $\text{pr}_2^{-1}(\text{Dom } f) \subseteq \text{Dom}(f \circ \text{pr}_2)$ (?). $\square$

11.3 Purity of the polar locus

This section proves that on an integral, locally Noetherian scheme whose local rings are all regular, the locus where a fixed element of the function field fails to be regular is of *pure codimension one*: every point of the locus is a specialisation of a codimension-one point of the locus (??). The classical route runs through normality and the Krull-domain intersection $R = \bigcap_{\text{ht } \mathfrak{p}=1} R_{\mathfrak{p}}$; the proof given here is elementary, replacing the integral-closedness input by a two-element “swap” argument valid in any regular local ring of dimension ≥ 2 .

Lemma 1234 (Swap lemma). *Let A be a domain, and let $u, v, b, x \in A$ with $u \neq 0$ and v regular modulo (u) (i.e. $vs \in (u)$ implies $s \in (u)$). If $ux \in (b)$ and $vx \in (b)$, then $x \in (b)$.*

Proof. If $b = 0$, then $ux = 0$ and $u \neq 0$ force $x = 0 \in (0)$. Assume $b \neq 0$ and write $ux = bs$, $vx = bt$. Then

$$b(ut) = u(vx) = v(ux) = b(vs),$$

and cancelling b (a nonzero element of a domain) gives $ut = vs$. Thus $vs \in (u)$, so $s \in (u)$ by regularity of v modulo (u) : write $s = us'$. Then $ux = bus'$, and cancelling u gives $x = bs' \in (b)$. $\square$

Theorem 1235 (Swap pairs exist in regular local rings of dimension ≥ 2). *Let $(A, \mathfrak{m})$ be a regular local (Noetherian) ring of Krull dimension ≥ 2 . Then there are $u, v \in \mathfrak{m}$ with $u \neq 0$ and v regular modulo (u) .*

Proof. In a regular local ring the minimal number of generators $\mu(\mathfrak{m})$ equals the Krull dimension, so $\mu(\mathfrak{m}) \geq 2$. Choose $u \in \mathfrak{m} \setminus \mathfrak{m}^2$ (??); then $u \neq 0$ since $0 \in \mathfrak{m}^2$. The quotient $A' = A/(u)$ is again a regular local ring, with $\mu(\mathfrak{m}') = \mu(\mathfrak{m}) - 1 \geq 1$ (??); being regular local it is a domain (??), and $\mathfrak{m}' \neq 0$ because $\mu(\mathfrak{m}') \geq 1$. Pick a nonzero $\bar{v} \in \mathfrak{m}'$ and lift it to $v \in A$; then $v \in \mathfrak{m}$ (a unit would have a unit image, but $\bar{v} \in \mathfrak{m}'$), and v is regular modulo (u) : if $vs \in (u)$ then $\bar{v}\bar{s} = 0$ in the domain A' with $\bar{v} \neq 0$, so $\bar{s} = 0$, i.e. $s \in (u)$. $\square$

Theorem 1236 (Height bound for colon primes of principal ideals). *Let A be a Noetherian domain, $b, y \in A$, and let $Q = ((b) : y)$ be a prime ideal. If the localisation A_Q is a regular local ring, then $\text{ht}(Q) \leq 1$.*

Proof. Suppose $\text{ht}(Q) \geq 2$; then $\dim A_Q = \text{ht}(Q) \geq 2$ (the primes of A_Q are the primes of A inside Q), and A_Q is a domain (a localisation of a domain). Write $\varphi : A \rightarrow A_Q$ for the localisation map, injective since A is a domain.

Step 1: $\varphi(y) \notin (\varphi(b))$. Otherwise $\varphi(y) = \varphi(b) \cdot c/s$ with $s \notin Q$; cross-multiplying and using injectivity gives $sy \in (b)$ in A , i.e. $s \in ((b) : y) = Q$, a contradiction.

Step 2: the maximal ideal of A_Q is the colon $((\varphi b) : \varphi y)$. For $r \in Q$ we have $ry \in (b)$, so $\varphi(r)\varphi(y) \in (\varphi b)$; since the maximal ideal of A_Q is generated by $\varphi(Q)$, this gives $\mathfrak{m}_{A_Q} \subseteq ((\varphi b) : \varphi y)$. Conversely the colon is a proper ideal by Step 1 (it does not contain 1), hence is contained in the maximal ideal.

Step 3: contradiction by the swap lemma. By ?? there is a swap pair $u, v \in \mathfrak{m}_{A_Q}$, $u \neq 0$, v regular modulo (u) . By Step 2, $u \cdot \varphi(y) \in (\varphi b)$ and $v \cdot \varphi(y) \in (\varphi b)$; the swap lemma ?? gives $\varphi(y) \in (\varphi b)$, contradicting Step 1. $\square$

Theorem 1237 (Existence of a height-one pole prime). *Let R be a Noetherian domain all of whose localisations at primes are regular local rings. Let $a, b \in R$ with $b \neq 0$, and let $\mathfrak{p}$ be a prime containing the denominator ideal $((b) : a)$. Then there is a prime $\mathfrak{q} \subseteq \mathfrak{p}$ of height exactly 1 with $((b) : a) \subseteq \mathfrak{q}$.*

Proof. Work in the localisation $A = R_{\mathfrak{p}}$, a Noetherian domain, with injective localisation map $\varphi : R \rightarrow A$; primes of A correspond to primes of R contained in $\mathfrak{p}$, compatibly with inclusions and heights.

Step 1: $\varphi(a) \notin (\varphi b)$. Otherwise, cross-multiplying as in Step 1 of ??, some $s \notin \mathfrak{p}$ has $sa \in (b)$, i.e. $s \in ((b) : a) \subseteq \mathfrak{p}$ — a contradiction. Hence the class of $\varphi(a)$ in $A/(\varphi b)$ is a nonzero element of a finitely generated module over the Noetherian ring A .

Step 2: an associated prime. The annihilator of a nonzero element of a finitely generated module over a Noetherian ring is contained in an associated prime of that module, and an associated prime is by definition the annihilator of some element. So there is a prime $Q \subseteq A$, the annihilator of the class of some $y \in A$ in $A/(\varphi b)$ — that is, $Q = ((\varphi b) : y)$ — with $((\varphi b) : \varphi a) \subseteq Q$.

Step 3: $\text{ht}(Q) = 1$. The localisation A_Q is a localisation of R at the prime $Q \cap R$, hence regular by hypothesis; ?? gives $\text{ht}(Q) \leq 1$. Moreover $\varphi(b) \in Q$ (as $\varphi(b)y \in (\varphi b)$) and $\varphi(b) \neq 0$, so $Q \neq 0$, and in a domain a nonzero prime has height ≥ 1 (the chain $0 \subsetneq Q$). Hence $\text{ht}(Q) = 1$.

Step 4: descent to R . Let $\mathfrak{q} = Q \cap R$, a prime of R contained in $\mathfrak{p}$ (as Q is contained in the maximal ideal of A , whose contraction is $\mathfrak{p}$). The correspondence of primes preserves heights, so $\text{ht}(\mathfrak{q}) = 1$. Finally, for $r \in ((b) : a)$: $\varphi(r)\varphi(a) \in (\varphi b)$, so $\varphi(r) \in ((\varphi b) : \varphi a) \subseteq Q$, i.e. $r \in \mathfrak{q}$. $\square$

Theorem 1238 (Purity of the polar locus). *Let X be an integral, locally Noetherian scheme all of whose stalks $\mathcal{O}_{X,x}$ are regular local rings. Let $h \in K(X)$ and let $P \in X$ be a point at which h is not regular, i.e. h does not lie in the image of $\mathcal{O}_{X,P} \rightarrow K(X)$. Then there is a point $z \in X$ with*

$$z \rightsquigarrow P, \quad \text{coheight}(z) = 1, \quad h \notin \text{im}(\mathcal{O}_{X,z} \rightarrow K(X)),$$

where $z \rightsquigarrow P$ means $P \in \overline{\{z\}}$.

Proof. Choose an affine open $U \ni P$, $U \cong \text{Spec } R$ with $R = \Gamma(U, \mathcal{O}_U)$; R is a Noetherian domain, its fraction field is $K(X)$, and for every point $w \in U$ with prime $\mathfrak{w}$ the stalk $\mathcal{O}_{X,w}$ is the localisation $R_{\mathfrak{w}}$. In particular every localisation of R at a prime is a regular local ring, by the hypothesis on stalks. Let $\mathfrak{p}$ be the prime of P and write $h = a/b$ with $a, b \in R$, $b \neq 0$.

Step 1 (regularity as a colon condition). For a prime $\mathfrak{q} \subseteq R$,

$$\frac{a}{b} \in \text{im}(R_{\mathfrak{q}} \rightarrow K(X)) \iff ((b) : a) \not\subseteq \mathfrak{q}.$$

($\Rightarrow$) If $a/b = c/s$ with $s \notin \mathfrak{q}$, then $sa = bc$ in R (cross-multiply in the fraction field and use injectivity of $R \rightarrow K(X)$), so $s \in ((b) : a)$ and the colon is not contained in $\mathfrak{q}$. ($\Leftarrow$) If $s \in ((b) : a) \setminus \mathfrak{q}$, write $sa = bc$; then $a/b = c/s \in R_{\mathfrak{q}}$.

Step 2 (the height-one prime). Since h is not regular at P , Step 1 (applied to $\mathfrak{q} = \mathfrak{p}$, whose localisation is $\mathcal{O}_{X,P}$) gives $((b) : a) \subseteq \mathfrak{p}$. By ?? there is a prime $\mathfrak{q} \subseteq \mathfrak{p}$ of height 1 with $((b) : a) \subseteq \mathfrak{q}$.

Step 3 (the point). Let $z \in U$ be the point of $\mathfrak{q}$. Then $z \rightsquigarrow P$: inclusions of primes correspond to specialisations in $\text{Spec } R \cong U$, and specialisations in the open subspace U are specialisations in X . The stalk $\mathcal{O}_{X,z}$ is $R_{\mathfrak{q}}$, of Krull dimension $\text{ht}(\mathfrak{q}) = 1$, so $\text{coheight}(z) = 1$ by ??. Finally h is not regular at z by Step 1, since $((b) : a) \subseteq \mathfrak{q}$. $\square$

11.4 Two substeps of Milne's Lemma 3.3

Milne's Lemma 3.3 asserts that the indeterminacy locus of a rational map from a nonsingular variety to a group variety is empty or of pure codimension one (the statement is isolated as ?? below). This section proves two of its substeps: the topological slice argument producing the

auxiliary point u of Milne's formula $f(x) = \Phi(x, u) f(u)$, and the Krull-theoretic fact that the zero locus of a regular function is of pure codimension one.

Theorem 1239 (Slice witness on the self-product). *Let X be a scheme, smooth and geometrically irreducible over the field k . Let $V \subseteq X \times_k X$ be open, $D \subseteq X$ a dense open, $x \in X$ a point, and suppose some point p_0 of the fibre $\mathrm{pr}_1^{-1}\{x\}$ lies in V . Then there is a point p with*

$$\mathrm{pr}_1(p) = x, \quad p \in V, \quad \mathrm{pr}_2(p) \in D.$$

Proof. The fibre is irreducible. The projection pr_1 is a base change of the structure morphism of X , hence geometrically irreducible (geometric irreducibility is stable under base change) and open (??); the preimage of an irreducible subset under an open morphism with irreducible geometric fibres is irreducible. Applied to $\{x\}$, the fibre $F = \mathrm{pr}_1^{-1}\{x\}$ is irreducible.

Both opens meet the fibre. V meets F at p_0 . For $\mathrm{pr}_2^{-1}(D)$: pick $u \in D$ (dense opens of the nonempty X are nonempty). Over the one-point base $\mathrm{Spec} k$ there is a point q of $X \times_k X$ with $\mathrm{pr}_1(q) = x$ and $\mathrm{pr}_2(q) = u$: the residue fields $\kappa(x)$ and $\kappa(u)$ are extensions of k , their tensor product $\kappa(x) \otimes_k \kappa(u)$ is a nonzero ring, and any prime of it gives such a point. Thus $q \in F \cap \mathrm{pr}_2^{-1}(D)$.

Conclusion. An irreducible set is preirreducible: two open sets each meeting F meet F simultaneously. Applied to V and $\mathrm{pr}_2^{-1}(D)$ this produces the desired $p \in F \cap V \cap \mathrm{pr}_2^{-1}(D)$. $\square$

Theorem 1240 (Slice witness for the difference map). *In the situation of ??, with X moreover geometrically irreducible over k , X reduced and G separated: if the difference map Φ is defined at some point p_0 with $\mathrm{pr}_1(p_0) = x$, then there is a point p with $\mathrm{pr}_1(p) = x$ at which $f \circ \mathrm{pr}_1$ is defined.*

Proof. Apply ?? with $V = \mathrm{Dom}(\Phi)$ (open, met by the fibre at p_0) and $D = \mathrm{Dom}(f)$ (dense open): this yields p over x with $p \in \mathrm{Dom}(\Phi)$ and $\mathrm{pr}_2(p) \in \mathrm{Dom}(f)$. By the domain reflection ??, $f \circ \mathrm{pr}_1$ is defined at p . $\square$

Theorem 1241 (Height-one transport of Krull's Hauptidealsatz). *Let R be a Noetherian domain, $t \in R$ nonzero, and $\mathfrak{p}$ a prime containing t . Then there is a prime $\mathfrak{q} \subseteq \mathfrak{p}$ of height exactly 1 with $t \in \mathfrak{q}$.*

Proof. The ideal (t) is contained in $\mathfrak{p}$, so among the primes between (t) and $\mathfrak{p}$ there is one, say $\mathfrak{q}$, minimal over (t) (the set of primes containing (t) and contained in $\mathfrak{p}$ is nonempty and closed under intersections of chains). By Krull's principal ideal theorem, a prime minimal over a principal ideal in a Noetherian ring has height ≤ 1 . And $t \in \mathfrak{q}$ with $t \neq 0$ gives $\mathfrak{q} \neq 0$, so in the domain R the chain $0 \subsetneq \mathfrak{q}$ gives $\mathrm{ht}(\mathfrak{q}) \geq 1$. $\square$

Lemma 1242 (Regularity propagates to generisations). *Let X be an irreducible scheme and $z \rightsquigarrow P$ a specialisation. Then*

$$\mathrm{im}(\mathcal{O}_{X,P} \rightarrow K(X)) \subseteq \mathrm{im}(\mathcal{O}_{X,z} \rightarrow K(X)) :$$

an element of the function field regular at P is regular at every generisation z of P .

Proof. Since z is a generisation of P , every open set containing P contains z ; taking germs therefore defines a canonical homomorphism $\mathcal{O}_{X,P} \rightarrow \mathcal{O}_{X,z}$, compatible with the two maps into the common stalk at the generic point. Hence the image of $\mathcal{O}_{X,P}$ in $K(X)$ factors through that of $\mathcal{O}_{X,z}$. $\square$

Theorem 1243 (The zero locus of a regular function is pure codimension one). *Let X be an integral, locally Noetherian scheme with regular stalks. Let $P \in X$ and let $s \in \mathcal{O}_{X,P}$ lie in the maximal ideal, with nonzero image $t \in K(X)$. Then there is a point $z \rightsquigarrow P$ with $\text{coheight}(z) = 1$ and an element $s' \in \mathcal{O}_{X,z}$, again in the maximal ideal, with the same image t in $K(X)$.*

Proof. *Step A:* t^{-1} is not regular at P . Suppose t^{-1} were the image of some $u \in \mathcal{O}_{X,P}$. Then $us \in \mathfrak{m}_P$, so $1 - us$ is a unit of the local ring $\mathcal{O}_{X,P}$; but its image in $K(X)$ is $1 - t^{-1}t = 0$, and a ring homomorphism into the nonzero ring $K(X)$ sends units to units — a contradiction.

Step B: *purity.* By ?? applied to t^{-1} , there is $z \rightsquigarrow P$ with $\text{coheight}(z) = 1$ and t^{-1} not regular at z .

Step C: t is regular at z . By ??, t , being regular at P , is regular at the generisation z : there is $s' \in \mathcal{O}_{X,z}$ with image t .

Step D: s' lies in the maximal ideal. Otherwise s' is a unit of $\mathcal{O}_{X,z}$, and the image of s'^{-1} in $K(X)$ is t^{-1} , making t^{-1} regular at z — contradicting Step B. $\square$

11.5 Smooth varieties: charts, reducedness, regular stalks, discrete valuation rings

This section records the local structure of a scheme smooth over a field: standard-smooth charts, freeness and rank of the Kähler differentials through localisation, the cotangent computation at rational points, reducedness, regularity of all stalks, integrality of the self-product, and the discrete valuation rings at codimension-one points.

Theorem 1244 (Standard-smooth charts). *Let $\bar{k}$ be an algebraically closed field and X a scheme smooth over $\bar{k}$. Every point $z \in X$ has an affine open neighbourhood V , lying over an affine open $U \subseteq \text{Spec } \bar{k}$, such that the induced ring map $\Gamma(U) \rightarrow \Gamma(V)$ is standard smooth: a quotient $\Gamma(U)[x_1, \dots, x_n]/(f_1, \dots, f_c)$ whose Jacobian determinant $\det(\partial f_i/\partial x_j)_{i,j \leq c}$ is invertible.*

Proof. Smoothness is local on source and target, and a smooth ring map is, locally on the source, standard smooth (the local structure theorem for smooth morphisms, Stacks 00T7): choose an affine chart of X around z and shrink it accordingly. $\square$

Lemma 1245 (Kähler differentials of a standard-smooth algebra are free). *Let S be a standard-smooth algebra over a ring R . Then $\Omega_{S/R}$ is a free S -module.*

Proof. Write $S = R[x_1, \dots, x_n]/(f_1, \dots, f_c)$ with invertible Jacobian minor. In the conormal sequence of this presentation the classes of $f_1, \dots, f_c$ map to a basis of the summand $\bigoplus_{i \leq c} S dx_i$ (invertibility of the Jacobian determinant), so $\Omega_{S/R}$ is free with basis the images of $dx_{c+1}, \dots, dx_n$ (Stacks 00T7). $\square$

Lemma 1246 (Freeness of Kähler differentials localises). *Let S be an R -algebra with $\Omega_{S/R}$ a free S -module, and let S_M be a localisation of S at a multiplicative set M . Then $\Omega_{S_M/R}$ is a free S_M -module.*

Proof. Kähler differentials localise: $\Omega_{S_M/R} \cong S_M \otimes_S \Omega_{S/R}$ (Stacks 00RT). The base change of a free module is free. $\square$

Lemma 1247 (Rank of the localised Kähler differentials). *Let S be a nonzero standard-smooth R -algebra of relative dimension n , and S_M a nonzero localisation of S . Then $\Omega_{S_M/R}$ is free of rank n over S_M .*

Proof. $\Omega_{S/R}$ is free of rank n over S : the basis exhibited in ?? has $n-c$ elements for a presentation on n' variables and c relations with $n = n' - c$ — the relative dimension. Localisation is a base change, so $\Omega_{S_M/R} \cong S_M \otimes_S \Omega_{S/R}$ (??) is free of the same rank (well-defined since $S_M \neq 0$). $\square$

Theorem 1248 (Cotangent dimension at a rational point). *Let R be a ring and $(S, \mathfrak{m}, \kappa)$ a local R -algebra, formally smooth over R , with $\Omega_{S/R}$ free of rank $n \in \mathbb{N}$. If the composite $R \rightarrow S \rightarrow \kappa$ is bijective, then*

$$\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = n.$$

Proof. Since $R \rightarrow \kappa$ is an isomorphism of rings, κ is formally smooth over R , and $\Omega_{\kappa/R} = 0$ (every R -derivation of κ kills the image of R , which is everything).

Consider the conormal map of the surjection $S \twoheadrightarrow \kappa$ with kernel $\mathfrak{m}$ (Stacks 00RU):

$$c : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \kappa \otimes_S \Omega_{S/R}, \quad c(\bar{f}) = df \otimes 1.$$

Injectivity. Because κ is formally smooth over R and $S \rightarrow \kappa$ is surjective, the conormal map admits a retraction (formal smoothness of the quotient provides compatible liftings to the square-zero extensions $S/\mathfrak{m}^2 \twoheadrightarrow \kappa$, splitting c); in particular c is injective.

Surjectivity. The conormal sequence $\mathfrak{m}/\mathfrak{m}^2 \rightarrow \kappa \otimes_S \Omega_{S/R} \rightarrow \Omega_{\kappa/R} \rightarrow 0$ is exact, and $\Omega_{\kappa/R} = 0$, so c is surjective.

Hence c is an isomorphism of κ -vector spaces, and the right-hand side, the base change of a free rank- n module, has dimension n . $\square$

Lemma 1249 (Sections of the point over a field). *Let k be a field and U a nonempty open of $\text{Spec } k$. Then $\Gamma(\text{Spec } k, U) \cong k$ as rings.*

Proof. $\text{Spec } k$ has a single point, so the only nonempty open is the whole space, whose ring of sections is k . $\square$

Theorem 1250 (Regularity at a rational closed point of a standard-smooth algebra). *Let k be a field, S a nonzero standard-smooth k -algebra, $\mathfrak{m} \subseteq S$ a maximal ideal, and $S_{\mathfrak{m}}$ a localisation of S at $\mathfrak{m}$. If the residue map $k \rightarrow \kappa(\mathfrak{m})$ is bijective, then $S_{\mathfrak{m}}$ is a regular local ring.*

Proof. S is of finite type over k , hence Noetherian, and so is the localisation $S_{\mathfrak{m}}$. Fix n with S standard smooth of relative dimension n (every standard-smooth algebra has a presentation with a well-defined relative dimension). By ?? and ??, $\Omega_{S_{\mathfrak{m}}/k}$ is free of rank n . The localisation of a (formally) smooth algebra is formally smooth, so ?? applies and gives $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = n$ for the maximal ideal of $S_{\mathfrak{m}}$. On the other side, $\dim S_{\mathfrak{m}} \geq n$ by the height lower bound at maximal ideals (??). So $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = n \leq \dim S_{\mathfrak{m}}$, and ?? concludes. $\square$

Theorem 1251 (Regularity at every closed point over an algebraically closed field). *Let $\bar{k}$ be an algebraically closed field and S a nonzero standard-smooth $\bar{k}$ -algebra. For every maximal ideal $\mathfrak{m} \subseteq S$, the localisation $S_{\mathfrak{m}}$ is a regular local ring.*

Proof. By Zariski's lemma the residue field $S/\mathfrak{m}$, a finitely generated $\bar{k}$ -algebra that is a field, is a finite — hence algebraic — extension of $\bar{k}$; as $\bar{k}$ is algebraically closed, the residue map $\bar{k} \rightarrow \kappa(\mathfrak{m})$ is bijective. Apply ??. $\square$

Theorem 1252 (Standard-smooth algebras over an algebraically closed field are reduced). *Let $\bar{k}$ be an algebraically closed field and S a standard-smooth $\bar{k}$ -algebra. Then S is reduced.*

Proof. The zero ring is reduced; assume $S \neq 0$. A ring is reduced as soon as its localisations at all maximal ideals are reduced (a nonzero nilpotent remains nonzero in some such localisation, since its annihilator is a proper ideal). Each localisation $S_{\mathfrak{m}}$ is regular local (??), hence a domain (??), hence reduced. $\square$

Theorem 1253 (A scheme smooth over an algebraically closed field is reduced). *Let $\bar{k}$ be an algebraically closed field and X a scheme smooth over $\bar{k}$. Then X is reduced.*

Proof. Reducedness is checked on stalks. Fix $z \in X$ and choose a standard-smooth chart $V \ni z$ over $U \subseteq \text{Spec } \bar{k}$ (??); identifying $\Gamma(U) \cong \bar{k}$ (??, U is nonempty as it contains the image of z) makes $\Gamma(V)$ a standard-smooth $\bar{k}$ -algebra, which is reduced by ??. The stalk $\mathcal{O}_{X,z}$ is a localisation of $\Gamma(V)$, and localisations of reduced rings are reduced. $\square$

Theorem 1254 (Regularity of all stalks of a smooth variety). *Let $\bar{k}$ be an algebraically closed field and X an integral scheme, smooth, geometrically irreducible, separated and locally of finite type over $\bar{k}$. Then the stalk $\mathcal{O}_{X,z}$ at every point $z \in X$ (closed or not) is a regular local ring.*

Proof. Choose a standard-smooth chart $V \ni z$ over U (??) and identify $\Gamma(U) \cong \bar{k}$ (??), making $\Gamma(V)$ a nonzero standard-smooth $\bar{k}$ -algebra. The stalk $\mathcal{O}_{X,z}$ is the localisation of $\Gamma(V)$ at the prime of z . Since $\bar{k}$ is perfect, the smooth-prime regularity theorem ?? applies at the (arbitrary) prime of z and gives regularity of the localisation. $\square$

Theorem 1255 (Integrality of the self-product). *Let $\bar{k}$ be an algebraically closed field and X an integral scheme, smooth and geometrically irreducible over $\bar{k}$. Then the self fibre product $X \times_{\bar{k}} X$ is an integral scheme.*

Proof. The structure morphism of $X \times_{\bar{k}} X$ is the composite of the projection pr_1 (a base change of the smooth structure morphism of X , hence smooth) with the structure morphism of X ; so $X \times_{\bar{k}} X$ is smooth over $\bar{k}$ and therefore reduced (??). It is also irreducible: the structure morphism of X is geometrically irreducible and universally open (being smooth), and the fibre product of the irreducible X with a scheme whose structure morphism is geometrically irreducible and universally open is irreducible (the projection is open with irreducible fibres onto an irreducible base). Reduced and irreducible is integral. $\square$

Theorem 1256 (Matsumura's regular-sequence criterion). *Let $(A, \mathfrak{m}, \kappa)$ be a regular local Noetherian ring and $f_1, \dots, f_c \in \mathfrak{m}$ elements whose classes in $\mathfrak{m}/\mathfrak{m}^2$ are κ -linearly independent. Then $f_1, \dots, f_c$ is a regular sequence in A .*

Proof. Induction on c ; the empty sequence is regular. Let $c \geq 1$.

The head is a nonzerodivisor. A is a domain (??), and $[f_1] \neq 0$ in $\mathfrak{m}/\mathfrak{m}^2$ (linear independence), so $f_1 \notin \mathfrak{m}^2$; in particular $f_1 \neq 0$, hence f_1 is a nonzerodivisor.

The quotient is regular local. Since $f_1 \in \mathfrak{m} \setminus \mathfrak{m}^2$, the quotient $A' = A/(f_1)$ is again a regular local ring, with minimal generator count $\mu(\mathfrak{m}') = \mu(\mathfrak{m}) - 1$ (??; note $\mu(\mathfrak{m}) \geq 1$ because $f_1 \in \mathfrak{m} \setminus \mathfrak{m}^2$ forces $\mathfrak{m} \neq 0$).

Descent of linear independence. Let $\pi : \mathfrak{m}/\mathfrak{m}^2 \rightarrow \mathfrak{m}'/\mathfrak{m}'^2$ be the map induced by $A \twoheadrightarrow A'$ (well defined: $\mathfrak{m}$ maps onto $\mathfrak{m}'$ and $\mathfrak{m}^2$ into $\mathfrak{m}'^2$; the residue fields of A and A' agree). π is surjective, and its kernel is the κ -line spanned by $[f_1]$: an element $x \in \mathfrak{m}$ has $\pi([x]) = 0$ exactly when $x \in (f_1) + \mathfrak{m}^2$, i.e. $x = af_1 + m$ with $m \in \mathfrak{m}^2$; writing $a = a_0 + a'$ with $a' \in \mathfrak{m}$ gives $[x] = \bar{a}_0[f_1]$, and conversely. Now suppose $\sum_{i \geq 2} \bar{g}_i \pi([f_i]) = 0$ with $\bar{g}_i \in \kappa$. Lifting the coefficients to A and applying the kernel computation, $\sum_{i \geq 2} \bar{g}_i [f_i] \in \kappa[f_1]$, so some relation $\bar{b}[f_1] - \sum_{i \geq 2} \bar{g}_i [f_i] = 0$ holds in $\mathfrak{m}/\mathfrak{m}^2$; linear independence of $[f_1], \dots, [f_c]$ forces all coefficients, in particular every $\bar{g}_i$, to vanish. Hence the classes of $f_2, \dots, f_c$ in $\mathfrak{m}'/\mathfrak{m}'^2$ are κ -linearly independent.

Conclusion. By the inductive hypothesis applied to the regular local Noetherian ring A' , the images of $f_2, \dots, f_c$ form a regular sequence in A' . Together with the nonzerodivisor f_1 this makes $f_1, f_2, \dots, f_c$ a regular sequence in A : the successive quotients $A/(f_1, \dots, f_i) \cong A'/(f_2, \dots, f_i)$ agree, so each f_{i+1} acts as a nonzerodivisor on the corresponding quotient, and the total quotient $A/(f_1, \dots, f_c) \neq 0$ since A' is local and the images lie in its maximal ideal. $\square$

Theorem 1257 (Smooth and codimension one implies a discrete valuation ring). *Let $\bar{k}$ be an algebraically closed field and X an integral scheme, smooth, geometrically irreducible, separated and locally of finite type over $\bar{k}$. For every point $z \in X$ with $\text{coheight}(z) = 1$, the stalk $\mathcal{O}_{X,z}$ is a discrete valuation ring.*

Proof. The stalk $A = \mathcal{O}_{X,z}$ is Noetherian (X is locally of finite type over the field $\bar{k}$, hence locally Noetherian) and regular local (??), of Krull dimension $\dim A = \text{coheight}(z) = 1$ (??). Regularity in dimension one says $\dim_{\kappa} \mathfrak{m}/\mathfrak{m}^2 = 1$, so by Nakayama the maximal ideal is principal, generated by any lift of a basis vector, and $\mathfrak{m} \neq 0$ (else $\dim A = 0$). A regular local ring is a domain (??); and a Noetherian local domain whose maximal ideal is principal and nonzero is a discrete valuation ring (Stacks 00PD). $\square$

11.6 The indeterminacy locus and Milne's Theorem 3.1

Definition 1258 (Indeterminacy locus). The *indeterminacy locus* of a rational map $f : X \dashrightarrow Y$ is the closed subset

$$Z(f) = X \setminus \text{Dom}(f).$$

The map f is represented by a (unique, for X reduced and Y separated) total morphism if and only if $Z(f) = \emptyset$.

Definition 1259 (Codimension-one-indeterminacy-free rational maps). A rational map $f : X \dashrightarrow Y$ is *codimension-one-indeterminacy-free* if every point $x \in X$ with $\text{coheight}(x) = 1$ lies in $\text{Dom}(f)$ — equivalently, if no codimension-one point of X lies in $Z(f)$.

Definition 1260 (The Milne 3.3 indeterminacy property). A rational map $f : X \dashrightarrow G$ has the *Milne 3.3 indeterminacy property* if its indeterminacy locus is empty or of pure codimension one, in the following precise sense: either $Z(f) = \emptyset$, or every $x \in Z(f)$ satisfies

$$\exists z \in Z(f), \quad \text{coheight}(z) = 1 \quad \text{and} \quad x \in \overline{\{z\}}.$$

(The requirement $z \in Z(f)$ is essential: on a variety every non-generic point is a specialisation of *some* codimension-one point, so without it the second alternative would be nearly vacuous, and in particular too weak to combine with the codimension- ≥ 2 bound of ??.)

Milne's Lemma 3.3 asserts that every rational map over an algebraically closed field $\bar{k}$ from a nonsingular $\bar{k}$ -variety to a group $\bar{k}$ -variety has this property; this is ??, proved in §?? from the substeps developed here (??, ??, ??, ??, ??, ??) together with the row argument, the germ-range spreading criterion and the codimension-one transport of §??. The consumers in §?? nevertheless take the property as an explicit hypothesis.

Theorem 1261 (Milne's Theorem 3.1: indeterminacy in codimension ≥ 2). *Let $\bar{k}$ be an algebraically closed field, X a nonsingular $\bar{k}$ -variety (an integral scheme, smooth, geometrically irreducible, separated and locally of finite type over $\bar{k}$) and Y a complete $\bar{k}$ -variety (an integral scheme, proper and geometrically irreducible over $\bar{k}$). Let $f : X \dashrightarrow Y$ be a rational map over $\bar{k}$. Then every point of $Z(f)$ has coheight at least 2:*

$$\forall z \in Z(f), \quad 2 \leq \text{coheight}(z).$$

Proof. Let $z \in Z(f)$ and suppose $\text{coheight}(z) \leq 1$.

Case $\text{coheight}(z) = 0$. Then z admits no strict generisation, i.e. it is maximal in the specialisation preorder. The generic point η of the irreducible sober space X generizes every point, so maximality forces $z = \eta$. But the generic point lies in every nonempty open subset, in particular in the dense open $\text{Dom}(f)$ — contradicting $z \in Z(f)$.

Case $\text{coheight}(z) = 1$. The stalk $\mathcal{O}_{X,z}$ is a discrete valuation ring (??) whose fraction field is the function field $K(X)$ (the stalk of an integral scheme injects into the stalk at the generic point, and at a codimension-one point of a Noetherian integral scheme the fraction field of the stalk is $K(X)$). Consider the commutative square

$$\begin{array}{ccc} \text{Spec } K(X) & \xrightarrow{f_\eta} & Y \\ \downarrow & & \downarrow \\ \text{Spec } \mathcal{O}_{X,z} & \longrightarrow & \text{Spec } \bar{k} \end{array}$$

whose left leg is induced by the inclusion $\mathcal{O}_{X,z} \subseteq K(X)$ and whose bottom is the structure morphism of $\text{Spec } \mathcal{O}_{X,z}$ (the square commutes because f is a rational map over $\bar{k}$: composing f_η with the structure morphism of Y gives the structure morphism of $\text{Spec } K(X)$, by ?? applied to the over- $\bar{k}$ equation). Since $Y \rightarrow \text{Spec } \bar{k}$ is proper and $\mathcal{O}_{X,z}$ is a valuation ring with fraction field $K(X)$, the valuative criterion of properness produces a lift $L : \text{Spec } \mathcal{O}_{X,z} \rightarrow Y$ with $L|_{\text{Spec } K(X)} = f_\eta$, compatible with the structure morphisms.

Because Y is locally of finite type over $\bar{k}$ (properness includes finite type), the morphism L , defined at the level of the local ring at z , spreads out to a morphism $g : V \rightarrow Y$ on an open neighbourhood $V \ni z$: in an affine chart of Y around the image of the closed point of $\text{Spec } \mathcal{O}_{X,z}$, the finitely many coordinate germs of L are defined on a common neighbourhood of z . The partial map (V, g) — its domain is dense because X is irreducible — has generic-point morphism equal to the restriction of L to the generic point, i.e. f_η ; by the function-field correspondence (??) its class is f . Hence $z \in \text{Dom}(f)$ — contradicting $z \in Z(f)$. $\square$

Corollary 1262 (Rational maps to complete varieties are codimension-one-free). *In the situation of ??, f is codimension-one-indeterminacy-free.*

Proof. A point of coheight 1 cannot lie in $Z(f)$, whose points have coheight ≥ 2 by ??. $\square$

11.7 Cohen–Macaulay equidimensionality and kernel generation

The transport step of Milne’s Lemma 3.3 (§??) rests on a self-contained package of commutative algebra over a Noetherian local ring $(R, \mathfrak{m})$: Ischebeck’s depth bound, the resulting Cohen–Macaulay equidimensionality, the relation $\dim R/\mathfrak{p} + 1 = \dim R$ at a height-one prime, and the fact that the kernel of a surjection of regular local rings is generated by $\dim B - \dim A$ elements. These feed the codimension count that pins the pole point of the difference map to codimension one.

Lemma 1263 (The quotient dimension is the coheight). *For a prime $\mathfrak{p}$ of a commutative ring R ,*

$$\dim(R/\mathfrak{p}) = \text{coheight}(\mathfrak{p}) \quad \text{in } \text{Spec } R.$$

Proof. The prime spectrum of $R/\mathfrak{p}$ is order-isomorphic to the closed set $V(\mathfrak{p}) = \{\mathfrak{q} : \mathfrak{p} \subseteq \mathfrak{q}\}$, i.e. the up-set $\{\mathfrak{q} : \mathfrak{p} \leq \mathfrak{q}\}$ of $\mathfrak{p}$ in $\text{Spec } R$, via $\mathfrak{q}/\mathfrak{p} \mapsto \mathfrak{q}$. Krull dimension is invariant under order

isomorphism, and the coheight of $\mathfrak{p}$ is by definition the Krull dimension of that up-set. Hence $\dim(R/\mathfrak{p}) = \text{coheight}(\mathfrak{p})$. $\square$

Lemma 1264 (Depth drop along a regular element). *Let $(R, \mathfrak{m})$ be a Noetherian local ring, M a finite nontrivial R -module, and $x \in \mathfrak{m}$ an M -regular element. Then*

$$\text{depth } M \leq \text{depth}(M/xM) + 1.$$

Proof. Multiplication by x is injective on M (as x is M -regular), so

$$0 \longrightarrow M \xrightarrow{x} M \longrightarrow M/xM \longrightarrow 0$$

is a short exact sequence. The short-exact-sequence depth estimate (??) bounds the depth of the middle term by $\text{depth}(M/xM) + 1$; the middle term is M , giving the claim. $\square$

Theorem 1265 (Ischebeck's bound). *Let $(R, \mathfrak{m})$ be a Noetherian local ring, M a finite nontrivial R -module, and $\mathfrak{p} \in \text{Ass}(M)$ an associated prime. Then*

$$\text{depth } M \leq \text{coheight}(\mathfrak{p}) \quad (= \dim R/\mathfrak{p}, \text{ ??}).$$

Proof. It suffices to show $n \leq \text{coheight}(\mathfrak{p})$ whenever $n \leq \text{depth } M$, by induction on n (uniformly in M and $\mathfrak{p}$). The case $n = 0$ is trivial. Suppose $n + 1 \leq \text{depth } M$. Since $\text{depth } M \geq 1$, the ideal $\mathfrak{m}$ contains an M -regular element x . Write $\mathfrak{p} = \text{ann}(m)$ for some $m \neq 0$ (as $\mathfrak{p} \in \text{Ass}(M)$). Then $x \notin \mathfrak{p}$: otherwise $xm = 0$, contradicting M -regularity of x at $m \neq 0$.

Let $T = \{u \in M : \mathfrak{p} \cdot u = 0\}$ be the $\mathfrak{p}$ -torsion submodule; then $m \in T$. We claim $T \not\subseteq xM$. Otherwise every $t \in T$ is xt' with $t' \in T$ (from $\mathfrak{p}$ -torsion of xt' and M -regularity of x), so $T \subseteq \mathfrak{m}T$, and Nakayama (T finite) forces $T = 0$, contradicting $0 \neq m \in T$. Pick $t \in T \setminus xM$. Its class $\bar{t} \in M/xM$ is nonzero and is killed by $\mathfrak{p}$ (as $t \in T$) and by x (as $xt \in xM$), so $\text{ann}(\bar{t}) \supseteq \mathfrak{p} + (x)$. Choose an associated prime $\mathfrak{q} \in \text{Ass}(M/xM)$ with $\text{ann}(\bar{t}) \subseteq \mathfrak{q}$. Then $\mathfrak{p} \subseteq \mathfrak{q}$, while $x \in \mathfrak{q}$ and $x \notin \mathfrak{p}$; hence $\mathfrak{p} \subsetneq \mathfrak{q}$.

By ??, $n \leq \text{depth}(M/xM)$, so the inductive hypothesis at $(M/xM, \mathfrak{q})$ gives $n \leq \text{coheight}(\mathfrak{q})$. The strict specialisation $\mathfrak{p} \subsetneq \mathfrak{q}$ in $\text{Spec } R$ yields $\text{coheight}(\mathfrak{q}) + 1 \leq \text{coheight}(\mathfrak{p})$. Therefore $n + 1 \leq \text{coheight}(\mathfrak{q}) + 1 \leq \text{coheight}(\mathfrak{p})$. $\square$

Theorem 1266 (Cohen–Macaulay unmixedness). *Let $(R, \mathfrak{m})$ be a Cohen–Macaulay Noetherian local ring and $\mathfrak{p} \in \text{Ass}(R)$. Then $\text{coheight}(\mathfrak{p}) = \dim R$; equivalently $\dim(R/\mathfrak{p}) = \dim R$.*

Proof. Always $\text{coheight}(\mathfrak{p}) \leq \dim R$. Conversely, since R is Cohen–Macaulay, $\dim R = \text{depth } R$ (??); and $\text{depth } R \leq \text{coheight}(\mathfrak{p})$ by Ischebeck's bound (??) applied to $M = R$ (finite, nontrivial, with $\mathfrak{p} \in \text{Ass } R$). Equality follows, and the quotient form is ??. $\square$

Theorem 1267 (Associated primes of a Cohen–Macaulay ring are minimal). *In a Cohen–Macaulay Noetherian local ring R , every associated prime of R is a minimal prime.*

Proof. Let $\mathfrak{q} \in \text{Ass}(R)$ and suppose $\mathfrak{q}$ is not minimal. Then $\mathfrak{q}$ strictly contains a minimal prime $\mathfrak{q}_0$. The strict specialisation gives $\text{coheight}(\mathfrak{q}) + 1 \leq \text{coheight}(\mathfrak{q}_0) \leq \dim R = \text{coheight}(\mathfrak{q})$ (the last equality by ??); since $\text{coheight}(\mathfrak{q}) \leq \dim R$ is finite, this is impossible. $\square$

Theorem 1268 (Equidimensionality at a height-one prime). *In a Cohen–Macaulay Noetherian local ring R , a prime $\mathfrak{p}$ of height 1 satisfies $\dim(R/\mathfrak{p}) + 1 = \dim R$.*

Proof. By ?? it suffices to prove $\text{coheight}(\mathfrak{p}) + 1 = \dim R$.

($\leq$). $\mathfrak{p}$ is not minimal (its height is $1 \neq 0$), so it strictly contains a minimal prime $\mathfrak{q}_0$; the strict specialisation gives $\text{coheight}(\mathfrak{p}) + 1 \leq \text{coheight}(\mathfrak{q}_0) \leq \dim R$.

($\geq$). The associated primes of R are finite in number and, by ??, minimal. Now $\mathfrak{p}$ is not contained in any associated prime: if $\mathfrak{p} \subseteq \mathfrak{q}$ with $\mathfrak{q}$ associated, then $\mathfrak{q}$ is minimal, so $\mathfrak{q} \subseteq \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{q}$ has height 0, contradicting $\text{ht } \mathfrak{p} = 1$. Hence $\mathfrak{p}$ is not contained in the union of the associated primes, and prime avoidance produces $y \in \mathfrak{p}$ lying in no associated prime; as the zerodivisors of R are exactly that union, y is a nonzerodivisor. Then $\mathfrak{p}$ is minimal over (y) : if a prime $\mathfrak{q}$ satisfies $(y) \subseteq \mathfrak{q} \subseteq \mathfrak{p}$, then $\mathfrak{q}$ contains the nonzerodivisor y , so $\mathfrak{q}$ is not a minimal prime of R (minimal primes consist of zerodivisors) and has height ≥ 1 ; were $\mathfrak{q} \subsetneq \mathfrak{p}$, then $\text{ht } \mathfrak{p} \geq \text{ht } \mathfrak{q} + 1 \geq 2$, contradicting $\text{ht } \mathfrak{p} = 1$; so $\mathfrak{q} = \mathfrak{p}$. Consequently $\mathfrak{p}$ is a minimal prime of $\text{ann}(R/(y)) = (y)$, hence $\mathfrak{p} \in \text{Ass}(R/(y))$. Ischebeck's bound (??) for the module $R/(y)$ gives $\text{depth}_R(R/(y)) \leq \text{coheight}(\mathfrak{p})$, and the depth drop (??) for the R -regular y gives $\text{depth } R \leq \text{depth}(R/(y)) + 1$. Since R is Cohen–Macaulay, $\dim R = \text{depth } R \leq \text{coheight}(\mathfrak{p}) + 1$. $\square$

Lemma 1269 (A prime of maximal coheight is minimal). *In a commutative ring R of finite Krull dimension, a prime $\mathfrak{p}$ with $\text{coheight}(\mathfrak{p}) = \dim R$ is a minimal prime.*

Proof. If $\mathfrak{p}$ is not minimal, it strictly contains a minimal prime $\mathfrak{q}_0$; the strict specialisation gives $\text{coheight}(\mathfrak{p}) + 1 \leq \text{coheight}(\mathfrak{q}_0) \leq \dim R = \text{coheight}(\mathfrak{p})$, impossible for the finite $\text{coheight}(\mathfrak{p})$. $\square$

Lemma 1270 (Equal-dimension surjection has trivial kernel). *Let $\varphi : B \twoheadrightarrow A$ be a surjection of regular local Noetherian rings with equal embedding dimensions ($\dim A = \dim B$). Then $\ker \varphi = 0$.*

Proof. The quotient $B/\ker \varphi \cong A$ is a domain, so $\ker \varphi$ is prime. By ??, $\text{coheight}(\ker \varphi) = \dim(B/\ker \varphi) = \dim A = \dim B$ (using that embedding dimension equals Krull dimension for regular local rings). By ??, $\ker \varphi$ is a minimal prime of B . Since B , being regular local, is a domain (??), its only minimal prime is 0; hence $\ker \varphi = 0$. $\square$

Lemma 1271 (Cotangent detection of a kernel element). *Let $\varphi : B \twoheadrightarrow A$ be a surjection of Noetherian local rings whose target has strictly smaller embedding dimension than the source. Then some $x \in \ker \varphi$ does not lie in $\mathfrak{m}_B^2$.*

Proof. Suppose $\ker \varphi \subseteq \mathfrak{m}_B^2$. Take a minimal generating set of $\mathfrak{m}_A$ and lift each generator to $\mathfrak{m}_B$. These lifts generate $\mathfrak{m}_B$ modulo $\mathfrak{m}_B^2$: for $u \in \mathfrak{m}_B$, $\varphi(u)$ is an A -combination of the generators of $\mathfrak{m}_A$; lifting the coefficients to B and subtracting yields $u \equiv (\text{combination of the lifts}) \pmod{\ker \varphi}$, and $\ker \varphi \subseteq \mathfrak{m}_B^2$. By Nakayama the lifts then generate $\mathfrak{m}_B$, so the embedding dimension of B is at most that of A — contradicting the hypothesis. $\square$

Theorem 1272 (Kernels of surjections of regular local rings). *Let $\varphi : B \twoheadrightarrow A$ be a surjection of regular local Noetherian rings. Then $\ker \varphi$ is generated by $\dim B - \dim A$ elements: there is a list $x_1, \dots, x_r$ with $(x_1, \dots, x_r) = \ker \varphi$ and $r + \dim A = \dim B$.*

Proof. Induct downwards on $n = \dim B$ (the embedding dimension). If $\dim A = \dim B$, then $\ker \varphi = 0$ by ?? and the empty list works. Otherwise $\dim A < \dim B$; by ?? choose $x \in \ker \varphi \setminus \mathfrak{m}_B^2$. The quotient $B/(x)$ is again regular local, of dimension $\dim B - 1$ (??), and φ descends to a surjection $\bar{\varphi} : B/(x) \twoheadrightarrow A$. By the inductive hypothesis $\ker \bar{\varphi}$ is generated by a list of $\dim(B/(x)) - \dim A = \dim B - 1 - \dim A$ elements; lifting these to B and prepending x generates $\ker \varphi$ (using $\ker \bar{\varphi} = \ker \varphi/(x)$ and $(x) \subseteq \ker \varphi$). The count is $1 + (\dim B - 1 - \dim A) + \dim A = \dim B$. $\square$

Theorem 1273 (Krull codimension count). *Let $(R, \mathfrak{m})$ be a Cohen–Macaulay Noetherian local ring, J a prime of height 1, and $x_1, \dots, x_r$ a list such that the only prime containing $(x_1, \dots, x_r) + J$ is $\mathfrak{m}$. Then $\dim R \leq r + 1$.*

Proof. Pass to $C = R/J$, a local ring whose maximal ideal is the image of $\mathfrak{m}$. The images $\bar{x}_i$ generate an ideal $(\bar{x}_1, \dots, \bar{x}_r)$ whose only containing prime is the maximal ideal of C : a prime of C over it pulls back to a prime of R that contains J (as it comes from C) and the x_i , hence contains $(x_1, \dots, x_r) + J$, so equals $\mathfrak{m}$ by hypothesis and its image is the maximal ideal of C . Therefore $\dim(C/(\bar{x}_1, \dots, \bar{x}_r)) \leq 0$, and Krull’s height theorem in C bounds $\dim C \leq \dim(C/(\bar{x}_1, \dots, \bar{x}_r)) + r \leq r$. By equidimensionality at the height-one prime J (??), $\dim R = \dim(R/J) + 1 = \dim C + 1 \leq r + 1$. $\square$

Theorem 1274 (Transported codimension count). *Let R be a commutative ring with ideals $\mathfrak{d}, \mathfrak{q}, \mathfrak{r}$, where $\mathfrak{q}$ is prime of height 1 and $\mathfrak{r}$ is a minimal prime over $\mathfrak{d} + \mathfrak{q}$. Let $B' = R_{\mathfrak{r}}$ and $\varphi : B' \rightarrow A'$ a surjection of regular local rings that kills the image of $\mathfrak{d}$. Then $\dim A' \leq 1$.*

Proof. By kernel generation (??), write $\ker \varphi = (x_1, \dots, x_r)$ with $r + \dim A' = \dim B'$. The prime $\mathfrak{q}$ maps to a height-one prime $J' = \mathfrak{q}B'$ of B' (localisation at $\mathfrak{r} \supseteq \mathfrak{q}$ preserves the height of $\mathfrak{q}$). The maximal ideal $\mathfrak{m}_{B'}$ is the only prime of B' over $(x_1, \dots, x_r) + J'$: such a prime contracts to a prime of R that contains $\mathfrak{d}$ (as φ kills $\mathfrak{d}$ and $(x_1, \dots, x_r) = \ker \varphi$) and $\mathfrak{q}$, and is contained in $\mathfrak{r}$; minimality of $\mathfrak{r}$ over $\mathfrak{d} + \mathfrak{q}$ forces the contraction to be $\mathfrak{r}$, so the prime is $\mathfrak{r}B' = \mathfrak{m}_{B'}$. The Krull codimension count (??) gives $\dim B' \leq r + 1$; cancelling the finite r in $r + \dim A' = \dim B' \leq r + 1$ yields $\dim A' \leq 1$. $\square$

11.8 Milne’s Lemma 3.3: rows, diagonal triviality, and the codimension-one transport

We complete the proof of Milne’s Lemma 3.3. Fix an algebraically closed field $\bar{k}$, a nonsingular geometrically-irreducible $\bar{k}$ -variety X (smooth, integral, reduced, separated, locally of finite type over $\bar{k}$, with integral self-product $Y := X \times_{\bar{k}} X$), and a group $\bar{k}$ -variety G (a separated group scheme, locally of finite type, with separated structure morphism). Let $f : X \dashrightarrow G$ be a rational map over $\bar{k}$, $\Phi = f(x)f(y)^{-1}$ its difference rational map (??) with maximal representative Φ_0 , $\delta : X \rightarrow Y$ the diagonal $x \mapsto (x, x)$, and $e \in G$ the unit point. The three pillars are: the row argument (substep 2, hard direction), the germ-anchored spreading criterion (substep 3), and the diagonal–pole transport (substep 4b) built on §?? and the polar purity of §??.

Lemma 1275 (Representative agreement of the difference map). *The maximal representative Φ_0 of the difference map and the explicit pairing representative — the partial map $(f \circ \text{pr}_1, f \circ \text{pr}_2)$ post-composed with δ_G , defined on $\text{pr}_1^{-1}(\text{Dom } f) \cap \text{pr}_2^{-1}(\text{Dom } f)$ — agree as honest morphisms on the intersection of their domains. Consequently, for a test scheme T and morphisms j, q into the two domains whose composites into Y coincide,*

$$j \circ \Phi_0 = q \circ [(f \circ \text{pr}_1, f \circ \text{pr}_2) \circ \delta_G].$$

Proof. The pairing representative (??) represents Φ : its class is $(f \circ \text{pr}_1, f \circ \text{pr}_2)$, whose post-composition with δ_G is Φ (??). Two representatives of one rational map out of a reduced source into a separated target agree on any dense open contained in both domains; as both are morphisms over $\bar{k}$, the separatedness of the target upgrades this to agreement on the entire intersection of their domains. Factoring j and q through that intersection — their common composite into Y lands there — and restricting the agreement gives the stated identity. $\square$

Lemma 1276 (The evaluation formula). *Let $a, b : T \rightarrow \text{Dom } f$ be two morphisms into the domain of f whose composites to X agree over k , so that the pair $\langle a, b \rangle : T \rightarrow Y$ is well-formed, and let $j : T \rightarrow \text{Dom } \Phi_0$ factor $\langle a, b \rangle$ through the domain of Φ_0 . Then*

$$j \circ \Phi_0 = \langle f \circ a, f \circ b \rangle \circ \delta_G,$$

i.e. Milne's $\Phi(a, b) = f(a) f(b)^{-1}$ as an identity of honest morphisms.

Proof. The pair $\langle a, b \rangle$ factors pointwise through the domain of the pairing representative (both coordinates land in $\text{Dom } f$); call the factorisation q . Applying the representative agreement (??) to j and q gives $j \circ \Phi_0 = q \circ [(f \circ \text{pr}_1, f \circ \text{pr}_2) \circ \delta_G]$. The pairing computes coordinatewise — each projection of $G \times_{\bar{k}} G$ recovers $f \circ \text{pr}_i$ restricted along the corresponding coordinate — so the composite equals $\langle f \circ a, f \circ b \rangle \circ \delta_G$. $\square$

Theorem 1277 (Diagonal triviality). *At any diagonal point $\delta(z) = (z, z)$ with $z \in \text{Dom } f$ at which Φ_0 is defined, $\Phi_0(z, z) = e$. Equivalently, any factorisation of the diagonal $\text{Dom } f \rightarrow Y$ through $\text{Dom } \Phi_0$ composes with Φ_0 to the constant unit morphism.*

Proof. Apply the evaluation formula (??) with $a = b = \text{id}_{\text{Dom } f}$: the diagonal factors as $\langle \iota, \iota \rangle$, and Φ_0 on it equals $\langle f, f \rangle \circ \delta_G$, which is $f \cdot f^{-1}$. In the hom-group this is the constant unit morphism $g \cdot g^{-1} = e$. Reading it off at the point z gives $\Phi_0(z, z) = e$. $\square$

Lemma 1278 (The diagonal maps into the domain of the difference map). *For $z \in \text{Dom } f$, the diagonal point $\delta(z) = (z, z)$ lies in $\text{Dom } \Phi$.*

Proof. Both coordinates of (z, z) equal $z \in \text{Dom } f$, so $(z, z) \in \text{pr}_1^{-1}(\text{Dom } f) \cap \text{pr}_2^{-1}(\text{Dom } f) \subseteq \text{Dom } \Phi$ by ?? $\square$

Theorem 1279 (The generic image specialises to the unit). *If a point $P \in \text{Dom } \Phi_0$ specialises to a diagonal point $\delta(z)$ with $z \in \text{Dom } f$, then $\Phi_0(P) \rightsquigarrow e$. In particular, applied to the generic point η_Y , the generic image $\gamma = \Phi(\eta_Y)$ satisfies $\gamma \rightsquigarrow e$.*

Proof. The diagonal point $\delta(z)$ lies in $\text{Dom } \Phi_0$ (??) with $\Phi_0(\delta(z)) = e$ (??). The specialisation $P \rightsquigarrow \delta(z)$ inside the open $\text{Dom } \Phi_0$ pushes forward along the continuous map Φ_0 to $\Phi_0(P) \rightsquigarrow \Phi_0(\delta(z)) = e$. $\square$

Lemma 1280 (Jacobson density of closed points). *Every nonempty locally closed subset of a scheme locally of finite type over a field contains a point closed in the ambient scheme (in particular, so does every nonempty open subset).*

Proof. A scheme locally of finite type over a field is a Jacobson space, in which the closed points are very dense: for any locally closed subset O the closure of O equals the closure of $O \cap \{\text{closed points}\}$. If a nonempty O contained no ambient-closed point, then $O \cap \{\text{closed points}\} = \emptyset$, so $\overline{O} = \overline{\emptyset} = \emptyset$, contradicting $O \neq \emptyset$. $\square$

Theorem 1281 (The row argument: substep 2, hard direction). *If x_0 is a closed point of X and the difference map Φ is defined at the diagonal point (x_0, x_0) , then f is defined at x_0 .*

Proof. As x_0 is closed and $\bar{k}$ is algebraically closed, there is a $\bar{k}$ -rational point at x_0 . The row through x_0 (first coordinate frozen, $y \mapsto (x_0, y)$) meets $\text{Dom } \Phi$ at (x_0, x_0) , so its preimage of $\text{Dom } \Phi$ is a nonempty open; intersecting with the dense open $\text{Dom } f$ (as X is irreducible) and applying Jacobson density (??) gives a closed — hence $\bar{k}$ -rational — point $u \in \text{Dom } f$ with $(x_0, u) \in \text{Dom } \Phi$.

Let $\sigma_u : v \mapsto (v, u)$ be the row through u . Then $x_0 \in V_1 := \sigma_u^{-1}(\text{Dom } \Phi_0)$, an open subset, and $\sigma_u(x_0) = (x_0, u) \in \text{Dom } \Phi_0$. Define the honest morphism (Milne's $\psi(v) = \Phi(v, u) f(u)$)

$$\psi = m_G \circ \langle \Phi_0 \circ \sigma_u|_{V_1}, f(u) \rangle : V_1 \longrightarrow G.$$

On the dense open $W := V_1 \cap \text{Dom } f$ the inclusion factors as the pair $\langle \text{incl}, u \rangle$, so the evaluation formula (??) gives $\Phi_0(v, u) = f(v) f(u)^{-1}$ there; the scheme-level reconstruction identity (??), $(f(v) f(u)^{-1}) f(u) = f(v)$, then shows $\psi|_W = f|_W$. Hence the partial map (V_1, ψ) represents f , and $x_0 \in V_1 \subseteq \text{Dom } f$. No smooth descent and no unique-factorisation input are used. $\square$

Theorem 1282 (The abstract spreading criterion). *Let $F : Y \dashrightarrow Z$ be a rational map of S -schemes with Y integral and Z locally of finite type over S . Suppose the generic morphism $\text{Spec } K(Y) \rightarrow Z$ of F factors through a morphism $\phi : \text{Spec } \mathcal{O}_{Y,P} \rightarrow Z$ compatible with the structure morphisms (ϕ followed by $Z \rightarrow S$ equals $\text{Spec } \mathcal{O}_{Y,P} \rightarrow Y \rightarrow S$, and $\text{Spec } K(Y) \rightarrow \text{Spec } \mathcal{O}_{Y,P} \xrightarrow{\phi} Z$ is the generic morphism of F). Then $P \in \text{Dom } F$.*

Proof. The morphism ϕ , defined on the local ring at P , spreads out — using that Z is locally of finite type over S and that Y , being integral, is germ-injective at every point — to a partial map ψ_P defined at P with the prescribed local behaviour. Both ψ_P and any representative g of F restrict at the generic point to the generic morphism of F : for ψ_P by the specialisation-compatibility of its stalk-to-scheme morphism together with the hypothesis, for g by definition. Hence ψ_P and g represent the same rational map, so ψ_P represents F and $P \in \text{Dom } \psi_P \subseteq \text{Dom } F$. This is the scheme-theoretic form of Milne's criterion “ Φ is defined at (x, x) iff $\text{Im}(\mathcal{O}_{G,e}) \subseteq \mathcal{O}_{Y,(x,x)}$ ”. $\square$

Theorem 1283 (The germ-range criterion: substep 3). *Let $F : Y \dashrightarrow Z$ be a rational map over an affine base S , with Y integral and Z locally of finite type over S , and let $V \subseteq Z$ be an affine open containing the generic image $\gamma = F(\eta_Y)$. Write $\Lambda : \Gamma(Z, V) \rightarrow K(Y)$ for the generic germ pullback (germ at γ , pulled through the stalk map of F , evaluated at the generic point). If Λs lies in the image of $\mathcal{O}_{Y,P} \rightarrow K(Y)$ for every section $s \in \Gamma(Z, V)$, then $P \in \text{Dom } F$.*

Proof. The hypothesis says Λ corestricts to a ring map $\alpha : \Gamma(Z, V) \rightarrow \mathcal{O}_{Y,P}$ (using that $\mathcal{O}_{Y,P} \rightarrow K(Y)$ is injective, Y integral). Then $\text{Spec } \alpha$ followed by $V \hookrightarrow Z$ is a morphism $\phi : \text{Spec } \mathcal{O}_{Y,P} \rightarrow Z$. We check the hypotheses of ??: (i) the generic factorisation — precomposing ϕ with $\text{Spec}(\mathcal{O}_{Y,P} \rightarrow K(Y))$ recovers the generic morphism of F , because α agrees with Λ after the injection; (ii) compatibility over S — both ϕ followed by $Z \rightarrow S$ and $\text{Spec } \mathcal{O}_{Y,P} \rightarrow Y \rightarrow S$ induce the same ring map at P , verified after the injective $\mathcal{O}_{Y,P} \rightarrow K(Y)$ using the over- S compatibility of F at the generic point. ?? gives $P \in \text{Dom } F$. $\square$

Theorem 1284 (Definedness gives germ-range membership: substep 3, easy direction). *Let $F : Y \dashrightarrow Z$ with Y integral, and let V be an affine open containing the generic image. If a representative g of F is defined at a point Q with value $g(Q) \in V$, then for every section $s \in \Gamma(Z, V)$ the generic germ pullback Λs lies in the image of $\mathcal{O}_{Y,Q} \rightarrow K(Y)$.*

Proof. The germ of s at $g(Q)$, pulled back through the stalk map of g at Q and the stalk isomorphism of the domain inclusion, is an element of $\mathcal{O}_{Y,Q}$; its image in $K(Y)$ is exactly Λs (both are computed from the stalk pullback along the generic specialisation). Hence $\Lambda s \in \text{im}(\mathcal{O}_{Y,Q} \rightarrow K(Y))$. $\square$

Theorem 1285 (Pole existence where the map is undefined). *Let $F : Y \dashrightarrow Z$ be over an affine base with Y integral, locally Noetherian and with regular stalks, Z locally of finite type over the base, and V an affine open containing the generic image. If F is not defined at P , then some section $s \in \Gamma(Z, V)$ has generic germ pullback Λs non-regular at a coheight-one point $w \rightsquigarrow P$.*

Proof. By the contrapositive of ??, some s has $\Lambda s \notin \text{im}(\mathcal{O}_{Y,P} \rightarrow K(Y))$ — a pole of Λs at P . By polar purity (??) the pole propagates to a coheight-one generisation: there is $w \rightsquigarrow P$ with $\text{coheight}(w) = 1$ and $\Lambda s \notin \text{im}(\mathcal{O}_{Y,w} \rightarrow K(Y))$. $\square$

Lemma 1286 (Chart image point of a closed immersion). *Let $g : X \rightarrow Y$ be a closed immersion, $U \subseteq Y$ an affine open, and $\mathfrak{r}$ a prime of $\Gamma(Y, U)$ containing the kernel of the section map $g^\sharp : \Gamma(Y, U) \rightarrow \Gamma(X, g^{-1}U)$. Then the point of U corresponding to $\mathfrak{r}$ lies in the image of g .*

Proof. Since g is a closed immersion, $g^\sharp$ is surjective. As $\ker g^\sharp \subseteq \mathfrak{r}$, the image $\mathfrak{r} \cdot \Gamma(X, g^{-1}U)$ is a prime whose chart point z satisfies $g(z) = \mathfrak{r}$: the contraction of that prime along the surjection $g^\sharp$ recovers $\mathfrak{r}$ (surjectivity plus $\ker g^\sharp \subseteq \mathfrak{r}$). $\square$

Theorem 1287 (The transport kernel). *Let Y be an integral scheme with regular stalks, $g : X \rightarrow Y$ a closed immersion with X having regular stalks, $x_0 \in X$, and $h \in K(Y)$ a function with a pole at a coheight-one point $w \rightsquigarrow g(x_0)$. Then some $z \rightsquigarrow x_0$ with $\dim \mathcal{O}_{X,z} \leq 1$ still carries the pole: $h \notin \text{im}(\mathcal{O}_{Y,g(z)} \rightarrow K(Y))$.*

Proof. Work in an affine chart $U \ni P := g(x_0)$, with ring $R = \Gamma(Y, U)$. Let $\mathfrak{d} = \ker(R \rightarrow \mathcal{O}_{X,x_0})$ be the diagonal ideal, with $\mathfrak{d} \leq \mathfrak{p}$ (the prime of P) since a section invertible at P has nonzero germ at x_0 ; and let $\mathfrak{q}$ be the prime of w , with $\mathfrak{q} \leq \mathfrak{p}$ from $w \rightsquigarrow P$. Pick a minimal prime $\mathfrak{r}$ of $\mathfrak{d} + \mathfrak{q}$ with $\mathfrak{r} \leq \mathfrak{p}$. Since $\ker(R \rightarrow \Gamma(X, g^{-1}U)) \subseteq \mathfrak{d} \subseteq \mathfrak{r}$, the chart point $\tilde{z}$ of $\mathfrak{r}$ is $g(z)$ for a point z of X (?); moreover $z \rightsquigarrow x_0$ (as $\mathfrak{r} \leq \mathfrak{p}$ and g is a closed embedding) and $w \rightsquigarrow \tilde{z}$ (as $\mathfrak{q} \leq \mathfrak{r}$). The prime $\mathfrak{q}$ has height one (its stalk $\mathcal{O}_{Y,w}$ has dimension $\text{coheight}(w) = 1$). At the stalk $B' = \mathcal{O}_{Y,\tilde{z}} = R_{\mathfrak{r}}$ (regular local), the closed-immersion stalk surjection $\mathcal{O}_{Y,\tilde{z}} \twoheadrightarrow \mathcal{O}_{X,z}$ kills the image of $\mathfrak{d}$; the transported codimension count (??) gives $\dim \mathcal{O}_{X,z} \leq 1$. Finally, h non-regular at w and $w \rightsquigarrow \tilde{z}$ force h non-regular at $\tilde{z} = g(z)$ (??, contrapositive). $\square$

Lemma 1288 (The generic-value bridge). *For a partial map $g : Y \dashrightarrow Z$ out of an integral scheme, defined at the generic point η_Y , the generic morphism of g 's rational-map class sends the closed point of $\text{Spec } K(Y)$ to the value $g(\eta_Y)$.*

Proof. The generic morphism factors as $\text{Spec } K(Y) \rightarrow \text{Spec } \mathcal{O}_{Y,\eta} \rightarrow \text{Dom } g \rightarrow Z$; the closed point of $\text{Spec } K(Y)$ maps to η_Y under the first two arrows, and g sends the point η_Y of its domain to $g(\eta_Y)$. $\square$

Lemma 1289 (Coheight strictness). *For points of a scheme, a nontrivial specialisation $a \rightsquigarrow b$ with $a \neq b$ gives $\text{coheight}(a) + 1 \leq \text{coheight}(b)$. Consequently, on an irreducible scheme every point other than the generic point has coheight at least 1.*

Proof. A nontrivial specialisation $a \rightsquigarrow b$ is a strict relation $b < a$ in the specialisation order, so the coheight — the supremum length of specialisation chains descending from a point — satisfies $\text{coheight}(a) + 1 \leq \text{coheight}(b)$. For the corollary, apply this to $\eta \rightsquigarrow z$ (the generic point η generises every point) for $z \neq \eta$: $\text{coheight}(z) \geq \text{coheight}(\eta) + 1 \geq 1$. $\square$

Theorem 1290 (The codimension-one transport: substep 4b). *If a closed point x_0 of X is outside $\text{Dom } f$, then some point z outside $\text{Dom } f$ with $\text{coheight}(z) = 1$ specialises to x_0 .*

Proof. Write $Y = X \times_{\bar{k}} X$, $\delta : X \rightarrow Y$ the diagonal. All stalks of Y and X are regular (?), Y is integral (?), and δ is a closed immersion (X is separated over $\bar{k}$).

Front end. If $\delta(x_0) \in \text{Dom } \Phi$ then $x_0 \in \text{Dom } f$ (??, contrapositive); so $\delta(x_0) \notin \text{Dom } \Phi$. Take an affine open $V \ni e$ at the unit point. The generic image $\gamma = \Phi(\eta_Y)$ lies in V : $\gamma \rightsquigarrow e$ (??, with the generic-value identification ??) and V is open about e . By pole existence (??), some section s of V has generic germ pullback with a pole at a coheight-one $w \rightsquigarrow \delta(x_0)$.

Transport. The transport kernel (??) moves the pole to $\delta(z)$ for some $z \rightsquigarrow x_0$ with $\dim \mathcal{O}_{X,z} \leq 1$.

Back end. If f were defined at z , then $\delta(z) \in \text{Dom } \Phi$ (??) with $\Phi(\delta(z)) = e \in V$ (??), so the germ pullback of s would be regular at $\delta(z)$ (??) — contradicting the pole. Hence $z \notin \text{Dom } f$.

Finally $z \neq \eta_X$ (the generic point lies in $\text{Dom } f$), so $\text{coheight}(z) = 1$: $\dim \mathcal{O}_{X,z} \leq 1$ gives $\text{coheight}(z) \leq 1$, and $z \neq \eta_X$ gives $\text{coheight}(z) \geq 1$ (??). $\square$

Theorem 1291 (Milne’s Lemma 3.3). *Let $\bar{k}$ be an algebraically closed field, X a nonsingular geometrically-irreducible $\bar{k}$ -variety, and G a group $\bar{k}$ -variety; let $f : X \dashrightarrow G$ be a rational map over $\bar{k}$. Then f has the Milne 3.3 indeterminacy property (??): $Z(f)$ is empty or of pure codimension one.*

Proof. If $Z(f) = \emptyset$ we are in the first alternative. Otherwise fix $x \in Z(f)$; we produce $z \in Z(f)$ of coheight one with $x \in \overline{\{z\}}$. The generic point of X lies in $\text{Dom } f$, hence not in $Z(f)$. Choose an affine, topologically Noetherian window $W \ni x$, and let $Z_0 = Z(f) \cap W$ with its finitely many irreducible components; let C be the component through x . By Jacobson density on the locally closed set $C \setminus (\text{other components})$ (??) pick a point $x_0 \in C$, closed in X , avoiding the other components; then $x_0 \in Z(f)$.

The codimension-one transport (??) yields $z \in Z(f)$ with $z \rightsquigarrow x_0$ and $\text{coheight}(z) = 1$. The closure of z in W is an irreducible subset of Z_0 through x_0 , hence lies in a component; that component is C , since x_0 avoids the others. Thus the generic point ζ of C generises z . Now $\zeta \in Z(f)$ is not the generic point of X , so $\text{coheight}(\zeta) \geq 1$ (??); and coheight strictness along $\zeta \rightsquigarrow z$ forces $\zeta = z$ (otherwise $\text{coheight}(\zeta) + 1 \leq \text{coheight}(z) = 1$, impossible). Therefore $x \in C = \overline{\{\zeta\}} = \overline{\{z\}}$, giving the second alternative. $\square$

11.9 The extension theorem

Lemma 1292 (Morphisms are determined by their rational-map class). *Let X be a reduced scheme, Y a separated scheme, and $g_1, g_2 : X \rightarrow Y$ morphisms inducing the same rational map. Then $g_1 = g_2$.*

Proof. Equality of the rational-map classes provides a dense open $W \subseteq X$ with $g_1|_W = g_2|_W$. The locus where g_1 and g_2 agree is the preimage of the diagonal of Y under $(g_1, g_2) : X \rightarrow Y \times Y$, a closed subscheme since Y is separated. It contains the dense open W , so its underlying closed subset is all of X ; a closed subscheme of a reduced scheme with full underlying space is the whole scheme (its ideal sheaf consists of nilpotents). Hence $g_1 = g_2$. $\square$

Theorem 1293 (Extension from an empty indeterminacy locus). *Let X be a reduced scheme, Y a separated scheme, and $f : X \dashrightarrow Y$ a rational map with $Z(f) = \emptyset$. Then there is a unique morphism $g : X \rightarrow Y$ whose rational map is f .*

(The emptiness hypothesis cannot be weakened to codimension-one-freeness: the projection $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ away from a point has indeterminacy a single closed point of codimension two — so it is codimension-one-free — yet admits no regular extension.)

Proof. Since $Z(f) = \emptyset$, the maximal representative (??) of f is defined on $\text{Dom}(f) = X$; it is a total morphism g representing f . Uniqueness is ?? $\square$

Theorem 1294 (Milne’s Theorem 3.2: extension to an abelian variety). *Let $\bar{k}$ be an algebraically closed field, X a nonsingular $\bar{k}$ -variety (as in ??), and A an abelian variety over k : a group scheme, proper, smooth and geometrically irreducible over $\bar{k}$. Let $f : X \dashrightarrow A$ be a rational map*

over $\bar{k}$ which has the Milne 3.3 indeterminacy property (??). Then there is a unique morphism $g : X \rightarrow A$ whose rational map is f .

Proof. Step 0: A is a complete variety. The scheme A is reduced by ?? (it is smooth over the algebraically closed $\bar{k}$). It is irreducible: $\text{Spec } \bar{k}$ has a single point, and a scheme whose structure morphism to a one-point base is geometrically irreducible is irreducible. Reduced and irreducible is integral; together with properness and geometric irreducibility, A satisfies the target hypotheses of ?. The scheme A is also separated (its structure morphism is separated, being proper, and $\text{Spec } \bar{k}$ is affine, hence separated).

Step 1: $Z(f) = \emptyset$. By the Milne 3.3 property, either $Z(f) = \emptyset$ and we are done, or every point of the nonempty $Z(f)$ specialises from a point $z \in Z(f)$ with $\text{coheight}(z) = 1$. The second alternative is impossible: by ?? every point of $Z(f)$ has coheight at least 2. Hence $Z(f)$ has no points at all, i.e. it is empty.

Step 2: extend. X is reduced and A separated, so ?? produces the unique morphism g representing f . $\square$

Chapter 12

The Jacobian datum

Throughout this chapter k is a field and C is the challenge curve: a scheme over k , proper, smooth of relative dimension one, and geometrically irreducible. The degree-zero relative Picard functor $\mathrm{Pic}_{C/k}^0 : \mathrm{Over}(\mathrm{Spec} k)^{\mathrm{op}} \rightarrow \mathrm{CommGrp}$ was constructed in ???. This chapter introduces the *Jacobian datum* — a representing object for $\mathrm{Pic}_{C/k}^0$ together with the certificates its construction provides — and develops what follows formally from it: the group structure and universal property of the representing object, its uniqueness up to canonical isomorphism, separatedness (every group scheme over a field is separated), the homogeneity supplied by translations, and, given a proper irreducible *Abel source* mapping onto it, properness and geometric irreducibility.

12.1 The representability datum

Definition 1295 (Jacobian datum). A *Jacobian datum* for C consists of:

1. a scheme J over k ;
2. a representation of the degree-zero relative Picard functor by J : a family of bijections $\mathrm{Hom}_{\mathrm{Over}(\mathrm{Spec} k)}(T, J) \cong \mathrm{pic}^0(C, T)$, natural in the test object T ;
3. the certificate that the structure morphism of J is locally of finite type;
4. the certificate that the structure morphism of J is quasi-compact.

The two certificates come from the construction of the representing object and are carried as part of the datum rather than re-derived.

Definition 1296 (The group structure of the representing object). For a Jacobian datum d with representing object J , the scheme J is a group object of $\mathrm{Over}(\mathrm{Spec} k)$, with multiplication, unit and inversion induced by the commutative-group structure of $\mathrm{pic}^0(C, -)$ through the representation.

Proof. Transport the group structure across the natural bijections of ???: each $\mathrm{Hom}(T, J)$ becomes a group, naturally in T — precomposition with any $T' \rightarrow T$ is a group homomorphism because the restriction maps of $\mathrm{Pic}_{C/k}^0$ are. By the Yoneda lemma the natural transformation $\mathrm{Hom}(-, J) \times \mathrm{Hom}(-, J) \rightarrow \mathrm{Hom}(-, J)$ given by the pointwise products is induced by a unique morphism $\mu : J \otimes J \rightarrow J$, and similarly for the unit $1 \rightarrow J$ and the inversion $J \rightarrow J$; the group axioms hold for these morphisms because they hold pointwise on T -points. $\square$

Lemma 1297 (The universal property, consumer form). *Let d be a Jacobian datum with representing object J . For every test object T there is a bijection*

$$\theta_T : \text{Hom}_{\text{Over}(\text{Spec } k)}(T, J) \cong \text{pic}^0(C, T),$$

natural against restriction: for $f : T' \rightarrow T$ and $g : T \rightarrow J$,

$$\theta_{T'}(g \circ f) = f^*(\theta_T(g)),$$

where f^ is the restriction map of $\text{Pic}_{C/k}^0$ (??).*

Proof. This is the representation carried by the datum (??), with its naturality read element-wise. $\square$

Definition 1298 (Uniqueness of the representing object up to canonical isomorphism). Let d, d' be two Jacobian data for the same curve, with representing objects J, J' and universal bijections θ, θ' . There is a canonical isomorphism $\sigma : J \cong J'$ in $\text{Over}(\text{Spec } k)$, characterised by the intertwining property

$$\theta'_T(\sigma \circ f) = \theta_T(f) \quad \text{for every } f : T \rightarrow J.$$

Proof. Let $u = \theta_J(\text{id}_J) \in \text{pic}^0(C, J)$ be the universal class of d , and set $\sigma = \theta_J^{-1}(u) : J \rightarrow J'$. For $f : T \rightarrow J$, naturality of θ' (??) gives

$$\theta'_T(\sigma \circ f) = f^*(\theta'_J(\sigma)) = f^*(u) = f^*(\theta_J(\text{id}_J)) = \theta_T(\text{id}_J \circ f) = \theta_T(f).$$

Symmetrically there is $\sigma' = \theta_J^{-1}(u') : J' \rightarrow J$, where $u' = \theta_{J'}(\text{id}_{J'})$, with $\theta_T(\sigma' \circ f') = \theta'_T(f')$ for every $f' : T \rightarrow J'$. The two composites are identities: e.g. $\theta_J(\sigma' \circ \sigma) = \theta'_J(\sigma) = u = \theta_J(\text{id}_J)$, and θ_J is injective, so $\sigma' \circ \sigma = \text{id}_J$; likewise the other composite. Uniqueness of an isomorphism with the intertwining property follows from injectivity of θ' at $T = J$. $\square$

12.2 Functoriality of the datum in the curve

Throughout this section X, Y and Z are challenge curves over k (proper, smooth of relative dimension one, geometrically irreducible), with Jacobian data d_X, d_Y, d_Z and representing objects J_X, J_Y, J_Z .

Theorem 1299 (The Yoneda embedding for group objects). *Let $\mathcal{C}$ be a cartesian monoidal category. Sending a group object G to the presheaf of groups $\text{Hom}(-, G)$ is a fully faithful embedding of the category of group objects of $\mathcal{C}$ into presheaves of groups on $\mathcal{C}$. If a presheaf of groups F is representable, with representing object J , then the transported group structure makes J a group object and $\text{Hom}(-, J) \cong F$ as presheaves of groups.*

Definition 1300 (Pullback of degree-zero classes as a morphism of group schemes). Let $g : X \rightarrow Y$ be a morphism of curves. The composite natural transformation

$$\text{Hom}(-, J_Y) \cong \text{Pic}_{Y/k}^0 \xrightarrow{g^*} \text{Pic}_{X/k}^0 \cong \text{Hom}(-, J_X)$$

of presheaves of groups — the outer isomorphisms being the universal properties of the two data, as isomorphisms of presheaves of groups (Theorem ??), and the middle map the degree-zero curve transport (??) — is induced, by full faithfulness of the Yoneda embedding for group objects, by a unique morphism of group objects

$$g^* : J_Y \longrightarrow J_X$$

in $\text{Over}(\text{Spec } k)$: the pullback of degree-zero line-bundle classes, realized on the representing objects.

Theorem 1301 (Functor laws of the pullback morphism). $\text{id}_X^* = \text{id}_{J_X}$ (both slots taken by the same datum d_X), and for $g: X \rightarrow Y$ and $g': Y \rightarrow Z$,

$$(g' \circ g)^* = g^* \circ g'^*: J_Z \longrightarrow J_X.$$

Proof. The Yoneda embedding for group objects is faithful (Theorem ??), so it suffices to compare the conjugated transformations of presheaves of groups. For the identity, the middle map is the identity transformation (??) and the universal isomorphism cancels against its inverse. For the composite, the middle map of $(g' \circ g)^*$ is the composite of the two middle maps (??), and the inner universal isomorphism of d_Y cancels against its inverse between the two conjugates. $\square$

Lemma 1302 (The universal-element characterization). Let $g: X \rightarrow Y$, let T be a test object and $f: T \rightarrow J_Y$. Under the universal bijections θ^X, θ^Y of the two data (??),

$$\theta_T^X(g^* \circ f) = g^*(\theta_T^Y(f)),$$

the right-hand transport being the degree-zero pullback of ?? : on functor points, $g^*: J_Y \rightarrow J_X$ is pullback of degree-zero classes.

Proof. Evaluate the defining natural transformation of Definition ?? at T on the element f : the first isomorphism sends f to $\theta_T^Y(f)$, the middle map transports it to $g^*(\theta_T^Y(f))$, and the last isomorphism is $(\theta_T^X)^{-1}$; composing with θ_T^X gives the display. $\square$

12.3 Group schemes are separated

The diagonal argument below is categorical and uses no finiteness hypothesis. In particular, it proves a stronger form than the standard statement for group schemes locally of finite type over a field.

Theorem 1303 (The diagonal of a group object is a pullback of the unit). Let G be a group object in a cartesian monoidal category, with unit $e: 1 \rightarrow G$ and difference morphism $\delta = \text{pr}_1 \cdot \text{pr}_2^{-1}: G \otimes G \rightarrow G$ (??). Then the square

$$\begin{array}{ccc} G & \xrightarrow{\Delta = \langle \text{id}, \text{id} \rangle} & G \otimes G \\ \downarrow & & \downarrow \delta \\ 1 & \xrightarrow{e} & G \end{array}$$

is a pullback: the diagonal of G is the base change of the unit along the difference map.

Proof. Commutativity. In the hom-group $\text{Hom}(G, G)$: $\delta \circ \Delta = (\text{pr}_1 \circ \Delta) \cdot (\text{pr}_2 \circ \Delta)^{-1} = \text{id} \cdot \text{id}^{-1} = 1$, the constant unit, which is the composite through the terminal object.

Universality. Let $s_1: T \rightarrow 1$ and $s_2: T \rightarrow G \otimes G$ satisfy $\delta \circ s_2 = e \circ s_1$. The right-hand side is the unit of $\text{Hom}(T, G)$, and the left-hand side is $(\text{pr}_1 \circ s_2) \cdot (\text{pr}_2 \circ s_2)^{-1}$; hence $\text{pr}_1 \circ s_2 = \text{pr}_2 \circ s_2 =: h$. The morphism $h: T \rightarrow G$ satisfies $\Delta \circ h = \langle h, h \rangle = s_2$ (equal components) and $t_G \circ h = s_1$ (morphisms to the terminal object agree). For uniqueness, any m with $\Delta \circ m = s_2$ has $m = \text{pr}_1 \circ \Delta \circ m = \text{pr}_1 \circ s_2 = h$. $\square$

Theorem 1304 (A group scheme with closed unit section is separated). Let S be a scheme and G a group object of $\text{Over}(S)$. If the unit section $e: S \rightarrow G$ is a closed immersion of schemes, then the structure morphism $G \rightarrow S$ is separated. No finiteness hypothesis is required.

Proof. Apply the forgetful functor $\text{Over}(S) \rightarrow \text{Sch}$ to the pullback square of ??; it preserves the square and its pullback property (fibre products in $\text{Over}(S)$ are computed on underlying schemes), and the monoidal product $G \otimes G$ becomes the fibre product $G \times_S G$ with Δ the diagonal morphism of G over S . Thus the diagonal $\Delta_{G/S}$ is a base change of the closed immersion e , and closed immersions are stable under base change; a morphism whose diagonal is a closed immersion is separated. $\square$

Theorem 1305 (Group schemes over a field are separated). *Every group scheme G over a field K is separated over K .*

Proof. The unit section $e : \text{Spec } K \rightarrow G$ of a group scheme over a field is a closed immersion: its image is a K -rational point, and a section of a K -scheme at a rational point is a closed immersion. Apply ??. $\square$

Corollary 1306 (The representing object of a Jacobian datum is separated). *For every Jacobian datum d with representing object J , the structure morphism of J is separated.*

Proof. J is a group scheme over k (?); apply ??. $\square$

12.4 Translations

Group schemes are homogeneous: right translation by a point is an automorphism, so a point-local property holding at one rational point spreads to every rational point. This section provides the translation isomorphisms and the transport statements, first for a group object G in a cartesian monoidal category $\mathcal{C}$ (with the hom-group calculus of ??), then on underlying schemes. For a point $g : 1 \rightarrow G$, the *right translation* $\rho_g : G \cong G$ is the isomorphism $\rho_g = \mu \circ \langle \text{id}_G, g \circ t_G \rangle$ (“multiply on the right by g ”), with inverse the right translation by g^{-1} .

Lemma 1307 (Composition with a right translation). *For morphisms $f : X \rightarrow G$ and points $g : 1 \rightarrow G$ of a group object G ,*

$$\rho_g \circ f = f \cdot \hat{g} \quad \text{and} \quad \rho_g^{-1} \circ f = f \cdot \hat{g}^{-1},$$

computed in the hom-group $\text{Hom}(X, G)$, where $\hat{g} = g \circ t_X$ is the constant morphism at g .

Proof. Unfolding ρ_g : $\rho_g \circ f = \mu \circ \langle \text{id}, g \circ t_G \rangle \circ f = \mu \circ \langle f, g \circ t_X \rangle = f \cdot \hat{g}$, using $t_G \circ f = t_X$ (morphisms into the terminal object). The second identity follows from the first applied to $\rho_g^{-1} = \rho_{g^{-1}}$, or directly: unfolding the inverse translation gives $\mu \circ \langle f, \iota \circ g \circ t_X \rangle = f \cdot \hat{g}^{-1}$. $\square$

Definition 1308 (The translation carrying one point to another). For points $x, y : 1 \rightarrow G$ of a group object G , the *point translation*

$$\tau_{x,y} = \rho_y \circ \rho_x^{-1} : G \cong G$$

is the automorphism “multiply on the right by $x^{-1}y$ ”.

Lemma 1309 (The point translation carries x to y). *For points $x, y : 1 \rightarrow G$, $\tau_{x,y} \circ x = y$.*

Proof. By ??, computed in the hom-group $\text{Hom}(1, G)$, where the constant $\hat{g}$ at a point g is g itself ($t_1 = \text{id}$): $\rho_x^{-1} \circ x = x \cdot x^{-1} = 1$, and then $\rho_y \circ (\rho_x^{-1} \circ x) = 1 \cdot y = y$. $\square$

Definition 1310 (Translations of the underlying scheme). Let S be a scheme, G a group scheme over S , and x, y two sections $S \rightarrow G$. The *translation isomorphism* $\tau_{x,y} : G \cong G$ of underlying schemes is the image of the point translation ?? under the forgetful functor from schemes over S to schemes; its inverse is the image of the inverse point translation.

Lemma 1311 (Translations commute with the structure morphism). *In the situation of ??, composing $\tau_{x,y}$ with the structure morphism $G \rightarrow S$ gives the structure morphism: $\tau_{x,y}$ is an automorphism of G over S .*

Proof. The point translation is a morphism in the category of schemes over S , and the forgetful functor remembers the commutation with the structure morphisms. $\square$

Lemma 1312 (Translations act on points as prescribed). *In the situation of ??, for every point $s \in S$,*

$$\tau_{x,y}(x(s)) = y(s)$$

on underlying topological spaces.

Proof. The equation $\tau_{x,y} \circ x = y$ of ?? holds in $\text{Over}(S)$, hence on underlying schemes, hence on underlying topological spaces; evaluate at s . $\square$

Lemma 1313 (Transport of the smooth locus). *Let $f : X \rightarrow S$ be a morphism locally of finite presentation and $e : X \rightarrow X$ an open immersion with $f \circ e = f$. Then for every $z \in X$,*

$$e(z) \in \text{SmoothLocus}(f) \iff z \in \text{SmoothLocus}(f).$$

Proof. Smoothness of f at a point is local on X : composing with the open immersion e , the morphism f is smooth at $e(z)$ if and only if $f \circ e$ is smooth at z (an open immersion is smooth, identifies local rings, and smoothness is a stalk-local property of the morphism). Now rewrite $f \circ e = f$. $\square$

Lemma 1314 (Transport of stalk reducedness). *Let $e : X \rightarrow Y$ be an open immersion of schemes and $z \in X$. Then $\mathcal{O}_{X,z}$ is reduced if and only if $\mathcal{O}_{Y,e(z)}$ is reduced.*

Proof. An open immersion induces an isomorphism on stalks, and reducedness is invariant under ring isomorphism. $\square$

Lemma 1315 (Transport of irreducibility). *Let $e : X \rightarrow Y$ be an isomorphism of schemes and $t \subseteq Y$ a subset. Then $e^{-1}(t)$ is irreducible if and only if t is irreducible.*

Proof. An isomorphism of schemes is a homeomorphism of underlying spaces; irreducibility is a topological property, preserved and reflected by homeomorphisms (the image of an irreducible set under a continuous map is irreducible, applied to e and e^{-1}). $\square$

Corollary 1316 (Homogeneity of the representing object). *Let d be a Jacobian datum with representing object J , and let $x, y : \text{Spec } k \rightarrow J$ be two k -points. With the group structure of ??, the translation $\tau_{x,y}$ is an automorphism of J over k with $\tau_{x,y} \circ x = y$; on the underlying scheme it commutes with the structure morphism and carries the point of x to the point of y ; and it preserves and reflects membership in the smooth locus of the structure morphism (when the latter is locally of finite presentation), reducedness of stalks, and irreducibility of subsets.*

Proof. Instantiate ??–?? at the group scheme J : the translation exists and carries x to y (??, ??); its underlying isomorphism commutes with the structure morphism (??) and acts on points as prescribed (??); the three transport lemmas (??, applicable by ??; ??; ??) give the invariances. $\square$

12.5 The Abel source: properness and geometric irreducibility

Definition 1317 (Abel source). Let d be a Jacobian datum with representing object J . An *Abel source* for d consists of:

1. a scheme D over k ;
2. a certificate that D is proper over k ;
3. a certificate that D is geometrically irreducible over k ;
4. an *Abel morphism* $a : D \rightarrow J$ over k ;
5. a certificate that a is surjective on underlying topological spaces.

This is a conditional interface: its definition asserts neither the existence of D nor the existence of a . The intended instance takes D to be a proper scheme of effective divisors of large degree on C and a to be the Abel map.

Theorem 1318 (The representing object is universally closed). *Let d be a Jacobian datum admitting an Abel source. Then the structure morphism of the representing object J is universally closed.*

Proof. Let $a : D \rightarrow J$ be the Abel morphism. Since a is a morphism over k , the composite of a with the structure morphism of J is the structure morphism of D , which is universally closed (D is proper). Surjectivity and closedness are stable under base change, so it suffices to see: if $g \circ f$ is universally closed and f is surjective, then g is universally closed. Indeed, after any base change, for a closed subset Z of the source of the (base-changed) g' , $g'(Z) = (g' \circ f')(f'^{-1}(Z))$ by surjectivity of the base-changed f' , and the right-hand side is closed since $g' \circ f' = (g \circ f)'$ is a closed map. $\square$

Theorem 1319 (The representing object is proper). *Let d be a Jacobian datum admitting an Abel source. Then the structure morphism of the representing object J is proper.*

Proof. Properness is the conjunction of separatedness, universal closedness and finite type. Separatedness is ??; universal closedness is ??; the structure morphism is locally of finite type by the datum's certificate (??), and quasi-compact because universally closed morphisms are quasi-compact; locally of finite type and quasi-compact is finite type. $\square$

Theorem 1320 (The representing object is geometrically irreducible). *Let d be a Jacobian datum admitting an Abel source. Then the structure morphism of the representing object J is geometrically irreducible: for every field extension K/k , the base change J_K is irreducible.*

Proof. Fix a field extension K/k and a fibre product $Z = J \times_k K$ with projections to J and $\text{Spec } K$. The base change $D_K = D \times_k K$ is irreducible by the Abel source's certificate. The comparison morphism $D_K \rightarrow Z$ induced by the Abel morphism a is, by pasting of pullback squares, a base change of a along $Z \rightarrow J$ (the square with legs $D_K \rightarrow Z \rightarrow \text{Spec } K$ over $D \rightarrow J \rightarrow \text{Spec } k$ decomposes into two pullbacks); surjectivity is stable under base change, so $D_K \rightarrow Z$ is surjective. A continuous surjective image of an irreducible space is irreducible, so Z is irreducible. $\square$

12.6 Base change of the representability datum

Fix a field extension $k \rightarrow L$, write $\sigma : \operatorname{Spec} L \rightarrow \operatorname{Spec} k$ and $C_L = C \times_{\operatorname{Spec} k} \operatorname{Spec} L$. We transport a Jacobian datum for C to one for C_L and compare the two group structures on the base-changed representing object; this is the datum-level mechanism behind ??.

Definition 1321 (Base change of a Jacobian datum). A Jacobian datum d for C , with representing object J , base-changes to a Jacobian datum d_L for C_L with representing object $J_L = J \times_{\operatorname{Spec} k} \operatorname{Spec} L$. Its universal bijections are obtained from the pullback–base-change adjunction $\operatorname{Hom}_{\operatorname{Over}(\operatorname{Spec} L)}(T, J_L) \cong \operatorname{Hom}_{\operatorname{Over}(\operatorname{Spec} k)}(\sigma_! T, J)$ followed by d ’s representation and the base-field comparison of the degree-zero Picard functor $\operatorname{pic}^0(C_L, T) \cong \operatorname{pic}^0(C, \sigma_! T)$; the locally-of-finite-type and quasi-compactness certificates transport by stability of these properties under base change.

Lemma 1322 (Base change respects the group structure). *Let d be a Jacobian datum for C . The group-object structure that base change induces on J_L — as the image of the group object J under the finite-product-preserving base-change functor — coincides with the group-object structure carried by the base-changed datum d_L (??) through its representation.*

Proof. A group-object structure is determined by its multiplication morphism, so it suffices to compare those. The multiplication induced by a finite-product-preserving functor on the image of a group object is, through the pullback–base-change adjunction, exactly the multiplication that the adjunction representation $\operatorname{Hom}(T, J_L) \cong \operatorname{Hom}(\sigma_! T, J)$ transports from J . The representation carried by d_L factors that same adjunction through the base-field comparison of pic^0 , which is an isomorphism of presheaves of *groups*; being a homomorphism at each test object it preserves the pointwise products, so the two multiplications agree. $\square$

Definition 1323 (Datum-level base-change isomorphism). Let d be a Jacobian datum for C and d_L a Jacobian datum for C_L . There is a canonical isomorphism of group schemes over L

$$J_L \cong d_L.J,$$

where the left-hand side carries the group structure induced by base change. It is the canonical isomorphism of representing objects (??) for the two Jacobian data of C_L — the base-changed datum of ?? and the given d_L — which is a homomorphism of group objects by ?? together with that intertwining property. Specialised to the produced Jacobian data it is the datum-level form of ??.